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SpaceCDF — A Course Textbook in Concurrent Spacecraft Design

Companion textbook for the SpaceCDF 40-hour Concurrent Design Facility programme

SpaceCDF

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Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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How to use this book

This textbook is the SpaceCDF course volume. It accompanies the 40-hour Concurrent Design Facility programme and is organised in four parts:

1. **Per-session course chapters** — one chapter per course session (Sessions 1.1 – 5.4). Each contains learning objectives, course chapters, exercises, expected reading, and supporting figures.
2. **Per-role background briefings** — a single chapter giving every CDF position the decisions, formulas, charts, pitfalls, and references they need from Day 1.
3. **Position appendix** — short-form per-position cards kept for quick reference at a console.
4. **Verification appendix** — a first-principles derivation and plot for every formula students use, with cross-method sanity checks. Suitable as the worked verification trail at PDR.

This textbook is paired with two companion volumes:

- the **SpaceCDF Course Workbook** — worksheets and exercises that students complete alongside the chapters in Part 1, plus the operational drills that exercise Part 2 (CubeSat Operations); and
- the **SpaceCDF Instructor Book** — simulator setup, anomaly scenarios, breakpoints, and instructor protocols, intended for the simulator operator and the instructor coach (not for trainees on console).

> **Expected reading before any session.** *NASA SEH* §1 – §3 — https://www.nasa.gov/reference/systems-engineering-handbook/. NPR 7123.1D §3 — https://nodis3.gsfc.nasa.gov/. *NASA CubeSat 101* — https://www.nasa.gov/wp-content/uploads/2017/03/nasa_csli_cubesat_101_508.pdf.

Acknowledgement — Generative AI (AIG)

This textbook was produced with the assistance of generative AI as part of the SpaceCDF Concurrent Design Facility workflow. The design-loop convergence, agent rationales, embedded figures and document rendering are generated by the SpaceCDF backend (Python · matplotlib · python-docx · WeasyPrint) guided by ECSS, NASA SEH and SMAD references. Editorial framing, worked examples, and pedagogical commentary remain owned by the SpaceCDF teaching team. *Appendix D — How This Textbook Was Generated* describes the pipeline in detail.

Attribution follows the AIG (Assisted by Generative AI) framework — Peters (2023), Logos IA-EN, CC BY-NC-SA 4.0 — https://mpeters.uqo.ca/en/logos-ia-en-peters-2023/

Any course deliverable that incorporates content from this textbook — or from any document exported by SpaceCDF — must carry the AIG badge and a short note describing how generative AI was used. Refusing to use AI where it is permitted is fine; what is not acceptable is using it without saying so.

Agents, Positions, and the SpaceCDF Architecture

Two distinct concepts run through this textbook and they are sometimes conflated in industry literature. Disentangling them is essential before the first design loop.

A **position** is a *human role* assigned to a person sitting at a console in the Concurrent Design Facility — for example *Power Engineer*, *AOCS Lead*, or *Mission Analyst*. SpaceCDF defines fifteen positions, each with scoped editing rights on a subset of the design parameters.

A **design agent** is a *computational solver* that runs in the background and converges a particular subsystem or analysis. The current release of SpaceCDF ships twenty agents grouped in two tiers:

Table 0.1 — SpaceCDF design agents grouped by parametric sizing (Tier 1) and cross-system analysis (Tier 2).

Tier	Purpose	Agents
Tier 1 — Parametric sizing	Compute the design point of one subsystem from inputs supplied by the others	orbit · link · data · power · thermal · AOCS · propulsion · mass · structure
Tier 2 — Cross-system analysis	Score, check or audit the converged design	cost · risk · reliability · debris · radiation · sustainability · volume · systems · conflicts · community · TRL

Each agent publishes three things to the design state when it runs: its **parameters** (numbers with units, margin and confidence), its **rationale** (a paragraph of textbook-grade reasoning), and a set of **assumptions** that downstream agents and reviewers can interrogate. Several agents also emit *structured intermediates* — the orbit agent exposes a ΔV breakdown, the link agent exposes a full waterfall, the thermal agent exposes hot- and cold-case node temperatures, the reliability agent exposes a FMECA, and so on. Most exports surface these intermediates as tables and figures alongside the bare numbers.

When a human position assignment changes a value, that parameter becomes *sticky* in the design state and the agents do not overwrite it during subsequent design loops. This is the formal handover between the human operator and the computational solver, and it is the reason a CDF session converges so quickly: the agents do the iterative arithmetic while the positions do the engineering judgement.

Design Reviews and the SpaceCDF Exports

A review gate is not merely a meeting. It is a *document set* whose existence and contents are agreed in advance, whose generation is traceable to a converged design state, and whose acceptance moves the project from one ECSS phase to the next. SpaceCDF treats each review gate as the natural anchor of an export bundle.

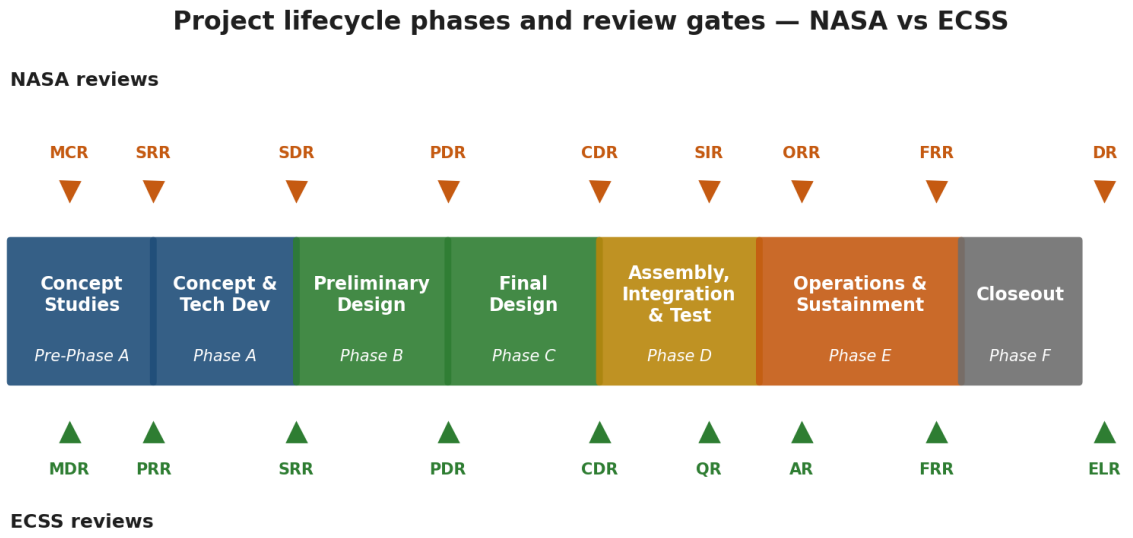
Table 0.2 — Review gates mapped to ECSS phases and the SpaceCDF export bundles produced at each.

Review	ECSS phase	Exported bundle	Principal contents
MCR	Pre-Phase A	Mission Concept Report	Stakeholders, MoP/TPM, alternatives, ConOps draft
SRR	Phase A	SRR Design-Review Pack (Word + Excel + Markdown)	Mass / power / data / cost budgets with margin, orbit and architecture figures, compliance matrix, equipment list, TRL roll-up, risk matrix, cost P-curve, WBS bar
SDR	Phase A/B	System Design Review Pack	SRR contents plus block diagrams, derived requirements, preliminary trade studies
PDR	Phase B	PDR Design-Review Pack	SRR + SDR contents plus full trade-study record, verification matrix, interface control documents
CDR	Phase C	CDR Design-Review Pack	PDR contents plus test plan, AIT procedures, qualification status

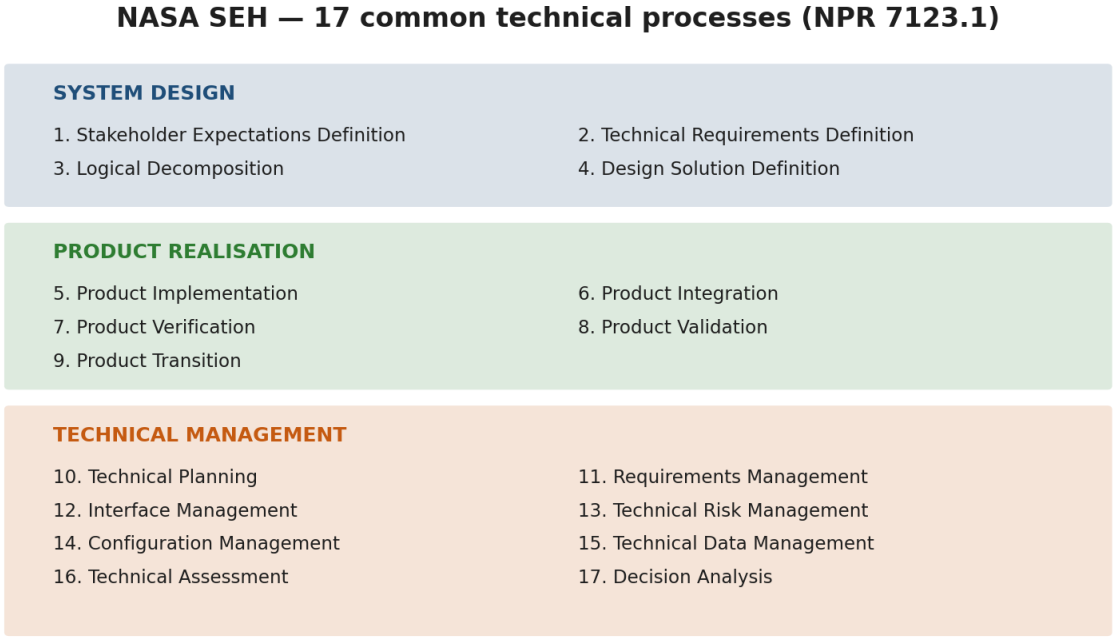
In addition to the design-review packs, SpaceCDF generates a family of *data item descriptions* (DIDs) in editable Word format: the Mission Requirements Document (MRD), Concept of Operations (ConOps), Verification Plan (VP), Systems Engineering Management Plan (SEMP), Risk Management Plan (RMP), Interface Requirements Document (IRD), Test Plan, and the Bill of Materials. Specialist reports cover the thermal analysis, the end-of-life deorbit plan, the Remote Sensing Space Systems Act (RSSSA) licence application, and the verification test programme.

Part 1 — CubeSat Design

Chapter 1.1 — Introduction to Space Mission Design



Lifecycle phases and review gates.



NASA SEH 17 processes pyramid.

> **Expected reading before this session.** NASA SEH §1 – §3 (≈ 60 min) — https://www.nasa.gov/reference/systems-engineering-handbook/. NPR 7123.1D §3 (≈ 30 min).

Prerequisites: None (engineering background assumed) **References:** - NASA, Systems Engineering Handbook Rev 2 (SP-2016-6105), 2016 - NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020 - ECSS, ECSS-M-ST-10C Rev.1 -- Space Project Management, 2009 - ESA, "20 Years of Concurrent Design at ESA", 2018

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the purpose and structure of a Concurrent Design Facility (CDF) and contrast it with sequential design 2. Explain the System-V model and its two sides (decomposition + integration) 3. List the 17 common technical processes defined in NPR 7123.1D and explain their recursive application 4. Map NASA lifecycle phases (Pre-A through F) to ECSS phases and identify key review gates 5. Identify the role of each CDF engineering position and their parameter ownership

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1. What is a Concurrent Design Facility?

1.1 The Problem with Sequential Design

Consider the question: How is spacecraft design traditionally done?

Traditional spacecraft design follows a **sequential (waterfall) approach**: one discipline completes its work, documents it, and passes the design to the next discipline. This method dominated the space industry from the 1960s through the 1990s. Its limitations are well documented:

- **Interface mismatches** discovered late in integration, requiring costly redesign
- **Budget overruns** in mass, power, and cost not caught until formal reviews
- **Long iteration cycles** -- months between design reviews with limited inter-discipline feedback
- **Knowledge silos** -- critical design rationale locked in individual engineers' notes
- **Rework costs** -- the cost of correcting an error increases by roughly an order of magnitude per lifecycle phase (the "1-10-100 rule")

The aerospace industry's response was **concurrent engineering (CE)**, also called Integrated Product Development (IPD) or Integrated Concurrent Engineering (ICE). The core idea: bring all disciplines together to work simultaneously on a shared parametric model, resolving conflicts in real-time rather than discovering them in review.

> **Industry Practice:** Boeing's Phantom Works implemented concurrent engineering for the X-32 Joint Strike Fighter demonstrator in the late 1990s, reducing design iteration time from weeks to hours. JPL's Team X has been performing rapid mission assessments since 1995, completing Pre-Phase A studies in as little as one week. ESA's CDF, established in November 1998 at ESTEC (Noordwijk, Netherlands), became the most widely replicated model for space-specific concurrent design.

1.2 ESA's CDF -- The Reference Model

ESA's Concurrent Design Facility pioneered the approach for space missions and has been widely replicated at space agencies worldwide (DLR, CNES, JAXA, CSA, and over 60 universities).

Table 1.1.1 — Operating parameters of the ESA Concurrent Design Facility, from team size to track record.

Parameter	Value
Established	November 1998, ESTEC, Noordwijk, NL
Team size	15--25 domain specialists per study

Parameter	Value
Session format	3--4 hour focused design sessions, typically 2 per week
Study duration	3--6 weeks for a complete Phase 0/A mission assessment
Tooling	Originally Excel-based IDM; now Open Concurrent Design Tool (OCDT)
Output	Complete mission feasibility assessment: mass, power, cost, risk, schedule
Track record	200+ studies completed by 20th anniversary (2018)
Cost savings	Studies completed in ~1/5 the time and ~1/3 the cost of traditional approach

[Source: ESA CDF official documentation; Bandecchi et al., "The ESA/ESTEC Concurrent Design Facility", ESA Bulletin No. 107, 2001]

1.3 Key Principles of Concurrent Design

Shared parametric model. All disciplines read from and write to a common data model. When the power engineer changes the solar array area, the structures engineer immediately sees the mass impact, and the thermal engineer sees the changed radiator view factor. This eliminates the "frozen interface document" problem.

Scoped parameter ownership. Each engineering position "owns" a defined set of parameters. Only the power engineer can modify EPS parameters; only the AOCS engineer can modify attitude control parameters. This prevents conflicting edits while maintaining concurrent access.

Real-time conflict detection. When parameter changes create conflicts (e.g., mass budget exceeded, link margin negative), the system flags them immediately. The systems engineer arbitrates.

Design convergence through iteration. A CDF study typically converges through 3--5 major iterations, with each session refining the design toward closure. The parametric model propagates changes automatically, so the team sees the system-level impact of every decision.

1.4 SpaceCDF -- A Web-Based CDF

SpaceCDF implements the CDF concept as a web-based tool accessible from any browser:

Table 1.1.2 — SpaceCDF web-based features that implement core Concurrent Design Facility capabilities.

Feature	Implementation
Shared model	Real-time synchronisation via WebSocket; all participants see live updates
Parameter ownership	15 engineering positions with scoped editing rights
Design convergence	20 background design agents (9 Tier 1 sizing, 11 Tier 2 analysis) that propagate parametric relationships
Conflict detection	Automated constraint engine with 187 inter-parameter connections
Document generation	ECSS-compliant exports (Word, PDF) with full traceability
Review gates	Automated gate criteria evaluation for MCR, SRR, SDR, PDR, CDR

Discussion point. *What advantages does concurrent design offer over sequential? What risks does it introduce (e.g., groupthink, premature convergence)?*

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2. The System-V Model

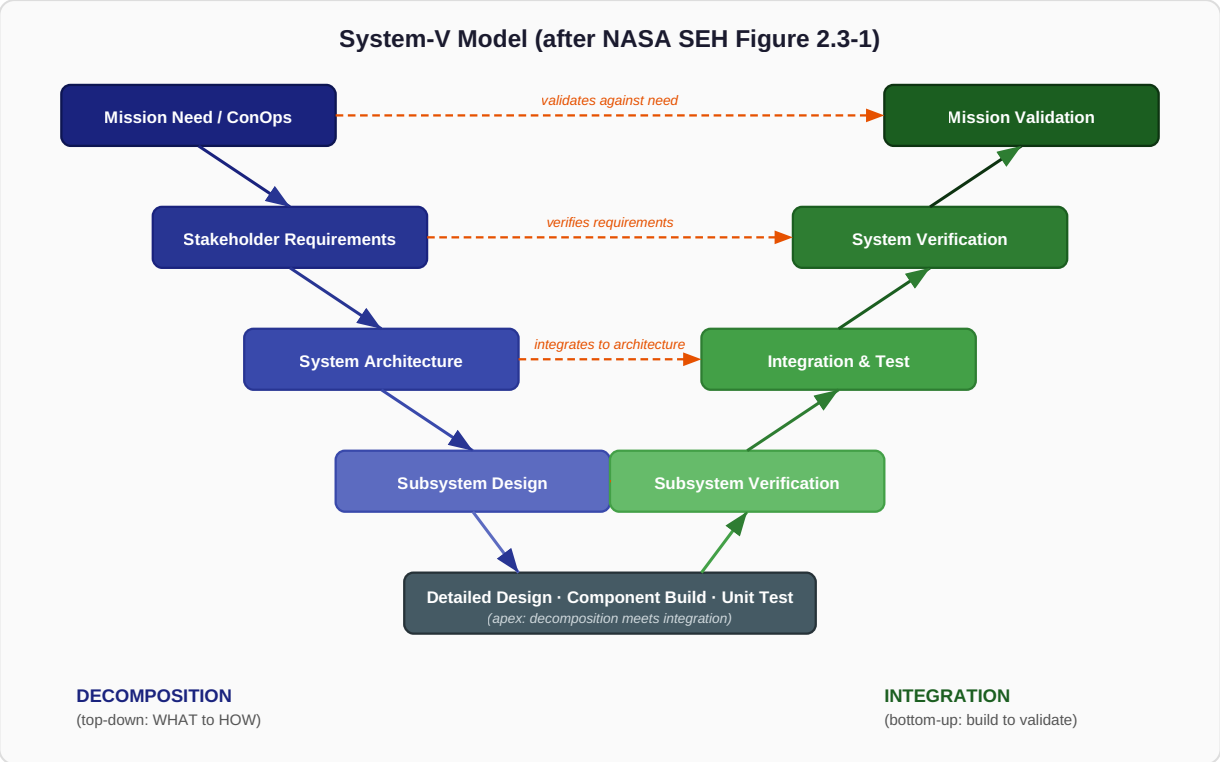
2.1 Origin and Purpose

The "Vee" (V) model is the foundational framework for systems engineering. It appears in NASA SEH Section 2.3 (Figure 2.3-1), in ECSS-E-ST-10C, and in ISO/IEC 15288. It is not a project schedule -- it is a **logical model** showing the relationship between decomposition (breaking the problem down) and integration (building and verifying the solution).

The V-model answers two fundamental questions: 1. **Left side (top-down):** How do we decompose the mission need into buildable components? 2. **Right side (bottom-up):** How do we verify that what we built satisfies the original need?

The key insight is **horizontal traceability**: every element on the left side has a corresponding verification or validation activity on the right side, connected at the same level of abstraction.

2.2 V-Model Diagram



2.3 Left Side: Top-Down Decomposition

The left side of the V answers "how do we break the problem down into solvable pieces?"

Table 1.1.3 — Decomposition levels of the System-V model from mission need to component selection.

Level	Activity	Key Question	Output
1. Mission Need			

Level	Activity	Key Question	Output
	Define the problem and concept of operations	What problem are we solving? For whom?	ConOps, problem statement, MoEs
2. Stakeholder Requirements	Capture needs of all stakeholders	What must the system achieve? (WHAT, not HOW)	Stakeholder requirements baseline
3. System Architecture	Define segments, elements, interfaces	How is the system structured?	System architecture, ICDs
4. Subsystem Design	Design each subsystem to meet allocated requirements	How does each subsystem work?	Subsystem specifications
5. Component Selection	Choose or design lowest-level components	Which hardware/software fulfils each need?	Component specifications, BoM

2.4 Right Side: Bottom-Up Integration

The right side answers "how do we verify that what we built satisfies the original need?"

Table 1.1.4 — Integration levels of the System-V model with corresponding verification methods.

Level	Activity	Key Question	Verification Method
5. Component Verification	Test each component against its spec	Does each part work?	Inspection, test
4. Subsystem Verification	Integrate and test each subsystem	Do subsystems meet allocated requirements?	Test, analysis
3. Integration & Test	Assemble the system and test interfaces	Do the subsystems work together?	Test, demonstration
2. System Verification	Verify system against requirements	Does the system meet all requirements?	Test, analysis, inspection, demonstration
1. Mission Validation	Validate against the original need	Does it solve the original problem?	Operations, user acceptance

2.5 Horizontal Traceability

The orange dashed arrows in the diagram represent **horizontal traceability** -- the principle that every element on the left must have a corresponding verification/validation activity on the right:

- Each **requirement** (left) has a **verification method** (right): analysis, test, inspection or review of design, plus the demonstration that NASA practice adds (ATIR(D))
- Each **design decision** (left) has a **test or analysis** (right) to confirm it works
- This traceability is captured in the **Requirements Verification Matrix (RVM)**

> **Key Equation:** The traceability completeness metric is:
>
> $$\mathrm{T_c} = \frac{\mathrm{N_{verified}}}{\mathrm{N_{total_requirements}}} \times 100\%$$
>
> A target of $\mathrm{T_c} = 100\%$ is required at CDR. At PDR, $\mathrm{T_c} \geq 90\%$ is typical.

2.6 Iteration and the SE Engine

The V is not a single pass. Real design iterates -- requirements change as design constraints are discovered, and design parameters feed back to refine requirements. This iterative loop is captured in the concept of the **"SE Engine"** (NASA SEH Section 2.1): the 17 common technical processes applied recursively at every level of the system hierarchy.

[Source: NASA SEH Section 2.3, Figure 2.3-1; INCOSE Systems Engineering Handbook, 4th Edition, Section 3.2]

Exercise. *Sketch the V-model from memory and label each level. Add the horizontal traceability arrows and name one verification method for each level.*

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3. The 17 Common Technical Processes

3.1 Overview

NASA's NPR 7123.1D defines 17 processes that apply recursively at every level of the system hierarchy. They are grouped into three categories, collectively called the **"SE Engine"**. These processes are not sequential -- they execute concurrently and iteratively, which is precisely what a CDF facilitates.

[Source: NPR 7123.1D Chapter 3; NASA SEH Section 2.1, Figure 2.1-1]

3.2 System Design Processes (1--4)

These processes decompose the problem into a solution. They correspond to the left side of the V-model.

Table 1.1.5 — NPR 7123.1D System Design processes for decomposing problem into solution.

#	Process	Key Activity	Key Output	NASA SEH Reference
1	**Stakeholder Expectations Definition**	Elicit needs, define ConOps, establish MoEs	Stakeholder requirements baseline	Section 4.1
2	**Technical Requirements Definition**	Write "shall" statements, define MoPs and TPMs	Technical requirements baseline	Section 4.2
3	**Logical Decomposition**	Functional analysis, behavioural modelling, derived requirements	Functional architecture	Section 4.3
4	**Design Solution Definition**	Trade studies, select among alternatives, produce baseline design	Design solution baseline	Section 4.4

3.3 Product Realisation Processes (5--9)

These processes build, verify, and deliver the solution. They correspond to the right side of the V-model.

Table 1.1.6 — NPR 7123.1D Product Realisation processes for building, verifying, and delivering the solution.

#	Process	Key Activity	Key Output	NASA SEH Reference
5				Section 5.1

#	Process	Key Activity	Key Output	NASA SEH Reference
	Product Implementation	Make, buy, code, or reuse lowest-level products	Hardware/software products	
6	**Product Integration**	Assemble per integration plan, verify interfaces	Integrated system	Section 5.2
7	**Product Verification**	Confirm product meets technical requirements (ATIR(D))	Verification evidence	Section 5.3
8	**Product Validation**	Confirm product meets stakeholder expectations	Validation evidence	Section 5.4
9	**Product Transition**	Deliver, deploy, hand over to operations	Operational system	Section 5.5

3.4 Technical Management Processes (10--17)

These processes manage the engineering work and apply throughout the lifecycle.

Table 1.1.7 — NPR 7123.1D Technical Management processes governing engineering work across the lifecycle.

#	Process	Key Activity	Governing Document
10	**Technical Planning**	SEMP, subsidiary plans, WBS	NPR 7123.1D Section 3.10
11	**Requirements Management**	Baselining, traceability, change control	NPR 7123.1D Section 3.11
12	**Interface Management**	ICDs, IRDs, internal + external interfaces	NPR 7123.1D Section 3.12
13	**Technical Risk Management**	Identification, assessment, mitigation per NPR 8000.4	NPR 8000.4B
14	**Configuration Management**	Baselines, CM plan, CCBs	NPR 7120.5F Chapter 4
15	**Technical Data Management**	Data rights, retention, dissemination	NPR 7123.1D Section 3.15
16	**Technical Assessment**	TPMs, reviews, EVM, health checks	NPR 7123.1D Section 3.16
17	**Decision Analysis**	Structured alternative selection (trade studies)	NASA SEH Section 6.8

3.5 Recursion Across the System Hierarchy

A critical property of the 17 processes is that they apply **recursively** at every level of the system hierarchy:

```
Mission Level:     17 processes applied (mission-level requirements, ConOps, validation)
  System Level:    17 processes applied (system requirements, architecture, verification)
    Subsystem Level: 17 processes applied (subsystem requirements, design, testing)
      Component Level: 17 processes applied (component specs, procurement, acceptance)
```

At each level, the same processes execute but with different scope, detail, and duration. In a CDF environment, multiple levels are often worked concurrently -- the systems engineer manages the mission and system levels while subsystem engineers work their respective domains.

> **Industry Practice:** On the James Webb Space Telescope (JWST), requirements management (Process 11) was applied at five levels of the system hierarchy simultaneously: Observatory, Optical Telescope Element, Integrated Science Instrument Module, individual instruments (NIRSpec, MIRI, NIRCam, FGS), and components (detectors, mechanisms). Each level had its own requirements baseline, traceability matrix, and verification plan. The total requirements count exceeded 10,000.

Exercise. *Map each of the 17 processes to a feature in SpaceCDF. Which processes does the tool support directly? Which require human judgment? Complete Part A of Worksheet 1.1.*

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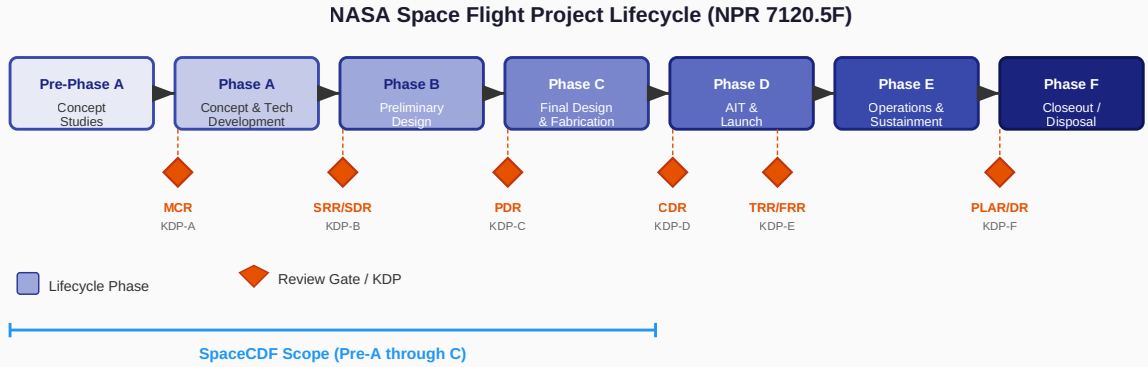
4. Lifecycle Phases and Review Gates

4.1 NASA Lifecycle Phases

NASA defines seven lifecycle phases for space flight projects, governed by NPR 7120.5F. Each phase has specific objectives, activities, and exit criteria evaluated at formal review gates called Key Decision Points (KDPs).

[Source: NPR 7120.5F Chapter 2; NASA SEH Chapter 3]

4.2 Lifecycle Phase Diagram



4.3 Phase Details

Table 1.1.8 — NASA lifecycle phases Pre-A through F with primary activities, exit reviews, and outputs.

Phase	Name	Primary Activity	Exit Review	Key Outputs
Pre-A	Concept Studies	Identify need, explore feasibility, develop ConOps	MCR	Mission concept, preliminary ConOps, feasibility assessment
A	Concept & Technology Development	Develop requirements, mature technology to TRL 4+	SRR/SDR	Requirements baseline, technology development plan
B			PDR	

Phase	Name	Primary Activity	Exit Review	Key Outputs
	Preliminary Design & Tech Completion	Preliminary design, close preliminary budgets, TRL 6+		Preliminary design, budget allocations, test plans
C	Final Design & Fabrication	Detailed design, build hardware, TRL 8	CDR	Detailed drawings, as-built documentation, flight hardware
D	Assembly, Integration & Test, Launch	System AIT, environmental testing, launch	TRR, ORR, FRR	Tested flight system, launch readiness
E	Operations & Sustainment	Operate the mission, produce data products	PLAR	Science data, operational lessons
F	Closeout	Decommission, passivate, dispose, lessons learned	DR	Disposal confirmation, final report

Key Decision Points (KDPs): These are go/no-go authority decisions between phases, lettered A through F. They are distinct from technical reviews -- KDPs are management decisions informed by technical review outcomes. For projects exceeding $250M (USD), KDP-C requires a Joint Confidence Level (JCL) analysis per NPR 7120.5F.

4.4 ECSS Lifecycle Phases

ECSS defines a similar but not identical lifecycle. The phase letters conveniently align, but the entry/exit criteria and review content differ.

[Source: ECSS-M-ST-10C Rev.1, Section 5.3]

Table 1.1.9 — ECSS lifecycle phases aligned with NASA equivalents and notable scope differences.

ECSS Phase	Name	NASA Equivalent	Key Difference
0	Mission Analysis / Needs Identification	Pre-A	ECSS Phase 0 includes formal mission analysis
A	Feasibility	A	ECSS-A focuses on demonstrating feasibility
B	Preliminary Definition (B1/B2)	B	ECSS splits into B1 (system) and B2 (detailed)
C	Detailed Definition	C	Similar scope
D	Qualification & Production	D	ECSS-D emphasises qualification models
E	Utilisation	E	Similar scope
F	Disposal	F	ECSS-F explicitly addresses passivation

4.5 Review Gates -- What Each Checks

Table 1.1.10 — Purpose, key question, and required evidence for each major design review gate.

Review	Full Name	Key Question	Required Evidence
MCR	Mission Concept Review	Is the mission need justified? Is space the right answer?	Problem statement, stakeholder analysis, alternatives analysis, ConOps draft
SRR	System Requirements Review	Are requirements complete, consistent, traceable, and verifiable?	Requirements baseline, traceability matrix, risk register, preliminary ConOps
SDR	System Design Review	Does the system architecture satisfy requirements?	System architecture, functional decomposition, interface definitions
PDR	Preliminary Design Review	Does the preliminary design meet all requirements?	Design description, budget status (mass/power/cost), risk mitigation plans
CDR	Critical Design Review	Is the design complete and ready for fabrication?	Detailed drawings, analysis results, test plans, all budgets closed with positive margins
TRR	Test Readiness Review	Is the system ready for formal testing?	Test procedures, facility readiness, acceptance criteria
FRR	Flight Readiness Review	Is everything ready for launch?	All tests passed, waivers documented, launch procedures verified

> **Industry Practice:** RADARSAT-2 (MDA/CSA, launched 2007) went through all NASA-equivalent review gates over a 6-year development. The CDR alone involved over 400 review items and 50 reviewers across 3 weeks. The mission exceeded its 7-year design life, operating for over 15 years -- a testament to thorough verification at each gate.

Exercise. *In SpaceCDF, go to the Gate Review tab and examine the MCR exit criteria. Which criteria are auto-evaluated by the tool? Which require manual review by an engineer?*

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5. CDF Engineering Positions

5.1 Position Overview

In a CDF study, each engineering position owns a set of parameters and is responsible for their domain's design decisions. SpaceCDF supports 15 positions:

Table 1.1.11 — The fifteen SpaceCDF engineering positions with their owned parameters and interfaces.

Position	Responsibility	Key Owned Parameters	Primary Interfaces
Systems Engineer	Overall architecture, budgets, margins, conflict resolution	Mass margin, power margin, system-level budgets	All positions
Mission Analyst	Orbit design, coverage analysis, ground station access	Altitude, inclination, RAAN, eclipse fraction, contact time	Comms, Payload, Ground
Payload Lead	Instrument performance, data generation rates	GSD, spectral bands, data rate, FOV, SNR	Mission Analyst, Power, AOCS, Comms
Power Engineer			

Position	Responsibility	Key Owned Parameters	Primary Interfaces
	Solar arrays, batteries, EPS architecture, duty cycling	SA area, battery capacity, bus voltage, DoD	All (power is a universal constraint)
AOCS Engineer	Attitude sensors, actuators, pointing modes	Pointing accuracy, slew rate, wheel momentum, sensor suite	Payload, Structures, Propulsion
Thermal Engineer	Temperature control, radiators, heaters, coatings	Max/min temps, radiator area, heater power, thermal margin	Structures, Power, Payload
Comms Engineer	Link budget, transponder, antenna, spectrum licensing	Link margin, data rate, frequency band, EIRP, G/T	Mission Analyst, Payload, Ground
Propulsion Engineer	Delta-V budget, thruster selection, propellant mass	Isp, propellant mass, total impulse, thrust vector	Mission Analyst, AOCS, Structures
Structures Engineer	Primary structure, mechanisms, launch loads	Structure mass, natural frequency, margin of safety	All (mass is a universal constraint)
Cost Engineer	WBS, cost estimating relationships, schedule	Total cost, per-subsystem cost, launch cost, risk-adjusted cost	Systems Engineer, all subsystems
Compliance Engineer	Standards, frequency licensing, export control	Standard applicability, ITAR/EAR status, filing status	All
User Representative	End-user needs, data product requirements	Data format, latency, accessibility, coverage	Payload, Comms, Ground
Mission Operations	Ground ops concept, staffing, automation	Contact schedule, automation level, anomaly procedures	Comms, Ground, Systems
Ground Segment	Ground stations, data processing pipeline	Station network, processing latency, archive capacity	Comms, Mission Ops, User Rep
Software Engineer	Flight software, FDIR, TC/TM interfaces	FSW architecture, command dictionary, FDIR rules	OBC, AOCS, Comms, Payload

5.2 Interface Conflicts

The most productive CDF sessions are those where **interface conflicts** are identified and resolved. Common conflict patterns:

Table 1.1.12 — Common CDF interface conflicts with positions involved and resolution strategies.

Conflict	Positions Involved	Resolution Approach
Mass budget exceeded	Systems, Structures, all subsystems	Re-allocate margins, descope payload, select lighter components
Power budget exceeded	Systems, Power, Payload, Comms	Adjust duty cycles, increase SA area, reduce payload power
Pointing requirement vs actuator capability	Payload, AOCS, Structures	Relax pointing requirement, add fine steering mirror, improve isolation
Downlink capacity vs data generation	Comms, Payload, Ground	Add ground stations, increase TX power, reduce data rate, add compression

Conflict	Positions Involved	Resolution Approach
Volume constraint vs thermal rejection	Structures, Thermal, Payload	Redesign radiator location, add deployable radiator, reduce heat dissipation

Discussion point. *Which positions would interact most frequently in your mission? Where do you expect the most difficult trade-offs?*

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6. Session Summary

Table 1.1.13 — Session summary of key takeaways for CDF, V-model, processes, phases, gates, and positions.

Topic	Key Takeaway
CDF	Concurrent multi-discipline design resolves conflicts in real-time; SpaceCDF implements this as a web-based tool
System-V	Left side decomposes (need to components); right side integrates (build to validate); horizontal traceability links them
17 Processes	Recursive at every system level; grouped as Design (1--4), Realisation (5--9), Management (10--17)
Lifecycle Phases	Pre-A through F with KDP gates; ECSS phases approximately align; SpaceCDF covers Pre-A through C
Review Gates	Each gate answers a specific question with required evidence; auto-evaluated where possible in SpaceCDF
Positions	Each owns a parameter domain; conflicts arise at interfaces; the systems engineer arbitrates

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1U Worked Example: UniSat-1

Throughout this course, we use a second running example alongside the 3U EO CubeSat: **UniSat-1**, a 1U CubeSat technology demonstrator designed by a university team. This is the simplest realistic spacecraft design.

Mission: Demonstrate a novel MEMS-based magnetometer for space weather monitoring from LEO.

Why 1U? The 1U form factor (100 x 100 x 113.5 mm, up to 2.00 kg) is the smallest standard CubeSat and the entry point for many university and educational missions. It forces extreme design discipline -- every gram, every milliwatt, and every cubic centimetre matters.

Table 1.1.14 — UniSat-1 baseline parameters for a 1U CubeSat MEMS magnetometer demonstrator.

Parameter	Value
Form factor	1U (100 x 100 x 113.5 mm)
Mass limit	2.00 kg (CDS Rev 14)
Target mass	1.0 kg
Orbit	400 km circular, 51.6 deg (ISS rideshare)

Parameter	Value
Design lifetime	6 months
Payload	MEMS magnetometer (50 g, 0.2 W, < 1 kbps)
Comms	UHF 437 MHz, 9600 bps
Power	~2 W orbit average (body-mounted solar cells)
AOCS	Passive magnetic (permanent magnet + hysteresis rods)
Propulsion	None (natural deorbit in ~1 year)
Estimated cost	50--200 kEUR
Development time	6--12 months

UniSat-1 illustrates that a meaningful space mission can be accomplished with just five subsystems (EPS, OBC, Comms, Structure, Payload), no active attitude control, no propulsion, and no thermal hardware beyond surface coatings. As the course progresses, each session will show how the same design processes apply to UniSat-1, but with radically simpler solutions at every step.

Discussion point. *How does the CDF process differ when the team has only 5 subsystems instead of 8--9? Which engineering positions are still needed, and which can be combined?*

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7. Tool Exercise

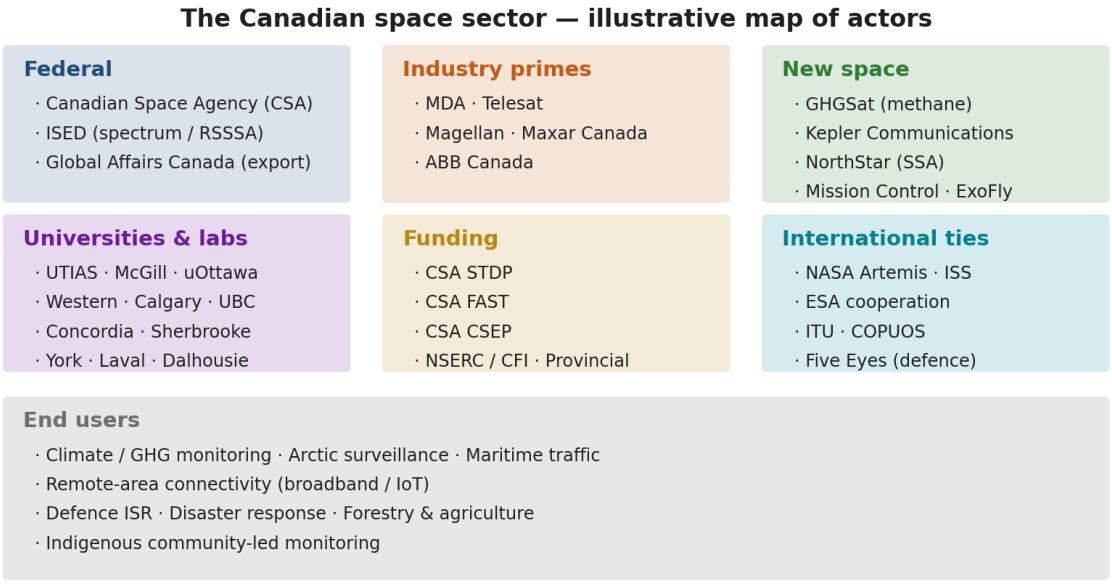
1. Open SpaceCDF and navigate through the workflow steps (Step 1 Mission Need · Step 2 Concept · Step 3 Requirements · Step 4 Design)
2. On the Design Dashboard, identify which KPI cards correspond to which engineering positions
3. Go to the Positions tab and review the key questions for your assigned position
4. Go to the Gate Review tab and examine the MCR exit criteria

Complete Worksheet 1.1: Map the 17 processes to SpaceCDF features and identify lifecycle phases for given activities.

References

1. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
2. NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
3. NASA, NPR 7120.5F -- NASA Space Flight Program and Project Management Requirements, 2021
4. ECSS, ECSS-M-ST-10C Rev.1 -- Space Project Management, 2009
5. INCOSE, Systems Engineering Handbook, 4th Edition, 2015
6. Bandecchi, M. et al., "The ESA/ESTEC Concurrent Design Facility", ESA Bulletin No. 107, 2001
7. Wertz, J.R. et al., Space Mission Engineering: The New SMAD, Microcosm Press, 2011

Chapter 1.2 — The Canadian Space Ecosystem & Regulatory Environment



Representative stakeholder map for Canadian missions.

> **Expected reading before this session.** NASA SEH §4.1 (≈ 30 min).

Prerequisites: Chapter 1.1 **References:** - Government of Canada, Canadian Space Agency Act (S.C. 1990, c. 13) - Government of Canada, Remote Sensing Space Systems Act (RSSSA, S.C. 2005, c. 45) - Government of Canada, Radiocommunication Act (R.S.C., 1985, c. R-2) - ISED Canada, Spectrum Management and Telecommunications - Government of Canada, Export and Import Permits Act (EIPA) - NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 - ITU, Radio Regulations, Edition 2020

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the Canadian Space Agency's mandate and organisational structure 2. Map the Canadian space industry ecosystem -- prime contractors, SMEs, universities, and funding programs 3. Identify the regulatory bodies that govern space activities in Canada (CSA, ISED, GAC, DND) 4. Explain the RSSSA licensing process for remote sensing satellites 5. Describe the spectrum licensing process through ISED and ITU coordination 6. Identify Canadian export control obligations under EIPA, ITAR, and EAR

Part 1: The Canadian Space Ecosystem

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1. The Canadian Space Agency

1.1 Mandate and History

The Canadian Space Agency (CSA) was established by the *Canadian Space Agency Act* (S.C. 1990, c. 13) with the mandate to "promote the peaceful use and development of space, to advance the knowledge of space through science and to ensure that space science and technology provide social and economic benefits for Canadians."

Table 1.2.1 — Major Canadian space milestones from Alouette 1 to the 2026 pause of the Lunar Gateway and the repurposing of Canadarm3.

Milestone	Year	Significance
Alouette 1	1962	Canada becomes the 3rd nation to build and design a satellite
Hermes (CTS)	1976	World's first high-power direct broadcast satellite (CSA/NASA)
Canadarm	1981	Robotic arm for Space Shuttle; established Canada's space robotics dominance
CSA established	1990	Canadian Space Agency Act creates the agency
RADARSAT-1	1995	First Canadian Earth observation satellite; C-band SAR
Canadarm2	2001	ISS robotic arm; continues robotics legacy
RADARSAT-2	2007	Commercial SAR satellite (MDA); RSSSA applies
RADARSAT Constellation Mission (RCM)	2019	Three C-band SAR satellites; maritime surveillance, disaster management
Canadarm3	2019–	Begun for the Lunar Gateway; as of late 2026 the CSA is repurposing Canadarm3 technologies for lunar-surface operations under Artemis (announced August 2026)
Lunar Gateway participation	2019–2026	Canada committed as partner; in March 2026 NASA paused Gateway "in its current form" (widely reported as a cancellation) in favour of a lunar surface base

[Source: CSA, "Canada's Space Milestones", www.asc-csa.gc.ca]

1.2 CSA Organisational Focus Areas

The CSA is headquartered in Saint-Hubert, Quebec, and organises its activities around three strategic priorities:

Table 1.2.2 — CSA's three strategic priorities with associated activities and flagship programs.

Priority	Activities	Key Programs
Space Exploration	ISS operations, Artemis lunar exploration (Canadarm3 technologies repurposed for lunar-surface operations), astronaut programme	Lunar Exploration Accelerator Program (LEAP)
Space Utilisation	Earth observation, satellite communications, GNSS augmentation	RADARSAT, SmartEarth, eHealth

Priority	Activities	Key Programs
Space Science & Technology	Astronomy, planetary science, technology development	CASTOR, JWST/NIRISS, space health

1.3 Canadian Space Strategy (2019)

The *Exploration, Imagination, Innovation: A New Space Strategy for Canada* (2019) committed $2.05B over 24 years and established five priorities: 1. Ensure astronaut health and well-being in deep space 2. Develop and use advanced AI and robotics in space 3. Connect rural and remote Canadians through next-generation satellite communications 4. Use space data and technology to manage natural resources sustainably 5. Position the Canadian space sector for the future

> **Industry Practice:** The $150M Lunar Exploration Accelerator Program (LEAP) specifically targets Canadian SMEs and universities to develop technologies for lunar surface science, health, AI, and robotics. This "pull-through" model -- where a strategic government investment creates opportunities for industry -- is a hallmark of the Canadian space ecosystem.

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2. Canadian Space Industry Map

2.1 Industry Structure

The Canadian space industry generated approximately $2.7B in revenue (2022) and employs ~12,000 people. It is structured in tiers:

Table 1.2.3 — Four-tier structure of the Canadian space industry with key players and revenue share.

Tier	Role	Key Players	Revenue Share
Tier 1: Primes	System integration, large platforms, constellation operations	MDA Space, Telesat, SFL (U of T)	~60%
Tier 2: Subsystem/ Component	Subsystem design, specialised components, ground systems	Honeywell Canada, ABB, Magellan Aerospace, NGC Aerospace	~25%
Tier 3: SMEs & Startups	Niche technologies, applications, data analytics	GHGSat, NorthStar, Kepler Communications, Wyvern	~10%
Tier 4: Academia	R&D, training, technology incubation	U of T (SFL), York U, UBC, U of Calgary, Western	~5%

2.2 Key Canadian Space Capabilities

Canada has world-leading capabilities in several domains:

Table 1.2.4 — Canadian world-leading space capabilities, principal companies, and notable missions.

Capability	Key Player(s)	Notable Missions	Global Standing
Space robotics	MDA Space	Canadarm, Canadarm2, Dextre, Canadarm3	World leader

Capability	Key Player(s)	Notable Missions	Global Standing
SAR Earth observation	MDA Space, CSA	RADARSAT-1/2/RCM	Top 3 globally
Satellite communications	Telesat	Anik fleet, Telesat Lightspeed (LEO constellation)	Pioneer in GEO DBS
Small satellite systems	SFL (U of T)	MOST, NEOSSat, GHGSat-D, BRITE	Top university lab globally
GHG monitoring	GHGSat	GHGSat constellation (methane)	World's only commercial GHG monitoring constellation
Optical instruments	ABB, Honeywell	JWST/FGS-NIRISS, OSIRIS-REx laser altimeter	Tier 1 instrument builder
Star trackers/AOCS	NGC Aerospace/ Honeywell	Numerous missions	Widely used sensors

2.3 Funding Programs

Canadian space activities are funded through several mechanisms:

Table 1.2.5 — Canadian federal funding programs available for space activities and their typical values.

Program	Administered By	Scope	Typical Value
Space Technology Development Program (STDP)	CSA	TRL advancement for flight heritage	$100K--$3M per project
Class Grants and Contributions	CSA	Space science, awareness, STEM	Variable
LEAP	CSA	Lunar technology development	Up to $3M per project
IRAP	NRC	SME innovation support	$50K--$1M
SIF	ISED	Strategic innovation projects	$10M--$100M+
DRDC contracts	DND	Dual-use technology	Classified--$10M
NSERC / CFI	Federal	Academic research infrastructure	$50K--$5M

> **Industry Practice:** GHGSat (Montreal) progressed from a CSA-funded technology demonstrator (GHGSat-D "Claire", launched 2016) to a commercial constellation of 12+ satellites monitoring methane emissions from individual industrial facilities. This trajectory -- government seed funding enabling commercial scale-up -- is the canonical Canadian space industry growth model.

Discussion point. *For your mission concept, which industry tier would you engage? What CSA funding programs might apply? What capabilities would need to be sourced internationally?*

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3. Comparison with Other National Models

Understanding Canada's space ecosystem requires comparing it with partner agencies:

Table 1.2.6 — Comparison of CSA, NASA, ESA, and JAXA on budget, staff, launch capability, and industry model.

Parameter	Canada (CSA)	USA (NASA)	Europe (ESA)	Japan (JAXA)
Annual budget (approx.)	~$300M CAD	~$25B USD	~$7.5B EUR	~$1.5B USD
Staff	~670	~18,000	~2,300	~1,500
Launch capability	None (uses partners)	SLS, commercial (SpaceX, ULA)	Ariane 6, Vega-C	H3, Epsilon
Key strength	Robotics, SAR, instruments	Full spectrum	Science, EO, navigation	Rockets, science, ISS (Kibo)
Industry model	Government seed + commercial scale	Government prime + commercial	Juste retour (geographic return)	Government-led + commercial

Canada's lack of indigenous launch capability means all Canadian satellites launch on foreign vehicles, which has implications for export control (Section 5 below).

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4. Canadian Space in the International Context

4.1 International Partnerships

Canada participates in space through multilateral and bilateral agreements:

Table 1.2.7 — Canada's international space partnerships and the contributions and benefits of each.

Partnership	Canadian Contribution	Canadian Benefit
ISS	Canadarm2, Dextre, SPDM	Astronaut access, research time, technology demonstrations
Artemis (Lunar Gateway until its March 2026 pause)	Canadarm3, repurposed for lunar-surface operations (2026)	Astronaut flights, lunar science access
Copernicus (ESA)	Data access agreement	Free access to Sentinel data for Canadian users
COSPAS-SARSAT	Canadian ground stations, MEOSAR payloads	Search and rescue capability for Canadian territory
Five Eyes	Intelligence sharing, dual-use technology	Access to allied space intelligence capabilities

4.2 The Canadian Niche Strategy

Canada has historically pursued a "niche excellence" strategy: rather than attempting full-spectrum space capability, Canada develops world-class expertise in specific domains (robotics, SAR, instruments) and trades access to these capabilities for partnership in larger programs.

Key Equation: The "partnership leverage ratio" -- informal but useful -- describes how Canada trades capability for access:

$$\mathrm{L} = \frac{\mathrm{V}_{\mathrm{access}}}{\mathrm{C}_{\mathrm{contribution}}}$$

Where $\mathrm{V}_{\mathrm{access}}$ is the value of program access obtained and $\mathrm{C}_{\mathrm{contribution}}$ is the cost of Canada's contribution. For the ISS, Canada's Canadarm2/Dextre contribution (~$2B CAD over 30 years) provides access to a $150\mathrm{B} + \mathrm{program}, \mathrm{yielding}\, \mathrm{L}$ \approx 75$. This is the highest leverage ratio of any ISS partner.

Part 2: Regulatory Environment

Expected reading before this part. ITU Radio Regulations (Edition of 2024), Article 21 — Terrestrial and space services sharing frequency bands above 1 GHz; ISED CPC-2-6-02 — Procedure for the Submission of Applications for Spectrum Licences for Space Stations (Issue 7, April 2026); Remote Sensing Space Systems Act (RSSSA) — https://laws-lois.justice.gc.ca/eng/acts/R-5.4/.

5. Regulatory Framework Overview

Space activities in Canada are governed by multiple federal bodies. There is no single "Space Act" -- instead, regulation is distributed across several statutes:

Table 1.2.8 — Canadian federal regulatory bodies, governing statutes, and their mission-design relevance.

Regulatory Body	Statute	What It Regulates	Relevance to Mission Design
CSA	Canadian Space Agency Act	Space program coordination, policy	Mission approval, funding
ISED	Radiocommunication Act	Radio spectrum allocation, licensing	Frequency licensing for all RF transmissions
ISED	Remote Sensing Space Systems Act (RSSSA)	Operation of remote sensing satellites	Licensing for any satellite that images Earth
GAC	Export and Import Permits Act (EIPA)	Export control of strategic goods	Export permits for satellite technology
GAC	Controlled Goods Program	Access to controlled (ITAR/CG) goods	Personnel security clearances
DND/DRDC	National Defence Act	Dual-use technology, military space	Classification, security requirements
Transport Canada	(No current statute)	Launch operations (future)	Canada has no domestic launch regulation yet

6. Remote Sensing Space Systems Act (RSSSA)

6.1 Purpose and Scope

The RSSSA (S.C. 2005, c. 45) regulates the operation of remote sensing satellites from Canada. It was enacted primarily in response to the commercialisation of RADARSAT-2 (the first Canadian commercial EO satellite) and addresses national security and foreign policy concerns related to Earth imagery.

[Source: Government of Canada, RSSSA, https://laws-lois.justice.gc.ca/eng/acts/r-5.4/]

6.2 Who Needs a Licence?

An RSSSA licence is required for any person or entity that: - **Operates** a remote sensing space system from Canada, or - **Controls** the collection or distribution of remote sensing data from a Canadian system

This applies regardless of where the satellite is manufactured or launched. If it is operated from Canadian soil, RSSSA applies.

6.3 Licence Categories and Conditions

Table 1.2.9 — The two RSSSA licence categories and typical conditions imposed on each.

Licence Type	Scope	Typical Conditions
System Operating Licence	Operate a remote sensing satellite system	Data handling plan, priority access for Canadian government, shutter control provisions
Interoperability Arrangements	Share data with foreign entities	Approval for each foreign government or commercial partner

Key conditions typically imposed:

Table 1.2.10 — Typical RSSSA licence conditions covering priority access, shutter control, and data handling.

Condition	Description	Rationale
Priority access	Government of Canada can request priority imaging of any area	National security
Shutter control	Government can restrict imaging of specified areas	National security, diplomatic relations
Data disposition	Approved plan for data storage, distribution, and destruction	Prevent unauthorised access
System disposal	Plan for end-of-life decommissioning	Space debris mitigation
Raw data protection	Restrictions on distribution of unprocessed data	Prevent circumvention of controls

6.4 Implications for CubeSat and University Missions

Even small CubeSat missions with cameras must comply with RSSSA if they produce images of Earth. Key considerations:

Table 1.2.11 — Mission-design parameters that trigger RSSSA applicability for CubeSat missions.

Parameter	Threshold	Implication
GSD	No formal threshold in the Act	All resolutions require licensing

Parameter	Threshold	Implication
Orbit type	Any orbit imaging Earth	LEO, MEO, GEO all in scope
Data distribution	Any distribution of imagery	Must have approved data handling plan
Foreign participation	Any foreign access to data	Requires interoperability arrangement

> **Industry Practice:** MDA's RADARSAT-2 operates under an RSSSA licence that includes shutter control provisions. During the 2010 Haiti earthquake response, the Government of Canada exercised priority access to direct RADARSAT-2 to image the disaster zone within hours. This demonstrates both the operational utility and the regulatory mechanism of RSSSA.

Discussion point. *Does your mission concept involve any remote sensing of Earth? If so, what RSSSA conditions might apply?*

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7. Spectrum Management and Licensing

7.1 Why Spectrum Licensing Matters

Every satellite that transmits or receives radio signals requires spectrum authorisation. This is not optional -- unauthorised RF transmission is illegal in all ITU member states and can result in harmful interference to other systems.

The process involves two levels: 1. **National:** ISED (Innovation, Science and Economic Development Canada) issues the radio licence 2. **International:** ITU (International Telecommunication Union) coordinates the frequency filing to prevent interference with other national administrations

7.2 The ITU Framework

The International Telecommunication Union allocates spectrum globally through the Radio Regulations (RR), updated at World Radiocommunication Conferences (WRC) every 3--4 years.

Table 1.2.12 — Core ITU spectrum-management concepts and their relevance to satellite missions.

ITU Concept	Description	Relevance
Frequency allocation	Spectrum divided among radio services (Fixed Satellite, Mobile, Earth Exploration Satellite, etc.)	Determines which bands are available for your mission type
ITU Regions	Region 1 (Europe/Africa), Region 2 (Americas), Region 3 (Asia/Pacific)	Canada is in Region 2; allocations may differ by region
Coordination	Bilateral/multilateral agreements to avoid interference	Required before a new satellite system can operate
Notification	Formal filing with the ITU Radiocommunication Bureau (BR)	Records the satellite system in the Master International Frequency Register (MIFR)

7.3 Common Satellite Frequency Bands

Table 1.2.13 — Common satellite frequency bands with typical uses and key technical characteristics.

Band	Frequency Range	Typical Use	Key Characteristics
UHF	400--450 MHz	CubeSat TT&C, AIS, ADS-B	Simple antennas, low data rate (1--9.6 kbps), crowded
S-band	2.0--2.3 GHz	TT&C (uplink + downlink)	Standard for housekeeping; moderate antenna gain required
X-band	8.025--8.4 GHz	EO payload data downlink	Higher data rates (10--150 Mbps); requires tracking antenna on ground
Ka-band	26.5--40 GHz	High-throughput data downlink, inter-satellite links	Very high data rates (100+ Mbps); rain attenuation significant
L-band	1.5--1.6 GHz	GNSS, mobile satellite service	Navigation, SAR beacons
C-band	3.7--4.2 / 5.9--6.4 GHz	Legacy GEO comms (DBS), SAR (RADARSAT)	Being repurposed for 5G in some countries

7.4 The Licensing Process (Canada)

The ISED spectrum licensing process for a satellite system follows these steps:

Table 1.2.14 — Sequential steps and durations of the ISED spectrum-licensing process for a satellite system.

Step	Activity	Duration	Documents
1	Pre-application consultation with ISED	2--4 weeks	Preliminary technical description
2	Formal application submission	--	ISED application form, technical annex
3	ISED technical review	3--6 months	Interference analysis, coordination requirements
4	ITU filing (Advance Publication Information -- API)	--	ITU API filing (submitted by ISED on Canada's behalf)
5	ITU coordination with affected administrations	2--7 years	Coordination agreements
6	ITU notification and recording in MIFR	--	MIFR entry
7	ISED licence issuance	--	Radio licence

Key Equation: The link budget equation governs whether a given frequency allocation provides sufficient performance. The received signal power is:

$$\mathrm{P_r} = \mathrm{P_t} + \mathrm{G_t} + \mathrm{G_r} - \mathrm{L_{fs}} - \mathrm{L_{atm}} - \mathrm{L_{misc}}$$

Where all values are in dB: - $\mathrm{P_t}$ = transmit power (dBW) - $\mathrm{G_t}$ = transmit antenna gain (dBi) - $\mathrm{G_r}$ = receive antenna gain (dBi) - $\mathrm{L_{fs}} = 20\log_{10}(4\pi d/\lambda)$ = free-space path loss (dB) - $\mathrm{L_{atm}}$ = atmospheric loss (dB) - $\mathrm{L_{misc}}$ = miscellaneous losses (pointing, polarisation, etc.)

The link margin is then:

$$\mathrm{M} = \mathrm{P_r} - \mathrm{P_{r,\,min}} = (\mathrm{E_b}/\mathrm{N_0})_{\mathrm{received}} - (\mathrm{E_b}/\mathrm{N_0})_{\mathrm{required}}$$

> A positive margin of at least 3 dB is typically required; 6 dB for critical links.

7.5 CubeSat-Specific Considerations

Many CubeSat operators use amateur radio frequencies (VHF/UHF) during development but must transition to commercial bands for operational missions, especially if there is any commercial data distribution.

Table 1.2.15 — CubeSat spectrum-licensing approaches comparing amateur, UHF, S-band, and X-band paths.

Approach	Licensing Path	Limitations
Amateur radio (VHF/UHF)	IARU coordination; national amateur licence	No commercial use; data must be freely available; limited power/bandwidth
Commercial UHF	ISED commercial licence; ITU coordination	Low data rate; crowded band; longer licensing timeline
S-band TT&C	ISED licence; SFCG coordination	Standard path; moderate timeline (6--12 months)
X-band payload	ISED licence; ITU coordination	Higher data rate; requires tracking ground station

> **Industry Practice:** Kepler Communications (Toronto) initially operated its first pathfinder satellites on Ku-band for store-and-forward IoT data relay. Obtaining Ku-band spectrum rights required coordination with incumbent GEO operators, a process that took over 2 years. The lesson: begin spectrum licensing as early as Phase A.

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8. Export Control

8.1 The Export Control Landscape

Satellite technology is controlled goods in most countries due to dual-use (civil/military) applications. Canadian space companies and universities must navigate three overlapping regimes:

Table 1.2.16 — Three overlapping export-control regimes (EIPA, ITAR, EAR) and their key control lists.

Regime	Administered By	Scope	Key List
EIPA (Canada)	Global Affairs Canada (GAC)	Canadian-origin goods and technology	Export Control List (ECL), Groups 1--7
ITAR (USA)	US State Department (DDTC)	US-origin defence articles and services	US Munitions List (USML), Category XV (spacecraft)
EAR (USA)	US Commerce Department (BIS)	US-origin dual-use goods and technology	Commerce Control List (CCL), ECCN 9Axx

8.2 Canadian Export Controls (EIPA)

The Export and Import Permits Act requires an export permit for goods on Canada's Export Control List (ECL). Space-relevant ECL groups:

Table 1.2.17 — Space-relevant groups within Canada's Export Control List with example controlled items.

ECL Group	Description	Examples
Group 1	Dual-use goods (Wassenaar Arrangement)	Radiation-hardened electronics, space-qualified components
Group 2	Munitions (defence goods)	Military satellite systems, encrypted comms
Group 4	Nuclear-related (NSG)	RTGs, nuclear propulsion components
Group 6	Missile technology (MTCR)	Complete rocket systems, guidance systems, propulsion
Group 7	Chemical/biological weapons	Not typically space-relevant

8.3 US Export Controls (ITAR and EAR)

Any mission using US-origin components (which includes nearly all commercial space electronics) must comply with US export controls:

ITAR (International Traffic in Arms Regulations): - USML Category XV covers spacecraft and related articles - Requires a Technical Assistance Agreement (TAA) or Manufacturing License Agreement (MLA) - Applies to US-origin defence articles regardless of where they are located - Violation penalties: up to $1M per violation and 20 years imprisonment

EAR (Export Administration Regulations): - Controls dual-use items (commercial space components, software, technology) - Uses Export Control Classification Numbers (ECCNs) - Requires a licence for export to most countries (except Canada for many items under bilateral agreements) - The US-Canada Defence Production Sharing Agreement provides some exemptions

8.4 Practical Implications for Mission Design

Table 1.2.18 — Mission-design decisions and the export-control obligations each one triggers.

Design Decision	Export Control Implication
Using US-origin reaction wheels	EAR licence may be required; ITAR if military-grade
Collaborating with a non-Five Eyes university	TAA required for ITAR-controlled technical data
Launching on a non-US rocket	ITAR Technical Assistance Agreement (TAA) for US components on foreign launcher
Open-sourcing satellite software	Must verify no controlled algorithms (e.g., encryption above EAR thresholds)
Hosting a foreign payload	May require export permit under EIPA and/or ITAR approval
Publishing detailed design in a paper	Technical data review required if ITAR-controlled components discussed

> **Industry Practice:** MDA's export of RADARSAT-2 ground segment technology to international partners required extensive EIPA permitting and ITAR compliance review (since US-origin components were embedded in the system). The process took 18+ months and required dedicated compliance staff. Even university CubeSat projects using US-origin COTS components must track their ECCN classifications.

Discussion point. *Does your mission concept involve any US-origin components? Any international partners? What export control obligations might apply?*

Session Summary

Table 1.2.19 — Session summary of Canadian space ecosystem and regulatory takeaways.

Topic	Key Takeaway
CSA mandate	Promote peaceful use of space; niche excellence strategy (robotics, SAR, instruments)
Industry structure	4-tier ecosystem; government seed funding enables commercial scale-up
RSSSA	Any remote sensing satellite operated from Canada requires a licence; includes shutter control
Spectrum licensing	Two-level process (ISED national + ITU international); start in Phase A; 2+ year timeline
Export control	Three overlapping regimes (EIPA, ITAR, EAR); nearly all missions encounter US controls
Key lesson	Regulatory compliance is not optional -- it shapes design decisions from the earliest phases

References

1. Government of Canada, Canadian Space Agency Act (S.C. 1990, c. 13)
2. Government of Canada, Remote Sensing Space Systems Act (S.C. 2005, c. 45)
3. Government of Canada, Radiocommunication Act (R.S.C., 1985, c. R-2)
4. Government of Canada, Export and Import Permits Act (R.S.C., 1985, c. E-19)
5. ISED, Spectrum Management and Telecommunications
6. ITU, Radio Regulations, Edition 2020
7. CSA, Exploration, Imagination, Innovation: A New Space Strategy for Canada, 2019
8. CSA, State of the Canadian Space Sector Report, 2022
9. US Department of State, ITAR (22 CFR 120-130)
10. US Department of Commerce, EAR (15 CFR 730-774)

Chapter 1.3 — International Standards for Space Systems

Orbital lifetime vs altitude with FCC/IADC limits.

Orbital lifetime vs altitude with FCC/IADC limits.

> **Expected reading before this session.** NASA SEH §6.5 — Decision Analysis (≈ 30 min).

Prerequisites: Sessions 1.1--1.2 **References:** - ECSS, ECSS System -- Description, Implementation and General Requirements (ECSS-S-ST-00C), 2020 - NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 - CDS, Interface Definition Document for Launch Vehicle Adapters, Rev 14, 2022 - IADC, Space Debris Mitigation Guidelines (IADC-02-01 Rev 3), 2021 - ISO, ISO 24113:2023 -- Space Debris Mitigation Requirements, 2023 - UNOOSA, COPUOS Guidelines for the Long-term Sustainability of Outer Space Activities, 2019 - ESA, Space Debris Mitigation Compliance Verification Guidelines (ESSB-HB-U-002), 2023

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the ECSS standard framework and its hierarchical structure (Management, Engineering, Product Assurance, Sustainability) 2. Compare ECSS and NASA standards and identify equivalences 3. Explain the CDS Rev 14 interface standard and its role in launch vehicle integration 4. Apply space debris mitigation guidelines (IADC, ISO 24113, FCC 5-year rule) to mission design 5. Describe the role of COPUOS and the Outer Space Treaty framework 6. Use SpaceCDF's compliance tracking features

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Part 1: The ECSS Framework

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1. Why Standards Matter in Space Engineering

1.1 The Cost of Non-Compliance

Space missions operate in an environment where failures are catastrophic, repair is impossible, and the consequences of interference or debris affect all space users. Standards exist to:

1. **Ensure mission safety and reliability** -- by codifying lessons learned from decades of space operations
2. **Enable interoperability** -- by defining common interfaces, data formats, and protocols
3. **Reduce cost** -- by providing proven design approaches rather than re-inventing solutions
4. **Satisfy regulatory requirements** -- by demonstrating compliance with national and international obligations
5. **Facilitate technology transfer** -- by establishing a common engineering language

> **Industry Practice:** The loss of the Mars Climate Orbiter (1999) is the canonical example of what happens without rigorous interface standards. Lockheed Martin's ground software produced thruster

force data in pound-force seconds, while NASA's navigation software expected Newton-seconds. The spacecraft entered the Martian atmosphere at 57 km altitude instead of the planned 226 km and was destroyed. Total loss: $327.6M. This failure directly led to NASA's strengthening of Process 12 (Interface Management) in NPR 7123.1D. The lesson: standards and interface control are not bureaucratic overhead -- they are mission-critical.

1.2 The Major Standard Frameworks

Three major standard frameworks govern space activities globally:

Table 1.3.1 — Comparison of ECSS, NASA, and ISO standard frameworks by scope and document count.

Framework	Scope	Typical Adopter	Document Count
ECSS (European Cooperation for Space Standardization)	Full lifecycle, all disciplines	ESA, European industry, CSA (partial)	144 standards + 56 handbooks
NASA Technical Standards	NASA programs and projects	NASA centres, US contractors	NPR/NPD/NASA-STD series
ISO TC 20/SC 14	Space systems and operations	International (reference standard)	~40 standards

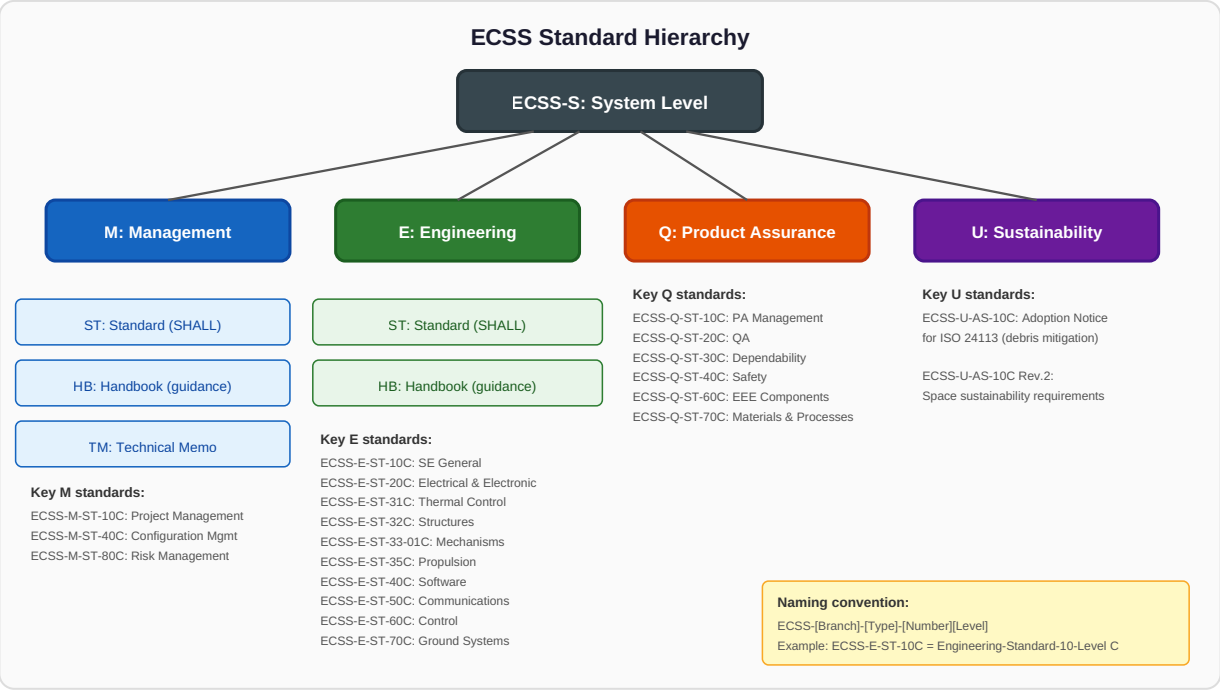
These frameworks are not mutually exclusive. Many missions (including Canadian ones) adopt a tailored combination: ECSS for systems engineering and product assurance, NASA standards for specific technical domains, and ISO for debris mitigation.

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2. ECSS Standard Hierarchy

2.1 Structure

The ECSS system organises standards into four branches, each with three levels:



2.2 The Four Branches

Table 1.3.2 — The four ECSS branches (M, E, Q, U) with scope and an example standard for each.

Branch	Code	Scope	Example Standard
Management	M	Project management, planning, reviews, configuration management, risk management	ECSS-M-ST-10C Rev.1 (Space Project Management)
Engineering	E	Technical disciplines: systems, structures, thermal, EPS, comms, AOCS, software, ground	ECSS-E-ST-10C (Systems Engineering General Requirements)
Product Assurance	Q	Quality, reliability, safety, EEE components, materials, contamination	ECSS-Q-ST-10C (Product Assurance Management)
Sustainability	U	Space debris mitigation, end-of-life disposal, space environment protection	ECSS-U-AS-10C Rev.2 (adoption of ISO 24113)

2.3 Document Types

Table 1.3.3 — ECSS document types (Standard, Handbook, Technical Memorandum) and their obligation level.

Type	Code	Meaning	Obligation
Standard	ST	Contains "shall" requirements -- mandatory when invoked	Must comply or formally request a waiver
Handbook	HB	Contains guidance, best practices, worked examples	Advisory -- not mandatory but strongly recommended
Technical Memorandum	TM	Contains background information, state of the art	Informative only

2.4 Tailoring

A critical concept in ECSS (and all standard frameworks) is **tailoring** -- the process of selecting which standards and which requirements within those standards apply to a given project.

ECSS-S-ST-00C defines three tailoring levels:

Table 1.3.4 — ECSS tailoring levels with descriptions and typical mission classes of application.

Tailoring Level	Description	Typical Application
Full application	All requirements of the invoked standard apply	Large ESA missions (Sentinel, JUICE)
Partial application	Selected clauses apply; others are waived with rationale	Medium missions, CSA projects
Not applicable	Standard is not invoked for this project	CubeSat missions, technology demonstrators

The tailoring rationale is documented in the **Product Assurance and Safety Plan (PASP)** and the **Systems Engineering Management Plan (SEMP)**.

Key Equation: The cost of standards compliance scales non-linearly with mission class. A rough relationship observed in ESA studies:

$$C_{\text{compliance}} \approx k \cdot M_{SC}^{0.7} \cdot N_{\text{standards}}^{0.3}$$

Where M_{SC} is the spacecraft dry mass (kg), $N_{\text{standards}}$ is the number of invoked standards, and k is a constant dependent on the organisation's maturity. For a 10 kg CubeSat invoking 5 standards, compliance costs are roughly 5--10% of mission cost. For a 1000 kg satellite invoking 40 standards, compliance costs can reach 15--20%.

3. Key ECSS Engineering Standards

3.1 Systems Engineering (ECSS-E-ST-10C)

This is the master engineering standard, equivalent to NASA's NPR 7123.1D. It defines: - System engineering processes and activities - Requirements engineering (writing, verification, traceability) - Functional analysis and decomposition - Interface management - Configuration management (technical aspects)

Table 1.3.5 — Mapping ECSS-E-ST-10C systems engineering concepts to their NASA equivalents.

ECSS-E-ST-10C Concept	NASA Equivalent	Key Difference
System requirements specification	Technical requirements baseline	ECSS requires a formal Requirements Specification Document (RSD)
Functional analysis	Logical decomposition (Process 3)	Similar scope
Verification matrix	Requirements verification matrix	ECSS uses a formal DRD (Document Requirements Definition)
Design justification file	Design solution baseline	ECSS requires a DJF that traces every design decision

3.2 Key Subsystem Standards

Table 1.3.6 — Key ECSS engineering subsystem standards with NASA equivalents across disciplines.

Standard	Discipline	Key Requirements	NASA Equivalent
ECSS-E-ST-20C	Electrical & Electronic	EPS design, grounding, EMC, harness	NASA-STD-4003A (EEE), NASA-HDBK-4001
ECSS-E-ST-31C	Thermal Control	Thermal design, analysis, testing	GSFC-STD-7000A (GEVS), thermal sections
ECSS-E-ST-32C	Structures	Structural design, factors of safety, testing	NASA-STD-5001B, GEVS (GSFC-STD-7000A)
ECSS-E-ST-33-01C	Mechanisms	Mechanism design, testing, lubrication	NASA-STD-5017
ECSS-E-ST-35C	Propulsion	Propulsion system design, testing	--
ECSS-E-ST-40C	Software		NASA-STD-8739.8

Standard	Discipline	Key Requirements	NASA Equivalent
		SW development, verification, FDIR	
ECSS-E-ST-50C	Communications	Comms system design, link budget, protocols	CCSDS standards
ECSS-E-ST-60C	Control	AOCS design, pointing, navigation	--
ECSS-E-ST-70C	Ground Systems	Ground segment design, operations	CCSDS standards

3.3 Structural Design Requirements (Example)

To illustrate the depth of ECSS standards, consider the structural design requirements from ECSS-E-ST-32C:

Table 1.3.7 — Selected structural design requirements from ECSS-E-ST-32C with values and rationale.

Requirement	Value	Rationale
Factor of Safety (FoS) -- Yield	≥ 1.25	Prevent permanent deformation under limit loads
Factor of Safety -- Ultimate	≥ 1.5	Prevent structural failure under ultimate loads
Qualification loads	$1.25\times$ limit loads	Demonstrate margin beyond expected environment
First natural frequency (axial)	>25 Hz typical (launcher-dependent)	Avoid coupling with launcher modes
First natural frequency (lateral)	>10 Hz typical (launcher-dependent)	Avoid coupling with launcher modes

Key Equation: The margin of safety (MoS) is defined as:

$$\text{MoS} = \frac{\sigma_{\text{allowable}}}{\text{FoS} \times \sigma_{\text{applied}}} - 1$$

A positive MoS ($\text{MoS} > 0$) indicates the structure meets the requirement. A MoS of 0.0 means the structure exactly meets the requirement with no margin to spare.

Example: If the allowable stress is 280 MPa, the applied stress is 150 MPa, and the FoS is 1.5:

$$\text{MoS} = \frac{280}{1.5 \times 150} - 1 = \frac{280}{225} - 1 = 0.244$$

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4. NASA Standards Comparison

4.1 NASA Technical Standards Architecture

NASA's standards are organised differently from ECSS. Key document types:

Table 1.3.8 — NASA technical-standard document types (NPD, NPR, STD, HDBK, SP) and their obligation.

Type	Code	Example	Obligation
NASA Policy Directive	NPD	NPD 8700.1 (Safety and Mission Assurance)	Mandatory for all NASA programs
NASA Procedural Requirement	NPR	NPR 7123.1D (SE Processes)	Mandatory; defines processes and requirements
NASA Technical Standard	NASA-STD	NASA-STD-5001B (Structural Design)	Mandatory when invoked
NASA Handbook	NASA-HDBK	NASA-HDBK-4001 (EEE Parts)	Advisory guidance
Special Publication	SP	SP-2016-6105 (SEH)	Reference text

4.2 Cross-Reference Table

Table 1.3.9 — Cross-reference of ECSS, NASA, and ISO standards across major engineering domains.

Domain	ECSS Standard	NASA Standard	ISO Standard
Systems Engineering	ECSS-E-ST-10C	NPR 7123.1D, SP-2016-6105	ISO 15288
Project Management	ECSS-M-ST-10C	NPR 7120.5F	ISO 21500
Configuration Management	ECSS-M-ST-40C	NPR 7120.5F Ch. 4	ISO 10007
Risk Management	ECSS-M-ST-80C	NPR 8000.4B	ISO 31000
Structural Design	ECSS-E-ST-32C	NASA-STD-5001B	--
Debris Mitigation	ECSS-U-AS-10C	NASA-STD-8719.14A	ISO 24113
Cleanliness	ECSS-Q-ST-70-01C	NASA-SN-C-0005	ISO 14644
Software	ECSS-E-ST-40C	NASA-STD-8739.8	ISO 12207

Part 2: Launch Interfaces, Debris Mitigation & International Law

> **Expected reading before this part.** Cal Poly, *CubeSat Design Specification (1U – 12U)*, Rev. 14.1 (CP-CDS-R14.1, February 2022), §4 — CubeSat dispenser interface and §5 — dispenser options; Wertz & Larson, *Space Mission Analysis and Design*, 3rd ed., Ch. 18 — Launch systems.

5. CDS Rev 14 -- Launch Vehicle Interface Standard

5.1 What is the CDS?

The **Cubesat Design Specification (CDS)**, maintained by the California Polytechnic State University (Cal Poly) and updated by the CubeSat community, defines the mechanical, electrical, and operational interfaces between CubeSats and their deployment systems (P-PODs, ISIPOD, etc.).

[Source: CDS Rev 14, 2022, available via Cal Poly CubeSat Program]

5.2 Key CDS Requirements

Table 1.3.10 — Key CDS Rev 14 mechanical, electrical, and operational requirements for CubeSat compliance.

Parameter	CDS Rev 14 Requirement	Rationale
Unit dimensions	$100 \times 100 \times 113.5$ mm per U	Standard deployer rail spacing
Mass per U	≤ 2.0 kg (more only by deployer waiver; check the ICD)	Deployer spring mechanism limits
Rail material	Hard-anodised aluminium (7075 or 6061-T6)	Deployer contact surface compatibility
Centre of gravity	Within 2 cm of geometric centre	Deployment dynamics, tumble rate control
Deployables	Must be constrained during launch; no protrusion beyond CubeSat envelope	Protect adjacent payloads in deployer
Separation springs	Prohibited on CubeSat (deployer provides)	Standardised deployment mechanism
RF silence	No RF transmission until 30 minutes after deployment	Avoid interference with launch vehicle
Deployment switches	Minimum 1 per deployable; 2 for redundancy	Prevent premature deployment
Battery charge state	Fully charged (recommended); charging from deployer not available	No power interface with deployer
Propulsion	If present: must be inhibited by 3 independent inhibits; no toxic propellants	Safety of primary payload and deployer

5.3 Form Factors

Table 1.3.11 — CubeSat form factors from 1U to 16U with dimensions, mass limits, and applications.

Form Factor	Dimensions (mm)	Mass Limit	Typical Applications
1U	100 x 100 x 113.5	2.0 kg	Technology demonstrators, IoT nodes
1.5U	100 x 100 x 170.2	3.0 kg	Enhanced technology demonstrators
2U	100 x 100 x 227.0	4.0 kg	Simple instruments, store-and-forward
3U	100 x 100 x 340.5	6.0 kg	Standard EO/science missions
6U	100 x 226.3 x 366.0	12.0 kg	Advanced EO, communications
12U	226.3 x 226.3 x 340.5	24.0 kg	High-performance missions
16U	226.3 x 226.3 x 454.0	32.0 kg	Near-microsatellite capability

5.4 Rideshare Launch Interfaces

For non-CubeSat smallsats (microsatellites 10--200 kg), the interface standard depends on the launch provider:

Table 1.3.12 — Rideshare launch-vehicle interface standards across SpaceX, Rocket Lab, Arianespace, and ISRO.

Launch Provider	Interface Standard	Adapter Type	Key Document
SpaceX (Rideshare)	ESPA-class	15" or 24" ESPA port	SpaceX Rideshare User's Guide
Rocket Lab (Electron)	Custom separation system	Rocket Lab-provided	Electron Payload User's Guide
Arianespace (Vega-C)	ASAP-S	Multi-payload adapter	Vega-C User Manual
ISRO (PSLV)	Custom adapter	ISRO-provided	PSLV User Manual

> **Industry Practice:** Planet Labs' SuperDove constellation (150+ satellites) uses a standardised 3U-plus form factor that is CDS-compliant but extends the standard with a custom deployer arrangement for batch deployment. Each SuperDove satellite (mass ~5.8 kg) carries a multispectral imager with 8 bands and ~3m GSD. The standardisation of the bus design enabled a manufacturing rate of 2+ satellites per week -- only possible because the CDS provides a stable interface baseline.

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6. Space Debris Mitigation

6.1 The Debris Problem

As of 2025, there are approximately: - **36,500+** tracked objects larger than 10 cm in Earth orbit - **1,000,000+** estimated objects 1--10 cm (untracked, lethal to spacecraft) - **130,000,000+** estimated objects 1 mm -- 1 cm (can damage components)

The debris population is growing due to collisions (the Kessler Syndrome), anti-satellite tests (e.g., Chinese ASAT 2007: 3,500+ tracked fragments), and the rapid growth of mega-constellations.

[Source: ESA Space Debris Office, Annual Report 2024; NASA ODPO Orbital Debris Quarterly News]

6.2 Debris Mitigation Guidelines Hierarchy

Table 1.3.13 — Hierarchy of debris-mitigation documents from voluntary IADC guidelines to binding regulations.

Level	Document	Scope	Status
International voluntary	IADC Space Debris Mitigation Guidelines (Rev 3, 2021)	Global best practice	Advisory; adopted by COPUOS
International standard	ISO 24113:2023	Requirements for debris mitigation	Standard; invoked by many agencies
European standard	ECSS-U-AS-10C Rev.2	ECSS adoption notice for ISO 24113	Mandatory for ESA missions
US regulation	FCC 47 CFR 25.114(d)(14)	Post-mission disposal for US-licensed satellites	Legally binding for FCC licensees
NASA standard	NASA-STD-8719.14A	NASA process for limiting orbital debris	Mandatory for NASA missions
ESA requirement	ESA/ADMIN/IPOL(2023)2	Clean Space requirements	

Level	Document	Scope	Status
			Mandatory for ESA missions (2023+)

6.3 Key Debris Mitigation Requirements

Table 1.3.14 — Key quantitative debris-mitigation requirements covering disposal, passivation, and casualty risk.

Requirement	Source	Value	Notes
Post-mission disposal	IADC, ISO 24113	≤ 25 years	Voluntary guideline; being tightened
Post-mission disposal	FCC (2024+)	≤ 5 years	Legally binding for FCC-licensed sats
Probability of successful disposal	ISO 24113	≥ 0.9	Must demonstrate reliability of deorbit mechanism
Casualty risk on re-entry	NASA-STD-8719.14A	$\leq 1:10{,}000$ per event	Drives material selection and design-for-demise
Collision avoidance probability	IADC	$< 10^{-4}$ per year (cumulative)	Drives orbit selection and manoeuvre capability
Passivation	ISO 24113, ECSS-U-AS-10C	All stored energy sources depleted at EOL	Batteries, pressure vessels, wheels, RF

6.4 Post-Mission Disposal Options

Table 1.3.15 — Post-mission disposal options with applicable orbits, mechanisms, and re-entry timescales.

Method	Applicable Orbit	Mechanism	Time to Re-entry
Atmospheric drag (natural)	LEO < 600 km	Natural orbital decay	Months to years
Atmospheric drag (augmented)	LEO < 700 km	Drag sail, drag tether	Weeks to years
Propulsive deorbit	LEO	Thrusters lower perigee to ~200 km	Days
Graveyard orbit	GEO	Raise orbit ~300 km above GEO	Indefinite (not re-entry)
Heliocentric disposal	Beyond GEO	Escape Earth orbit	N/A

Key Equation: The orbital lifetime of a satellite in LEO due to atmospheric drag is approximately:

$$\tau \approx \frac{\mathrm{C_D} \cdot \mathrm{A} \cdot \rho \cdot \mathrm{a}^2}{2\mathrm{m}} \cdot (\text{complex integral})$$

A simpler rule-of-thumb for circular orbits:

$$\tau_{\text{years}} \approx \frac{\mathrm{h} - 200}{30} \cdot \frac{\mathrm{m/A}}{50}$$

Where h is altitude in km, m is mass in kg, and A is the cross-sectional area in m^2. This is very approximate; actual lifetime depends on solar activity (which modulates atmospheric density at high altitudes), drag coefficient, and orbit eccentricity. Use NASA's DAS (Debris Assessment Software) or ESA's DRAMA tool for accurate predictions.

More precisely, the ballistic coefficient is:

$$B = \frac{m}{C_D \cdot A}$$

Lower B (lighter, larger area) means faster re-entry. A 3U CubeSat ($m \approx 4$ kg, $A \approx 0.03$ m^2, $C_D \approx 2.2$) at 400 km has $B \approx 61$ kg/m^2 and will re-enter within ~1--3 years depending on solar cycle.

6.5 Practical Implications for Mission Design

Table 1.3.16 — Mission-design decisions and their associated debris-mitigation impacts.

Design Decision	Debris Mitigation Impact
Orbit altitude selection	Altitudes > 600 km require active deorbit; > 700 km strongly discouraged for non-manoeuvrable spacecraft
Propulsion system	Required for altitudes > 500--600 km to meet 25-year (or 5-year FCC) rule
Passivation design	Must design battery disconnect, pressure relief, wheel spin-down circuits
Material selection	Aluminium structures preferred for design-for-demise; titanium and carbon fibre survive re-entry
Collision avoidance	Manoeuvre capability or conjunction assessment service (e.g., 18th SDS, ESA SSA) needed

Industry Practice: OneWeb (648 satellites at 1200 km) carries propulsion on every satellite specifically for end-of-life deorbit, since natural decay from 1200 km would take centuries. Each satellite carries sufficient propellant for multiple collision avoidance manoeuvres plus a complete deorbit burn to lower perigee below 300 km. The deorbit operation takes approximately 3 months per satellite.

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7. International Space Law and COPUOS

7.1 The UN Space Treaties

The legal framework for space activities is established by five UN treaties negotiated under the Committee on the Peaceful Uses of Outer Space (COPUOS):

Table 1.3.17 — The five UN space treaties with year, key provisions, and ratification counts.

Treaty	Year	Key Provisions	Ratification
Outer Space Treaty (OST)	1967	Space is free for exploration; no national appropriation; states responsible for national activities; liability for damage	114 parties (incl. Canada)
Rescue Agreement	1968	Return astronauts and space objects	99 parties

Treaty	Year	Key Provisions	Ratification
Liability Convention	1972	Launching state liable for damage on Earth (absolute) and in space (fault-based)	98 parties
Registration Convention	1976	States must register space objects with UN	72 parties
Moon Agreement	1979	Moon and celestial bodies are "common heritage of mankind"	18 parties (NOT US, Russia, China)

7.2 Key Legal Principles for Mission Design

Table 1.3.18 — Key legal principles from the UN space treaties and their mission-design implications.

Principle	Source	Implication
State responsibility	OST Art. VI	The Government of Canada is internationally responsible for all Canadian space activities, including private/university missions
Authorisation and supervision	OST Art. VI	Canada must authorise and continuously supervise all non-governmental space activities (this is why RSSSA exists)
Liability	Liability Convention	Canada (as launching state) is liable for damage caused by Canadian satellites; absolute liability on Earth, fault-based in space
Registration	Registration Convention	Canada must register all space objects with the UN; CSA maintains the Canadian registry
Non-contamination	OST Art. IX	Must avoid harmful contamination of space and celestial bodies (planetary protection)
Due regard	OST Art. IX	Must conduct activities with "due regard" for other states' interests (debris mitigation)

7.3 COPUOS Long-Term Sustainability Guidelines (2019)

In 2019, COPUOS adopted 21 guidelines for the long-term sustainability of outer space activities. These are voluntary but politically significant:

Table 1.3.19 — The four COPUOS Long-Term Sustainability guideline categories with counts and key points.

Guideline Category	Count	Key Points
Policy and regulatory	7	Adopt national regulatory frameworks, register space objects, share SSA data
Safety of operations	4	Conjunction assessment, collision avoidance, re-entry risk assessment
International cooperation	4	Share debris mitigation best practices, coordinate spectrum use
Scientific and technical	6	Improve debris models, develop removal technology, research space weather effects

[Source: A/74/20, Report of COPUOS, 2019, Annex II]

8. SpaceCDF Compliance Features

SpaceCDF tracks compliance through the **Compliance Engineer** position:

Table 1.3.20 — SpaceCDF compliance-tracking features and their level of automation.

Feature	What It Tracks	Automation Level
Standard applicability matrix	Which ECSS/NASA/ISO standards apply to this mission	Semi-automated (suggests based on mission type)
Debris mitigation compliance	Post-mission lifetime, passivation plan, casualty risk	Automated (calculates from orbit and mass)
RSSSA checklist	Licence requirements for Canadian remote sensing missions	Manual (checklist with guidance)
Spectrum filing status	ISED/ITU filing progress	Manual (status tracking)
Export control classification	ECCN/USML classification of key components	Manual (per-component)
CDS compliance	Mechanical/electrical interface compliance with CDS Rev 14	Semi-automated (checks mass, dimensions, CG)

Exercise. *In SpaceCDF, navigate to the Compliance panel. Review the debris mitigation compliance status for your mission. What orbit altitude would be needed to comply with the FCC 5-year rule without propulsion? Use the tool's orbital lifetime calculator.*

Session Summary

Table 1.3.21 — Session summary of ECSS, NASA, CDS, debris mitigation, and international space law.

Topic	Key Takeaway
ECSS framework	Four branches (M, E, Q, U); three document types (ST, HB, TM); tailoring is essential
NASA standards	Organised as NPD/NPR/NASA-STD/NASA-HDBK; broadly equivalent to ECSS
CDS Rev 14	Defines CubeSat mechanical/electrical/operational interfaces; compliance is required for rideshare launch
Debris mitigation	IADC 25-year rule; FCC 5-year rule (2024+); passivation required; drives orbit and propulsion design
International law	OST establishes state responsibility; Canada liable for all Canadian space activities
COPUOS	21 sustainability guidelines (2019); voluntary but increasingly influential
Compliance in SpaceCDF	Tracked through the Compliance Engineer position; automated where possible

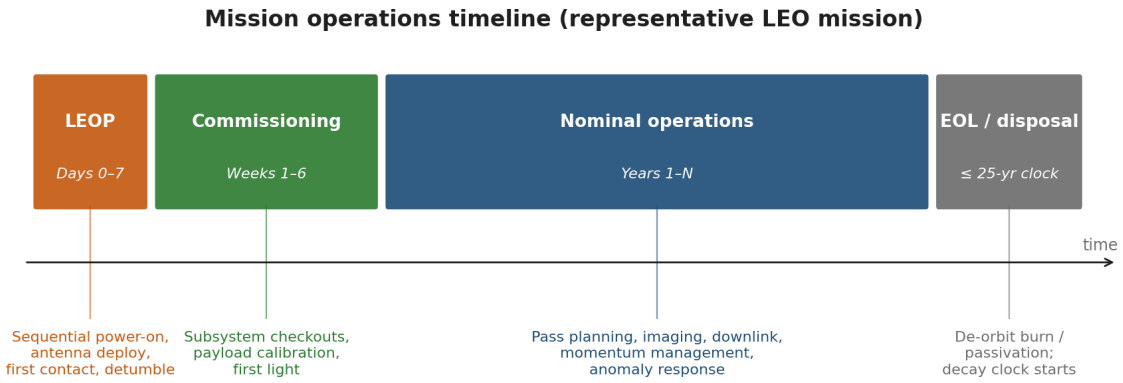
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8. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
9. NASA, NASA-STD-8719.14A -- Process for Limiting Orbital Debris, 2019
10. UNOOSA, A/74/20 -- Report of COPUOS, 2019
11. UNOOSA, Outer Space Treaty, 1967
12. FCC, 47 CFR 25.114 -- Satellite Applications

Chapter 1.4 — Mission Needs, Stakeholder Analysis & Trade Studies



Mission operations timeline (LEOP through disposal).

> **Expected reading before this session.** NASA SEH Appendix S — ConOps (≈ 45 min).

Prerequisites: Sessions 1.1--1.3 **References:** - NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016 -- Sections 4.1, 4.4, 6.8 - NASA, NPR 7123.1D -- Sections 3.2.1 (Process 1), 3.2.4 (Process 4), 3.5.8 (Process 17) - ECSS, ECSS-E-ST-10C -- Section 5.1 (Requirements Engineering) - ECSS, ECSS-M-ST-10C Rev.1 -- Section 5.3 (Review Process) - Wertz, J.R. et al., Space Mission Engineering: The New SMAD, Microcosm Press, 2011 -- Chapters 1--3 - NASA SEH Appendix C -- How to Write a Good Requirement - NASA SEH Appendix S -- ConOps Outline

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Learning Objectives

By the end of this session, participants will be able to: 1. Write a clear problem statement that defines the mission need without prescribing a solution 2. Identify and categorise mission stakeholders using a structured analysis matrix 3. Define mission objectives with measurable success criteria and trace them through the MoE/MoP/TPM hierarchy 4. Structure and execute a trade study using weighted scoring methods (Decision Analysis -- Process 17) 5. Evaluate space vs non-space alternatives objectively and document the decision rationale 6. Design a Concept of Operations (ConOps) including mission phases, operational modes, and data flow 7. Use SpaceCDF Steps 1--2 and the Trade Studies tab to capture and evaluate mission concepts

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Part 1: Mission Need & Stakeholder Analysis

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1. The Problem Statement

1.1 Why Start with the Problem?

NASA SEH Section 4.1.1 states: *"Clearly define the problem before solving it."*

The single most common failure mode in mission design is not technical -- it is **solving the wrong problem**. Teams jump to solutions ("we need a 6U CubeSat at 500 km") before articulating the actual need ("agricultural monitoring in sub-Saharan Africa requires 10 m resolution multispectral imagery every 5 days"). This premature commitment to a solution:

- Closes off potentially superior alternatives (commercial data purchase, aerial surveys, ground sensors)
- Introduces unnecessary constraints that drive up cost and risk
- Makes it impossible to evaluate mission success (success against what?)
- Violates the fundamental SE principle of **WHAT before HOW**

[Source: NASA SEH Section 4.1; NASA SEH Appendix C -- "How to Write a Good Requirement"]

1.2 Structure of a Good Problem Statement

A rigorous problem statement answers four questions:

Table 1.4.1 — Four-question structure for a rigorous problem statement with worked agricultural-monitoring examples.

Question	Content	Example
What is the problem?	The capability gap -- what is missing or inadequate	"Lack of timely, affordable multispectral imagery at sufficient resolution and revisit"
Who is affected?	Stakeholders and end users who suffer from the gap	"Agricultural monitoring agencies in sub-Saharan Africa"
What is the impact?	Consequence of not solving it -- quantified if possible	"Delayed or inaccurate crop assessments affecting food aid allocation for 300M people"
What constraints exist?	Budget, schedule, political, regulatory, or technical limitations	"Budget: <$10M total mission cost; timeline: 3 years to first data"

1.3 Good vs Bad Problem Statements

Example -- GOOD:

> "Agricultural monitoring agencies in sub-Saharan Africa lack timely, affordable access to multispectral imagery at sufficient resolution (≤ 10 m) and revisit rate (≤ 5 days) to support crop yield prediction and food security planning. Current Sentinel-2 data provides 10 m resolution but only 5-day revisit at the equator, and cloud cover reduces usable observations to approximately 60%. The consequence is delayed or inaccurate crop assessments affecting food aid allocation for 300 million people. Budget constraint: total mission cost less than $10M; first data delivery within 3 years."

Example -- BAD:

> "We need a 6U CubeSat with a multispectral imager at 500 km SSO."

The bad example prescribes a solution (6U CubeSat, SSO), specifies a design parameter (500 km), and says nothing about the actual problem. It is a **solution statement**, not a problem statement.

1.4 The WHAT vs HOW Principle

At the mission need level, everything should describe **WHAT** is needed, not **HOW** to achieve it:

Table 1.4.2 — Contrasting WHAT-level needs against premature HOW-level design prescriptions.

WHAT (correct at this stage)	HOW (premature at this stage)
"10 m resolution imagery"	"Use a 15 cm aperture telescope"
"Daily revisit at equator"	"Deploy a Walker delta constellation of 12 satellites"
"Data within 6 hours of acquisition"	"Use X-band downlink at 150 Mbps to 3 ground stations"
"Total cost under $10M"	"Use COTS components exclusively"
"5-year operational lifetime"	"Select radiation-hardened components"

The HOW column is not wrong -- these are all reasonable design decisions. But they belong in later phases (Phase A/B), not in the problem statement. Fixing the HOW prematurely eliminates design freedom and may lead to suboptimal solutions.

Discussion point. *Why is it harmful to specify HOW at this stage? What design options does it close off?*

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2. Stakeholder Identification & Analysis

2.1 Who is a Stakeholder?

NASA SEH Section 4.1.2 defines stakeholders as "all parties who have a legitimate interest in the system throughout its lifecycle." This is broader than just the end user -- it includes everyone who affects or is affected by the mission.

2.2 Stakeholder Categories for Space Missions

Table 1.4.3 — Stakeholder categories for space missions with typical needs and constraints for each.

Category	Examples	Typical Needs	Typical Constraints
End Users	Scientists, farmers, shipping companies, emergency responders	Data quality, resolution, latency, format, accessibility	Data rights, training, infrastructure
Operators	Mission control centre, ground station operators	Operability, automation level, staffing, training	Budget for operations, personnel availability
Sponsors / Funders	Space agency (CSA), university, commercial investor, DND	Return on investment, schedule adherence, risk profile	Budget cap, political timelines, reporting requirements
Regulatory	ISED, ITU, FCC, export control (GAC)	Compliance with spectrum, debris, RSSSA, export regulations	Licensing timelines, filing requirements
Launch Provider	SpaceX, Rocket Lab, ISRO, Arianespace	CDS compliance, mass/volume limits, schedule, payment	Interface requirements, launch window, manifesting
Ground Segment	KSAT, SSC, SatNOGS, own stations	Frequency compatibility, data volume, contact time	Station availability, geographic coverage

Category	Examples	Typical Needs	Typical Constraints
Data Consumers	Archives (e.g., EODMS), APIs, partner agencies	Data format (NetCDF, GeoTIFF), metadata standards, timeliness	Format standards, distribution agreements
General Public	Taxpayers (for government-funded missions)	Value for money, transparency, societal benefit	Political expectations, public engagement
Orbital Environment	Other satellite operators, debris community	Debris mitigation, spectrum cleanliness, collision avoidance	IADC guidelines, ISO 24113, FCC 5-year rule

2.3 Stakeholder Analysis Matrix

For each stakeholder, capture five attributes:

Table 1.4.4 — Five attributes captured per stakeholder in the analysis matrix with scoring scales.

Attribute	Description	Scale
Name / Role	Who are they?	Text
Needs	What do they require from the mission?	Text (specific, measurable where possible)
Constraints	What limitations do they impose?	Text (mandatory vs desirable)
Priority	How critical is satisfying this stakeholder?	Primary (must satisfy) / Secondary (should satisfy) / Tertiary (nice to have)
Influence	How much power do they have over the mission?	High / Medium / Low

> **Industry Practice:** For the RADARSAT Constellation Mission (RCM), the stakeholder analysis identified over 20 distinct stakeholder groups, including DND (maritime surveillance), Environment Canada (sea ice monitoring), Agriculture and Agri-Food Canada (crop monitoring), Natural Resources Canada (forestry), Public Safety Canada (disaster response), and international partners. Each stakeholder had different priority imaging modes, coverage requirements, and data latency needs. The systems engineering challenge was to design a 3-satellite constellation that satisfied all these needs within a single system architecture -- a classic example of multi-stakeholder optimisation.

Exercise. *Complete Part A of Worksheet 1.4 -- identify at least 4 stakeholders for your team's mission and fill in the analysis matrix.*

---

3. Mission Objectives and the MoE/MoP/TPM Hierarchy

3.1 From Need to Objectives

Objectives translate the problem statement into specific, testable goals. Each objective must have:

Table 1.4.5 — Required attributes for each mission objective with example entries.

Attribute	Description	Example
Text	Clear statement of what the mission will achieve	"Provide 10 m GSD multispectral imagery over the target region"
Priority	Primary (mission fails without) or secondary (desirable)	Primary
Measurable criterion	How you know the objective is met -- a number with a unit	"GSD $\leq$ 10 m at nadir, 4+ spectral bands, revisit $\leq$ 5 days"
Type	Category of objective	Observation, communication, navigation, science, technology demonstration

3.2 Writing Good Objectives

Table 1.4.6 — Examples of well-formed mission objectives explaining why each meets the criteria.

Good Objective	Why It Is Good
"Provide 10 m GSD multispectral imagery with 4+ bands for the target region (30S--30N), with $\leq$ 5-day revisit and $\leq$ 24 h data latency"	Specific (10 m, 4 bands, geographic scope), measurable (all criteria have numbers + units), relevant to agriculture, achievable with CubeSat technology
"Achieve 99.5% AIS ship detection probability in the North Atlantic within 30 minutes of ship transmission"	Specific (AIS, North Atlantic), measurable (99.5%, 30 min), relevant to maritime safety

Table 1.4.7 — Examples of poorly written objectives with diagnoses of their flaws.

Bad Objective	Why It Is Bad
"Take pictures from space"	Not specific, not measurable, no criterion for success
"Build a 3U CubeSat"	This is a solution, not an objective -- it says nothing about what the mission should accomplish
"Demonstrate new technology"	What technology? What does "demonstrate" mean? How do you know it succeeded?

3.3 The MoE / MoP / TPM Hierarchy

These three measures form a chain from operational need to tracked design parameter:

[Source: NASA SEH Section 4.1.4, Section 4.2.4, Section 6.7.3]

Table 1.4.8 — MoE, MoP and TPM measures distinguished by scope, owner and evaluation phase.

Measure	What It Measures	Who Sets It	Example	When Evaluated
MoE (Measure of Effectiveness)	How well the system satisfies the operational need	Users / stakeholders	"% of crop assessments delivered within 5 days of acquisition"	Phase E (operations)
MoP (Measure of Performance)	Technical performance of the system	Systems engineer	"GSD $\leq$ 10 m at nadir"	Phase C/D (verification)

Measure	What It Measures	Who Sets It	Example	When Evaluated
TPM (Technical Performance Measure)	Design parameter tracked over time	Design team	"Current best estimate of imager mass vs allocation (1.5 kg target)"	All phases (continuous)

The hierarchy flows downward:

```
Stakeholder Need
  -> MoE (operational effectiveness)
    -> Mission Objective
      -> MoP (technical performance)
        -> Technical Requirement ("shall" statement)
          -> TPM (tracked design parameter)
```

Key Equation: Measures of Effectiveness often involve probability and time, connecting to system-level performance:

$$\mathrm{MoE} = \mathrm{P}_{\mathrm{detection}} \times \mathrm{P}_{\mathrm{data_delivery}} \times \mathrm{f}(\mathrm{T}_{\mathrm{latency}})$$

For an Earth observation mission:

$$\mathrm{MoE}_{\mathrm{coverage}} = \frac{\mathrm{A}_{\mathrm{imaged_per_revisit}}}{\mathrm{A}_{\mathrm{total_target}}} \times (1 - \mathrm{P}_{\mathrm{cloud}})$$

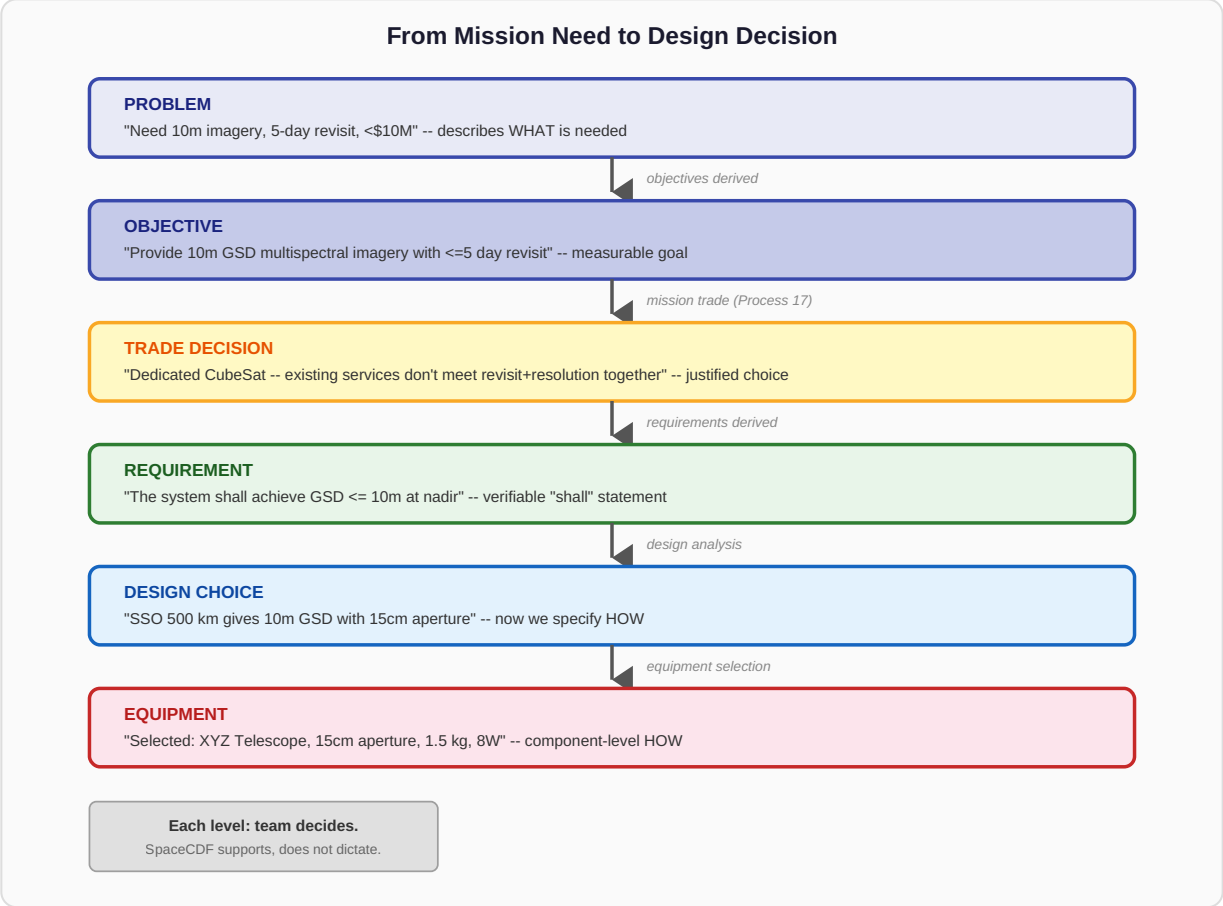
Where $\mathrm{A}_{\mathrm{imaged_per_revisit}}$ depends on swath width and orbit ground track spacing, $\mathrm{A}_{\mathrm{total_target}}$ is the target area, and $\mathrm{P}_{\mathrm{cloud}}$ is the cloud cover probability.

Exercise. *For one of your objectives, trace the full hierarchy from stakeholder need through MoE, MoP, and TPM. Complete Part B of Worksheet 1.4.*

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4. From Need to Design: The Flow

Show how the mission need flows through to design decisions without the need itself prescribing the design:



At each level in this hierarchy, the design team makes decisions informed by analysis and trade studies. The tool supports this process with parametric calculations and automated constraint checking, but the team retains decision authority.

Part 2: Trade Study Methodology

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5. Decision Analysis Framework

5.1 Process 17: Decision Analysis

NASA SEH Process 17 (Decision Analysis) provides the framework for structured alternative evaluation. Every major design decision should follow this process to ensure decisions are auditable, traceable, and defensible.

[Source: NASA SEH Section 6.8, Process 17: Decision Analysis]

5.2 The Five Elements of a Rigorous Trade Study

Table 1.4.9 — Five elements of a rigorous trade study with common pitfalls to avoid.

Element	Description	Pitfall to Avoid
1. Decision statement	What are we deciding? Framed as a question.	Too broad ("What should we build?") or too narrow ("Which reaction wheel?") for the current phase

Element	Description	Pitfall to Avoid
2. Alternatives	What are the options? Minimum 3, including "do nothing"	Straw-man alternatives designed to lose; failure to include obvious options
3. Criteria	What matters? Performance, cost, schedule, risk, compliance	Missing a critical criterion; criteria that overlap (double-counting)
4. Weightings	How important is each criterion? Normalised to sum to 1.0	All criteria weighted equally (means nothing matters more than anything else); weights set after seeing scores
5. Scoring	How well does each alternative perform?	Inconsistent scoring scale; anchoring bias; group conformity

5.3 Weighting Methods

Table 1.4.10 — Comparison of pairwise, swing, direct and AHP weighting methods and their limitations.

Method	Procedure	Best When	Limitations
Pairwise comparison	Compare criteria two at a time; count wins. Weight = wins / total comparisons	Small number of criteria (< 8)	Circular preferences possible; does not capture magnitude
Swing weighting	For each criterion, assess the value of moving from worst to best case. Normalise.	Criteria have different units and scales	Requires clear understanding of worst/best cases
Direct assignment	Team discusses and agrees on weights	Quick preliminary assessment	Subjective; dominated by vocal team members
AHP (Analytic Hierarchy Process)	Pairwise comparison on a 1--9 ratio scale; compute eigenvector	Complex decisions with many stakeholders	Mathematically complex; consistency check needed

Key Equation (Pairwise Comparison): For n criteria, there are $\frac{n(n-1)}{2}$ pairwise comparisons. The raw weight of criterion i is:

$$w_i^{raw} = \sum_{j \neq i} p_{ij}$$

Where $p_{ij} = 1$ if criterion i is preferred over j, 0.5 for a tie, and 0 if j is preferred. The normalised weight is:

$$w_i = \frac{w_i^{raw}}{\sum_{k=1}^{n} w_k^{raw}}$$

5.4 Scoring Methods

Table 1.4.11 — Quantitative, qualitative and threshold scoring methods with usage guidance.

Type	Description	Example	When to Use
Quantitative	Numeric value normalised to 0--1	Mass: 5 kg scores 0.8, 10 kg scores 0.4	When hard numbers exist

Type	Description	Example	When to Use
Qualitative	Verbal rating mapped to number	"Excellent" = 1.0, "Good" = 0.75, "Fair" = 0.5, "Poor" = 0.25	When only expert judgment is available
Threshold (go/no-go)	Pass/fail gate applied before scoring	"TRL ≥ 6 required" -- below threshold is eliminated	Non-negotiable requirements

5.5 Weighted Score Calculation

For each alternative a across n criteria:

Key Equation:

$$\mathrm{S(a)} = \sum_{\mathrm{c}=1}^{\mathrm{n}} \mathrm{w_c} \cdot \mathrm{s(a,c)}$$

Where $\mathrm{w_c}$ is the normalised weight of criterion c and $\mathrm{s(a,c)}$ is the normalised score (0--1) of alternative a on criterion c.

For "higher is better" criteria:

$$\mathrm{s(a,c)} = \frac{\mathrm{v(a,c)} - \mathrm{v_{min}(c)}}{\mathrm{v_{max}(c)} - \mathrm{v_{min}(c)}}$$

For "lower is better" criteria:

$$\mathrm{s(a,c)} = 1 - \frac{\mathrm{v(a,c)} - \mathrm{v_{min}(c)}}{\mathrm{v_{max}(c)} - \mathrm{v_{min}(c)}}$$

5.6 Sensitivity Analysis

A trade study result is only meaningful if it is **robust** -- i.e., the ranking does not change with small perturbations in weights or scores. Sensitivity analysis tests this:

Table 1.4.12 — Weight, score and threshold sensitivity-analysis methods with what each reveals.

Method	Procedure	What It Reveals
Weight perturbation	Vary each weight by $\pm 20\%$, re-normalise, re-score	Which criteria are "swing" criteria that could flip the outcome
Score perturbation	Vary each score by ± 1 level, re-calculate	Which scores are the most uncertain and impactful
Threshold analysis	For each criterion, find the weight at which the 2nd-place alternative overtakes the 1st	How much the weights would need to change to reverse the decision

Industry Practice: For the Iridium NEXT constellation (66 operational + 6 on-orbit spares), the initial trade study for the constellation architecture compared Walker Delta, Walker Star, and hybrid configurations across 12 criteria including global coverage, revisit time, inter-satellite link geometry, launch cost, and orbital debris risk. Sensitivity analysis revealed that the ranking was robust for all weight perturbations up to $\pm 30\%$, giving high confidence in the selected Walker Star configuration at 780 km altitude.

Exercise. *Practice the pairwise comparison method by weighting 4 criteria for a lunch restaurant choice: taste, cost, healthiness, speed. Then score 3 options. This builds the skill on a low-stakes example before applying it to mission design.*

---

6. Space vs Non-Space Alternatives

6.1 The Most Important Trade Study

Before committing to building a satellite, teams must honestly evaluate whether existing solutions meet the need. This is the most important trade study in the mission lifecycle because it determines whether the entire project should proceed.

6.2 Alternative Categories

Table 1.4.13 — Seven alternative data-source categories compared on strengths and weaknesses.

Category	Examples	Strengths	Weaknesses
Existing free data	Copernicus Sentinel-2, Landsat 8/9	Zero acquisition cost, proven quality, long archive	Fixed resolution/revisit, no tasking control
Commercial data purchase	Planet (SuperDove), Maxar (WorldView), ICEYE (SAR)	High resolution, fast tasking, no development risk	Ongoing cost, data rights limitations, vendor dependency
Aerial (drones/ aircraft)	Survey drones, P-3 Orion, Twin Otter	Very high resolution (< 1 m), flexible scheduling	Limited area coverage, weather dependent, regulatory constraints
Ground sensors	IoT networks, weather stations, tide gauges	Continuous monitoring, low per-unit cost	Point measurements only, no spatial coverage
Dedicated satellite	New CubeSat, SmallSat, or microsatellite	Full control, custom instrument, IP ownership	High cost ($2--50M), development risk, 2--5 year schedule
Constellation	Multiple dedicated satellites	Global coverage, short revisit (< 1 day)	Much higher cost, operational complexity
Hosted payload	Payload on another operator's bus	Lower cost, shared bus risk	Compromised orbit, limited control, schedule dependency

6.3 Decision Criteria: When is Space Justified?

Space is likely NOT the right answer when: - Existing free data (Sentinel-2, Landsat) meets resolution and revisit needs - Coverage requirement is local (drones or aircraft are cheaper and higher resolution) - Data rate is very low (ground sensors with cellular/satellite IoT backhaul suffice) - Budget does not support minimum viable satellite cost (~$2M for a basic 3U CubeSat) - Technology readiness is too low for the available schedule

Space is likely justified when: - Global or wide-area coverage is needed simultaneously - Persistent monitoring is required (24/7 or very frequent revisit < 5 days) - User needs data ownership and control over acquisition scheduling - No existing service provides the required measurement type (e.g., specific

wavelength, polarisation) - Regulatory or sovereignty requirements demand national control over the sensor

6.4 Constellation Sizing

For missions requiring short revisit, the number of satellites drives cost:

Key Equation: For a Walker Delta constellation at altitude h with half-swath angle η, the number of orbital planes P and satellites per plane S needed for revisit time $\mathrm{T_{rev}}$ is approximately:

$$\mathrm{P} \times \mathrm{S} \geq \frac{2\pi \mathrm{R_E}}{\mathrm{v_{ground}} \times \mathrm{T_{rev}} \times \tan(\eta)}$$

Where $\mathrm{v_{ground}} \approx 7.1$ km/s (ground track velocity at 500 km) and η is related to the instrument swath width W by $\eta = \mathrm{W}/(2\mathrm{R_E})$ for small angles.

More practically, constellation cost scales sub-linearly due to learning curves:

$$\mathrm{C_{total}} = \mathrm{C_1} \times \sum_{\mathrm{i}=1}^{\mathrm{N}} \mathrm{i}^{\log_2(\mathrm{L})}$$

Where $\mathrm{C_1}$ is the first unit cost, N is the number of satellites, and L is the learning curve factor (typically 0.90--0.95 for small batches, 0.85 for large batches like Planet's SuperDove).

---

7. Documenting the Trade Decision

Every trade study must produce an **auditable decision record**. NASA SEH Process 17 requires documentation of:

Table 1.4.14 — Eight elements required in an auditable Process 17 trade-decision record.

Element	Content	Purpose
Decision statement	What was decided	Clarity
Alternatives considered	All options evaluated, including rejected ones	Completeness
Criteria and weights	What mattered and how much	Transparency
Scoring rationale	Why each alternative received its score	Auditability
Result and recommendation	The selected alternative and its total score	Decision record
Sensitivity analysis	Robustness of the result	Confidence
Risks of selected option	What could go wrong with the chosen approach	Risk awareness
Responsible person and date	Who decided and when	Accountability

Industry Practice: The Canadian Hydrographic Service evaluated commercial SAR data (ICEYE, Capella) versus a dedicated satellite for Arctic maritime surveillance. The trade study documented 8 alternatives across 11 criteria, with weights set by a stakeholder panel including DND, Transport Canada, and CCG. The decision to use RCM data supplemented by commercial SAR tasking -- rather than building a new satellite -- saved an estimated $100M while meeting 90% of the maritime domain awareness requirements.

Part 3: Concept of Operations

8. Mission Architecture

8.1 Three Segments of a Space Mission

Every space mission comprises three segments that must be designed together:

Table 1.4.15 — Three space-mission segments listing components and key design drivers.

Segment	Components	Key Design Drivers
Space Segment	Platform (bus): EPS, AOCS, OBC, thermal, structure, propulsion. Payload: instrument(s). Communications: TT&C + payload data link.	Mass, power, volume, orbit, lifetime
Ground Segment	Ground Operations: commanding, telemetry, orbit determination. Payload Data Centre: data reception, processing (L0-L1-L2-L3), archive.	Contact time, data volume, processing capacity
User Segment	Data products and services. APIs, portals, archives. Training and documentation.	Latency, format, accessibility, user capacity

8.2 Data Interfaces Between Segments

Table 1.4.16 — Data interfaces between space, ground and user segments with bands and content.

Interface	Direction	Band/Protocol	Content
TM/TC	Space <-> Ground Ops	S-band (typical)	Housekeeping telemetry, telecommands
Payload data	Space -> Data Centre	X-band or Ka-band	Science/imagery data (high volume)
Orbit/TLE	Ground Ops -> Data Centre	Network	Geolocation metadata for data products
Data products	Data Centre -> Users	Internet/API	Processed imagery (L2/L3), analytics

8.3 Interactive Architecture Diagram

SpaceCDF provides a **drag-and-drop architecture diagram editor** in the ConOps tab:

Table 1.4.17 — SpaceCDF architecture-diagram symbol legend mapping icons to segment types.

Symbol	Type	Represents
Satellite (blue)	`satellite`	Space segment (spacecraft + payload)
Ground Station (green)	`groundStation`	Ground receiving station with antenna
Processing (cyan)	`processing`	Data processing, MCC, archive
User (amber)	`user`	End user / data consumer
Sensor (orange)	`sensor`	Ground sensor, IoT device, in-situ instrument
GNSS/External (purple)	`gnss`	External system (GNSS, relay sat, other constellation)

Exercise. *In the ConOps tab, build your mission architecture diagram. Add all segments, label all connections with data type and frequency band.*

---

9. Mission Phases and Operational Modes

9.1 Operational Mission Phases

Table 1.4.18 — Typical CubeSat mission phases with duration, activities and key risks.

Phase	Duration (typical CubeSat)	Activities	Key Risks
LEOP	1--3 days	Deployment, antenna deploy, first contact, initial health check	Deployment failure, tumbling, no contact
Commissioning	2--4 weeks	Subsystem checkout, calibration, first light, orbit determination	Anomalies, calibration issues
Nominal Ops	Months to years	Routine acquisition, downlink, orbit maintenance	Component degradation, anomalies
Extended Ops	Beyond design life	Continued operations with degraded performance	Solar array degradation, propellant depletion
Disposal	Days to months	Passivation, deorbit manoeuvre or natural decay	Failure of deorbit system

9.2 Operational Modes and Power Budgets

Each mode defines which subsystems are active, the pointing configuration, power demand, and data flow:

Table 1.4.19 — Operational modes for a 3U CubeSat listing active subsystems, pointing, power and data flow.

Mode	Subsystems Active	Pointing	Power (3U typical)	Data Flow
Safe	EPS, OBC, TTC (beacon), AOCS (coarse)	Sun-pointing	~1--2 W	Beacon only
Idle	EPS, OBC, AOCS (standby), TTC (beacon)	Inertial hold	~2 W	Health TM periodic
Science/Imaging	+ Payload, AOCS (fine pointing)	Nadir or target	~6 W	Instrument -> OBC storage
Downlink	+ TTC (full TX power)	Ground station track	~8 W	OBC -> TX -> GS
Eclipse	EPS (battery), OBC, TCS (heaters), AOCS	Inertial hold	~3 W (battery)	None
Orbit Maintenance	+ Propulsion	Thrust direction	~7 W	Manoeuvre TM

9.3 Duty Cycling and Power Analysis

CubeSats have limited power generation (7--25 W for a 3U with deployable panels). Not all modes can run simultaneously. The orbit timeline determines what can happen when:

Typical 95-minute orbit at 500 km SSO: - 60 min sunlight, 35 min eclipse - ~10 min imaging per orbit (~10% duty cycle) - ~8 min downlink per pass (1--2 passes/day over a single ground station) - ~42 min idle - 35 min eclipse (battery-powered)

> **Key Equations -- Power Budget:**
>
> Orbit-average power:
>
> $$\mathrm{P_{avg}} = \sum_{\mathrm{i}} \mathrm{P_{mode,\,i}} \times \mathrm{DC_i}$$
>
> Where $\mathrm{DC_i}$ is the duty cycle (fraction of orbit) for each mode.
>
> Solar array sizing:
>
> $$\mathrm{P_{SA}} = \mathrm{P_{sunlight}} + \frac{\mathrm{P_{eclipse}} \times \mathrm{t_{eclipse}}}{\mathrm{t_{sunlight}} \times \eta_{\mathrm{charge}}}$$
>
> Where $\eta_{\mathrm{charge}} \approx 0.9$ (battery charge efficiency).
>
> Battery sizing:
>
> $$\mathrm{E_{battery}} = \frac{\mathrm{P_{eclipse}} \times \mathrm{t_{eclipse}}}{\mathrm{DoD} \times \eta_{\mathrm{discharge}}}$$
>
> Where DoD is the maximum depth of discharge (typically 0.2--0.3 for long life, up to 0.5 for short missions) and $\eta_{\mathrm{discharge}} \approx 0.95$.

[Source: Wertz et al.; ECSS-E-ST-20C power budget methodology]

---

10. Data Flow Pipeline

10.1 End-to-End Data Flow

```
Instrument -> Onboard Storage -> Downlink -> Ground Reception
  -> Processing (L0 -> L1 -> L2) -> Archive -> User Delivery
```

Table 1.4.20 — End-to-end data-flow stages with their key parameter and sizing driver.

Stage	Key Parameter	Sized By
Data generation	GB/day	Payload data rate $\times$ imaging duty cycle
Onboard storage	GB	Must hold ≥ 1 day of data ($2\times$ for margin)
Downlink per pass	GB/pass	Link data rate $\times$ contact time per pass
Processing	hours	Algorithm complexity, compute infrastructure
User delivery	hours	Archive API, network bandwidth

10.2 Data Budget Balance

> **Key Equation:** For the system to be sustainable, daily downlink capacity must equal or exceed daily data generation:
>
> $$\mathrm{C_{downlink}} = \mathrm{R_{data}} \times \mathrm{N_{passes}} \times \mathrm{t_{contact}} \times \eta_{\mathrm{protocol}} \geq \mathrm{G_{daily}}$$
>
> Where: - $\mathrm{R_{data}}$ = data link rate (Mbps) - $\mathrm{N_{passes}}$ = number of ground station passes per day - $\mathrm{t_{contact}}$ = average contact time per pass (seconds) - η_{protocol} = protocol overhead factor (~0.8 for CCSDS framing) - $\mathrm{G_{daily}}$ = daily data generation (Mb)
>
> If $\mathrm{C_{downlink}} < \mathrm{G_{daily}}$, data accumulates on board and storage eventually fills. Solutions: - Increase $\mathrm{R_{data}}$ (higher TX power, higher frequency band, better antenna) - Increase $\mathrm{N_{passes}}$ (more ground stations, polar ground station for SSO) - Increase $\mathrm{t_{contact}}$ (higher altitude increases contact time but worsens resolution) - Decrease $\mathrm{G_{daily}}$ (lower duty cycle, on-board compression, selective downlink)

SpaceCDF exercise: *Check the Data Budget on the Dashboard. Does your design balance? If not, which parameter would you change first?*

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11. ConOps Tool Exercise

1. Navigate to the **ConOps** tab in SpaceCDF
2. Review and edit the **mission architecture diagram** -- add all segments and connections
3. Edit the **mission phases**: adjust durations for your mission type
4. Review the **operational modes**: are the right modes defined? Adjust power values.
5. Check the **data flow pipeline**: does downlink capacity balance data generation?

Complete Worksheet 1.4, Parts E and F: Document your ConOps outline and calculate the orbit-average power budget.

1U Worked Example: UniSat-1

Trade Study: Why 1U instead of 2U or 3U for UniSat-1?

The UniSat-1 team must justify the choice of a 1U form factor for their MEMS magnetometer technology demonstration. This is a classic Process 17 (Decision Analysis) exercise.

Decision statement: "What CubeSat form factor best supports a MEMS magnetometer technology demonstration within the university's budget and schedule constraints?"

Alternatives:

Table 1.4.21 — UniSat-1 form-factor alternatives compared on mass, volume, cost and development time.

Alternative	Mass Limit	Internal Volume	Typical Cost	Dev Time
1U	2.00 kg	~1000 cm^3	50--200 kEUR	6--12 months
2U	4.00 kg	~2000 cm^3	100--400 kEUR	12--18 months
3U	6.0 kg	~3000 cm^3	200--800 kEUR	18--24 months

Criteria and scoring:

Table 1.4.22 — Weighted trade-study scoring of 1U, 2U and 3U for the UniSat-1 magnetometer demonstrator.

Criterion	Weight	1U	2U	3U	Rationale
Cost	0.35	1.0	0.5	0.2	University budget is 150 kEUR total
Schedule	0.25	1.0	0.6	0.3	Must launch within 12 months
Payload fits	0.20	0.8	1.0	1.0	MEMS sensor is 50 g, 0.2 W -- fits easily in 1U
Design simplicity	0.10	1.0	0.7	0.5	Smaller team, fewer subsystems
Data return	0.10	0.5	0.7	1.0	More volume allows better comms, but 9600 bps is sufficient for < 1 kbps payload
Weighted Total		**0.91**	**0.67**	**0.50**	

Result: 1U wins decisively. The MEMS magnetometer payload (50 g, 0.2 W, < 1 kbps) has no need for the extra volume, mass, or power that 2U/3U would provide. The additional cost and schedule of a larger bus are unjustified.

Sensitivity check: Even if cost weight drops from 0.35 to 0.15 (and schedule from 0.25 to 0.15, redistributing to data return), 1U still wins (0.76 vs 0.72 vs 0.73 for 1U, 2U and 3U). The winner is robust, but the runner-up is not: 3U overtakes 2U once data return dominates.

Key lesson: Do not over-design the bus for a simple payload. The 1U form factor imposes healthy constraints that force the team to focus on the mission objective rather than adding unnecessary capability.

Session Summary

Table 1.4.23 — Session summary: key takeaways for mission needs, stakeholder analysis and trade studies.

Topic	Key Takeaway
Problem statement	Defines WHAT is needed, not HOW to solve it; answers what, who, impact, constraints
Stakeholder analysis	Identify all parties with legitimate interest; capture needs, constraints, priority, influence
Objectives	Must be specific, measurable, and traceable to stakeholder needs via MoE/MoP/TPM chain
Trade study structure	5 elements: decision, alternatives, criteria, weights, scores; must include sensitivity analysis
Space vs non-space	Existing services may already meet the need -- evaluate honestly before committing to build
Decision documentation	Every trade decision must have auditable rationale per Process 17
ConOps	

Topic	Key Takeaway
	Three segments (space, ground, user); operational modes drive power budget via duty cycling
Data pipeline	Daily downlink capacity must equal or exceed daily data generation

References

1. NASA, Systems Engineering Handbook (SP-2016-6105 Rev 2), 2016
2. NASA, NPR 7123.1D -- Systems Engineering Processes and Requirements, 2020
3. ECSS, ECSS-E-ST-10C -- System Engineering General Requirements, 2009
4. Wertz, J.R. et al., Space Mission Engineering: The New SMAD, Microcosm Press, 2011
5. NASA SEH Appendix C -- How to Write a Good Requirement
6. NASA SEH Appendix S -- ConOps Outline
7. Saaty, T.L., "The Analytic Hierarchy Process", McGraw-Hill, 1980

Chapter 2.1 — The System-V and Requirements Engineering

> **Expected reading before this session.** NASA SEH Appendix C — How to Write a Good Requirement (≈ 30 min); ECSS-E-ST-10C §5.5.

Prerequisites: Week 1 complete (mission need, stakeholder analysis, ConOps defined) **SpaceCDF Tab:** Requirements

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References

- NASA, *Systems Engineering Handbook* (NASA/SP-2016-6105 Rev 2), 2016, Ch. 4.2 & Appendix C
- ECSS, *ECSS-E-ST-10-06C: Technical Requirements Specification*, 2009, Sec. 5
- ECSS, *ECSS-E-ST-10C Rev.1: Space Engineering -- System Engineering General Requirements*, 2017, Sec. 5.2
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 3--4
- INCOSE, *Systems Engineering Handbook*, 5th ed., 2023, Ch. 2.3.5

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the System-V lifecycle model and how requirements flow down the left side while verification flows up the right side
2. Write requirements that satisfy the SMART criteria and express WHAT, not HOW
3. Classify requirements by type: functional, performance, interface, constraint, and environmental
4. Construct a three-level requirement hierarchy (mission, system, subsystem) with bidirectional traceability
5. Assign verification methods (ATIR(D)) to requirements and justify each choice
6. Use SpaceCDF's Requirements tab to generate, validate, and manage requirements

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1. The System-V Model — Requirements Perspective

The **System-V** (or "Vee model") was introduced in Chapter 1.1 §2 as the canonical systems-engineering lifecycle (see the System-V figure in Chapter 1.1). This section revisits the V from a *requirements-engineering perspective*: each level on the left branch decomposes the *requirements* one level finer, and each level on the right branch contains the *verification or validation activity* that closes the requirement at that level.

[Source: NASA SEH Fig. 3-1; ECSS-E-ST-10C Rev.1 Fig. 5-1; INCOSE SEH 5th ed. Fig. 2.7]

Key insight: the V is not a timeline — it is a *correspondence map*. The left branch asks "what must be done?" at increasing levels of detail; the right branch asks "did we do it?" at corresponding levels. Each horizontal row is a matched pair (requirement ↔ verification).

Phase Mapping to the V

Table 2.1.1 — Mapping of V-model levels to decomposition activities, verification activities and NASA project phases.

V-Level	Left Branch (Decomposition)	Right Branch (Verification)	NASA Phase
Top	Stakeholder needs & objectives	Operational validation (does the mission deliver value?)	Pre-A / E
Mission	Mission-level requirements	Mission-level V&V (commissioning, in-orbit checkout)	A / E
System	System-level requirements	System integration & test (end-to-end)	B / D
Subsystem	Subsystem requirements	Subsystem qualification & acceptance test	C / D
Bottom	Implementation (detailed design, build)	Component acceptance	C-D

Real Mission Example: Planet SuperDove

Planet's SuperDove constellation (2020--present) illustrates the V-model in commercial practice:

Table 2.1.2 — Planet SuperDove example showing requirements paired with verification at each V level.

Level	Requirement (Left)	Verification (Right)
Stakeholder	"Daily global coverage at <5 m for agricultural analytics"	In-orbit validation: coverage maps, user feedback
Mission	"Revisit <= 1 day at +-60deg latitude"	Constellation coverage simulation confirmed
System	"GSD <= 3.7 m at 475 km SSO"	End-to-end imaging test from orbit
Subsystem	"Telescope aperture >= 90 mm, 8 spectral bands"	Instrument calibration on ground + in-orbit

[Source: Planet Labs, "Planet Imagery Product Specifications," 2023]

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2. What is a Requirement?

[Source: NASA SEH Sec. 4.2; ECSS-E-ST-10-06C Sec. 5.2]

A requirement is a **formal, verifiable statement of what the system must do or how well it must perform**. It is expressed as a "shall" statement with a measurable threshold.

The Shall Convention

Requirements use contractually precise language:

Table 2.1.3 — Shall, should and will keywords and their contractual force in requirement statements.

Keyword	Meaning	Contractual Force
"Shall"	Mandatory requirement -- must be met	Binding

Keyword	Meaning	Contractual Force
"Should"	Goal or guideline -- desired but not mandatory	Advisory
"Will"	Statement of fact or intent -- not a requirement	Informational

Example:

> "The system **shall** achieve a ground sample distance of 10 m or better at nadir from the operational orbit."

This is testable: GSD can be computed from optical geometry and verified from calibration imagery.

Requirement Types

Table 2.1.4 — Five requirement types (functional, performance, interface, constraint, environmental) with examples.

Type	Definition	Example
Functional	What the system must *do* (a capability)	"The system shall acquire multispectral imagery of the target area"
Performance	How *well* it must do it (quantified)	"The system shall achieve GSD <= 10 m at nadir"
Interface	Boundary agreements between elements	"The EPS shall provide 5.0 V +/- 0.25 V regulated bus to all subsystems"
Constraint	Non-negotiable external limits	"The system total mass shall not exceed 6.0 kg"
Environmental	Survival conditions	"The system shall survive launch loads of 9 g axial and 4 g lateral"

[Source: NASA SEH Appendix C, Table C-1; ECSS-E-ST-10-06C Sec. 5.2.2]

Requirements vs. Design Choices

Table 2.1.5 — Side-by-side examples separating WHAT requirements from HOW design choices.

Requirement (WHAT)	Design Choice (HOW)
"The system shall achieve GSD <= 10 m"	"Use a 150 mm aperture telescope"
"The system shall survive launch loads"	"Use aluminium 7075-T6 structure"
"The system shall provide >= 3 dB link margin"	"Use S-band with 2 W transmitter"
"The system shall deorbit within 5 years of EOL"	"Include cold gas propulsion system"
"The system shall operate for >= 3 years"	"Use rad-tolerant components"

Key principle: Requirements constrain the solution space without specifying the solution. This preserves design freedom for trade studies.

---

3. The SMART Framework

While NASA Appendix C provides a detailed quality checklist, the SMART acronym serves as a rapid quality screen:

Table 2.1.6 — SMART criteria with test questions to screen requirement quality.

Letter	Meaning	Test Question
S	Specific	Does it address exactly one concern? Is it unambiguous?
M	Measurable	Does it have a numeric threshold with units? Can it be verified?
A	Achievable	Is it technically feasible with current or near-term technology?
R	Relevant	Does it trace to a stakeholder need or higher-level objective?
T	Traceable	Can you identify its parent (where it came from) and its children (how it will be verified)?

NASA SEH Appendix C: Full Quality Checklist

[Source: NASA SEH Appendix C -- "How to Write a Good Requirement"]

The full NASA checklist adds criteria beyond SMART:

1. **Single requirement per statement** -- no compound "and" requirements
2. **Positive form** -- state what the system SHALL DO, not what it shall NOT do
3. **Active voice** -- "The system shall..." not "It is required that..."
4. **No implementation** -- avoid naming specific hardware, software, or methods
5. **Verifiable** -- must be provable by analysis, test, inspection, or demonstration
6. **No TBDs** -- every threshold must have a value (even if it changes later)
7. **Consistent** -- no contradictions with other requirements in the set
8. **Bounded** -- tolerance or range specified where appropriate

Common Anti-Patterns

Table 2.1.7 — Common requirement anti-patterns with diagnoses and improved rewrites.

Anti-Pattern	Problem	Better Version
"The spacecraft shall operate at 500 km altitude"	Prescribes orbit (design choice)	"The system shall provide global coverage with revisit <= 5 days at +/-60 deg latitude"
"The spacecraft shall use triple-junction GaAs solar cells"	Prescribes technology	"The EPS shall generate >= 15 W EOL with <= 0.5 m^2 array area"
"The system shall use AX.25 protocol"	Prescribes implementation	"The TTC system shall provide reliable command reception with BER <= 10^{-6} "
"The spacecraft shall be a 3U CubeSat"	Prescribes form factor	"The system shall have total mass <= 6 kg and fit within a standard 3U deployer envelope"

Exception: Interface requirements ARE specific because they define contractual boundaries:

> "The EPS shall provide 28 V +/- 2 V regulated bus voltage to all subsystems."

This is acceptable because bus voltage is a negotiated interface agreement, not a unilateral design choice.

---

4. Requirement Hierarchy and Traceability

Requirements exist at multiple levels, each decomposed from the level above. Every requirement must trace bidirectionally.

Hierarchy Structure

```
Stakeholder Need: "Timely agricultural monitoring for food security"
  |
  v derives
Mission Requirement (MR):
  MR-001: "The system shall provide multispectral imagery with
           GSD <= 10 m and revisit <= 5 days over target region"
  |
  v decomposes into
System Requirements (SR):
  SR-PL-001:   "The payload shall achieve GSD <= 10 m at nadir
                from the operational orbit"
  SR-AOCS-001: "The AOCS shall provide pointing accuracy <= 0.1 deg
                (3-sigma) during imaging"
  SR-LINK-001: "The comms system shall downlink >= 5 GB/day"
  |
  v derives
Subsystem Requirements (SSR):
  SSR-PL-001a:   "The telescope aperture shall be >= 80 mm"
  SSR-AOCS-001a: "The star tracker shall provide <= 5 arcsec
                  attitude knowledge (3-sigma)"
  SSR-LINK-001a: "The X-band transmitter shall provide >= 2 W RF power"
```

Level Definitions

Table 2.1.8 — Definitions of mission, system, subsystem and component requirement levels and who verifies each.

Level	Scope	Written By	Verified By
Mission	What the mission achieves (MoE-derived)	Systems engineer + stakeholders	Operational validation
System	What the system must do (MoP-derived)	Systems engineer	System-level integration & test
Subsystem	What each subsystem must provide	Subsystem engineers	Subsystem qualification & acceptance
Component	What each component must meet	Component engineers	Component acceptance testing

Traceability Rules

Every requirement must trace in three directions:

1. **Upward:** to its parent requirement or objective (WHY does this exist?)
2. **Downward:** to child requirements or design parameters (HOW is it decomposed?)
3. **Horizontally:** to a verification method (HOW will it be confirmed?)

This three-axis traceability is captured in the **Requirements Traceability Matrix (RTM)**.

[Source: NASA SEH Sec. 6.2 (Process 11: Requirements Management); ECSS-E-ST-10C Sec. 5.2]

Splitting Compound Requirements

A compound requirement that addresses multiple concerns must be split for independent verification:

Table 2.1.9 — Compound requirements rewritten as independently verifiable split statements.

Compound (Poor)	Split (Correct)
"The system shall provide 10 m GSD with 5-day revisit and 24 h latency"	MR-001: "GSD <= 10 m" + MR-002: "Revisit <= 5 days" + MR-003: "Latency <= 24 h"
"The EPS shall provide positive power margin in all modes including eclipse"	SR-PWR-001: "Positive margin in sunlit modes" + SR-PWR-002: "Positive margin in eclipse" + SR-PWR-003: "Battery DOD <= 30% in worst-case eclipse"

Rationale: Each split requirement can be verified independently. If GSD passes but revisit fails, the team knows exactly what to fix without ambiguity.

---

5. Verification Methods -- ATIR(D)

[Source: ECSS-E-ST-10-02C Rev.1 §5.2.2; NASA SEH Sec. 5.3]

Every requirement must have an assigned verification method. ECSS-E-ST-10-02C Rev.1 §5.2.2.1 defines four methods -- test, analysis, review of design and inspection -- and counts demonstration as part of test; NASA practice lists demonstration as a separate fifth method. This book abbreviates the set as ATIR(D), with the D in parentheses for that reason:

Table 2.1.10 — Analysis, Test, Inspection, Review-of-design and Demonstration verification methods with usage guidance.

Method	Code	What It Proves	When Used
Analysis	A	Requirement satisfied by mathematical model, simulation, or similarity	Phase B--C; when testing is impractical or cost-prohibitive
Test	T	Requirement satisfied by physical measurement	Phase C--D; when proof requires hardware in a controlled environment
Inspection	I	Requirement confirmed by visual or physical examination	Physical characteristics (dimensions, labels, markings, mass)
Review of design	R		

Method	Code	What It Proves	When Used
		Requirement confirmed by examination of drawings, schematics or documentation	Design features that can be established from the records (e.g., use of approved parts, redundancy present)
Demonstration	D	Requirement confirmed by operational exercise	Functional requirements proven by executing procedures

Verification Method Selection Guide

Table 2.1.11 — Recommended verification method by requirement type with rationale.

Requirement Type	Typical Method	Rationale
Mass <= X kg	**I** (weigh it)	Direct physical measurement
Pointing <= Y deg	**A** (simulation) + **T** (TVAC/HITL)	Analysis first, confirmed by test
Link margin >= 3 dB	**A** (link budget analysis)	Full test requires satellite in orbit
Survival at launch loads	**T** (vibration test)	Must physically prove structural integrity
Data latency <= 24 h	**D** (end-to-end ops exercise)	Pipeline is procedural -- demonstrate it works
Operating temperature range	**T** (thermal vacuum test)	Thermal environment must be simulated
Software FDIR logic	**D** (fault injection test)	Demonstrate correct autonomous response

Verification by Project Phase

Table 2.1.12 — Verification activities mapped to NASA project phases B through E.

Phase	Verification Activities
B	Analysis verification (models, simulations, budgets, trade studies)
C	Design verification (detailed analysis, breadboard testing, qualification)
D	Acceptance testing (flight hardware), system integration, environmental test
E	In-orbit validation (commissioning, calibration, operational checkout)

---

6. Worked Example: 3U Earth Observation CubeSat

Worked Example -- Deriving Requirements for a 3U EO CubeSat

Stakeholder need: "Monitor crop health across the Canadian prairies with weekly updates."

Step 1 -- Mission requirement: MR-001: "The system shall provide NDVI-capable imagery with GSD <= 10 m and revisit <= 7 days over 49--54 deg N, 100--115 deg W."

Step 2 -- System requirements (decomposed): - SR-PL-001: "The payload shall acquire imagery in at least RED (630--690 nm) and NIR (760--900 nm) bands." - SR-AOCS-001: "The AOCS shall provide pointing accuracy <= 0.5 deg (3-sigma) during imaging." - SR-LINK-001: "The comms system shall

downlink >= 2 GB per day to a Canadian ground station." - SR-PWR-001: "The EPS shall provide positive power margin in all operational modes."

Step 3 -- Subsystem requirements (derived): - SSR-PL-001a: "The telescope aperture shall be >= 80 mm." [Derived from GSD + altitude] - SSR-AOCS-001a: "The star tracker shall provide <= 30 arcsec attitude knowledge." [Derived from pointing budget] - SSR-LINK-001a: "The X-band transmitter shall provide >= 2 W RF output power." [Derived from link budget]

Step 4 -- Verification assignment:

Table 2.1.13 — Verification method, phase and rationale for sample 3U EO CubeSat requirements.

Requirement	Method	Phase	Rationale
MR-001 (GSD + revisit)	A + D	B + E	Analysis during design; demonstrated from orbit
SR-PL-001 (spectral bands)	T	D	Spectral calibration in lab
SR-AOCS-001 (pointing)	A + T	B + D	Simulation then hardware-in-loop test
SR-LINK-001 (downlink)	A	B	Link budget analysis
SSR-PL-001a (aperture)	I	D	Physical measurement

1U Worked Example: UniSat-1

Worked Example -- Deriving Requirements for a 1U Technology Demonstrator

Stakeholder need: "Demonstrate the feasibility of a MEMS-based magnetometer for space weather monitoring in LEO."

Step 1 -- Mission requirements:

ID	Requirement	Type	Rationale
MR-001	"The system shall measure the local magnetic field vector with resolution <= 10 nT at 1 Hz sampling rate for a minimum of 6 months."	Performance	Minimum science return for technology validation
MR-002	"The system total mass shall not exceed 2.00 kg."	Constraint	CDS Rev 14, 1U mass limit
MR-003	"The system shall downlink at least 1 MB of magnetometer data per day."	Performance	Sufficient for statistical analysis of sensor performance
MR-004	"The system shall operate for a minimum of 6 months in LEO."	Performance	Minimum mission duration for seasonal variation coverage

Step 2 -- System requirements (decomposed): - SR-PWR-001: "The EPS shall provide orbit-average power >= 2 W to all subsystems." - SR-PWR-002: "The battery shall provide >= 10 Wh capacity." - SR-LINK-001: "The comms system shall provide a downlink data rate >= 9600 bps." - SR-LINK-002: "The comms system shall achieve link margin >= 3 dB at 10 deg minimum elevation." - SR-OBC-001: "The

OBC shall consume <= 0.5 W average power." - SR-STR-001: "The structure shall comply with CDS Rev 14, 1U envelope (100 x 100 x 113.5 mm)."

Step 3 -- Subsystem requirements (derived): - SSR-PWR-001a: "Body-mounted solar cells shall generate >= 2 W orbit-average power accounting for eclipse and geometry." [Derived from power budget] - SSR-LINK-001a: "The UHF transmitter shall provide >= 0.5 W RF output at 437 MHz." [Derived from link budget]

Step 4 -- Verification assignment:

Requirement	Method	Phase	Rationale
MR-001 (magnetometer performance)	T + D	D + E	Ground calibration then in-orbit demonstration
MR-002 (mass <= 2.00 kg)	I	D	Weigh the flight unit
SR-PWR-001 (orbit-avg power)	A	B	Power budget analysis
SR-LINK-001 (9600 bps downlink)	A + D	B + E	Link budget analysis; demonstrated from orbit
SR-STR-001 (CDS compliance)	I	D	Physical measurement against CDS template

Key difference from 3U: The 1U requirement set is much smaller (~15--20 requirements vs ~40--60 for a 3U EO mission). There are no pointing requirements, no imaging requirements, and no propulsion requirements. This makes the verification campaign significantly simpler and cheaper.

---

7. SpaceCDF Exercise

Instructions

1. Navigate to the **Requirements** tab in SpaceCDF
2. Click **"Generate from Objectives"** -- the AI agent generates SMART requirements from your mission objectives and ConOps
3. For each generated requirement: - Review the SMART validation badges (green = pass, amber = warning, red = fail) - Check: does it say WHAT not HOW? Is it a single concern? - **Accept**, **Edit**, or **Reject** each one
4. Use the **Level filter** (Mission / System / Subsystem) to view by hierarchy level
5. Use the **Type filter** (Functional / Performance / Interface / Constraint) to classify
6. Navigate to the **V&V Matrix** sub-panel to assign verification methods (A/T/I/D) to each accepted requirement

Exercise Tasks

1. Generate requirements for your team's mission
2. Identify at least one requirement that specifies HOW (implementation) -- rewrite it as WHAT
3. Split any compound requirements into individual testable statements
4. Classify 5 requirements by type (functional, performance, interface, constraint, environmental)
5. For 5 key requirements, assign verification method (A/T/I/D) and target phase (B/C/D/E)
6. Complete Worksheet 2.1

Key Equations Reference

> **There are no equations in requirements engineering per se**, but the following relationship governs traceability completeness:

$$\text{Coverage} = \frac{\text{N}_{\text{verified requirements}}}{\text{N}_{\text{total requirements}}} \times 100\%$$

> Target: 100% coverage -- every requirement must have at least one assigned verification method and one parent traceability link.

Session Summary

Table 2.1.14 — Session summary: key takeaways for the System-V and requirements engineering.

Topic	Key Takeaway
System-V model	Left branch decomposes requirements; right branch integrates and verifies at matching levels
Shall statements	Formal, verifiable, single-concern statements
WHAT not HOW	Requirements preserve design freedom; design choices come later in trade studies
SMART	Specific, Measurable, Achievable, Relevant, Traceable
Requirement types	Functional, performance, interface, constraint, environmental
Hierarchy	Mission -> System -> Subsystem with bidirectional traceability (up, down, horizontal)
Splitting	One concern per requirement for independent verification
ATIR(D)	Analysis, Test, Inspection, Review of design (+ Demonstration) -- assigned per requirement per phase

Chapter 2.2 — Functional Decomposition and Allocation

> **Expected reading before this session.** NASA SEH §4.3 (≈ 30 min); ECSS-E-ST-10C §5.6.

Prerequisites: Chapter 2.1 (requirements defined and validated) **SpaceCDF Tab:** Functions

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References

- NASA, *Systems Engineering Handbook* (NASA/SP-2016-6105 Rev 2), 2016, Sec. 4.3 (Process 3: Logical Decomposition)
- ECSS, *ECSS-E-ST-10C Rev.1: System Engineering General Requirements*, 2017, Sec. 5.3
- INCOSE, *Systems Engineering Handbook*, 5th ed., 2023, Ch. 2.3.5.3
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 2
- Blanchard & Fabrycky, *Systems Engineering and Analysis*, 5th ed., 2010, Ch. 3

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the purpose of functional decomposition in bridging requirements to design
2. Decompose mission objectives into a hierarchical function tree (3+ levels)
3. Categorise functions by type (observe, communicate, navigate, point, power, protect, store, process, support)
4. Allocate functions to subsystem domains, including multi-allocation for shared responsibilities
5. Derive requirements from functional analysis
6. Identify and resolve coverage gaps (leaf functions without linked requirements)
7. Use SpaceCDF's function tree editor to build and validate a function architecture

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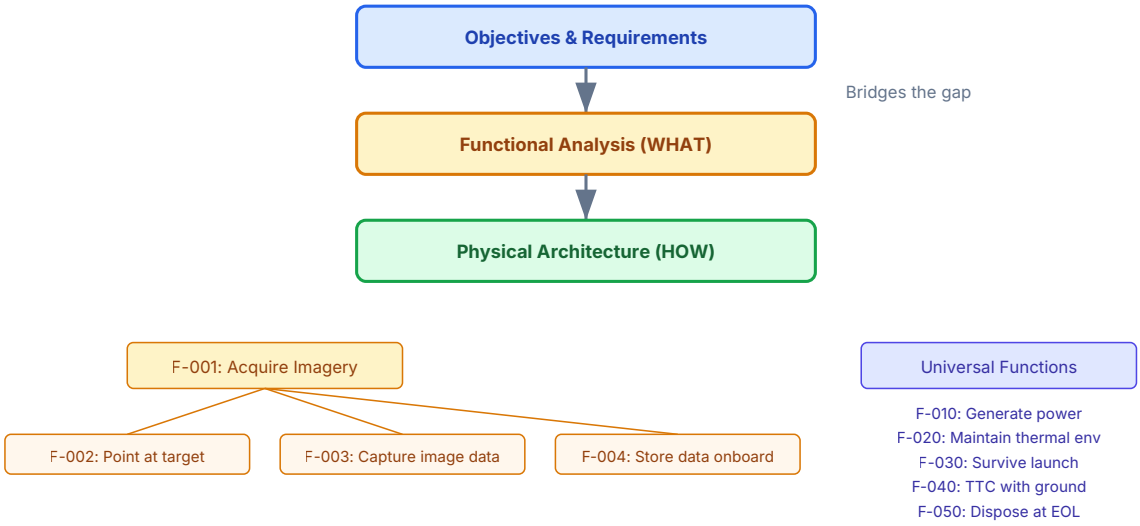
1. What is Functional Decomposition?

[Source: NASA SEH Sec. 4.3 -- Process 3: Logical Decomposition]

Functional decomposition answers: **"What must the system DO to satisfy the requirements?"**

It creates a bridge between requirements (WHAT the system must achieve) and physical design (HOW it will be built). Functions describe *actions* the system must perform, without specifying the physical hardware or software that will perform them.

The Decomposition Flow



Function Types

Every spacecraft function falls into one of these categories:

Table 2.2.1 — Nine spacecraft function types with characteristic verb and example actions.

Type	Verb	Description	Example
Observe	Sense/measure	Acquire data from the environment	Capture multispectral imagery, receive AIS signals
Communicate	Transfer	Move information between elements	Downlink science data, uplink telecommands
Navigate	Determine/control position	Know or change the orbit	Perform orbit maintenance manoeuvre
Point	Orient	Control spacecraft attitude	Nadir pointing during imaging, target tracking
Power	Generate/store/ distribute	Provide electrical energy	Solar power generation, battery charge management
Protect	Maintain environment	Keep components within limits	Thermal control, radiation shielding
Store	Record	Buffer data for later use	Onboard solid-state data recording
Process	Transform	Convert data between forms	On-board image compression, L0 framing
Support	Provide structure	Mechanical integrity	Survive launch loads, deploy mechanisms

Universal Functions

Regardless of mission type, **every spacecraft** must perform these five functions:

1. **Generate electrical power** (solar, RTG, battery)

2. **Maintain thermal environment** (passive/active thermal control)
3. **Survive the launch environment** (structural loads, vibration, shock)
4. **Communicate with ground for TTC** (telemetry, tracking, command)
5. **Dispose of the spacecraft at end of life** (deorbit, graveyard, passivation)

SpaceCDF auto-generates these universal functions for every mission.

---

2. Mission-Type-Specific Function Trees

Different mission types produce fundamentally different primary function trees. The universal functions remain the same; it is the mission-specific branch that differentiates.

Earth Observation (Optical/SAR)

```
F-001: Acquire imagery of target area
  +-- F-002: Point instrument at target
  +-- F-003: Capture image data (expose detector / SAR pulse)
  +-- F-004: Store acquired data onboard
  +-- F-005: Downlink data to ground station
```

Real example -- Planet SuperDove: F-001 decomposes into 8-band pushbroom acquisition, on-board radiometric correction, lossless compression, and X-band downlink. Each sub-function traces to specific SuperDove requirements (3.7 m GSD, 8 spectral bands, 200 Mbps downlink).

Communications Relay (Store-and-Forward or Bent-Pipe)

```
F-001: Relay communications between users
  +-- F-002: Receive uplink signal from user terminal
  +-- F-003: Process and route data (store-and-forward or bent-pipe)
  +-- F-004: Transmit downlink signal to destination terminal or gateway
```

Real example -- Astrocast (IoT): F-001 decomposes into L-band receive (from IoT devices), on-board message deduplication and store-and-forward, and UHF/S-band downlink to gateway stations. Each message is <= 160 bytes.

[Source: Astrocast, "Astrocast Network Overview," astrocast.com]

AIS/IoT Receiver (Passive)

```
F-001: Receive and process signals of interest
  +-- F-002: Receive AIS/IoT signals (passive -- no uplink transmission)
  +-- F-003: Decode and validate messages
  +-- F-004: Store processed data onboard
  +-- F-005: Downlink to ground for distribution
```

Ground Segment Functions

Missions that include ground-side processing need ground-domain functions:

```
F-G01: Receive satellite data at ground station
F-G02: Process data pipeline (L0 -> L1 -> L2 products)
F-G03: Archive and distribute data products to users
F-G04: Operate mission control centre (planning, commanding, monitoring)
```

SpaceCDF supports allocation to ground domains: `ground_station`, `ground_processing`, `ground_sensor`.

---

3. Allocation to Subsystems

Each function must be **allocated** to one or more responsible subsystem domains. This allocation defines system boundaries and, critically, identifies **interfaces** wherever a function is shared.

Allocation Rules

Table 2.2.2 — Typical subsystem allocation for common functions and whether multi-allocation applies.

Function	Typical Allocation	Multi-allocation?
Acquire imagery	Payload	No (single owner)
Point at target	AOCS	No
Store data	OBC/Data Handling	No
Downlink data	Comms + AOCS	**Yes** -- comms for RF chain, AOCS for antenna pointing
Generate power	EPS	No
Maintain thermal	Thermal	No
Relay communications	Payload + Comms	**Yes** -- boundary depends on architecture
Survive launch	Structure	No
Dispose at EOL	Propulsion (or Ops)	No (or multi if drag-sail)

Multi-Allocation and Interface Identification

When a function is allocated to more than one subsystem, it creates an **interface** that must be explicitly managed.

Example -- "Downlink data to ground station":

Table 2.2.3 — Multi-allocation of the downlink function across Comms, AOCS and Data Handling with resulting interfaces.

Responsible Subsystem	Contribution	Interface Created
Comms (Link)	Transponder, antenna, modulation, RF chain	Comms <-> AOCS (antenna pointing)
AOCS	Antenna pointing towards ground station during pass	Comms <-> Data Handling (packet routing)
Data Handling	Data packaging, prioritisation, CCSDS framing	

Responsible Subsystem	Contribution	Interface Created
		Data Handling <-> AOCS (pass scheduling)

SpaceCDF supports multi-allocation by entering comma-separated domains.

Discussion point. *For each multi-allocated function in your design, where should the system boundary be drawn? Who "owns" the function, and who is a "contributor"?*

Derived Requirements from Functions

Each allocated function generates **derived requirements** -- requirements not explicitly stated by stakeholders but necessary for the function to work:

Table 2.2.4 — Derived requirements generated by representative functions with derivation logic.

Function	Derived Requirement	Derivation Logic
"Point instrument at target"	"AOCS shall provide <= 0.1 deg pointing accuracy during imaging"	From GSD requirement + optical geometry
"Store data onboard"	"OBDH shall provide >= 32 GB solid-state storage"	From data rate x orbit period x missed-pass margin
"Downlink within daily contact window"	"TX shall provide >= 50 Mbps effective data rate"	From daily data volume / total contact time
"Generate power in all modes"	"SA shall produce >= 15 W EOL"	From worst-case sunlit power demand + recharge

[Source: NASA SEH Sec. 4.3.3 -- derived requirements from functional allocation]

---

4. Performance Criteria and Coverage Analysis

Each leaf function (a function with no sub-functions) must have **performance criteria** -- quantitative thresholds that define "how well" the function must be performed:

Table 2.2.5 — Quantitative performance criteria assigned to representative leaf functions.

Function	Performance Criteria
Acquire imagery	GSD <= 10 m; SNR >= 100:1; >= 4 spectral bands
Point at target	Accuracy <= 0.1 deg; stability <= 0.01 deg/s; slew rate >= 1 deg/s
Downlink data	Link margin >= 3 dB; daily throughput >= 5 GB
Generate power	Positive margin in all modes; battery DOD <= 30%
Maintain thermal env	All components within operating range with >= 5 degC margin

Performance criteria form the quantitative basis for subsystem-level requirements.

Coverage Analysis

Every leaf function must trace to at least one requirement. If it does not, there is a **coverage gap** -- the function is defined but has no verification path.

SpaceCDF shows coverage status with badges:

- **Green badge:** Function has linked requirements (covered)
- **Amber badge:** "No requirements" -- coverage gap detected
- **Red badge:** Function conflicts with existing requirements

Real Mission Example: CAPSTONE Coverage Gap

NASA's CAPSTONE mission (2022, Advanced Space) experienced a coverage gap during development: the "maintain attitude during trajectory correction manoeuvre" function initially had no formal pointing requirement during burns. This gap was identified during functional review and led to the derivation of SSR-AOCS-007: "The AOCS shall maintain pointing accuracy <= 2 deg during all propulsive manoeuvres."

[Source: Advanced Space, "CAPSTONE Mission Overview," 2022]

---

5. SpaceCDF Function Tree Exercise

Instructions

1. Navigate to the **Functions** tab in SpaceCDF
2. The tool auto-generates functions based on your selected mission type
3. **Review the generated tree:** Are the functions appropriate for YOUR mission? - For comms missions: should show relay/receive/transmit (not multispectral imagery) - For EO missions: should show acquire/point/store/downlink - For technology demos: should show demonstrate/characterise/report
4. **Edit functions:** - Click **edit** to change name, domain allocation, or performance criteria - Click **+sub** to add sub-functions (decompose further) - Click **x** to remove inappropriate functions
5. **Add ground segment functions** if your mission includes ground processing
6. **Check multi-allocation:** For any function allocated to multiple domains, verify the interface is captured
7. **Coverage check:** Are there any leaf functions (amber badge) without linked requirements? Derive requirements for them.

Exercise Tasks

1. Build a function tree with at least 3 levels of decomposition
2. For each leaf function, write one derived requirement with a measurable threshold
3. Create a function-to-requirement traceability table (minimum 8 entries)
4. Identify at least one multi-allocated function and define the interface boundary
5. Complete Worksheet 2.2

Worked Example: 3U EO CubeSat Function Tree

> **Function Tree for Agricultural Monitoring CubeSat**

``` F-001: Acquire multispectral imagery +-- F-002: Point telescope at target area (AOCS) +-- F-003: Expose detector and capture image frames (Payload) +-- F-004: Compress and store image data (OBC) +-- F-005: Downlink stored data to ground station (Comms + AOCS)

F-010: Generate electrical power (EPS) +-- F-011: Convert solar energy to electrical (SA) +-- F-012: Store energy for eclipse (Battery) +-- F-013: Regulate and distribute power (EPS board)

F-020: Maintain thermal environment (Thermal) +-- F-021: Reject waste heat from electronics (Radiator) +-- F-022: Maintain battery temperature during eclipse (Heater)

F-030: Survive launch environment (Structure) F-040: Communicate with ground for TTC (Comms) F-050: Dispose at end of life (Propulsion/Ops) ```

**Derived requirements from F-002 (Point telescope at target):** - SSR-AOCS-001: "The AOCS shall achieve <= 0.5 deg pointing accuracy during imaging mode" - SSR-AOCS-002: "The AOCS shall achieve pointing stability <= 0.01 deg/s during imaging"

**Multi-allocation for F-005 (Downlink data):** - Comms subsystem: owns RF chain (transponder, antenna, modulation) - AOCS subsystem: provides antenna pointing towards ground station - Interface: AOCS must receive pass-schedule triggers from OBC to initiate slew to ground station pointing

## Session Summary

Table 2.2.6 — Chapter 2.2 session summary of topics and key takeaways.

| Topic | Key Takeaway |
|---|---|
| Functional decomposition | Bridges requirements (WHAT) to design (HOW) by identifying system actions |
| Function types | Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support |
| Universal functions | Power, thermal, launch survival, TTC, disposal -- every spacecraft needs all five |
| Mission-specific trees | EO: acquire/point/store/downlink; Comms: receive/route/transmit; AIS: receive/decode/store |
| Allocation | Each function assigned to subsystem domain(s); multi-allocation creates interfaces |
| Derived requirements | Functions generate new requirements not stated by stakeholders |
| Coverage analysis | Every leaf function must trace to at least one requirement -- no gaps allowed |
```

Chapter 2.3 — Orbit Selection and Mission Architecture

> **Expected reading before this session.** Vallado §2 — Keplerian orbital elements (≈ 45 min); ECSS-U-AS-10C Rev.2 — Space Debris Mitigation Requirements (skim).

Prerequisites: Sessions 2.1--2.2 (requirements and functions defined) **SpaceCDF Tabs:** Mission Architecture, Orbit Trade Advisor

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References

- Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed., 2013, Ch. 2--6
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 5--7
- Curtis, *Orbital Mechanics for Engineering Students*, 4th ed., 2020, Ch. 2--4
- NASA, *Systems Engineering Handbook*, 2016, Sec. 4.4 (Process 4: Design Solution Definition)
- ECSS, *ECSS-U-AS-10C Rev.2: Space Debris Mitigation Requirements*, 2023
- IADC, *Space Debris Mitigation Guidelines*, IADC-02-01 Rev 3, 2021
- FCC, *Report and Order FCC 22-74: Space Innovation*, 2022

Learning Objectives

By the end of this session, participants will be able to:

1. Define and compute the six Keplerian orbital elements
2. Calculate orbital period, velocity, and eclipse fraction for circular orbits
3. Compute the Sun-synchronous inclination using J2 precession
4. Evaluate orbit trade-offs across altitude, coverage, lifetime, radiation, and debris compliance
5. Apply the Hohmann transfer ΔV equations for orbit raising and deorbit
6. Use SpaceCDF's orbit trade advisor with mission-appropriate scoring weights

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1. Keplerian Orbital Elements

Six parameters (the *classical orbital elements*) fully describe the size, shape, and orientation of an orbit, plus the satellite's position on it.

[Source: Vallado, Ch. 2; Curtis, Ch. 2]

Six Keplerian orbital elements with in-place labels

Figure 2.3.1 — The six Keplerian orbital elements (a, e, i, Ω, ω, ν) shown on a single Earth-centred orbit, with each element labelled in place next to the geometric feature it describes.

Table 2.3.1 — Six Keplerian orbital elements with symbols, physical meaning, and typical CubeSat ranges.

Element	Symbol	Physical Meaning	Typical Range
Semi-major axis	a	Size of the orbit	6571--42164 km (LEO to GEO)
Eccentricity	e	Shape (0 = circle, 0 < e < 1 = ellipse)	0--0.001 for LEO circular
Inclination	i	Tilt of orbit plane from equatorial plane	0--180 deg
RAAN	Ω	Orientation of ascending node in equatorial plane	0--360 deg
Argument of perigee	ω	Orientation of perigee within orbit plane	0--360 deg
True anomaly	ν	Satellite position along the orbit	0--360 deg

For **circular LEO** missions (the most common CubeSat orbit type), the key design parameters reduce to three: **altitude** (h), **inclination** (i), and **LTAN** (local time of ascending node, for Sun-synchronous orbits).

Key Equations

Key Equations -- Orbital Mechanics Fundamentals

Orbital period (circular orbit):

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}$$

where $a = R_E + h$ is the semi-major axis (m), $\mu = 3.986 \times 10^{14}$ m^3/s^2 is Earth's gravitational parameter, and $R_E = 6371$ km.

Orbital velocity (circular):

$$v = \sqrt{\frac{\mu}{a}}$$

Eclipse fraction (maximum, circular orbit, cylindrical shadow):

$$f_{\text{eclipse}} = \frac{1}{\pi}\arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right)$$

Sun-synchronous inclination (J2-driven RAAN precession = 360 deg/year):

$$\cos(i) = -\frac{2\dot{\Omega}_{\text{req}}\, a^{7/2}}{3R_E^2\, J_2\, \sqrt{\mu}}$$

where $J_2 = 1.0826 \times 10^{-3}$, $\dot{\Omega}_{\text{req}} = \frac{2\pi}{365.25 \times 86400}$ rad/s.

2. Worked Examples: Orbital Computations

Worked Example -- 500 km Circular LEO

Given: $\mathrm{h} = 500$ km, circular orbit.

Step 1 -- Semi-major axis: $\mathrm{a} = 6371 + 500 = 6871$ km $= 6.871 \times 10^6$ m

Step 2 -- Orbital period: $\mathrm{T} = 2\pi\sqrt{\frac{(6.871\times10^6)^3}{3.986\times10^{14}}} = 2\pi\sqrt{8.137\times10^5} = 2\pi \times 902.1 =$ **5668 s** $\approx$ **94.5 min**

Step 3 -- Orbital velocity: $\mathrm{v} = \sqrt{\frac{3.986\times10^{14}}{6.871\times10^6}} = \sqrt{5.802\times10^7} =$ **7617 m/s** $\approx$ 7.62 km/s

Step 4 -- Eclipse fraction (maximum):

$\mathrm{f} = \frac{1}{\pi}\arccos\left(\frac{\sqrt{6871^2 - 6371^2}}{6871}\right) = \frac{1}{\pi}\arccos\left(\frac{2573}{6871}\right) = \frac{1}{\pi}\arccos(0.3745) = \frac{1}{\pi}\times 68.0° =$ **0.378** (37.8%)

Step 5 -- Eclipse and sunlight duration: $\mathrm{t_{eclipse}} = 94.5 \times 0.378 =$ **35.7 min**;

$\mathrm{t_{sun}} = 94.5 - 35.7 =$ **58.8 min**

Step 6 -- Sun-synchronous inclination: $\cos(\mathrm{i}) = -\frac{2\times1.991\times10^{-7}\times(6.871\times10^6)^{3.5}}{3\times(6.371\times10^6)^2\times1.0826\times10^{-3}\times\sqrt{3.986\times10^{14}}}$

Numerator: $2\cdot1.991\times10^{-7}\cdot(6.871\times10^6)^{3.5} \approx 3.386\times10^{17}$; Denominator: $3\cdot(6.371\times10^6)^2\cdot1.0826\times10^{-3}\cdot\sqrt{3.986\times10^{14}} \approx 2.632\times10^{18}$

$\cos(\mathrm{i}) \approx -3.386/26.32 = -0.1286 \Rightarrow \mathrm{i} \approx$ **97.4 deg**

3. Orbit Selection Trade-Offs

The orbit is the single most impactful early design decision. It cascades to every subsystem.

Table 2.3.2 — Lower-altitude versus higher-altitude LEO trade across GSD, lifetime, drag, and link.

Parameter	Lower Altitude (300--400 km)	Higher Altitude (600--800 km)
GSD	Better (shorter range to target)	Worse (longer range)
Orbital lifetime	Short (1--5 years, natural decay)	Long (25--100+ years)
Debris compliance	Easy (FCC 5-year rule met naturally)	May require active deorbit propulsion
Radiation (TID)	Lower (2--5 krad/yr)	Higher (10--20 krad/yr above 700 km)
Launch cost	Lower $\Delta\mathrm{V}$ to orbit	Slightly higher
Link budget	Better (shorter slant range)	Worse (longer path loss)
Coverage/swath	Narrower per pass	Wider per pass
Atmospheric drag	Significant (limits lifetime)	Negligible above ~700 km
Eclipse fraction	~35--38%	~33--35%

Orbit Types and Applications

Table 2.3.3 — Orbit type catalogue mapping altitude and inclination to mission applications and flight examples.

Orbit	Altitude	Inclination	Best For	Real Example
LEO (non-SSO)	300--600 km	Any	Technology demos, ISS deployment	Many CubeSats
SSO	400--800 km	97--99 deg	EO (consistent solar illumination)	Planet SuperDove (475 km)
ISS orbit	~410 km	51.6 deg	ISS-deployed CubeSats	NanoRacks deployments
MEO	2000--20200 km	Various	Navigation	GPS (20200 km)
GEO	35786 km	0 deg	Comms, weather	Anik F2 (Telesat)
HEO (Molniya)	500--40000 km	63.4 deg	High-latitude coverage	Meridian (Russia)
Lunar NRHO	1500--70000 km	Lunar	Cislunar operations	CAPSTONE (NASA/Advanced Space)

Debris Compliance Rules

Table 2.3.4 — Post-mission disposal rules from IADC, FCC, ECSS, and ISED authorities.

Rule	Requirement	Applies To
IADC guideline (2021)	Post-mission disposal within 25 years	International (voluntary but expected)
FCC rule (2024+)	Post-mission disposal within **5 years**	All FCC-licensed LEO satellites
ECSS-U-AS-10C Rev.2	Compliance with IADC + ESA Zero Debris Charter	ESA missions
ISED (Canada)	Currently 25-year rule; tightening under review	Canadian-licensed satellites

Critical altitude boundaries for FCC 5-year compliance:

Table 2.3.5 — Altitude bands mapped to natural lifetime and FCC five-year deorbit compliance options.

Altitude	Natural Lifetime	FCC Compliant?	Action Needed
< 450 km	< 5 years	Yes	None (natural decay)
450--550 km	5--20 years	Marginal	Drag augmentation may suffice
550--650 km	20--50 years	**No**	Active deorbit (propulsion or drag sail)
> 700 km	> 100 years	**No**	Propulsion mandatory; ESA Zero Debris zone

[Source: IADC-02-01 Rev 3, Sec. 5.3.2; FCC 22-74; ECSS-U-AS-10C Rev.2]

Perturbation Effects

Table 2.3.6 — Orbit perturbations ranked by cause, effect, and resulting design impact.

Perturbation	Cause	Effect on Orbit	Design Impact
J_2 **(Earth oblateness)**	Equatorial bulge	RAAN precession, argument of perigee drift	Enables Sun-synchronous; frozen orbits at $\omega = 90°$
Atmospheric drag	Residual atmosphere	Semi-major axis decay, eventual re-entry	Limits lifetime below ~600 km; drives propulsion need
Solar radiation pressure	Photon momentum	Eccentricity oscillations, orbit perturbation	Significant for large A/m ratio (drag sails, large SA)
Third-body (Moon/Sun)	Gravitational pull	Long-period oscillations in e, i	Significant for GEO and HEO; negligible for LEO
Magnetic field	Lorentz force on charged S/C	Very small drag-like effect	Negligible for most missions

---

4. Ground Coverage and Revisit

Key Equations -- Ground Coverage

Swath width (nadir-pointing sensor with half-cone angle θ):

$$W_{swath} = 2h\tan(\theta)$$

Ground track spacing (for a single satellite in LEO): The Earth rotates ~22.9 deg per orbit (for $T \approx 95$ min). At the equator, this corresponds to:

$$\Delta_{lon} = \frac{360°}{T_{sidereal}} \times T_{orbit} \approx 22.9° \approx 2550 \text{ km (at equator)}$$

Revisit time (approximate, single satellite):

$$t_{revisit} \approx \frac{\Delta_{lon}}{W_{swath}} \times T_{orbit}$$

Constellation revisit (N identical satellites in same plane):

$$t_{revisit,constellation} \approx \frac{t_{revisit,single}}{N}$$

Example -- SuperDove constellation: Planet operates ~200 SuperDove satellites. With a ~24 km swath at 475 km and multiple orbital planes, the constellation achieves daily global revisit -- a dramatic improvement over a single satellite's ~7-day revisit.

Ground Station Contact Geometry

Contact time per pass depends on the minimum elevation angle ϵ_{min} (typically 5--10 deg):

$$\mathrm{t_{contact}} \approx \frac{\mathrm{T}}{\pi} \arccos\left(\frac{\cos(\rho)}{\cos(\epsilon_{\min})}\right) - \text{geometric correction}$$

Simplified rule of thumb for LEO at 500 km, $\epsilon_{\min} = 10°$: - Maximum pass duration: ~10 min (overhead pass) - Average pass duration: ~6--7 min - Passes per day (mid-latitude station): ~4--6

Constellation Design

A single LEO satellite sees any given point on Earth only briefly and infrequently; sustained coverage or short revisit requires a **constellation** — multiple satellites whose orbits are chosen so their coverage footprints overlap in space and time. The dominant design pattern is the **Walker constellation**, specified by the notation $\mathrm{i{:}t/p/f}$, where i is the common inclination, t the total number of satellites, p the number of equally-spaced orbital planes, and f the inter-plane phasing factor (which staggers the satellites between adjacent planes so the gaps in one plane are covered by the next). Choosing i, altitude, p, and t trades coverage against cost: more planes improve cross-track coverage and resilience to a single launch failure but multiply launch and station-keeping cost; higher altitude widens each footprint (fewer satellites for global coverage) at the price of higher launch energy, longer slant ranges, and slower deorbit. Constellations also demand a **phasing and station-keeping budget**: differential drag, $\mathrm{J_2}$ precession, and injection errors must be managed so the satellites hold their relative geometry over the mission life.

Constellation Coverage and Revisit

For a constellation the single-satellite revisit equations above are divided, to first order, by the number of satellites contributing coverage at the target latitude (see the constellation-revisit relation in §4). The useful figures of merit are **percentage coverage** (fraction of the target region in view of at least one satellite at any instant), **mean and maximum revisit time** (average and worst-case gap between successive views of a point), and **n-fold coverage** (how many satellites see a point simultaneously — important for navigation and for multi-look or stereo imaging). Coverage is strongly latitude-dependent: a Walker constellation at inclination i gives best coverage near latitude i and leaves a polar gap above it, which is why true global coverage often mixes inclined planes with one or more near-polar planes. The SuperDove example above (≈200 satellites across multiple planes, daily global revisit) illustrates the headline trade: revisit shortens roughly in proportion to satellite count, but so does cost and constellation-management complexity.

Downlinks and Data Return

Coverage of the *target* is only half the problem; the data must also reach the ground. Each satellite can only downlink when it is in view of a **ground station**, so a constellation's effective data-return capacity is set by the combination of per-satellite downlink rate, the number and geographic spread of ground stations, and the resulting contact minutes per orbit. Common strategies are a **distributed ground-station network** (e.g. commercial ground-station-as-a-service providers, or high-latitude stations such as Svalbard/Inuvik that see many consecutive polar passes), **store-and-forward** operation (the payload buffers data on board and dumps it over a station), and on-board **data reduction** (compression, on-board processing, or tasking to capture only what is needed). The downlink contact budget — total available contact time per day versus data generated per day — is a first-order sizing check for any constellation and is exactly the kind of budget closure SpaceCDF performs.

Inter-Satellite Links and Space-to-Space Backhaul

When a satellite cannot reach a ground station quickly enough — or when low-latency delivery is required — data can be relayed **satellite-to-satellite** over an **inter-satellite link (ISL)** rather than waiting for a ground pass. Two architectures dominate. **Cross-links within a LEO constellation** (typically optical, increasingly RF for smaller satellites) let a satellite hand its data to a neighbour that is over a ground station, so the

constellation behaves as a mesh and any node can reach the ground within seconds (the approach used by large broadband constellations). **Relay through a higher-orbit asset** uses a GEO data-relay satellite (e.g. NASA's TDRS or ESA's EDRS "SpaceDataHighway") that is permanently visible to both the LEO satellite and a fixed ground terminal, providing near-continuous backhaul without a dense LEO ground network. ISLs add mass, power, and pointing-accuracy demands (optical cross-links need arc-second-class pointing between fast-moving platforms) and a new layer of network and timing complexity, so they are justified mainly where latency, continuous contact, or independence from a global ground network is worth the cost.

---

5. Hohmann Transfer and Deorbit

> **Key Equations -- Hohmann Transfer**
>
> The minimum-energy transfer between two circular orbits of radii r_1 and r_2:

$$\Delta \mathrm{V}_1 = \sqrt{\frac{\mu}{\mathrm{r}_1}}\left(\sqrt{\frac{2\mathrm{r}_2}{\mathrm{r}_1+\mathrm{r}_2}}-1\right)$$

$$\Delta \mathrm{V}_2 = \sqrt{\frac{\mu}{\mathrm{r}_2}}\left(1-\sqrt{\frac{2\mathrm{r}_1}{\mathrm{r}_1+\mathrm{r}_2}}\right)$$

$$\Delta \mathrm{V}_{\mathrm{total}} = |\Delta \mathrm{V}_1| + |\Delta \mathrm{V}_2|$$

> **Worked Example -- Deorbit from 600 km to 200 km (re-entry perigee)**
>
> $\mathrm{r}_1 = 6971$ km, $\mathrm{r}_2 = 6571$ km
>
> $\Delta \mathrm{V}_1 = \sqrt{\frac{3.986\times10^5}{6971}}\left(\sqrt{\frac{2\times6571}{6971+6571}}-1\right)$
> $= 7.561\times(\sqrt{0.9705}-1) = 7.561\times(-0.01498) = -0.1133$ km/s
>
> $\Delta \mathrm{V}_{\mathrm{total}} \approx$ **113 m/s** (only the first burn needed to lower perigee)

This deorbit $\Delta \mathrm{V}$ is a critical input to the propulsion sizing in Chapter 3.2.

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6. Radiation Environment

[Source: ECSS-E-ST-10-04C Rev.1]

Total Ionising Dose (TID) by Orbit

Table 2.3.7 — Annual total ionising dose by orbit regime with electronics class guidance.

Orbit	TID (krad/year behind 2mm Al)	Electronics Class
ISS (410 km, 51.6 deg)	2--5	Commercial COTS OK
SSO 500 km	5--10	Commercial / rad-tolerant
SSO 800 km	10--20	Rad-tolerant recommended

Orbit	TID (krad/year behind 2mm Al)	Electronics Class
MEO 2000 km	50--100	Rad-hard required
GEO 35786 km	10--20	Rad-hard required

Rule of thumb: Below 600 km in LEO, commercial COTS electronics can survive 3-year missions with modest shielding (2--3 mm Al equivalent). Above 600 km, radiation becomes a significant design driver and component cost escalates.

South Atlantic Anomaly (SAA)

At 200--600 km altitude over South America, trapped protons from the inner Van Allen belt dip to lower altitudes. The SAA causes: - Single-event upsets (SEU) in memory - Single-event latch-ups (SEL) in CMOS - Increased background noise in optical detectors

Mitigation: error-correcting memory (EDAC), watchdog timers, latch-up protection circuits.

1U Worked Example: UniSat-1

Orbit Selection: ISS Orbit (400 km, 51.6 deg) for Rideshare

UniSat-1 selects the ISS orbit not by optimisation but by **access**: deployment from the ISS via NanoRacks or a similar deployer is the cheapest and most accessible launch opportunity for a university 1U CubeSat.

Worked Example -- Orbital Parameters at 400 km, 51.6 deg

Step 1 -- Semi-major axis: $\mathrm{a} = 6371 + 400 = 6771$ km $= 6.771 \times 10^{6}$ m

Step 2 -- Orbital period: $\mathrm{T} = 2\pi\sqrt{\frac{(6.771 \times 10^{6})^{3}}{3.986 \times 10^{14}}} = 2\pi \times 882.4 = 5545$ s $\approx$ **92.4 min**

Step 3 -- Orbital velocity: $\mathrm{v} = \sqrt{\frac{3.986 \times 10^{14}}{6.771 \times 10^{6}}} =$ **7672 m/s** $\approx$ 7.67 km/s

Step 4 -- Eclipse fraction (maximum):
$\mathrm{f} = \frac{1}{\pi}\arccos\left(\frac{\sqrt{6771^{2} - 6371^{2}}}{6771}\right) = \frac{1}{\pi}\arccos(0.3386) = \frac{1}{\pi} \times 70.2° \approx$ **0.390** (39.0%)

Step 5 -- Eclipse and sunlight duration: $\mathrm{t_{eclipse}} = 92.4 \times 0.390 \approx$ **36 min**; $\mathrm{t_{sun}} = 92.4 - 36 =$ **56 min**

Why this orbit works for UniSat-1:

Table 2.3.8 — UniSat-1 mission-fit assessment for the 400 km, 51.6 degree ISS orbit.

Factor	400 km / 51.6 deg	Impact on UniSat-1
Launch cost	Lowest (ISS resupply rideshare)	Fits university budget
Orbital lifetime	~1 year (natural decay)	Exceeds 6-month mission; compliant with FCC 5-year rule without propulsion
Radiation	Low (~2--3 krad/yr behind 2 mm Al)	COTS electronics safe for 6-month mission
Inclination	51.6 deg	Adequate latitude coverage for space weather science

Factor	400 km / 51.6 deg	Impact on UniSat-1
Eclipse	~36 min (~39% of orbit)	Manageable with 10 Wh battery
Ground contacts	Mid-latitude stations: ~4--6 passes/day	Sufficient for 9600 bps UHF downlink

What this orbit does NOT provide: - Sun-synchronous lighting (not needed -- magnetometer is not an optical instrument) - Polar coverage (acceptable -- 51.6 deg covers the majority of magnetic field variation) - Long lifetime (acceptable -- 6-month design life is well within ~1-year natural lifetime)

No propulsion trade: At 400 km, atmospheric drag causes natural re-entry within approximately 1 year (depending on solar activity and ballistic coefficient). UniSat-1 therefore needs no propulsion system for either orbit maintenance or debris compliance. This eliminates an entire subsystem -- a major simplification for a 1U mission.

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7. SpaceCDF Orbit Trade Exercise

Instructions

1. Navigate to the **Mission Architecture** tab in SpaceCDF
2. Open the **Orbit Trade Advisor** panel
3. Enter your mission parameters: - GSD target (optical missions) or "N/A" for non-optical - Revisit target (days) - Mission lifetime (years) - Latitude band of interest - Cost ceiling (MEUR) - Aperture diameter (if known from optical sizing)
4. Click **"Compute Orbit Trade"**
5. Review the scored candidates: - Which orbit scores highest overall? - Is it debris-compliant (FCC 5-year rule)? - What is the natural orbital lifetime?
6. Click **"Use"** on your preferred orbit to populate the design parameters
7. Verify orbit fields updated in the dashboard (look for "Set from advisor" badge)

Discussion Points

- Does the best-scoring orbit match your engineering intuition?
- What happens when you change scoring weights (prioritise GSD over cost, or vice versa)?
- For non-optical missions (comms, AIS), does the orbit scoring still make physical sense?
- If your selected orbit is above 600 km, what deorbit strategy will you use?
- Complete Worksheet 2.3

Session Summary

Table 2.3.9 — Session summary: key takeaways for orbit selection and mission architecture.

Topic	Key Takeaway
Keplerian elements	Six parameters define the orbit; for LEO: altitude + inclination + LTAN
Key formulae	$T = 2\pi\sqrt{a^3/\mu}$; $v = \sqrt{\mu/a}$; eclipse fraction from geometry

Topic	Key Takeaway
Sun-synchronous	Requires specific inclination (~97 deg at 500 km) using J_2 precession
Perturbations	J_2 enables SSO; drag limits lifetime; SRP affects high-A/m satellites
Trade-offs	Lower altitude = better GSD and link but shorter lifetime and more drag
Debris rules	FCC 5-year, IADC 25-year; above ~550 km requires active deorbit
Hohmann transfer	$\Delta \mathrm{V}_{\mathrm{deorbit}} \approx 100$ m/s from 600 km; critical propulsion input
Radiation	Below 600 km: COTS OK; above: rad-tolerant/hard needed; SAA causes SEU
Cascade	Orbit choice affects every subsystem -- highest-leverage design decision

Chapter 2.4 — Mission Architecture -- Segments, Interfaces, and Budgets

> **Expected reading before this session.** NASA SEH §4.4 (≈ 30 min).

Prerequisites: Sessions 2.1--2.3 (requirements, functions, orbit selected) **SpaceCDF Tabs:** Mission Architecture, System Architecture, Interfaces, Dashboard

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References

- NASA, *Systems Engineering Handbook*, 2016, Sec. 4.4 (Process 4) & Sec. 6.3 (Process 12: Interface Management)
- ECSS, *ECSS-E-ST-10-24C: Interface Management*, 2015
- ECSS, *ECSS-E-HB-10-02A: Verification Guidelines*, 2010, Sec. 5.2 (Mass Margins)
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 10--11
- ECSS, *ECSS-E-ST-20C: Electrical and Electronic*, 2021 (Power Budgets)

Learning Objectives

By the end of this session, participants will be able to:

1. Decompose a space mission into its constituent segments (space, ground, launch, user)
2. Construct a system architecture block diagram with subsystem boundaries
3. Identify and classify all subsystem-to-subsystem interfaces using an N-squared matrix
4. Write formal interface requirements for critical subsystem pairs
5. Construct a mass budget with ECSS margin policy and a mode-based power budget
6. Interpret SpaceCDF's dashboard KPIs and budget displays

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1. Mission Segment Decomposition

[Source: NASA SEH Sec. 4.4; ECSS-E-ST-10C Sec. 5.4]

Every space mission decomposes into segments, each with distinct functions and interfaces:

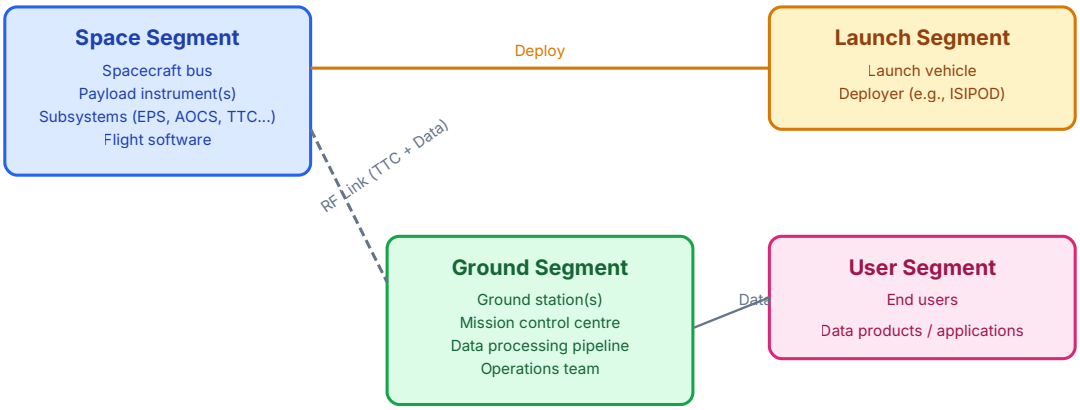
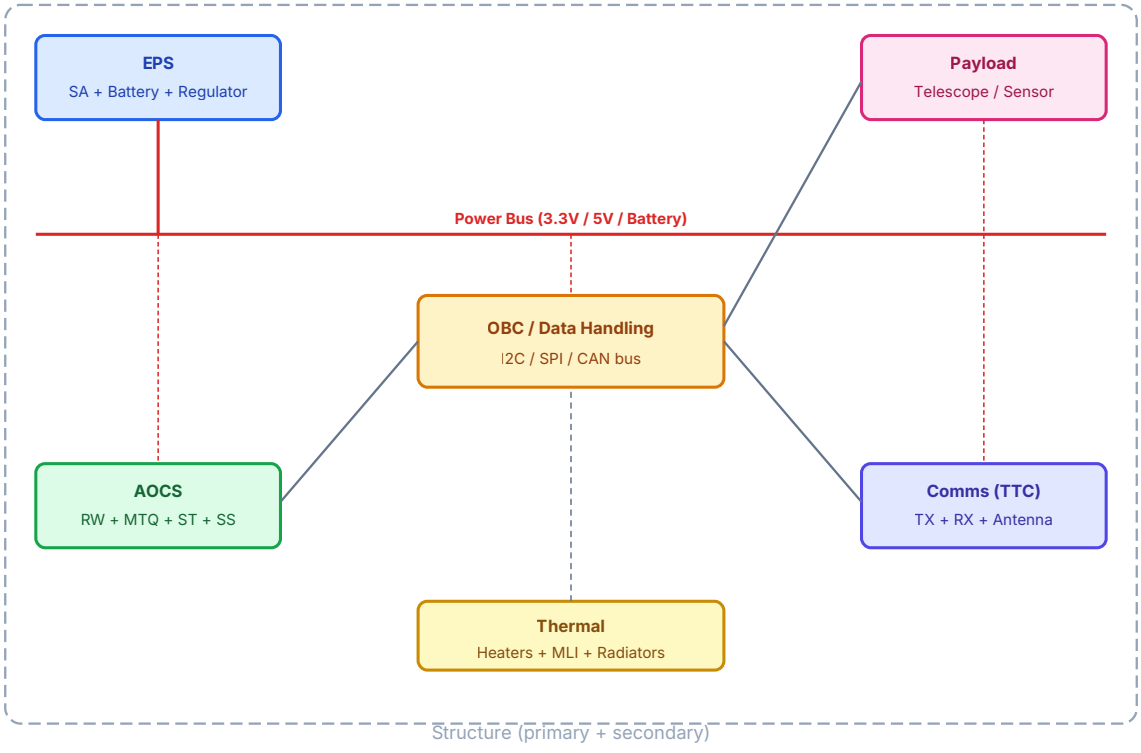


Table 2.4.1 — Mission segment decomposition listing elements and inter-segment interfaces.

Segment	Elements	Key Interfaces
Space	Spacecraft bus, payload, flight software	To ground (RF), to launch (mechanical/electrical)
Ground	Ground station, MCC, data processing	To space (RF), to user (network)
Launch	Launch vehicle, deployer, adapter	To space (mechanical, electrical inhibits)
User	End users, applications, data consumers	To ground (data products)

2. System Architecture Block Diagram

The system block diagram shows the internal architecture of the space segment -- all subsystems and their data/power/mechanical connections.



Subsystem Roles (CubeSat Reference)

Table 2.4.2 — CubeSat subsystem reference with abbreviations, primary functions, and typical mass fractions.

Subsystem	Abbreviation	Primary Function	Typical Mass Fraction
Payload	PL	Mission-specific sensing or communication	25--40%
Electrical Power System	EPS	Generate, store, distribute electrical power	15--25%
Attitude & Orbit Control	AOCS	Determine and control attitude; orbit knowledge	8--15%
Communications (TTC)	LINK / TTC	Telemetry downlink, telecommand uplink, data downlink	5--10%
On-Board Computer	OBC / C&DH	Command execution, data handling, flight software	2--5%
Thermal Control	TCS	Maintain all components within temperature limits	1--5%
Structure	STR	Mechanical support, launch load path, CDS compliance	15--25%
Propulsion	PROP	Orbit manoeuvres, deorbit (if required)	0--15%
Harness	HAR	Electrical interconnections	3--7%

---

3. The N-Squared Interface Matrix

[Source: NASA SEH Sec. 6.3 -- Process 12: Interface Management; ECSS-E-ST-10-24C]

Interface problems are the leading cause of integration failures. NASA SEH states: "Most system failures can be traced back to interface problems."

N-Squared Matrix Structure

The N^2 matrix is a standard systems engineering tool for mapping all subsystem interactions:

- **Diagonal cells:** Subsystems (EPS, AOCS, Comms, Thermal, Structure, Propulsion, OBC, Payload)
- **Off-diagonal cells:** Interface between the row subsystem and the column subsystem
- **Upper triangle:** Outputs from row to column (data, power, commands)
- **Lower triangle:** Outputs from column to row

For 8 subsystems: $8 \times 7/2 = 28$ potential interface pairs. A typical CubeSat has 18--22 active interfaces.

N² interface matrix example for a CubeSat with six subsystems on the diagonal and representative inter-subsystem interfaces in off-diagonal cells.

N² matrix example.

Interface Types

Table 2.4.3 — Six interface type categories with symbols and representative CubeSat examples.

Type	Symbol	Description	Example
Mechanical	M	Physical attachment, loads, alignment tolerances	Payload mounting to structure face
Electrical	E	Power connections, bus voltage, switched lines	EPS 5V bus to all subsystems
Thermal	T	Heat transfer paths, thermal coupling, conduction	Transponder waste heat to radiator panel
Data	D	Digital bus (I^2C, SPI, UART, CAN, RS-422)	OBC commands to AOCS controller
RF	R	Electromagnetic coupling or intentional RF paths	TX emissions coupling into payload receiver
Optical	O	Light paths, field-of-view clearance, stray light	Star tracker FOV clearance from solar array

Common CubeSat Interface Concerns

Table 2.4.4 — Common CubeSat interface pairs with interface types and dominant design concerns.

Interface Pair	Types	Key Concern
EPS <-> AOCS	E	Bus voltage compatibility; reaction wheel peak power draw
EPS <-> Comms	E	TX peak power demand (~6--10 W); switched line allocation
EPS <-> Thermal	E, T	SA thermal coupling; radiator vs SA area competition on external faces
EPS <-> Payload	E	Peak power switching; duty cycle coordination
Structure <-> AOCS	M	Reaction wheel and star tracker mounting alignment; vibration isolation
Structure <-> Payload	M, O	Payload alignment stability; optical FOV clearance
OBC <-> AOCS	D	Attitude data for payload pointing; mode transition commands
OBC <-> Comms	D	Telemetry packet routing; telecommand distribution
OBC <-> Payload	D	Science data acquisition trigger; instrument commanding
Comms <-> Payload	R	**EMC:** TX conducted/radiated emissions vs payload receiver sensitivity
Comms <-> AOCS	R, O	Antenna pattern vs star tracker FOV; antenna pointing coordination
Thermal <-> Payload	T	Detector cooling requirement; operating temperature limits
AOCS <-> Payload	M	Reaction wheel micro-vibration vs payload pointing stability (jitter)

Conflict Detection and Resolution

Interface conflicts arise when two subsystems have incompatible requirements at their shared boundary.

Severity Classification:

Table 2.4.5 — Interface conflict severity classification with examples and required engineering actions.

Severity	Description	Example	Required Action
Critical	Design cannot close without resolution	EMC: TX radiation prevents payload operation	Must resolve before PDR
Major	Significant impact on design margin	Radiator area competes with SA area	Mitigation plan required by PDR
Minor	Manageable with minor adjustment	Star tracker FOV partially blocked by antenna stow	Accommodation analysis

Resolution Options:

1. **Relocate:** Move a component to avoid the conflict (e.g., star tracker to different face)
2. **Shield/Isolate:** Add EMC shielding, vibration isolators, thermal insulation
3. **Time-Division:** Schedule conflicting activities to avoid simultaneity (e.g., no TX during imaging)
4. **Accept Risk:** Document residual risk and margin impact in the risk register

Writing Interface Requirements

For each significant interface, write formal requirements:

Example -- EPS <-> All Subsystems:

```
IR-PWR-001: "The EPS shall provide regulated bus voltages of
             3.3 V +/- 0.1 V and 5.0 V +/- 0.25 V to all subsystems."
IR-PWR-002: "Each subsystem shall not exceed its allocated power
             draw without EPS coordination."
```

Example -- Comms <-> Payload (EMC):

```
IR-EMC-001: "TX conducted emissions shall be below -60 dBm in the
             payload receiver band (1.5-1.6 GHz) during imaging mode."
IR-EMC-002: "TX and payload acquisition shall not operate simultaneously
             unless IR-EMC-001 is verified by test."
```

---

4. Engineering Budgets: Mass and Power

[Source: ECSS-E-HB-10-02A Sec. 5.2]

Engineering budgets are the quantitative backbone of the design. They answer: **"Will this design close?"**

Mass Budget

Key Equations -- Mass Budget

Mass margin:

$$\text{Margin}_{\%} = \frac{\text{M}_{\text{allocation}} - \text{M}_{\text{MEV}}}{\text{M}_{\text{allocation}}} \times 100\%$$

where MEV = Maximum Expected Value = CBE + maturity margins.

Status thresholds: Green: > 20% | Amber: 10--20% | Red: < 10% | Exceeded: < 0%

Table 2.4.6 — Mass budget terminology defining CBE, MEV, equipment margin, and system margin.

Term	Definition
CBE (Current Best Estimate)	Best estimate of actual mass based on current knowledge
MEV (Maximum Expected Value)	CBE + equipment maturity margin = worst-case expected mass
Equipment maturity margin	Applied per component based on design maturity (TRL)
System margin	Applied at system level as management reserve

ECSS Margin Policy by Phase

[Source: ECSS-E-HB-10-02A Sec. 5.2]

Table 2.4.7 — ECSS equipment and system mass margin allocations by project phase.

Phase	Equipment Margin	System Margin	Compound
0/A (concept)	20%	20%	~44%
B1 (preliminary)	10%	20%	~32%
B2 (detailed)	5%	15%	~21%
C/D (build/test)	3%	10%	~13%
E (as-built)	0%	5%	~5%

Worked Example -- 3U CubeSat Mass Budget (Phase A)

Table 2.4.8 — Worked Phase A mass budget for a 3U CubeSat against a 6 kg launcher allocation.

Subsystem	CBE (kg)	Equip. Margin (20%)	MEV (kg)
Payload	1.50	0.30	1.80
EPS	0.75	0.15	0.90
AOCS	0.55	0.11	0.66
Comms (TTC)	0.25	0.05	0.30
OBC	0.08	0.02	0.10
Thermal	0.05	0.01	0.06
Structure	0.35	0.07	0.42
Harness	0.15	0.03	0.18
Dry Total	**3.68**		**4.42**
System Margin (20%)			**0.88**
Dry MEV			**5.30**

Subsystem	CBE (kg)	Equip. Margin (20%)	MEV (kg)
Propellant			0.00
Wet Mass			**5.30**
Launcher Allocation			**6.00** (3U limit)
Mass Margin			**0.70 kg (11.7%)** -- Amber

Mode-Based Power Budget

The power budget is computed **per operational mode** because not all subsystems draw power simultaneously:

Table 2.4.9 — Per-subsystem power demand across safe, idle, imaging, downlink, and eclipse modes.

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC	1.0	1.0	1.0	1.0	1.0
AOCS	0.5	1.0	3.0	2.0	0.5
Payload	0	0	5.0	0	0
Comms (TX)	0.5	0.5	0.5	6.0	0
Thermal	0.5	0.5	0.5	0.5	2.0
Total	**2.5**	**3.0**	**10.0**	**9.5**	**3.5**

Orbit-Average Power

Key Equations -- Orbit-Average Power

$$\mathrm{P}_{\mathrm{avg}} = \sum_{\mathrm{modes}} (\mathrm{P}_{\mathrm{mode}} \times \mathrm{duty}_{\mathrm{mode}})$$

Example for 95-min orbit (60 min sunlight, 35 min eclipse):

Table 2.4.10 — Orbit-average power calculation combining mode power with duty cycle fractions.

Mode	Power (W)	Duty (%)	Contribution (W)
Idle	3.0	45%	1.35
Imaging	10.0	10%	1.00
Downlink	9.5	8%	0.76
Eclipse	3.5	37%	1.30
Total		**100%**	**4.41 W**

Other Budget Types (Preview)

These budgets are developed in detail in the following chapters:

Table 2.4.11 — Preview of link, pointing, delta-V, and data budgets with governing equations.

Budget	Key Equation	Session
Link	$\text{Margin} = \text{EIRP} - \text{FSPL} + \text{G/T} - \text{k} - 10\log_{10}(\text{R}_\text{b}) - \text{E}_\text{b}/\text{N}_0$	3.3
Pointing	$\theta_{\text{total}} = \sqrt{\sum \theta_\text{i}^2}$ (RSS)	3.2
ΔV	$\Delta\text{V} = \text{I}_\text{sp} \cdot \text{g}_0 \cdot \ln(\text{m}_0/\text{m}_\text{f})$ (Tsiolkovsky)	3.2
Data	$\text{Daily Downlink} \geq \text{Daily Generation}$	3.3

1U Worked Example: UniSat-1

Simplified Architecture: Only 5 Subsystems

UniSat-1 demonstrates that a 1U CubeSat can be built with a radically simplified architecture compared to a 3U EO mission. Several subsystems are eliminated entirely:

Table 2.4.12 — Comparison of 3U EO and 1U UniSat-1 subsystems showing eliminations and rationale.

Subsystem	3U EO CubeSat	UniSat-1 (1U)	Rationale for Elimination
EPS	Required	**Required**	Always needed
OBC	Required	**Required**	Always needed (minimal: MSP430 or ARM Cortex-M class)
Comms	S-band + X-band	**UHF only**	9600 bps is sufficient for < 1 kbps payload data
Structure	Required	**Required**	Always needed (ISIS 1U or Pumpkin Rev C)
Payload	Telescope (complex)	**MEMS magnetometer** (simple: 50 g, 0.2 W)	Defines the mission
AOCS	Star tracker + RWs + MTQs	**Eliminated**	Passive magnetic only (permanent magnet + hysteresis rods, treated as structure, not a subsystem)
Thermal	MLI + heaters	**Eliminated**	Passive coatings only; 400 km LEO thermal environment is benign for 6-month mission
Propulsion	May be needed	**Eliminated**	400 km orbit decays naturally in ~1 year; no orbit maintenance needed
Harness	Significant	**Minimal**	Only 3--4 board-to-board connections via PC/104 stack

N-squared matrix -- drastically reduced:

For UniSat-1, the interface matrix shrinks from 28 potential pairs (8 subsystems) to just 10 pairs (5 subsystems). The active interfaces are:

Table 2.4.13 — Reduced UniSat-1 interface set with five subsystems and ten active pairs.

Interface Pair	Types	Key Concern
EPS <-> OBC	E, D	Power regulation, housekeeping telemetry
EPS <-> Comms	E	TX peak power (~0.5 W -- small relative to budget)
EPS <-> Payload	E	Payload switching (0.2 W -- trivial)
OBC <-> Comms	D	TM/TC packet routing
OBC <-> Payload	D	Magnetometer data acquisition (I2C or SPI)
Structure <-> All	M	Mounting, CDS rail compliance

Mass budget (Phase A):

Subsystem	CBE (g)	Equip. Margin (20%)	MEV (g)
Payload (MEMS magnetometer)	50	10	60
EPS (board + battery + body-mounted cells)	250	50	300
OBC (MSP430/Cortex-M board)	30	6	36
Comms (UHF transceiver + antenna)	80	16	96
Structure (1U frame)	200	40	240
Passive magnetic AOCS (magnet + rods)	30	6	36
Harness	50	10	60
Dry Total	**690**		**828**
System Margin (20%)			**166**
Dry MEV			**994**
CDS Allocation			**1330**
Mass Margin			**336 g (25.3%)** -- Green

Power budget (all modes use the same simple duty cycle):

Mode	Power (W)	Duty (%)	Contribution (W)
Idle (OBC + beacon)	0.7	50%	0.35
Science (+ magnetometer)	0.9	10%	0.09
Downlink (+ UHF TX)	1.2	5%	0.06
Eclipse (OBC only)	0.5	35%	0.18
Orbit Average		**100%**	**0.68 W**

With ~2 W available from body-mounted cells, the power margin is substantial (~1.3 W, or 66%). This is one of the luxuries of a simple payload on a 1U bus.

Key architectural insight: The UniSat-1 architecture is so simple that a small team (3--5 people) can design, build, test, and operate it. This makes it ideal for a university or educational programme. The CDF process still applies -- but the sessions are shorter and the trade space is narrower.

Budget Types and Where They Live

This session deliberately treats **mass** and **power** in detail because they are the two *system-level* budgets: every subsystem contributes to them, they close (or fail to close) at the spacecraft level, and they are owned by the systems engineer. A handful of other budgets sit at the same system level — **cost**, **ΔV (propellant)**, **schedule**, and, on more demanding missions, **mass-properties/CG** and **volume/accommodation** — because they too are integrated across the whole vehicle and gate the feasibility of the architecture.

The budgets the reviewer often asks about — **link budget**, **data (downlink) budget**, **pointing budget**, **thermal budget**, **momentum/desaturation budget** — are *subsystem-level* budgets. Each is owned by one discipline and is the natural subject of that discipline's chapter rather than the architecture session: the link and data budgets belong to Communications (Chapter 3.3), the pointing budget to AOCS (Chapter 3.2), the thermal budget to Thermal (Chapter 3.1), and so on. They are no less important — a mission can be killed by a link budget as surely as by a mass budget — but they are computed *within* a subsystem against an allocation handed down from the system level.

The distinction that matters in practice is **roll-up versus allocation**. System-level budgets *roll up*: subsystems report their mass, power, and cost, and the systems engineer sums them, applies margin, and checks against the launch and bus limits. Subsystem-level budgets are *allocated down*: the system level sets a pointing requirement, a downlink-volume requirement, or a thermal limit, and the responsible engineer must close their budget within that allocation, escalating to a system-level trade only if they cannot. A useful rule of thumb: if a number is meaningful only after summing across the whole spacecraft, it is a system-level budget and belongs here; if it is computed and closed inside one subsystem against a handed-down allocation, it is a lower-level budget and belongs in that subsystem's chapter. Volume is the common edge case — it is tracked at the system level (does everything fit inside the CDS envelope?) but closed largely through subsystem-level packaging.

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5. SpaceCDF Exercise

Instructions

1. **System Architecture tab:** Review or edit the subsystem block diagram for your mission
2. **Interfaces tab:** Review the N^2 matrix - Click on 3 interface cells to examine types and concerns - For any red-bordered cell (conflict), use the "Resolve Conflict" workflow - Write interface requirements for your most critical pair
3. **Dashboard:** Examine all KPI cards: - Mass margin: green/amber/red? - Power margin per mode - Link margin (if computed) - Cost vs ceiling
4. **Budget Breakdown:** Open per-subsystem mass and power charts
5. Complete Worksheet 2.4

Discussion Questions

- Which budget is tightest (closest to zero or negative margin)?
- What single design change would most improve the tightest budget?
- How does the ECSS margin policy affect your design freedom at Phase A vs Phase C?

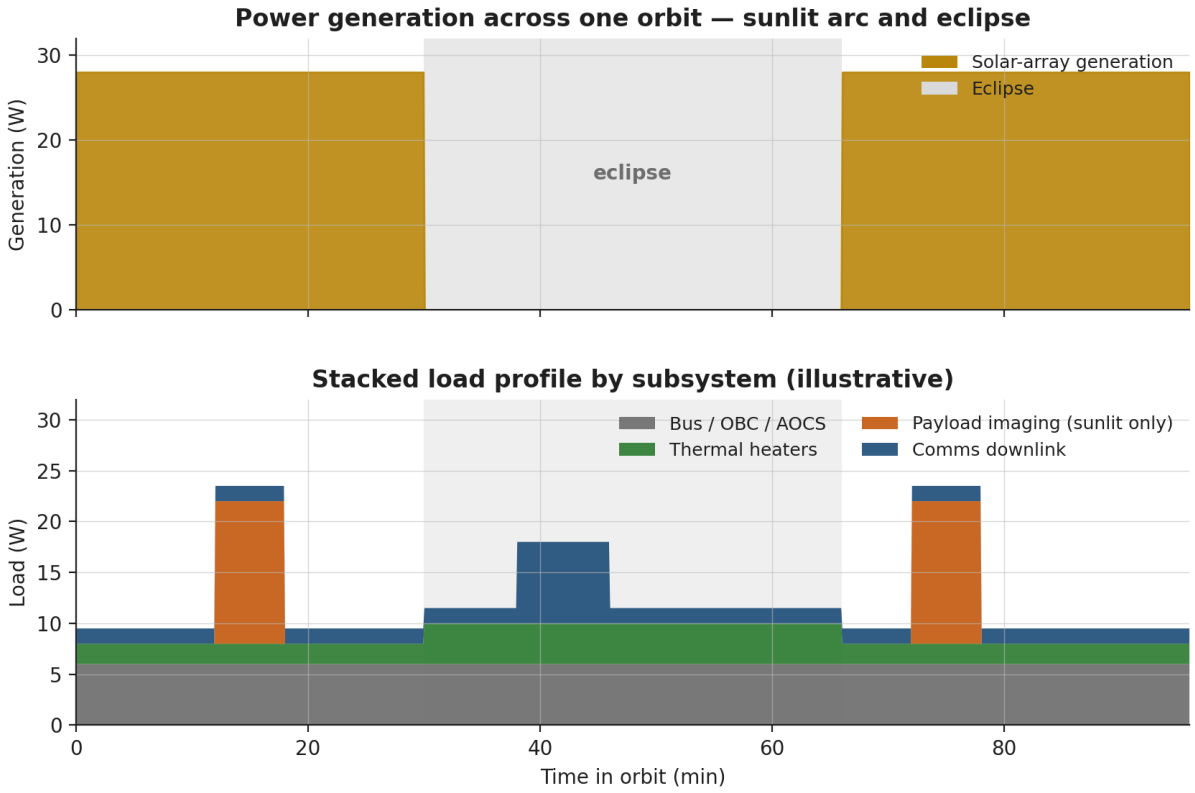
- Which interface pair is most likely to cause integration problems?

Session Summary

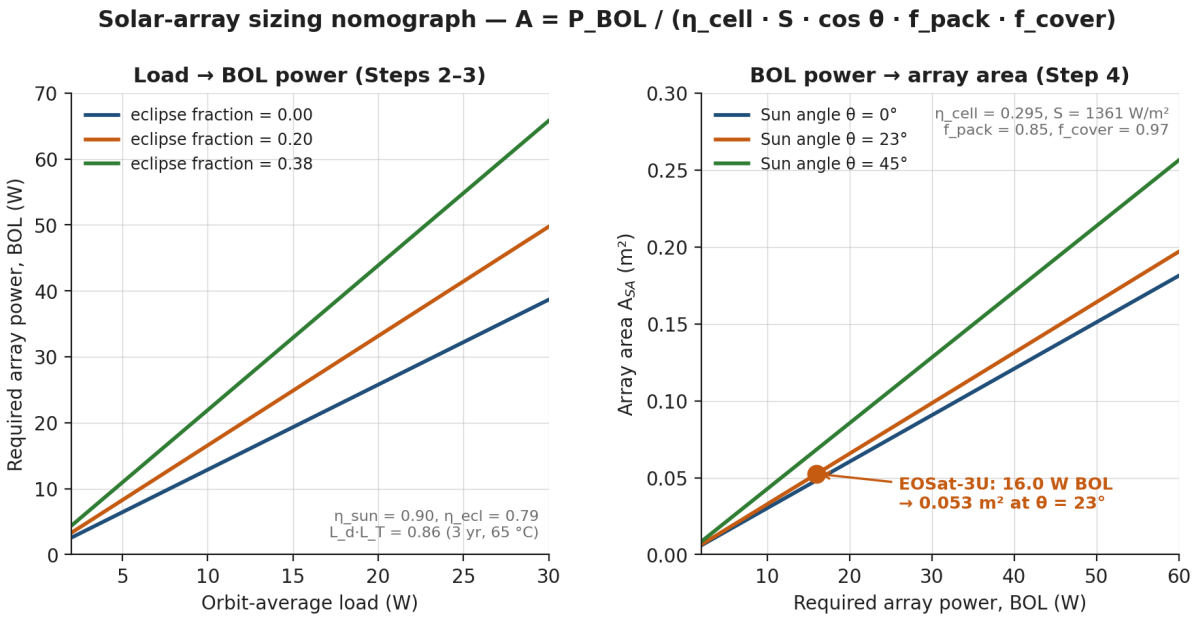
Table 2.4.14 — Session summary: key takeaways for mission segments, interfaces and budgets.

Topic	Key Takeaway
Mission segments	Space, Ground, Launch, User -- each with defined interfaces
System architecture	Block diagram shows subsystems, data buses, power distribution
N^2 matrix	Maps all 28 potential interface pairs; 6 types (M, E, T, D, R, O)
Interface conflicts	Severity classification (critical/major/minor); 4 resolution options
Interface requirements	Formal boundary agreements; must be verifiable
Mass budget	CBE + equipment margin + system margin = MEV; compare to allocation
Power budget	Mode-based; duty cycling gives orbit-average; SA must cover peak + recharge
ECSS margins	Decrease with maturity: ~44% at Phase A to ~13% at Phase C/D
Budget closure	Negative margin = design does not close; reduce demand or increase allocation

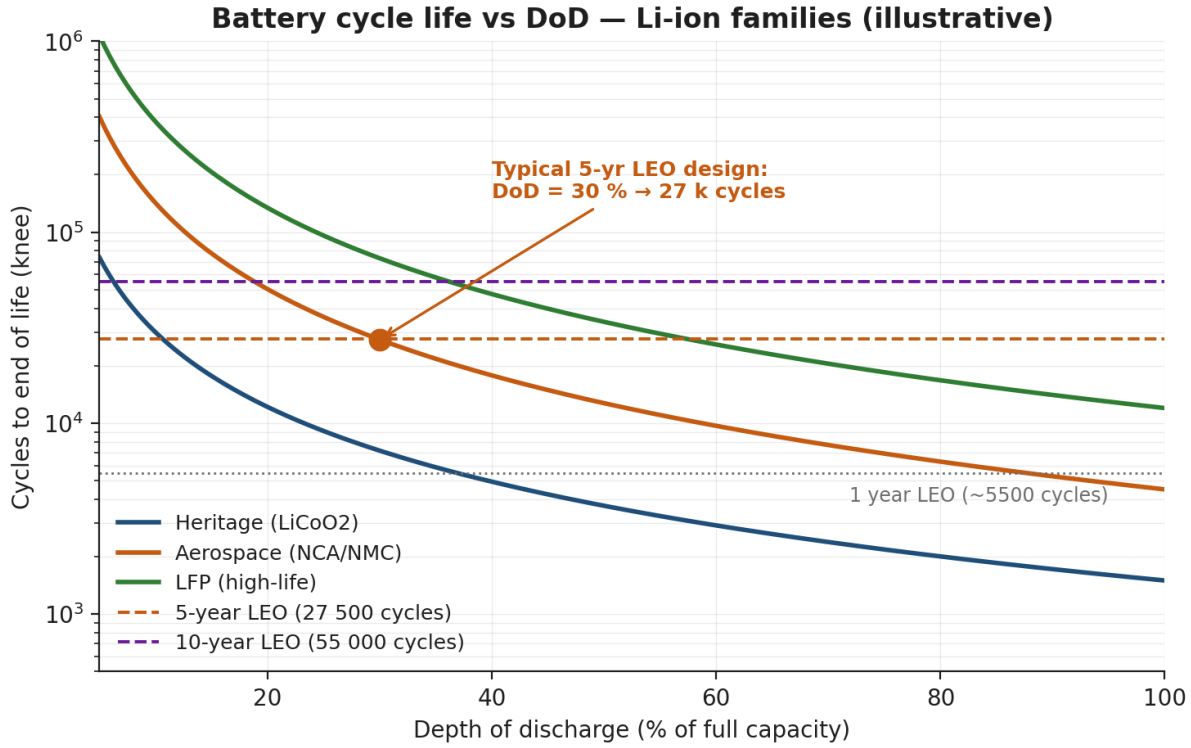
Chapter 3.1 — Power System and Thermal Control Design



Power generation and stacked load profile.



Solar-array sizing nomograph.



Battery cycle life vs DoD.

> **Expected reading before this session.** ECSS-E-ST-20C §5 — power (≈ 60 min); ECSS-E-ST-31C §5 — thermal; Patel, *Spacecraft Power Systems*, Ch. 4; Gilmore, *Spacecraft Thermal Control Handbook*, Ch. 1.

Prerequisites: Sessions 2.1--2.4 (requirements, functions, orbit, architecture defined) **SpaceCDF Tabs:** Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

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References

- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 11.4 (EPS), Ch. 11.5 (Thermal)
- ECSS, *ECSS-E-ST-20C: Electrical and Electronic*, 2021
- ECSS, *ECSS-E-ST-31C: Thermal Control*, 2020
- Patel, *Spacecraft Power Systems*, 2005, Ch. 3--8
- Gilmore, *Spacecraft Thermal Control Handbook, Vol. 1*, 2002
- GomSpace, *P31u EPS Datasheet*, 2023
- MMA Design, *HaWK Solar Array Datasheet*, 2023
- Spectrolab, *30% Triple-Junction Solar Cell Datasheet*, 2020
- Ratnakumar et al., *Lithium-Ion Batteries for Space*, NASA JPL, 2003
- Gilmore, *Spacecraft Thermal Control Handbook, Vol. 2: Cryogenics*, 2003

Learning Objectives

By the end of this session, participants will be able to:

1. Size a solar array from mission power demand, eclipse profile, and degradation
2. Size a battery from eclipse energy demand, depth-of-discharge, and cycle-life requirements
3. Compute orbit-average power using duty cycle analysis
4. Explain EPS architecture (DET vs PPT, MPPT, bus regulation) and articulate the physics of each
5. Perform first-order thermal balance analysis (hot case and cold case) with full radiative derivation
6. Select thermal control methods and apply ECSS thermal margins
7. Explain MLI construction, heat pipe operation, and heater sizing
8. Verify power and thermal budgets in SpaceCDF

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1. Electrical Power System Architecture

The EPS is the "utility company" of the spacecraft. It must continuously supply regulated power to all subsystems through every operational mode, including eclipse. Unlike terrestrial power systems, spacecraft EPS cannot draw from a grid -- the solar array, battery, and power conditioning electronics must form a fully self-contained, autonomous energy system with zero maintenance for the mission lifetime.

EPS Block Diagram

CubeSat EPS architecture: solar array → BCR → battery and regulated bus, with PCDU providing both switched (fused) loads to non-essential subsystems and unswitched loads to always-on essentials.

CubeSat EPS architecture: solar array → BCR → battery and regulated bus, with PCDU providing both switched (fused) loads to non-essential subsystems and unswitched loads to always-on essentials.

Solar Cell Physics

A photovoltaic cell converts photon energy into electrical energy via the photovoltaic effect. When a photon with energy $\mathrm{E} = \mathrm{h}\nu \geq \mathrm{E}_\mathrm{g}$ (where E_g is the semiconductor bandgap) strikes the cell, it promotes an electron from the valence band to the conduction band, creating an electron-hole pair. The built-in electric field at the p-n junction sweeps carriers to opposite terminals, producing a voltage (V_oc) and current (I_sc).

Single-junction vs multi-junction cells:

A single-junction cell (e.g., silicon, $\mathrm{E}_\mathrm{g} = 1.12$ eV) can only absorb photons with $\mathrm{E} \geq \mathrm{E}_\mathrm{g}$. Photons with $\mathrm{E} < \mathrm{E}_\mathrm{g}$ pass through unabsorbed; photons with $\mathrm{E} \gg \mathrm{E}_\mathrm{g}$ lose excess energy as heat (thermalisation loss). The theoretical maximum efficiency for a single-junction cell under AM0 (space) illumination is approximately 31% (Shockley-Queisser limit).

Multi-junction (MJ) cells stack two or three p-n junctions of different semiconductor materials, each tuned to absorb a different portion of the solar spectrum:

Table 3.1.1 — Comparison of single, dual, and triple-junction solar cell technologies for space.

Cell Technology	Structure	Bandgaps (eV)	AM0 Efficiency	Temp Coefficient	Flight Heritage
Monocrystalline Si	Single junction	1.12	16--18%	-0.45 %/ degC	Extensive (ISS, many LEO)

Cell Technology	Structure	Bandgaps (eV)	AM0 Efficiency	Temp Coefficient	Flight Heritage
GaAs single-junction	Single junction	1.42	22--24%	-0.21 %/degC	Moderate
InGaP/GaAs dual-junction	2-junction	1.86 / 1.42	26--28%	-0.20 %/degC	Moderate
InGaP/GaAs/Ge triple-junction	3-junction (lattice-matched)	1.86 / 1.42 / 0.67	28--30%	-0.19 %/degC	Extensive (>90% of modern S/C)
InGaP/GaAs/InGaAs IMM	3-junction (inverted metamorphic)	1.86 / 1.42 / 1.0	32--33%	-0.18 %/degC	Growing (SolAero ZTJ, Spectrolab XTJ Prime)

[Source: Spectrolab XTJ Prime datasheet; SolAero ZTJ datasheet; Green et al., "Solar Cell Efficiency Tables," Progress in Photovoltaics, v.62, 2024]

Why triple-junction GaAs dominates space: The AM0 solar spectrum (above the atmosphere) is richer in UV and blue photons than the AM1.5 terrestrial spectrum. Triple-junction cells capture this energy across three bandgaps. The top cell (InGaP, $\mathrm{E_g} = 1.86$ eV) absorbs blue/UV, the middle cell (GaAs, $\mathrm{E_g} = 1.42$ eV) absorbs visible, and the bottom cell (Ge, $\mathrm{E_g} = 0.67$ eV) absorbs near-IR. Each junction contributes voltage in series, while current is limited by the lowest-current junction (current matching constraint).

Temperature effects: In orbit, solar cells operate at 40--80 degC depending on mounting and orbit. Cell efficiency decreases with temperature due to increased intrinsic carrier concentration (which reduces $\mathrm{V_{oc}}$). The temperature coefficient for triple-junction GaAs is approximately -0.19 %/degC relative -- meaning a cell rated at 29.5% at 28 degC drops to approximately 28.5% at 80 degC. This must be included in power budget calculations:

$$\eta_{\mathrm{cell}}(\mathrm{T}) = \eta_{\mathrm{ref}} \times [1 + \beta(\mathrm{T} - \mathrm{T_{ref}})]$$

where $\beta \approx -0.0019$ /degC for triple-junction GaAs and $\mathrm{T_{ref}} = 28$ degC (standard test conditions).

Degradation mechanisms: Solar cells degrade in orbit due to: - **Radiation damage:** Energetic protons and electrons (trapped in the Van Allen belts and from solar particle events) displace atoms in the crystal lattice, creating recombination centres that reduce minority carrier lifetime and thus current. Degradation is characterised as equivalent 1 MeV electron fluence. Typical rates: 2--3%/year in LEO, 5--8%/year in MEO (through proton belt), 1--2%/year in GEO. - **UV darkening:** Ultraviolet radiation darkens cover glass adhesive over time, reducing transmission. - **Micrometeoroid erosion:** Gradual pitting of cover glass reduces optical transmission. - **Electrostatic discharge (ESD):** In GEO or polar orbits, differential charging can cause arcing between cells, permanently damaging interconnects.

Cover glass (typically ceria-doped borosilicate, 100--150 um thick) mitigates radiation and UV effects. The combined degradation factor is:

$$\mathrm{L_d} = (1 - \delta)^{\mathrm{n}}$$

where $\delta = 0.025$ (2.5%/year for triple-junction GaAs in LEO with standard cover glass) and n = mission lifetime in years.

Body-Mounted vs Deployable Solar Arrays

Body-mounted cells are bonded directly to the spacecraft's external panels. They are the simplest and most reliable option (no deployment mechanism, no hinges, no drive motors) but are severely area-limited.

Table 3.1.2 — Trade between body-mounted, fixed deployable, and tracking solar arrays.

Mounting	Advantages	Disadvantages	Typical Power
Body-mounted	No mechanism risk, no power for tracking, low mass, low cost	Limited area (satellite surface only), poor illumination geometry for nadir-pointing S/C, high cell temperature	1U: ~2 W, 3U: ~7 W, 6U: ~12 W
Fixed deployable	2--4x more area, better sun angle, lower cell temp (radiative cooling from back side)	Deployment mechanism (single point of failure), aerodynamic drag increase, structural dynamics	3U: ~15--25 W, 6U: ~30--48 W
Tracking deployable	Optimal sun incidence ($\cos\theta \approx 1$), maximum power	SADM (Solar Array Drive Mechanism) adds mass/cost/complexity, continuous power for motor	Large S/C: 100+ W

Real mission examples: - **Planet SuperDove (3U+):** Body-mounted + two fixed deployable wings. ~25 W BOL. The deployables are spring-hinged panels that fold against the 3U body during launch and deploy after ejection from the P-POD. - **Asteria (6U, JPL):** Dual deployable arrays, 48 W BOL. Used MMA Design HaWK panels with triple-junction GaAs cells. - **ISS CubeSats (various 1U):** Body-mounted only, 2--3 W. Adequate for simple sensor missions with low duty cycles.

Architecture Types

Table 3.1.3 — EPS bus architecture comparison across DET, MPPT, unregulated, and regulated topologies.

Architecture	How It Works (Physics)	Efficiency	Complexity	Typical Use
DET (Direct Energy Transfer)	SA connects directly to bus; a shunt regulator diverts excess current to a resistor bank (dissipated as heat) when SA output exceeds load. No series regulator between SA and bus. Voltage varies with illumination.	80--85% (shunt losses)	Low	Heritage GEO spacecraft, some CubeSats
PPT / MPPT (Peak Power Tracking)	A DC-DC converter (typically boost or buck-boost) between SA and bus continuously adjusts its input impedance to operate the SA at its maximum power point (MPP). Uses perturb-and-observe or incremental conductance algorithm. Extracts 10--15% more power than DET, especially at off-nominal temperatures and end-of-life.	90--95%	Medium	Most modern CubeSats
Unregulated bus	Battery connects directly to the power bus through protection FETs only. Bus voltage equals battery voltage (varies from 3.0 V at empty to 4.2 V at full per cell, or 6.0--8.4 V for 2S configuration). Subsystems must tolerate voltage variation.	Highest (no regulator losses)	Lowest	Very simple CubeSats, 1U missions

Architecture	How It Works (Physics)	Efficiency	Complexity	Typical Use
Regulated bus	DC-DC converters create fixed voltage rails (3.3 V, 5 V, 12 V) from battery/SA input. Subsystems see constant voltage regardless of battery state. Adds ~5--10% losses in regulators but greatly simplifies subsystem design.	Good (85--90% overall)	Medium	Most CubeSat COTS EPS (GomSpace P31u, Endurosat, AAC Clyde)

MPPT physics: A solar cell's I-V curve has a distinct "knee" where the maximum power ($\mathrm{P} = \mathrm{I} \times \mathrm{V}$) is extracted. At open-circuit ($\mathrm{I} = 0$), voltage is maximum but power is zero. At short-circuit ($\mathrm{V} = 0$), current is maximum but power is zero. The MPP sits at approximately 75--80% of V_{oc} and 90--95% of I_{sc}. Temperature shifts the I-V curve (higher T moves V_{oc} left), so the MPP moves. An MPPT controller dynamically tracks this point, typically updating every 0.1--1 s. Common CubeSat MPPT converters achieve 95--97% tracking efficiency.

Battery charge regulation: The EPS must prevent overcharging (which causes lithium plating, gas generation, and thermal runaway in Li-ion cells) and overdischarging (which causes copper dissolution from the negative current collector, permanently damaging the cell). Modern CubeSat EPS boards implement: - **CC-CV charging:** Constant-current charging until cell voltage reaches 4.2 V, then constant-voltage taper until current drops below C/20 - **Under-voltage lockout:** Bus disconnect when cell voltage falls below 3.0 V (or 2.8 V for emergency) - **Cell balancing:** For multi-cell series configurations (2S or higher), passive or active balancing circuits ensure cells remain within 50 mV of each other - **Temperature cutoffs:** Charging inhibited below 0 degC and above 45 degC (Li-ion charging below 0 degC causes lithium plating)

CubeSat standard: Most commercial EPS boards (GomSpace P31u, Endurosat, AAC Clyde) use MPPT + regulated bus with 3.3 V and 5 V rails, plus an unregulated battery rail (6.0--8.4 V for 2S Li-ion).

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2. Solar Array Sizing

> **Key Equations -- Solar Array Sizing (Full Derivation)**
>
> **Step 1: Orbit-average power demand:**

$$\mathrm{P}_{\mathrm{avg}} = \sum_{\mathrm{modes}} \mathrm{P}_{\mathrm{mode}} \times \mathrm{f}_{\mathrm{duty,mode}}$$

> This is computed from the ConOps mode table. For each mode (imaging, downlink, eclipse/safe, idle), multiply the mode power by the fraction of orbit spent in that mode.
>
> **Step 2: SA end-of-life power requirement:**
>
> During sunlight, the SA must simultaneously: (a) power all sunlit loads and (b) recharge the battery for the upcoming eclipse. The recharge power accounts for battery charge/discharge efficiency:

$$\mathrm{P}_{\mathrm{SA,EOL}} = \mathrm{P}_{\mathrm{peak,sunlight}} + \frac{\mathrm{P}_{\mathrm{eclipse}} \times \mathrm{t}_{\mathrm{eclipse}}}{\mathrm{t}_{\mathrm{sunlight}} \times \eta_{\mathrm{path,eclipse}}}$$

> where $\eta_{\mathrm{path,eclipse}} = \eta_{\mathrm{charge}} \times \eta_{\mathrm{discharge}} \times \eta_{\mathrm{regulator}}$.

For a typical CubeSat EPS: $\eta_{\text{charge}} \approx 0.92$ (Li-ion coulombic efficiency $\times$ charge regulator), $\eta_{\text{discharge}} \approx 0.95$ (battery internal resistance losses), $\eta_{\text{regulator}} \approx 0.90$ (DC-DC converter). Combined: $\eta_{\text{path,eclipse}} \approx 0.79$.

A simpler approximation uses $\eta_{\text{charge}} \approx 0.90$ as a lumped path efficiency, which is common in textbooks but slightly optimistic.

Step 3: Account for degradation and temperature:

$$\mathrm{P_{SA,BOL}} = \frac{\mathrm{P_{SA,EOL}}}{\mathrm{L_d} \times \mathrm{L_T}}$$

where: - $\mathrm{L_d} = (1-\delta)^{\mathrm{n}}$ is the radiation degradation factor ($\delta = 0.025$/yr for TJ GaAs in LEO) - $\mathrm{L_T} = 1 + \beta(\mathrm{T_{cell}} - \mathrm{T_{ref}})$ is the temperature derating factor ($\beta = -0.0019$/degC, $\mathrm{T_{ref}} = 28$ degC, $\mathrm{T_{cell}}$ typically 60--80 degC in LEO)

For a cell operating at 65 degC: $\mathrm{L_T} = 1 + (-0.0019)(65 - 28) = 1 - 0.070 = 0.930$ (7% power loss from temperature).

Step 4: Compute SA area:

$$\mathrm{A_{SA}} = \frac{\mathrm{P_{SA,BOL}}}{\eta_{\text{cell}} \times \mathrm{S} \times \cos(\theta) \times \mathrm{f_{pack}} \times \mathrm{f_{cover}}}$$

where: - $\eta_{\text{cell}} = 0.295$ (triple-junction GaAs efficiency at AM0, 28 degC -- STC rating) - $\mathrm{S} = 1361$ W/m^2 (solar constant at 1 AU, per Kopp & Lean 2011) - θ = sun incidence angle (0 deg for ideal tracking; for body-mounted, use orbit-averaged $\cos\theta$) - $\mathrm{f_{pack}} = 0.85$--0.90 (cell packing factor -- fraction of panel area covered by cells; gaps exist for cell interconnects, edge clearance, and harness routing) - $\mathrm{f_{cover}} = 0.97$ (cover glass transmission loss, typically 2--4%)

Step 5: SA mass:

$$\mathrm{m_{SA}} = \mathrm{A_{SA}} \times \sigma_{\text{SA}} + \mathrm{m_{mechanism}}$$

where σ_{SA} = areal density of the panel: - Body-mounted (cells on Al substrate): $\sigma \approx 2.0$--2.5 kg/m^2 - Rigid deployable (cells on Al honeycomb/CFRP panel): $\sigma \approx 1.5$--2.0 kg/m^2 - Flexible deployable (roll-out or fold-out): $\sigma \approx 0.8$--1.2 kg/m^2 - $\mathrm{m_{mechanism}}$ = hinge, spring, hold-down mechanism: typically 0.1--0.3 kg per panel for CubeSats

Worked Example -- 3U EO CubeSat (SuperDove-class) Solar Array

Given: $\mathrm{P_{peak,sunlight}} = 10.0$ W (imaging mode), $\mathrm{P_{eclipse}} = 3.5$ W, $\mathrm{t_{eclipse}} = 35$ min, $\mathrm{t_{sunlight}} = 60$ min, mission lifetime = 3 years, cell temperature = 65 degC, single deployable panel with fixed sun angle $\theta = 23$ deg (average over orbit for SSO with body-fixed panel).

Step 2: Recharge power (using detailed path efficiency): $\eta_{\text{path}} = 0.92 \times 0.95 \times 0.90 = 0.786$

$$\mathrm{P_{recharge}} = \frac{3.5 \times 35}{60 \times 0.786} = \frac{122.5}{47.2} = 2.60 \text{ W}$$

$$\mathrm{P_{SA,EOL}} = 10.0 + 2.60 = 12.60 \text{ W}$$

Step 3: BOL accounting for 3-year degradation + temperature: $\mathrm{L_d} = (1 - 0.025)^3 = 0.9269$

$$L_T = 1 + (-0.0019)(65 - 28) = 0.930$$

$$P_{SA,BOL} = \frac{12.60}{0.9269 \times 0.930} = \frac{12.60}{0.862} = 14.62 \text{ W}$$

Step 4: SA area: $A_{SA} = \frac{14.62}{0.295 \times 1361 \times \cos(23°) \times 0.85 \times 0.97}$

$$= \frac{14.62}{0.295 \times 1361 \times 0.921 \times 0.85 \times 0.97}$$

$$= \frac{14.62}{305.0} = 0.0479 \text{ m}^2$$

This is approximately 22 cm x 22 cm -- achievable with a single deployable panel on a 3U CubeSat (the MMA HaWK panel provides up to 0.06 m^2 per wing).

Step 5: SA mass (deployable, rigid): $m_{SA} = 0.0479 \times 1.8 + 0.15 = 0.086 + 0.15 = 0.24$ kg (panel + mechanism)

Comparison to Planet SuperDove: The actual SuperDove uses body-mounted cells plus two deployable wings, achieving ~25 W BOL. Our calculation (14.6 W BOL from one panel) is consistent -- SuperDove's higher power supports continuous imaging plus S-band downlink simultaneously.

CubeSat SA Power Reference

Table 3.1.4 — CubeSat solar array power reference by form factor and deployment configuration.

Configuration	1U	3U	6U	12U
Body-mounted only	~2 W	~7 W	~12 W	~20 W
Single deployable	~4 W	~15 W	~30 W	~55 W
Dual deployable	--	~25 W	~48 W	~100 W
Quad deployable	--	--	~80 W	~180 W

[Source: GomSpace, ISIS, MMA Design vendor datasheets; ASTERIA 6U confirmed 48 W BOL; Dove/ SuperDove confirmed ~25 W]

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3. Battery Sizing

Li-ion Cell Chemistry and Physics

All modern spacecraft batteries use lithium-ion (Li-ion) chemistry. During discharge, lithium ions migrate from the graphite anode (negative electrode) through an organic electrolyte and polymer separator to the lithium metal oxide cathode (positive electrode), while electrons flow through the external circuit doing work. During charging, the process reverses.

Common cathode chemistries used in space:

Table 3.1.5 — Li-ion cathode chemistries compared for voltage, energy density, and cycle life.

Chemistry	Cathode	Nominal Voltage	Energy Density	Cycle Life (30% DOD)	Thermal Stability	Space Heritage
	Cobalt oxide	3.7 V		~10,000	Moderate	

Chemistry	Cathode	Nominal Voltage	Energy Density	Cycle Life (30% DOD)	Thermal Stability	Space Heritage
LCO ($LiCoO_2$)			150--200 Wh/kg			ISS, many CubeSats (18650 cells)
NMC ($LiNiMnCoO_2$)	Nickel-manganese-cobalt	3.7 V	170--250 Wh/kg	~5,000	Moderate	Growing heritage
LFP ($LiFePO_4$)	Iron phosphate	3.2 V	90--120 Wh/kg	~50,000	Excellent	Niche (when cycle life is paramount)
NCA ($LiNiCoAlO_2$)	Nickel-cobalt-aluminium	3.6 V	200--260 Wh/kg	~3,000	Lower	Limited space heritage

[Source: Ratnakumar et al., NASA JPL; Saft VES-16 space cell datasheet; Samsung SDI 18650 specifications]

Cell form factors:

- **18650 cylindrical:** 18 mm diameter, 65 mm long. The workhorse of CubeSat missions. Common cells: Samsung 25R (2500 mAh, 20A continuous), Panasonic NCR18650B (3350 mAh, moderate rate), Sony VTC6 (3000 mAh, high rate). Energy: 9--12 Wh per cell.
- **Pouch cells:** Custom dimensions, higher energy density (~250 Wh/kg) but require external structural support. Used in some 6U+ missions and all large spacecraft. GomSpace NanoPower BPX uses pouch cells.
- **Prismatic cells:** Rigid case, intermediate between cylindrical and pouch. Used in some mission-specific designs.

Cell configuration:

Cells are arranged in series (S) to increase voltage and parallel (P) to increase capacity: - **1S (single cell):** 3.0--4.2 V bus. Used for very simple 1U CubeSats. - **2S (two in series):** 6.0--8.4 V bus. Standard for most CubeSats (matches GomSpace P31u default). - **2S2P (two series, two parallel):** 6.0--8.4 V bus, double capacity. Used for higher-energy missions (e.g., GomSpace BP4 pack: 4 cells, 2S2P, ~38 Wh).

Protection circuits: Every flight battery pack includes: - **Cell voltage monitoring:** Per-cell voltage measurement to detect over/under-voltage - **Over-current protection:** Current-sense resistors + MOSFET switches to disconnect at overcurrent (prevents short-circuit thermal runaway) - **Temperature monitoring:** Thermistors on each cell; inhibit charging below 0 degC and above 45 degC - **Heater circuit:** Kapton heater on battery pack, thermostatically controlled, to maintain cells above minimum charging temperature during eclipse

Capacity fade model: Li-ion cells lose capacity over time due to solid electrolyte interphase (SEI) growth on the anode. A simplified calendar + cycling fade model:

$$\mathrm{C}(\mathrm{t}, \mathrm{N}) = \mathrm{C}_0 \times (1 - \alpha\sqrt{\mathrm{t}}) \times (1 - \beta \cdot \mathrm{N} \cdot \mathrm{DOD}^{\gamma})$$

where C_0 = initial capacity, t = time (years), N = number of cycles, $\alpha \approx 0.02$ (calendar fade), $\beta \approx 3 \times 10^{-6}$, $\gamma \approx 2.1$ (cycling fade). For LEO at 30% DOD, this gives approximately 5--8% capacity loss per year -- consistent with flight data from ISS battery replacements and CubeSat fleet telemetry.

Key Equations -- Battery Sizing

Required battery energy:

$$\mathrm{E_{bat}} = \frac{\mathrm{P_{eclipse}} \times \mathrm{t_{eclipse}}}{\mathrm{DOD} \times \eta_{\mathrm{discharge}}}$$

where: - DOD = maximum depth of discharge - $\eta_{\mathrm{discharge}} = 0.95$ (discharge efficiency, accounting for internal resistance $\mathrm{I^2R}$ losses) - $\mathrm{t_{eclipse}}$ in hours

Battery mass:

$$\mathrm{m_{bat}} = \frac{\mathrm{E_{bat}}}{\mathrm{e_{specific}}}$$

where $\mathrm{e_{specific}} = 150\text{--}200$ Wh/kg for packaged Li-ion 18650 cells (cell-level energy density is higher, but packaging adds ~30% mass).

Cycle life vs DOD relationship (Li-ion 18650, LCO chemistry):

DOD	Typical Cycle Life	Suitable Mission Duration	Eclipses over the mission duration (15/day)
80%	~500 cycles	< 1 month	450
50%	~2,000 cycles	< 4 months	1,800
30%	~10,000 cycles	1--2 years	5,475
20%	~30,000 cycles	3--5 years	10,950
10%	~100,000 cycles	> 7 years	38,325

Design rule of thumb: For a multi-year LEO mission, start with 20--30% DOD. For a short technology demonstration (< 6 months), 40--50% DOD is acceptable and significantly reduces battery size/mass/cost.

Worked Example -- Battery for 3U EO CubeSat (SuperDove-class)

Given: $\mathrm{P_{eclipse}} = 3.5$ W, $\mathrm{t_{eclipse}} = 35$ min $= 0.583$ h, $\mathrm{DOD} = 0.25$ (conservative for 3-year mission), $\eta = 0.95$.

Step 1 -- Required battery energy: $\mathrm{E_{bat}} = \frac{3.5 \times 0.583}{0.25 \times 0.95} = \frac{2.04}{0.2375} = 8.59$ Wh

Step 2 -- Apply ECSS margin (20% at Phase A/B): $\mathrm{E_{bat,spec}} \geq 8.59 \times 1.20 = 10.3$ Wh. Specify **minimum 10 Wh**, ideally **20 Wh** for operational flexibility.

Step 3 -- Verify cycle life: 3-year mission at 15 orbits/day = 16,425 eclipses. At 25% DOD, Li-ion 18650 cells provide ~20,000 cycles. **Margin = 22%. Pass.**

Including capacity fade: after 3 years, ~15--20% capacity loss from calendar + cycling aging. Effective DOD increases to $0.25/0.82 = 0.30$ -- still within the 10,000-cycle regime. **Acceptable with monitoring.**

Step 4 -- Battery mass: $\mathrm{m_{bat}} = \frac{20}{170} = 0.118$ kg (using 170 Wh/kg for packaged 18650 cells)

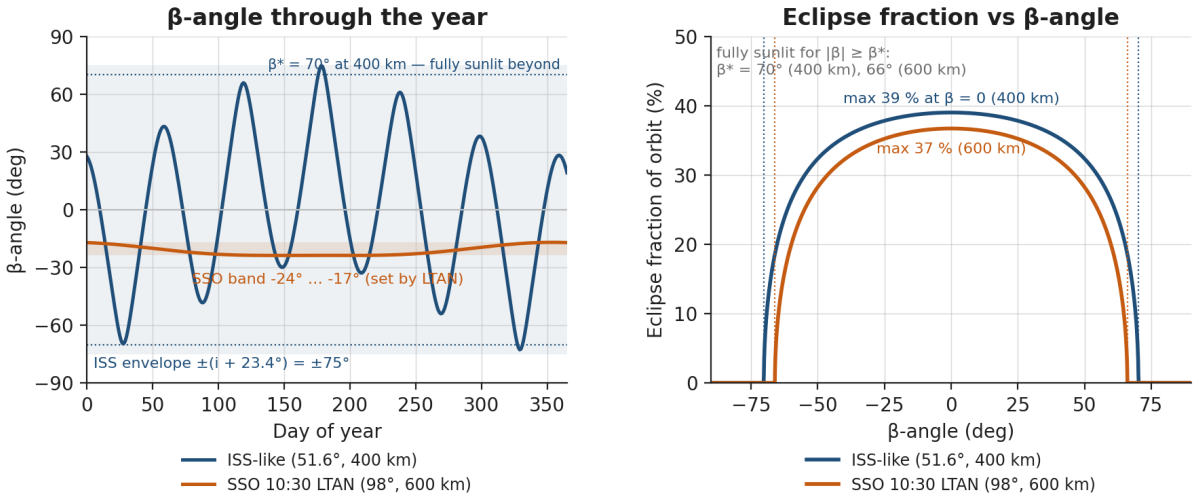
Step 5 -- Battery volume: Two 18650 cells in 2S1P: $2 \times \pi \times (9\text{mm})^2 \times 65\text{mm} \approx 33$ mL, fitting easily within a 3U stack.

Comparison to Planet SuperDove: SuperDove carries approximately 20 Wh in a 2S2P configuration (4 cells), consistent with our sizing. The actual operating DOD is estimated at 10--15% per eclipse, giving substantial cycle-life margin for the multi-year constellation replenishment cadence.

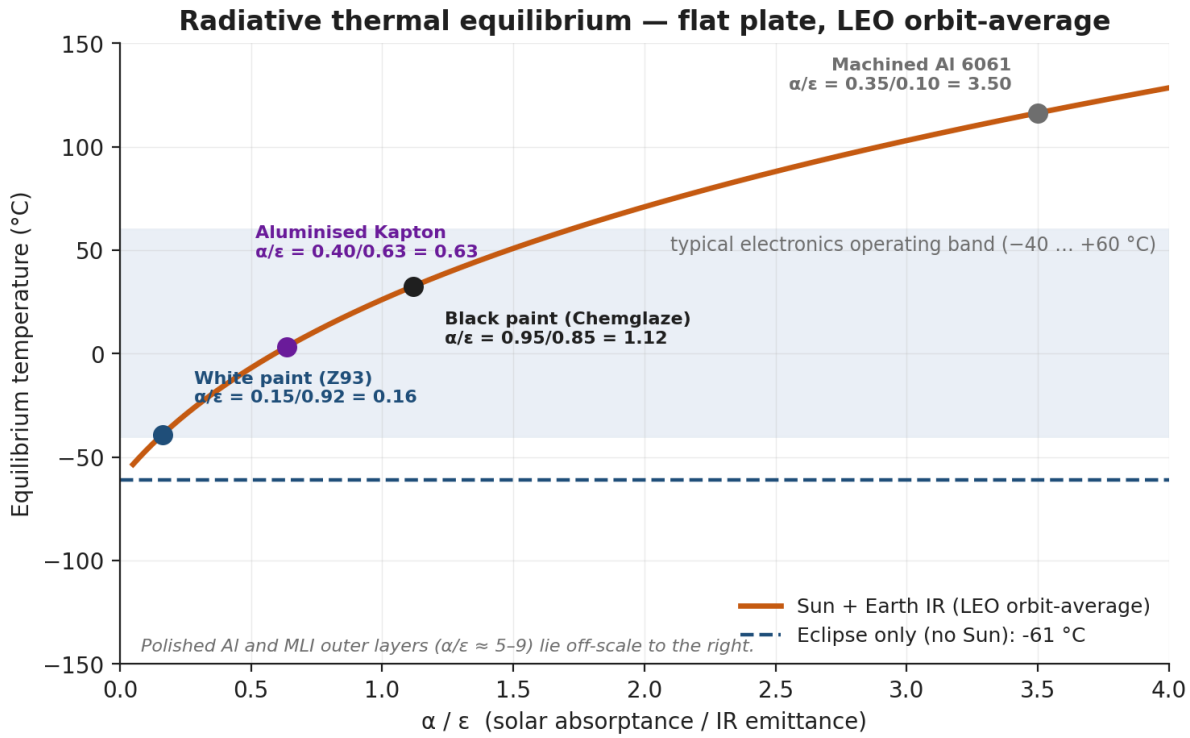
Failure modes to watch for: - **Thermal runaway:** If a cell is overcharged, mechanically damaged, or experiences an internal short, the exothermic decomposition of the cathode material can lead to thermal runaway (self-heating > heat dissipation), potentially reaching 600+ degC. Mitigation: cell-level fuses, per-cell voltage monitoring, thermal cutoffs. - **Lithium plating:** Charging below 0 degC causes metallic lithium to deposit on the anode surface rather than intercalating into graphite. This is irreversible, reduces capacity, and can cause internal shorts. Mitigation: battery heater + thermostat + software lockout. - **Capacity imbalance:** In series configurations, the weakest cell limits the pack. If cells age at different rates, the weakest cell hits under-voltage lockout while others still have capacity. Mitigation: cell balancing circuits, matched cell lots.

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4. Thermal Control System



β-angle envelope and eclipse fraction analytics.



Radiative thermal equilibrium vs α/ε.

[Source: ECSS-E-ST-31C; Gilmore, Ch. 1--4]

The Physics of Spacecraft Thermal Control

Spacecraft thermal control is fundamentally different from terrestrial thermal engineering because **there is no convection in vacuum**. The only heat transfer mechanisms are:

1. **Conduction:** Heat flow through solid material, governed by Fourier's law: $\dot{Q} = -kA\frac{dT}{dx}$. Critical within the spacecraft structure and between components and mounting surfaces.
2. **Radiation:** Heat transfer via electromagnetic radiation, governed by the Stefan-Boltzmann law: $\dot{Q} = \varepsilon\sigma AT^4$. This is the **only** mechanism for rejecting heat to the environment.

There is no convective cooling. A component that overheats cannot be cooled by a fan. All waste heat must be conducted to a radiating surface and then radiated to space. This is the central constraint of spacecraft thermal design.

Thermal Environment in LEO

A spacecraft in LEO experiences four thermal inputs and one thermal sink:

Table 3.1.6 — LEO thermal environment sources with fluxes, directions, and variability.

Source	Flux	Direction	Variability
Direct solar	$S = 1361 \pm 1$ W/m^2 (at 1 AU)	Sun-facing surfaces only	Seasonal (± 3.3 % due to Earth's orbital eccentricity: $S_{perihelion} = 1414$ W/m^2 in January, $S_{aphelion} = 1322$ W/m^2 in July)
Earth albedo	$\alpha_E \times S \approx 0.30 \times 1361 \approx 408$ W/m^2	Earth-facing surfaces (nadir)	

Source	Flux	Direction	Variability
			Varies with cloud cover, surface type (0.06 for ocean to 0.80 for fresh snow); orbit-average range 0.25--0.35
Earth infrared	$q_{IR} \approx 240$ W/m^2 (orbit average)	Earth-facing surfaces (nadir)	Range 200--270 W/m^2 depending on latitude, season, cloud cover
Internal dissipation	$Q_{int} = P_{dissipated}$	From electronics waste heat	Varies with operational mode; nearly all electrical power eventually becomes heat
Deep space (sink)	$T_{space} \approx 2.7$ K (CMB)	Zenith-facing radiator surfaces	Effectively 0 K for engineering purposes

View Factors

The fraction of a surface's radiative "view" that sees each thermal source is critical for accurate thermal modelling. For a nadir-pointing spacecraft in LEO:

$$F_{Earth} = \frac{1}{1 + (h/R_E)^2 + 2(h/R_E)}$$

where h = altitude (km) and $R_E = 6371$ km. For a 500 km orbit: $F_{Earth} = 1/(1 + (500/6371)^2 + 2 \times 500/6371) = 1/1.163 = 0.860$. The nadir face sees 86% Earth and 14% deep space. The zenith face sees 100% deep space (assuming no S/C self-shadowing). Side faces see a mix.

At higher altitudes, F_{Earth} decreases: at 800 km it is 0.79, at 35,786 km (GEO) it is only 0.023 -- which is why GEO thermal design is dominated by solar flux and internal dissipation, not Earth IR/albedo.

Thermal Balance Equation -- Full Derivation

> **Key Equations -- Thermal Equilibrium**
>
> At steady state, the absorbed heat equals the radiated heat:

$$Q_{in} = Q_{out}$$

> **Absorbed heat (expanded):**

$$Q_{in} = \alpha_s \cdot A_{sun} \cdot S + \alpha_s \cdot A_{alb} \cdot F_{alb} \cdot \alpha_E \cdot S + \varepsilon \cdot A_{IR} \cdot F_{IR} \cdot q_{IR} + Q_{int}$$

> The four terms are, in order: direct solar, Earth albedo, Earth IR, and internal dissipation.
>
> **Radiated heat:**

$$Q_{out} = \varepsilon \cdot \sigma \cdot A_{rad} \cdot T^4$$

> where: - α_s = solar absorptance of surface coating (dimensionless, 0--1) - ε = infrared emittance of surface coating (dimensionless, 0--1) - $\sigma = 5.670 \times 10^{-8}$ W/m^2/K^4 (Stefan-Boltzmann constant) -

A_{sun}, A_{alb}, A_{IR}, A_{rad} = projected areas for each flux (m^2) - F_{alb}, F_{IR} = view factors to Earth for albedo and IR surfaces - T = equilibrium temperature (K)

Note on α_s vs ε: Solar absorptance (α_s) is measured over the solar spectrum (0.2--2.5 um, peak at 0.5 um visible). Infrared emittance (ε) is measured over the thermal IR spectrum (3--50 um, peak at ~10 um for room-temperature objects). These are **different spectral ranges**, so $\alpha_s \neq \varepsilon$ for most real surfaces. This decoupling is the basis of all passive thermal control: by choosing the α_s/ε ratio, the designer controls the equilibrium temperature.

Solving for equilibrium temperature:

$$T = \left(\frac{Q_{absorbed} + Q_{internal}}{\varepsilon \sigma A_{rad}} \right)^{1/4}$$

This equation is valid only for a single isothermal node (lumped-parameter model). For multi-node thermal models (which are needed for any real spacecraft), the heat balance is solved simultaneously for all nodes using numerical methods (Thermal Desktop, ESATAN, or similar thermal analysis software).

Hot Case and Cold Case

Table 3.1.7 — Hot case versus cold case conditions and the components they stress.

Case	Conditions	Design Concern
Hot case	Maximum solar exposure ($S = 1414$ W/m^2, perihelion), all subsystems active (max Q_{int}), worst sun angle (max A_{sun}), EOL coatings (α_s at maximum -- increased by UV darkening), max albedo (0.35)	Components exceed maximum operating temperature
Cold case	Eclipse (no solar), minimum power dissipation (safe mode, min Q_{int}), BOL coatings (α_s at minimum -- fresh white paint), minimum Earth IR (200 W/m^2), deep space view	Components fall below minimum operating temperature

Design philosophy: The thermal engineer designs to keep all components within their qualified temperature range under both worst-case hot and worst-case cold conditions, with ECSS-mandated margins applied.

Surface Coatings -- The Passive Thermal Toolbox

Table 3.1.8 — Spacecraft surface coatings with absorptance, emissivity, and use cases.

Coating	α_s (BOL)	α_s (EOL, 5 yr LEO)	ε	α_s/ε (BOL)	Use Case
White paint (AZ-93, S13G-LO)	0.14--0.20	0.25--0.35	0.89--0.92	0.16--0.22	Radiator surfaces (stay cool; low solar absorption, high IR emission)
Black paint (Aeroglaze Z306)	0.95	0.95	0.89	1.07	Internal surfaces (maximise radiative exchange between components)

Coating	α_s (BOL)	α_s (EOL, 5 yr LEO)	ε	α_s/ε (BOL)	Use Case
Gold tape (2 mil Kapton + VDA)	0.22--0.25	0.25--0.30	0.03--0.05	5.0--7.5	MLI outer layer, thermal isolation
Bare aluminium (polished)	0.10--0.15	0.15--0.20	0.03--0.05	2.5--4.0	Reflective surfaces, low emissivity
Alodine (chromate conversion on Al)	0.35--0.40	0.40--0.50	0.12--0.16	2.5--3.0	Moderate thermal control, structural surfaces
Anodised aluminium (clear or black)	0.30--0.50	0.35--0.55	0.75--0.86	0.4--0.65	CubeSat external structure (standard finish per CDS)
MLI blanket (effective)	0.05--0.15	0.10--0.20	0.02--0.05	~3	Thermal isolation of sensitive components
Solar cells (with cover glass)	0.75--0.92	0.78--0.92	0.80--0.85	~1.0	SA surfaces (high absorption is unavoidable; cells get hot)
OSR (Optical Solar Reflector)	0.05--0.08	0.08--0.12	0.78--0.80	0.06--0.10	High-performance radiators (large S/C, GEO)

Key design insight: A surface with low α_s/ε (e.g., white paint: 0.2) stays cool because it reflects most solar energy but efficiently radiates thermal IR. A surface with high α_s/ε (e.g., gold tape: 6.0) stays warm because it absorbs solar energy but barely radiates. This is why white paint is used on radiators and gold/MLI is used for insulation.

Coating degradation: UV radiation darkens most white paints over time, increasing α_s while leaving ε nearly unchanged. This means α_s/ε increases, and the surface gets hotter at EOL. Note carefully which end of life is conservative for which case: BOL white paint has the *lowest* α_s, so the **cold case** must be evaluated with BOL coatings and the **hot case** with EOL coatings, once α_s has increased. This is a common source of confusion -- BOL coatings give a colder cold case, EOL coatings give a hotter hot case, and both must be checked.

MLI (Multi-Layer Insulation) -- Construction and Physics

MLI blankets are the most common thermal insulation on spacecraft. They work by minimising both radiation and conduction heat transfer through multiple reflective layers separated by low-conductance spacers.

Construction (typical MLI blanket): 1. **Outer cover:** 1 mil (25 um) aluminised Kapton (VDA -- Vapour Deposited Aluminium on one side). Provides mechanical protection and low solar absorptance. 2. **Inner reflective layers:** 10--20 layers of 0.25 mil (6 um) double-aluminised Mylar (DAM). Each layer reflects IR radiation, and the vacuum gaps between layers have zero convection. 3. **Spacer material:** Dacron or Nomex netting between each DAM layer, preventing conductive contact between adjacent reflective sheets. 4. **Inner cover:** 1 mil aluminised Kapton, protecting the inner layers.

Effective emissivity: An ideal MLI blanket with N reflective layers has an effective emissivity of:

$$\varepsilon_{\mathrm{eff}} = \frac{1}{2/\varepsilon_{\mathrm{inner}} + (\mathrm{N}-1)(2/\varepsilon_{\mathrm{layer}} - 1)}$$

For 20 layers of DAM ($\varepsilon_{\text{layer}} = 0.03$): $\varepsilon_{\text{eff}} \approx 0.002$. In practice, real MLI achieves $\varepsilon_{\text{eff}} = 0.01$--$0.03$ due to seams, penetrations (harness, mounting), and edge effects. The ratio of actual to theoretical performance is typically 2--5x worse.

CubeSat MLI challenges: CubeSats have limited surface area and many penetrations (connectors, antennas, sensors, solar cells), making it difficult to achieve good MLI performance. Most CubeSats in LEO do not use MLI -- they rely on the moderate thermal environment (Earth IR provides a "warm floor") and surface coatings. MLI becomes essential for deep-space CubeSats (e.g., MarCO, CAPSTONE) or missions with sensitive payloads (IR detectors, laser systems).

Heater Sizing

When passive thermal control cannot prevent a component from falling below its minimum temperature (typically during eclipse or safe mode), electrical heaters are required.

> **Key Equations -- Heater Sizing**
>
> **Required heater power** (to maintain minimum temperature during worst cold case):

$$\mathrm{P}_{\text{heater}} = \varepsilon \sigma \mathrm{A}_{\text{rad}} \mathrm{T}_{\text{min}}^4 - \mathrm{Q}_{\text{environment,cold}} - \mathrm{Q}_{\text{internal,cold}}$$

> where T_{min} is the minimum allowable temperature of the component (converted to Kelvin).
>
> **Heater types for CubeSats:** - **Kapton foil heaters:** Etched-foil resistance elements laminated between Kapton sheets. Flexible, thin (0.2--0.5 mm), lightweight (2--10 g each). Typical power: 0.5--5 W per heater. Bond directly to component surface with pressure-sensitive adhesive. - **Cartridge heaters:** Cylindrical, inserted into drilled holes. Higher power density but heavier and less common on CubeSats.
>
> **Thermostat control:** Simple bimetallic thermostats (e.g., Honeywell Klixon) switch heaters on/off at set temperatures (e.g., on at -5 degC, off at +5 degC). Mass: ~2 g each. For higher reliability, software-controlled heaters using temperature sensor feedback and EPS switches are preferred on modern CubeSats -- but this requires OBC to be running, which may not be the case in safe mode.

Heat Pipes and Thermal Straps

Heat pipes are passive two-phase heat transfer devices that transport large amounts of thermal energy with very small temperature differences. They are widely used on larger spacecraft and are beginning to appear on 6U+ CubeSats.

How a heat pipe works: 1. Working fluid (ammonia, methanol, or water) evaporates at the hot end (evaporator), absorbing latent heat 2. Vapour travels through the hollow pipe core to the cold end (condenser) 3. At the condenser, vapour releases latent heat and condenses back to liquid 4. Liquid returns to the evaporator via capillary action in a wick structure (sintered metal, axial grooves, or screen mesh) 5. The process is continuous, passive (no moving parts, no power), and can transport 10--100 W across 20--50 cm with < 5 degC temperature difference

Thermal conductance of a heat pipe: Effective thermal conductivity is 10,000--100,000 W/m/K (compared to copper at 400 W/m/K and aluminium at 237 W/m/K). A 6 mm diameter ammonia heat pipe can transport ~30 W over 30 cm with < 3 degC gradient.

Thermal straps are flexible conductive links (braided copper, graphite fibre, or pyrolytic graphite sheet) used to conduct heat between components that cannot be rigidly connected (e.g., across a hinge or between a vibration-isolated payload and the spacecraft bus). Typical conductance: 0.5--5 W/K.

Table 3.1.9 — Heat transport methods compared by conductance, mass, and CubeSat applicability.

Heat Transport Method	Conductance	Mass	Power	Orientation Sensitivity	CubeSat Use
Aluminium conduction	~0.5--2 W/K per path	Part of structure	0 W	None	Always (inherent)
Copper thermal strap	1--5 W/K	10--50 g	0 W	None	Occasional (6U+)
Heat pipe (ammonia)	5--50 W/K	20--100 g	0 W	Gravity-dependent (must test in relevant orientation)	Rare (6U+, some 3U)
Pumped fluid loop	50--500 W/K	500+ g	5--20 W	None	Large S/C only

Thermal Control Methods Summary

Table 3.1.10 — Passive and active thermal control methods with mass impact and design parameters.

Method	Type	Mass Impact	Typical Use	Key Design Parameter
Surface coatings	Passive	Negligible	Always -- select α_s/ε ratio per face	α_s/ε ratio
MLI blankets	Passive	0.5--2.0 kg/m ^{2}	Insulate sensitive components from environment	Number of layers, ε_{eff}
Radiators	Passive	Part of structure	Reject internal waste heat to deep space	Radiator area, ε, view to space
Heaters	Active	0.005--0.02 kg each	Maintain minimum temp during eclipse/safe mode	Power, thermostat set point
Heat pipes	Passive	0.02--0.10 kg each	Transport heat from source to radiator	Working fluid, $\mathrm{Q}_{\max}$, orientation
Thermal straps	Passive	0.01--0.05 kg each	Flexible conductive link across joints/hinges	Conductance (W/K)
Louvers	Active	0.1--0.5 kg	Variable-conductance radiators (rare on CubeSats)	Open/close temperature range

ECSS Thermal Margins

[Source: this book's teaching convention; the standards are discussed below the table]

Table 3.1.11 — Thermal margins for qualification, acceptance, and operating temperature limits (this book's convention).

Phase	Hot Margin (above predicted max)	Cold Margin (below predicted min)
Qualification	Predicted + 15 degC	Predicted - 15 degC

Phase	Hot Margin (above predicted max)	Cold Margin (below predicted min)
Acceptance	Predicted + 10 degC	Predicted - 10 degC
Operating	Predicted + 5 degC	Predicted - 5 degC

These margins ensure that thermal model uncertainties (typically ± 5 -- 10 degC for simplified models, ± 2 -- 5 degC for detailed correlated models) do not cause in-orbit temperature exceedances.

How this relates to the standards. ECSS-E-ST-31C does not tabulate these values. Its Figure 3-1 defines the chain from the predicted temperature, through the uncertainty, to the acceptance and qualification temperatures; it leaves the margins to the system authority and notes that uncertainties below 10 K are not normally applied before a thermal-balance test. ECSS-E-ST-10-03C (Tables 5-2 and 5-4) adds an acceptance margin of 5 degC and a qualification margin of 5 degC on top of that range, and GSFC-STD-7000B §2.6 tests a proto-flight unit at the maximum and minimum predicted temperatures ± 10 degC for at least 4 cycles. The proto-flight thermal-vacuum level of Chapter 4.2 (predicted ± 10 degC) follows those standards.

Worked Example -- 3U CubeSat Full Thermal Analysis

Hot case (sunlit, all systems active, perihelion, EOL coatings):

Simplified single-node model. 3U CubeSat, nadir-pointing.

- Sun-facing area ($+\mathrm{Z}$, zenith-facing 3U panel): $\mathrm{A}_{\mathrm{sun}} = 0.034$ m^2
- SA surfaces (sun-facing): $\alpha_{\mathrm{s}} = 0.85$, $\varepsilon = 0.82$ (solar cell properties)
- Nadir face ($-\mathrm{Z}$): $\mathrm{A}_{\mathrm{nadir}} = 0.034$ m^2, anodised Al ($\alpha_{\mathrm{s}} = 0.45$ EOL, $\varepsilon = 0.82$)
- Side faces (4x): $\mathrm{A}_{\mathrm{side}} = 4 \times 0.01$ m^2 = 0.04 m^2, anodised Al
- Internal dissipation (imaging mode): $\mathrm{Q}_{\mathrm{int}} = 10.0$ W

$\mathrm{Q}_{\mathrm{solar}} = 0.85 \times 0.034 \times 1414 = 40.9$ W (ouch -- but much of this is captured by the SA and converted to electricity, so the net thermal input from solar cells is $\mathrm{Q}_{\mathrm{solar,thermal}} = \alpha_{\mathrm{s}} \times \mathrm{A} \times \mathrm{S} \times (1 - \eta_{\mathrm{cell}}) = 0.85 \times 0.034 \times 1414 \times 0.705 = 28.8$ W)

$\mathrm{Q}_{\mathrm{albedo}} = 0.45 \times 0.034 \times 0.35 \times 1414 = 7.6$ W

$\mathrm{Q}_{\mathrm{Earth\ IR}} = 0.82 \times 0.034 \times 270 = 7.5$ W (nadir face, hot case Earth IR)

$\mathrm{Q}_{\mathrm{int}} = 10.0$ W (essentially the whole electrical load ultimately becomes heat inside the spacecraft -- only the few tenths of a watt radiated away as RF leaves by any other route)

Total $\mathrm{Q}_{\mathrm{in}} \approx 28.8 + 7.6 + 7.5 + 10.0 = 53.9$ W

Total radiating area (all 6 faces, minus SA area which is a net absorber): $\mathrm{A}_{\mathrm{rad}} \approx 0.066$ m^2 (accounting for partial Earth blockage on nadir face)

Average emissivity: $\varepsilon_{\mathrm{avg}} \approx 0.82$

$$\mathrm{T}_{\mathrm{hot}} = \left(\frac{53.9}{0.82 \times 5.67 \times 10^{-8} \times 0.066} \right)^{0.25} = \left(\frac{53.9}{3.07 \times 10^{-9}} \right)^{0.25}$$

$$= (1.756 \times 10^{10})^{0.25} = 364 \text{ K} = +91 \text{ degC}$$

This exceeds most component limits! However, this simplified calculation overestimates temperature because it treats the satellite as a single isothermal node and includes solar cell thermal absorption on the zenith face. In practice: - The zenith face (solar cells) runs hotter than the bus - Internal

components are conductively coupled to all faces, including the cold nadir face - A multi-node model typically predicts peak internal temperatures of +40 to +55 degC for this scenario

Thermal engineer's response: If the single-node calculation exceeds 60 degC, a detailed multi-node thermal model is required. The simplified calculation is a screening tool, not a design tool.

Cold case (eclipse, safe mode, aphelion, BOL coatings):

- No solar flux, no albedo
- Earth IR only: $Q_{\text{Earth IR}} = 0.82 \times 0.034 \times 200 = 5.58$ W (cold case: 200 W/m^2)
- Internal dissipation (safe mode): $Q_{\text{int}} = 1.5$ W (OBC + heater)
- Total $Q_{\text{in}} = 5.58 + 1.5 = 7.08$ W

$$T_{\text{cold}} = \left(\frac{7.08}{0.82 \times 5.67 \times 10^{-8} \times 0.070}\right)^{0.25}$$

$$= \left(\frac{7.08}{3.26 \times 10^{-9}}\right)^{0.25} = (2.172 \times 10^{9})^{0.25} = 216 \text{ K} = -57 \text{ degC}$$

This is too cold for Li-ion batteries (min -20 degC operating, min 0 degC charging) and most COTS electronics (min -40 degC).

Action: Add battery heater. To maintain battery at $T_{\text{min}} = -10$ degC = 263 K:

Need additional heat input: $Q_{\text{heater}} = \varepsilon\sigma A_{\text{rad}} T_{\text{min}}^4 - Q_{\text{other}}$

$= 0.82 \times 5.67 \times 10^{-8} \times 0.070 \times 263^4 - 7.08 = 3.26 \times 10^{-9} \times 4.78 \times 10^{9} - 7.08 = 15.6 - 7.08 = 8.5$ W

Problem: 8.5 W heater in eclipse exceeds the battery capacity. **Resolution:** This is an isothermal whole-spacecraft calculation. In reality, the battery is inside the bus, partially insulated by the structure and surrounding boards. A targeted heater of 0.5--1.0 W directly on the battery pack, with some MLI wrapping, is typically sufficient to keep the battery above -10 degC in a 35-minute eclipse. The structure's thermal mass (aluminium at $c_p = 900$ J/kg/K) provides significant thermal inertia -- the 5 kg 3U itself, losing about 7 W net, cools by only about 3.3 degC in a 35-minute eclipse (estimate below).

Transient check: $\Delta T = \frac{Q_{\text{net}} \times t}{m \times c_p} = \frac{(7.08 - 0) \times 35 \times 60}{5.0 \times 900} = \frac{14,868}{4500} = 3.3$ degC per 35-min eclipse

Wait -- in eclipse $Q_{\text{out}} > Q_{\text{in}}$: net cooling rate = $\varepsilon\sigma A_{\text{rad}} T^4 - Q_{\text{in,eclipse}}$. At $T = 293$ K (20 degC):

$Q_{\text{out}} = 0.82 \times 5.67 \times 10^{-8} \times 0.070 \times 293^4 = 3.26 \times 10^{-9} \times 7.37 \times 10^{9} = 24.0$ W

$Q_{\text{net cooling}} = 24.0 - 7.08 = 16.9$ W

$\Delta T = \frac{16.9 \times 2100}{5.0 \times 900} = \frac{35,490}{4500} = 7.9$ degC drop in 35 minutes

So from +20 degC, the satellite cools to about +12 degC after one eclipse -- well within limits. **No heater needed for the bus; battery heater only if battery is thermally isolated from bus.**

ECSS margin check -- Payload CCD: Predicted maximum temperature of payload CCD = 42 degC. - Operating limit = 50 degC. Margin = 50 - 42 = 8 degC > 5 degC. **Pass.** - Qualification test: must test at 42 + 15 = 57 degC. If qualification limit is 60 degC: **Pass.**

Predicted minimum temperature of battery = -8 degC during worst eclipse. - Operating limit = -10 degC. Margin = -8 - (-10) = 2 degC < 5 degC. **Fail -- inadequate margin.** - **Action:** Add heater (0.5 W survival heater with thermostat set to -5 degC), or add MLI around battery pack.

1U Worked Example: UniSat-1

Power Sizing: Body-Mounted Only

UniSat-1 uses body-mounted solar cells on all five sun-exposed faces (the sixth face mounts the deployment switch interface). With no deployable panels, the power system is simpler, lighter, and cheaper -- but severely power-limited.

Worked Example -- UniSat-1 Solar Array Sizing

Given: Body-mounted cells on 5 faces of a 1U (100 x 100 mm each). ISS orbit: 400 km, 51.6 deg inclination, 92.4 min period, 56 min sunlight, 36 min eclipse. $P_{eclipse} = 0.5$ W (OBC only), $P_{peak,sunlight} = 1.2$ W (science + downlink overlap avoided by scheduling). Mission lifetime = 6 months. Cell temperature = 55 degC (body-mounted cells run cooler on 1U due to better thermal coupling to bus mass).

Step 1 -- Effective illuminated area: At any given time in LEO with passive magnetic attitude (slow tumble ~1 deg/s), on average only ~1.5 faces are well-illuminated. Effective average area: $A_{eff} \approx 1.5 \times (0.10 \times 0.10) = 0.015$ m^2

Step 2 -- SA BOL power (with temperature derating): $L_T = 1 + (-0.0019)(55 - 28) = 0.949$

$P_{SA,BOL} = \eta_{cell} \times L_T \times S \times A_{eff} \times f_{pack} = 0.295 \times 0.949 \times 1361 \times 0.015 \times 0.80 = 4.57$ W (illuminated peak)

Step 3 -- Orbit-average power available:
$P_{avg,avail} = P_{SA,BOL} \times \frac{t_{sun}}{T} \times \eta_{EPS} = 4.57 \times \frac{56}{92.4} \times 0.85 = 2.35$ W

After 6-month degradation ($(1 - 0.025)^{0.5} = 0.987$): $P_{avg,EOL} = 2.35 \times 0.987 = 2.32$ W

Step 4 -- Power demand (orbit-average): From Chapter 2.4: $P_{avg,demand} = 0.68$ W.

Power margin: $2.32 - 0.68 = 1.64$ W (**71% margin**). Even with conservative geometry assumptions, the power budget closes comfortably.

Note on body-mounted vs tumbling: The key uncertainty in 1U body-mounted power is the attitude. With passive magnetic stabilisation, the satellite aligns roughly with Earth's magnetic field, providing more predictable illumination than a random tumble. However, the effective area varies significantly around the orbit as the B-field direction changes with latitude. The 1.5-face average is conservative. A Monte Carlo simulation of illumination geometry over many orbits typically yields a more optimistic 1.7--1.9 effective face average.

Worked Example -- UniSat-1 Battery Sizing

Given: $P_{eclipse} = 0.5$ W, $t_{eclipse} = 36$ min $= 0.60$ h, $DOD = 0.50$ (acceptable for 6-month mission: ~2,740 cycles), $\eta = 0.95$.

$E_{bat} = \frac{0.5 \times 0.60}{0.50 \times 0.95} = \frac{0.30}{0.475} = 0.63$ Wh

With margin: specify minimum **10 Wh** (standard GomSpace NanoPower P31u battery pack -- this is the smallest available COTS battery with flight heritage).

Cycle count check: 6 months at 15 orbits/day = 2,740 eclipses. At 50% DOD, Li-ion cells comfortably survive > 2,000 cycles. **Pass.**

Actual operating DOD: With 10 Wh battery and 0.63 Wh per eclipse demand, actual DOD = 0.63 / 10 = **6.3%** per eclipse. At this DOD, cycle life exceeds 100,000 cycles. **Battery degradation is negligible** over the 6-month mission.

Key insight: The 10 Wh battery is massively oversized for the actual eclipse demand. This is common in 1U missions -- the minimum COTS battery available provides far more capacity than needed. The excess capacity provides excellent margin and enables recovery from anomalies (multiple missed sunlit periods).

Thermal: Passive Only

UniSat-1 uses no heaters, no MLI, and no active thermal control. This is justified by three factors:

1. **Low altitude (400 km):** Strong Earth IR flux (~240 W/m ^{2}) provides a warm floor, preventing extreme cold cases
2. **Short mission (6 months):** No long-term coating degradation to worry about
3. **Tolerant components:** COTS electronics typically operate from -20 degC to +60 degC; the 400 km LEO thermal environment stays within -10 degC to +45 degC for a 1U with standard aluminium/ anodised surfaces

Quick Thermal Check -- UniSat-1 Cold Case (Transient)

Worst eclipse, all subsystems off except OBC (0.5 W internal dissipation): - Earth IR absorbed: $\varepsilon \times \mathrm{A_{nadir}} \times \mathrm{q_{IR}} = 0.85 \times 0.01 \times 240 = 2.04$ W - Internal dissipation: 0.5 W - Total heat in: 2.54 W

Steady-state temperature (if eclipse were infinite):
$\mathrm{T_{steady}} = (2.54/(0.85 \times 5.67 \times 10^{-8} \times 0.045))^{0.25} = 185$ K $= -88$ degC

This looks alarming -- but the eclipse is only 36 minutes, and the thermal mass prevents the satellite from reaching steady state.

Transient analysis: Starting at $\mathrm{T_0} = +15$ degC (288 K) at eclipse entry:

Net cooling rate at 288 K:
$\mathrm{Q_{rad,out}} = 0.85 \times 5.67 \times 10^{-8} \times 0.045 \times 288^4 = 2.17 \times 10^{-9} \times 6.88 \times 10^9 = 14.9$ W

Net cooling: $14.9 - 2.54 = 12.4$ W

Temperature drop: $\Delta \mathrm{T} = \frac{\mathrm{Q_{net}} \times \mathrm{t}}{\mathrm{m} \times \mathrm{c_p}} = \frac{12.4 \times 36 \times 60}{1.0 \times 900} = \frac{26,784}{900} = 30$ degC

But this is a linear approximation -- as the satellite cools, the radiation rate drops as T^4 , so the actual cooling slows. A more accurate estimate gives $\Delta \mathrm{T} \approx 20$ -- 25 degC, resulting in a minimum temperature of about -5 to -10 degC.

Conclusion: The 1U at 400 km reaches approximately -5 to -10 degC during worst-case eclipse, starting from a warm sunlit entry. This is within COTS operating limits (-20 degC to +60 degC for most components) and within battery operating range (-20 degC to +60 degC for discharge). **No heaters needed.** The ISS orbit's comparable eclipse (36 min vs 35 min for SSO) and the strong Earth IR flux at 400 km make passive thermal control viable for a 1U mission.

6U Reference Design: EOSAT-1 Power and Thermal, As Flown

The 1U and 3U examples above are sizing exercises. EOSAT-1 -- the 6U EO spacecraft used as the reference mission throughout this book -- is a *completed* design, so its numbers can be read back rather than estimated. Every figure below comes from the flight configuration (`configs/eosat1/subsystems/eps.yaml`, `tcs.yaml`) and the sizing derivation that produced it, and each hardware value is tied to a named flight part rather than to a margin target. The temperature coefficient ($\beta = -0.0020$/degC) and degradation rate (2.75 %/yr) below are therefore the flight values, not the generic -0.0019/degC and 2.5 %/yr used in the sizing exercises above.

Step 1 -- the plane-of-array constant. Rather than carry η_{cell}, f_{pack} and f_{cover} as separate factors at the end of the sizing chain, the design collapses the whole loss stack into a single delivered flux K in W/m^2 of cell area:

$$\text{K} = \text{S} \times \eta_{\text{cell}} \times \text{f}_{\text{inh}} \times [1 + \beta(\text{T} - \text{T}_{\text{ref}})] \times \eta_{\text{MPPT}} \times \eta_{\text{harness}}$$

$$\text{K}_{\text{BOL}} = 1361 \times 0.295 \times 0.85 \times [1 - 0.0020(65 - 28)] \times 0.97 \times 0.95$$

$$= 401.5 \times 0.85 \times 0.926 \times 0.97 \times 0.95 = 291.2 \ \text{W/m}^2$$

Applying three years of triple-junction GaAs degradation at 2.75 %/yr:

$$\text{K}_{\text{EOL}} = 291.2 \times (1 - 0.0275)^3 = 291.2 \times 0.9198 = \mathbf{267.9} \ \text{W/m}^2$$

That number is the design's single most useful sanity check. Endurosat publish 19.2 W in LEO from one 0.0715 m^2 panel of their 6U deployable array, which is $19.2/0.0715 = 268$ W/m^2 delivered. The loss stack and a flight vendor's datasheet agree to within 0.1 W/m^2 -- so if a CubeSat power budget disagrees with the hardware, the error is almost always in the *area*, not in the physics.

Step 2 -- cell area. EOSAT-1 carries body-mounted cells plus two single-axis-driven deployable wings:

Surface	Cell area (m^2)	Note
$+\text{X}$ / $-\text{X}$ (366 x 226 faces)	0.060 each	73 % packing
$+\text{Y}$ / $-\text{Y}$ (366 x 100 faces)	0.030 each	82 % packing
$+\text{Z}$ (226 x 100 face)	0.018	80 % packing
$-\text{Z}$	0.000	imager aperture + radiator -- no cells
Body total	**0.198**	
$+\text{Y}$ wing / $-\text{Y}$ wing	0.120 each	Endurosat 6U deployable, 2 x (209 x 342 mm) at ~84 % packing
Deployed total	**0.438**	

Wing rating at $\beta = 0$, BOL: $2 \times 0.120 \times 291.2 = 69.9$ W -- against the MMA eHaWK 6U+ published rating of "starts at 72 W". Peak whole-spacecraft generation is 88.6 W. Over the year β stays between -20.8 and -31.2 deg, and because the solar-array drive tracks the Sun about one axis only, generation follows $\cos\beta$.

Orbit-average generation at EOL is therefore **45.1 W** at the worst-generation beta (-31.2 deg, the largest $|\beta|$ and the shortest eclipse) and 47.0 W at the longest-eclipse beta (-20.8 deg); at BOL the same two figures are 49.1 W and 51.2 W.

Step 3 -- the load sheet. The PCDU has **nine** power lines, indices 0--8. Two are non-switchable; **every switchable line boots OFF** and must be commanded on:

Idx	Line	Rated draw	Switchable?
0	`obc`	6.0 W (3 W OBC + 3 W PDU/EPS housekeeping)	No -- always on
1	`ttc_rx`	1.5 W (1.2 W S-band idle/Rx + ~0.3 W UHF beacon Rx)	No -- always on
2	`ttc_tx`	5.0 W at the LOW tier, 11.0 W at the HIGH tier	Yes, default OFF
3	`payload`	5.5 W imaging, 2.4 W idle	Yes, default OFF
4	`fpa_cooler`	15.0 W	Yes, default OFF
5	`htr_bat`	8.0 W (billed at actual thermostat duty)	Yes, default OFF
6	`htr_obc`	4.0 W (billed at actual thermostat duty)	Yes, default OFF
7	`aocs_wheels`	12.0 W line rating; the suite publishes 4 W COARSE / 9 W NOMINAL / 15 W FINE	Yes, default OFF
8	`prop`	2 W armed/firing + 12 W catalyst-bed heater	Yes, default OFF; last in the load-shed order

Note the ordering that a real datasheet enforces and a guessed budget does not: **imaging costs more than idling** (5.5 W vs 2.4 W, both from the Simera Sense TriScape100 datasheet), and the transmitter is a 5--11 W DC part with 0.5--2 W of RF output, not a "20 W transmitter" -- no CubeSat radio draws that.

Step 4 -- does the budget close? The acceptance criterion is that orbit-average generation must exceed orbit-average load with all nominal systems on *including* AOCS, judged at nadir pointing rather than sun pointing:

Profile	Gen (W)	Load (W)	Margin (W)	Ratio
quiescent	45.1	19.7	+25.4	2.29
nominal (payload STANDBY, cooler off)	**45.1**	**22.7**	**+22.4**	**1.98**
nominal imaging	45.1	26.4	+18.7	1.71
manoeuvre	45.1	25.8	+19.4	1.75
worst (abuse case)	45.1	46.6	-1.4	0.97

The table is at 3-year EOL and the worst-generation beta (-31.2 deg), from `tools/power_budget.py` re-run on 2026-11-03 with the thermal network and the FPA cooler off in STANDBY (it now runs only to prepare for and during imaging). At the longest-eclipse beta (-20.8 deg) generation is 47.0 W and the nominal profile closes at 2.05x.

The "worst" row is deliberately negative: it holds FINE pointing for a whole orbit, runs the cooler continuously, images through eclipse and dumps at the HIGH tier on every pass. A 6U that sustained all of

that at once would need an array nobody can build. The correct engineering answer is to **duty-cycle the imaging chain** -- three ~90 s captures per orbit with roughly 20 min of cooler around each -- which is what the "nominal imaging" row models. That is the lesson the exercise exists to teach.

Cross-check the margin against flight hardware: the NanoAvionics M6P 6U bus publishes 12.5--17.5 W of payload power available while sun-tracking, and the EOSAT-1 margin with imaging duty-cycled in, +18.7 W, sits just above the top of that range (the +22.4 W nominal figure leaves the cooler off). A 6U design showing +40 W of payload headroom is not well designed; it is mis-specified.

Worked Example -- EOSAT-1 Eclipse Energy and Battery Sizing

Given: 513 km SSO (mean altitude), 97.39 deg inclination, LTAN 10:00, 15.24 rev/day so $\mathrm{T} = 86400/15.24 = 5669$ s $= 1.575$ h. Battery: GomSpace NanoPower BPX class, 8S-1P, **100 Wh**, 3.5 Ah, 28.8 V nominal on the bus model (33.6 V at 100 % SoC, 24.0 V at 0 %, the datasheet values), charge efficiency 0.94, CV taper from 90 % SoC.

Step 1 -- eclipse duration (bounding case, $\beta = 0$. A 10:00 LTAN orbit never reaches it: EOSAT-1's β stays between -20.8 and -31.2 deg, and the simulator gives 33.2--35.2 min of eclipse on every orbit, so the 35.6 min below is conservative by about 0.4 min):

$$\mathrm{f_{ecl}} = \frac{1}{\pi}\arccos\left(\frac{\sqrt{\mathrm{h}(2\mathrm{R_E}+\mathrm{h})}}{(\mathrm{R_E}+\mathrm{h})\cos\beta}\right) = \frac{1}{\pi}\arccos\left(\frac{\sqrt{513\times 13255}}{6884}\right)$$

$= \frac{1}{\pi}\arccos(2607.6/6884) = \frac{1}{\pi}\arccos(0.3788) = \frac{67.74^\circ}{180^\circ} = 0.376$

Eclipse $= 0.376 \times 5669 = 2133$ s $= 35.6$ min; sunlight $= 3536$ s $= 58.9$ min.

Step 2 -- eclipse load. Payload, cooler and TX are off; OBC 6.0 + `ttc_rx` 1.5 + `aocs_wheels` 12.0 = 19.5 W, plus the battery heater at roughly 50 % thermostat duty ($8.0 \times 0.5 = 4.0$ W) = **23.5 W**.

Step 3 -- eclipse energy and DoD: $\mathrm{E_{ecl}} = 23.5 \times (2133/3600) = 23.5 \times 0.5925 = 13.9$ Wh

$\mathrm{DOD} = 13.9/100 = \mathbf{13.9}\%$ per eclipse.

Step 4 -- recharge check. Energy that must go back into the pack: $13.9/0.94 = 14.8$ Wh, over 0.982 h of sunlight $= 15.1$ W of dedicated charge power.

Generation is collected only in sunlight, so the sunlit-average generation, taking the worst-generation 45.1 W together with the bounding eclipse, is $45.1 \times 1.575/0.982 = 72.3$ W. The sunlit load is $(22.7 \times 1.575 - 13.9)/0.982 = (35.8 - 13.9)/0.982 = 22.3$ W (22.7 W is the nominal orbit-average load, payload in STANDBY with the cooler off), leaving $72.3 - 22.3 = 50.0$ W of charge power against the 15.1 W required. **Recharge margin 3.3x. Closes.**

Step 5 -- cycle life. A 3-year mission at 15.24 orbits/day is $3 \times 365 \times 15.24 = 16{,}688$ eclipses. At 13.9 % DoD, Li-ion cells are in the $> 30{,}000$-cycle regime. **Pass with a factor of ~2 on cycles.**

Step 6 -- why 100 Wh and not 30 Wh. The eclipse demand alone would be met by a 30 Wh pack at 46 % DoD. The 100 Wh pack is sized by the *contingency* case -- survive a lost sunlit period, ride out a failed deployment, and keep the bus above the load-shed voltage while FDIR works -- not by the nominal eclipse. Protection levels follow, all from one table (`simulation.yaml` `eps.bus_protection`): the undervoltage flag and the in-model load shed fire at 28.4 V (about 46 % SoC on the $24.0 + 9.6 \times \mathrm{SoC}$ battery curve) and strip `payload` , `fpa_cooler` , `ttc_tx` ,

`aocs_wheels`, `prop` in that order, with a 0.5 V restore hysteresis; FDIR EPS safe mode is at 27.7 V (about 39 %), and the FDIR SoC ladder at 30 / 20 / 10 %.

Thermal: what is actually active. EOSAT-1 is passive apart from three heater circuits and one cooler:

Circuit	Power	Control	Telemetry
Battery heater	8.0 W	Thermostat ON +5 degC / OFF +10 degC	`tcs.htr_battery` 0x040A
OBC heater	4.0 W	Thermostat ON -10 degC / OFF -5 degC (survival heater: idle in nominal operations)	`tcs.htr_obc` 0x040B
Thruster catalyst-bed heater	12.0 W	MANUAL by default; AUTO holds 190/230 degC	`tcs.htr_thruster` 0x041B
FPA cooler	15.0 W	Setpoint-commanded, -20 to 0 degC	`tcs.cooler_fpa` 0x040C

The battery node has $\mathrm{C} = 800$ J/K and $\tau = 2667$ s, so $\mathrm{UA} = \mathrm{C}/\tau = 800/2667 = 0.30$ W/K and the 8 W heater buys $8.0/0.30 = 26.7$ K of authority -- enough that the thermostat genuinely cycles rather than sitting permanently on or permanently off. The catalyst bed is $\mathrm{C} = 50$ J/K, $\tau = 1200$ s, $\mathrm{UA} = 0.042$ W/K, so 12 W reaches the 180 degC propulsion arming interlock in about 18 min, so the ~20 min preheat costs $12\ \mathrm{W} \times 20\ \mathrm{min} = 4.0$ Wh of preheat energy per burn.

Worked Example -- The FPA Cooler Is a Cryocooler, Not a Peltier Stack

The focal-plane cooler is a **miniature Stirling-class machine**, and getting that classification right is the whole point of the example.

Given: FPA node $\mathrm{C} = 100$ J/K, $\tau = 120$ s, so $\mathrm{UA} = 100/120 = 0.833$ W/K. Internal sink $+12$ degC (285.15 K). Cooler input $\mathrm{P} = 15.0$ W. Setpoint -15 degC.

Step 1 -- what lift is demanded? -15 degC from a $+12$ degC sink is a 27 K lift, and the node leaks $\mathrm{UA} \times \Delta\mathrm{T} = 0.833 \times 27 = 22.5$ W back in. The machine must therefore pump 22.5 W using 15 W of input: $\mathrm{COP} = 1.50$.

Step 2 -- can a thermoelectric do that? Goldsmid's optimum-current result for a single stage is

$$\mathrm{COP}_{\max} = \frac{\mathrm{T_c}}{\Delta\mathrm{T}} \cdot \frac{\mathrm{g} - \mathrm{T_h}/\mathrm{T_c}}{\mathrm{g}+1}, \qquad \mathrm{g} = \sqrt{1 + \mathrm{Z}\bar{\mathrm{T}}}$$

With Bi_2Te_3 at $\mathrm{ZT} \approx 0.73$ this gives $\mathrm{COP}_{\max} = 0.88$ at a 27 K lift -- **less than the 1.50 required.** A Peltier stack cannot hold this focal plane. The word "Peltier" was the error, not the mission.

Step 3 -- what class of machine can? The Carnot limit at this lift is $\mathrm{COP}_{\mathrm{Carnot}} = \mathrm{T_c}/\Delta\mathrm{T} = 258.15/27 = 9.56$, so $\mathrm{COP} = 1.50$ is $1.50/9.56 = 15.7\%$ of Carnot. Miniature Stirling and free-piston space coolers -- the machines that actually fly on small-satellite EO detectors -- reach 10--25 % of Carnot at near-ambient lifts. The model is that machine, at **23 % of Carnot**.

Step 4 -- solve for the achievable floor. Authority *degrades with lift*, which is exactly what a flat "40 K max lift" model failed to capture. Balance the heat pumped against the heat leaking in, with ΔT unknown:

$$UA \cdot \Delta T + Q_{load} = f_{Carnot} \cdot P \cdot \frac{T_h - \Delta T}{\Delta T}$$

Idle ($Q_{load} = 0$), sink 285.15 K, $f_{Carnot} \times P = 0.23 \times 15 = 3.45$:

$$0.833\,\Delta T^2 + 3.45\,\Delta T - 3.45 \times 285.15 = 0$$

$$\Delta T = \frac{-3.45 + \sqrt{3.45^2 + 4 \times 0.833 \times 983.8}}{2 \times 0.833} = \frac{-3.45 + 57.36}{1.666} = 32.4 \text{ K}$$

Floor $= 12 - 32.4 = -20.4$ **degC.** The -15 degC setpoint holds with **5.4 K of margin**; the -20 degC end of the command envelope only just holds.

Step 5 -- imaging costs margin. Add the 2.0 W detector/ROIC dissipation:

$0.833\,\Delta T^2 + 5.45\,\Delta T - 983.8 = 0 \Rightarrow \Delta T = 31.2$ K, floor $= -19.2$ degC.

So -20 degC holds at idle and **fails while imaging** -- a 1 K difference that only exists because the cooler is modelled as a real machine.

Step 6 -- a warm bus eats the margin entirely. At a $+20$ degC sink: $\Delta T = 32.8$ K, floor $= -12.8$ degC. The nominal -15 degC setpoint is now **out of reach**, and the cooler will run continuously without ever reaching its target.

Design consequence: the margin is finite, spendable and *visible*. It is published live as `tcs.fpa_cooler_floor_c` (0x0424, the achievable floor) and `tcs.fpa_cooler_margin_k` (0x0425, setpoint minus floor -- ≤ 0 means the setpoint is unreachable), mirrored by the payload at 0x064A / 0x064B. The setpoint itself is commanded as a uint16 in **centi-kelvin**, so -15 degC is 258.15 K $= 25815$ cK $=$ `0x64D7`.

The FPA envelope that everything else derives from: commandable setpoint range -20 to 0 degC (default -15), measured control performance 0.5 K at 3-sigma, uncooled equilibrium $+12.5$ degC, imaging-readiness limit -3 degC (a *science* limit, not a health limit), health warning band $[-22, +14]$ degC and alarm band $[-25.5, +17.5]$ degC.

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5. Real Mission Examples

Planet SuperDove EPS

Table 3.1.12 — Planet SuperDove EPS and thermal parameters with design rationale.

Parameter	Value	Design Rationale
Form factor	3U+, ~5 kg	Flock constellation; P-POD compatible
SA configuration	Body-mounted + two deployable wings	~25 W BOL needed for continuous imaging + S-band downlink
SA cells		Standard space-grade, 29.5% AM0

Parameter	Value	Design Rationale
	Triple-junction GaAs (Spectrolab or SolAero)	
Battery	Li-ion, ~20 Wh (2S2P 18650)	Supports ~3.5 W eclipse load for 35 min at < 20% DOD
Bus voltage	Unregulated 6.0--8.4 V (2S) + regulated 3.3 V, 5 V	Standard CubeSat EPS architecture
Peak demand	~18 W (imaging mode)	Multi-spectral imager + star tracker + reaction wheels + OBC
Orbit	475 km SSO, ~94 min period	Optimal for EO: sun-synchronous for consistent lighting
Eclipse	~35 min max, ~3 W demand	OBC + AOCS only during eclipse; no imaging or downlink
Thermal	Body-mounted radiator panels + battery heater	Passive thermal control; battery heater for eclipse charging margin

[Source: Planet Labs conference presentations; Salas et al., "SuperDove Constellation," SSC 2021]

CAPSTONE Thermal Design

NASA's CAPSTONE (12U, 25 kg) operates in a near-rectilinear halo orbit (NRHO) around the Moon with extreme thermal cycling:

- Perilune: strong Earth/Moon IR + solar
- Apolune: deep space cold, long shadow periods (up to 12+ hours)
- Thermal control: MLI wrapping (10-layer DAM blankets), heaters on propulsion lines (to prevent propellant freezing), passive radiator panels with white paint (AZ-93)
- Battery heaters: 5 W total, thermostatically controlled, critical for survival during long eclipses
- Operating temperature range: -20 degC to +50 degC for electronics; propulsion lines maintained above +5 degC

[Source: Advanced Space, "CAPSTONE Design Overview," SmallSat Conference 2022]

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6. SpaceCDF Exercise

Instructions

1. **Run the design** in SpaceCDF if not already converged
2. **Dashboard** -- review power KPIs: - Is power margin positive in **all** modes (sunlight, eclipse, safe)? - What is the orbit-average power demand? - What SA configuration was selected (body/deployable)?
3. **Timing Budget** card -- review mode durations: - Does the duty cycle match your ConOps modes? - Is eclipse time consistent with your orbit calculation from Chapter 2.3?
4. **Engineering Budgets** -- review the power waterfall: - Which subsystem is the largest power consumer in each mode? - Where could power be reduced if the budget is tight?
5. **Parametric** tab -- review thermal predictions: - Hot case temperature prediction - Cold case temperature prediction - Any exceedances?

Worksheet 3.1 Tasks

1. Size the solar array for your mission (show all 5 calculation steps, including temperature derating)
2. Size the battery (show calculation with DOD justification and cycle-life verification)
3. Compute orbit-average power using duty cycle table
4. Identify hot case and cold case conditions for your orbit
5. Check thermal margins against ECSS requirements
6. Identify the dominant thermal concern for your mission (hot case or cold case) and propose a mitigation

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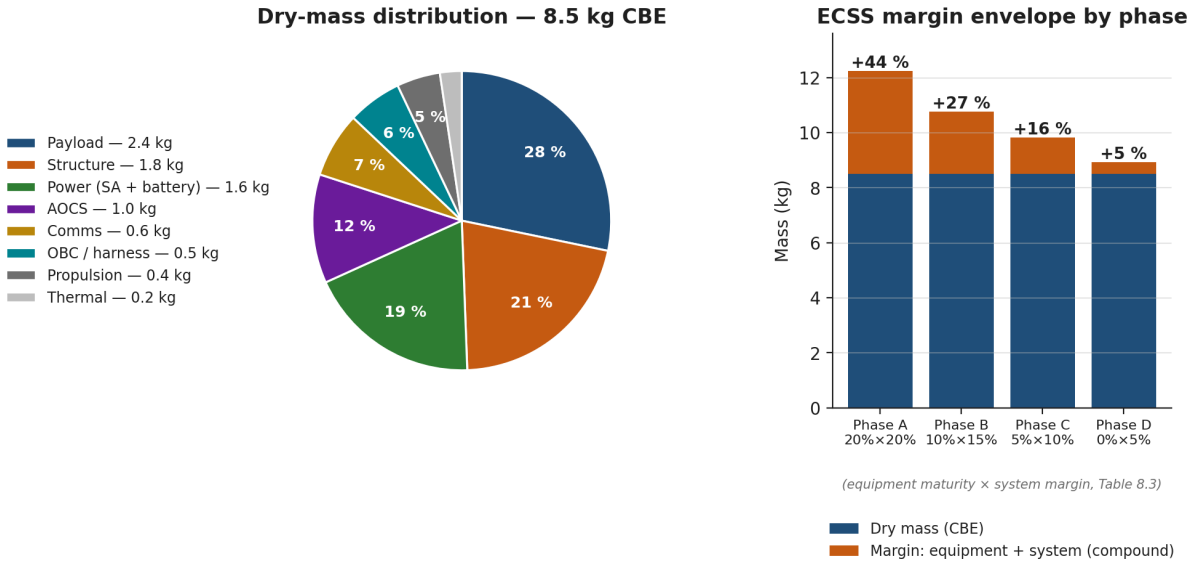
Session Summary

Table 3.1.13 — Session summary: key takeaways for power and thermal control design.

Topic	Key Takeaway
Solar cell physics	Triple-junction GaAs (InGaP/GaAs/Ge) achieves 28--30% AM0; multi-junction stacking captures broader solar spectrum
Temperature effects	Cells lose ~0.19%/degC (relative); operating at 65 degC costs ~7% power vs STC
Degradation	Radiation damage: $(1-\delta)^{n}$, $\delta \approx 2.5\,\%$/yr LEO; cover glass mitigates proton/electron damage
EPS architecture	MPPT + regulated bus is CubeSat standard; MPPT extracts 10--15% more power than DET
SA sizing	$P_{SA} = P_{peak} + P_{recharge}$; derate for degradation $(1-\delta)^{n}$ and temperature L_T
SA area	$A = P_{BOL} / (\eta \cdot S \cdot \cos\theta \cdot f_{pack} \cdot f_{cover})$
Battery chemistry	Li-ion (LCO/NMC): 150--200 Wh/kg packaged; 3.0--4.2 V per cell; CC-CV charging
Battery sizing	$E = P_{ecl} \cdot t_{ecl} / (DOD \cdot \eta)$; DOD 20--30% for multi-year LEO
Battery failure modes	Thermal runaway (overcharge), lithium plating (cold charge), capacity imbalance (series cells)
SA power reference	Body-mounted: 2--12 W; single deploy: 4--30 W; dual deploy: 25--48 W
Thermal physics	No convection in vacuum; radiation ($\varepsilon\sigma T^4$) is only mechanism for heat rejection
Thermal balance	$Q_{in} = Q_{out}$; solve for $T = (Q/\varepsilon\sigma A)^{1/4}$ (single-node); multi-node for real design
α_s/ε ratio	Low ratio = cold surface (radiator); high ratio = warm surface (insulation)
MLI construction	VDA Kapton outer + 10--20 DAM layers + Dacron spacers; $\varepsilon_{eff} = 0.01$ -- 0.03
Heater sizing	$P_{heater} = \varepsilon\sigma A T_{min}^4 - Q_{env} - Q_{int}$; Kapton foil heaters, thermostat control
Thermal margins	ECSS: +/-5 degC operating, +/-10 degC acceptance, +/-15 degC qualification
Transient effects	Thermal mass (mc_p) prevents reaching steady state during short eclipses; 1U at 400 km cools ~20--25 degC in 36 min eclipse

Topic	Key Takeaway
Plane-of-array constant	Collapse the loss stack into one number: $\mathrm{K} = 291.2$ W/m^2 BOL, **267.9 W/m^2 at 3-yr EOL** for TJ GaAs; cross-checks a vendor datasheet to 0.1 W/m^2
6U reality check	0.438 m^2 of cells $\rightarrow$ 88.6 W peak, 45.1 W orbit-average EOL at the worst beta against a 22.7 W nominal load (+22.4 W, 1.98x); +40 W of payload headroom on a 6U means the budget is wrong
Power lines	Nine lines, indices 0--8 (`obc` , `ttc_rx` , `ttc_tx` , `payload` , `fpa_cooler` , `htr_bat` , `htr_obc` , `aocs_wheels` , `prop`); only the first two are non-switchable and **every switchable line boots OFF**
FPA cooling	A 27 K lift at $\mathrm{COP} = 1.5$ is beyond any single-stage Peltier ($\mathrm{COP}_{\max} = 0.88$); it needs a **miniature Stirling-class cryocooler** at ~23 % of Carnot, whose authority *falls* as the lift grows

Chapter 3.2 — Attitude & Orbit Control (AOCS) and Propulsion



Mass distribution and ECSS margin policy.

> **Expected reading before this session.** ECSS-E-ST-60-30C — AOCS (≈ 60 min); Wertz, *Spacecraft Attitude Determination and Control*, Ch. 1 – 2.

Prerequisites: Sessions 2.1--2.4 and 3.1 (requirements, orbit, power defined) **SpaceCDF Tabs:** Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

References

- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 11.1 (ADCS)
- Sidi, *Spacecraft Dynamics and Control*, 1997, Ch. 4--9
- Markley & Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, 2014
- ECSS, *ECSS-E-ST-60-10C: Control Performance*, 2008
- ECSS, *ECSS-E-ST-60-20C Rev.2: Star sensor terminology and performance specification*, 2019
- Wertz, *Space Mission Analysis and Design*, 3rd ed., 1999, Ch. 11 (ADCS)
- Hughes, *Spacecraft Attitude Dynamics*, 1986
- Blue Canyon Technologies, *XACT ADCS Datasheet*, 2023
- CubeSpace, *ADCS Product Catalogue*, 2023

Learning Objectives

By the end of this session, participants will be able to:

1. Explain the distinction between attitude determination and attitude control and the physics of each sensor/actuator type
2. Select AOCS hardware architecture based on pointing requirements
3. Compute a pointing error budget using root-sum-square (RSS) combination
4. Calculate disturbance torques from gravity gradient, aerodynamic drag, solar radiation pressure, and residual magnetic dipole
5. Size actuators (reaction wheels, magnetorquers) for the computed disturbance environment with worked equations
6. Explain the physics of momentum storage, saturation, and desaturation
7. Compare CMGs vs reaction wheels and articulate when each is appropriate
8. Verify AOCS design against requirements using SpaceCDF's pointing budget tool

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1. AOCS Fundamentals

The AOCS performs two distinct functions:

Table 3.2.1 — Functional split between attitude determination and control with associated sensor and actuator hardware.

Function	Purpose	Hardware
Attitude Determination (AD)	Know the spacecraft's orientation relative to a reference frame	Sensors: star tracker, sun sensors, magnetometer, gyroscope, Earth sensor, GPS receiver
Attitude Control (AC)	Change or maintain the spacecraft's orientation	Actuators: reaction wheels, magnetorquers, thrusters, control moment gyros

The attitude state: A spacecraft's attitude is its orientation in 3D space relative to a reference frame (typically J2000 Earth-centred inertial, or the local-vertical/local-horizontal frame for nadir-pointing missions). The attitude is described by a rotation (e.g., quaternion, direction cosine matrix, or Euler angles) plus angular velocity $\vec{\omega}$. Euler's equation of rotational motion governs the dynamics:

$$\mathbf{I}\dot{\vec{\omega}} + \vec{\omega} \times (\mathbf{I}\vec{\omega}) = \vec{\mathrm{T}}_{\mathrm{external}} + \vec{\mathrm{T}}_{\mathrm{control}}$$

where $\mathbf{I}$ is the spacecraft inertia tensor, $\vec{\mathrm{T}}_{\mathrm{external}}$ is the sum of disturbance torques, and $\vec{\mathrm{T}}_{\mathrm{control}}$ is the actuator torque. The $\vec{\omega} \times (\mathbf{I}\vec{\omega})$ term is the gyroscopic coupling -- it means that rotating about one axis can induce motion about other axes if the inertia tensor is not spherically symmetric.

Attitude Sensors -- How They Work

Star Trackers

A star tracker is the most accurate attitude sensor available, providing absolute attitude knowledge to arcsecond-level accuracy.

How it works: 1. A CMOS or CCD detector (typically 1024x1024 to 2048x2048 pixels) images a patch of sky through a wide-angle lens (typically 15--25 deg FOV) 2. The onboard processor detects bright point sources (stars) in the image, computing their centroid positions to sub-pixel accuracy using Gaussian fitting 3. The

processor matches the observed pattern of star positions against an onboard star catalogue (typically Hipparcos or Tycho-2, containing 3,000--10,000 stars, stored in a k-d tree or hash table for fast lookup) 4. The "lost in space" algorithm identifies the star pattern without any prior attitude knowledge (first acquisition), typically taking 1--5 seconds 5. Once identified, the processor computes a quaternion rotation from the catalogue (inertial) frame to the camera (body) frame 6. In tracking mode, the processor tracks known stars frame-to-frame, providing updates at 1--10 Hz with 3--10 arcsec accuracy (1-sigma, per axis)

Key specifications:

Table 3.2.2 — Star tracker performance specifications compared between CubeSat and large-spacecraft classes.

Parameter	Typical CubeSat Star Tracker	Large S/C Star Tracker
Accuracy (1-sigma, boresight)	5--15 arcsec	0.5--3 arcsec
Accuracy (1-sigma, roll)	30--100 arcsec	5--20 arcsec
FOV	10--15 deg circular	15--25 deg circular
Update rate	2--5 Hz	5--20 Hz
Sensitivity	Stars to magnitude 6--7	Stars to magnitude 8--10
Mass	50--350 g	1--5 kg
Power	0.5--2 W	5--15 W
Products	Blue Canyon NST (350g), Sinclair SS-411 (90g), CubeSpace CubeStar (50g)	Sodern Hydra, Leonardo AA-STR

[Source: Blue Canyon NST datasheet; CubeSpace CubeStar datasheet; ECSS-E-ST-60-20C]

Exclusion zones: Star trackers cannot operate when bright objects are in or near the FOV: - **Sun:** Exclusion angle typically 25--45 deg (direct sunlight saturates the detector and can cause permanent damage to some sensor types) - **Earth (illuminated limb):** Exclusion angle typically 25--35 deg (Earth's brightness overwhelms star signals) - **Moon:** Exclusion angle typically 10--15 deg - **Stray light:** Internal reflections from nearby spacecraft structure can create false stars

Implication for mission design: A nadir-pointing spacecraft in LEO always has the Earth within ~65 deg of one hemisphere. The star tracker must be mounted on a face that never points toward Earth (typically the zenith or anti-velocity face). If the Sun is near the orbital plane (beta angle near 0), the star tracker may be periodically blinded. Two star trackers mounted on different faces provide redundancy and eliminate single-axis exclusion zone gaps.

Why star trackers are the most accurate: Other sensors measure vectors to specific objects (Sun, Earth, magnetic field) -- each provides only 2 of the 3 attitude degrees of freedom (direction but not roll around that vector). A star tracker measures multiple star directions simultaneously, providing a full 3-axis attitude fix from a single measurement. The accuracy is limited by optical diffraction, centroiding noise, and catalogue accuracy -- all of which are extremely well characterised.

Sun Sensors

Sun sensors determine the direction to the Sun, providing 2-axis attitude information (the Sun vector in the body frame). They are simple, reliable, radiation-tolerant, and low-power.

Coarse sun sensors (photodiode-based): A coarse sun sensor consists of one or more photodiodes behind a mask or window. The photocurrent is proportional to the cosine of the incidence angle:

$$I = I_0 \cos(\theta)$$

A set of 6 coarse sun sensors (one per face of the spacecraft) determines the Sun direction to 2--5 deg accuracy by comparing the photocurrents from each face. The face with the highest current is sun-facing; the ratio between adjacent faces gives the angle.

Fine sun sensors (linear array or quadrant detector): A fine sun sensor uses a slit mask above a linear photodiode array (similar to a miniature sundial). Sunlight passes through the slit and illuminates a specific position on the array, which is proportional to the incidence angle. Fine sun sensors achieve 0.1--1.0 deg accuracy.

Table 3.2.3 — Coarse, fine, and digital sun-sensor accuracy, mass, power, and example products.

Type	Accuracy	FOV	Mass	Power	Products
Coarse (photodiode)	2--5 deg	~hemisphere	1--5 g	< 1 mW	NewSpace NCSS-SA05, Solar MEMS nanoSSOC-A60
Fine (analog, slit+array)	0.1--0.5 deg	60--120 deg	5--30 g	10--50 mW	NewSpace NFSS-411, Solar MEMS nanoSSOC-D60
Digital (APS detector + slit)	0.01--0.05 deg	60--120 deg	30--50 g	50--200 mW	TNO micro digital sun sensor

[Source: NewSpace Systems NFSS-411 datasheet; Solar MEMS nanoSSOC datasheets]

Why every spacecraft needs sun sensors: Sun sensors are the only sensor guaranteed to work in any attitude, at any rate, in any environment (LEO, GEO, deep space). They are the primary sensor for safe mode: when the spacecraft tumbles, the OBC reboots, or the star tracker is blinded, the sun sensors can determine the Sun direction and enable the spacecraft to orient its solar arrays for power generation. Without sun sensors, a spacecraft that enters safe mode may not recover power.

Magnetometers

A magnetometer measures the local geomagnetic field vector $\vec{B}$ in the body frame. By comparing the measured $\vec{B}$ to a model of Earth's magnetic field (IGRF -- International Geomagnetic Reference Field) at the known orbit position, the spacecraft attitude can be determined.

How it works: - **Fluxgate magnetometer:** A ferromagnetic core is periodically driven into saturation by an excitation coil. The presence of an external magnetic field creates an asymmetry in the saturation waveform, which is detected by a sense coil. Three orthogonal fluxgate elements provide the 3-axis field vector. Resolution: 1--10 nT. Accuracy: $\pm$200--500 nT (limited by spacecraft residual magnetic dipole contamination). - **Magnetoresistive (AMR/GMR):** Thin-film magnetic sensors. Smaller and cheaper than fluxgates but noisier and less stable. Common in COTS CubeSat magnetometers.

Attitude determination from magnetometers: - A single magnetometer measurement provides the magnetic field direction in the body frame (2 DOF, analogous to a sun sensor providing the Sun direction) - Comparing with the IGRF model gives attitude, but accuracy is limited by: - IGRF model uncertainty (~100 nT in LEO) - Spacecraft residual magnetic dipole contamination (can be 1000+ nT close to electronics) - Only 2 DOF per measurement (no roll determination around the field vector) - Magnetometer-only attitude determination: ~5--10 deg accuracy - Magnetometer + sun sensor (combined): ~1--3 deg accuracy (the two independent vectors provide a full 3-axis solution via TRIAD or q-method)

Residual dipole contamination: Every electronic circuit creates a magnetic field. Current loops, permanent magnets in motors/speakers, magnetised structural components, and battery cells all contribute to the spacecraft's residual magnetic dipole. This contaminates the magnetometer reading. Mitigation: mount the magnetometer on a deployable boom (10--30 cm from the bus), use magnetically clean design practices (twisted pairs, balanced current loops), and perform a residual dipole calibration in orbit by comparing magnetometer readings during eclipse (no solar array current) vs sunlit.

Table 3.2.4 — Typical CubeSat magnetometer range, resolution, accuracy, mass, and example products.

Parameter	Typical CubeSat Magnetometer
Range	$\pm$60,000 nT (sufficient for LEO: B ~20,000--50,000 nT)
Resolution	10--100 nT
Accuracy (after calibration)	200--1000 nT
Mass	5--30 g
Power	10--50 mW
Products	NewSpace NMAG, PNI RM3100, Honeywell HMC5883L

Gyroscopes (Rate Sensors)

Gyroscopes measure angular velocity $\vec{\omega}$ rather than absolute attitude. They provide high-bandwidth rate information for control loops and bridge the gap between star tracker updates.

Types used on CubeSats: - **MEMS gyroscope:** Vibrating structure (tuning fork or ring) whose Coriolis force is proportional to rotation rate. Small, cheap (< 50 EUR per axis), low power. Bias drift: 1--10 deg/hr. Products: InvenSense MPU-6050, Analog Devices ADIS16265. - **MEMS IMU (6-axis):** Combined 3-axis accelerometer + 3-axis gyroscope. Mass: 5--15 g. Very common in CubeSats. Products: Sensonor STIM300 (high-end), InvenSense ICM-20948 (COTS). - **Fibre optic gyroscope (FOG):** Much better stability (bias drift 0.01--1 deg/hr) but larger, heavier (200+ g), and more expensive. Used on high-end CubeSats and small satellites.

Why gyroscopes drift: MEMS gyroscopes have a non-zero bias (a constant offset in the measured rate even when stationary) that changes with temperature and time. Integrating angular rate to get attitude ($\theta = \int \omega \, \mathrm{dt}$) accumulates this bias error linearly. A 1 deg/hr bias drift means the attitude estimate drifts 1 deg per hour. Star trackers provide absolute attitude corrections that reset this drift -- the combination of star tracker (low rate, absolute) + gyroscope (high rate, relative) via a Kalman filter is the standard approach for high-performance attitude determination.

GPS Receivers for Orbit Determination

GPS receivers determine the spacecraft's **position and velocity** (orbit determination), not attitude (though multi-antenna GPS can provide coarse attitude).

How it works in LEO: - GPS satellites orbit at ~20,200 km altitude; LEO spacecraft orbit below them at 300--800 km - GPS signals travel downward through the ionosphere to the LEO receiver - The receiver must use a specialised correlator that handles Doppler shifts up to $\pm$40 kHz (LEO relative velocity ~7.5 km/s vs GPS satellite velocity ~3.9 km/s) - Typical accuracy: 5--20 m position, 0.1--0.5 m/s velocity (single frequency, C/A code) - Dual-frequency GPS with carrier phase: sub-meter position accuracy

Why GPS matters for AOCS: Accurate orbit knowledge is needed for: - Nadir pointing (must know where "down" is, which requires knowing position) - Ground target tracking (must know position to compute

pointing angles) - IGRF evaluation (magnetometer attitude determination needs position input) - Orbit manoeuvre planning

Table 3.2.5 — Typical CubeSat GPS receiver position, velocity, time accuracy and mass-power figures.

Parameter	Typical CubeSat GPS Receiver
Position accuracy	5--20 m (C/A code)
Velocity accuracy	0.1--0.5 m/s
Time accuracy	100 ns
Altitude limit	Typically 600 km (must verify ITAR/COCOM limits removed)
Mass	15--30 g
Power	0.5--1.0 W
Products	SkyFox Labs piNAV-NG, NovAtel OEM719, Hemisphere V200

[Source: SkyFox Labs piNAV datasheet; NovAtel OEM7 specifications]

AOCS Architecture Selection by Pointing Requirement

Table 3.2.6 — AOCS architecture selection mapped to pointing requirement, sensors, actuators, mass, power, and cost.

Pointing Requirement	Architecture	Sensors	Actuators	Mass	Power	Cost
> 5 deg	Passive magnetic	None (or 1 magnetometer)	Permanent magnet + hysteresis rods	~0.05 kg	0 W	~2 kEUR
2--5 deg	B-dot detumble + magnetic pointing	3-axis magnetometer + coarse sun sensors	3-axis magnetorquers	~0.10 kg	0.2 W	~8 kEUR
0.1--2 deg	Active 3-axis (RW + sensors)	Fine sun sensors + magnetometer + (optional gyro)	3--4 RW + 3 MTQ for desaturation	~0.50 kg	2--4 W	~35 kEUR
< 0.1 deg	Fine pointing (RW + star tracker)	Star tracker + fine sun sensors + magnetometer + gyro	4 RW + 3 MTQ	~0.80 kg	3--5 W	~55 kEUR
< 0.01 deg	Very fine pointing	Dual star trackers + MEMS gyro + fine sun sensors + magnetometer	4 RW + 3 MTQ	~1.20 kg	4--6 W	~80 kEUR

Real mission examples:

Table 3.2.7 — Real CubeSat missions with their pointing requirements and AOCS architectures.

Mission	Form Factor	Pointing Req	Architecture	AOCS Mass	Key Sensor
Astrocast (3U, IoT)	3U	~5 deg	MTQ + sun sensors + magnetometer	0.1 kg	Sun sensors
Planet SuperDove (3U+, EO)	3U+	~0.1 deg	4 RW + ST + 3 MTQ + 6 sun sensors	0.8 kg	Blue Canyon NST star tracker
CAPSTONE (12U, cislunar nav)	12U	~0.05 deg	4 RW + ST + sun sensors + IMU	~1.0 kg	Star tracker + IMU
ASTERIA (6U, exoplanet)	6U	~0.003 deg (10 arcsec)	4 RW + ST + fine guidance sensor	~1.2 kg	Custom fine guidance camera

[Source: Pong et al., "ASTERIA: Achieving 10-arcsecond Pointing on a 6U CubeSat," SSC 2018]

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2. Disturbance Torques

In LEO, four external torques disturb the spacecraft attitude. The AOCS must counteract them continuously. Understanding the source and magnitude of each disturbance is essential for sizing actuators.

Gravity Gradient Torque

Physics: A spacecraft in orbit experiences a non-uniform gravitational field -- the side closer to Earth is pulled more strongly than the far side. For an elongated body, this differential pull creates a torque that tends to align the long axis with the local vertical (nadir direction). This is the principle behind gravity gradient stabilisation.

> **Key Equations -- Gravity Gradient Torque**

$$\mathrm{T_{gg}} = \frac{3\mu}{2\mathrm{a}^3}|\mathrm{I_z} - \mathrm{I_x}|\sin(2\theta)$$

> where: - $\mu = 3.986 \times 10^{14}$ $\mathrm{m}^3/\mathrm{s}^2$ (Earth's gravitational parameter) - $\mathrm{a} = \mathrm{R_E} + \mathrm{h}$ (semi-major axis in metres) - $\mathrm{I_z}$, $\mathrm{I_x}$ are principal moments of inertia about the maximum and minimum axes (kg m^2) - θ = angle between the long axis and the local vertical - Worst case occurs at $\theta = 45$ deg, where $\sin(2\theta) = 1$
>
> For a body with $\mathrm{I_z} \approx \mathrm{I_x}$ (a cube), $\mathrm{T_{gg}} \approx 0$ -- this is why 1U CubeSats experience minimal gravity gradient torque.

Aerodynamic Torque

Physics: At LEO altitudes (200--600 km), residual atmospheric molecules collide with the spacecraft surface. The force acts through the centre of pressure (cp), which generally does not coincide with the centre of mass (cm). The offset creates a torque.

> **Key Equations -- Aerodynamic Torque**

$$\mathrm{T_{aero}} = \frac{1}{2}\rho \mathrm{v}^2 \mathrm{C_D} \mathrm{A_{ref}}\, \mathrm{d_{cp-cm}}$$

where: - ρ = atmospheric density (kg/m^3) -- varies by orders of magnitude with altitude, solar activity (F10.7), and local time: - 300 km: $\rho \approx 2 \times 10^{-11}$ (solar min) to 3×10^{-10} (solar max) - 400 km: $\rho \approx 4 \times 10^{-12}$ to 1×10^{-11} - 500 km: $\rho \approx 6 \times 10^{-13}$ to 5×10^{-12} - 600 km: $\rho \approx 1 \times 10^{-13}$ to 8×10^{-13} - v = orbital velocity (~7.6 km/s at 500 km) - $C_D \approx 2.0$--2.3 (molecular flow drag coefficient; 2.2 is standard for flat plates in free molecular flow) - A_{ref} = cross-sectional area perpendicular to velocity (m^2) - d_{cp-cm} = offset between centre of pressure and centre of mass (m); typically 0.5--5 cm for CubeSats depending on deployable configuration

Solar Radiation Pressure Torque

Physics: Sunlight carries momentum. When photons strike a surface, they transfer momentum ($p = E/c$ for absorption, $p = 2E/c$ for specular reflection). The resulting force acts through the centre of solar pressure, which may not coincide with the cm.

Key Equations -- SRP Torque

$$T_{SRP} = \frac{S}{c} A_s (1 + q)\, d_{sp-cm}$$

where: - $S = 1361$ W/m^2 (solar constant at 1 AU) - $c = 3 \times 10^8$ m/s (speed of light) - $S/c = 4.54 \times 10^{-6}$ N/m^2 (solar radiation pressure at 1 AU) - A_s = illuminated area (m^2) - q = surface reflectance (0 for perfect absorber, 1 for perfect specular reflector) - d_{sp-cm} = offset between solar pressure centre and centre of mass (m)

Note: SRP torque is tiny in LEO compared to aero and magnetic torques. It becomes the dominant disturbance in GEO and deep space (where there is no atmosphere and the magnetic field is weak).

Residual Magnetic Dipole Torque

Physics: A spacecraft with a net magnetic dipole moment $\vec{M}$ (from current loops in wiring, permanent magnets in reaction wheel motors, magnetised ferromagnetic components) interacts with Earth's magnetic field $\vec{B}$ to produce a torque:

Key Equations -- Magnetic Dipole Torque

$$\vec{T}_{mag} = \vec{M} \times \vec{B}$$

Magnitude: $T_{mag} = M \cdot B \cdot \sin(\alpha)$

where: - M = spacecraft residual magnetic dipole moment (A m^2) - B = local geomagnetic field strength (T): - LEO (400--600 km): $B \approx 2$--5×10^{-5} T (varies with latitude; strongest near poles, weakest near equator) - GEO (35,786 km): $B \approx 1 \times 10^{-7}$ T - α = angle between $\vec{M}$ and $\vec{B}$

Typical CubeSat residual dipole moments:

Source	Dipole Moment (A m^2)	Notes
Reaction wheel motor	0.005--0.02 per wheel	Permanent magnets in brushless motor
Solar array wiring	0.001--0.01	Current loops from SA to EPS

Source	Dipole Moment (A m^2)	Notes
Battery cells	0.001--0.005	Nickel in cell casing
Unshielded cables	0.005--0.05	Depends on routing and length
Total (typical 3U)	**0.01--0.10**	Varies significantly with design

Why magnetic torque dominates for CubeSats: COTS electronics are not designed for magnetic cleanliness. Short wiring runs create small but unbalanced current loops. Reaction wheel motors contain permanent magnets. The result is a residual dipole of 0.01--0.1 A m^2, which in a 30 uT field produces 3×10^{-7} to 3×10^{-6} N m -- often larger than gravity gradient or SRP torques.

Worked Example -- Disturbance Torques for 3U CubeSat at 500 km (SuperDove-class)

Spacecraft properties: 3U (100 x 100 x 340 mm), mass = 5 kg, $I_z = 0.035$ kg m^2 (long axis), $I_x = 0.007$ kg m^2 (short axis), $A_{ref} = 0.034$ m^2 (3U face), $d_{cp-cm} = 0.02$ m (deployable panels offset cm from geometric centre).

Gravity gradient (worst case, $\theta = 45$ deg): $T_{gg} = \frac{3 \times 3.986 \times 10^{14}}{2 \times (6871 \times 10^3)^3} \times |0.035 - 0.007| \times 1$

$= \frac{1.196 \times 10^{15}}{6.494 \times 10^{20}} \times 0.028 = 1.84 \times 10^{-6} \times 0.028 = 5.2 \times 10^{-8}$ N m

Aerodynamic (at 500 km, solar minimum, $\rho \approx 6 \times 10^{-13}$ kg/m^3):

$F_{aero} = 0.5 \times 6 \times 10^{-13} \times 7617^2 \times 2.2 \times 0.034 = 1.30 \times 10^{-6}$ N

$T_{aero} = F_{aero} \times d_{cp-cm} = 1.30 \times 10^{-6} \times 0.02 = 2.6 \times 10^{-8}$ N m

Note: at solar maximum ($\rho \approx 5 \times 10^{-12}$), this increases by ~8x to 2.1×10^{-7} N m.

Solar radiation pressure: $F_{SRP} = \frac{1361}{3 \times 10^8} \times 0.034 \times 1.5 = 2.31 \times 10^{-7}$ N

$T_{SRP} = F_{SRP} \times d_{sp-cm} = 2.31 \times 10^{-7} \times 0.02 = 4.6 \times 10^{-9}$ N m

Residual magnetic dipole ($M = 0.05$ A m^2, $B = 3 \times 10^{-5}$ T):

$T_{mag} = 0.05 \times 3 \times 10^{-5} = 1.5 \times 10^{-6}$ N m

Summary:

Source	Torque (N m)	Rank	Notes
Gravity gradient	5.2×10^{-8}	3rd	Small because 3U is not very elongated
Aerodynamic (solar min)	2.6×10^{-8}	4th	Increases 8x at solar max
Solar radiation pressure	4.6×10^{-9}	5th	Negligible at LEO distances
Residual magnetic dipole	1.5×10^{-6}	**1st**	**Dominates by >10x**
Total (worst-case sum)	$\approx 1.6 \times 10^{-6}$		Conservative estimate
Total (RSS)	$\approx 1.5 \times 10^{-6}$		More realistic (uncorrelated sources)

Key finding: The residual magnetic dipole dominates for CubeSats due to COTS electronics and short wiring runs. **Magnetic cleanliness matters.** Reducing the residual dipole from 0.05 to 0.01 A m^2 (achievable with careful wire routing, twisted pairs, and degaussing) would reduce the total disturbance by 5x.

Disturbance torques on a 3U CubeSat at 500 km, log scale.

Figure 3.2.1 — Order-of-magnitude comparison of the four disturbance torques on a 3U CubeSat at 500 km SSO (data from the worked example above, sourced from Chapter 3.2 §2). The residual-magnetic-dipole torque dominates by an order of magnitude, justifying why "magnetic cleanliness" is the AOCS designer's first design-cleanup target.

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3. Attitude Actuators -- Physics and Sizing

Reaction Wheels -- Physics of Angular Momentum Storage

How reaction wheels work:

A reaction wheel is a flywheel (typically a brass or steel ring, 20--200 g for CubeSats) spun by a brushless DC motor. By Newton's third law, changing the wheel's angular momentum produces an equal and opposite torque on the spacecraft:

$$\vec{\mathrm{H}}_{\mathrm{total}} = \vec{\mathrm{H}}_{\mathrm{spacecraft}} + \vec{\mathrm{H}}_{\mathrm{wheels}} = \mathrm{constant}$$

If the wheel speeds up ($\Delta \mathrm{H}_{\mathrm{wheel}} > 0$), the spacecraft receives an equal and opposite angular momentum change ($\Delta \mathrm{H}_{\mathrm{SC}} = -\Delta \mathrm{H}_{\mathrm{wheel}}$), causing it to rotate. The control torque is:

$$\mathrm{T}_{\mathrm{control}} = \frac{\mathrm{dH}_{\mathrm{wheel}}}{\mathrm{dt}} = \mathrm{I}_{\mathrm{wheel}} \cdot \dot{\omega}_{\mathrm{wheel}}$$

where $\mathrm{I}_{\mathrm{wheel}}$ is the wheel's moment of inertia and $\dot{\omega}_{\mathrm{wheel}}$ is the wheel's angular acceleration.

Momentum storage: The maximum angular momentum a wheel can store is:

$$\mathrm{H}_{\mathrm{max}} = \mathrm{I}_{\mathrm{wheel}} \times \omega_{\mathrm{max}}$$

For a Blue Canyon RW210: $\mathrm{I}_{\mathrm{wheel}} \approx 1.5 \times 10^{-5}$ kg m ^{2} , $\omega_{\mathrm{max}} \approx 6000$ RPM $= 628$ rad/s, giving $\mathrm{H}_{\mathrm{max}} = 0.0094$ N m s ≈ 10 mN m s.

Saturation: As disturbance torques act on the spacecraft, the reaction wheels absorb angular momentum. Over time, the wheel speed increases until it reaches ω_{max} (saturation). At saturation, the wheel can no longer absorb momentum in that direction, and control authority is lost. The time to saturation from zero speed is:

$$\mathrm{t}_{\mathrm{sat}} = \frac{\mathrm{H}_{\mathrm{max}}}{\mathrm{T}_{\mathrm{disturbance}}} = \frac{10 \times 10^{-3}}{1.5 \times 10^{-6}} = 6667 \ \mathrm{s} \approx 111 \ \mathrm{minutes}$$

This is approximately 1.2 orbits -- so the wheels would saturate after about one orbit without desaturation. This is why magnetorquers are essential companions to reaction wheels.

The zero-crossing problem: When a reaction wheel passes through zero speed (reversing direction), the static friction in the bearings creates a "dead zone" where the wheel cannot produce smooth, continuous torque. This causes a brief loss of control authority and increased jitter. Mitigations: - **Bias momentum:** Operate all wheels with a positive bias speed (e.g., 500 RPM), so they never cross zero during normal operations - **4-wheel pyramid configuration:** The skewed geometry means individual wheels reverse less frequently - **Lubrication:** Space-rated bearings use solid or vapour-deposited lubricants (MoS $_2$, Braycote) that minimise static friction

Jitter: Reaction wheel imbalance (mass asymmetry in the flywheel) creates vibrations at the spin frequency and its harmonics. For imaging missions, this jitter degrades image quality. Jitter amplitude depends on

wheel speed, imbalance mass, and the spacecraft's structural transfer function. Typical CubeSat reaction wheel jitter: 5--20 arcsec at the payload, depending on isolation.

Reaction Wheel Configurations

Table 3.2.8 — Comparison of three-orthogonal, three-plus-skew, and four-wheel pyramid reaction-wheel configurations.

Configuration	Description	Pros	Cons	Use
3 orthogonal	One wheel per body axis (X, Y, Z)	Minimum mass, simple control	No redundancy; single wheel failure = loss of 1-axis control	Low-cost missions with short lifetime
3 + 1 skew	3 orthogonal + 1 on a skew axis (e.g., [1,1,1] direction)	Single-fault tolerant; the skew wheel + remaining 2 provide 3-axis control	Slightly more complex control law distribution	**Standard for CubeSats**
4-wheel pyramid	4 wheels tilted ~20--30 deg from body axes, symmetrically arranged	Optimal torque/momentum distribution; single-fault tolerant; reduced zero-crossings	More complex mounting, heavier	High-performance missions, agile S/C

The **distribution matrix** maps wheel torques to body-frame torques:

$$\vec{\mathrm{T}}_{\mathrm{body}} = \mathbf{D} \cdot \vec{\mathrm{T}}_{\mathrm{wheels}}$$

For a 4-wheel pyramid with cant angle β:

$$\boldsymbol{D} = \begin{bmatrix} \cos\beta & 0 & -\cos\beta & 0 \\ 0 & \cos\beta & 0 & -\cos\beta \\ \sin\beta & \sin\beta & \sin\beta & \sin\beta \end{bmatrix}$$

Key Equations -- Reaction Wheel Sizing

Torque requirement (to counteract disturbances + provide slewing capability):

$$\mathrm{T}_{\mathrm{RW,min}} \geq \mathrm{k} \times \mathrm{T}_{\mathrm{disturbance,total}}$$

where $\mathrm{k} \geq 2$ is the control margin factor (typically 2--5 to provide adequate control bandwidth and slewing performance).

Momentum storage requirement (accumulation between desaturation cycles):

$$\mathrm{H}_{\mathrm{required}} = \mathrm{T}_{\mathrm{disturbance}} \times \frac{\mathrm{t}_{\mathrm{desat}}}{2}$$

where t_{desat} is the time between magnetorquer desaturation events (typically one half-orbit to one orbit). The factor of 1/2 accounts for the average (sinusoidal disturbance torques average to half their peak over a quarter orbit).

Slew rate (for agile/imaging missions):

$$\dot{\theta}_{max} = \frac{H_{RW,max}}{I_{axis}}$$

For a 3U CubeSat with RW210 ($H = 10$ mN m s) and $I_{axis} = 0.035$ kg m^2: $\dot{\theta}_{max} = 0.010/0.035 = 0.286$ rad/s $= 16.4$ deg/s -- more than adequate for target-to-target slewing.

Slew time for a given angle (acceleration-limited, trapezoidal profile):

$$t_{slew} = 2\sqrt{\frac{\theta_{slew} \cdot I_{axis}}{T_{RW}}}$$

For a 90 deg (1.57 rad) slew with RW210 ($T = 1.0$ mN m) and $I = 0.035$: $t_{slew} = 2\sqrt{\frac{1.57 \times 0.035}{0.001}} = 2\sqrt{54.95} = 14.8$ s

Worked Example -- Reaction Wheel Sizing for 3U CubeSat (SuperDove-class)

Given: $T_{disturbance} = 1.5 \times 10^{-6}$ N m (from Section 2), desaturation interval = 1 orbit (5670 s), pointing requirement = 0.1 deg, slew requirement = 90 deg in < 60 s.

Torque requirement: $T_{RW,min} = 3 \times 1.5 \times 10^{-6} = 4.5 \times 10^{-6}$ N m $= 0.0045$ mN m

This is a very low torque requirement. The minimum available CubeSat wheel (RW-0.01 at 0.23 mN m) exceeds this by 50x. The sizing driver is actually the slew rate and momentum storage, not the disturbance rejection torque.

Momentum storage: $H_{required} = 1.5 \times 10^{-6} \times \frac{5670}{2} = 4.25 \times 10^{-3}$ N m s $= 4.25$ mN m s

Slew time check with candidate wheels:

Product	Torque (mN m)	Momentum (mN m s)	Mass (g)	90 deg Slew (s)	Momentum Margin	Manufacturer
RW-0.01	0.23	1.0	30	30.9 s	-3.25 mN m s (FAIL)	Hyperion
RW210	1.0	10	55	14.8 s	+5.75 mN m s (135%)	Blue Canyon
RW3-1.0	1.0	15	50	14.8 s	+10.75 mN m s (253%)	CubeSpace
RW400	4.0	40	120	7.4 s	+35.75 mN m s (841%)	Blue Canyon

The RW-0.01 fails the momentum storage requirement (would saturate in < 1 orbit). The RW210 (10 mN m s) provides 135% margin and 14.8 s slew time. **Selected: RW210 (or CubeSpace RW3-1.0).**

> **Configuration:** 4 wheels (3+1 skew) for single-fault tolerance. Total AOCS actuator mass: 4 x 55 g = 220 g.

Magnetorquers -- Physics of Magnetic Torque Generation

How magnetorquers work:

A magnetorquer (MTQ) is simply a coil of wire (or a ferromagnetic rod wrapped with wire). When current flows through the coil, it creates a magnetic dipole moment:

$$\vec{m} = N \cdot I \cdot A \cdot \hat{n}$$

where N = number of turns, I = current (A), A = coil cross-sectional area (m^2), and $\hat{n}$ is the coil normal direction.

This magnetic dipole interacts with Earth's geomagnetic field $\vec{B}$ to produce a torque:

$$\vec{T}_{MTQ} = \vec{m} \times \vec{B}$$

The torque is **perpendicular** to both the dipole moment and the magnetic field. This has a critical implication: **a magnetorquer cannot produce torque parallel to the local magnetic field vector.** At any instant, only 2 of 3 axes can be torqued. Over a full orbit, as the field direction rotates, all 3 axes become accessible -- but not simultaneously.

Magnetorquer types for CubeSats:

Table 3.2.9 — Magnetorquer technology comparison across dipole moment, mass, power, and form factor.

Type	Dipole Moment	Mass	Power	Form Factor	Products
Air-core coil (PCB trace)	0.01--0.05 A m^2	1--5 g	0.1--0.3 W	Flat PCB, integrates into solar panel substrate	ZARM Technik MTC-1, custom
Air-core rod (wound wire)	0.05--0.50 A m^2	10--30 g	0.2--0.5 W	Cylindrical, 60--100 mm long	CubeSpace CubeMAG, NewSpace NTQS
Ferromagnetic core rod	0.2--5.0 A m^2	20--100 g	0.3--1.0 W	Cylindrical with mu-metal core, 60--100 mm long	ZARM Technik MTQ-1, ISIS iMTQ

The ferromagnetic core concentrates the magnetic flux, providing 5--20x more dipole moment per unit current than an air-core coil of the same size. However, ferromagnetic cores can retain residual magnetism after power-off, contributing to the spacecraft's residual magnetic dipole.

Why magnetorquers cannot point (only detumble and desaturate):

The torque $\vec{T} = \vec{m} \times \vec{B}$ is always perpendicular to $\vec{B}$. This means: 1. You cannot generate torque about the $\vec{B}$ direction at any given instant 2. The achievable torque direction changes continuously as the spacecraft orbits (because $\vec{B}$ rotates) 3. Pointing control requires torque in any direction at any time -- magnetorquers cannot provide this

Magnetorquers are excellent for: - **Detumbling:** The B-dot controller ($\vec{m} = -k\dot{\vec{B}}$) brakes spacecraft rotation by opposing the change in the measured B-field. Works regardless of attitude, needs no attitude knowledge, and needs no sensor but the magnetometer. **It is also slow, and this surprises people.** The

available torque is $\mathrm{m} \times \mathrm{B}$ with m capped by the coil (1.6 A m^2 per axis on our 6U reference) and $\mathrm{B} \approx 3 \times 10^{-5}$ T, so the braking torque is of order 5×10^{-5} N m -- and only the component perpendicular to $\vec{\mathrm{B}}$ does any work. Detumbling a 6U from a 14 deg/s post-separation tumble takes on the order of **3.5 x 10^3 s**, dropping below a 0.5 deg/s exit guard at about 1170 s. Any claim that a B-dot detumble completes in a couple of minutes should be treated as a red flag. - **Desaturation:** Systematically dumping momentum from reaction wheels by applying the correct dipole moment: $\vec{\mathrm{m}} = -\mathrm{k}(\vec{\mathrm{H}}_{\mathrm{wheel}} \times \hat{\mathrm{B}})$ - **Coarse pointing** (2--10 deg): Possible over time using model-predictive control that plans the dipole commands over a full orbit, exploiting the field rotation. But accuracy is limited.

Key Equations -- Magnetorquer Sizing

Desaturation torque:

$$\mathrm{T}_{\mathrm{MTQ}} = \mathrm{m}_{\mathrm{dipole}} \times \mathrm{B} \times \sin(\alpha)$$

Average torque over an orbit (accounting for varying α): $\mathrm{T}_{\mathrm{MTQ,avg}} \approx 0.7 \times \mathrm{m}_{\mathrm{dipole}} \times \mathrm{B}_{\mathrm{avg}}$

Desaturation time (to dump one wheel from full momentum):

$$\mathrm{t}_{\mathrm{dump}} = \frac{\mathrm{H}_{\mathrm{wheel}}}{\mathrm{T}_{\mathrm{MTQ,avg}}}$$

Design requirement: $\mathrm{t}_{\mathrm{dump}} < \mathrm{t}_{\mathrm{shadow}}$ (must complete desaturation during the portion of the orbit where the field geometry is favourable).

Worked Example -- Magnetorquer Sizing for 3U CubeSat

Given: RW210 momentum = 10 mN m s, $\mathrm{B}_{\mathrm{avg}} = 3 \times 10^{-5}$ T.

Option 1: CubeMAG rod ($\mathrm{m} = 0.2$ A m^2): $\mathrm{T}_{\mathrm{MTQ,avg}} = 0.7 \times 0.2 \times 3 \times 10^{-5} = 4.2 \times 10^{-6}$ N m

$\mathrm{t}_{\mathrm{dump}} = \frac{10 \times 10^{-3}}{4.2 \times 10^{-6}} = 2381$ s ≈ 40 min

This is approximately 42% of one orbit period. **Acceptable** -- desaturation can be scheduled once per orbit during the non-imaging portion.

Disturbance rejection check: The MTQ average torque (4.2×10^{-6} N m) is 2.8x the total disturbance torque (1.5×10^{-6} N m). The MTQ can dump momentum faster than it accumulates. **Pass.**

Configuration: 3 MTQ rods, one per body axis (X, Y, Z). This ensures torque can be generated about any two axes at any given time. Total MTQ mass: 3 x 30 g = 90 g.

Control Moment Gyroscopes (CMGs) vs Reaction Wheels

CMGs are an alternative momentum exchange device used on large, agile spacecraft. A CMG consists of a spinning flywheel mounted on a gimbal. Instead of changing the wheel speed (as in a reaction wheel), the CMG changes the direction of the angular momentum vector by rotating the gimbal. This produces a gyroscopic output torque:

$$\mathrm{T}_{\mathrm{CMG}} = \mathrm{H}_{\mathrm{wheel}} \times \dot{\delta}$$

where $\mathrm{H}_{\mathrm{wheel}}$ is the constant wheel momentum and $\dot{\delta}$ is the gimbal rate.

The torque amplification effect: For a CMG wheel spinning at high speed ($H = 1\text{--}100$ N m s), even a moderate gimbal rate ($\dot{\delta} = 1$ rad/s) produces a large output torque ($T = 1\text{--}100$ N m). This is orders of magnitude more than a reaction wheel of similar mass. CMGs are "torque machines"; reaction wheels are "momentum machines."

Table 3.2.10 — Reaction wheel versus single-gimbal control moment gyroscope on torque, complexity, and use cases.

Parameter	Reaction Wheel	CMG (Single Gimbal)
Output torque	$T = I_w \dot{\omega}_w$ (low, 0.001--10 N m)	$T = H_w \dot{\delta}$ (high, 0.1--1000+ N m)
Control complexity	Simple (speed command)	Complex (gimbal singularity avoidance)
Mass efficiency	Lower torque/kg	Higher torque/kg (10--100x)
Failure mode	Bearing wear, motor failure	Gimbal lock (singularity), bearing wear
Typical use	CubeSats, small satellites, non-agile	Large agile satellites, ISS, Earth observation with rapid retargeting
CubeSat status	Standard (many COTS products)	Emerging (Honeybee Robotics microCMG, some research prototypes)

When to use CMGs: - Spacecraft requiring rapid slewing (> 3 deg/s for large spacecraft) - Large moments of inertia where reaction wheel torque is insufficient - Missions requiring frequent retargeting (e.g., video from orbit, rapid revisit EO)

When to use reaction wheels: - CubeSats and small satellites (adequate torque, simpler control, more COTS options) - Missions with modest agility requirements (< 10 deg/s slew for CubeSats) - Cost-constrained missions (CMGs are significantly more expensive)

The ISS uses four 4600 kg CMGs, each storing 3500 N m s of momentum, to maintain attitude without propellant. The Pleiades Neo Earth observation satellite uses CMGs for rapid retargeting between imaging strips.

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4. Pointing Error Budget

[Source: ECSS-E-ST-60-10C]

The pointing error budget combines all independent error sources using root-sum-square (RSS) to determine the total pointing uncertainty. This is a statistical combination assuming errors are uncorrelated and normally distributed.

ECSS pointing performance taxonomy:

Table 3.2.11 — ECSS pointing performance taxonomy defining absolute, relative, and mean performance errors.

Term	Definition	Measured Over
APE (Absolute Performance Error)	Total error between actual pointing and commanded pointing	Single measurement

Term	Definition	Measured Over
RPE (Relative Performance Error)	Variation in pointing over a short time (jitter/stability)	Measurement window (e.g., integration time)
MPE (Mean Performance Error)	Systematic bias in pointing	Long-term average

For most CubeSat missions, APE is the primary requirement. RPE matters for long-exposure imaging (e.g., ASTERIA's 10-arcsec stability over 20-minute exposures).

Key Equations -- Pointing Error Budget (RSS)

$$\theta_{\text{APE}} = \sqrt{\theta_{\text{sensor}}^2 + \theta_{\text{actuator}}^2 + \theta_{\text{alignment}}^2 + \theta_{\text{thermal}}^2 + \theta_{\text{jitter}}^2 + \theta_{\text{orbit}}^2 + \theta_{\text{timing}}^2}$$

The result must satisfy:

$$\theta_{\text{APE}} \leq \theta_{\text{requirement}}$$

Error Source Definitions and Physics

Table 3.2.12 — Pointing error sources with their physics and typical star-tracker and sun-sensor magnitudes.

Source	Description	Physics	Typical Values (Star Tracker)	Typical Values (Sun Sensor)
Sensor accuracy	Intrinsic measurement noise of attitude sensor	Photon noise, centroiding error, optical distortion	3--15 arcsec (0.001--0.004 deg)	0.5--2 deg
Actuator resolution	Minimum controllable torque step; control loop dead band	Motor cogging torque, driver quantisation, control bandwidth	2--5 arcsec (0.001 deg)	N/A (MTQ: 1--5 deg)
Alignment knowledge	Misalignment between sensor boresight and payload boresight; measured during I&T	Mechanical tolerances, shimming, bonding accuracy, measurement uncertainty	30--60 arcsec (0.01--0.02 deg)	0.5 deg
Thermal distortion	Structural deformation with temperature changes; orbital thermal cycling	CTE mismatch between materials, temperature gradients across structure	10--30 arcsec (0.003--0.01 deg)	0.1 deg
Jitter	High-frequency vibration from reaction wheels, mechanisms	Wheel imbalance forces at spin frequency and harmonics, structural resonances	5--20 arcsec (0.001--0.006 deg)	N/A
Orbit knowledge	Uncertainty in satellite position (affects nadir	GPS accuracy, propagation error between GPS fixes	1--5 arcsec (< 0.001 deg)	0.05 deg

Source	Description	Physics	Typical Values (Star Tracker)	Typical Values (Sun Sensor)
	pointing vector computation)			
Timing	Time-stamping error between sensor read and actuator command	Clock synchronisation, interrupt latency, bus communication delay	1--3 arcsec (< 0.001 deg)	0.01 deg

Worked Example -- Pointing Budget for 3U EO CubeSat (Star Tracker + RW)

Error Source	Value (deg)	Value (arcsec)	Value ^{2} (deg^2)	Notes
Star tracker accuracy	0.003	10.8	9.0×10^{-6}	Blue Canyon NST, 1-sigma boresight
Reaction wheel resolution	0.001	3.6	1.0×10^{-6}	Motor cogging + control dead band
Alignment knowledge	0.020	72	4.0×10^{-4}	**Dominant** -- shimmed to 1 arcmin
Thermal distortion	0.010	36	1.0×10^{-4}	Al structure, 40 degC orbital range
RW jitter	0.005	18	2.5×10^{-5}	At 3000 RPM, no isolation mount
Orbit knowledge (GPS)	0.001	3.6	1.0×10^{-6}	GPS fix every 10 s
Timing error	0.001	3.6	1.0×10^{-6}	< 1 ms timestamp sync
RSS Total	$\sqrt{5.37 \times 10^{-4}}$ = **0.023 deg**	**83 arcsec**		

Requirement: 0.1 deg (3-sigma) -- this is typical for a 5 m GSD imager at 500 km (where 0.1 deg corresponds to ~870 m pointing error on ground, or ~175 pixels for a 5 m GSD sensor).

Margin: 0.1 - 0.023 = 0.077 deg (77% margin) -- **comfortable**.

Key insight: Alignment knowledge (0.020 deg = 72 arcsec) dominates the budget at 74% of the RSS. Improving the star tracker accuracy from 10.8 to 3 arcsec would change the RSS total from 83.4 to 82.8 arcsec -- negligible improvement. **Budget-driven design** means investing effort in the dominant term: better alignment calibration (e.g., on-orbit calibration using ground targets) would have far more impact than upgrading any sensor.

To achieve < 0.01 deg (36 arcsec) pointing, the alignment must be improved to < 0.005 deg (18 arcsec), which requires precision optical alignment during I&T and/or on-orbit alignment calibration.

5. Momentum Management and Desaturation

Disturbance torques cause angular momentum to accumulate in reaction wheels over time. Without management, wheels saturate and lose control authority. Understanding this cycle is essential for AOCS design.

The momentum lifecycle:

1. **Accumulation:** External disturbance torques (gravity gradient, aero, SRP, magnetic) act on the spacecraft body. The control loop commands the reaction wheels to counteract these torques, absorbing the angular momentum. Wheel speed increases (or decreases) at a rate of $\dot{\mathrm{H}} = \mathrm{T}_{\mathrm{disturbance}}$.
2. **Monitoring:** The OBC monitors wheel speeds. When any wheel exceeds a threshold (typically 80% of maximum), a desaturation manoeuvre is triggered.
3. **Desaturation:** The OBC activates the magnetorquers to generate a torque that opposes the stored wheel momentum. The algorithm computes the optimal dipole command:

$$\vec{\mathrm{m}}_{\mathrm{cmd}} = -\mathrm{k}_{\mathrm{d}}(\vec{\mathrm{H}}_{\mathrm{wheel}} \times \hat{\mathrm{B}})$$

where k_{d} is the desaturation gain and $\hat{\mathrm{B}}$ is the unit magnetic field vector. This produces a torque $\vec{\mathrm{T}} = \vec{\mathrm{m}} \times \vec{\mathrm{B}}$ that is in the direction to reduce $\vec{\mathrm{H}}_{\mathrm{wheel}}$.

1. **Completion:** Wheel speeds return to near-zero (or bias speed). The cycle repeats.

Desaturation constraints: - MTQs can only generate torque **perpendicular** to the local magnetic field vector. They cannot dump momentum parallel to $\vec{\mathrm{B}}$. - Near the magnetic equator, $\vec{\mathrm{B}}$ is nearly horizontal (north-pointing, i.e. roughly along-track in a polar orbit). MTQs can effectively dump momentum about the pitch and yaw axes but not roll. - Near the magnetic poles, $\vec{\mathrm{B}}$ is nearly vertical. MTQs can dump pitch and roll but not yaw. - Over a full orbit, the field direction rotates sufficiently to dump all three axes. But at any instant, one axis is poorly controllable. - A multi-pass desaturation strategy (spreading the dump over a full orbit) is more efficient than a single-point dump.

Typical desaturation frequency for CubeSats: - At 500 km with 1.5 uN m total disturbance and 10 mN m s wheels: one desaturation per orbit (every ~95 minutes) - Duration: **20--50 minutes** per cycle. This is the number students most often get wrong: the worked sizing above gives $\mathrm{t}_{\mathrm{dump}} = \mathrm{H}/\mathrm{T}_{\mathrm{MTQ,avg}} \approx 2400$ s for a single wheel at a favourable field geometry, and a real dump is slower still because the field angle α swings through poor geometry and more than one wheel has to come down. A dump that "takes ten minutes" is a dump that has not finished. Flight software therefore carries a **timeout** on the manoeuvre, holds an **imaging inhibit** for its duration, and returns to the *pre-desaturation* mode on completion rather than dropping to a default - Power: 0.5--1.5 W during desaturation (3 MTQ rods active) - During desaturation, pointing accuracy degrades slightly (the MTQ torques perturb the attitude). Imaging should be inhibited during desaturation.

Wheel Configurations -- 3+1 Redundancy

The standard 4-wheel configuration provides full 3-axis control with one spare: - **3 wheels** in the body X, Y, Z axes provide minimum control - **4th wheel** on a skew axis (typically [1,1,1] normalised, or cant angle 20--30 deg from each axis) provides redundancy and enhanced torque distribution - If one wheel fails, the remaining three (including the skew wheel) maintain 3-axis control with reduced but adequate authority

4-wheel torque envelope: With 4 wheels in a pyramid configuration, the maximum torque in any body direction is:

$$\mathrm{T}_{\max,\mathrm{body}} = \sqrt{2} \cdot \mathrm{T}_{\mathrm{wheel}} \approx 1.41 \times \mathrm{T}_{\mathrm{wheel}}$$

for the optimal distribution. This is better than the 3-orthogonal configuration where $\mathrm{T}_{\max,\mathrm{body}} = \mathrm{T}_{\mathrm{wheel}}$ along any axis.

Worked through on the 6U reference. EOSAT-1 flies a **four-wheel pyramid** (Sinclair RW-0.03 class: 0.030 N m s each, 5500 rpm maximum) plus three magnetorquers saturating at 1.6 A m^2 per axis. On the determination side it carries **two star trackers** (ST1/ST2, selectable) and **three single-axis gyroscopes -- one per axis, with no redundant unit** -- feeding a six-state multiplicative extended Kalman filter (attitude error plus gyro bias). Gyro bias is calibrated **on board**, not uploaded from the ground.

The design consequence of the pyramid is worth stating explicitly, because it is the reason to pay for the fourth wheel: fine pointing is retained on **three** wheels. A single wheel failure degrades performance -- the mode reports itself as degraded -- but does not cost the mission its imaging capability. Only when fewer than three wheels are available does fine pointing become unachievable. Measured fine-pointing error on this design is 0.019--0.021 deg, comfortably inside a 0.1 deg requirement.

1U Worked Example: UniSat-1

Passive Magnetic Attitude Stabilisation

UniSat-1 does not have an active AOCS. Instead, it uses **passive magnetic stabilisation** -- the simplest and cheapest attitude control method, requiring zero power and minimal mass.

How it works -- physics:

1. **Permanent magnet:** A small bar magnet (typically AlNiCo or NdFeB, ~10--20 g, dipole moment $\mathrm{M}_{\mathrm{p}} = 0.1$ -- 1.0 A m^2) is embedded along one body axis (say, the Z-axis). In Earth's magnetic field $\vec{\mathrm{B}}$, the magnet experiences a restoring torque:

$$\vec{\mathrm{T}}_{\mathrm{restoring}} = \vec{\mathrm{M}}_{\mathrm{p}} \times \vec{\mathrm{B}}$$

This torque acts to align the magnet axis with the local field direction, analogous to a compass needle aligning with magnetic north. The restoring torque is maximum when the magnet is perpendicular to the field ($\alpha = 90$ deg) and zero when aligned ($\alpha = 0$).

For $\mathrm{M}_{\mathrm{p}} = 0.5$ A m^2 and $\mathrm{B} = 3 \times 10^{-5}$ T: $\mathrm{T}_{\max} = 0.5 \times 3 \times 10^{-5} = 1.5 \times 10^{-5}$ N m. This is ~10x larger than any environmental disturbance torque, ensuring stable alignment.

1. **Hysteresis rods:** Two or more strips of magnetically soft material (e.g., Permalloy, HyMu-80, ~5--10 g each, dimensions ~60 x 1 x 1 mm) are mounted perpendicular to the permanent magnet. As the satellite oscillates around the field-aligned equilibrium, the external field component along the hysteresis rod alternates, driving the rod material around its B-H hysteresis loop. The area enclosed by the hysteresis loop represents energy dissipated per cycle as heat in the rod material. This extracts kinetic energy from the satellite's oscillation, damping it over time.

The energy dissipated per oscillation cycle is:

$$E_{\text{dissipated}} = V_{\text{rod}} \times \oint H\,dB$$

where V_{rod} is the rod volume and $\oint H\,dB$ is the area of the hysteresis loop. For HyMu-80 material, typical energy density is ~100 J/m^3 per cycle.

Damping time constant: From initial tumble (~10 deg/s after deployment) to settled oscillation (~1--5 deg amplitude), the damping process typically takes hours to days, depending on the hysteresis rod material, volume, and the initial tumble rate.

Performance:

Table 3.2.13 — Passive magnetic stabilisation versus active reaction-wheel-plus-star-tracker AOCS.

Parameter	Passive Magnetic	Active (RW + ST)
Pointing accuracy	~10--15 deg (to local B-field)	< 0.1 deg (to inertial frame)
Settling time	Hours to days after deployment	Minutes after mode transition
Residual tumble rate	~1--5 deg/s (damped from initial ~10 deg/s)	< 0.01 deg/s
Power	0 W	3--5 W
Mass	~30--50 g	500--800 g
Cost	~2 kEUR	~55 kEUR
Failure modes	Demagnetisation (radiation, temperature)	Motor failure, bearing wear, software bugs

Why this works for UniSat-1:

The MEMS magnetometer payload does not require accurate pointing. In fact, it benefits from being in a slowly rotating/tumbling state because this provides magnetic field measurements across multiple directions, improving the scientific data quality. The magnetometer can measure the field vector regardless of spacecraft orientation.

No pointing budget needed: Since there is no payload pointing requirement, there is no need for a pointing error budget. This eliminates the star tracker, reaction wheels, magnetorquers (as actuators), sun sensors, and gyroscopes -- removing the most expensive and power-hungry subsystem from the design.

Disturbance environment for 1U at 400 km:

Source	Torque (N m)	Calculation	Notes
Gravity gradient	~ 1×10^{-8}	$\frac{3\mu}{2a^3}\Delta I;\ \Delta I \approx 0.001$ kg m^2 for 1U	Small because nearly cubic shape ($I_z \approx I_x$)
Aerodynamic	~ 3×10^{-8}	$\frac{1}{2}\rho v^2 C_D A d;$ $\rho(400\text{km}) \approx 5 \times 10^{-12}$	Higher ρ at 400 km than 500 km
			Small area

Source	Torque (N m)	Calculation	Notes
Solar radiation pressure	~ 1×10^{-9}	$\frac{S}{c} A(1+q)d;\ A = 0.01$ m^2	
Permanent magnet (restoring)	~ 1×10^{-5}	$M_p \times B \times \sin(\alpha)$	**Dominant** -- this IS the control torque

The permanent magnet restoring torque (~10^{-5} N m) is three orders of magnitude larger than all disturbances combined. This ensures the satellite remains approximately field-aligned.

Limitation: Passive magnetic stabilisation provides alignment to the *local* magnetic field, which rotates as the satellite orbits. The satellite does not point at nadir, the Sun, or any fixed direction. For missions requiring Earth-pointing or Sun-tracking, active AOCS is mandatory. For UniSat-1's magnetometer mission, this is not a limitation -- it is a feature.

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6. Propulsion System Design

When Propulsion is Required

Table 3.2.14 — Mission scenarios requiring propulsion with typical delta-V and execution timeline.

Need	Typical ΔV	Example Scenario	Timeline
Orbit maintenance (drag compensation)	5--15 m/s per year	LEO below 400 km in solar maximum	Continuous low-thrust
Deorbit (active disposal)	50--150 m/s	Active disposal from > 600 km (FCC 5-year rule)	End of mission
Collision avoidance	1--5 m/s per event	Conjunction avoidance, 2--5 events per year for LEO	On-demand, within hours
Constellation phasing	10--50 m/s	Spreading satellites into operational orbit slots	Weeks to months
Orbit raising	50--200 m/s	Transfer from deployment orbit to operational orbit	Weeks to months
Station-keeping	1--10 m/s per year	Maintain orbit altitude and phase	Periodic

When NO Propulsion is Needed *for Disposal*

Note the scope of this list carefully. It says nothing about whether a spacecraft should carry propulsion at all -- it says when propulsion is not needed **for end-of-life disposal**:

- **Orbit < 500 km:** Natural atmospheric decay provides FCC 5-year deorbit compliance (depends on ballistic coefficient and solar activity)
- **Low-cost technology demonstration:** Limited lifetime acceptable, no orbit maintenance needed
- **Constellation using differential drag for phasing** (e.g., Planet SuperDove adjusts its cross-section area to create differential drag, enabling free phasing manoeuvres)

- **Budget-constrained missions** where propulsion cost/risk/mass exceeds benefit

A spacecraft that satisfies every line above may still need propulsion for *debris avoidance* and *orbit maintenance*, which are operational needs and not disposal needs. Our 6U reference spacecraft is the instructive borderline case: at 513 km it sits just above the first line, so it needs propulsion for those operational needs *and* for a modest end-of-life disposal burn. It is worked through at the end of this section.

The Tsiolkovsky Rocket Equation -- Physics

The rocket equation is the fundamental relationship governing all propulsive manoeuvres. It derives from conservation of momentum: the momentum of the exhaust equals the momentum change of the spacecraft.

> **Key Equations -- Tsiolkovsky Rocket Equation**
>
> Starting from $\mathrm{F} = \dot{\mathrm{m}}\mathrm{v_e}$ (thrust = mass flow rate x exhaust velocity) and integrating:

$$\Delta \mathrm{V} = \mathrm{v_e} \ln\left(\frac{\mathrm{m_0}}{\mathrm{m_f}}\right) = \mathrm{I_{sp}} \cdot \mathrm{g_0} \cdot \ln\left(\frac{\mathrm{m_0}}{\mathrm{m_f}}\right)$$

> Rearranged for propellant mass:

$$\mathrm{m_{propellant}} = \mathrm{m_{dry}} \times \left(\mathrm{e}^{\Delta \mathrm{V}/(\mathrm{I_{sp}} \cdot \mathrm{g_0})} - 1\right)$$

> where: - $\mathrm{v_e} = \mathrm{I_{sp}} \times \mathrm{g_0}$ = effective exhaust velocity (m/s) - $\mathrm{I_{sp}}$ = specific impulse (s) -- the "fuel efficiency" of the thruster. Physically: how many seconds a thruster can produce 1 N of thrust from 1 kg of propellant under standard gravity. Higher $\mathrm{I_{sp}}$ = less propellant needed. - $\mathrm{g_0} = 9.80665$ m/s^2 -- standard gravitational acceleration (conversion factor) - $\mathrm{m_0}$ = initial (wet) mass (kg) = $\mathrm{m_f} + \mathrm{m_{propellant}}$ - $\mathrm{m_f}$ = final (dry) mass (kg) = spacecraft mass after all propellant is consumed
>
> **The tyranny of the rocket equation:** The propellant mass grows exponentially with $\Delta \mathrm{V}/\mathrm{v_e}$. For $\Delta \mathrm{V} = \mathrm{v_e}$ (one exhaust velocity worth of $\Delta \mathrm{V}$), 63% of the initial mass must be propellant. For $\Delta \mathrm{V} = 2\mathrm{v_e}$, 86% must be propellant. This is why high-$\mathrm{I_{sp}}$ systems are so valuable for large $\Delta \mathrm{V}$ missions -- they move the $\mathrm{v_e}$ in the denominator, dramatically reducing the mass ratio.

CubeSat Propulsion Technologies -- Physics and Comparison

Cold Gas Propulsion

Physics: Compressed gas (N_2, xenon, R-236fa refrigerant, or other) is stored in a tank at 1--30 MPa. When a valve opens, gas expands through a nozzle, converting thermal/pressure energy to kinetic energy. No combustion, no heating, no chemical reaction.

Thrust: $\mathrm{F} = \dot{\mathrm{m}}\mathrm{v_e} + (\mathrm{p_e} - \mathrm{p_a})\mathrm{A_e}$. For a small converging nozzle, $\mathrm{v_e} \approx \sqrt{2\mathrm{c_p}\mathrm{T_0}}$ where $\mathrm{T_0}$ is the tank temperature. For N_2 at 300 K: $\mathrm{v_e} \approx 500$ -- 750 m/s, giving $\mathrm{I_{sp}} \approx 50$ -- 75 s.

Advantages: Simplest system (no ignition, no power except valve solenoid), fast response (ms-level valve opening), high reliability, no plume contamination concerns.

Disadvantages: Low $\mathrm{I_{sp}}$ (large propellant mass for given $\Delta \mathrm{V}$), bulky high-pressure tank, limited total impulse.

Products: VACCO MiPS (R-236fa, $I_{sp} = 40$ s, 4 thrusters, 0.3 kg dry), VACCO ArgoMoon (Xe cold gas), Bradford ECAPS cold gas.

Missions using cold gas: MarCO (6U, JPL -- used R-236fa cold gas for trajectory correction and attitude control en route to Mars), many ISS-deployed CubeSats for collision avoidance.

Resistojet / Warm Gas

Physics: Similar to cold gas, but the propellant is electrically heated before expansion through the nozzle. Heating increases the gas temperature T_0, which increases the exhaust velocity ($v_e \propto \sqrt{T_0}$). Common propellants: butane (C_4H_{10}), water (H_2O), ammonia (NH_3).

Butane propulsion: Butane is stored as a liquid at its saturation pressure (~2 atm at 20 degC). When heated to 200--400 degC and expanded through a nozzle, it achieves $I_{sp} \approx 80$ -- 100 s. The liquid storage is much denser than compressed gas, allowing more propellant in a smaller tank.

Advantages: Higher I_{sp} than cold gas (~2x), dense liquid storage, moderate complexity.

Disadvantages: Requires electrical power for heating (5--15 W during firing), lower thrust than cold gas, potential for nozzle clogging if propellant decomposes.

Products: Busek BGT-X5 (butane, $I_{sp} = 80$ s), NanoAvionics EPSS (butane, $I_{sp} = 85$ s), Pale Blue water resistojet ($I_{sp} = 70$ -- 80 s).

Green Monopropellant

Physics: A liquid propellant is injected into a catalyst bed where it decomposes exothermically, producing hot gases that expand through a nozzle. "Green" propellants are alternatives to hydrazine (N_2H_4) that are less toxic and easier to handle.

Table 3.2.15 — Green and legacy monopropellants compared on specific impulse, density, toxicity, TRL, and heritage.

Propellant	Chemical	I_{sp} (s)	Density (kg/m^3)	Toxicity	TRL	Heritage
Hydrazine (N_2H_4)	Monopropellant, Shell 405 catalyst	220--230	1010	**Extremely toxic** (carcinogen)	9	50+ years, thousands of missions
AF-M315E (ASCENT)	HAN-based ionic liquid	235--250	1460	Low toxicity	8	GPIM demo (2019, NASA)
LMP-103S	ADN-based ionic liquid	225--235	1240	Low toxicity	8	PRISMA demo (2010, SSC), SkySat
HTP (H_2O_2 90%)	Hydrogen peroxide + silver catalyst	150--165	1400	Moderate (oxidiser)	7	Various small satellites

[Source: Masse et al., "GPIM AF-M315E Propulsion System," AIAA 2019; Anflo et al., "Flight Demonstration of LMP-103S," AIAA 2011]

Advantages: High thrust (0.1--1 N for CubeSat systems), good I_{sp} (220+ s), proven technology (heritage from hydrazine systems).

Disadvantages: Requires catalyst preheating (2--10 W, 10--30 min warmup), higher system mass (tank + catalyst bed + valves + feed system), propellant handling safety requirements (even "green" propellants require PPE), higher cost ($100K+ for flight units).

Products: Aerojet MPS-130 (AF-M315E, 1 N thrust, $\mathrm{I_{sp}} = 235$ s, 3 kg system mass), Bradford HPGP (LMP-103S, 1 N, $\mathrm{I_{sp}} = 230$ s).

Electric Propulsion -- Electrospray (FEEP)

Physics: Field Emission Electric Propulsion (FEEP) uses a strong electric field (~10^9 V/m) at the tip of a needle or along the edge of a slit to ionise and extract metal atoms (typically indium or gallium) or ionic liquid droplets. The ions are accelerated by an electric field to high velocities (10--50 km/s), producing very high $\mathrm{I_{sp}}$.

How it works (indium FEEP): 1. Solid indium is heated to just above its melting point (157 degC) to form a liquid reservoir 2. Capillary action draws liquid indium to an array of sharp emitter tips 3. A high voltage (1--10 kV) between the emitter tips and an extractor grid creates an intense electric field at the tip apex 4. The field ionises individual indium atoms via field evaporation 5. The ions are accelerated through the extractor grid, creating a beam of In^+ ions at 20--40 km/s 6. A neutraliser (typically a carbon nanotube or thermionic emitter) emits electrons to neutralise the beam and prevent spacecraft charging

Performance: $\mathrm{I_{sp}} = 500$--5000 s (adjustable by varying acceleration voltage), thrust = 0.01--1 mN, power = 20--60 W.

Advantages: Extremely high $\mathrm{I_{sp}}$ (minimal propellant consumption), no pressurised tanks, compact solid propellant storage, precise thrust control (useful for formation flying).

Disadvantages: Very low thrust (months-long burn times for significant $\Delta \mathrm{V}$), requires significant power (20--60 W for ~0.5 mN thrust), plume contamination from metal ions (indium deposition on surfaces), limited heritage.

Products: Enpulsion NANO R3 (indium FEEP, 0.35 mN, $\mathrm{I_{sp}} = 1000$--5000 s, 0.9 kg dry mass, < 40 W), Accion TILE (ionic liquid electrospray, 0.1 mN, $\mathrm{I_{sp}} = 1500$ s).

Missions using FEEP/electrospray: LISA Pathfinder (ESA, precision formation flying, used colloid thrusters), SSTL NovaSAR-1 (Enpulsion NANO for orbit maintenance).

Electric Propulsion -- Hall Effect Thruster

Physics: A Hall effect thruster uses crossed electric and magnetic fields to ionise a neutral propellant gas (xenon, krypton, or iodine) and accelerate the resulting ions to high velocity.

How it works: 1. Neutral propellant gas (Xe or I_2) is injected into an annular discharge channel 2. A radial magnetic field (from permanent magnets or electromagnets) traps electrons in a Hall current loop, preventing them from reaching the anode 3. The trapped electrons collide with neutral gas atoms, ionising them 4. The ions, being much heavier, are not significantly deflected by the magnetic field and are accelerated axially by the electric field (100--500 V) between anode and cathode 5. An external cathode (hollow cathode or RF cathode) provides electrons for beam neutralisation and to sustain the discharge

Performance: $\mathrm{I_{sp}} = 800$--3000 s, thrust = 1--50 mN for CubeSat-scale systems, power = 50--300 W.

Advantages: Higher thrust than FEEP (N range for larger systems), excellent $\mathrm{I_{sp}}$, well-proven technology (GEO station-keeping heritage: Aerojet PPS-1350, Busek BHT-200).

Disadvantages: Requires significant power (> 50 W for CubeSat-scale), heavy cathode assembly, xenon storage requires high-pressure tanks (100--300 bar), channel erosion limits lifetime.

Products: Exotrail ExoMG-nano (40 mN, $I_{sp} = 800$ s, 1.5 kg, 60 W), Busek BHT-200 (13 mN, $I_{sp} = 1370$ s, 1.0 kg, 200 W), Enpulsion MICRO (Hall thruster, iodine propellant, 1 mN, $I_{sp} = 1000$ s).

Iodine propulsion: Iodine (I_2) is emerging as an alternative to xenon for Hall thrusters and gridded ion engines. Iodine is solid at room temperature (stored without a pressure vessel), has a density of 4940 kg/m^3 (vs xenon at ~1600 kg/m^3 at 100 bar), and has similar atomic mass (127 vs 131). The Busek BIT-3 (iodine Hall thruster) and ThrustMe NPT30-I2 have demonstrated iodine propulsion in orbit.

No Propulsion -- Passive Deorbit Strategies

For missions below ~500 km where propulsion is not needed for operations, passive deorbit can satisfy the FCC 5-year or IADC 25-year guidelines:

Table 3.2.16 — Passive deorbit methods spanning natural drag, drag sails, and electrodynamic tethers.

Method	Mechanism	Mass	Volume	Effectiveness	TRL
Atmospheric drag (natural)	Below 500 km, atmospheric drag naturally decays the orbit	0	0	Depends on ballistic coefficient and solar cycle	9
Drag sail	Deployable membrane increases cross-section area by 10--100x	0.1--0.5 kg	0.25--1U	Very effective above 600 km	7--8
Electrodynamic tether	Electrodynamic tether interacts with geomagnetic field to decelerate	0.2--0.5 kg	0.5U	Moderate effectiveness; depends on orbital inclination	6--7

[Source: Cranfield Icarus drag sail, 0.1 kg; NanoSail-D2, NASA; InflateSail, SSC]

Propulsion System Comparison Table

Table 3.2.17 — CubeSat propulsion technology trade across Isp, thrust, propellant mass, dry mass, power, and cost.

Parameter	Cold Gas (N_2)	Warm Gas (Butane)	Green Monoprop (AF-M315E)	Electrospray (FEEP)	Hall Effect (Xe)
I_{sp} (s)	40--75	80--100	230--250	500--5000	800--3000
Thrust	10--100 mN	5--50 mN	0.1--1 N	0.01--1 mN	1--50 mN
Propellant mass (100 m/s, 5 kg S/C)	0.93 kg	0.60 kg	0.22 kg	0.043 kg	0.034 kg
(I_{sp} assumed for that row)	60 s	90 s	235 s	1200 s	1500 s
System dry mass	0.3 kg	0.5 kg	3.0 kg	0.9 kg	1.5 kg
Total system mass	**1.23 kg**	**1.10 kg**	**3.22 kg**	**0.94 kg**	**1.53 kg**
Burn time (100 m/s)	Minutes	Minutes	Seconds	**Months**	Days--weeks
Power during firing		5--15 W	2--10 W (preheat)	20--60 W	50--300 W

Parameter	Cold Gas (N_2)	Warm Gas (Butane)	Green Monoprop (AF-M315E)	Electrospray (FEEP)	Hall Effect (Xe)
	< 1 W (valve)				
Complexity	Low	Low-medium	High	Medium	High
Cost	~15 kEUR	~30 kEUR	~120 kEUR	~50 kEUR	~80 kEUR
TRL	9	7--8	7--8	7--8	6--8 (CubeSat)

Worked Example -- Propellant Mass Comparison for 100 m/s Deorbit

Scenario: 3U CubeSat, $m_{dry} = 5.0$ kg, deorbit from 600 km ($\Delta V = 113$ m/s).

Cold gas ($I_{sp} = 60$ s, $v_e = 589$ m/s):
$m_{prop} = 5.0 \times (e^{113/589} - 1) = 5.0 \times (e^{0.192} - 1) = 5.0 \times 0.212 =$ **1.06 kg**

Total system: 1.06 + 0.3 = 1.36 kg = **23% of 6 kg CubeSat mass limit**

Green monopropellant ($I_{sp} = 235$ s, $v_e = 2305$ m/s):
$m_{prop} = 5.0 \times (e^{113/2305} - 1) = 5.0 \times (e^{0.0490} - 1) = 5.0 \times 0.0502 =$ **0.251 kg**

Total system: 0.251 + 3.0 = 3.25 kg = **54% of mass limit** (system is heavy even though propellant is light)

Electrospray (FEEP) ($I_{sp} = 1200$ s, $v_e = 11,772$ m/s):
$m_{prop} = 5.0 \times (e^{113/11772} - 1) = 5.0 \times (e^{0.00960} - 1) = 5.0 \times 0.00965 =$ **0.048 kg**

Total system: 0.048 + 0.9 = 0.95 kg = **16% of mass limit** (lightest total, but takes months to execute)

Trade-off summary: The electrospray system is the lightest overall because the high I_{sp} minimises propellant mass, and the dry mass is moderate. Cold gas uses the most propellant but has the lowest dry mass. Green monopropellant is dominated by its heavy feed system. **The optimal choice depends on the mission timeline:** if deorbit must happen quickly (days), cold gas or monoprop; if months are acceptable, electric propulsion wins on mass.

Tsiolkovsky propellant-mass ratio vs ΔV for five propulsion technologies.

Figure 3.2.2 — Propellant mass ratio (m_p/m_d) vs required ΔV from the Tsiolkovsky rocket equation $\Delta V = I_{sp} g_0 \ln(m_0/m_f)$, plotted for five common CubeSat propulsion technologies. The 113 m/s guide is the deorbit ΔV from 600 km circular to a 200 km perigee (worked example above). At low ΔV the propellant mass differences across technologies are modest; above ~500 m/s the "tyranny of the rocket equation" begins to bite and high-I_{sp} electric propulsion becomes the only feasible choice.

6U Reference Design: EOSAT-1 Propulsion, As Flown

EOSAT-1 carries a **1U green-monopropellant blowdown module**, and it is worth being precise about what it is for, because the wrong answer is the intuitive one.

Natural decay is not enough. From 513 km at a ballistic coefficient of 77.9 kg/m^2, natural decay takes **5.0 yr** at mean solar activity and **14.2 yr** at solar minimum, so on its own it fails the 5-year disposal rule. EOSAT-1 therefore ends its mission with a retrograde **perigee-lowering burn of 34.3 m/s**, taking the perigee

from 513 km to **390 km**, flown as six burns of 5.72 m/s. From the resulting 513 x 390 km orbit, re-entry takes **1.5 yr** at mean activity and **4.2 yr** at solar minimum -- inside the rule at both. (Method: the project's drag integration with the CIRA-72 exponential atmosphere, solar minimum modelled as 0.35 times that table.) The burn size is a single tangential impulse at the 513 km point, which becomes the apogee:

$$\Delta \mathrm{V} = \sqrt{\frac{\mu}{\mathrm{r}_1}} - \sqrt{\mu\left(\frac{2}{\mathrm{r}_1} - \frac{2}{\mathrm{r}_1 + \mathrm{r}_\mathrm{p}}\right)} = 7605.4 - 7571.1 = 34.3 \ \mathrm{m/s}$$

with $\mathrm{r}_1 = \mathrm{R}_\mathrm{E} + 513$ km and $\mathrm{r}_\mathrm{p} = \mathrm{R}_\mathrm{E} + 390$ km. After the burns the spacecraft is passivated: the remaining propellant is depleted, the propulsion inhibit is latched, the wheels are stopped, and the charge regulator and the transmitters are switched off (simulator card NOM-014, End-of-Life Disposal).

> **Model uncertainty -- why margins and model choice matter.** The 4.2-year figure is only as good as the atmosphere behind it. Re-run the same 390 km perigee with NRLMSIS 2.1 and hold solar minimum for years, and re-entry takes **7.2 years** -- outside the rule. A perigee of **360 km**, i.e. **42.8 m/s**, would meet 5 years even under the sustained-F10.7 = 70 bound, but it does not close: it does not fit in the 66.9 m/s tank on top of the active-phase budget. Neither model is "the" answer: a scaled exponential table and a full empirical model held at a sustained minimum bracket the real future atmosphere, and the gap between them (4.2 against 7.2 years) is larger than the rule leaves to spare. That is why a disposal plan states its atmospheric model and solar-activity assumption, carries margin on the $\Delta \mathrm{V}$ and on the reserve, and is re-planned against the observed solar cycle before the burn.

It is also a debris-avoidance and orbit-maintenance system. Those are ongoing operational needs that drag does not satisfy, and no amount of natural decay substitutes for a collision-avoidance manoeuvre executed inside a few hours of a conjunction warning.

Parameter	Value
Propellant	LMP-103S / ADN-class green monopropellant, 0.400 kg
Tank	560 cm^3, isothermal blowdown 22.0 $\rightarrow$ 9.33 bar (2.36:1)
Thrust	0.33 N at BOL falling to 0.14 N at depletion
I_{sp}	220 s at BOL falling to 196 s (208 s mean)
Total $\Delta \mathrm{V}$	**66.9 m/s**
Dry mass (module)	1.15 kg of a 12.0 kg dry spacecraft; 12.40 kg wet at BOL
Preheat	12 W catalyst-bed heater for ~20 min (4.0 Wh) before arming is permitted
Thruster axis	$+\mathrm{X}$ body, mounted on the $-\mathrm{X}$ face

Check the $\Delta \mathrm{V}$ with Tsiolkovsky. Using the mean $\mathrm{I}_{\mathrm{sp}} = 208$ s, $\mathrm{v}_\mathrm{e} = 208 \times 9.80665 = 2040$ m/s, $\mathrm{m}_0 = 12.40$ kg, $\mathrm{m}_\mathrm{f} = 12.00$ kg:

$$\Delta \mathrm{V} = 2040 \times \ln(12.40/12.00) = 2040 \times \ln(1.03333) = 2040 \times 0.032790 = 66.9 \ \mathrm{m/s}$$

And check that it is enough. Over a two-year active phase plus disposal: drag make-up 15.1 (at 513 km and design solar activity) + semi-major-axis and LTAN biasing 3.0 + six collision-avoidance events at 0.1 m/s each = 0.6 + commissioning and calibration 1.0 + eccentricity control 1.5 + **end-of-life disposal 34.3** = 55.5 m/s, plus a 5 % Phase-C margin (2.8) and a 10 % operations and dispersion reserve (5.8) = **64.1 m/s required**

against 66.9 m/s available, +4.4 % (+2.6 % against the 65.8 m/s obtained by integrating the blowdown $\mathrm{I_{sp}}$ curve). That is a real, closing propulsion budget, and a thin one: the disposal burn is its largest line, so at least 38 m/s is held in reserve for it from mid-mission and is not available for drag make-up.

The AOCS coupling is deliberate, not an oversight. The single thruster fires along $+\mathrm{X}$, which in the nominal nadir attitude is prograde. A *retrograde* burn therefore requires a deliberate 180 deg yaw first, and the arming interlocks enforce the attitude: the manoeuvre is only permitted in NOMINAL, FINE_POINT or INERTIAL with attitude error under 2 deg, catalyst bed above 180 degC, and tank pressure between 6 and 24 bar. The 5 mm thrust-offset produces a 1.65 mN m disturbance torque at BOL, which the wheels must absorb throughout the burn.

A burn moves the mission, not just the spacecraft. The delivered $\Delta\mathrm{V}$ is applied to the orbit as a single impulse at the burn mid-point and the element set is then **re-fitted**, so ground-track timings, pass windows and eclipse entries all shift. The scaling to remember:

$$\Delta\mathrm{a} = \frac{2\mathrm{a}^2\mathrm{v}}{\mu}\,\Delta\mathrm{V} \quad \Rightarrow \quad 1\ \mathrm{m/s} \rightarrow \Delta\mathrm{a} = +1.800\ \mathrm{km}$$

and the resulting period change walks the ground track: at $+24$ h after the burn, a 1 m/s prograde impulse has shifted pass timings by -33.5 **s**. Pass plans built before the burn are stale after it, which is an operations lesson as much as an astrodynamics one.

What the model deliberately does not do: thrust-vector control, propulsive disposal, plane changes or direct inclination/LTAN control, orbit raising, phasing or rendezvous, closed-loop autonomous orbit control, real conjunction screening, and slosh or CG migration. Those are taught, not simulated -- and knowing which is which is part of reading any simulation honestly.

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7. SpaceCDF Exercise

Instructions

1. **Architecture tab (AOCS):** Select the AOCS architecture appropriate for your pointing requirement - Review the derived requirements that appear - Check that the selected hardware fits within your power and mass budgets
2. **Pointing Budget** card on the Dashboard: - Review the RSS error tree - Identify the largest contributor - Verify the total is within your pointing requirement
3. **Dashboard AOCS KPI:** - Pointing accuracy achieved - Margin to requirement - AOCS power demand by mode
4. **Equipment Browser (if time permits):** - Browse reaction wheels: compare mass, torque, momentum, cost - Browse star trackers: compare accuracy, mass, FOV, exclusion zones - Note that star tracker exclusion zones constrain mounting face options

Worksheet 3.2 Tasks

1. Select AOCS architecture and justify based on pointing requirement
2. Calculate all 4 disturbance torques for your orbit and spacecraft configuration
3. Size the reaction wheel (torque, momentum storage, and slew time)
4. Size the magnetorquer for desaturation (verify dump time < 1 orbit)
5. Complete the pointing error budget table (RSS of all 7 sources)
6. Verify margin to pointing requirement and identify the dominant error source

Session Summary

Table 3.2.18 — Session summary: key takeaways for AOCS and propulsion design.

Topic	Key Takeaway
AD vs AC	Determination = know orientation (sensors); Control = change orientation (actuators)
Star trackers	Pattern-match stars against catalogue; 3--15 arcsec accuracy; exclusion zones (Sun 25--45 deg, Earth 25--35 deg); most accurate sensor
Sun sensors	Coarse (photodiode, 2--5 deg) or fine (slit+array, 0.1 deg); essential for safe mode; every S/C needs them
Magnetometers	Measure $\vec{B}$ field; compare with IGRF model; 5--10 deg accuracy alone; residual dipole contamination is key error source
GPS receivers	Orbit determination (position/velocity), not attitude; 5--20 m accuracy in LEO; needed for nadir pointing computation
Gyroscopes	Measure angular rate $\vec{\omega}$; MEMS drift 1--10 deg/hr; combined with star tracker via Kalman filter for high-bandwidth estimation
Architecture selection	Driven by pointing requirement: passive magnetic for > 5 deg; MTQ for 2--5 deg; RW+ST for < 0.1 deg
Disturbance torques	Gravity gradient, aero, SRP, magnetic dipole; **magnetic dipole dominates for CubeSats**
Reaction wheel physics	$H = I_w \omega_w$; $T = dH/dt$; saturation occurs when $\omega \to \omega_{max}$; zero-crossing jitter
RW sizing	Torque $\geq 2\times$ disturbance; momentum $\geq$ half-orbit accumulation; check slew time
Magnetorquer physics	$\vec{T} = \vec{m} \times \vec{B}$; torque always perpendicular to $\vec{B}$; cannot point, only detumble/desaturate
MTQ sizing	Desaturation torque must exceed disturbance accumulation rate; dump time < 1 orbit
CMGs vs RWs	CMGs: torque amplification ($T = H\dot{\delta}$), for large/agile S/C; RWs: simpler, cheaper, standard for CubeSats
Pointing budget	RSS of 7 independent sources; alignment knowledge typically dominates; improve the dominant term
Budget-driven design	Investing in the smallest error source has negligible impact on total; focus on the dominant term
Redundancy	4-wheel (3+1 skew or pyramid) provides single-fault tolerance; distribution matrix maps wheels to body torques; fine pointing survives on **three** wheels, degraded but capable
Detumble timescale	B-dot is slow: ~3.5 x 10^3 s for a 6U from 14 deg/s, below a 0.5 deg/s guard at ~1170 s; a "two-minute detumble" is a modelling error
6U reference AOCS	Four-wheel pyramid (0.030 N m s, 5500 rpm), 3 MTQ at 1.6 A m^2, two star trackers, three single-axis gyros with no redundant unit, 6-state MEKF, on-board gyro bias calibration

Topic	Key Takeaway
Momentum management	MTQs dump momentum against Earth's $\vec{B}$-field; ~once per orbit; **20--50 min**, with a timeout and an imaging inhibit; returns to the pre-desat mode
Tsiolkovsky equation	$m_{prop} = m_{dry} \times (e^{\Delta V/(I_{sp} g_0)} - 1)$; exponential growth with $\Delta V / v_e$
Cold gas	I_{sp} 40--75 s; simple, fast, reliable; heavy propellant penalty for $\Delta V > 30$ m/s
Green monoprop	I_{sp} 230--250 s; high thrust (0.1--1 N); heavy feed system; AF-M315E and LMP-103S flight-proven
Electrospray (FEEP)	I_{sp} 500--5000 s; minimal propellant; months-long burns; 20--60 W power; indium or ionic liquid
Hall thruster	I_{sp} 800--3000 s; moderate thrust (1--50 mN); 50--300 W; iodine emerging as Xe alternative
Propulsion trades	High-I_{sp}: less propellant, more dry mass, long burns; Low-I_{sp}: more propellant, lighter system, fast burns
When to skip propulsion	Disposal only: below 500 km (natural deorbit); tech demo; differential drag constellation. Debris avoidance and orbit maintenance are separate needs -- EOSAT-1 at 513 km flies a 1U green monoprop, 66.9 m/s, for those *and* for a 34.3 m/s end-of-life perigee-lowering burn (64.1 m/s needed in all)
Disposal from 513 km	Natural decay 5.0 yr (mean) / 14.2 yr (solar minimum) fails the 5-year rule; a 34.3 m/s burn to a 390 km perigee gives 1.5 / 4.2 yr; NRLMSIS 2.1 at a sustained minimum gives 7.2 yr, so the model and the margin matter

Chapter 3.3 — Communications and Link Budget Design

> **Expected reading before this session.** CCSDS 401.0 / PUS §1 (≈ 30 min); Pratt et al., *Satellite Communications*, Ch. 4; ITU Radio Regulations (skim).

Prerequisites: Sessions 2.1--3.2 (requirements, orbit, power, AOCS defined) **SpaceCDF Tabs:** Link Budget, Spectrum Selector, Equipment Browser (Comms)

References

- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 13 (Communications)
- ECSS, *ECSS-E-ST-50-05C: Radio Frequency and Modulation*, 2011
- CCSDS, *131.0-B-4: TM Synchronization and Channel Coding*, 2023
- ITU, *Radio Regulations*, 2020 (Articles 5 and 22)
- Maral & Bousquet, *Satellite Communications Systems*, 6th ed., 2020, Ch. 5
- Roddy, *Satellite Communications*, 4th ed., 2006, Ch. 4--6
- Haykin, *Communication Systems*, 5th ed., 2009
- Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed., 2001
- IARU, *Amateur Satellite Frequency Coordination*, 2023

Learning Objectives

By the end of this session, participants will be able to:

1. Construct a complete link budget from first principles (in decibels)
2. Compute free space path loss for any frequency and slant range
3. Select an appropriate frequency band based on data rate, licensing, and equipment availability
4. Explain the physics of each antenna type and compute antenna gain and beamwidth
5. Choose modulation and coding scheme based on required $\mathrm{E_b/N_0}$ and spectral efficiency
6. Explain the physical basis for coding gain and why FEC is essential for space links
7. Size an antenna (gain, beamwidth, mass) for the selected frequency
8. Determine data throughput and verify the data budget closes
9. Identify ground station options and compute contact geometry
10. Use SpaceCDF's link budget tool and spectrum selector

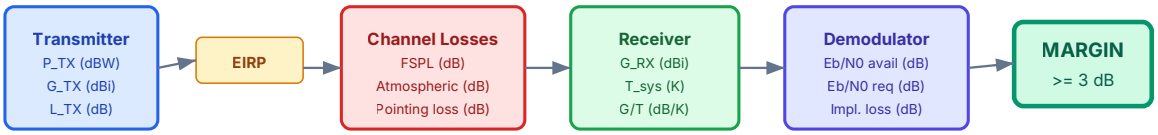
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1. The Link Budget Concept

The link budget is the accounting statement for the communication link. Every gain and every loss from transmitter to receiver is tallied in **decibels (dB)** to determine whether the link "closes" -- meaning the received signal is strong enough to decode with acceptable error rate.

The fundamental question: **does the received signal have enough energy per bit, relative to the noise, to achieve the required bit error rate?** This is quantified by $\mathrm{E_b/N_0}$ -- the ratio of energy per information bit to noise spectral density.

Link Budget Flow



Decibel Refresher

All link budget terms are in decibels to convert multiplication/division into addition/subtraction:

Table 3.3.1 — Decibel conversions between watts, milliwatts, and ratios with worked examples.

Conversion	Formula	Example
Watts to dBW	$\mathrm{P_{dBW} = 10 log_{10}\left(P_W\right)}$	2 W = +3.0 dBW
Milliwatts to dBm	$\mathrm{P_{dBm} = 10 log_{10}\left(P_{mW}\right)}$	1 W = +30 dBm
dBW to Watts	$\mathrm{P_W = 10^{P_{dBW}/10}}$	-3 dBW = 0.5 W
Ratio to dB	$\mathrm{G_{dB} = 10 log_{10}\left(G\right)}$	Gain of 100 = 20 dB
dBW vs dBm	dBm = dBW + 30	0 dBW = 30 dBm

Common power values:

Table 3.3.2 — Common transmitter power values expressed in watts, dBW, and dBm.

Power	dBW	dBm
0.1 W	-10.0	+20.0
0.5 W	-3.0	+27.0
1 W	0.0	+30.0
2 W	+3.0	+33.0
5 W	+7.0	+37.0
10 W	+10.0	+40.0

2. Complete Link Budget Equation

> **Key Equations -- Link Budget (dB form)**

EIRP (Effective Isotropic Radiated Power):

$$\mathrm{EIRP} = \mathrm{P_{TX}} + \mathrm{G_{TX}} - \mathrm{L_{TX}} \quad (\mathrm{dBW})$$

EIRP represents the power that an isotropic antenna would need to radiate to produce the same signal strength in the direction of maximum antenna gain. It combines the transmitter power, antenna gain (directivity), and cable/filter losses.

Free Space Path Loss:

$$\mathrm{FSPL} = 20\log_{10}\left(\frac{4\pi \mathrm{d}}{\lambda}\right) = 20\log_{10}\left(\frac{4\pi \mathrm{df}}{\mathrm{c}}\right) \quad (\mathrm{dB})$$

where d = slant range (m), f = frequency (Hz), $\mathrm{c} = 3 \times 10^8$ m/s.

Physical interpretation: FSPL is not "energy absorption" -- no energy is lost. It is the geometric spreading of radiated power over a sphere of radius d. An isotropic antenna radiates equally in all directions, so the power per unit area at distance d is $\mathrm{P}/(4\pi \mathrm{d}^2)$. The λ^2 dependence arises because a receive antenna's effective aperture scales as $\mathrm{A_{eff}} = \mathrm{G}\lambda^2/(4\pi)$ -- at higher frequencies the receive antenna "captures" less of the available flux (unless the antenna is physically larger).

Receiver figure of merit:

$$\mathrm{G/T} = \mathrm{G_{RX}} - 10\log_{10}\left(\mathrm{T_{sys}}\right) \quad (\mathrm{dB/K})$$

$\mathrm{G/T}$ is the single most important parameter of any receive station. It combines the antenna gain (signal capture) with the system noise temperature (noise floor). A high $\mathrm{G/T}$ means good sensitivity.

System noise temperature $\mathrm{T_{sys}}$ includes: - Antenna noise temperature $\mathrm{T_A}$ (depends on what the antenna "sees": sky ~10--50 K at zenith, ~150--300 K looking at Earth/ground) - Feed/cable losses: $\mathrm{T_{feed}} = \mathrm{T_{physical}}(\mathrm{L} - 1)$ where L = loss factor - LNA noise temperature: $\mathrm{T_{LNA}} = \mathrm{T_0}(\mathrm{F} - 1)$ where F = noise figure, $\mathrm{T_0} = 290$ K - Subsequent stages (reduced by LNA gain)

Rule of thumb: $\mathrm{T_{sys}} \approx 100$ -- 200 K for a professional ground station with cryogenic or low-noise LNA; $\mathrm{T_{sys}} \approx 400$ -- 800 K for an amateur station with COTS LNA.

Carrier-to-noise density ratio:

$$\mathrm{C/N_0} = \mathrm{EIRP} - \mathrm{FSPL} - \mathrm{L_{atm}} - \mathrm{L_{point}} - \mathrm{L_{pol}} + \mathrm{G/T} - \mathrm{k} \quad (\mathrm{dBHz})$$

where $\mathrm{k} = -228.6$ dBW/K/Hz (Boltzmann constant: $\mathrm{k} = 1.381 \times 10^{-23}$ J/K).

Energy per bit to noise density:

$$\mathrm{E_b/N_0} = \mathrm{C/N_0} - 10\log_{10}\left(\mathrm{R_b}\right) \quad (\mathrm{dB})$$

where $\mathrm{R_b}$ = data rate (bps). This is the fundamental quality metric: each bit needs a certain amount of energy ($\mathrm{E_b}$) relative to the noise floor ($\mathrm{N_0}$) to be correctly demodulated.

Link margin:

$$\text{Margin} = \text{E}_\text{b}/\text{N}_{0,\,\text{available}} - \text{E}_\text{b}/\text{N}_{0,\,\text{required}} - \text{L}_\text{implementation} \quad (\text{dB})$$

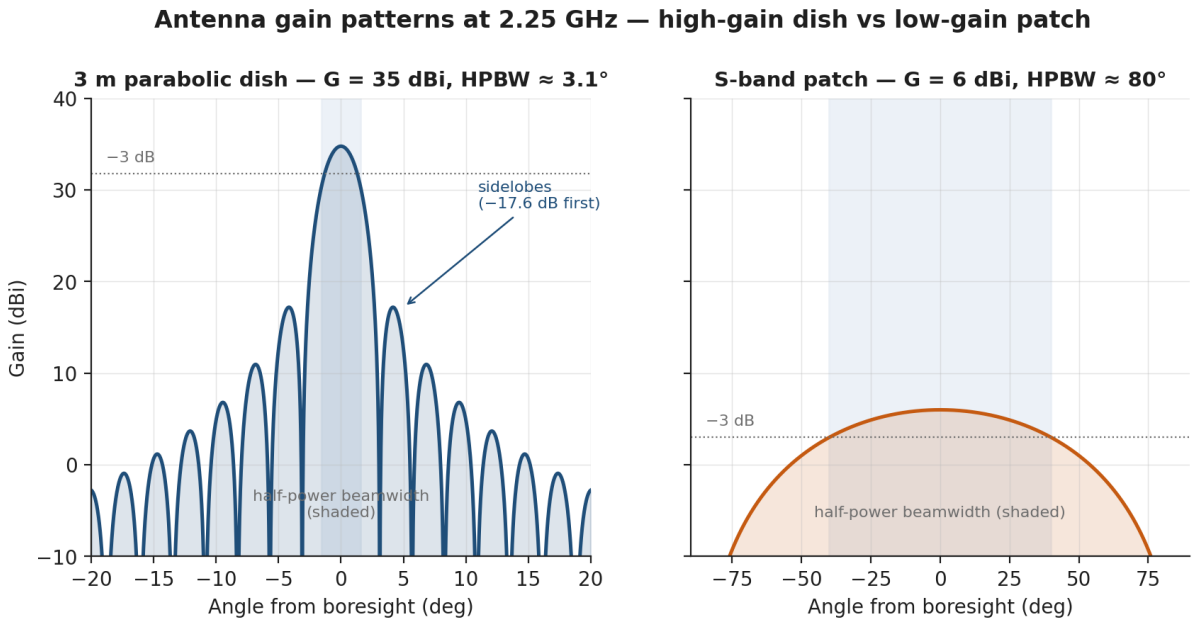
> **Requirement:** Margin $\geq$ 3 dB for Phase B+ (per ECSS-E-ST-50-05C). This 3 dB margin covers: - Transmitter power variation (aging, temperature) - Antenna gain uncertainties - Atmospheric scintillation - Pointing error variations - Ground station performance variation

Complete Link Budget Table

Table 3.3.3 — Eighteen-line decibel link-budget structure from transmit power through EIRP, path loss, and final margin.

Line	Parameter	Formula / Typical Value	Unit	Physical Meaning
1	TX Power	P_TX (e.g., 2 W = +3.0)	dBW	RF power from amplifier output
2	TX Antenna Gain	G_TX (e.g., +6.0 for patch)	dBi	Directivity relative to isotropic
3	TX Line Losses	L_TX (cables, filters: -1.5)	dB	Ohmic loss in RF cables and connectors
4	**EIRP**	$= \text{P}_\text{TX} + \text{G}_\text{TX} - \text{L}_\text{TX}$	dBW	Effective radiated power
5	Free Space Path Loss	$\text{FSPL} = 20\log_{10}(4\pi\text{df}/\text{c})$	dB	Geometric spreading + aperture effect
6	Atmospheric Loss	L_atm (-0.3 to -3.0)	dB	Molecular absorption (O_2, H_2O)
7	Pointing Loss	L_point (-0.5 to -3.0)	dB	Signal reduction from antenna mispointing
8	Polarisation Loss	L_pol (-0.1 to -3.0)	dB	Mismatch between TX and RX polarisation
9	RX Antenna Gain	G_RX (e.g., +35 for 3 m dish)	dBi	Ground antenna directivity
10	System Noise Temp	T_sys (e.g., 150 K = 21.8 dBK)	dBK	Total noise temperature of receive chain
11	**G/T**	$= \text{G}_\text{RX} - 10\log_{10}(\text{T}_\text{sys})$	dB/K	Receiver figure of merit
12	Boltzmann Constant	$\text{k} = -228.6$	dBW/K/Hz	Thermal noise power spectral density
13	**C/N** $_0$	$= \text{EIRP} - \text{FSPL} + \text{G/T} - \text{k} - \text{L}_\text{losses}$	dBHz	Signal-to-noise density ratio
14	Data Rate	$10\log_{10}(\text{R}_\text{b})$	dBbps	Information throughput
15	$\text{E}_\text{b}/\text{N}_0$ **available**	$= \text{C}/\text{N}_0 - 10\log_{10}(\text{R}_\text{b})$	dB	Available energy per bit
16	$\text{E}_\text{b}/\text{N}_0$ required	From modulation/coding selection	dB	Minimum needed for target BER
17	Implementation Loss	Typically 1.5--2.5 dB	dB	Real vs ideal demodulator performance
18	**LINK MARGIN**	$= \text{E}_\text{b}/\text{N}_{0,\,\text{avail}} - \text{E}_\text{b}/\text{N}_{0,\,\text{req}} - \text{L}_\text{impl}$	dB	Must be $\geq$ 3 dB

3. Antenna Types -- Physics and Selection



Antenna patterns: high-gain dish vs patch.

The antenna converts guided RF energy (in cables/waveguides) into radiated electromagnetic waves (and vice versa for receive). The key performance parameters are:

- **Gain** (G): How much more signal the antenna concentrates in its beam direction compared to an isotropic radiator. Higher gain = narrower beam = more signal but requires more precise pointing.
- **Beamwidth** ($\theta_{3\mathrm{dB}}$): The angular width of the main beam between the -3 dB (half-power) points.
- **Polarisation:** Linear (vertical/horizontal), circular (RHCP/LHCP), or elliptical.

Fundamental trade-off: $\mathrm{G} \propto 1/\theta_{3\mathrm{dB}}^2$. A higher-gain antenna has a narrower beam and requires more precise pointing. For a spacecraft without accurate pointing, a low-gain omnidirectional antenna is mandatory. For a spacecraft with 0.1 deg pointing, a high-gain directional antenna is feasible.

Antenna Types for Spacecraft

Table 3.3.4 — Spacecraft antenna types compared on gain, beamwidth, pointing tolerance, mass, and underlying physics.

Antenna Type	Gain (dBi)	Beamwidth	Pointing Needed?	Mass (CubeSat)	Physics
Monopole/ Dipole	0--2	~omnidirectional (toroidal pattern)	No	5--20 g (deployable wire)	Current distribution on a wire radiates; $\lambda/4$ monopole on ground plane or $\lambda/2$ dipole
Turnstile	0--3	Hemispherical	No	10--30 g	Two crossed dipoles fed 90 deg apart; produces circular polarisation
	5--8	60--90 deg		10--50 g	

Antenna Type	Gain (dBi)	Beamwidth	Pointing Needed?	Mass (CubeSat)	Physics
Patch (microstrip)			Loose (~10 deg)		Resonant conducting patch on dielectric substrate over ground plane; $\lambda/2$ patch at resonance
Patch array (2x2, 4x4)	10--18	15--40 deg	Moderate (~5 deg)	50--200 g	Multiple patch elements with corporate feed network; gain = $\mathrm{G_{element}} + 10\log_{10}(\mathrm{N})$
Horn	15--25	10--25 deg	Yes (~2 deg)	100--500 g	Flared waveguide; smooth aperture illumination gives low side lobes
Parabolic reflector	25--45	1--5 deg	Yes (< 1 deg)	500 g -- 5 kg	Feed at focal point illuminates parabolic dish; $\mathrm{G} = \eta_\mathrm{a}(\pi \mathrm{D}/\lambda)^2$
Phased array	15--35	Electronically steered, 1--30 deg	Electronic (no mechanical)	200 g -- 2 kg	Array of elements with individual phase shifters; beam steered by adjusting relative phases; no moving parts

Monopole/dipole physics: A quarter-wave monopole ($\mathrm{L} = \lambda/4$) is the simplest antenna. At 437 MHz (UHF), $\lambda = 0.686$ m, so a monopole is 17.2 cm long -- easily deployable from a CubeSat using a spring-loaded tape measure or nitinol wire. The radiation pattern is omnidirectional in the azimuthal plane (perpendicular to the wire) and null along the wire axis. Gain is approximately 0--2 dBi depending on the ground plane size.

Patch antenna physics: A microstrip patch antenna consists of a conducting patch (typically rectangular, $\mathrm{L} \approx \lambda/2$ at the desired frequency) printed on a dielectric substrate (e.g., Rogers RT/duroid, $\varepsilon_\mathrm{r} = 2.2$--$10.2$, thickness 1--3 mm) above a ground plane. The patch resonates at the frequency where its length equals $\lambda_\mathrm{eff}/2$ (where $\lambda_\mathrm{eff} = \lambda_0/\sqrt{\varepsilon_\mathrm{r}}$). At S-band (2.25 GHz), a patch is approximately 40 x 40 mm -- small enough to mount on a CubeSat face. Gain is typically 5--8 dBi with a hemispherical pattern. Circular polarisation is achieved by feeding two orthogonal modes 90 deg apart (dual-feed or corner-truncated patch).

Parabolic reflector physics: A parabolic dish focuses incoming plane waves to its focal point (for receive) or collimates spherical waves from a feed at the focus into a parallel beam (for transmit). The gain depends on the dish diameter D and the wavelength λ:

> **Key Equations -- Antenna**
>
> **Gain of a parabolic antenna:**

$$\mathrm{G} = \eta_\mathrm{a}\left(\frac{\pi \mathrm{D}}{\lambda}\right)^2$$

In dBi: $\mathrm{G_{dBi}} = 10\log_{10}\left[\eta_\mathrm{a}\left(\frac{\pi \mathrm{D}}{\lambda}\right)^2\right]$

where $\eta_\mathrm{a} \approx 0.55$ -- 0.65 (aperture efficiency, accounting for illumination taper, spillover, blockage, and surface errors), D = antenna diameter (m), $\lambda = \mathrm{c}/\mathrm{f}$.

Half-power beamwidth (HPBW):

$$\theta_{3\mathrm{dB}} \approx \frac{70\lambda}{\mathrm{D}} \quad (\text{degrees})$$

Patch antenna gain (single element): typically 5--8 dBi, beamwidth ~60--90 deg

Patch array gain (N elements): $\mathrm{G_{array}} = \mathrm{G_{element}} + 10\log_{10}(\mathrm{N})$

Pointing loss (when antenna is mispointed by angle $\Delta\theta$):

$$\mathrm{L_{point}} \approx -12\left(\frac{\Delta\theta}{\theta_{3\mathrm{dB}}}\right)^2 \quad (\mathrm{dB})$$

This shows that a narrower beam (smaller $\theta_{3\mathrm{dB}}$) is more sensitive to pointing errors.

Worked example -- antenna gain at different bands for a 30 cm dish:

Band	Frequency	λ (mm)	D/λ	Gain (dBi)	HPBW (deg)	Pointing req
S-band	2.25 GHz	133	2.25	14.4	31	~5 deg
X-band	8.4 GHz	35.7	8.40	25.8	8.3	~1 deg
Ka-band	26 GHz	11.5	26.0	35.7	2.7	~0.3 deg

(All three rows use the same aperture efficiency $\eta_\mathrm{a} = 0.55$ and $\theta_{3\mathrm{dB}} = 70\lambda/\mathrm{D}$; e.g. S-band $\mathrm{G} = 0.55(\pi \times 2.25)^2 = 27.5 = 14.4$ dBi.)

This table illustrates the fundamental link between frequency, antenna size, gain, and pointing requirement. At Ka-band, even a small dish provides high gain -- but the beamwidth is so narrow that sub-degree pointing accuracy is essential. This is why Ka-band CubeSat links require star tracker + reaction wheel AOCS.

Phased array physics: A phased array consists of multiple antenna elements (patch, dipole, or slot) arranged in a grid. Each element has an individual phase shifter (and sometimes amplitude control). By adjusting the relative phase between elements, the beam can be electronically steered without moving the antenna. The beam direction θ_s for element spacing d and phase increment $\Delta\phi$:

$$\sin(\theta_\mathrm{s}) = \frac{\Delta\phi \cdot \lambda}{2\pi \mathrm{d}}$$

Advantages: no moving parts, fast beam steering (microseconds), multiple simultaneous beams possible. Disadvantages: complex, expensive, high power consumption for active arrays, scan loss at wide angles ($\cos\theta_\mathrm{s}$ factor). Phased arrays are used on Starlink satellites (Ka-band), Iridium NEXT (L-band), and are emerging for CubeSats in Ka-band.

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4. Free Space Path Loss by Band

FSPL increases with both frequency and distance. At a fixed slant range, higher-frequency bands lose more signal -- but this is offset by the ability to use smaller, higher-gain antennas at higher frequencies.

Key Equations -- FSPL (expanded form)

$$\mathrm{FSPL}\ (\mathrm{dB}) = 20\log_{10}(4\pi) + 20\log_{10}(\mathrm{d}) + 20\log_{10}(\mathrm{f}) - 20\log_{10}(\mathrm{c})$$

$$= 21.98 + 20\log_{10}(\mathrm{d_m}) + 20\log_{10}(\mathrm{f_{Hz}}) - 169.54$$

Practical form (with km and GHz):

$$\mathrm{FSPL}\ (\mathrm{dB}) = 92.45 + 20\log_{10}(\mathrm{d_{km}}) + 20\log_{10}(\mathrm{f_{GHz}})$$

Slant Range Geometry

The slant range d from a LEO spacecraft to a ground station depends on the orbit altitude h and the elevation angle ε above the horizon:

$$\mathrm{d} = \mathrm{R_E}\left[\sqrt{\left(\frac{\mathrm{R_E}+\mathrm{h}}{\mathrm{R_E}}\right)^2 - \cos^2(\varepsilon)} - \sin(\varepsilon)\right]$$

For typical LEO orbits:

Table 3.3.5 — Slant range from LEO altitudes versus ground-station elevation angle.

Altitude	Elevation 90 deg (nadir)	Elevation 30 deg	Elevation 10 deg	Elevation 5 deg
400 km	400 km	739 km	1439 km	1804 km
500 km	500 km	909 km	1695 km	2077 km
600 km	600 km	1075 km	1932 km	2328 km
800 km	800 km	1395 km	2366 km	2783 km

(Every entry is the equation above evaluated with $\mathrm{R_E} = 6371$ km -- e.g. at 500 km and 10 deg: $6371[\sqrt{1.07848^2 - \cos^2 10^\circ} - \sin 10^\circ] = 6371 \times 0.26597 = 1695$ km. The low-elevation slant range is roughly 3.4x the nadir range at 500 km, not 2.6x; under-estimating it is one of the most common link-budget errors.)

Design rule: Always compute the link budget at the **minimum elevation angle** (worst case), typically 5--10 deg. Below 5 deg, atmospheric losses increase sharply, ground clutter enters the antenna sidelobes, and the link is generally unusable.

FSPL by Band and Geometry

Table 3.3.6 — Free-space path loss by band at nadir and at 10 deg elevation slant range.

Band	Centre Frequency	FSPL at 500 km (nadir)	FSPL at 1695 km (10 deg elev)	Difference
VHF	146 MHz	129.7 dB	140.3 dB	10.6 dB

Band	Centre Frequency	FSPL at 500 km (nadir)	FSPL at 1695 km (10 deg elev)	Difference
UHF	437 MHz	139.2 dB	149.8 dB	10.6 dB
S-band	2250 MHz	153.5 dB	164.1 dB	10.6 dB
X-band	8200 MHz	164.7 dB	175.3 dB	10.6 dB
Ka-band	26 GHz	174.7 dB	185.3 dB	10.6 dB

(Values verified against ITU-R P.525: $\mathrm{FSPL} = 32.45 + 20\log_{10}(\mathrm{d_{km}}) + 20\log_{10}(\mathrm{f_{MHz}})$*.)*

FSPL vs slant range for five RF bands, verified against ITU-R P.525.

Figure 3.3.1 — FSPL across the five operationally-relevant RF bands as a function of slant range, computed exactly from $20\log_{10}(4\pi \mathrm{d}/\lambda)$ *and verified against ITU-R P.525. The two vertical guides mark the 500 km nadir slant range and a 1300 km mid-pass slant range (which for a 500 km orbit corresponds to about 17° elevation; the worst-case slant at 10° elevation is 1695 km). Use this chart to read off path loss for any pass geometry without recomputing.*

The elevation angle penalty: Going from nadir to 10 deg elevation increases FSPL by ~10.6 dB (distance increases by 3.39x; FSPL scales as $20\log_{10}(3.39) = 10.6$ dB). This is a large loss -- more than a factor of eleven in received power -- and is why contact time at high elevations is much more valuable than contact time at low elevations.

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5. Frequency Band Selection and Licensing

[Source: ITU Radio Regulations, Articles 5 and 22; ISED RSS-SAT; FCC Part 25; IARU Satellite Frequency Coordination]

Band selection is a **design constraint** that affects data rate, antenna size, atmospheric losses, equipment availability, licensing cost, and data policy. The choice of band is one of the earliest and most consequential decisions in mission design.

Band Comparison Table

Table 3.3.7 — RF band comparison spanning allocation, bandwidth, data rate, rain fade, licensing, and equipment.

Band	Frequency Range	Allocation	Max BW	Practical Data Rate	Atmospheric Loss (10 deg)	Rain Fade	Licensing	Equipment Availability
VHF	144--146 MHz	Amateur	15 kHz	< 9.6 kbps	0.1 dB	None	IARU coord (free, 3--6 mo)	Many COTS, low cost
UHF	435--438 MHz	Amateur	20 kHz	< 19.2 kbps	0.2 dB	Negligible	IARU coord (free, 3--6 mo)	Many COTS, low cost
					0.5 dB	< 0.5 dB		

Band	Frequency Range	Allocation	Max BW	Practical Data Rate	Atmospheric Loss (10 deg)	Rain Fade	Licensing	Equipment Availability
S-band	2200--2290 MHz	Space research/ EES	5 MHz	0.1--10 Mbps			ISED/FCC ($30--45K, 6--12 mo)	Many COTS
X-band	8025--8400 MHz	EES	375 MHz	10--400 Mbps	1.0 dB	1--3 dB	ISED/FCC + ITU ($50--80K, 12+ mo)	Growing COTS
Ka-band	25.5--27.0 GHz	EES/FSS	1.5 GHz	100--2000+ Mbps	2--5 dB	3--15 dB (location-dependent)	Complex ITU ($100K+, 18+ mo)	Emerging COTS

EES = Earth Exploration Satellite; FSS = Fixed Satellite Service

Atmospheric attenuation physics: The atmosphere absorbs and scatters RF energy. Molecular oxygen has a strong absorption line at 60 GHz (used for inter-satellite links where atmospheric penetration is not needed). Water vapour absorbs at 22.2 GHz (near Ka-band). At S-band and below, atmospheric losses are minimal (< 1 dB). At Ka-band, losses of 2--5 dB are typical at 10 deg elevation, and rain fade can add 3--15 dB in tropical regions.

Rain fade: Raindrops scatter and absorb RF energy. The effect scales approximately as f^2 -- negligible below 4 GHz, significant above 10 GHz, and severe above 20 GHz. Rain fade is characterised by the rain rate (mm/hr) and the path length through rain. In tropical regions, rain rates of 50+ mm/hr can cause 10+ dB additional loss at Ka-band. Mitigation: adaptive data rate (reduce rate during rain), site diversity (multiple ground stations), power control (increase TX power during fade). For Ka-band links, a rain fade margin of 5--10 dB is typically included.

Amateur band regulations (IARU): - Non-commercial, educational, and experimental use only - Open data policy: all transmissions must be unencrypted and the protocol must be published - No commercial data or imagery downlink - Coordination through IARU (International Amateur Radio Union): free but takes 3--6 months - Very popular for university CubeSats (low cost, no license fees, large amateur community provides free ground station support via SatNOGS network)

Commercial band licensing: - Requires national filing (FCC in US, ISED in Canada, Ofcom in UK, CNES/ANFR in France) + ITU coordination for frequencies above 1 GHz - Costs: $30--45K for S-band (typical), $50--80K for X-band, $100K+ for Ka-band - Timeline: 6--12 months for S-band, 12--18 months for X-band, 18+ months for Ka-band - Commercial data can be encrypted and proprietary - Bandwidth allocation may be limited by the national authority

Band-selection decision tree for a CubeSat downlink, mapping data-rate requirement to UHF / S-band / X-band / Ka-band with licensing, antenna size, and EIRP considerations.

Band-selection decision tree for a CubeSat downlink, mapping data-rate requirement to UHF / S-band / X-band / Ka-band with licensing, antenna size, and EIRP considerations.

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6. Modulation and Coding -- Physics and Selection

[Source: CCSDS 131.0-B-4; Sklar, Ch. 7--8; Haykin, Ch. 10; DVB-S2 standard]

Modulation -- How Information Becomes RF

Modulation encodes digital data onto an RF carrier by varying the carrier's amplitude, frequency, or phase.

Phase Shift Keying (PSK): The most common modulation family for space communications. The carrier phase is shifted by discrete amounts to represent different bit patterns:

- **BPSK (Binary PSK):** 2 phase states (0 deg and 180 deg). Each symbol carries 1 bit. Spectral efficiency: 1 bps/Hz. Most robust to noise.
- **QPSK (Quadrature PSK):** 4 phase states (0, 90, 180, 270 deg). Each symbol carries 2 bits. Spectral efficiency: 2 bps/Hz. Requires the same $\mathrm{E_b/N_0}$ as BPSK but doubles the data rate for the same bandwidth. **The standard choice for most space links.**
- **8PSK:** 8 phase states, 3 bits/symbol, spectral efficiency 3 bps/Hz. Requires ~3.5 dB more $\mathrm{E_b/N_0}$ than QPSK for the same BER.
- **16APSK:** 16 states (amplitude + phase), 4 bits/symbol, spectral efficiency 4 bps/Hz. Requires ~7 dB more $\mathrm{E_b/N_0}$ than QPSK.

Frequency Shift Keying (FSK): The carrier frequency is shifted between discrete values. Less spectrally efficient than PSK but more tolerant of amplifier nonlinearity. Variants: - **FSK:** Simple frequency switching. - **GFSK (Gaussian FSK):** Gaussian filter smooths frequency transitions, reducing spectral spreading. Used by many CubeSat UHF radios (AX.25 protocol at 9600 bps). - **MSK (Minimum Shift Keying):** A special case of FSK with minimum frequency deviation that still allows coherent detection. Constant envelope (important for nonlinear amplifiers). Spectral efficiency ~1 bps/Hz.

Why QPSK is preferred over BPSK for space links: QPSK transmits 2 bits per symbol while requiring the same energy per bit as BPSK. This means QPSK achieves twice the data rate for the same bandwidth and the same $\mathrm{E_b/N_0}$ performance. The only additional complexity is that the receiver must resolve 4 phase states instead of 2, which is trivial for modern digital receivers.

Forward Error Correction (FEC) -- Coding Gain

FEC adds redundant bits to the data stream before transmission. The receiver uses these redundant bits to detect and correct bit errors without retransmission. The improvement in $\mathrm{E_b/N_0}$ requirement (compared to uncoded) is called the **coding gain**.

Why coding is essential for space links: Space links operate at very low received signal power (femtowatts to picowatts). Without coding, the required $\mathrm{E_b/N_0}$ for BER 10^{-6} is 10.5 dB. With LDPC coding, this drops to 2.0 dB -- a coding gain of 8.5 dB. This is equivalent to either: increasing the TX power by 7x, or increasing the antenna diameter by 2.7x, or reducing the data rate by 7x. Coding achieves the same benefit at the cost of only a few watts of digital processing power.

Code types used in space communications:

Table 3.3.8 — Forward-error-correction code types showing rate, coding gain, decoding complexity, and standards.

Code Type	Code Rate	Coding Gain at BER 10^{-6}	Decoding Complexity	Standard	Use
Convolutional (K=7)	1/2	~5.5 dB	Low (Viterbi decoder, hardware)	CCSDS	Legacy telecommand, AX.25
Convolutional + RS (concatenated)	~0.44	~7.5 dB	Medium	CCSDS	Standard CCSDS telemetry (many heritage missions)
Turbo code	1/2 to 1/6	~8--10 dB	High (iterative decoder)	CCSDS	Deep space links (Mars, Jupiter)
LDPC (Low-Density Parity Check)	1/2	~8.5 dB	Medium-High (iterative BP decoder)	CCSDS, DVB-S2	Modern space downlinks, **recommended for CubeSats**
LDPC	3/4	~6.5 dB	Medium-High	CCSDS, DVB-S2	Balanced performance and throughput
LDPC	7/8	~5.0 dB	Medium-High	DVB-S2	Maximum throughput, strong signal

[Source: CCSDS 131.0-B-4; DVB-S2 ETSI EN 302 307]

Code rate r: The ratio of information bits to total transmitted bits. A rate-1/2 code transmits 1 information bit for every 2 transmitted bits (50% overhead). This means the channel data rate must be $\mathrm{R}_{\mathrm{channel}} = \mathrm{R}_{\mathrm{info}}/\mathrm{r}$ -- a rate-1/2 code requires twice the channel bandwidth for a given information rate.

Complete Modulation + Coding Table

Table 3.3.9 — Modulation-and-coding combinations with required Eb/N0, spectral efficiency, and typical use.

Modulation + Coding	$\mathrm{E_b}/\mathrm{N_0}$ Required (BER 10^{-6})	Spectral Efficiency (bps/Hz)	Typical Use	Implementation
GMSK uncoded	10.5 dB	~1.0	AX.25 amateur, legacy	Simple radio IC
BPSK uncoded	10.5 dB	1.0	Legacy telecommand	Simple
QPSK uncoded	10.5 dB	2.0	Simple telemetry	Moderate
QPSK + conv (r=1/2, K=7)	5.0 dB	1.0	Standard CCSDS TM	Hardware Viterbi
QPSK + conv + RS (concat)	3.0 dB	0.88	Heritage CCSDS	Hardware
QPSK + LDPC (r=1/2)	2.0 dB	1.0	High-efficiency downlink	FPGA-based
	4.0 dB	1.5		FPGA-based

Modulation + Coding	E_b/N_0 Required (BER 10^{-6})	Spectral Efficiency (bps/Hz)	Typical Use	Implementation
QPSK + LDPC (r=3/4)			**Balanced -- recommended**	
QPSK + LDPC (r=7/8)	5.5 dB	1.75	Bandwidth-limited	FPGA-based
8PSK + LDPC (r=3/4)	6.5 dB	2.25	High-rate downlink	FPGA-based
16APSK + LDPC (r=3/4)	8.5 dB	3.0	Maximum throughput	FPGA-based

Design guidance for CubeSats: - **UHF amateur:** GMSK or AFSK (AX.25 protocol) at 1.2--9.6 kbps channel rate. Add convolutional coding if link margin is tight. - **S-band:** QPSK + LDPC (r=1/2 or r=3/4). This is the sweet spot: 6.5--8.5 dB coding gain with manageable complexity. Most COTS S-band CubeSat transmitters (Endurosat, NanoAvionics, AAC Clyde) support DVB-S2 or CCSDS LDPC natively. - **X-band / Ka-band:** 8PSK or 16APSK + LDPC for maximum spectral efficiency when bandwidth is limited.

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7. Ground Stations

The ground station is half of the communication link. Its performance (G/T) directly determines the achievable data rate. Upgrading the ground station is often the cheapest way to improve link performance (compared to upgrading the spacecraft transmitter or antenna).

Ground Station Types

Table 3.3.10 — Ground-station types compared on antenna size, G/T, cost, and access model.

Type	Antenna	G/T (S-band)	Cost	Availability	Examples
Amateur (SatNOGS)	10--15 dBi Yagi	-15 to -10 dB/K	Free (volunteer-operated)	Global network, 200+ stations	SatNOGS network
University	2--3 m dish	+10 to +15 dB/K	$50--200K (build)	Limited availability	Many universities have S-band stations
Commercial (small)	3--5 m dish	+15 to +25 dB/K	$500/pass or $5--20K/month	KSAT Lite, AWS Ground Station	KSAT, Amazon, Leaf Space
Commercial (large)	5--13 m dish	+25 to +35 dB/K	$1000/pass or $20--50K/month	SSC, KSAT, ATLAS	SSC (Esrange), KSAT (Svalbard), ATLAS (Fairbanks)
Deep Space Network	34--70 m dish	+45 to +60 dB/K	NASA-funded only	DSN (3 sites globally)	Goldstone, Canberra, Madrid

[Source: KSAT Lite pricing 2024; AWS Ground Station pricing; SSC SmallSat ground segment]

Contact geometry and pass duration:

A LEO satellite is in view of a ground station for a limited time per orbit. The pass duration depends on the orbit altitude and the maximum elevation angle:

$$t_{pass} \approx \frac{2}{n} \arccos\left(\frac{\cos(\varepsilon_{max})}{\cos(\varepsilon_{min})}\right)$$

For a simplified estimate:

Table 3.3.11 — LEO pass duration and daily contact count versus orbit altitude at 10 deg minimum elevation.

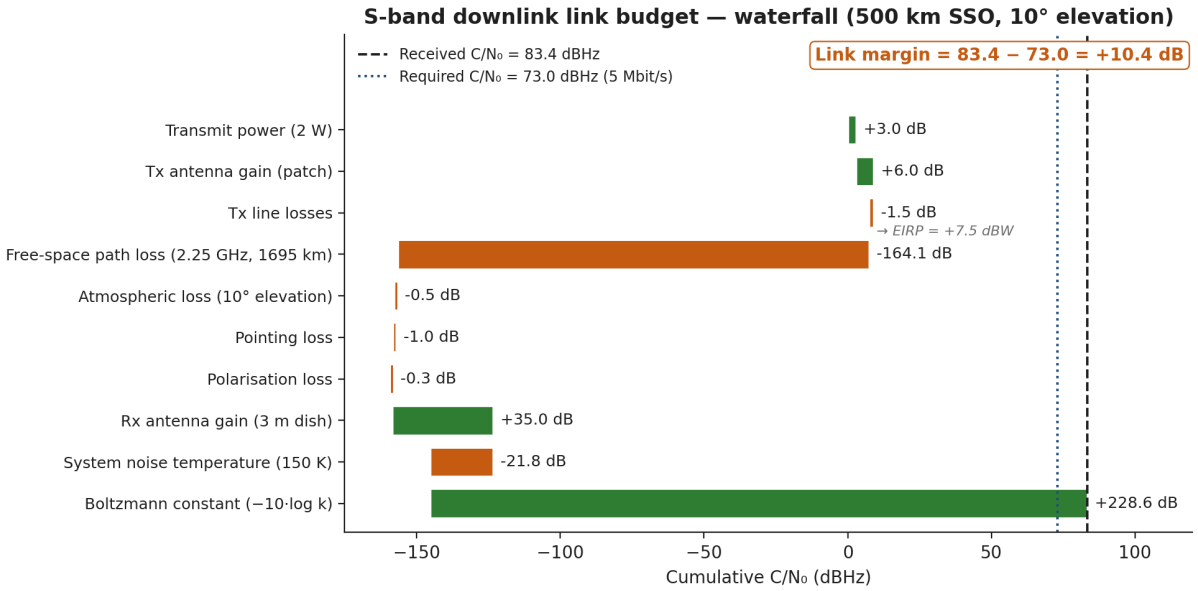
Altitude	Min Elevation	Max Pass Duration	Typical Usable Duration	Passes/Day (mid-latitude)
400 km	10 deg	~7 min	~5 min	3--4
500 km	10 deg	~8 min	~6 min	4--5
600 km	10 deg	~9 min	~7 min	4--5
800 km	10 deg	~11 min	~9 min	5--6

The "usable duration" is shorter than the total pass because the link only closes above the minimum elevation angle, and the first/last 30--60 seconds are used for signal acquisition and link setup.

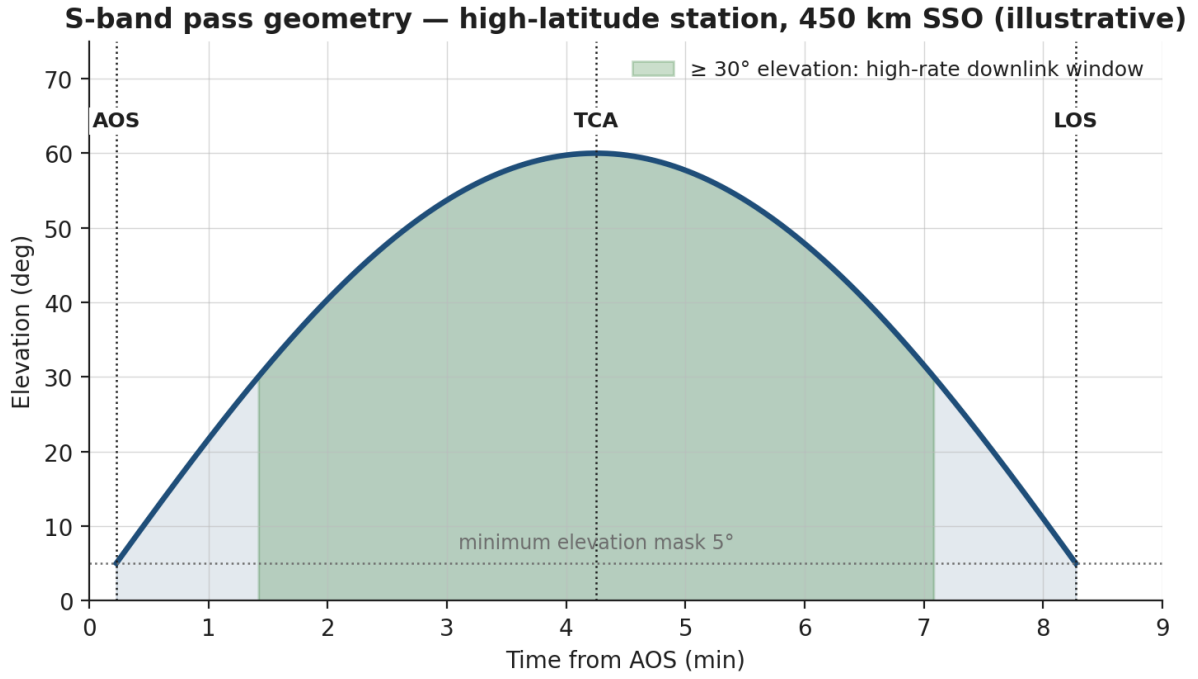
Ground station selection criteria: - **Location:** Polar stations (Svalbard at 78 degN, McMurdo at 78 degS) see every orbit of a polar/SSO satellite, providing 12+ contacts per day. Mid-latitude stations see only 3--5 passes. Equatorial stations see even fewer passes of polar satellites. - **G/T:** Determines the achievable data rate. A 3 dB improvement in G/T allows doubling the data rate. - **Licensing:** Must be compatible with the spacecraft frequency allocation - **Cost:** Ranges from free (SatNOGS, university) to $1000+/pass (commercial) - **Reliability:** SLA (service-level agreement) for commercial stations; university stations may have limited operator availability

EIRP (ground station transmit, for uplink): For telecommand uplink, the ground station transmits to the spacecraft. Ground station EIRP is typically 40--60 dBW for commercial stations, sufficient for robust command links even with low-gain spacecraft receive antennas.

8. Worked Examples: Complete Link Budgets



S-band downlink link-budget waterfall.



Iqaluit S-band pass geometry.

3U EO CubeSat -- S-band Downlink

Worked Example -- S-band Downlink for 3U EO CubeSat (SuperDove-class)

Scenario: 500 km SSO, S-band (2250 MHz), 1 Mbps downlink, 10 deg minimum elevation (slant range 1695 km, from the geometry table above), 3 m ground station dish ($T_{sys} = 150$ K, $G_{RX} = 35$ dBi at S-band).

Line	Parameter	Value	Unit	Calculation/Source
1	TX Power (2 W)	+3.0	dBW	COTS S-band TX (Endurosat)
2	TX Antenna Gain (patch)	+6.0	dBi	Single-element S-band patch
3	TX Line Losses	-1.5	dB	15 cm cable + connector
4	**EIRP**	**+7.5**	dBW	$3.0 + 6.0 - 1.5$
5	FSPL (2250 MHz, 1695 km)	-164.1	dB	$92.45 + 20\log_{10}(1695) + 20\log_{10}(2.25) = 92.45 + 64.58 + 7.04$
6	Atmospheric Loss	-0.5	dB	S-band, 10 deg elevation
7	Pointing Loss	-1.0	dB	5 deg mispoint, 80 deg beamwidth patch
8	Polarisation Loss (RHCP-RHCP)	-0.3	dB	Minor axial ratio mismatch
9	RX Antenna Gain (3 m dish)	+34.8	dBi	$10\log_{10}(0.6 \times (\pi \times 3.0/0.133)^2)$
10	System Noise Temp (150 K)	21.8	dBK	Professional LNA + sky temp
11	**G/T**	**+13.0**	dB/K	$34.8 - 21.8$
12	Boltzmann Constant	+228.6	dBW/K/Hz	$-\mathrm{k}$ in link equation
13	**C/N** $_0$	**+83.2**	dBHz	$7.5 - 164.1 - 0.5 - 1.0 - 0.3 + 13.0 + 228.6$
14	Data Rate (1 Mbps)	60.0	dBbps	$10\log_{10}(10^6)$
15	$\mathrm{E_b/N_0}$ **available**	**+23.2**	dB	$83.2 - 60.0$
16	$\mathrm{E_b/N_0}$ required (QPSK + LDPC r=3/4)	4.0	dB	From modulation/coding table
17	Implementation Loss	2.0	dB	Real demodulator vs ideal
18	**LINK MARGIN**	**+17.2**	dB	$23.2 - 4.0 - 2.0$

Result: Link closes with **17.2 dB margin** (requirement: >= 3 dB). **Pass with very large headroom.**

Design insight: A 17.2 dB margin at 1 Mbps is *much* more than required for a Phase B+ design. Real S-band CubeSat links typically run with 3--8 dB margin to leave room for adaptive data-rate increases and to handle worst-case ground-station performance. The team can comfortably increase the data rate:

Maximum data rate at 3 dB margin: $\mathrm{E_b/N_0}\ \text{available at max rate} = 4.0 + 2.0 + 3.0 = 9.0$ dB

$\mathrm{C/N_0} = 83.2$ dBHz, so $\mathrm{R_{b,\,max}} = 10^{(83.2\,-\,9.0)/10} = 10^{7.42} \approx$ **26 Mbps** at 3 dB margin.

However, the 5 MHz spectral allocation in the S-band space-research segment limits a practical QPSK link to about $\eta_s \cdot \mathrm{B} = 1.5 \cdot 5 = 7.5$ Mbps using LDPC r=3/4 (spectral efficiency ~ 1.5 bit/s/Hz including overhead and roll-off). So in practice, the *spectrum* and not the *link energy* is the constraint, and the team should design for around **5--6 Mbps** with comfortable margin.

1U Worked Example: UniSat-1

UHF Link Budget: 437 MHz at 9600 bps

UniSat-1 uses the UHF amateur band at 437 MHz with a ground station equipped with a 10 dBi Yagi antenna. This is the lowest-cost and simplest communication architecture available to CubeSat missions.

Worked Example -- UHF Downlink Link Budget for UniSat-1

Scenario: 400 km orbit, UHF (437 MHz), 9600 bps downlink (GMSK), 10 deg minimum elevation angle, amateur ground station with 10 dBi Yagi antenna, $T_{sys} = 600$ K (COTS LNA, coax cable losses, sky noise near horizon).

Slant range at 10 deg elevation: From 400 km altitude, the worst-case slant range at 10 deg elevation is 1439 km (geometry table, Section 4).

Line	Parameter	Value	Unit	Notes
1	TX Power (0.5 W)	-3.0	dBW	Standard CubeSat UHF radio
2	TX Antenna Gain (monopole)	0.0	dBi	Quarter-wave monopole, ~omnidirectional
3	TX Line Losses	-0.5	dB	Short cable to antenna
4	**EIRP**	**-3.5**	dBW	Low EIRP is the fundamental UHF challenge
5	FSPL (437 MHz, 1439 km)	-148.4	dB	$32.45 + 20\log_{10}(1439) + 20\log_{10}(437) = 32.45 + 63.16 + 52.81$
6	Atmospheric Loss	-0.3	dB	UHF atmospheric loss is minimal
7	Pointing Loss (omni antenna)	-0.5	dB	Omni pattern -- negligible
8	Polarisation Loss (linear-linear)	-3.0	dB	**Major loss** -- Faraday rotation in ionosphere rotates polarisation randomly
9	RX Antenna Gain (10 dBi Yagi)	+10.0	dBi	5-element Yagi, manually tracked
10	System Noise Temp (600 K)	27.8	dBK	Amateur station: warm LNA + cable loss
11	**G/T**	**-17.8**	dB/K	Low G/T is the ground station limitation
12	Boltzmann Constant	+228.6	dBW/K/Hz	
13	**C/N_0**	**+55.1**	dBHz	$-3.5 - 148.4 - 0.3 - 0.5 - 3.0 - 17.8 + 228.6$
14	Data Rate (9600 bps)	39.8	dBbps	
15	**E_b/N_0 available**	**+15.3**	dB	$55.1 - 39.8$
16	E_b/N_0 required (GMSK uncoded, BER 10^{-5})	9.6	dB	Per Sklar Ch.4 (≈ Q(sqrt(2·Eb/N0)) bound)
17	Implementation Loss	-2.0	dB	

Margin = 15.3 - 9.6 - 2.0 = +3.7 dB. The link closes uncoded, but with only 0.7 dB to spare against the 3 dB Phase B+ requirement.

But beware: the 3 dB Faraday-rotation polarisation loss is the *mean* loss; instantaneous fades in the ionosphere can be 10--20 dB deeper when the rotation aligns the satellite's linear polarisation exactly orthogonal to the ground antenna's. So although the *nominal* link closes, the **fade margin is the design driver**, and the team should still apply two of the three remedies below:

Remedy 1: Add FEC. Convolutional coding (r=1/2, K=7): - $\mathrm{E_b/N_0}$ required drops to **5.0 dB** (4.6 dB coding gain) - Channel rate remains 9600 bps, but useful throughput is 4800 bps (half is redundancy) - **Margin** = 15.3 - 5.0 - 2.0 = **+8.3 dB** -- close to, but not quite, headroom against a ~10 dB Faraday fade.

Remedy 2: Upgrade ground antenna to cross-Yagi (13 dBi) with circular polarisation: - RX gain: +13.0 dBi (was +10.0) -> +3.0 dB improvement - Polarisation loss: -0.5 dB (was -3.0 dB, now RHCP-to-RHCP) -> +2.5 dB improvement - Net improvement: +5.5 dB - New $\mathrm{C/N_0}$: 60.6 dBHz, $\mathrm{E_b/N_0}$ available: 20.8 dB - **Margin** = 20.8 - 5.0 - 2.0 = **+13.8 dB** (with FEC) -- adequate headroom for the deepest expected Faraday fades.

Remedy 3: Adaptive data rate. Drop to 1200 bps during the worst part of the polar pass. Trades throughput for reliability.

BER vs Eb/N0 for common modulations and a representative coded scheme.

Figure 3.3.2 — Theoretical bit-error-rate curves vs $\mathrm{E_b/N_0}$ for BPSK/QPSK ($\mathrm{P_b = \frac{1}{2}\,erfc(\sqrt{E_b/N_0})}$), 8-PSK, non-coherent FSK, and BPSK with concatenated Reed-Solomon + convolutional coding (5 dB coding gain). The annotation marks the $\mathrm{E_b/N_0 = 9.6}$ dB point at which uncoded BPSK reaches 10^{-5} BER — the standard reference quoted in most space-comms textbooks (e.g. Sklar Ch. 4). Use this chart to read off the required $\mathrm{E_b/N_0}$ for any target BER and modulation, and to quantify the coding gain.

Polarisation loss physics: At UHF (437 MHz), the ionosphere causes Faraday rotation of linearly polarised signals. The rotation angle depends on the total electron content (TEC) along the path and varies with time of day, solar activity, and signal path. If the satellite transmits linear polarisation and the ground station receives with linear polarisation, the polarisation planes may be orthogonal at times, causing up to complete signal loss (theoretically infinite loss, practically 20+ dB fades). Using circular polarisation on at least one end (preferably both) eliminates Faraday rotation loss, reducing it to ~0.3--0.5 dB axial ratio mismatch. **This is why circular polarisation is mandatory for reliable UHF satellite links.**

Final link budget summary (recommended: FEC + cross-Yagi, against deep fades):

Parameter	Value
TX power	0.5 W
TX antenna	Monopole (0 dBi)
Frequency	437 MHz
Channel rate	9600 bps
Useful throughput	4800 bps (with r=1/2 FEC)
Ground antenna	13 dBi cross-Yagi, RHCP
Nominal link margin	**+13.8 dB**
Margin against worst-case 10 dB Faraday fade	**+3.8 dB**

Data throughput for UniSat-1: At 4800 bps useful throughput, a 7-minute pass delivers: $\mathrm{V}_{\mathrm{pass}} = 4800 \times 420 \times 0.85 = 1.71$ Mbit $= 214$ kB per pass.

The 0.85 factor accounts for protocol overhead (packet headers, acknowledgements, retransmissions, link setup time).

With 4 passes/day: $\mathrm{V}_{\mathrm{daily}} = 856$ kB/day $\approx$ **0.86 MB/day**. The magnetometer generates < 1 kbps $\times$ 600 s/orbit $\times$ 15 orbits $= 9$ Mbit/day $= 1.13$ MB/day. This is **marginal** -- the team may need to prioritise data or add a second ground station. Using the SatNOGS network (200+ volunteer stations worldwide) could provide 10+ additional contacts per day at no cost.

Key lesson from UniSat-1 link budget: At UHF, the lower FSPL at 437 MHz ($20\log_{10}(2250/437) = 14.2$ dB less than S-band) almost exactly compensates for the low TX power (0.5 W vs 2 W = 6 dB), low antenna gain (0 dBi vs 6 dBi = 6 dB), and warmer system noise temperature of amateur stations (600 K vs 150 K = 6 dB). The result is a *nominal* link that closes, but the **fade margin against ionospheric Faraday rotation and rain/atmospheric scintillation is the real design driver**. FEC coding and circular polarisation are not strictly required to make the nominal link close — they are required to absorb the 10--20 dB fades that occur stochastically during a polar pass. Treat them as fade-margin investments, not as the only way to make the numbers work.

6U Reference Design: EOSAT-1 RF, As Flown

EOSAT-1 is the case where the link budget stops being the constraint, and that is itself the lesson.

Two transmitters, not one. This is the single most important structural fact about the RF design, and it is easy to miss because both chains carry "telemetry":

	S-band prime chain	UHF beacon transmitter
Band	2025 MHz up / 2200 MHz down	437.5 MHz down only
Antenna	Body-fixed patch, 3.0 dBi	Deployable tape-spring dipole, 0 dBi
RF power	1.0 / 2.0 / 5.0 W selectable (2.0 W nominal)	1.0 W
Rate	1 / 16 / 128 kbps ladder	1 kbps, fixed
Carries	Everything: real-time HK, playback, image dumps, command verification	HK SID 11 and nothing else
Ground asset	7--13 m polar dish, G/T +18.5 to +22.0 dB/K	University crossed Yagi, G/T -8.0 dB/K
Polarisation loss	0.3 dB (pointed circular pair)	3.0 dB (linear dipole vs circular, possibly tumbling)

The teaching point is what happens when the S-band PA is disabled. All S-band telemetry stops -- real-time HK, playback, dumps, verification, the lot. **The beacon keeps transmitting.** It shares none of the prime chain's single points of failure: not the S-band PA, not the switchable `ttc_tx` power line, not the deployable patch, not the transponder, not the uplink lock chain. It is billed to the *non-switchable* receive line precisely so that no operator can de-energise the lifeline with a power-line command. That is why a real spacecraft carries a second box, and it is why "the spacecraft has gone quiet" and "the spacecraft is dead" are different diagnoses.

Do not confuse the beacon *transmitter* with the S-band ladder's lowest rung, which is confusingly also called BEACON and also runs at 1 kbps. They are different radios on different bands, gated independently.

The rate ladder and what a rung costs. Three rungs, and the steps are exactly the $-10\log_{10}(\mathrm{R_b})$ term of the link budget:

Rung	Rate	Modulation	Coding	Coding gain	Step from previous
BEACON	1 kbps	BPSK	conv(7,1/2) + RS(255,223)	7.5 dB	--
LOW (routine ops)	16 kbps	BPSK	conv(7,1/2) + RS(255,223)	7.5 dB	$-10\log_{10}(16) = -12.04$ dB
HIGH (dumps only)	128 kbps	QPSK	RS(255,223) only	6.0 dB	$-10\log_{10}(8) = -9.03$ dB, plus 1.5 dB of lost coding gain $= -10.53$ dB

LOW is not an arbitrary choice: the shipped housekeeping generates 1378.5 bps of PUS packets, which is 1588 bps of channel after RS and frame overhead. A 1 kbps "low rate" is *below the generation rate* -- the real-time backlog could never drain. 16 kbps clears real-time HK at 9.7 % duty and still plays back 0.85 orbits of stored HK inside one pass.

Five ground stations, and every rung closes everywhere. The teaching baseline is the full five-station network. At 513 km the geometry is:

Elevation	Slant range	FSPL at 2200 MHz
5 deg	2111.7 km	165.79 dB
10 deg	1726.8 km	164.04 dB
30 deg	931.3 km	158.68 dB
90 deg	513.0 km	153.50 dB

Link margin at 5 deg elevation -- the worst geometry any station ever sees:

Station	G/T (dB/K)	BEACON	LOW	HIGH
Inuvik	22.0	59.0	47.0	36.5
Svalbard	21.5	58.5	46.5	36.0
Iqaluit	20.0	57.0	45.0	34.5
O'Higgins	20.0	57.0	45.0	34.5
Troll	18.5	55.5	43.5	**33.0**

Work one cell to see where those come from -- Iqaluit, LOW tier, 5 deg: EIRP $= 10.0$ dBW (at the nominal 2 W PA) $+3.0$ dBi patch $= 13.0$ dBW; atmospheric loss $0.25 \times \mathrm{cosec}(5°) = 2.87$ dB; polarisation 0.3 dB; G/T de-rated -1.5 dB below 10 deg, so $20.0 - 1.5 = 18.5$ dB/K.

$$\mathrm{C/N_0} = 13.0 - 165.79 - 2.87 - 0.3 + 18.5 + 228.6 = 91.1 \text{ dBHz}$$

$$\mathrm{E_b/N_0} = 91.1 - 10\log_{10}(16000) = 91.1 - 42.0 = 49.1 \text{ dB}$$

$$\text{Margin} = 49.1 - (9.6 - 7.5) - 2.0 = \mathbf{45.0} \text{ dB}$$

Read that table again. The worst case in the entire matrix -- the top rate, the weakest station, the lowest elevation -- still has **33.0 dB** of margin. At 513 km into a professional polar dish, a 2 W S-band link is not power-limited by anything. If you have been taught that a data rate is chosen by finding where the margin runs out, this is the more common real-world situation and it teaches a harder lesson: **the rate is chosen by policy, bandwidth, PA power, thermal cost and pass planning, not by a power cliff.** Any worked example that shows this spacecraft on a marginal link has an error in it.

That is also why the HIGH tier is gated on spacecraft mode and on elevation ≥ 10 deg rather than on a margin threshold. Modelling a fake cliff would have been easier to explain and less honest.

Elevation is worth more than rate ladders suggest. Going from 5 deg to 10 deg buys about 4.7 dB (1.75 dB of path loss, 1.43 dB of atmosphere, 1.5 dB of the G/T de-rate coming off). Going to zenith buys 16.4 dB. An operator waits for a better pass, not for a better rate.

One honest caveat on the absolute numbers: 10.0 dBW of EIRP at 2 W of RF implies about 7 dB of feed-and-antenna gain ahead of the 3.0 dBi patch, which no 6U actually has. The defensible absolute figure is 3.0 dBW conducted plus 3.0 dBi $=$ 6.0 dBW, i.e. every margin in the table is roughly 7 dB optimistic. It is quoted as-configured here because the number is shared with the RF link model and moving it is a coordinated two-file change; every rung still closes with 26+ dB either way, so nothing above depends on it. Flagging a known optimism in your own budget is normal engineering practice, not an admission of failure.

The beacon budget, side by side. UHF wins $20\log_{10}(2200/437.5) = 14.0$ dB of path loss and 0.3 dB of atmosphere, and gives back 28.0 dB of G/T (a 9 m dish at $+20.0$ dB/K against a crossed Yagi at -8.0 dB/K), 3.0 dB of spacecraft antenna gain and 2.7 dB of extra polarisation loss. The 12.0 dB gained by dropping from 16 to 1 kbps roughly cancels the 10.0 dB lost between the configured 10.0 dBW S-band transmit level and the beacon's 1 W (0 dBW), and both links use the same conv + RS coding. Net: the beacon closes on 1 W at 1 kbps with about **26 dB**, against about 45 dB for the S-band LOW rung at 16 kbps. Same physics, different everything else -- and the beacon still gets home when the prime chain does not.

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9. Data Budget

The data budget determines whether the communication system can deliver all mission data to the ground. Even if the link budget closes, the mission fails if the total data volume exceeds the downlink capacity.

> **Key Equations -- Data Budget**
>
> **Daily data generation:**

$$\mathrm{V_{gen}} = \mathrm{R_{payload}} \times \mathrm{t_{imaging}} \times \mathrm{N_{orbits}} \times \mathrm{f_{compression}}$$

> **Daily downlink capacity:**

$$\mathrm{V_{DL}} = \mathrm{R_{downlink}} \times \mathrm{t_{contact}} \times \mathrm{N_{passes}} \times \eta_{\mathrm{protocol}}$$

where $\eta_{\text{protocol}} = 0.80$--$0.90$ accounts for packet overhead, retransmissions, link setup time, and handshaking.

Data budget closure:

$$\text{V}_{\text{DL}} \geq \text{V}_{\text{gen}} \quad (\text{data budget closes})$$

Backlog clearance time: If $\text{V}_{\text{DL}} < \text{V}_{\text{gen}}$ per day, data accumulates in onboard storage. The backlog clearance time is:

$$\text{t}_{\text{clear}} = \frac{\text{V}_{\text{stored}}}{\text{V}_{\text{DL}} - \text{V}_{\text{gen}}}$$

If $\text{V}_{\text{DL}} < \text{V}_{\text{gen}}$, the backlog grows forever -- the mission cannot sustain its data generation rate.

Worked Example -- Data Budget for 3U EO CubeSat

Generation: 240 Mbps raw imaging data x 5 min/orbit x 15 orbits/day x 0.25 (4:1 JPEG2000 compression) = 240 x 300 x 15 x 0.25 = 270,000 Mbit/day = **33.75 GB/day**

Wait -- that's extremely high. Let's be more realistic about imaging time. Not every orbit has a target. Assume 4 imaging passes per day, 5 minutes each:

$\text{V}_{\text{gen}} = 240 \times 10^6 \times 300 \times 4 \times 0.25 = 72,000$ Mbit $= 9.0$ GB/day

Still high. Planet SuperDove images approximately 1--2 minutes per orbit over priority targets, compresses heavily (10:1+), and downlinks selectively.

Revised: 240 Mbps x 1 min/pass x 4 passes/day x 0.10 (10:1 compression) = 240 x 60 x 4 x 0.10 = 5,760 Mbit = **720 MB/day**

Downlink (S-band at 5 Mbps): 5 Mbps x 6 min/pass x 5 passes/day x 0.85 protocol efficiency.

Per pass: $5 \times 10^6 \times 360 \times 0.85 = 1.53 \times 10^9$ bit $= 1530$ Mbit $= 191$ MB

$\text{V}_{\text{DL}} = 1530 \times 5 = 7{,}650$ Mbit $= 956$ MB/day

Result: 956 MB/day > 720 MB/day. **Data budget closes** with 33% margin.

Sensitivity: If imaging duty cycle doubles (2 min/pass), generation rises to 1440 MB/day > 956 MB/day. Options: (a) X-band for higher data rate, (b) additional ground stations, (c) more aggressive compression, (d) onboard data prioritisation/selection.

Worked Example -- What One EOSAT-1 Image Costs in Pass Time

The courseware image product is 4 MB. At the HIGH tier:

$$\text{t}_{\text{dump}} = \frac{4 \times 1{,}048{,}576 \times 8}{128{,}000} = \frac{33{,}554{,}432}{128{,}000} = \mathbf{262}\ \text{s}$$

The average pass across the five-station network is 450 s, so **one image occupies 58 % of an average pass**. Two images do not fit in one pass. This is the number that makes pass planning the binding constraint rather than the link budget: the margin table above says the link will carry anything you ask of it, and the clock says you cannot ask for very much per pass.

> At the LOW tier the same image would take $262 \times 8 = 2097$ s -- roughly five passes -- which is why the HIGH tier exists and why it is gated rather than default.

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10. SpaceCDF Exercise

Instructions

1. **Spectrum Selector** (Dashboard): Select your license type and frequency band
2. **Link Budget** tab: - Enter TX power, antenna type, frequency, ground station parameters - Review the computed link margin - Verify it meets >= 3 dB requirement
3. **Compare** to your hand calculation from the worked example
4. **Equipment Browser:** Select a transponder and antenna that match your band choice - Note RF compatibility warnings (band mismatch between transponder and antenna)
5. Complete Worksheet 3.3

Discussion Questions

- What is the most impactful parameter in your link budget? (Usually FSPL or G/T)
- How does doubling the data rate affect link margin? (Reduces by 3 dB -- because $10\log_{10}(2) = 3$ dB)
- Could you use a lower-power transmitter and still close the link? What is the minimum TX power?
- Does your data budget close? If not, what is the cheapest fix? (Usually: more ground station passes, or lower imaging duty cycle)
- What is the effect of Faraday rotation on your UHF link? (If applicable)

Session Summary

Table 3.3.12 — Session summary: key takeaways for communications and link budget design.

Topic	Key Takeaway
Link budget	$\text{Margin} = \text{E}_\text{b}/\text{N}_{0,\,\text{avail}} - \text{E}_\text{b}/\text{N}_{0,\,\text{req}} - \text{L}_\text{impl} \geq 3$ dB
EIRP	$\text{EIRP} = \text{P}_\text{TX} + \text{G}_\text{TX} - \text{L}_\text{TX}$ -- transmitter's effective power in beam direction
FSPL physics	Geometric spreading ($1/\text{d}^2$) + aperture scaling (λ^2); not energy absorption
FSPL formula	$92.45 + 20\log_{10}(\text{d}_\text{km}) + 20\log_{10}(\text{f}_\text{GHz})$; increases 6 dB per doubling of frequency or distance
G/T	Receiver figure of merit: antenna gain minus noise temperature; most important ground station parameter
Antenna types	Monopole (0 dBi, omni) to parabolic (25--45 dBi, narrow beam); gain vs pointing trade-off
Antenna gain	$\text{G} = \eta_\text{a}(\pi\text{D}/\lambda)^2$; patch ~6 dBi; 3 m dish: ~34 dBi at S-band, ~46 dBi at X-band
Pointing loss	$-12(\Delta\theta/\theta_{3\text{dB}})^2$ dB; narrower beam = more sensitive to pointing errors

Topic	Key Takeaway
Modulation	QPSK: 2 bits/symbol, same $\mathrm{E_b/N_0}$ as BPSK but 2x spectral efficiency; **standard choice**
FEC coding	LDPC (r=3/4): $\mathrm{E_b/N_0} = 4.0$ dB, coding gain ~6.5 dB; **essential for space links**
Coding gain	5--10 dB improvement for free (just digital processing); equivalent to 3--10x power increase
UHF challenges	Low EIRP, Faraday rotation (use circular polarisation), high ground station noise; FEC mandatory
Band selection	UHF for < 9.6 kbps (free, amateur); S-band for < 10 Mbps; X-band for < 400 Mbps; Ka-band for > 400 Mbps
Rain fade	Negligible below 4 GHz; 1--3 dB at X-band; 3--15 dB at Ka-band; add margin or adaptive rate
Ground stations	G/T: amateur -15 dB/K, university +12 dB/K, commercial +25 dB/K; polar stations see every orbit
Data budget	Daily downlink capacity must exceed daily data generation; compression ratio is a key lever
Licensing	Amateur (free, IARU, 3--6 mo); S-band ($30--45K, 6--12 mo); X-band ($50--80K, 12+ mo)
Slant range	At 500 km the 10 deg slant is **1695 km, 3.4x** the nadir range -- worth 10.6 dB of extra path loss, not 8 dB
Two transmitters	An S-band prime chain plus a separate UHF beacon share no single points of failure; disabling the S-band PA kills all S-band TM while the beacon keeps reporting
Rate ladder	1 / 16 / 128 kbps; each rung costs exactly the $-10\log_{10}(\mathrm{R_b})$ term (12.0 dB then 10.5 dB including the coding change)
When margin is not the constraint	Every rung of EOSAT-1's ladder closes at every one of five stations with 33.0--75.4 dB; the rate is set by policy, bandwidth, PA power, thermal cost and pass planning, not by a power cliff
Pass planning	A 4 MB image at 128 kbps takes 262 s -- 58 % of an average 450 s pass; the clock binds long before the link does

Chapter 3.4 — Spacecraft Structure and On-Board Data Handling

> **Expected reading before this session.** Cal Poly CDS Rev 14 §3 — structures (≈ 30 min); ECSS-E-ST-32C §4; Sarafin, *Spacecraft Structures and Mechanisms*, Ch. 3 – 5; Sutton & Biblarz, *Rocket Propulsion Elements*, Ch. 1 — propulsion.

Prerequisites: Sessions 2.1--3.3 (all subsystem sizing complete) **SpaceCDF Tabs:** Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

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References

- Cal Poly, *CubeSat Design Specification (CDS) Rev 14.1*, February 2022
- Wertz, Everett & Puschell, *Space Mission Engineering: The New SMAD*, 2011, Ch. 11.3 (Structure), Ch. 17 (Propulsion)
- ECSS, *ECSS-E-ST-32C Rev.1: Structural General Requirements*, 2008
- ECSS, *ECSS-E-ST-35C: Propulsion General Requirements*, 2008
- Sarafin, *Spacecraft Structures and Mechanisms*, 1995
- Sutton & Biblarz, *Rocket Propulsion Elements*, 9th ed., 2017, Ch. 2--4
- Enpulsion, *NANO R3 Thruster Datasheet*, 2023
- VACCO, *MiPS Propulsion System Datasheet*, 2023
- Tyvak, *Structure Specifications*, 2023
- ECSS, *ECSS-E-ST-10-03C: Testing*, 2012

Learning Objectives

By the end of this session, participants will be able to:

1. Verify CubeSat Design Specification (CDS) compliance for 1U--12U form factors
2. Explain the structural load environment (quasi-static, vibration, shock) and its physical origins
3. Compute structural margin of safety for quasi-static launch loads
4. Estimate the fundamental frequency requirement and verify it against deployer specifications
5. Select OBC architecture based on processing, radiation, and interface requirements
6. Size onboard data storage from the data budget
7. Select flight hardware using SpaceCDF's equipment browser with budget tracking

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1. CubeSat Structure and CDS Compliance

[Source: Cal Poly CDS Rev 14.1, February 2022; ECSS-E-ST-32C]

CDS Dimensional Specifications

Table 3.4.1 — CubeSat Design Specification dimensions, maximum mass, and typical deployers for 1U through 12U.

Form Factor	Dimensions (mm)	Max Mass (kg)	Internal Volume (cm ^{3})	Typical Deployer
1U	100 x 100 x 113.5	2.0	~1000	ISIPOD, P-POD
1.5U	100 x 100 x 170.2	3.0	~1500	ISIPOD
2U	100 x 100 x 227.0	4.0	~2000	ISIPOD, P-POD
3U	100 x 100 x 340.5	6.0	~3000	P-POD, ISIPOD, NanoRacks NRCSD
6U	100 x 226.3 x 340.5	12.0	~6000	6U deployer (Exolaunch, D-Orbit)
12U	226.3 x 226.3 x 340.5	24.0	~12000	12U deployer (Exolaunch)

Structural Materials

CubeSat rail material: Aluminium 7075-T6

This is the most commonly specified structural aluminium alloy for CubeSat rails. The CDS mandates hard-anodised aluminium for the rails (the four load-bearing edges that slide along the deployer guide channels).

Table 3.4.2 — Mechanical and thermal properties of Al 7075-T6, Al 6061-T6, Ti-6Al-4V, and quasi-isotropic CFRP.

Property	Al 7075-T6	Al 6061-T6	Ti-6Al-4V	CFRP (quasi-isotropic)
Density (kg/m ^{3})	2810	2700	4430	1600
Yield strength σ_y (MPa)	503	276	880	N/A (use ultimate)
Ultimate strength σ_u (MPa)	572	310	950	500--800
Young's modulus E (GPa)	71.7	68.9	114	70--150 (direction-dependent)
CTE (ppm/degC)	23.6	23.1	8.6	0--2 (tuneable)
Thermal conductivity (W/m/K)	130	167	6.7	3--10

[Source: MMPDS / ASM Handbook; Hexcel HexPly datasheets]

Why Al 7075-T6 for rails: - High strength-to-weight ratio (superior to 6061-T6) - Hard anodisation provides a durable, low-friction surface finish for deployer guide channel contact (reduces galling, provides electrical insulation) - Good machinability - Extensive flight heritage (virtually every CubeSat ever launched) - CTE-matched to Al deployer structure (prevents differential thermal expansion binding)

Why NOT other materials for rails: - **Titanium:** Excellent strength but poor thermal conductivity (thermal hot spots), difficult to machine, risk of galling against aluminium deployer - **CFRP:** Cannot be anodised; CTE mismatch with Al deployer causes binding at temperature extremes; poor electrical conductivity (grounding/bonding issues) - **Stainless steel:** Too heavy; poor CTE match

Anodisation physics: Anodisation is an electrochemical process that grows a hard aluminium oxide (Al_2O_3) layer on the surface. Hard anodisation (Type III) produces a 25--75 um thick oxide layer with hardness of 60--70 HRC (harder than most steel). This layer provides: wear resistance against deployer contact, electrical insulation (prevents arcing between satellite and deployer), corrosion resistance, and

controlled surface optical properties ($\alpha_s \approx 0.3$--0.5, $\varepsilon \approx 0.8$--0.85 for clear anodise; $\alpha_s \approx 0.9$, $\varepsilon \approx 0.85$ for black anodise).

Key CDS Requirements

Table 3.4.3 — Key CDS requirements with physical rationale spanning rails, switches, RBF pin, CG, and frequency.

Requirement	Specification	Physical Rationale
Rail material	Hard anodised aluminium (7075-T6 or 6061-T6)	Wear resistance, CTE match to deployer, electrical isolation
Rail cross-section	8.5 x 8.5 mm minimum contact area	Adequate bearing area for launch loads; prevents rail yielding under quasi-static acceleration
Surface finish	All external surfaces anodised or non-outgassing coating	Prevent contamination of other payloads on launch vehicle (molecular outgassing deposits on optics)
Deployment switches	Minimum 1 on each accessible rail face (+X, -X)	Inhibit all spacecraft activity until fully deployed from deployer (prevents inadvertent deployment, RF emissions in fairing)
Remove Before Flight (RBF) pin	Required; physically disables all power systems	Final safety inhibit; removed at launch pad after integration; ensures zero RF emissions and zero deployment actuator current until intentional removal
Protrusions	None beyond rail envelope in stowed configuration	Ensures clean ejection from deployer; prevents snagging on guide rails or adjacent CubeSat
Centre of gravity	Within 2 cm of geometric centre (per deployer ICD)	Prevents wobble during deployment ejection; ensures all CubeSats eject with similar tip-off rates
Fundamental frequency	> 40 Hz first mode (typical deployer requirement)	Prevents dynamic coupling between satellite and launch vehicle structural modes (which cluster at 10--30 Hz)

PC/104 Stack Architecture

Most CubeSat avionics use the PC/104-compatible stack architecture, a heritage from the industrial embedded computing standard adapted for space:

Physical specifications: - **Board size:** 96 x 90 mm (standard) or 90 x 96 mm - **Connector:** 104-pin stack-through header (2 x 52 pins, 2.54 mm pitch) -- original PC/104 pinout carries power + I2C + SPI + UART + GPIO - **Stack spacing:** Typically 10--15 mm between boards (constrained by component height and connector mating height) - **Stack capacity:** 1U accommodates ~4 boards; 3U accommodates ~12 boards (340 mm / ~28 mm per board slot)

What rides on the stack: - EPS board (battery management, MPPT, power distribution) - OBC board (processor, memory, interfaces) - Communications board (UHF radio, or S-band transponder) - AOCS board (if integrated -- some vendors combine IMU + magnetorquer driver + RW interface on one PCB) - Payload interface board (ADC, sensor interfaces)

Mechanical concerns: - Solder joints are the weakest point; random vibration causes fatigue cracking at heavy component leads (especially tall electrolytic capacitors, large connectors, and crystal oscillators) - Board-to-board connectors must be properly preloaded (too loose = intermittent contact; too tight =

difficult assembly/disassembly during I&T) - Standoffs and spacers must be correctly torqued; Loctite 222 (low-strength threadlocker) is standard

Launch Load Environment

The launch environment subjects the satellite to loads from engine thrust, aerodynamic buffeting, stage separation, and pyrotechnic events. The satellite must survive all of these without structural failure or functional degradation.

Table 3.4.4 — Launch load environment categories with physical source, level, duration, and verification method.

Load Type	Physical Source	Typical Level	Duration	Frequency Range	Verification Method
Quasi-static acceleration	Engine thrust + aeroloading	6--12 g axial, 2--4 g lateral	Seconds to minutes	0 (static equivalent)	Analysis + sine vibration test
Sine vibration	Low-frequency vehicle dynamics	0.5--3 g (5--100 Hz)	Minutes	5--100 Hz	Sine sweep test (3 axes)
Random vibration	Acoustic noise + turbulent boundary layer	5--15 grms (20--2000 Hz)	60--120 s per axis	20--2000 Hz	Random vibration test (3 axes)
Shock	Pyrotechnic separation events (stage sep, fairing sep, deployer spring release)	500--2000 g at separation (high frequency)	< 10 ms	100--10,000 Hz	Shock response spectrum (SRS) test
Acoustic	Sound pressure from engine exhaust, aerodynamic noise	120--140 dB (20--10,000 Hz)	Minutes	20--10,000 Hz	Usually covered by random vib for CubeSats

[Source: ECSS-E-ST-10-03C; NASA GEVS (GSFC-STD-7000B); Falcon 9 Payload User's Guide; PSLV User's Guide]

Random vibration PSD profile (GEVS minimum workmanship level):

Table 3.4.5 — Random-vibration acceleration-spectral-density profile: the GSFC-STD-7000B (GEVS) minimum workmanship level, Table 2.4-4.

Frequency (Hz)	ASD Level (g^2 / Hz)	Notes
20	0.01	Low-frequency start
20--80	+3 dB/oct	Ramp up
80--500	0.04	Flat plateau
500--2000	-3 dB/oct	Roll off
2000	0.01	End of profile
Overall	**6.8 grms**	

Frequency (Hz)	ASD Level (g^2/Hz)	Notes
		Workmanship (acceptance) level; proto-flight is +3 dB, about 9.6 grms, for 1 min per axis

The flat region at 0.04 g^2/Hz from 80--500 Hz is where most structural damage occurs, because this is where PCB resonances and solder joint fatigue are excited. The binding levels for a given flight are set by the deployer or launch-vehicle user's guide; use this profile only where it envelops them.

What fails during vibration testing: 1. **Solder joints:** Heavy components (connectors, tall capacitors, transformers) with long lever arms crack at their solder joints. Mitigation: use surface-mount components, stake tall components with adhesive (Loctite 4860 or similar), use conformal coating. 2. **Deployable mechanisms:** Antenna hinges, solar panel hold-down mechanisms, and deployment springs can fail if not properly constrained. Mitigation: adequate preload on hold-down mechanisms, shock testing of pyrotechnic release devices. 3. **Optical components:** Lenses and mirrors can shift or crack if not properly mounted with strain-relief. Mitigation: RTV potting, flexure mounts. 4. **Wire harness:** Chafing against structure edges. Mitigation: edge radii > 1 mm, harness tie-downs every 50 mm, protective sleeving.

Structural Margin of Safety

Key Equations -- Structural Margin of Safety

$$\mathrm{MoS} = \frac{\sigma_{\text{allowable}}}{\sigma_{\text{design}} \times \mathrm{FoS}} - 1$$

where: - $\sigma_{\text{allowable}}$ = material yield or ultimate strength (MPa) - σ_{design} = computed stress under design loads (MPa) - FoS = factor of safety

Requirement: MoS $\geq$ 0 for all load cases.

Factors of safety (ECSS-E-ST-32C):

Table 3.4.6 — ECSS factors of safety on yield and ultimate stress for metals, composites, joints, and mechanisms.

Material / Joint	Yield FoS	Ultimate FoS	Rationale
Metallic (Al 7075-T6)	1.25	1.5	Standard structural metals
Composite (CFRP)	1.5	2.0	Higher variability in laminate properties
Bonded joints	1.5	2.0	Bond strength is highly process-dependent
Pressurised systems	1.5	2.0	Burst hazard to other payloads
Mechanisms (single-use)	--	2.0	Must work first time; no test opportunity

Design loads: The design load includes the quasi-static acceleration (from the launch vehicle user guide), multiplied by a dynamic amplification factor ($\mathrm{DAF} \approx 1.25\text{--}1.5$) if the satellite's natural frequency is near any launch vehicle forcing frequency.

Worked Example -- Axial Load on 3U CubeSat Rail (SuperDove-class)

Given: 3U CubeSat, mass = 5 kg, axial launch load = 9 g (Falcon 9 typical), 4 rails (load shared equally), rail cross-section = 8.5 x 8.5 mm, material = Al 7075-T6 ($\sigma_y = 503$ MPa, $\sigma_u = 572$ MPa).

Step 1 -- Design load per rail: $\mathrm{F} = \frac{\mathrm{m} \times \mathrm{n} \times \mathrm{g}_0}{4} = \frac{5 \times 9 \times 9.81}{4} = \frac{441.5}{4} = 110.4$ N

Step 2 -- Compressive stress: $\sigma = \frac{\mathrm{F}}{\mathrm{A}_{\mathrm{rail}}} = \frac{110.4}{8.5 \times 10^{-3} \times 8.5 \times 10^{-3}} = \frac{110.4}{7.225 \times 10^{-5}} = 1.53$ MPa

Step 3 -- Margin of safety (yield): $\mathrm{MoS}_y = \frac{503}{1.53 \times 1.25} - 1 = \frac{503}{1.91} - 1 = 262 \gg 0$ **Pass** (by a very large margin)

Step 4 -- Margin of safety (ultimate): $\mathrm{MoS}_u = \frac{572}{1.53 \times 1.5} - 1 = \frac{572}{2.30} - 1 = 248 \gg 0$ **Pass**

Key insight: For CubeSats, quasi-static axial stress on the rails is never the critical load case. The rails are massively over-designed for direct compression. The critical structural design drivers are usually: 1. **Stiffness** (fundamental frequency > 40 Hz) -- driven by internal board/component mounting, not rail strength 2. **Random vibration fatigue** on PCB solder joints -- the real failure mode 3. **Deployment mechanism reliability** -- spring force, latch engagement, alignment tolerances 4. **CG location** -- difficult to achieve with asymmetric payloads or propulsion tanks

Fundamental Frequency

Key Equations -- Fundamental Frequency (simplified beam model)

For a cantilevered beam (simplified CubeSat model, clamped at deployer interface):

$$\mathrm{f}_1 = \frac{1.875^2}{2\pi \mathrm{L}^2} \sqrt{\frac{\mathrm{EI}}{\rho \mathrm{A}_{\mathrm{cross}}}}$$

where E = Young's modulus (Pa), I = second moment of area (m^4), ρ = linear density (kg/m), $\mathrm{A}_{\mathrm{cross}}$ = cross-section area (m^2), L = length (m).

Requirement: $\mathrm{f}_1 > 40$ Hz (from deployer ICD). Some deployers require > 90 Hz (e.g., NanoRacks NRCSD).

For a 3U CubeSat modelled as a cantilevered Al box beam: - $\mathrm{L} = 0.34$ m, $\mathrm{E} = 72$ GPa, box wall thickness $\mathrm{t} = 1.5$ mm - $\mathrm{I} \approx \frac{\mathrm{b}^4 - (\mathrm{b} - 2\mathrm{t})^4}{12} = \frac{0.10^4 - 0.097^4}{12} = \frac{1.0 \times 10^{-4} - 8.853 \times 10^{-5}}{12} \approx 9.56 \times 10^{-7}$ m^4 - Linear mass: $\rho_{\mathrm{L}} = \mathrm{m}/\mathrm{L} = 5/0.34 = 14.7$ kg/m

$$\mathrm{f}_1 = \frac{1.875^2}{2\pi} \sqrt{\frac{\mathrm{EI}}{\rho_{\mathrm{L}} \mathrm{L}^4}} = \frac{3.516}{6.283} \sqrt{\frac{72 \times 10^9 \times 9.56 \times 10^{-7}}{14.7 \times 0.34^4}} = 0.560 \sqrt{3.50 \times 10^5} \approx 331 \text{ Hz}$$

This easily exceeds 40 Hz. The structure itself is very stiff. However, the actual first mode is usually determined by: (a) the heaviest internal component on its mounting bracket (e.g., a 350 g star tracker cantilevered on a bracket), or (b) a deployable mechanism in its stowed configuration (e.g., a folded solar panel constrained only by a hold-down pin). **These local modes, not the overall structural mode, are typically the design concern.**

Launch-Vehicle Interface and Deployables

The CubeSat does not bolt directly to the launch vehicle; it rides inside a deployer (such as an ISIPOD, NRCSD, or Exolaunch EXOpod) whose rails, guide channels, and pusher spring define the mechanical interface, while the electrical interface is limited to the deployment (separation) switches and the remove-

before-flight pin that keep the satellite electrically inhibited until it has cleared the rails. The applicable deployer ICD therefore governs the rail finish, the minimum number of inhibits (typically at least three independent inhibits on the RF transmitter and any propulsion or pyrotechnic function), and the stowed envelope, which no part of the satellite may violate before ejection. Deployable mechanisms -- solar arrays, antennas (UHF/VHF tape monopoles or dipoles), and gravity-gradient or magnetometer booms -- are held in their stowed position by a hold-down and release mechanism such as a burn-wire, frangibolt, or spring-loaded latch, which must survive the full launch load and shock environment without premature release. After ejection and a programmed inhibit timer, the release mechanism is actuated and the stored strain energy or a dedicated spring drives the appendage to its deployed configuration. Deployment is verified on the ground by repeated functional actuation (including hot/cold and post-vibration deployments to demonstrate margin) and confirmed in flight through deployment-status telemetry such as microswitches, continuity loops, or the characteristic change in attitude dynamics, since a partial or failed deployment can compromise power generation, communications, or attitude stability.

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2. On-Board Data Handling

OBC Architecture -- Processor Selection

The OBC is the spacecraft's brain, managing all data handling, commanding, telemetry generation, and FDIR (Fault Detection, Isolation, and Recovery). Processor selection involves a fundamental trade between radiation tolerance, processing power, power consumption, and cost.

Flight-Heritage Processors

Table 3.4.7 — Flight-heritage processors from MSP430 to RAD750 with clock, memory, radiation tolerance, and cost.

Processor	Architecture	Clock (MHz)	RAM	Rad Tolerance	Power (W)	TRL (Space)	Cost	Typical Use
TI MSP430	16-bit RISC	16--25	2--10 kB	Moderate (COTS, tested to 30 krad)	0.005--0.01	9	< 10 EUR	Ultra-low-power housekeeping, safe mode OBC
ARM Cortex-M4 (STM32F4)	32-bit ARM	168	192 kB + ext	Low-moderate (COTS, 10--30 krad)	0.05--0.20	8--9	10--20 EUR	**Standard CubeSat OBC**, TM/TC handling, ADCS control loop
ARM Cortex-M7 (STM32H7)	32-bit ARM	400--480	1 MB + ext	Low-moderate (COTS, 10--20 krad)	0.1--0.5	7--8	15--30 EUR	Higher-performance CubeSat OBC, onboard image processing
ARM Cortex-A (Linux-	32/64-bit ARM	500--1200		Low (COTS, < 10 krad)	1--3	6--7	20--50 EUR	Payload processing, AI/ML

Processor	Architecture	Clock (MHz)	RAM	Rad Tolerance	Power (W)	TRL (Space)	Cost	Typical Use
capable, e.g., NXP i.MX6)			256 MB--1 GB DDR					inference, Linux OS
Xilinx Zynq (SoC: ARM + FPGA)	ARM Cortex-A9 + Artix-7 FPGA	667 + programmable	512 MB + FPGA fabric	Moderate (Zynq-7000) to High (Kintex radhard)	2--5	7--8	100--500 EUR	High-throughput data processing, SDR (software-defined radio), image compression
LEON3/4 (rad-hard SPARC)	32-bit SPARC	50--250	External	High (100--300 krad, SEL immune)	1--3	9	10K--50K EUR	ESA heritage missions, GEO, deep space
RAD750 (BAE Systems)	32-bit PowerPC	200	128 MB	Very high (1 Mrad, SEL immune)	5--10	9	200K+ EUR	NASA flagship missions (MRO, Curiosity, JWST)

[Source: ST Microelectronics STM32 datasheets; Xilinx Zynq-7000 datasheet; Cobham Gaisler LEON3 datasheet]

RTOS vs Bare-Metal vs Linux

Table 3.4.8 — Bare-metal, RTOS, and Linux flight-software approaches compared on determinism, footprint, and use case.

Approach	OS	Pros	Cons	When to Use
Bare-metal	None (custom event loop)	Minimum overhead, deterministic timing, smallest code size	Hard to maintain, no task isolation, no file system	Ultra-simple 1U missions (MSP430)
RTOS (FreeRTOS, ChibiOS, Zephyr)	Real-time OS	Deterministic scheduling, task isolation, mature ecosystem, small footprint (10--50 kB)	More complex than bare-metal; requires task priority design	**Standard for CubeSat C&DH**: ADCS loop, TM/TC, mode management
Linux (Yocto, Buildroot)	Full OS	Rich ecosystem (Python, networking, file system, device drivers), easy development	Non-deterministic (not suitable for hard real-time), large footprint (50+ MB), power-hungry processor	Payload data processing, AI/ML, onboard image analysis

Radiation effects on processors:

The space radiation environment causes two categories of effects:

1. **Total Ionising Dose (TID):** Accumulated radiation damage from trapped protons/electrons and solar particles. Measured in rad(Si) or gray. Causes threshold voltage shifts in CMOS transistors, increasing leakage current and eventually causing functional failure. - LEO (500 km, 51.6 deg): ~1--5 krad/year behind 2 mm Al shielding - LEO polar/SSO (800 km): ~5--10 krad/year - MEO (through proton belt): ~50--100 krad/year - GEO: ~10--30 krad/year
2. **Single Event Effects (SEE):** A single energetic particle (proton or heavy ion) deposits enough charge in a transistor to flip a bit (SEU -- Single Event Upset), latch a transistor (SEL -- Single Event Latchup, potentially destructive), or burn out a power device (SEB -- Single Event Burnout). - **SEU rate in LEO:** ~1--10 bit flips per day per GB of SRAM (highly variable with orbit and shielding) - **SEL mitigation:** Current-limiting resistors on power lines, latchup detection circuits, watchdog resets - **SEU mitigation:** Error Detection and Correction (EDAC) on memory (Hamming codes, TMR -- Triple Modular Redundancy)

CubeSat approach to radiation: Most CubeSats in LEO (< 600 km, < 3-year mission) use COTS processors with EDAC on memory and a watchdog timer. Total dose over a 3-year LEO mission is typically 3--15 krad, which most COTS ARM Cortex-M processors survive (tested and characterised, even if not guaranteed). For longer missions, higher orbits, or critical applications, rad-tolerant or rad-hard processors are needed.

Data Storage Sizing

Key Equations -- Data Storage

$$\mathrm{S_{required}} = \mathrm{V_{daily}} \times \mathrm{N_{days}} \times \mathrm{f_{safety}}$$

where $\mathrm{V_{daily}}$ = daily data generation, $\mathrm{N_{days}}$ = days between full downlinks (typically 1--3 for LEO), $\mathrm{f_{safety}} = 2$ (to handle missed passes, ground station outages, and safe mode periods).

Storage technologies:

Table 3.4.9 — Non-volatile storage technologies showing capacity, write speed, radiation tolerance, and CubeSat role.

Technology	Capacity	Write Speed	Radiation Tolerance	Power	CubeSat Use
NOR flash	4--256 MB	1--5 MB/s	Moderate (10--50 krad)	Low	Code storage, boot ROM, critical parameters
NAND flash (SLC)	1--128 GB	10--50 MB/s	Low-moderate (5--20 krad)	Low	**Primary data storage**
NAND flash (MLC/TLC)	32--512 GB	50--200 MB/s	Low (< 10 krad)	Low	Maximum capacity (use EDAC and scrubbing)
SD card (industrial)	4--128 GB	10--50 MB/s	Low (< 5 krad)	Very low	Budget missions (risk: wear levelling + radiation = data loss)
MRAM	1--64 MB				

Technology	Capacity	Write Speed	Radiation Tolerance	Power	CubeSat Use
		10--50 MB/s	High (> 100 krad)	Very low	Critical parameters, non-volatile log

Worked Example -- Storage for 3U EO CubeSat

Given: Daily generation = 720 MB (from Chapter 3.3 data budget), daily downlink = 480 MB, days to clear backlog = $720/480 = 1.5$ days.

$$S_{\text{required}} = 720 \times 3 \times 2 = 4320 \text{ MB} \approx \textbf{4.3 GB}$$

Specify: >= 8 GB NAND flash storage (next standard size). The GomSpace A3200 OBC includes 4 GB NAND flash; adding an 8 GB SD card or additional NAND chip provides adequate capacity.

For a high-resolution imager generating 4.5 GB/day, storage requirement is: $4500 \times 3 \times 2 = 27$ GB. Specify >= 32 GB flash storage.

Flight Software Functions

Table 3.4.10 — Flight-software functions with execution rate and criticality classification.

Function	Description	Typical Execution Rate	Criticality
Mode management	Transition between Safe, Idle, Imaging, Downlink, Eclipse modes based on state machine	Event-driven	**Critical** (incorrect transition = mission loss)
ADCS control loop	Read sensors (star tracker, gyro, magnetometer), compute attitude estimate (Kalman filter), command actuators (PID controller)	1--10 Hz	**Critical** (loss of pointing = loss of mission)
TM/TC handling	Generate CCSDS telemetry packets, parse and execute telecommands	1 Hz (TM), event-driven (TC)	**Critical** (loss of communication = loss of mission)
Data handling	Payload data acquisition, compression (JPEG2000, CCSDS 122.0), buffering, downlink queue management	On-demand	Important
FDIR	Fault Detection, Isolation, and Recovery: watchdog timer, over-current protection, sensor consistency checks, autonomous safe mode trigger	1--10 Hz	**Critical** (must detect and recover from faults autonomously)
Housekeeping	Monitor temperatures, voltages, currents, wheel speeds; log to non-volatile memory	0.1--1 Hz	Important
Scheduling	Time-tagged command execution: autonomous imaging over target, downlink preparation before ground pass, desat scheduling	1 Hz	Important

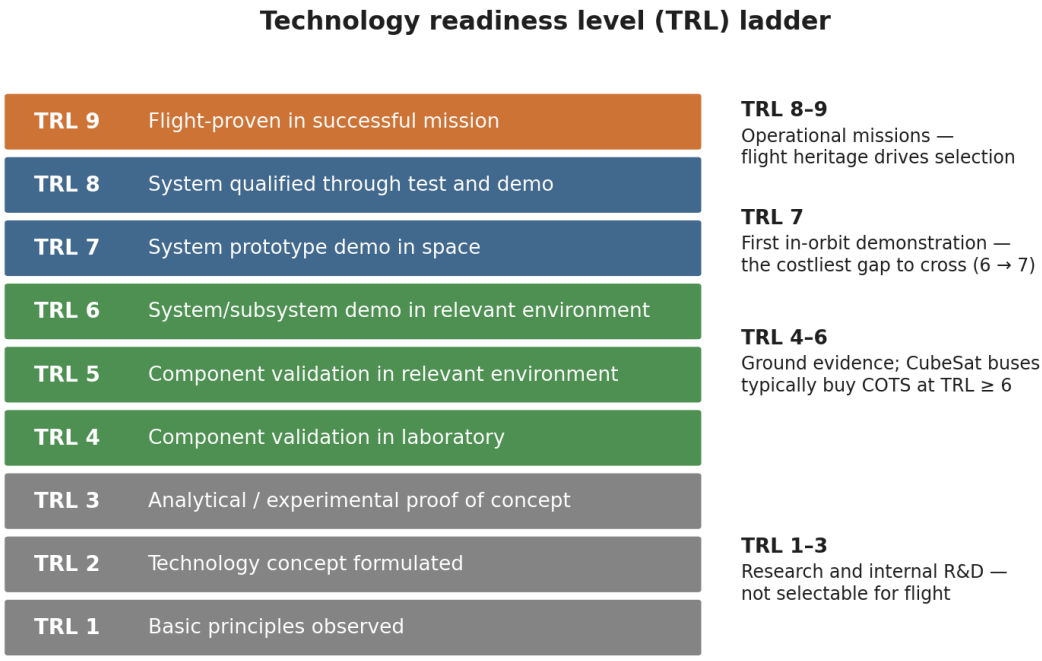
Function	Description	Typical Execution Rate	Criticality
Thermal control	Read temperature sensors, control heaters (on/off thermostat or PID)	0.1 Hz	Moderate (for heater-equipped missions)

Session Summary

Table 3.4.11 — Session summary: key takeaways for spacecraft structure and on-board data handling.

Topic	Key Takeaway
CDS compliance	Standard dimensions, Al 7075-T6 anodised rails (8.5 mm), deployment switches, RBF pin, CG limits
Structural materials	Al 7075-T6: $\sigma_y = 503$ MPa, $E = 72$ GPa; anodisation for wear/insulation; CFRP not suitable for rails
Launch loads	6--12 g axial QS, 6.8 grms random vib (20--2000 Hz; 9.6 grms proto-flight), 500--2000 g shock; PCB solder joints are the weak point
Structural MoS	$\mathrm{MoS} = \sigma_{\mathrm{allow}} / (\sigma_{\mathrm{design}} \times \mathrm{FoS}) - 1 \geq 0$; FoS 1.25 yield / 1.5 ultimate (metallic)
Frequency req	First mode > 40 Hz; CubeSat Al structures easily exceed this; local modes (PCBs, deployables) are the risk
OBC processors	MSP430 (0.01 W, safe mode), Cortex-M4 (0.2 W, standard CubeSat), Zynq (3 W, high-throughput), LEON3 (rad-hard, ESA)
RTOS vs Linux	FreeRTOS for real-time C&DH (standard); Linux for payload processing (data-intensive)
Radiation effects	TID: 1--10 krad/yr LEO; SEU: 1--10 bit flips/day/GB; mitigate with EDAC, watchdog, redundancy
Data storage	$S \geq 2\times$ daily generation; SLC NAND flash for primary storage; NOR flash for code/critical params

Chapter 4.1 — Equipment Selection & Bill of Materials



Technology Readiness Level ladder.

> **Expected reading before this session.** NASA TRL definitions (NPR 7123.1D Appendix E).

Prerequisites: Subsystems sized and components identified via parametric agents (preceding chapters).
References: ECSS-Q-ST-20C (Quality Assurance), ECSS-E-ST-10-24C (Interfaces), ITAR/EAR (22 CFR 120-130 / 15 CFR 730-774), CDS Rev 14.1, NASA SEH Rev 2 section 6.8

Learning Objectives

By the end of this session, participants will be able to: 1. Apply a structured make/buy/reuse decision framework to component selection 2. Evaluate COTS components against requirements using TRL and heritage criteria 3. Construct a Bill of Materials (BOM) with full traceability to requirements 4. Identify export control constraints (ITAR, EAR, Canadian Controlled Goods) for selected equipment 5. Verify interface compatibility (RF, electrical, mechanical) during component selection 6. Track cumulative mass, power, and cost budgets as selections are made

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1. The Make/Buy/Reuse Decision

Before selecting any hardware, the team must decide the procurement strategy for each subsystem. This is a fundamental systems engineering decision that affects cost, schedule, risk, and performance.

[Source: NASA SEH Rev 2 section 6.8 "Decision Analysis"; ECSS-M-ST-10C Rev.1 section 5.4]

Decision Framework

```
For each subsystem or component:
  1. Does a COTS product exist that meets requirements?
     -> Yes: BUY (lowest risk, fastest schedule)
     -> No: Continue
  2. Does flight-proven hardware from a previous mission exist?
     -> Yes: REUSE (low risk, may need delta-qualification)
     -> No: Continue
  3. Can the requirement be met by modifying existing hardware?
     -> Yes: MODIFY (moderate risk, moderate schedule)
     -> No: MAKE (highest risk, longest schedule, most expensive)
```

Make/Buy/Reuse Trade Matrix

Table 4.1.1 — Trade matrix comparing buy, reuse, modify, and make procurement strategies across cost, schedule, risk, performance, IP, and qualification.

Factor	Buy (COTS)	Reuse (Heritage)	Modify	Make (Custom)
Cost (NRE)	None	None-Low	Moderate	High
Cost (Recurring)	Vendor price	Reproduction cost	Vendor + delta	Full development
Schedule	4-16 weeks lead	8-24 weeks	12-36 weeks	12-48 months
Risk	Low (if TRL >= 7)	Low (flight-proven)	Medium	High
Performance	Fixed by vendor	Fixed by heritage	Tuneable	Fully customisable
IP ownership	Vendor retains	May be shared	Negotiated	Full ownership
Qualification	Vendor-provided data	Delta-qual only	Partial re-qual	Full qualification

Key Equations

Non-Recurring Engineering (NRE) Cost Estimate:

NRE = Labour_hours x Hourly_rate + Material_cost + Facility_cost + Testing_cost

For COTS: NRE is near zero (vendor absorbs development cost). For custom: NRE can be 3-10x the unit recurring cost.

Worked Example

Problem: A 6U CubeSat mission needs a fine sun sensor with 0.1 degree accuracy. Three options:

Table 4.1.2 — Three sun-sensor candidates with accuracy, mass, cost, TRL, and lead time for a worked make/buy decision.

Option	Type	Accuracy	Mass	Cost	TRL	Lead Time
Bradford SSOC-D60	COTS	0.1 deg	35 g	EUR 15K	9	8 weeks
In-house photodiode array	Custom	0.05 deg	20 g	EUR 8K + 400 hr NRE	4	12 months

Option	Type	Accuracy	Mass	Cost	TRL	Lead Time
Modified heritage sensor	Reuse	0.08 deg	40 g	EUR 6K + 80 hr delta-qual	7	16 weeks

Decision: The COTS option (Bradford SSOC-D60) meets the requirement, has TRL 9, lowest schedule risk, and total cost of EUR 15K vs EUR 8K + ~EUR 40K NRE for custom. **Buy.**

---

2. Technology Readiness Level (TRL) Assessment

TRL is the standard metric for technology maturity. It was developed by NASA in the 1970s and is now used universally in space programmes.

[Source: NASA NPR 7123.1D Appendix E; ECSS-E-HB-11A "Technology Readiness Level (TRL) Guidelines"]
[URL: https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/]

TRL Scale with Decision Criteria

Table 4.1.3 — Nine-level TRL scale defining evidence required and CubeSat selection guidance at each rung.

TRL	Definition	Evidence Required	CubeSat Decision
1	Basic principles observed	Published research	Do not select
2	Technology concept formulated	Analytical studies	Do not select
3	Experimental proof of concept	Lab measurements	Do not select
4	Component validated in lab	Breadboard tested	High risk -- avoid unless no alternative
5	Component validated in relevant environment	Engineering model tested	Acceptable for technology demonstrator missions
6	System/subsystem model demonstrated in relevant environment	Prototype in relevant environment	Acceptable with risk mitigation
7	System prototype demonstrated in operational environment	Prototype tested in space	Preferred minimum for CubeSats
8	Actual system completed and qualified	Flight-qualified hardware	Strong preference
9	Actual system flight-proven	Successful on-orbit operation	Lowest risk

TRL and Risk Relationship

Risk Reduction Factor (empirical):

Risk_factor = 10^(-(TRL - 1) / 3)

Table 4.1.4 — Empirical risk reduction factor mapping TRL level to probability of failure or redesign.

TRL	Risk Factor	Interpretation
3	0.22	High probability of failure/redesign

TRL	Risk Factor	Interpretation
5	0.046	Moderate risk
7	0.010	Low risk
9	0.002	Very low risk

This empirical relationship (a widely-cited empirical heuristic) illustrates why TRL >= 6 is the typical threshold for mission-critical components.

Real Mission Example: MarCO (Mars Cube One)

NASA's MarCO A and B (launched May 2018) were the first interplanetary CubeSats. They used: - Iris transponder (JPL, TRL 6 at selection -- first deep-space CubeSat radio) - COTS reaction wheels (Blue Canyon Technologies, TRL 9) - Custom deployable reflectarray antenna (JPL, TRL 5 at selection)

The custom antenna was the highest-risk item. It required extensive development testing and was the last component to reach TRL 6 for CDR. This delayed the schedule by 4 months but ultimately succeeded.

Lesson: If you must fly a low-TRL component, it will dominate your schedule and risk register.

[Source: "Mars Cube One (MarCO) -- Lessons Learned", A. Klesh et al., 33rd Annual Small Satellite Conference, 2019]

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3. Bill of Materials (BOM) Construction

The BOM is the definitive list of all hardware, software, and consumables required to build the satellite. It is the bridge between design and procurement.

BOM Structure

A properly structured BOM follows the WBS hierarchy:

```
BOM Level 0: Spacecraft (S/C-001)
  BOM Level 1: Subsystem (e.g., EPS-000)
    BOM Level 2: Assembly (e.g., EPS-SA-000 Solar Array Assembly)
      BOM Level 3: Component (e.g., EPS-SA-001 Solar Cell String)
        BOM Level 4: Part (e.g., EPS-SA-001-01 Azur 3G30C cell)
```

BOM Fields

Table 4.1.5 — Standard BOM field set with description and example value for each column.

Field	Description	Example
Item ID	Unique hierarchical identifier	EPS-BAT-001
Description	Component name and model	GomSpace NanoPower P31u
Manufacturer	Vendor name	GomSpace A/S
Part Number	Manufacturer part number	P31U-9-30
Quantity	Number required	1

Field	Description	Example
Unit Mass (g)	Per-item mass	94
Unit Power (W)	Peak / average power draw	0.5 / 0.2
Unit Cost (EUR)	Procurement cost	8,500
TRL	Technology readiness level	9
Heritage	Previous mission(s) flown	GOMX-3, GOMX-4
Lead Time	Procurement lead time	12 weeks
ECCN/USML	Export classification	EAR99
Status	Selected / Ordered / Received / Tested	Selected

SpaceCDF BOM Generation

In SpaceCDF, the BOM is built automatically from the Equipment Browser selections: 1. Each component selected in the Equipment Browser creates a BOM entry 2. Quantities are set using the quantity selector (e.g., x4 for reaction wheels) 3. The **Exports** tab generates a formatted BOM spreadsheet 4. The BOM includes both parametric estimates and actual COTS data for comparison

Key Equations

BOM Mass Total:

$$\mathrm{M_{BOM}} = \sum_{\mathrm{i}} \mathrm{m_i} \cdot \mathrm{q_i} + \mathrm{M_{harness}} + \mathrm{M_{fasteners}} + \mathrm{M_{margin}}$$

Where: - $\mathrm{m_i}$ = unit mass of component i - $\mathrm{q_i}$ = quantity of component i - $\mathrm{M_{harness}}$ = harness mass (typically 5–8 % of dry mass for CubeSats) - $\mathrm{M_{fasteners}}$ = mechanical fasteners, standoffs, thermal hardware (~3–5 %) - $\mathrm{M_{margin}}$ = system margin (typically 20 % at Phase A)

Worked Example

3U CubeSat BOM Summary:

Table 4.1.6 — Subsystem-level BOM rollup giving mass, power, and cost totals against 3U allocations with margin.

Subsystem	Components	Total Mass (g)	Total Power (W)	Total Cost (kEUR)
Structure	Rails, panels, fasteners	350	0	8.0
EPS	SA + battery + board	420	0.5	18.0
OBC	Flight computer + storage	80	1.2	10.0
AOCS	Star tracker + magnetorquers + RWs	550	4.5	45.0
TTC	S-band transceiver + patch antenna	180	8.0 (TX)	22.0

Subsystem	Components	Total Mass (g)	Total Power (W)	Total Cost (kEUR)
Thermal	MLI + heaters	60	2.0 (peak)	3.0
Payload	Multispectral imager	800	12.0	65.0
Harness	Cables, connectors	180	0	2.5
Total		**2620**	**28.2 (peak)**	**173.5**
Allocation (3U)		6000	40.0 (SA EOL)	250.0
Margin		3380 (56%)	11.8 (30%)	76.5 (31%)

1U Worked Example: UniSat-1

Complete Bill of Materials

UniSat-1's BOM is remarkably short -- only 8 line items plus harness. This simplicity is a major advantage for university teams with limited procurement experience.

UniSat-1 BOM (Phase B -- vendor quotes obtained):

Table 4.1.7 — Phase B UniSat-1 BOM with manufacturer, part number, quantity, mass, cost, TRL, ECCN, and lead time per item.

Item ID	Component	Manufacturer	Part Number	Qty	Unit Mass (g)	Unit Cost (kEUR)	TRL	ECCN	Lead (wks)
STR-001	1U CubeSat Structure	ISIS	ISIS-1U-STR	1	200	4.0	9	EAR99	8
EPS-001	NanoPower P31us (EPS + 10Wh battery)	GomSpace	P31US-10	1	200	12.0	9	EAR99	12
EPS-SA-001	Body-mounted GaAs solar cells	AzurSpace	3G30C	5	10	1.5	9	EAR99	10
OBC-001	Custom flight computer (Cortex-M)	In-house	UNISAT-OBC-01	1	30	3.0	5	EAR99	--
COM-001	UHF Transceiver	GomSpace	NanoCom AX100	1	55	8.0	8	EAR99	12
COM-ANT-001	UHF Deployable Antenna	Endurosat	UHF-ANT-S	1	25	2.5	8	EAR99	8
PL-001		In-house	UNISAT-MAG-01	1	50	5.0	4	EAR99	--

Item ID	Component	Manufacturer	Part Number	Qty	Unit Mass (g)	Unit Cost (kEUR)	TRL	ECCN	Lead (wks)
	MEMS Magnetometer Board								
AOCS-001	Passive magnetic kit (magnet + rods)	NewSpace	PMAG-1U	1	30	1.0	9	EAR99	6
HAR-001	Internal harness	Custom	--	1	50	1.0	N/A	EAR99	--
	TOTALS				**690 g**	**~44 kEUR**			

Cost summary (total mission, hardware + services):

Table 4.1.8 — UniSat-1 mission cost summary by WBS element including hardware, software, I&T, ground, launch, and management.

WBS Element	Cost (kEUR)	Notes
Hardware (BOM)	44	All COTS except OBC and payload
OBC software	5	Student labour (costed at stipend rate)
Payload calibration	3	University magnetometer lab
I&T	8	Assembly + vibration test (university facility)
Ground station	5	SatNOGS (free) + dedicated Yagi antenna purchase
Launch (ISS deploy)	15	NanoRacks 1U deployment fee
PM/SE/QA	5	Faculty supervision
TOTAL	**~85 kEUR**	

Comparison to 3U EO mission:

Table 4.1.9 — Side-by-side comparison of UniSat-1 against a 3U EO CubeSat on line items, cost, schedule, and team size.

Metric	UniSat-1 (1U)	3U EO CubeSat
BOM line items	9	~15--20
Hardware cost	~44 kEUR	~290 kEUR
Total mission cost	~85 kEUR	~490 kEUR
Development time	6--12 months	18--24 months
Team size	3--5 people	8--15 people

Export control: All UniSat-1 components are classified EAR99 (no licence required). There are no ITAR-controlled items because the mission uses no star trackers, no radiation-hardened processors, and no propulsion with ITAR-restricted technology. This is a significant advantage for international university collaborations.

Make/Buy/Reuse decisions:

Table 4.1.10 — Per-component make/buy/reuse decision and rationale for the UniSat-1 subsystem set.

Component	Decision	Rationale
Structure	Buy (COTS)	ISIS 1U frame is flight-proven, TRL 9, low cost
EPS	Buy (COTS)	GomSpace P31us is the de facto standard, TRL 9
OBC	Make (custom)	Minimal board using university lab; lower cost than COTS OBC for this simple application
Comms	Buy (COTS)	GomSpace AX100, TRL 8, well-documented
Payload	Make (custom)	Novel MEMS sensor -- this IS the technology demonstration
Passive AOCS	Buy (COTS)	Standard magnetic stabilisation kit

4. Export Control and Procurement

Export control is one of the most commonly overlooked aspects of CubeSat missions, especially for international teams. Violations carry severe penalties (criminal prosecution, programme cancellation).

[Source: ITAR -- 22 CFR Parts 120-130; EAR -- 15 CFR Parts 730-774; Canadian Controlled Goods Program -- Defence Production Act] [URL: https://www.pmddtc.state.gov/ddtc_public (US ITAR); https://www.bis.doc.gov (US EAR)]

Export Control Regimes

Table 4.1.11 — Major export control regimes with governing law, scope, and key CubeSat-relevant concerns.

Regime	Governing Law	Applies To	Key Concern for CubeSats
ITAR	US Arms Export Control Act	Defence articles (USML Categories IV, XI, XV)	Star trackers, rad-hard processors, some GPS receivers
EAR	Export Administration Act	Dual-use items (CCL)	Most COTS space components (ECCN 9A515)
Canadian CGP	Defence Production Act	Controlled goods in Canada	Handling US-origin ITAR items in Canadian facilities
Wassenaar	Wassenaar Arrangement	Multilateral export controls	Encryption, high-accuracy GNSS, imaging sensors

Component Classification Decision

```
Is the component on the US Munitions List (USML)?
  -> Yes: ITAR-controlled. Need DSP-5 license for export.
     Categories: IV (launch vehicles), XI (military electronics),
                 XV (spacecraft systems)
  -> No: Check Commerce Control List (CCL)
     Is it classified under ECCN 9A515 (spacecraft)?
       -> Yes: EAR-controlled. May need BIS license.
       -> No: Likely EAR99 (no license required for most destinations)
```

Common ITAR-Controlled CubeSat Components

Table 4.1.12 — Commonly ITAR-controlled CubeSat component types with the reason for control and EAR99 alternatives.

Component Type	Why Controlled	Alternative
Radiation-hardened processors	Military-grade rad tolerance	COTS processors with software mitigation
High-accuracy star trackers (< 1 arcsec)	Missile guidance applicability	Lower-accuracy models (> 5 arcsec)
Certain GPS/GNSS receivers	Above COCOM limits (>60,000 ft, >1000 kt)	Space-rated receivers with COCOM compliance
Propulsion systems (some)	Missile technology	Cold-gas or water-based systems
Encryption modules	Signals intelligence	Open-source encryption (may still need EAR review)

Procurement Workflow

```
1. Requirements -> Derive component specification
2. Market survey -> Identify candidate COTS products
3. Export classification -> Request ECCN from vendor
4. Trade study -> Score and rank candidates
5. Request for Quote (RFQ) -> Obtain pricing and lead times
6. Purchase Order (PO) -> Commit to procurement
7. Incoming inspection -> Verify against PO and datasheet
8. Integration -> Install and functionally test
```

Real Mission Example: Export Control Impact

The BRITE-Constellation (Austria/Canada/Poland) was a series of nanosatellites for stellar photometry. The Austrian BRITE satellites (UniBRITE and BRITE-Austria, built by UTIAS/SFL in Toronto) required careful export control management: - US-origin ITAR components required Technical Assistance Agreements (TAAs) - Controlled Goods registration required for the Canadian team - Export permits needed for shipping between Canada, Austria, and the US launch site - Total regulatory compliance effort: ~6 person-months and 12+ months lead time

Lesson: Start export classification immediately when components are identified. A single ITAR component can add 6-12 months to the schedule.

[Source: Sarda, K. et al., "BRITE-Constellation Mission and Spacecraft", AIAA/USU Small Satellite Conference, 2014]

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5. Interface Compatibility Verification

As components are selected, every interface must be verified for compatibility. The three interface categories are RF, electrical, and mechanical.

RF Chain Compatibility

The RF chain (transponder, cable, antenna) must be frequency-matched. This is the most common equipment incompatibility for CubeSat newcomers.

Table 4.1.13 — RF chain compatibility rules covering band, impedance, connector, and polarisation matching.

Rule	Correct Example	Incorrect Example
Band match	S-band transponder + S-band patch antenna	S-band transponder + X-band horn
Impedance match	50 ohm transponder + 50 ohm cable + 50 ohm antenna	Mixed impedances cause reflections
Connector match	SMA on transponder + SMA-SMA cable + SMA on antenna	SMA to N-type needs adapter (loss)
Polarisation	RHCP antenna + RHCP ground station	RHCP to LHCP causes > 20 dB cross-pol loss

Impedance Mismatch Loss:

$$\text{Return Loss (dB)} = -20\log_{10}|\Gamma|, \qquad \Gamma = \frac{Z_{\text{load}} - Z_{\text{source}}}{Z_{\text{load}} + Z_{\text{source}}}$$

For a 50 Ω source into a 75 Ω load:

$$\Gamma = \frac{75-50}{75+50} = 0.2, \quad \text{Return Loss} = -20\log_{10}(0.2) = 14 \text{ dB},$$

$$L_{\text{mismatch}} = -10\log_{10}(1 - |\Gamma|^2) = -10\log_{10}(1 - 0.04) = 0.18 \text{ dB}.$$

SpaceCDF checks RF compatibility automatically: if you select a transponder in one band and an antenna in another, a warning dialog appears.

Electrical Interface Verification

Table 4.1.14 — Electrical interface parameters with typical CubeSat values and the verification check required.

Parameter	Typical CubeSat	What to Verify
Bus voltage	3.3V, 5V, or unregulated (6-8.4V Li-ion)	Component input voltage range covers bus voltage
Peak current	Per switched line limit (typically 1-3A)	Component inrush current does not exceed limit
Data protocol	I2C, SPI, UART, CAN, RS-422	All devices on same bus use compatible protocol
Connector	PC/104, Hirose, Harwin	Physical connector type matches or adapter planned

Mechanical Interface Verification

Parameter	CDS 3U Requirement	Verification
Dimensions	100.0 +/- 0.1 mm x 100.0 +/- 0.1 mm x 340.5 +/- 0.5 mm	Caliper measurement
Rail profile	8.5 x 8.5 mm +/- 0.1 mm	Profile gauge
Component stack height	<= 83 mm internal width per U	3D model check
CG location	Within 2 cm of geometric centre	Mass properties measurement

---

6. Equipment Selection Exercise

Instructions

Part A: Equipment Selection

1. Open the **Equipment Browser** in SpaceCDF
2. For each **required** category (blue dot), select at least one component: - Start with EPS (batteries + solar panels + EPS board) - Then OBC, AOCS sensors/actuators, TTC, structure - Use quantity selectors (e.g., x4 for reaction wheels, x3 for magnetorquers)
3. Watch the **live budget bar** as you select -- keep mass under allocation
4. When selecting TTC: verify the spectrum band matches your earlier selection
5. After all selections, review the **Budget Breakdown** on the Dashboard

Part B: BOM Review and Export Check

1. Navigate to the **Exports** tab
2. Generate the **BOM** -- review all entries
3. For each component, check: Is the ECCN listed? Is it EAR99, ECCN 9A515, or ITAR?
4. Flag any components that may require export licences
5. Compute: BOM total mass vs parametric estimate vs launcher allocation

Component Trade Study

For at least one subsystem, select 2--3 candidate components and run a formal tabular trade:

1. Navigate to the **Trade Studies** tab
2. Load or create a "Component Selection Trade" study

3. Define criteria: mass, power, cost, TRL, heritage, performance
4. Score each candidate (1--5 scale)
5. Apply weights and compute weighted scores
6. Document the winner and rationale

Real Mission Example: Iridium NEXT Equipment Selection

Iridium NEXT (Thales Alenia Space, 2017--2019) serves as a large-scale example of rigorous equipment selection. For the phased-array antenna:

Table 4.1.15 — Iridium NEXT phased-array weighted trade study between Thales and Raytheon candidate antennas.

Criterion	Weight	Candidate A (Thales)	Candidate B (Raytheon)
Performance	0.30	4.5	4.0
Mass	0.20	3.5	4.0
Cost	0.20	3.0	4.5
TRL	0.15	5.0	4.0
Schedule	0.15	4.0	3.5
Weighted		**4.00**	**4.025**

The selection was ultimately Candidate A (Thales) due to contractual considerations beyond the numerical trade -- illustrating that trade studies inform but do not dictate decisions.

Worksheet 4.1 Tasks

1. Complete the full BOM table (component, manufacturer, part number, mass, power, cost, TRL, ECCN)
2. Document the make/buy/reuse decision for each subsystem with rationale
3. Compute total BOM mass vs parametric estimate vs allocation -- state margin
4. Identify any export-controlled components and note the required licensing action
5. Describe one interface incompatibility found during selection and how it was resolved

---

7. SpaceCDF Budget Closure Check

Instructions

After equipment selection, perform a final budget health check:

1. **Dashboard KPIs:** Record all margins - Mass margin (%) -- green/amber/red? - Power margin per mode (W) - Link margin (dB) - Cost vs ceiling (MEUR) - Pointing accuracy vs requirement (deg)
2. **Budget Comparison:** Compare parametric estimates to equipment-based totals
3. **Identify any negative margins** -- these must be resolved before proceeding to integration (Week 3)

If a Budget Does Not Close

Table 4.1.16 — Common fixes when a mass, power, link, cost, or pointing budget fails to close.

Budget	Common Fix	Impact	Typical Mass/Power/Cost Trade
Mass (negative)	Remove propulsion; select lighter components; reduce redundancy; move to larger form factor	Risk / performance trade	Removing propulsion saves 0.5--1.5 kg
Power (negative)	Add deployable SA; reduce payload duty cycle; select lower-power AOCS; schedule operations to avoid simultaneous loads	Cost / schedule trade	Deployable panel adds ~15 W but costs 0.3 kg + 25 kEUR
Link (negative)	Increase TX power; use higher-gain antenna; reduce data rate; upgrade coding; upgrade ground station	Mass / power trade	3 dB gain from coding is "free"; 3 dB from bigger antenna costs mass
Cost (over ceiling)	Use COTS instead of rad-hard; remove propulsion; reduce ground segment; use SatNOGS instead of commercial ground	Risk / capability trade	COTS vs rad-hard saves 10--100x on processor cost
Pointing (insufficient)	Upgrade star tracker; improve alignment calibration; add vibration isolation for RW; reduce thermal gradients	Cost / complexity trade	Alignment improvement is usually cheapest

1U Worked Example: UniSat-1

CDS Compliance for 1U Form Factor

The CubeSat Design Specification (CDS Rev 14) defines the 1U envelope:

Parameter	1U Specification	UniSat-1 Design	Compliance
Dimensions	100.0 x 100.0 x 113.5 mm	100.0 x 100.0 x 113.5 mm (ISIS 1U frame)	**Pass**
Maximum mass	2.0 kg (CDS Rev 14, ISIPOD)	1.0 kg target (50% margin to 2 kg limit)	**Pass**
Rail material	Hard anodised Al 7075-T6	Standard (part of ISIS frame, 7075-T6, Type III anodise)	**Pass**
Rail cross-section	8.5 x 8.5 mm minimum	Standard (ISIS: 8.5 x 8.5 mm)	**Pass**
Deployment switches	Min 1 per accessible face	2 switches (ISIS standard, on +X/-X rail faces)	**Pass**
RBF pin	Required	Included (ISIS standard, on -Z face)	**Pass**
CG offset	<= 2 cm from geometric centre	< 1 cm (symmetric PCB stack layout, battery centred)	**Pass**
Protrusions	None beyond rail envelope (stowed)	UHF monopole antenna stowed along rail (spring-loaded, within envelope)	**Pass**
Fundamental frequency	> 40 Hz first mode	~600 Hz (1U Al structure is extremely stiff)	**Pass**

Note on CDS mass limit: The CDS Rev 14 specifies 2.0 kg as the 1U deployer limit for the ISIPOD. However, many deployer providers (e.g., NanoRacks, Exolaunch) specify 1.33 kg for 1U. Always check the specific deployer ICD. UniSat-1 targets 1.0 kg, well within either limit.

No propulsion: At 400 km altitude, atmospheric drag provides natural deorbit. The ballistic coefficient for a 1U is:

$$\mathrm{BC} = \frac{\mathrm{m}}{\mathrm{C_D} \times \mathrm{A}} = \frac{1.0}{2.2 \times 0.01} = 45.5 \text{ kg/m}^2$$

This gives an orbital lifetime of approximately 8--14 months depending on solar activity (F10.7 index). At solar maximum (F10.7 > 200), the denser atmosphere deorbits the satellite in ~6 months. At solar minimum (F10.7 ~ 70), lifetime extends to ~18 months. Both are within the FCC 5-year rule and IADC 25-year guideline without any propulsion system.

OBC selection rationale: UniSat-1 uses a custom board based on the TI MSP430 (safe mode / housekeeping) + STM32F4 Cortex-M4 (main OBC). The MSP430 runs bare-metal firmware handling the watchdog, power monitoring, and safe-mode recovery. The STM32F4 runs FreeRTOS handling TM/TC, magnetometer data acquisition, and scheduling. Dual-processor architecture provides redundancy: if the STM32F4 fails (SEU, latchup), the MSP430 can maintain safe-mode operations and respond to ground commands.

Data storage: With ~1.13 MB/day of magnetometer data and a 4800 bps useful downlink (9600 bps channel), onboard storage is not a bottleneck. A 4 MB NOR flash is more than adequate (stores ~3.5 days of data as buffer). No NAND flash, no SD card needed.

Complete 1U Equipment List:

#	Category	Component	Mass (g)	Power (W)	Cost (kEUR)	Qty	Interface
1	Structure	ISIS 1U CubeSat structure (Al 7075-T6)	200	--	4.0	1	Mechanical
2	EPS	GomSpace NanoPower P31us (EPS board + 2S Li-ion battery, 10 Wh)	200	0.3 (quiescent)	12.0	1	I^2C
3	Solar cells	Body-mounted triple-junction GaAs cells (5 faces)	50	-- (generates power)	7.5	5	Direct to EPS
4	OBC	Custom MSP430 + STM32F4 board (with 4 MB NOR flash)	30	0.3	3.0	1	I^2C, SPI, UART
5	Comms	UHF transceiver (NanoCom AX100, 0.5 W TX, 9600 bps)	55	0.5 (TX) / 0.1 (RX)	8.0	1	SPI
6	Antenna	UHF monopole (deployable, $\lambda/4 = 17$ cm nitinol)	25	--	2.5	1	RF coax
7	Payload		50	0.2	5.0	1	SPI

#	Category	Component	Mass (g)	Power (W)	Cost (kEUR)	Qty	Interface
		MEMS magnetometer (custom PCB, PNI RM3100 sensor)					
8	AOCS (passive)	Permanent magnet (AlNiCo, 0.5 A m^2) + 2 hysteresis rods (HyMu-80)	30	0	1.0	1	None (passive)
9	Harness	Internal cables, connectors, PC/104 stack header	50	--	1.0	1	Various
	TOTAL		**690**	**~1.3 (peak TX)**	**~44**		

Mass budget:

Level	Mass (g)
CBE (Current Best Estimate)	690
+ 20% equipment margin	828
+ 20% system margin	994
MEV (Maximum Expected Value)	**994**
Deployer limit (ISIPOD)	2000
Margin to limit	**1006 g (50%)**

Power budget (worst case: TX mode in sunlight):

Load	Power (W)
OBC (STM32F4 + MSP430)	0.3
EPS quiescent	0.3
UHF TX	0.5
Magnetometer	0.2
Total peak	**1.3**
SA available (orbit avg, EOL)	2.3
Margin	**1.0 W (43%)**

Key insight: The entire UniSat-1 BOM is 6 COTS components plus 2 custom boards (OBC and magnetometer PCB). Total hardware cost is ~44 kEUR -- an order of magnitude less than a typical 3U mission. With labour, I&T, launch, and operations, the total mission cost is 80--150 kEUR. This demonstrates that a useful space mission can be built for less than the cost of a mid-range car.

Worked Example: Complete 6U EO CubeSat Equipment List (EOSAT-1)

This is the flight equipment list of the reference spacecraft used throughout the operations chapters: EOSAT-1, a 6U wide-field multispectral imager in a 513 km Sun-synchronous orbit (97.39 deg, LTAN 10:00). Every line is a real, current-production part class rather than a parametric placeholder, which is what makes the totals worth arguing about.

Category	Component	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Interface
Structure	6U frame, rails and internal brackets (Al 7075-T6, hard anodised)	1.80	--	8	1	Mechanical
Solar cells (body)	Body-mounted TJ GaAs, 0.198 m^2 over five faces (-Z carries the aperture and radiator)	0.35	--	6	1 set	Direct to EPS
Solar wings	Endurosat-class 6U double-deployable, 0.120 m^2 of cells per wing, single-axis SADA	1.07	--	10	2	Direct to EPS
EPS Board	6U EPS/PDU (MPPT, nine switchable/non-switchable lines, indices 0--8)	0.25	3.0 (EPS housekeeping share of the 6 W obc line)	8	1	I^2C
Battery	GomSpace NanoPower BPX, 8S-1P, 100 Wh, 3.5 Ah, 28.8 V nominal (24.0--33.6 V), Li-ion 18650	0.50	--	5	1	I^2C (telemetry)
OBC	6U flight computer, 20 GB payload store	0.10	3.0 (OBDH share of the obc line)	12	1	I^2C, SPI, UART
Reaction Wheel	Sinclair RW-0.03 class, 0.030 N m s at 5500 rpm, 2 mN m peak torque; pyramid, four-of-four	0.185	0.3 (avg, per wheel)	5	4	SPI
Magnetorquer		0.05	0.1	2	3	I^2C

Category	Component	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Interface
	NewSpace NCTR-M016 class rod, 1.6 A m^2 dipole					
Star Tracker	Blue Canyon NST class, 6 arcsec cross-boresight NEA (ST1 / ST2, cold-redundant)	0.35	1.5	25	2	SPI/UART
Gyro	Single-axis tactical-grade rate channel (Epson M-G370 / Sensonor STIM class), one per body axis, **no spare unit**	0.02	0.3	1.3	3	SPI
Sun Sensor	Coarse sun-sensor array (attitude acquisition and safe mode)	0.06	0.01	4	1 set	Analog/I^2C
Magnetometer	Three-axis magnetometer (MAG A / MAG B, redundant pair; B-dot detumble reference)	0.03	0.05	1	2	I^2C
GNSS	GPS receiver + patch antenna (orbit determination, time reference)	0.12	1.0	8	1	UART
Payload	Wide-field pushbroom (power, mass and envelope from the Simera Sense TriScape100; its 580 mm optics are not EOSAT-1's): 8192 px x 5.5 um, f = 84.5 mm, 29.856 deg full FOV -> 273 km swath, 33.4 m GSD at 513 km; 244 Hz line rate, 96 Mbps raw readout	1.45	5.5 (imaging) / 2.4 (idle)	60	1	LVDS/SPI

Category	Component	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Interface
	(sized at the earlier 450 km figure)					
FPA cooler	Miniature Stirling-class cryocooler on the focal-plane bracket (-15 C setpoint, ~23 % of Carnot)	0.50	15.0	15	1	Own EPS line (`fpa_cooler`)
Transmitter	Endurosat-class S-band transceiver (prime chain): 2 W RF max, 5 W DC at the LOW tier, 11 W DC at the HIGH tier, 1.2 W idle Rx	0.25	5.0 (LOW) / 11.0 (HIGH)	15	1	SPI
Antenna	S-band patch (body-fixed, RHCP)	0.06	--	3	1	RF coax
Beacon	**Separate** UHF whip beacon transmitter (1 W RF; carries HK SID 11 only, and survives the S-band PA being off)	0.16	1.0	6	1	UART / RF coax
Propulsion	1U green monopropellant blowdown module (LMP-103S/ADN class), single thruster on -X, 0.33 -> 0.14 N, Isp 220 -> 196 s	1.15 (dry)	12.0 (catbed preheat) / 2.0 (armed)	30	1	UART + latch valves
Thermal	MLI, heaters (battery 8 W, OBC 4 W), radiator, thermal straps	0.40	12.0 (both heaters, thermostat-controlled)	3	1 set	Analog
Harness	Cables, connectors, fasteners, stack hardware	1.00	--	5	1	Various
TOTAL (dry)		**12.00**	see note	**~290**		
Propellant	Green monoprop, 0.400 kg -> 66.9 m/s total ΔV	0.40	--	--	--	--
		12.40				

Category	Component	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Interface
TOTAL (wet, BOL)						

Power note. The column is per-unit demand, not a sum: EOSAT-1 never runs every line at once. The nine EPS lines (`0 obc`, `1 ttc_rx`, `2 ttc_tx`, `3 payload`, `4 fpa_cooler`, `5 htr_bat`, `6 htr_obc`, `7 aocs_wheels`, `8 prop`) are switched against the orbit-average budget: at end of life and worst beta the array delivers ~45 W orbit-average against a ~23 W nominal load (payload in STANDBY, cooler off), a margin of about +22 W. Add continuous imaging, a continuously running cooler and HIGH-tier dumps on every pass and the budget goes negative -- which is why imaging is duty-cycled (three ~90 s captures per orbit with the cooler stabilised around them) rather than run continuously.

Array note. Cell area is 0.198 m^2 body-mounted plus 2 x 0.120 m^2 on the wings, 0.438 m^2 deployed. The plane-of-array constant after the full loss stack (cell efficiency, packing and cover glass, hot-array temperature coefficient, MPPT, harness) is 292 W/m^2 BOL and **267.9 W/m^2 at three-year EOL** -- which is exactly the 268 W/m^2 a published 6U deployable delivers in LEO. Peak whole-spacecraft generation is 88.6 W, not the 120 W-plus that an invented array area would suggest.

Mass budget: - Dry CBE: 12.00 kg (including 1.15 kg of propulsion dry mass) - Propellant: 0.400 kg - **Wet mass at BOL: 12.40 kg** - CDS 6U reference limit: 12.0 kg -- EOSAT-1 sits *at* the top of the 6U class, so the dispenser ICD governs: check the qualified mass of the specific 6U dispenser before assuming the CDS baseline applies.

Teaching point. This is a Phase C/D flight list, not a Phase A estimate, so the 20 % equipment and system margins used earlier in this chapter have already been consumed by real part masses. A design that still shows 40 % of mass margin at this stage has not yet been costed against real hardware.

Cited References

#	Reference	URL
1	NASA Technology Readiness Levels	https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/
2	ECSS-Q-ST-20C Quality Assurance	https://ecss.nl/standard/ecss-q-st-20c-rev-2-quality-assurance-1-march-2023/
3	US DDTC (ITAR)	https://www.pmddtc.state.gov/
4	US BIS (EAR)	https://www.bis.doc.gov/
5	CubeSat Design Specification Rev 14.1	https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf
6	SMAD (Cost Modelling)	Wertz, Everett, Puschell (eds.), Space Mission Engineering, Microcosm 2011
7	MarCO Lessons Learned (Klesh et al.)	SSC19-WKII-07, 33rd Small Sat Conference, 2019

Session Summary

Table 4.1.17 — Session summary: key takeaways for equipment selection and the bill of materials.

Topic	Key Takeaway
Make/Buy/Reuse	Buy COTS first; Reuse heritage second; Make custom only when necessary
TRL	Minimum TRL 6 for mission-critical components; TRL 7+ preferred for CubeSats
BOM	Hierarchical list: Subsystem -> Assembly -> Component -> Part, with full traceability
Export control	Classify every component (EAR99 / ECCN / ITAR); start early -- delays programme
RF compatibility	Band, impedance, connector, polarisation must all match across RF chain
Electrical	Bus voltage, data protocol, connector type verified per component
Budget tracking	Live totals during selection; stop if allocation exceeded

Chapter 4.2 — Verification & Validation Matrix and Compliance

> **Expected reading before this session.** ECSS-E-ST-10-02C — Verification (≈ 60 min); NASA SEH §6.4.

Prerequisites: Chapter 4.1 (equipment selected, BOM constructed) **References:** ECSS-E-ST-10-02C Rev.1 (Verification), ECSS-E-ST-10-03C (Testing), NASA SEH Rev 2 sections 5.3-5.4, MIL-STD-1540E, MIL-STD-461G

Learning Objectives

By the end of this session, participants will be able to: 1. Distinguish between verification and validation using the ECSS/NASA definitions 2. Assign appropriate verification methods (ATIR(D)) to each requirement using a decision logic 3. Define environmental test levels derived from launch vehicle ICD specifications 4. Construct a compliance verification matrix mapping every requirement to its method, level, and phase 5. Identify when waivers are appropriate and document the waiver rationale

1. Verification vs Validation: Precise Definitions

[Source: ECSS-E-ST-10-02C Rev.1 section 4; NASA SEH Rev 2 section 5.3 (Process 7: Product Verification) and section 5.4 (Process 8: Product Validation)]

Table 4.2.1 — Verification versus validation contrasted by question, standard, performer, timing, methods, and output.

	Verification	Validation
Question	"Did we build the system right?"	"Did we build the right system?"
Checks against	Requirements (shall statements)	Stakeholder needs and operational expectations
Standard	ECSS-E-ST-10-02C	ECSS-E-ST-10-06C
Performed by	Engineering team	Users, operators, stakeholders
When	Throughout development (Phases B-D)	End of development + early operations (Phase D-E)
Methods	ATIR(D) (Analysis, Test, Inspection, Review of design, Demonstration)	Operational demonstration, user acceptance testing
Output	Verification Control Document (VCD)	Validation report

The Critical Distinction

A system can be **verified** (every requirement marked "pass") but still **fail validation** if the requirements were written incorrectly. This happens when: - Requirements were derived from misunderstood

stakeholder needs - The operational environment differs from assumptions - Edge cases were not captured in requirements

Example

- **Requirement:** "The system shall achieve GSD <= 10 m from 500 km altitude"
- **Verification (Test):** Calibration target imagery at 500 km -> measured GSD = 9.2 m -> **verified**
- **Validation:** Users attempt crop assessment with 9.2 m imagery -> "We can see field boundaries but cannot distinguish crop types; we needed GSD <= 5 m" -> **not validated**

The requirement was met, but the stakeholder need was not satisfied because the requirement was insufficiently stringent.

---

2. ATIR(D) Verification Methods in Detail

ECSS-E-ST-10-02C Rev.1 §5.2.2.1 defines four verification methods: test (including demonstration), analysis (including similarity), review of design and inspection. NASA practice lists demonstration separately as a fifth method. This book uses the full set, abbreviated ATIR(D) as in Chapter 2.1; the order in which the methods are described below does not matter.

[Source: ECSS-E-ST-10-02C Rev.1 §5.2.2; freely available from https://ecss.nl]

Inspection (I)

Definition: Visual or physical examination of the product, without operation or stimulation, to verify conformance to requirements.

When to use: - Physical characteristics (dimensions, mass, surface finish, labelling) - Manufacturing workmanship (solder joints, cable routing, MLI installation) - Markings, identification, and safety labels - Mechanical fit (deployer rail compliance)

Evidence produced: Inspection report with photographs, measurements, pass/fail checklist.

Examples for CubeSats: - Mass measurement on calibrated scale -> verifies "M_dry <= 4.0 kg" - Caliper measurement of rail profile -> verifies CDS dimensional compliance - Visual inspection of antenna stowage -> verifies no protrusions beyond deployer envelope - Workmanship inspection of PCB solder joints -> verifies IPC-A-610 Class 3

Analysis (A)

Definition: Mathematical models, simulations, statistical analysis, or heritage comparison demonstrate compliance with acceptable confidence.

When to use: - Physical testing is impossible or impractical (orbital lifetime, radiation total ionising dose) - Early in development (Phase B) before hardware exists - To demonstrate margin (link budget, thermal model, structural FEA) - Statistical parameters (reliability prediction, debris casualty risk) - Requirements involving orbital mechanics or mission-level performance

Evidence produced: Analysis report with model description, assumptions, inputs, results, uncertainty assessment, and sensitivity analysis.

Examples for CubeSats: - Orbital mechanics analysis -> verifies revisit time requirement - Thermal model (finite difference/finite element) -> verifies temperature within limits - Link budget calculation -> verifies margin >= 3 dB at worst case - Structural FEA -> verifies positive Margin of Safety under launch loads

> **Margin of Safety (structural):**
>
> MoS = (Allowable_stress / (Factor_of_Safety x Applied_stress)) - 1
>
> MoS must be >= 0 for all load cases. Factor_of_Safety: 1.25 (yield), 1.5 (ultimate) per ECSS-E-ST-32-10C.

Review of design (R)

Definition: Examination of design documentation -- drawings, schematics, software design, analyses already performed, process records -- to confirm that the design as documented satisfies the requirement. In ECSS-E-ST-10-02C this is a formal method in its own right, distinct from analysis: no new model is built; the evidence is the documented design.

When to use: - Requirements on the existence or content of a document, process or plan ("a verification plan shall be produced") - Design features that are evident from the drawings (number of deployment switches, presence of an RBF pin, connector pin-out) - Requirements already covered by an approved heritage design

Evidence produced: Review-of-design record citing the documents examined, signed by the reviewer; typically closed at a design review.

Examples for CubeSats: - Review of the stack schematic -> verifies "two independent deployment switches shall inhibit all power" - Review of the software architecture document -> verifies "the watchdog shall be implemented in hardware, independent of the main processor"

Demonstration (D)

Definition: The system is operated under controlled conditions to show that it performs as required, without requiring precise measurement of performance parameters. ECSS-E-ST-10-02C records a demonstration as a test; NASA lists it as a separate method.

When to use: - Functional requirements ("the system shall deploy the antenna within 30 minutes of separation") - Operational procedures ("the system shall respond to a mode change command within 5 seconds") - Software functional requirements ("FDIR shall autonomously switch to safe mode on loss of attitude")

Evidence produced: Demonstration report with procedure, observations, photographs/video, pass/fail.

Examples for CubeSats: - Antenna deployment test -> demonstrates deployment within time limit - Safe mode entry test -> demonstrates autonomous FDIR response - Ground station commanding test -> demonstrates full command chain

Test (T)

Definition: Physical hardware subjected to controlled conditions; measured performance compared quantitatively to requirement thresholds.

When to use: - Physical properties must be proven with quantitative measurement - Environmental survivability must be demonstrated (vibration, thermal-vacuum, EMC) - End-to-end functional chains must be proven under representative conditions - Performance must be measured, not just observed

Test types by purpose:

Table 4.2.2 — Four test types (development, qualification, acceptance, proto-flight) with purpose, level, phase, and load conditions.

Test Type	Purpose	Level	Phase	Loads/Conditions
Development	Explore design space, find problems early	Unit/BB	B	As needed
Qualification	Prove design withstands worst-case + margin	Unit/ System	C	Qual levels (MPE + margin)
Acceptance	Prove flight hardware is defect-free	System	D	Acceptance levels (MPE)
Proto-flight	Combined qual + acceptance (single unit)	System	C/D	Qual levels, acceptance duration

Proto-Flight Approach

For CubeSats, the **proto-flight** approach is standard because building a separate qualification unit is prohibitively expensive. The flight unit is tested to **qualification levels** but for **acceptance duration**.

[Source: ECSS-E-ST-10-03C §4.5.4; MIL-STD-1540E section 6.2.4]

Rationale: Qualification duration is longer and more stressful -- this is acceptable for a dedicated qualification model that will not fly. For proto-flight, the shorter duration reduces fatigue life consumption while still screening for workmanship defects.

Table 4.2.3 — Vibration and TVAC parameters compared between qualification, acceptance, and proto-flight test campaigns.

Parameter	Qualification	Acceptance	Proto-flight
Vibration level	MPE + 3 dB	MPE	MPE + 3 dB
Vibration duration	2 min/axis	1 min/axis	1 min/axis
TVAC temperature range	Predicted +/- 15 C	Predicted +/- 10 C	Predicted +/- 10 C
TVAC cycles	8	4	4
Sine sweep rate	2 oct/min	4 oct/min	4 oct/min

The proto-flight levels follow the standards. GSFC-STD-7000B §2.6 tests a proto-flight unit at the maximum and minimum predicted temperatures +/- 10 C, with at least 4 cycles; ECSS-E-ST-10-03C reaches the same 10 C with its 5 C acceptance and 5 C qualification margins on the predicted range. ECSS-E-ST-10-03C sweeps sine at 2 oct/min for qualification (Table 5-2) and 4 oct/min for acceptance (Table 5-4), and proto-flight takes the acceptance duration, hence 4 oct/min. The qualification and acceptance temperature columns are those of Table 3.1.11 (Chapter 3.1).

Standard environmental test sequence: inspection through functional, vibration, thermal-vacuum, EMC, and final functional verification before shipment.

Environmental test sequence for spacecraft verification.

3. Environmental Test Programme

[Source: ECSS-E-ST-10-03C (Testing); Launch vehicle Payload User's Guides; MIL-STD-1540E]

Standard Environmental Test Sequence

The test sequence is designed so that each test stresses the hardware in a specific way, and functional tests between environmental tests detect any damage. The levels below are representative teaching values: the binding levels for a given flight are set by the deployer or launch-vehicle user's guide.

Sine Vibration Test Levels

[Source: representative CubeSat rideshare envelope; the deployer or launch-vehicle user's guide sets the binding levels]

Sine vibration simulates the low-frequency launch vehicle structural modes (typically 5-100 Hz). These are quasi-static loads.

Table 4.2.4 — Sine vibration profile giving frequency range, acceleration level, sweep rate, and axes.

Frequency Range (Hz)	Level	Notes
5 - 100	1.25 g (0-peak)	Acceleration-controlled over the whole band; no displacement band
Sweep rate	4 octaves/minute (proto-flight)	2 oct/min is the qualification rate
Axes	3 (X, Y, Z)	Sequential, one axis at a time

> **Why there is no displacement band -- the crossover frequency:**
>
> For a sinusoid, acceleration and displacement amplitude are related by $\mathrm{a} = (2\pi \mathrm{f})^2 \cdot \mathrm{d}$, so the frequency at which a given acceleration corresponds to a given displacement is

$$\mathrm{f_c} = \frac{1}{2\pi}\sqrt{\frac{\mathrm{a}}{\mathrm{d}}}.$$

> For $\mathrm{a} = 1.25\ \mathrm{g} = 12.26$ m/s^2 and a typical shaker stroke limit $\mathrm{d} = 12.5$ mm $= 0.0125$ m:

$$\mathrm{f_c} = \frac{1}{2\pi}\sqrt{\frac{12.26}{0.0125}} = \frac{31.3}{6.283} \approx 4.98\ \mathrm{Hz} \approx 5\ \mathrm{Hz}.$$

> *Below about 5 Hz, holding 1.25 g would need more than 12.5 mm of displacement, so a profile reaching lower frequencies needs a displacement-controlled band. This profile starts at 5 Hz, so it is acceleration-controlled throughout.*

Random Vibration Test Levels

Random vibration simulates the broadband acoustic and mechanical environment during launch. Levels are specified as Power Spectral Density (PSD) in g^2/Hz.

[Source: GSFC-STD-7000B (GEVS) Table 2.4-4, component minimum workmanship levels; Table 2.4-1 for the proto-flight level and duration]

Table 4.2.5 — Random vibration PSD profile in g^2/Hz: the GEVS minimum workmanship level, with the proto-flight level.

Frequency (Hz)	PSD Level (g^2/Hz)	Notes
20	0.01	Start of profile
20 - 80	+3 dB/octave ramp	Rising
80 - 500	0.04 (flat)	Maximum level plateau
500 - 2000	-3 dB/octave ramp	Falling
2000	0.01	End of profile
Overall gRMS	**6.8 gRMS**	Workmanship (acceptance) level
Proto-flight	**+3 dB: 9.6 gRMS**	PSD doubled at every frequency; 1 min per axis, 3 axes

Computing Overall gRMS from PSD Profile:

$$g_{\mathrm{RMS}} = \sqrt{\int_{f_1}^{f_2} \mathrm{PSD}(f)\, df}$$

For a flat PSD region:

$$g_{\mathrm{RMS,flat}} = \sqrt{\mathrm{PSD}_{\mathrm{flat}} \cdot (f_2 - f_1)}$$

Example: $\mathrm{PSD} = 0.04\ g^2/\mathrm{Hz}$ from 80–500 Hz:

$$g_{\mathrm{RMS,flat}} = \sqrt{0.04 \cdot 420} = \sqrt{16.8} \approx 4.10\ g_{\mathrm{RMS}}.$$

The ramp sections and flat section are summed in quadrature:

$$g_{\mathrm{RMS,total}} = \sqrt{g^2_{\mathrm{RMS,ramp1}} + g^2_{\mathrm{RMS,flat}} + g^2_{\mathrm{RMS,ramp2}}}\,.$$

A sloped segment is a straight line in log-log axes, $\mathrm{PSD}(f) = P_1 (f/f_1)^n$ with $n = (\mathrm{dB/oct})/3.01$, and integrates exactly:

$$\int_{f_1}^{f_2} \mathrm{PSD}\, df = \frac{P_1 f_1}{n+1}\left[\left(\frac{f_2}{f_1}\right)^{n+1} - 1\right] \quad (n \neq -1), \qquad = P_1 f_1 \ln\frac{f_2}{f_1} \quad (n = -1).$$

For the profile above: ramp 20--80 Hz ($n = +1$): $\frac{0.01 \cdot 20}{2}(4^2 - 1) = 1.50\ g^2$; plateau 80--500 Hz: $0.04 \cdot 420 = 16.80\ g^2$; roll-off 500--2000 Hz ($n = -1$): $0.04 \cdot 500 \cdot \ln 4 = 27.73\ g^2$. Total $46.03\ g^2$, so

$$g_{\mathrm{RMS}} = \sqrt{46.03} = 6.78 \approx 6.8\ g_{\mathrm{RMS}}.$$

At proto-flight (+3 dB, PSD multiplied by 2), $g_{RMS} = \sqrt{92.06} = 9.59 \approx 9.6\ g_{RMS}$.

Thermal Vacuum Test Profile

Table 4.2.6 — Proto-flight thermal vacuum parameters including temperatures, cycles, vacuum, dwell, and ramp rate.

Parameter	Proto-flight Level	Notes
Hot case temperature	Predicted max + 10 C	e.g., if thermal model predicts +55 C -> test to +65 C
Cold case temperature	Predicted min - 10 C	e.g., if thermal model predicts -20 C -> test to -30 C
Number of cycles	4 (proto-flight)	Start and end at hot; functional test at each extreme
Vacuum level	< 10^-5 mbar	Space-representative vacuum
Dwell time at extreme	>= 1 hour	Allow thermal stabilisation (< 1 C/hr gradient)
Temperature transition rate	1-3 C/min	Limited by chamber capability
Functional test	Full performance at hot and cold extremes	Subset functional at intermediate transitions

TVAC Cycle Duration Estimate:

$$t_{cycle} = 2 \cdot \frac{\Delta T}{r_{ramp}} + 2\, t_{dwell} + 2\, t_{functional}$$

Example: $\Delta T = 95\ ^\circ C$ (from -30 to $+65\ ^\circ C$), $r_{ramp} = 2\ ^\circ C/min$, $t_{dwell} = 1$ hr, $t_{functional} = 2$ hr:

$$t_{cycle} = 2 \cdot \tfrac{95}{2}\ min + 2 \cdot 60\ min + 2 \cdot 120\ min = 95 + 120 + 240 = 455\ min \approx 7.6\ hr.$$

Total TVAC campaign (4 cycles + setup): $\approx 4 \cdot 7.6 + 8 = 38.4$ hr ≈ 5 working days.

EMC Testing

[Source: MIL-STD-461G; ECSS-E-ST-20-07C]

Table 4.2.7 — MIL-STD-461G EMC test set with standard reference and purpose for each conducted and radiated test.

Test	Standard	Purpose
CE102 Conducted Emissions	MIL-STD-461G	Measure spurious conducted emissions on power lines
RE102 Radiated Emissions	MIL-STD-461G	Measure unintentional RF radiation from satellite

Test	Standard	Purpose
CS101 Conducted Susceptibility	MIL-STD-461G	Verify immunity to power line transients
RS103 Radiated Susceptibility	MIL-STD-461G	Verify immunity to external RF fields

Note: EMC testing is not always required for CubeSats. It depends on the launch provider and co-manifested payloads. Check the rideshare ICD.

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4. Compliance Verification Matrix Construction

The Verification Matrix (also called Compliance Matrix or Verification Cross-Reference Matrix, VCRM) is the central document linking every requirement to its verification evidence.

[Source: ECSS-E-ST-10-02C Rev.1 section 6; DRD in Annex A]

Matrix Structure

Table 4.2.8 — Sample verification matrix linking requirement ID to method, phase, level, responsible team, facility, status, and evidence.

Req ID	Requirement Text	Method	Phase	Level	Responsible	Facility	Status	Evidence
SYS-001	M_dry <= 4.0 kg	I	D	System	Structures	Clean room	Planned	Mass measurement report
SYS-002	Survive 9.6 gRMS random (proto-flight)	T	C/D	System	Structures	Vibe lab	Planned	Vibration test report
SYS-003	GSD <= 10 m at 500 km	A	B	System	Payload	N/A	Planned	Optical analysis report
SYS-004	Link margin >= 3 dB	A	B	System	Comms	N/A	Planned	Link budget analysis
SYS-005	Antenna deploys < 30 min	D	D	System	Mechanisms	Clean room	Planned	Deployment demonstration

ATIR(D) Assignment Decision Logic

```
Is it a physical property (mass, dimensions, surface finish)?
  -> Inspection (I), Phase D, System level

Can it only be proven by quantitative measurement under controlled conditions?
  -> Test (T), Phase C/D, Unit or System level

Can it be demonstrated by operating the system (functional, not precision)?
  -> Demonstration (D), Phase D, System level
     (ECSS records this as a test without quantitative measurement)
```

```
Can it be shown with acceptable confidence through modelling or similarity?
  -> Analysis (A), Phase B/C, System level

Is it a document, a process, a plan, or a design feature evident from the drawings?
  -> Review of design (R), Phase B
```

Multiple Methods

Some requirements need **both** analysis (early confidence) and test (final proof):

Table 4.2.9 — Requirements that combine early-phase analysis with later test, listing both verification activities.

Requirement	Analysis (Phase B)	Test (Phase C/D)
Structural survival under launch loads	FEA with positive MoS	Vibration test -- no damage
Pointing accuracy <= 0.1 deg	Error budget analysis	On-orbit calibration measurement
Thermal within limits (-20 to +55 C)	Thermal model simulation	TVAC functional test at extremes
RF link margin >= 3 dB	Link budget calculation	RF compatibility test (if feasible)

In the matrix, list the **primary** method and note "confirmed by [secondary method] in Phase [X]".

Waiver Process

When a requirement cannot be verified by the prescribed method, a **waiver** is needed:

Table 4.2.10 — Waiver types (non-compliance, method deviation, tailoring) with usage and required approval level.

Waiver Type	When Used	Approval Level
Non-compliance	Requirement cannot be met; accept deviation	Project manager + customer
Method deviation	Different verification method used (e.g., analysis instead of test)	Systems engineer + quality
Tailoring	Standard requirement not applicable to this mission	Systems engineer

Waiver documentation must include: Requirement ID, deviation description, technical justification, risk assessment, and compensating measures.

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5. SpaceCDF V&V Matrix Exercise

Instructions

1. Navigate to the **V&V Matrix** tab in SpaceCDF
2. Requirements are auto-populated from the design
3. For each requirement, assign: - **Method:** A, T, I, R or D (use the decision logic above) - **Phase:** When will verification occur (B, C, D)? - **Level:** Unit, subsystem, or system level? - **Status:** All should be "Planned" at this stage

4. Use the **filter buttons** to view by method type: - Count: How many are Analysis? Test? Inspection? Review of design? Demonstration? - Are there any requirements with no method assigned?
5. Identify requirements that need **two methods** (analysis in Phase B + test in Phase D)
6. Discuss with your team: which tests are most critical? Which drive the test campaign schedule?

Worksheet 4.2 Tasks

1. Assign an ATIR(D) method to 10 key requirements (justify each choice)
2. Identify 3 requirements that need both analysis AND test
3. Specify the complete environmental test sequence for your mission
4. Determine which test levels apply from your selected launch vehicle PUG
5. Identify any requirements that may need waivers -- document the rationale

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Cited References

Table 4.2.11 — Cited references for Chapter 4.2 covering ECSS verification, testing, MIL-STD-1540E, and MIL-STD-461G.

#	Reference	URL
1	ECSS-E-ST-10-02C Rev.1 (Verification)	https://ecss.nl/standard/ecss-e-st-10-02c-rev-1-verification/
2	ECSS-E-ST-10-03C (Testing)	https://ecss.nl/standard/ecss-e-st-10-03c-testing/
3	NASA SEH Rev 2 sections 5.3-5.4	https://www.nasa.gov/reference/systems-engineering-handbook/
4	MIL-STD-1540E (Test Requirements for Launch/ Upper Stage/Space Vehicles)	https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36197
5	MIL-STD-461G (EMI/EMC)	https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35789
6	CubeSat Design Specification Rev 14.1	https://www.cubesat.org/s/CDS-REV14_1-2022-02-09.pdf

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Session Summary

Table 4.2.12 — Session 4.2 takeaways summarising V&V definitions, ATIR(D), proto-flight, environmental levels, and waivers.

Topic	Key Takeaway
V vs V	Verification = meets requirements (ATIR(D)); Validation = meets stakeholder needs
ATIR(D)	Analysis (models and similarity), Test (quantitative), Inspection (physical), Review of design (documents and drawings), Demonstration (functional)
Proto-flight	CubeSats: single flight unit tested to qualification levels, acceptance duration
Sine vibration	5-100 Hz, 1.25 g, 4 oct/min for proto-flight (2 oct/min for qualification), 3 axes, no displacement band; quasi-static launch loads

Topic	Key Takeaway
Random vibration	20-2000 Hz, GEVS workmanship 6.8 gRMS; proto-flight +3 dB (about 9.6 gRMS), 1 min/axis
TVAC	Proto-flight at predicted +/- 10 C (GSFC-STD-7000B / ECSS-E-ST-10-03C), 4 cycles, functional at each extreme; ~7.6 h per cycle
Compliance matrix	Per requirement: method + phase + level + responsible + status + evidence
Waivers	Document deviation, justification, risk, and compensating measures

Chapter 4.3 — Risk, Interfaces & FMECA

> **Expected reading before this session.** ECSS-M-ST-80C — Risk management (≈ 60 min); ECSS-Q-ST-30-02C — FMEA / FMECA.

Prerequisites: Sessions 4.1-4.2 (equipment selected, V&V methods assigned) **References:** ECSS-M-ST-80C (Risk Management), ECSS-Q-ST-30-02C (FMEA/FMECA), ECSS-E-ST-10-24C (Interface Management), NASA SEH Rev 2 section 6.4, NPR 8000.4B

Learning Objectives

By the end of this session, participants will be able to: 1. Construct and populate a risk register using a 5x5 likelihood-consequence matrix 2. Build an N-squared (N^2) interface matrix and identify interface conflicts 3. Perform a simplified FMEA/FMECA for critical subsystems 4. Identify single-point failures and compute series/parallel reliability 5. Classify risks and select appropriate mitigation strategies (accept, mitigate, transfer, avoid)

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1. Risk Management Process

[Source: ECSS-M-ST-80C (Risk Management); NASA SEH Rev 2 section 6.4 (Process 13: Technical Risk Management)] [URL: https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/]

The Four-Step Process

Risk management is continuous throughout the project lifecycle. The four steps are:

1. **Identify** -- What can go wrong? (Brainstorming, checklists, historical data, expert judgment)
2. **Assess** -- How likely is it? How severe would the consequence be? (Scoring)
3. **Mitigate** -- What can we do to reduce likelihood or consequence? (Strategy selection)
4. **Monitor** -- Is the risk increasing, decreasing, or stable? (Tracking, triggers, escalation)

Risk Sources

Table 4.3.1 — Risk source categories (technical, schedule, cost, programmatic, external) with example risks.

Source	Example Risks
Technical	Component failure, interface mismatch, performance shortfall, software defect
Schedule	Component delivery delay, test facility unavailable, regulatory delay
Cost	Component price increase, scope creep, test campaign overrun
Programmatic	Funding cut, personnel turnover, partner withdrawal
External	Launch delay/failure, spectrum interference, export control denial

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2. The 5x5 Risk Matrix

The 5x5 matrix is the standard risk scoring tool used across ESA, NASA, and commercial space programmes.

Likelihood Scale

Table 4.3.2 — Five-point likelihood scale with qualitative description and probability range.

Level	Likelihood	Description	Probability Range
1	Remote	Very unlikely; no precedent in similar missions	< 5%
2	Unlikely	Could happen but improbable; has occurred rarely	5 - 20%
3	Possible	Has happened on similar missions; credible scenario	20 - 50%
4	Likely	Expected to occur at least once during the mission	50 - 80%
5	Almost certain	Will almost certainly occur; multiple precedents	> 80%

Consequence Scale

Table 4.3.3 — Five-point consequence scale with technical, cost, and schedule impact thresholds.

Level	Consequence	Technical Impact	Cost Impact	Schedule Impact
1	Negligible	No performance impact	< 1% overrun	< 1 week slip
2	Minor	Minor performance degradation; workaround exists	1 - 5% overrun	1 - 4 week slip
3	Moderate	Significant performance loss; mission degraded	5 - 15% overrun	1 - 3 month slip
4	Major	Mission capability severely degraded; partial loss	15 - 30% overrun	3 - 6 month slip
5	Catastrophic	Mission failure; total loss	> 30% overrun	> 6 month slip or cancellation

Risk Score and Classification

> **Risk Score:**
>
> R = L x C
>
> Where L = Likelihood (1-5), C = Consequence (1-5)

5×5 risk matrix with severity (rows, 1–5) and likelihood (columns, 1–5). Green = low, yellow = medium, red = high risk classification.

5×5 risk matrix with severity (rows, 1–5) and likelihood (columns, 1–5). Green = low, yellow = medium, red = high risk classification.

Risk Action Levels

The risk matrix above maps the risk index (likelihood × consequence, 1–25) to four action bands:

- **Low (1–4):** accept and monitor.

- **Medium (5–9):** mitigate if cost-effective.
- **High (10–15):** active mitigation required.
- **Critical (16–25):** redesign or descope required.

CubeSat-Specific Risk Examples

Table 4.3.4 — Representative CubeSat risks scored on the 5x5 matrix with category and typical mitigation.

Risk	L	C	Score	Category	Typical Mitigation
Deployment mechanism failure (antenna, SA)	3	4	12	High	Redundant mechanisms; 100+ ground test cycles
Communication loss after separation	2	5	10	High	Beacon mode; timer-based antenna deploy; multiple GS
ADCS does not achieve pointing specification	3	3	9	Medium	Margin in pointing budget; on-orbit calibration plan
Power budget negative during eclipse	2	4	8	Medium	Conservative duty cycling; 20% battery margin
COTS component radiation failure (SEU/SEL)	2	4	8	Medium	Watchdog resets; latchup protection circuits; EDAC
Software bug causing spurious safe mode entry	4	2	8	Medium	Extensive software testing; staged upload; safe mode must work independently
Thermal exceedance (hot case)	2	3	6	Medium	Additional radiator area; duty cycle limit
Launch delay (vehicle failure)	3	2	6	Medium	Manifest on multiple vehicles; schedule buffer
Spectrum licensing delay	3	2	6	Medium	Start filing 12+ months before planned launch

Mitigation Strategies

Table 4.3.5 — Four mitigation strategies (accept, mitigate, transfer, avoid) with usage criteria and cost impact.

Strategy	Description	When to Use	Cost Impact
Accept	Risk is within tolerance; monitor only	Score 1-4 (Low)	None
Mitigate	Reduce L or C through design changes, testing, or procedures	Score 5-15	Moderate -- cost of mitigation
Transfer	Pass risk to another party (insurance, supplier warranty)	Financial risk; operational risk	Premium or contract cost
Avoid	Change design to eliminate the risk entirely	Score 16-25 (Critical)	May affect performance or cost

3. N-Squared (N^2) Interface Matrix

The N^2 matrix is the standard systems engineering tool for identifying and managing interfaces between subsystems. It was popularised by NASA and ESA CDF practice.

[Source: ECSS-E-ST-10-24C (Interface Management); NASA SEH Rev 2 section 6.3 (Process 12: Interface Management)]

What is an N^2 Matrix?

For a system with N subsystems, the N^2 matrix is an N x N grid where: - **Diagonal cells** contain the subsystem names - **Off-diagonal cells** contain the interfaces between subsystems - **Cell (i, j)** = outputs FROM subsystem i TO subsystem j (read across the row) - **Cell (j, i)** = outputs FROM subsystem j TO subsystem i (read down the column)

N^2 Matrix Example (6 Subsystems)

N^2 interface matrix for a CubeSat: the six subsystems sit on the diagonal and each off-diagonal cell carries the interface(s) flowing from the row subsystem to the column subsystem (power, data, mechanical).

N^2 interface matrix for a CubeSat: the six subsystems sit on the diagonal and each off-diagonal cell carries the interface(s) flowing from the row subsystem to the column subsystem (power, data, mechanical).

Interface Types

Colour-code the N^2 matrix by interface type:

Table 4.3.6 — Colour-coding convention for N-squared matrix entries by interface type with examples.

Type	Colour	Examples
Electrical power	Red	28V bus, 5V regulated, switched lines
Data	Blue	I2C, SPI, UART, CAN, RS-422
RF	Green	Coaxial cable, waveguide
Mechanical	Orange	Mounting bolts, thermal straps, alignment pins
Thermal	Yellow	Conductive paths, radiative coupling
Software	Purple	Command interfaces, telemetry packets, mode transitions

Interface Conflict Detection

An **interface conflict** exists when: - Subsystem A expects to send 28V but Subsystem B only accepts 5V - Subsystem A uses I2C but Subsystem B only has SPI - Subsystem A is in S-band but Subsystem B antenna is X-band - Subsystem A mounts on the +Z face but Structure has no mounting provision on +Z

SpaceCDF detects many of these automatically through the **constraint engine** and displays conflicts as warning badges on the Dashboard.

Worked Example: Detecting a Conflict

Problem: The OBC sends commands to the reaction wheels via I2C. The selected reaction wheel unit (RWP100 from Blue Canyon) uses RS-422. This is a data protocol mismatch.

Resolution options: 1. Select a different reaction wheel that supports I2C 2. Add an I2C-to-RS-422 bridge (additional component, mass, cost, failure point) 3. Select a different OBC that supports RS-422

In SpaceCDF: This conflict would appear as a warning in the Equipment Browser when the reaction wheel is selected, because the system checks data protocol compatibility.

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4. FMEA/FMECA

[Source: ECSS-Q-ST-30-02C (Failure Mode, Effects, and Criticality Analysis); ECSS-Q-ST-30C (Dependability)]
[URL: https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmeca-6-march-2009/]

Definitions

- **FMEA** (Failure Mode and Effects Analysis): Identifies failure modes, their causes, and effects on the system
- **FMECA** (Failure Mode, Effects, and Criticality Analysis): FMEA + criticality ranking

FMEA/FMECA Table Structure

Table 4.3.7 — FMECA table illustrating function, failure mode, cause, effects, severity, detection, and criticality for four components.

Item	Function	Failure Mode	Cause	Local Effect	System Effect	Severity	Detection	Compe Provisio
OBC	Process commands, run FSW	Processor lockup	SEU, firmware bug	No command processing	Loss of mission control	5	HK timeout	Watchd reset; redund OBC (if
Battery	Store energy	Cell short	Manufacturing defect	Reduced capacity	Shortened eclipse survival	3	Voltage monitoring	Cell bal margin capacit
Reaction Wheel	Provide torque	Bearing seizure	Lubrication failure	Loss of one axis control	Degraded pointing	4	Current anomaly	4-whee config (redund
Antenna	Radiate RF	Deployment failure	Mechanism jam	No antenna deployed	No communication	5	Beacon absence	Redund deployr burn wi spring

Criticality Categories

Table 4.3.8 — FMECA criticality categories from catastrophic to minor with required action.

Category	Definition	Action Required
1 (Catastrophic)	Single failure causes mission loss; no compensation	Redesign to add redundancy or accept with justification
2 (Critical)	Single failure causes significant degradation	Mitigate (redundancy, operational workaround)
3 (Major)	Single failure causes moderate degradation	Monitor; plan operational workaround

Category	Definition	Action Required
4 (Minor)	Single failure has negligible mission impact	Accept

Single-Point Failure (SPF) Analysis

A **single-point failure** is any single component whose failure alone causes loss of mission. Identifying SPFs is a critical output of the FMECA.

Typical CubeSat Single-Point Failures:

Table 4.3.9 — Common CubeSat single-point failures with reason for criticality and typical mitigation approach.

Component	Why it is an SPF	Common Mitigation
OBC (single processor)	No command processing -> no mission	Watchdog timer + autonomous safe mode + EDAC memory
Battery (single pack)	No stored energy -> no eclipse survival	Cell-level monitoring; conservative DoD limit (< 20%)
Antenna (non-redundant deploy)	No RF link -> no commanding/ telemetry	Redundant deployment mechanisms (burn wire + spring)
Solar array (deployment)	No power generation -> mission loss within hours	Redundant deployment; hinge spring + motor backup
EPS main board	No power distribution	Typically no mitigation (accepted SPF in CubeSats)

Key Equations: Reliability

Series Reliability (all components must work):

$$\mathrm{R_{series}} = \prod_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{R_i}$$

Example: 5 components each with $\mathrm{R} = 0.99$:

$$\mathrm{R_{series}} = 0.99^5 = 0.951.$$

Parallel Reliability (at least one must work — redundancy):

$$\mathrm{R_{parallel}} = 1 - \prod_{\mathrm{i}=1}^{\mathrm{n}} (1 - \mathrm{R_i})$$

For n identical redundant units each with R:

$$\mathrm{R_{parallel}} = 1 - (1 - \mathrm{R})^{\mathrm{n}}.$$

Example: 2 redundant deployment mechanisms each with $\mathrm{R} = 0.95$:

$$\mathrm{R}_{\text{parallel}} = 1 - (1 - 0.95)^2 = 1 - 0.0025 = 0.9975.$$

Mean Time Between Failures (MTBF):

$$\mathrm{MTBF} = \frac{\text{Total operating hours}}{\text{Number of failures}} = \frac{1}{\lambda}$$

Reliability over time (exponential model):

$$\mathrm{R}(t) = e^{-t/\mathrm{MTBF}} = e^{-\lambda t}$$

Example: MTBF = 50,000 hours, mission duration = 8,760 hours (1 year):

$$\mathrm{R} = e^{-8760/50000} = e^{-0.1752} = 0.839.$$

System Availability:

$$\mathrm{A} = \frac{\mathrm{MTBF}}{\mathrm{MTBF} + \mathrm{MTTR}}$$

where MTTR = Mean Time To Repair (or recover, for spacecraft).

For a spacecraft with MTBF = 50,000 hr and MTTR = 2 hr (safe mode recovery):

$$\mathrm{A} = \frac{50000}{50000 + 2} = 0.99996.$$

Worked Example: Reaction Wheel Redundancy

Problem: A fine-pointing mission requires 3-axis attitude control. Each reaction wheel has R = 0.98 over the 2-year mission.

Configuration A -- 3 wheels (no redundancy): R_system = R^3 = 0.98^3 = 0.941

Configuration B -- 4 wheels in 3-of-4 redundancy (1 spare): R_system = 4 x R^3 - 3 x R^4 ... Using the binomial: P(>= 3 working) = C(4,4) x R^4 + C(4,3) x R^3 x (1-R)^1 = 0.98^4 + 4 x 0.98^3 x 0.02 = 0.9224 + 4 x 0.9412 x 0.02 = 0.9224 + 0.0753 = 0.9977

Adding one spare wheel improves reliability from 0.941 to 0.998 -- a significant improvement for modest mass and cost increase.

Real Mission Example: Hitomi (ASTRO-H)

JAXA's Hitomi X-ray observatory was lost on 26 March 2016, just 37 days after launch, due to a cascading failure in the AOCS subsystem: 1. After a star-tracker dropout, the attitude-determination software mis-estimated the gyro (IRU) bias, giving an erroneous attitude estimate 2. The reaction wheels applied incorrect torques based on the bad estimate 3. Thrusters fired to "correct" the (phantom) attitude error, but an incorrectly uploaded thruster-control parameter set made them accelerate the spin 4. The satellite spun up beyond structural limits, and the extensible optical bench broke off

Root cause: Attitude-determination error (IRU bias mis-estimated) + incorrectly uploaded thruster-control parameters + inadequate FDIR logic + no cross-check between attitude sensors.

Lesson: FMECA must consider cascading failures and common-cause failures, not just single-component failures. Software errors are failure modes too.

[Source: JAXA Hitomi Investigation Report, 2016, available at https://global.jaxa.jp/press/2016/05/20160531_hitomi.html]

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5. Risk & Interface Exercise

Instructions

1. **Dashboard** -- Check the reliability score and conflict count
2. **Risk Register Construction** (Worksheet 4.3): - Identify **5 technical risks** for your mission design - Score each on the 5x5 matrix (L x C) - Define mitigation strategy for any scoring >= 10 - Assign an owner (which CDF position is responsible)
3. **N^2 Matrix** -- Build a simplified N^2 matrix for your mission: - List 6 subsystems on the diagonal - Fill in the interfaces (power, data, RF, mechanical) - Identify at least 2 interface conflicts
4. **SPF Analysis** -- Identify all single-point failures in your design: - For each SPF: Is it acceptable? If not, what redundancy is needed? - Compute the reliability improvement from adding redundancy to one SPF

Discussion Prompts

- "What is the highest-risk item in your design? What would it cost to mitigate?"
- "Do you have any interface conflicts that cannot be resolved without changing a component selection?"
- "Which single-point failure are you most concerned about? Would the customer accept it?"

Worksheet 4.3 Tasks

1. Complete the risk register (5 risks minimum, scored)
2. Build a 6x6 N^2 interface matrix with colour-coded interface types
3. Complete the FMECA table for 4 critical components
4. List all single-point failures with accept/mitigate decision
5. Calculate series reliability for your mission's critical chain

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Cited References

Table 4.3.10 — Cited references for Chapter 4.3 covering ECSS risk, FMECA, interface management, and the Hitomi investigation.

#	Reference	URL
1	ECSS-M-ST-80C (Risk Management)	https://ecss.nl/standard/ecss-m-st-80c-risk-management-31-july-2008/
2		

#	Reference	URL
	ECSS-Q-ST-30-02C (FMEA/FMECA)	https://ecss.nl/standard/ecss-q-st-30-02c-failure-mode-effects-and-criticality-analysis-fmeca-6-march-2009/
3	ECSS-E-ST-10-24C (Interface Management)	https://ecss.nl/standard/ecss-e-st-10-24c-interface-management/
4	NASA SEH Rev 2, section 6.4	https://www.nasa.gov/reference/systems-engineering-handbook/
5	JAXA Hitomi Investigation Report	https://global.jaxa.jp/press/2016/05/20160531_hitomi.html
6	ECSS-Q-ST-30C (Dependability)	https://ecss.nl/standard/ecss-q-st-30c-dependability/

Session Summary

Table 4.3.11 — Session 4.3 takeaways summarising risk process, 5x5 matrix, N^2, FMECA, SPFs, and reliability formulae.

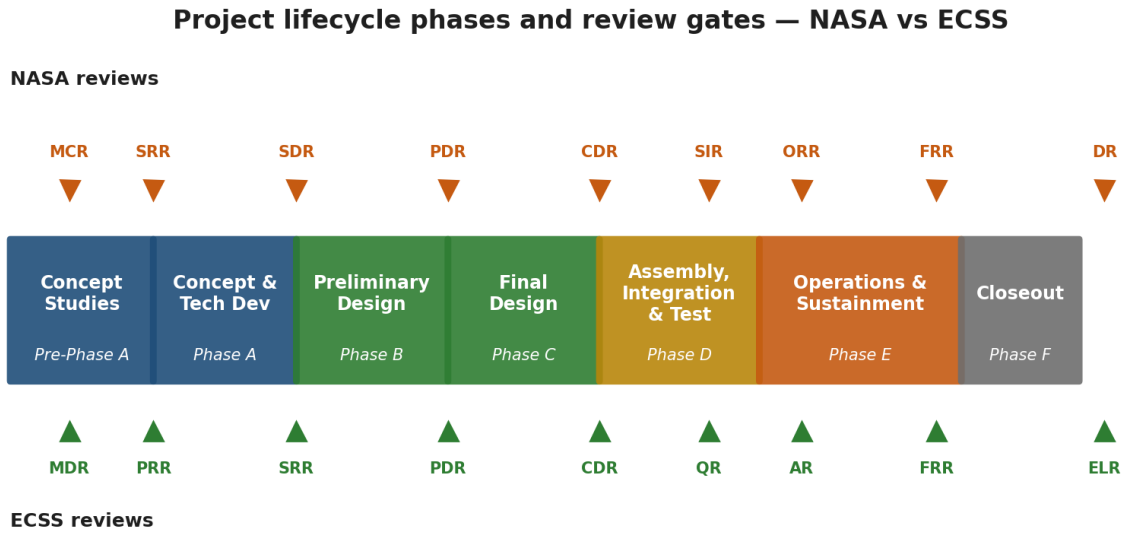
Topic	Key Takeaway
Risk process	Identify -> Assess -> Mitigate -> Monitor (continuous throughout lifecycle)
5x5 matrix	L x C = Risk Score; Low (1-4), Medium (5-9), High (10-15), Critical (16-25)
Strategies	Accept (low), Mitigate (medium-high), Transfer (financial), Avoid (critical)
N^2 matrix	N x N grid mapping all subsystem interfaces; colour-code by type
Interface conflicts	Protocol mismatch, voltage mismatch, band mismatch -- detect early
FMECA	Failure mode -> cause -> local effect -> system effect -> severity -> detection -> mitigation
SPF	Single-point failures must be identified, assessed, and accepted or mitigated
Reliability	R_series = Product(R_i); R_parallel = 1 - Product(1-R_i); R(t) = e^(-lambda*t)

Typical CubeSat project hierarchy with mission lead, systems-engineering lead, discipline leads, and project-management functions.

Typical CubeSat project hierarchy with mission lead, systems-engineering lead, discipline leads, and project-management functions.

Chapter 4.4 — Cost Estimation & Design Review

> **Expected reading before this session.** NPR 7120.5F — WBS (≈ 30 min).



Gate sequence at a glance.

> **Expected reading before the design review.** ECSS-M-ST-10C Rev.1 — Project planning and implementation, §4.4 (project phasing: the phases and their review gates) and §5.4 (≈ 60 min); ECSS-M-ST-10-01C — Organization and conduct of reviews (skim).

Prerequisites: Sessions 4.1-4.3 (equipment selected, V&V planned, risks assessed) **References:** SMAD, NASA Cost Estimating Handbook (CEH) v4.0, Aerospace Corp SSCM, NPR 7120.5F (WBS), ECSS-M-ST-60C (Cost Management), NPR 7123.1D Appendix G (Reviews)

Learning Objectives

By the end of this session, participants will be able to: 1. Estimate mission cost using parametric (CER), analogy, and bottom-up methods 2. Structure a Work Breakdown Structure (WBS) for a CubeSat project 3. Apply learning curve effects for constellation cost estimation 4. Assess cost risk using confidence levels (P50/P70/P80) 5. Prepare for and conduct a design review (SRR/PDR/CDR) with gate criteria 6. Present design evidence clearly and respond to review board questions

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1. Cost Estimation Methodologies

[Source: NASA CEH v4.0 Appendix C; Aerospace Corp SSCM] [URL: https://www.nasa.gov/offices/ocfo/references-and-tools/ (NASA CEH)]

Three Primary Methods

Table 4.4.1 — Parametric, analogy, and bottom-up cost methods with accuracy, applicable phase, and data needs.

Method	Description	Accuracy (1-sigma)	When to Use	Data Required
Parametric	Cost Estimating Relationships (CERs) from historical database	+/- 30-50%	Phase 0/A (few details known)	Mass, power, mission type
Analogy	Compare to similar past mission, adjust for differences	+/- 20-40%	Phase A/B (reference mission available)	Detailed knowledge of reference
Bottom-up	Sum actual vendor quotes + labour estimates + facilities	+/- 10-20%	Phase B/C (detailed design)	BOM, vendor quotes, labour rates

Parametric Cost Estimating Relationships (CERs)

CERs express cost as a function of technical parameters, derived from regression analysis of historical data.

> **General CER Form:**

$$\mathrm{Cost} = \mathrm{a} \cdot \mathrm{Parameter}^{\mathrm{b}} \cdot \mathrm{C}_{\mathrm{complexity}}$$

> Where: - a = coefficient (from regression) - $\mathrm{Parameter}$ = usually dry mass (kg), power (W), or data rate - b = exponent (typically 0.5–1.0) - $\mathrm{C}_{\mathrm{complexity}}$ = adjustment factor for mission difficulty

CER form per subsystem (illustrative)

CubeSat designers should derive a CER for each subsystem either from a modern public dataset (NASA NICM, ESA Cost Engineering, vendor parametric quotes) or by sampling completed missions in their own institution's portfolio. The general functional form is:

> Cost_subsystem = a × (Parameter)^b × Complexity_factor

The table below shows the *shape* of a typical CER per subsystem (coefficient and exponent are illustrative — derive a, b from the mission cohort and project context that applies to the current study). For an externally-calibrated dataset specific to small satellites, the Aerospace Corporation Small Satellite Cost Model (SSCM) referenced below is the recommended starting point.

Table 4.4.2 — Illustrative CER shape per subsystem, suitable as a starting structure for parametric cost rolls; numerical coefficients should be re-fitted to a current dataset before use in a real bid.

Subsystem	CER form (FY $K)	Parameter	Typical exponent
Structure	a × M_struct^b	Dry mass (kg)	~0.8
Thermal	a × M_therm^b	Dry mass (kg)	~0.6
EPS	a × M_EPS^b	Dry mass (kg)	~1.0

Subsystem	CER form (FY $K)	Parameter	Typical exponent
TT&C	a × M_TTC^b	Dry mass (kg)	~0.75
AOCS	a × M_AOCS^b	Dry mass (kg)	~0.85
Propulsion	a × M_prop^b	Dry mass (kg)	~0.75
Integration & Test	a × M_dry^b	Total dry mass (kg)	~0.9
Program Management	wrap factor on hardware cost	—	typically 10–15%
Systems Engineering	wrap factor on hardware cost	—	typically 12–18%

SSCM (Small Satellite Cost Model)

The Aerospace Corporation SSCM was specifically calibrated for satellites < 500 kg. It provides more accurate estimates for CubeSats than USCM.

[Source: Aerospace Corp, "Small Satellite Cost Model (SSCM)", available through Aerospace TOR]

Key SSCM adjustments for CubeSats: - COTS hardware costs are catalogue prices, not mass-based CERs - Labour costs dominate for university/small-team projects - NRE is heavily dependent on mission uniqueness - Standard CERs (designed for > 100 kg) over-predict by 2-6x for nano/micro class

CubeSat-Calibrated Pricing (SpaceCDF)

SpaceCDF uses a hybrid approach: COTS flat pricing for nano/micro class, CERs for larger spacecraft.

Table 4.4.3 — CubeSat COTS pricing compared with USCM CER predictions per subsystem.

Subsystem	CubeSat COTS Cost (kEUR)	USCM CER (kEUR, for CubeSat mass)	Ratio
EPS	15	60	CER 4x higher
AOCS (fine)	40	66	CER 1.6x higher
TTC (S-band)	20	25	Similar
OBC	10	12	Similar
Structure	8	3	CER lower (min order effect)
Payload	50 (variable)	300	CER 6x higher for COTS payloads

This confirms that for CubeSats, COTS pricing is more reliable than parametric CERs for hardware cost.

Worked Example: Parametric vs Bottom-Up

3U Earth observation mission:

Table 4.4.4 — Parametric versus bottom-up cost estimate for a 3U EO CubeSat broken down by WBS element.

WBS Element	Parametric (kEUR)	Bottom-Up (kEUR)	Notes
Structure	8	8.5	ISIS 3U structure

WBS Element	Parametric (kEUR)	Bottom-Up (kEUR)	Notes
EPS	18	16.2	GomSpace P31u + SA
OBC	12	10.0	NanoAvionics SatBus
AOCS	45	42.0	BCT XACT-15
TTC	22	20.5	Endurosat S-band
Thermal	5	3.0	Passive only
Payload	65	58.0	Simera Sense xScape
Harness	3	2.5	Custom cables
Hardware subtotal	**178**	**160.7**	
I&T (12%)	21	19.3	
Software (8%)	14	12.9	
PM/SE/MA (13%)	23	20.9	
Launch	200	195.0	SpaceX Transporter
Ground (5%)	9	8.0	SatNOGS + dedicated
Operations (3 yr)	45	40.0	0.5 FTE
TOTAL	**490**	**456.8**	

The parametric estimate is ~7% higher than bottom-up. This is expected: parametric includes inherent uncertainty and contingency.

---

2. Cost Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) is the hierarchical decomposition of all work required to complete the mission. It is the foundation for cost estimation, scheduling, and management.

[Source: NPR 7120.5F Appendix G; ECSS-M-ST-60C; NASA CEH v4.0 Appendix B]

Standard CubeSat WBS

```
WBS Level 1: Mission Total
  1.0  Programme Management                5%    (oversight, reviews, reporting)
  2.0  Systems Engineering                 5%    (budgets, interfaces, trade studies)
  3.0  Mission Assurance                   3%    (quality, reliability, parts)
  4.0  Payload                            20%    (instrument + calibration)
    4.1  Payload instrument hardware
    4.2  Payload software
    4.3  Payload calibration & characterisation
  5.0  Spacecraft Bus Hardware            27%    (all subsystems)
    5.1  Structure & mechanisms
    5.2  EPS (solar array + battery + board)
    5.3  AOCS (sensors + actuators)
    5.4  TTC (transponder + antenna)
```

```
    5.5  OBC & data handling
    5.6  Thermal control
    5.7  Propulsion (if applicable)
    5.8  Harness & cabling
  6.0  Integration & Test                12%    (assembly, env. testing)
    6.1  Assembly & integration
    6.2  Environmental test campaign
    6.3  Test facilities rental
  7.0  Software (Flight + Ground)         8%    (FSW, GSW, mission planning)
    7.1  Flight software
    7.2  Ground segment software
    7.3  Mission planning tools
  8.0  Launch Services                   10%    (vehicle, deployer, integration)
  9.0  Ground Segment                     5%    (antennas, MCS, networks)
  10.0 Operations (mission lifetime)      5%    (staff, consumables, maintenance)
```

Cost by Phase

The distribution of cost across lifecycle phases is important for budgeting:

Table 4.4.5 — Lifecycle phase cost share, key activities, and peak staffing level.

Phase	% of Total	Activities	Peak Staffing
0/A (Concept)	5-10%	Studies, trade-offs, requirements	Low
B (Preliminary Design)	15-20%	PDR, detailed analysis, long-lead procurement	Growing
C (Detailed Design)	30-40%	CDR, manufacturing, software development	Peak
D (Integration & Test)	20-25%	Assembly, environmental testing, commissioning	High
E (Operations)	10-20%	Routine operations, anomaly management	Low-steady
F (Disposal)	1-3%	Decommissioning, deorbit	Minimal

3. Learning Curve for Constellations

When building multiple identical units, the cost per unit decreases due to manufacturing efficiency, reduced test time, bulk purchasing, and labour learning.

Wright's Learning Curve

Nth Unit Cost:

$$C_N = C_1 \cdot N^b, \quad b = \frac{\ln(\text{learning rate})}{\ln 2}$$

Table 4.4.6 — Wright learning-rate exponents and 10th-unit cost percentages at 95%, 90%, and 85% learning rates.

Learning Rate	b	Cost of 10th Unit (% of 1st)
95 %		84 %

Learning Rate	b	Cost of 10th Unit (% of 1st)
	-0.074	
90 %	-0.152	70 %
85 %	-0.234	58 %

Total Cost for N Units (cumulative):

$$C_{total} = C_1 \cdot \sum_{i=1}^{N} i^{b}$$

or, to a continuous approximation,

$$C_{total} \approx C_1 \cdot \frac{N^{1+b}}{1+b}.$$

Simplified Rule of Thumb

At a **90% learning rate**, every time you **double** the number of units, the unit cost drops by **10%**.

Unit 1: EUR 800K Unit 2: EUR 720K (800 x 0.90) Unit 4: EUR 648K (720 x 0.90) Unit 8: EUR 583K (648 x 0.90) Unit 16: EUR 525K (583 x 0.90)

Worked Example: 20-Satellite Constellation

First unit cost (bus + payload + I&T): EUR 800K. Learning rate: 90%.

Table 4.4.7 — Cumulative cost for a 20-satellite constellation under a 90% learning curve.

Units	b	Avg Unit Cost	Total Hardware	Calc
1	-0.152	EUR 800K	EUR 800K	First unit
5	-0.152	EUR 626K	EUR 3,132K	5 x 800 x 5^(-0.152)
10	-0.152	EUR 564K	EUR 5,638K	Cumulative sum
20	-0.152	EUR 505K	EUR 10,100K	Cumulative sum

Total constellation estimate: - 20 satellites hardware: EUR 10.1M - 2 spare units (10%): EUR 1.0M - Launch (20 sats x EUR 200K rideshare): EUR 4.0M - Ground segment: EUR 1.0M - Operations (3 years): EUR 0.9M - PM/SE/MA (10%): EUR 1.7M - **Total: ~EUR 18.7M**

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4. Cost Risk and Confidence Levels

Point estimates are misleading. Every cost estimate has uncertainty. The standard practice is to express cost as a probability distribution.

[Source: NASA CEH v4.0 section 2.3; JPL parametric estimation practice]

Uncertainty by Cost Element

Table 4.4.8 — Cost element uncertainty distributions and one-sigma ranges by hardware, software, launch, ops, and I&T.

Cost Element	Distribution	Uncertainty (1-sigma)	Rationale
COTS hardware	Normal	+/- 10%	Known pricing from vendor quotes
Custom hardware	Lognormal	+/- 30%	Development uncertainty skews high
Software	Triangular	+/- 40%	Hardest to estimate; frequent overruns
Launch	Normal	+/- 15%	Published pricing; contract negotiation
Operations	Uniform	+/- 25%	Staffing level uncertainty
I&T	Lognormal	+/- 25%	Test anomalies cause schedule/cost growth

Confidence Levels

Table 4.4.9 — Confidence percentiles P50, P70, and P80 with meaning and typical project usage.

Percentile	Meaning	Use
P50	50% probability of being at or below this cost	Project baseline; "expected" cost
P70	70% probability	NASA standard commitment level (NPR 7120.5F)
P80	80% probability	Conservative planning; typical for proposals

Rule of Thumb for CubeSat Missions:

P70 ~ P50 x 1.2 P80 ~ P50 x 1.3

Example: If P50 = EUR 500K, then P80 ~ EUR 650K.

This approximation assumes moderate complexity and well-understood COTS hardware. For missions with custom payloads or new technology, use P80 ~ P50 x 1.5.

1U Worked Example: UniSat-1

Cost Breakdown: Simple WBS, Mostly COTS

UniSat-1's cost structure is fundamentally different from larger missions because (a) nearly all hardware is COTS, so NRE is near zero, and (b) the team is small and university-based, so labour costs are low.

UniSat-1 WBS Cost Estimate (Parametric vs Bottom-Up):

Table 4.4.10 — UniSat-1 WBS cost estimate comparing parametric and bottom-up totals with a P80 conservative figure.

WBS Element	Parametric (kEUR)	Bottom-Up (kEUR)	Notes
1.0 Programme Management	4	5	Faculty oversight, 0.1 FTE x 12 months

WBS Element	Parametric (kEUR)	Bottom-Up (kEUR)	Notes
2.0 Systems Engineering	3	3	Student team lead
3.0 Mission Assurance	1	1	Minimal QA for university mission
4.0 Payload	8	8	MEMS sensor PCB + calibration
5.0 Bus Hardware	39	38	See BOM from Chapter 4.1
5.1 Structure	4	4	ISIS 1U frame
5.2 EPS + SA	20	19.5	P31us + body-mounted cells
5.3 AOCS (passive)	1	1	Magnet + hysteresis rods
5.4 Comms (UHF)	11	10.5	AX100 + antenna
5.5 OBC	3	3	Custom Cortex-M board
6.0 I&T	8	8	University clean room + vibe test facility
7.0 Software	5	5	FSW (FreeRTOS) + GSW
8.0 Launch	15	15	NanoRacks 1U ISS deployment
9.0 Ground Segment	5	5	Yagi antenna + SatNOGS network
10.0 Operations (6 months)	3	3	Student operators, 0.2 FTE
TOTAL (P50)	**~91**	**~91**	
P80 (x 1.3)	**~118**	**~118**	Conservative estimate

Key cost observations for 1U missions:

1. **Hardware is cheap:** Total COTS hardware cost is ~36--44 kEUR. This is less than a single star tracker for a 3U mission.
2. **NRE is minimal:** Only the OBC and payload require custom development. NRE is estimated at ~8 kEUR (payload calibration + OBC board layout), compared to ~50--150 kEUR for custom payloads on larger missions.
3. **Launch cost is proportionally large:** At 15 kEUR, the launch represents ~16% of total cost. For a 3U mission at ~200 kEUR launch cost, launch is ~40% of total. The 1U launch cost is low in absolute terms but still a significant fraction.
4. **Labour dominates:** For a university team, the "free" student labour is the hidden cost. If students were costed at professional rates (~50 EUR/hr), the total labour cost would be ~100--200 kEUR, far exceeding the hardware cost. This is typical for educational missions.
5. **No cost drivers from complexity:** There is no AOCS software development, no deployable mechanism qualification, no propulsion system integration, no thermal vacuum testing of heaters -- all of which add 10--50 kEUR each on a 3U mission.

Learning curve applicability: If a university builds a series of 1U demonstrators (UniSat-1, UniSat-2, UniSat-3…), the 90% learning rate applies:

Table 4.4.11 — UniSat-1 follow-on builds showing hardware and total cost reduction across units under a 90% learning curve.

Unit	Hardware Cost (kEUR)	Total Cost (kEUR)
UniSat-1	44	85
UniSat-2	40	77
UniSat-4	36	69
UniSat-8	32	62

The asymptotic floor is dominated by launch cost (15 kEUR) and irreducible ground segment + operations costs (~8 kEUR), giving a minimum mission cost of ~40--50 kEUR for repeat builds.

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5. Design Review Process

Design reviews are formal decision gates where the project demonstrates readiness to proceed to the next lifecycle phase.

[Source: NPR 7123.1D Appendix G; ECSS-M-ST-10C Rev.1 section 6; NASA SEH Rev 2 section 3.7] [URL: https://www.nasa.gov/reference/systems-engineering-handbook/]

Review Sequence

Table 4.4.12 — Lifecycle review sequence (MCR, SRR, PDR, CDR, TRR, QR, FRR) with phase transition, key question, and entry criteria.

Review	Phase Transition	Key Question	Entry Criteria
MCR	Pre-A -> A	Is the mission need justified?	Problem statement, stakeholders, objectives defined
SRR	A -> B	Are requirements complete, consistent, traceable?	Requirements baselined, ConOps defined, feasibility shown
PDR	B -> C	Does preliminary design meet requirements with margin?	All budgets close, interfaces defined, risks identified
CDR	C -> D	Is detailed design complete and ready to build?	All drawings released, test plan approved, suppliers under contract
TRR	Pre-test	Is the system ready for environmental testing?	Assembly complete, procedures approved, facility booked
QR	Post-test	Has the system passed all tests?	All test reports approved, NCRs closed, V&V matrix complete
FRR	Pre-launch	Is everything ready for launch?	Shipping approved, launch manifest confirmed, operations ready

Gate Criteria for SRR (Phase A -> B)

Table 4.4.13 — SRR gate criteria with priority and required evidence for the A-to-B transition.

#	Criterion	Priority	Evidence
1	All Level 0/1 requirements baselined	Must pass	Requirements document signed
2	ConOps defined (all mission phases)	Must pass	ConOps document
3	Mission architecture trades completed	Must pass	Trade study reports with rationale
4	Feasibility confirmed (all budgets positive)	Must pass	Mass, power, link budgets with margin
5	Risk register established with mitigations	Must pass	Risk register with scores and plans
6	Preliminary V&V approach defined	Should pass	V&V matrix with methods assigned
7	Schedule and cost estimate (P50/P80)	Must pass	Cost estimate with WBS
8	Interface requirements identified	Should pass	N^2 matrix or ICD outline

Gate Criteria for PDR (Phase B -> C)

Table 4.4.14 — PDR gate criteria with priority and required evidence for the B-to-C transition.

#	Criterion	Priority	Evidence
1	All requirements allocated to subsystems	Must pass	Requirements allocation matrix
2	Preliminary design complete for all subsystems	Must pass	Design documents, block diagrams
3	All budgets close with >= 20% margin	Must pass	Mass, power, link, data, pointing budgets
4	Equipment selected (BOM)	Must pass	BOM with TRL, heritage, cost, lead time
5	All interfaces defined (N^2 matrix)	Must pass	Interface control documents
6	V&V matrix complete with methods assigned	Must pass	V&V matrix
7	Risk register updated; no Critical risks unmitigated	Must pass	Risk register
8	Test plan outline approved	Should pass	Environmental test plan
9	Software architecture defined	Should pass	Software design document
10	Cost estimate updated (bottom-up)	Must pass	Cost estimate with vendor quotes

Presentation Skills for Reviews

Do: - Lead with the conclusion ("Mass budget closes with 22% margin") - Show evidence, not just assertions ("Link budget analysis shows 4.2 dB margin at worst case") - Acknowledge risks honestly ("Antenna deployment is our highest risk at L3 x C4 = 12") - Answer questions directly; say "I don't know, we'll take an action" if needed

Do not: - Read slides aloud - Hide problems (review boards always find them) - Present analysis without assumptions stated - Skip backup slides -- have detailed data ready

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6. Cost & Review Exercise

Instructions

Part A: Cost Estimation

1. **Dashboard** -- Check the Cost KPI card (total cost in MEUR)
2. **Cost Breakdown** tab -- Review the breakdown by subsystem
3. **Exports** tab -- Generate BOM and sum COTS component costs
4. On Worksheet 4.4: - Fill in the WBS cost table using both parametric and bottom-up methods - Compute P50 and P80 estimates - If constellation: apply 90% learning curve

Part B: Design Review Preparation

1. Open the **Gate Review** tab in SpaceCDF
2. Check all criteria: which are Pass/Fail/Manual?
3. For any failing criteria, click **"Go fix"** and resolve
4. Prepare a 3-minute summary of your design for peer review

Worksheet 4.4 Tasks

1. Build a complete WBS cost table (parametric and bottom-up columns)
2. Compute P50 and P80 total cost estimates
3. If constellation: compute total cost with learning curve applied
4. Identify top 3 cost drivers and propose 20% cost reduction
5. List the gate criteria for SRR/PDR and assess your readiness

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Cited References

#	Reference	URL
1	NASA Cost Estimating Handbook v4.0	https://www.nasa.gov/offices/ocfo/references-and-tools/
2	SMAD (Cost)	Wertz, Everett, Puschell (eds.), Space Mission Engineering, Microcosm 2011
3	Aerospace Corp SSCM	https://www.aerospace.org/capabilities/small-satellite-cost-model
4	NPR 7120.5F (WBS)	https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7120&s=5F
5	ECSS-M-ST-60C (Cost Management)	https://ecss.nl/standard/ecss-m-st-60c-cost-and-schedule-management/
6	NPR 7123.1D (SE Processes) Appendix G	https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7123&s=1D
7	NASA SEH Rev 2, section 3.7	https://www.nasa.gov/reference/systems-engineering-handbook/

Session Summary

Table 4.4.15 — Session summary: key takeaways for cost estimation and design review.

Topic	Key Takeaway
Cost methods	Parametric (early, CERs), Analogy (reference mission), Bottom-up (vendor quotes)
CubeSat costs	COTS pricing often lower than CER predictions; use hybrid approach
WBS	Standard 10-element structure: PM, SE, MA, Payload, Bus, I&T, SW, Launch, Ground, Ops
CERs	Cost = a x M^b; USCM/SSCM databases; CubeSats need calibrated CERs
Learning curve	90% rate: each doubling of quantity reduces unit cost by 10%
Cost risk	P50 (baseline), P70 (NASA commitment), P80 (conservative); P80 ~ P50 x 1.3
Reviews	SRR, PDR, CDR: formal gates with exit criteria; must pass before proceeding
Presentation	Lead with conclusions, show evidence, acknowledge risks, answer directly

Appendix A — Engineering-Position Reference

This chapter is a per-role background briefing for first-time CubeSat designers in a CDF. For each engineering position students might take, it covers:

- **Decisions you own.** What you have to choose, in what order.
- **What you need to know first.** Inputs you cannot decide without.
- **Formulas, charts, and rules of thumb.** With references.
- **Common pitfalls.** What to watch for, especially first time.
- **References.** Live links to standards and texts.

The briefings are roughly 2–4 pages each. Each is self-contained; read the one for your assigned role before Day 2 of Week 2 (CDF intensive Day 1).

> **Design roles vs operations roles.** This appendix covers the *design* positions used in Part 1 — the people who choose the orbit, size the subsystems, and close the budgets. The companion *operations* control-room positions (Flight Director, SPACON, Mission Manager, and the rest) are in Appendix B. Some roles exist only in design (e.g. Mission Analyst, Cost & Schedule) and some only in operations (e.g. Flight Director, SPACON, Mission Manager). Several names appear in both lists — AOCS, Thermal, Communications/TT&C, Payload, Ground Segment — but the *activity* differs: in design you *size and select*; in operations you *watch*, *command*, *and respond* on the live spacecraft. Read this appendix for the design phase and Appendix B for the flight phase.
>
> **Expected reading.** Each role briefing is overnight reading the evening before students take their CDF positions. Read your own briefing in full and scan one adjacent one (e.g. Power Engineers should scan Thermal; AOCS should scan Structures).

Systems Engineer

You hold the budgets, the gate, and the authority to call a trade.

Decisions you own

1. **Margin policy for this study.** You set the per-budget margin targets for the current phase per ECSS-M-ST-10C — typically Phase A: +44 % mass, +25 % power, 6 dB link, +25 % Δv; tightening to +5 % / +5 % / 3 dB / +10 % at Phase D.
2. **Budget closure decisions.** When power is over budget, you decide whether the team trades a payload duty cycle, a battery capacity, an SA area, or an end-of-life lifetime.
3. **Cross-domain conflict resolution.** When AOCS wants more reaction-wheel mass and Thermal wants the same area for a radiator, you convene the affected positions and arbitrate.
4. **Gate-review readiness.** You judge whether the team is ready to enter PDR, and you sign the action items off.

What you need to know first

- **Mission objectives** (SE Process 1) — the *why* drives the budgets you're closing.
- **System-V model** and where each position sits on it (NASA SEH §2). See *Course Plan* Figure 2.2.

- **17 NASA SEH processes** — you'll touch most of them every day (NASA SEH §6).
- **The current ECSS phase the team is in** — which sets margin policy (ECSS-M-ST-10C §6).

Formulas, charts, rules of thumb

ECSS margin policy by phase (mass example, all margins reduce phase-by-phase):

Table A.1 — ECSS margin policy by lifecycle phase for mass, power, link, and delta-v allocations.

Phase	Mass	Power	Link	Δv
A	+44 %	+25 %	6 dB	+25 %
B	+24 %	+20 %	4 dB	+15 %
C	+13 %	+10 %	3 dB	+10 %
D	+5 %	+5 %	3 dB	+5 %

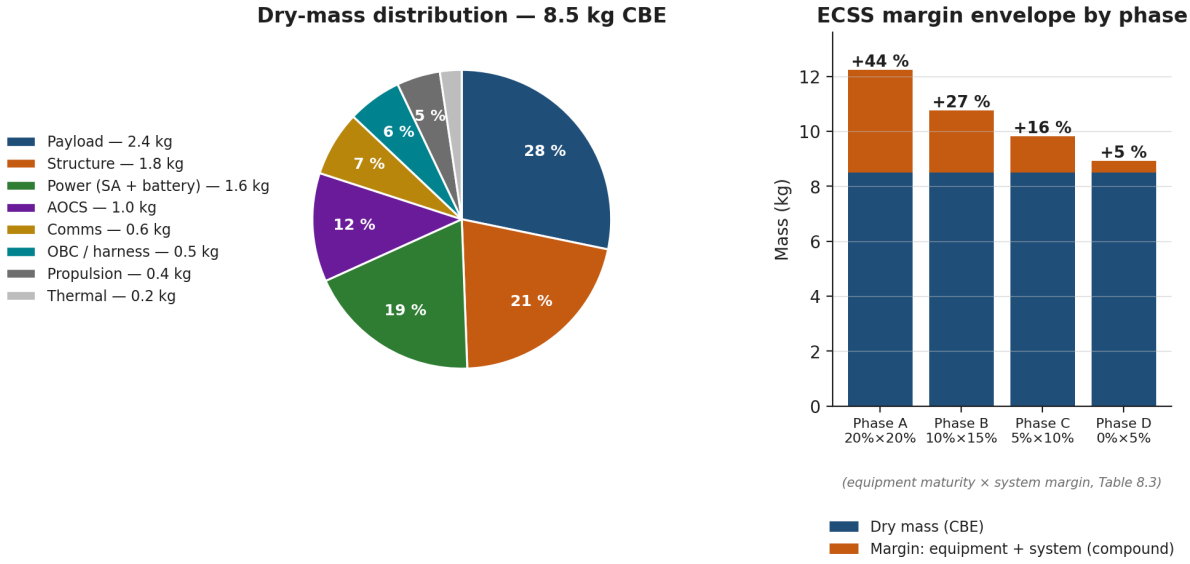


Figure SE.1 — Mass distribution and the ECSS margin schedule. Your job is to keep total mass under the appropriate column for the current phase.

Common pitfalls

> **Watch out.** Resolving conflicts by averaging the two domains' requests *without* re-baselining the affected requirements is the classic systems-engineering failure mode. Always update the requirement before accepting the trade.
>
> **Watch out.** Margin is not slack. Margin is uncertainty insurance. If you give the margin to anyone — to AOCS for an extra wheel, say — record it as a depletion of margin in the change log, not as a new allocation.

References

- NASA SEH (SP-2016-6105 Rev 2) — https://www.nasa.gov/reference/systems-engineering-handbook/

- ECSS-M-ST-10C — https://ecss.nl/
- Wertz et al., *Space Mission Engineering: The New SMAD* (2011), Ch. 2.
- Larson & Wertz, *Space Mission Analysis and Design* (SMAD), Ch. 1, 22.

Mission Analyst

You own the mission's relationship with physics: the orbit, the ground track, the lighting conditions, the contact times.

Decisions you own

1. **Orbit type and altitude.** LEO / SSO / MEO / GEO / HEO / lunar. Altitude to balance science return, lifetime, drag, radiation, and launch cost.
2. **Inclination.** Drives latitude coverage, ground-track repeat, and (with altitude) Sun-synchronous condition.
3. **LTAN / LTDN** for SSO missions — drives β-angle envelope and eclipse season.
4. **Ground-station network.** Where to downlink, latency, coverage gaps.
5. **Maneuver plan.** When to phase, how to maintain station, when to de-orbit.

What you need to know first

- The mission's **science target** — sun-glint avoidance? polar coverage? continuous Earth observation?
- **Latitude band** of interest (e.g. polar Arctic monitoring forces an inclined or polar orbit).
- **Lifetime target** and the regulatory framework (FCC 5-year / IADC 25-year).
- **Launch availability** — a target launcher constrains both inclination and altitude.

Formulas, charts, rules of thumb

Kepler period for a circular orbit (verified across known orbits in Figure MA.1 below):

$$T = 2\pi\sqrt{a^3/\mu}$$

where μ_Earth = 398 600.4418 km³/s² and a = R_Earth + h.

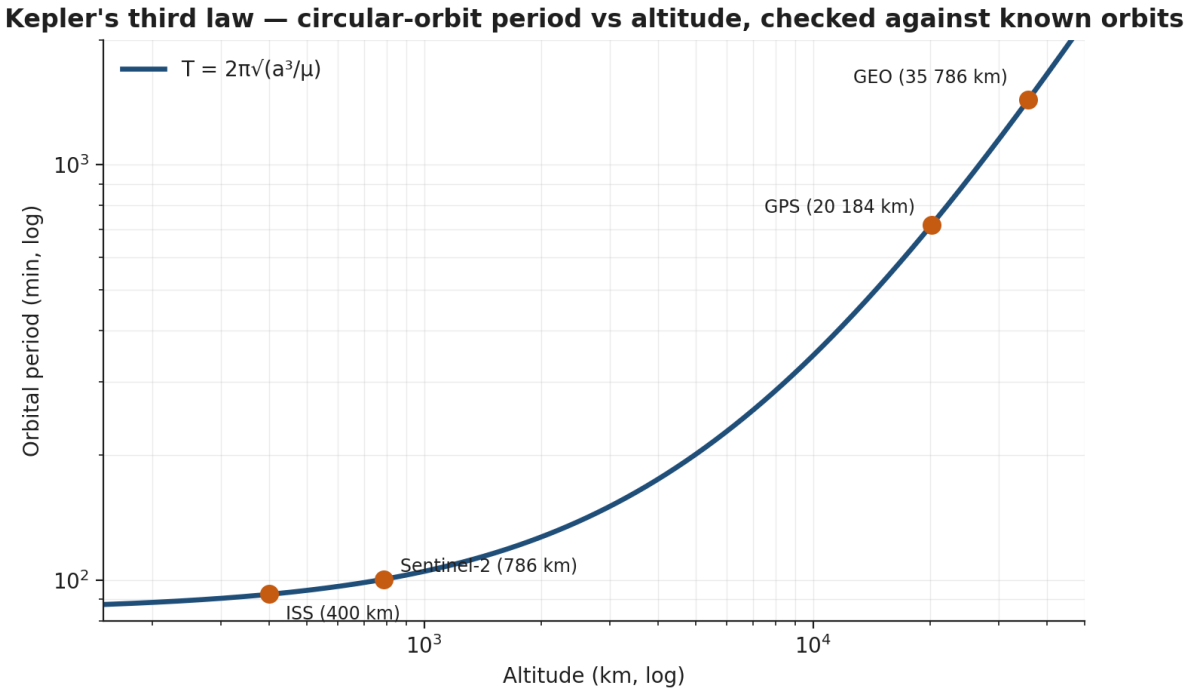


Figure MA.1 — Kepler's Third Law. Plotted markers (ISS-class, Sentinel-2, GPS, GEO) sit on the analytical curve to better than 0.5 % — which is the expected agreement once Earth oblateness is included separately.

Sun-synchronous condition — pick i so that J_2-driven nodal regression matches Earth's mean motion around the Sun (~0.9856 deg/day). At 600 km altitude, i ≈ 97.79°.

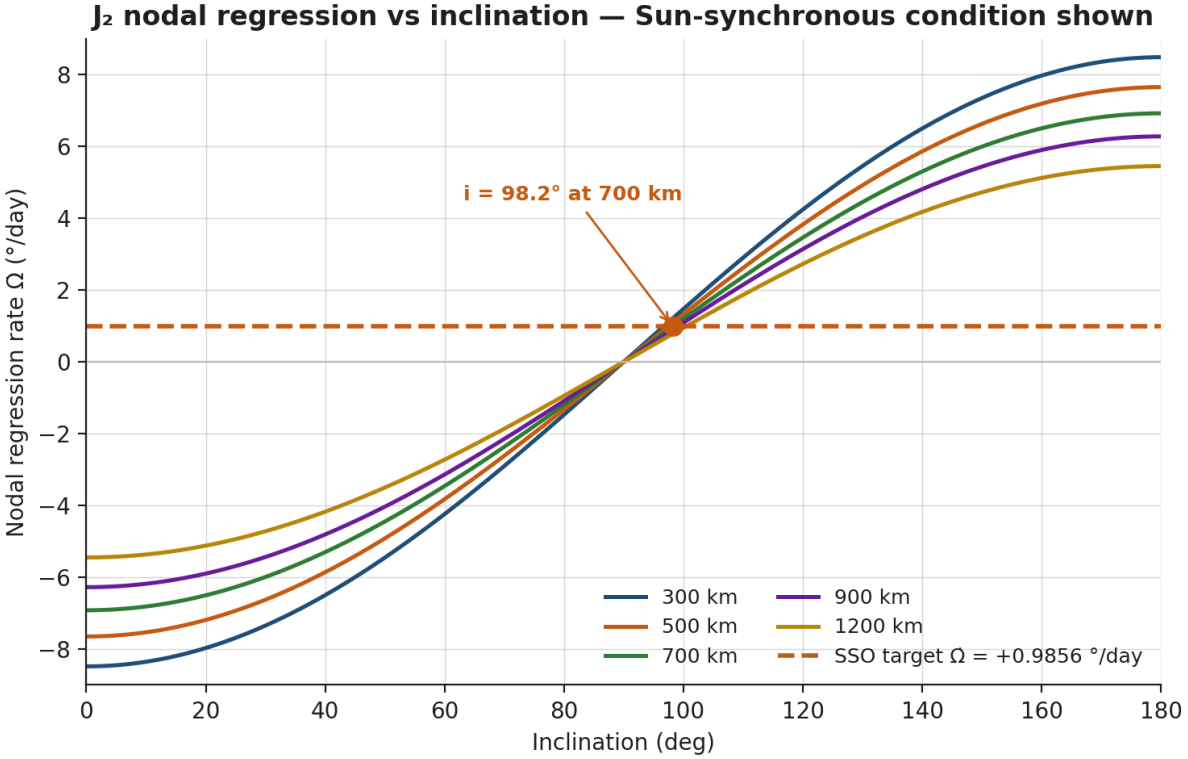


Figure MA.2 — J_2 nodal regression rate vs inclination. The dashed horizontal is the SSO target. The intersection with the 700-km curve gives the SSO inclination at that altitude.

Eclipse fraction for a circular orbit, valid for |β| < β* = arcsin(R/(R+h)):

$$f_{\text{eclipse}} = \frac{1}{\pi}\arccos\left(\frac{\sqrt{h^2+2Rh}}{(R+h)\cos\beta}\right)$$

For β > β*, the orbit is fully sunlit.

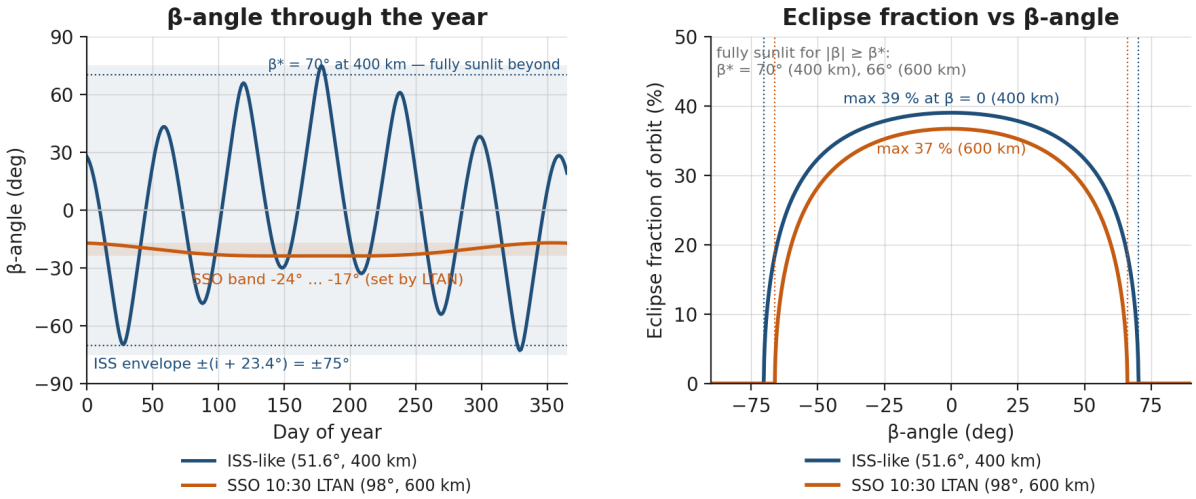


Figure MA.3 — β-angle envelope over a year and eclipse fraction analytical curves.

Orbital lifetime rule of thumb (drag-dominated LEO):

$$\tau_{\text{years}} \approx \frac{h-200}{30}\cdot\frac{m/A}{50}$$

Orbital lifetime vs altitude — FCC 5-yr and IADC 25-yr lines

Figure MA.4 — Orbital lifetime envelope vs altitude. Below ~600 km a 6U CubeSat (m/A ~ 50 kg/m²) decays inside 25 years; below ~ 425 km it can decay inside 5 years (the FCC rule).

Common pitfalls

> **Common mistake.** Picking 500 km thinking it gives a 25-year lifetime — it doesn't, for a typical 6U. Always confirm against the rule-of-thumb chart and an STK or GMAT propagation.
>
> **Watch out.** Sun-glint angle for ocean-colour imaging changes with both season (β) and longitude. If your mission cares about sun-glint, build a year-long β/Sun-azimuth heatmap before committing to LTAN.

References

- Curtis, *Orbital Mechanics for Engineering Students* (3rd ed.).
- Vallado, *Fundamentals of Astrodynamics and Applications* (4th ed.).
- SMAD (orbit selection).
- Wertz, *Mission Geometry: Orbit and Constellation Design and Management* (2001).
- AGI STK — https://www.agi.com/products/stk/.
- NASA GMAT (open source) — https://software.nasa.gov/software/GSC-17177-1.

Payload Engineer

You own the science. The mission exists to do what your payload does, and every other subsystem ultimately serves yours.

Decisions you own

1. **Payload concept & sensor type.** Optical / SAR / hyperspectral / radiometer / RF receiver / etc.
2. **Aperture & spectral bands.** Drives resolution, SNR, mass.
3. **Operating concept.** Pushbroom vs whiskbroom vs framing, continuous vs target-cued.
4. **Data rate & data volume per orbit.** This drives Comms and OBDH.
5. **Calibration plan.** On-orbit sources, ground references, cadence.

What you need to know first

- **Mission objective measurable** (MoP). For ocean colour you need certain bands at certain SNR; for SAR you need a certain ground-resolution and revisit; for AIS you need a certain detection probability.
- **Orbit altitude** — drives slant range, GSD, coverage.
- **Mass and power available** — typically 30–50 % of the platform.

Formulas, charts, rules of thumb

GSD diffraction limit for an unobstructed aperture:

$$\mathrm{GSD}_{\mathrm{diff}} = \frac{1.22 \lambda \mathrm{h}}{\mathrm{D}}$$

GSD pixel limit:

$$\mathrm{GSD}_{\mathrm{pix}} = \frac{\mathrm{p} \cdot \mathrm{h}}{\mathrm{f}}$$

where p is pixel pitch and f is focal length. The achievable GSD is the larger of the two.

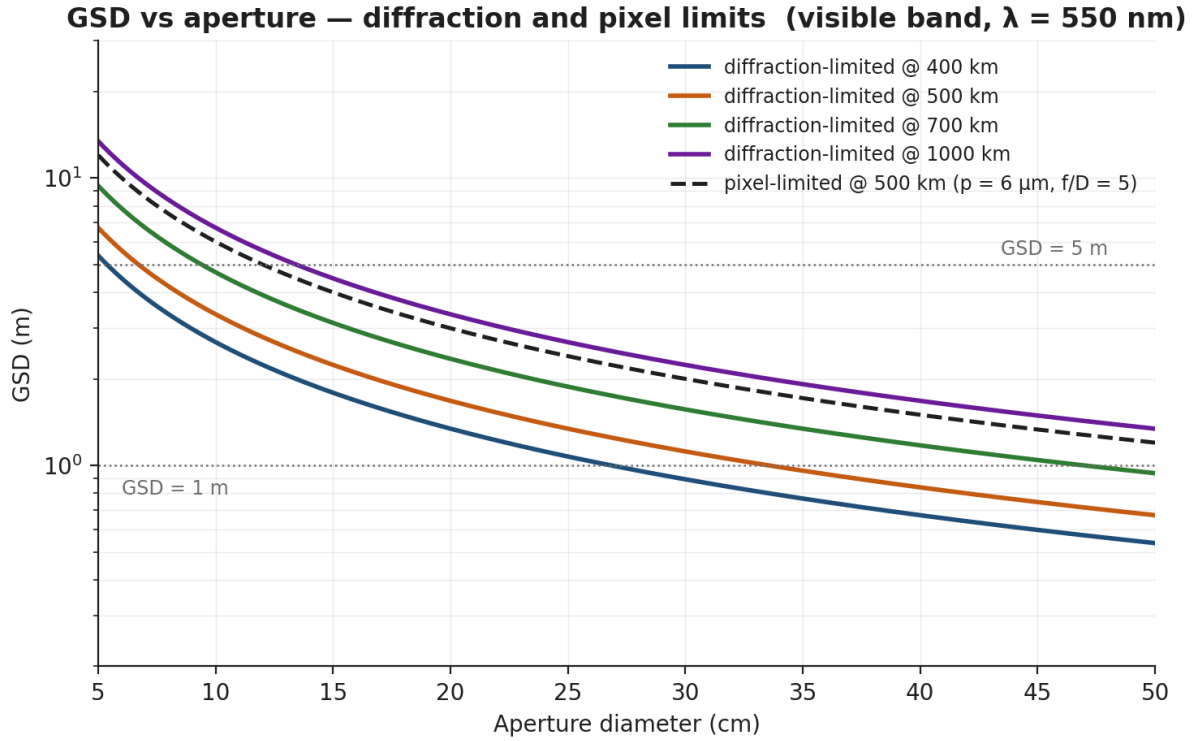


Figure PL.1 — GSD vs aperture for visible imagers (λ = 550 nm). The dashed line is the pixel-limit at 500 km for p = 6 µm and f/D = 5.

SAR antenna minimum length:

$$\mathrm{L}_{\mathrm{ant,\,min}} = 2 \cdot \delta_{\mathrm{azimuth}}$$

i.e. the antenna along-track length must exceed twice the desired azimuth resolution.

Common pitfalls

> **Common mistake.** Sizing the payload aperture for the diffraction limit while the detector pixels are too coarse to use it. Match the two — pixel-limited and diffraction-limited GSDs should be comparable.
>
> **Watch out.** Data volume per orbit grows with the aperture in two ways at once: more pixels, faster line rate. Check the daily download capacity (Figure PL.2) before committing.

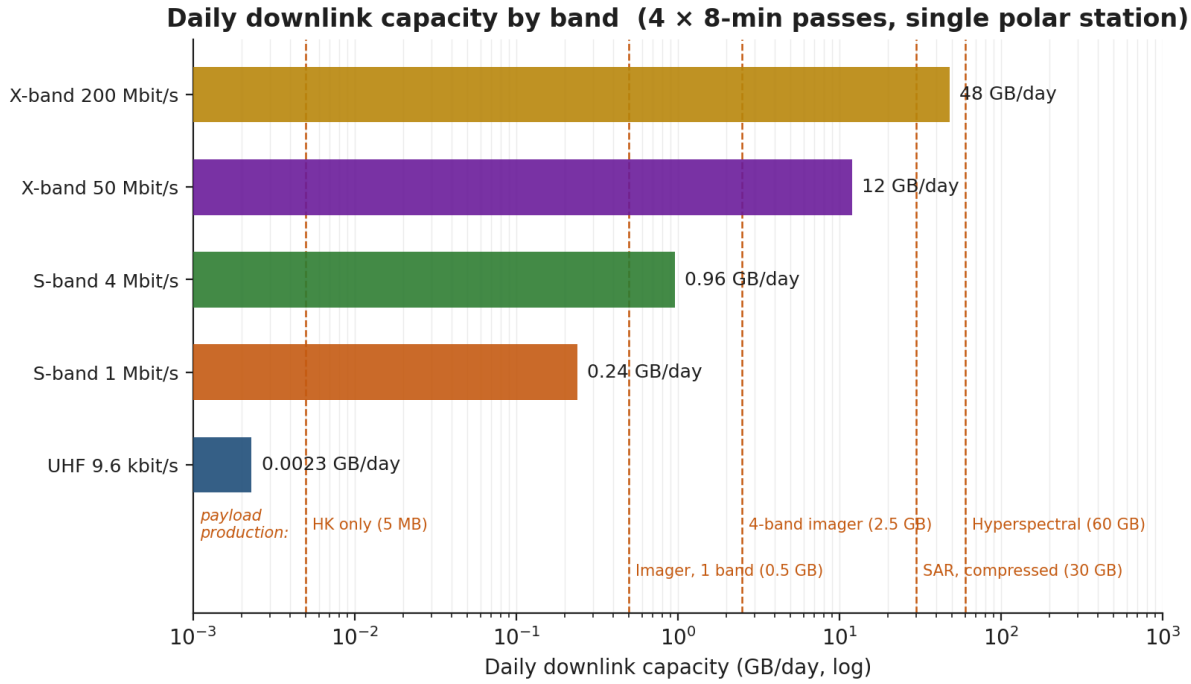


Figure PL.2 — Daily downlink capacity by band, against typical payload data volumes. UHF closes for housekeeping only; S-band for imaging; X-band for SAR / hyperspectral.

References

- SMAD (payload).
- Cracknell, *Introduction to Remote Sensing* (3rd ed.).
- Rees, *Physical Principles of Remote Sensing* (3rd ed.).
- Cumming & Wong, *Digital Processing of Synthetic Aperture Radar Data*.
- ECSS-E-ST-10-04C — Space environment for payload sizing — https://ecss.nl/.

Power Engineer

You own the orbit-by-orbit energy balance.

Decisions you own

1. **Solar-array architecture.** Body-mounted vs deployable; gimballed vs fixed; technology (triple-junction GaAs vs Si).
2. **Solar-array area** — at end-of-life, accounting for degradation, β-angle range, and pointing.
3. **Battery capacity and DoD.** Cycle life vs eclipse depth.
4. **Bus voltage and PCDU architecture.** Single-bus vs dual-bus; regulated vs unregulated; MPPT topology.
5. **Power-mode duty cycle.** Maps to ConOps.

What you need to know first

- **Orbit** — drives β-angle envelope and eclipse fraction.
- **Orbit-average load** from all subsystems (Watts).
- **Mission lifetime** — drives cycle count and EOL degradation.

Formulas, charts, rules of thumb

Solar-array sizing (first cut):

$$A_{SA} = \frac{P_{avg} \cdot \left(1 + \frac{f_{eclipse}}{f_{sun}\,\eta_{dis}}\right)}{S_0\,\eta_{cell}\cos\beta\,(1 - D_{deg})}$$

where S_0 = 1361 W/m², η_cell ≈ 0.30 for triple-junction GaAs, D_deg ≈ 10 % for 5-year LEO end-of-life.

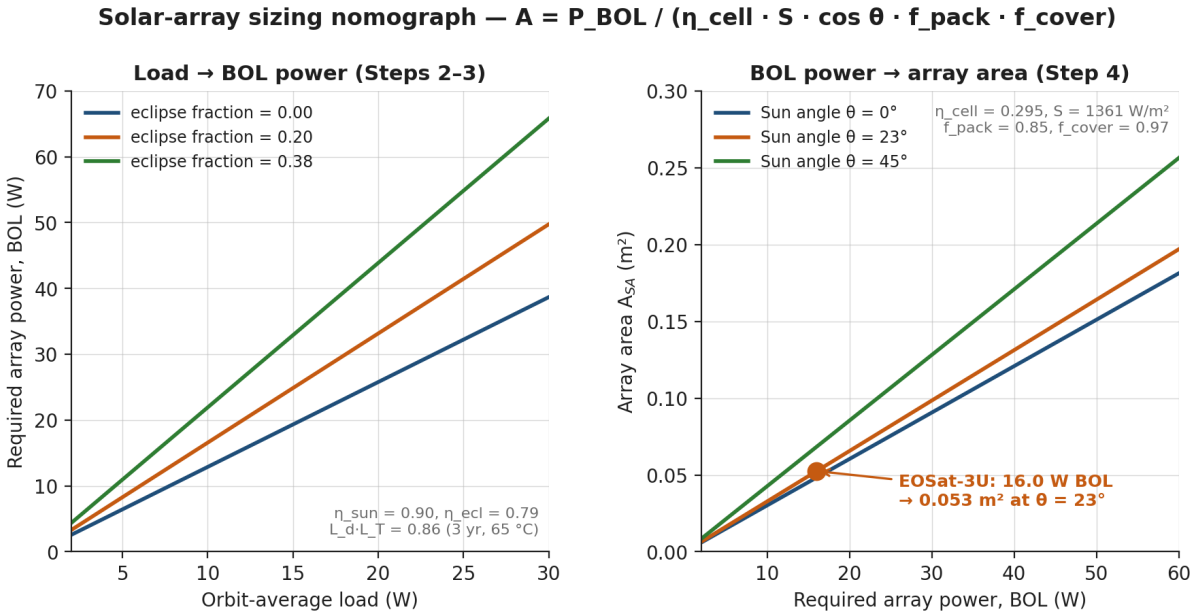


Figure PW.1 — Solar-array nomograph at η = 30 %, 10 % EOL degradation, β = 20°, η_dis = 85 %. A 6U at 15 W average needs ~870 cm² at end-of-life.

Battery sizing — usable energy:

$$E_{usable} = DoD \cdot E_{nominal} \quad ; \quad E_{nominal} = \frac{P_{avg} \cdot t_{eclipse}}{\eta_{dis} \cdot DoD}$$

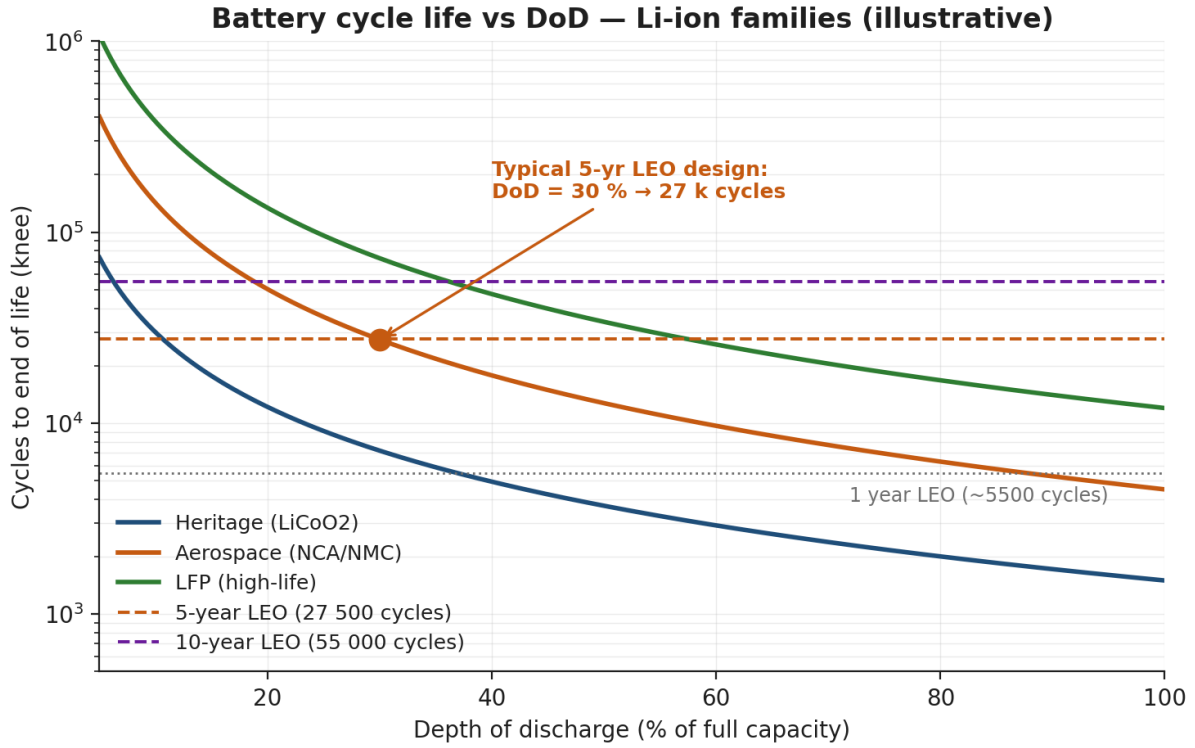


Figure PW.2 — Battery cycle life vs DoD for Li-ion families. A 30 % DoD design point gives the cycles needed for a 5-year LEO mission with margin. Heritage chemistry like $LiCoO_2$ requires deeper de-rating; LFP gives more headroom but is less mass-dense.

Power profile across one orbit — Figure PW.3 shows the typical generation/load picture.

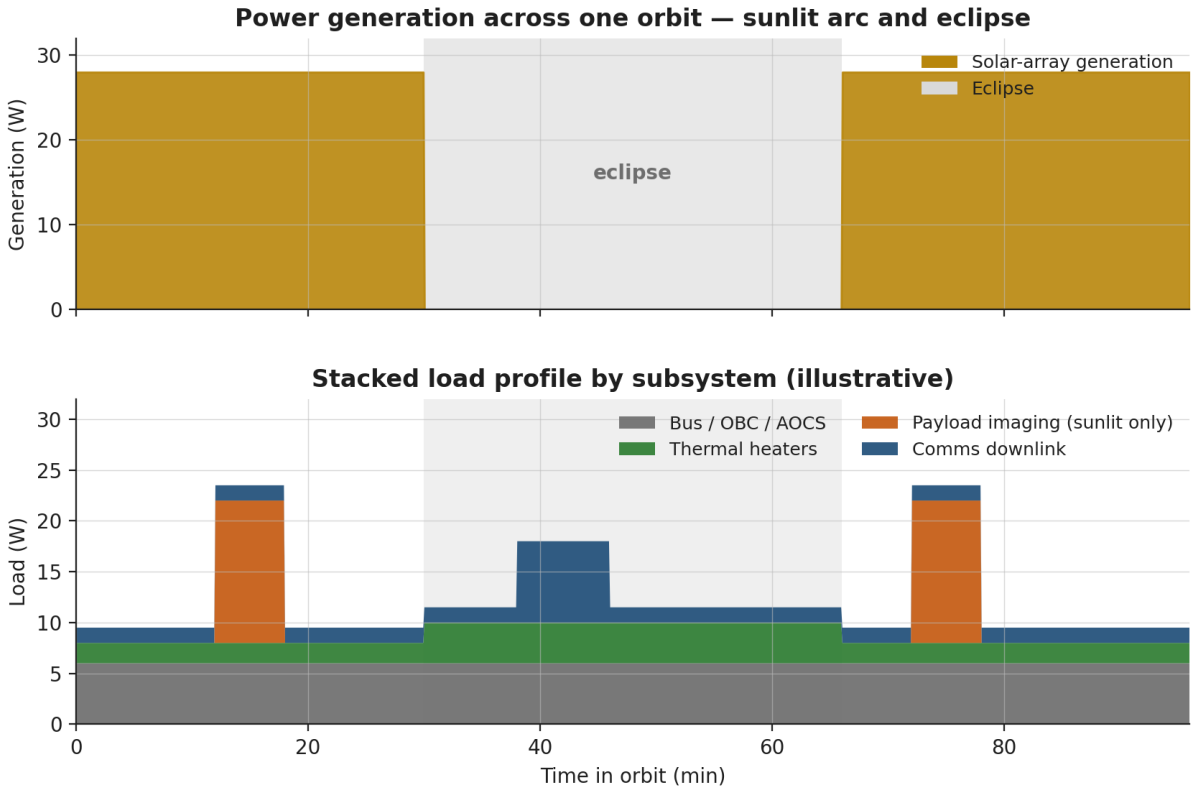


Figure PW.3 — Generation (top) and stacked load profile (bottom) across one orbit.

Common pitfalls

> **Watch out.** Sizing the array for begin-of-life will starve the battery at end-of-life. Always compute at EOL with realistic degradation (e.g. 10 % over 5 years for triple-junction GaAs in LEO).
>
> **Common mistake.** Forgetting that the array only generates when the cells see the Sun. β = 0° at season change means the array sees zero illumination at orbit noon — design for the worst case in the year.

References

- ECSS-E-ST-20C — *Electrical and electronic* — https://ecss.nl/
- SMAD (power).
- Patel, *Spacecraft Power Systems*.
- Brown, *Spacecraft Propulsion and Power Subsystems*.

AOCS Engineer

You own the spacecraft's pointing.

Decisions you own

1. **Pointing requirement** — knowledge vs control vs stability, per axis.
2. **Sensor suite** — Sun sensor / magnetometer / star tracker / GPS / IMU.
3. **Actuator suite** — magnetorquer / reaction wheel / monopropellant / cold-gas.
4. **Modes & transitions** — sun-acquisition, detumble, fine pointing, safe.
5. **Momentum management plan** — how often, with what.

What you need to know first

- **Pointing requirement** from Payload (e.g. 0.05° 1σ for a pushbroom imager).
- **Disturbance environment** at the chosen orbit (drag at 400 km vs 700 km vs SSO).
- **Spacecraft inertia tensor** — Structures gives you this.

Formulas, charts, rules of thumb

Pointing budget — RSS contributors:

$$\sigma_{\text{pointing}}^2 = \sigma_{\text{knowledge}}^2 + \sigma_{\text{control}}^2 + \sigma_{\text{alignment}}^2$$

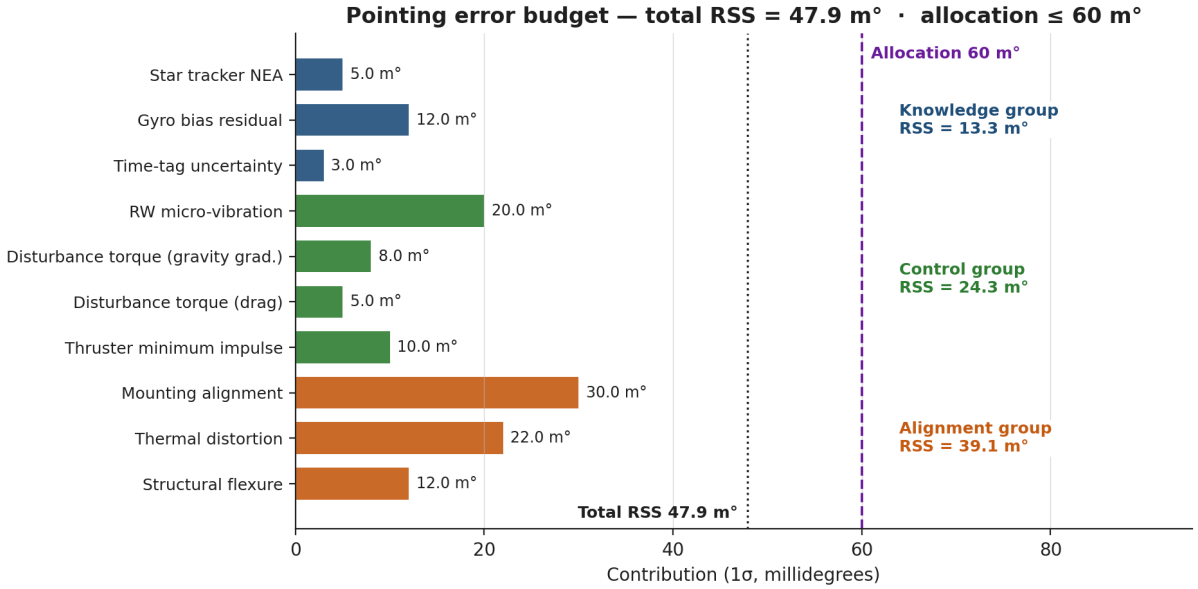


Figure AC.1 — Pointing-error budget tree. Knowledge, control, and alignment groups RSS into the total. The dashed line is the mission allocation.

Reaction-wheel sizing — momentum capacity:

$$\mathrm{H_{wheel}} \geq \mathrm{T_{disturbance}} \cdot \mathrm{t_{between\ desat}} \cdot \mathrm{k_{margin}}$$

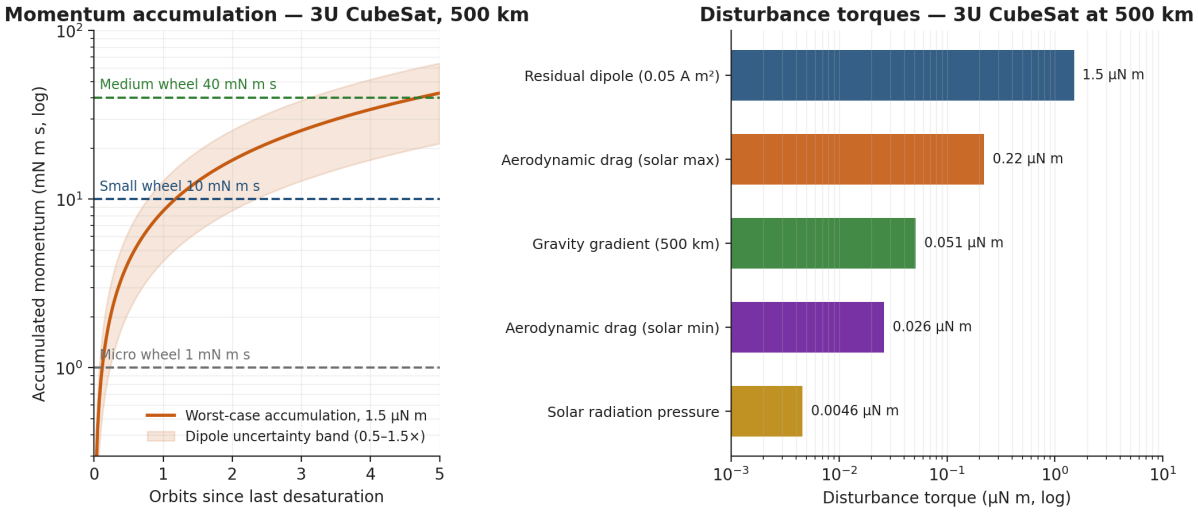


Figure AC.2 — Momentum accumulation per orbit (left) and disturbance-torque order-of-magnitude (right). At 500 km the residual magnetic dipole typically dominates with ~1.5 µN·m, accumulating ~ 10–40 mN·m·s over a few orbits.

Common pitfalls

> **Watch out.** Quaternion normalisation drift in flight software — even a 1e-6 drift per second adds up. Renormalise every cycle.
>
> **Common mistake.** Designing for fine pointing without sizing the wheel for desat unloading. Plan the desat strategy at the same time as the wheel.

References

- ECSS-E-ST-60-30C — *Satellite attitude and orbit control system (AOCS) requirements* — https://ecss.nl/.
- Markley & Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*.
- Sidi, *Spacecraft Dynamics and Control*.
- SMAD

Thermal Engineer

You keep every component inside its operating range across the hot, cold, and transient cases.

Decisions you own

1. **Hot-case / cold-case envelope** — define them in terms of β-angle, season, eclipse, internal heating.
2. **Surface finishes (α, ε)** for each external panel and radiator.
3. **MLI vs heaters** — passive insulation vs active control.
4. **Heat-rejection topology** — radiator location and area.
5. **Thermal interfaces** — conductive paths to and from each component.

What you need to know first

- **Operating temperature ranges** for every component (battery is the strictest, typically −5 to +25 °C charging).
- **Eclipse fraction & duration** from Mission Analyst.
- **Internal dissipation map** from Power.

Formulas, charts, rules of thumb

Radiative equilibrium for a flat plate:

$$T_{eq} = \left(\frac{\alpha_s S + \epsilon_{IR} \phi_{IR}}{\sigma \, \epsilon_{IR}} \right)^{1/4}$$

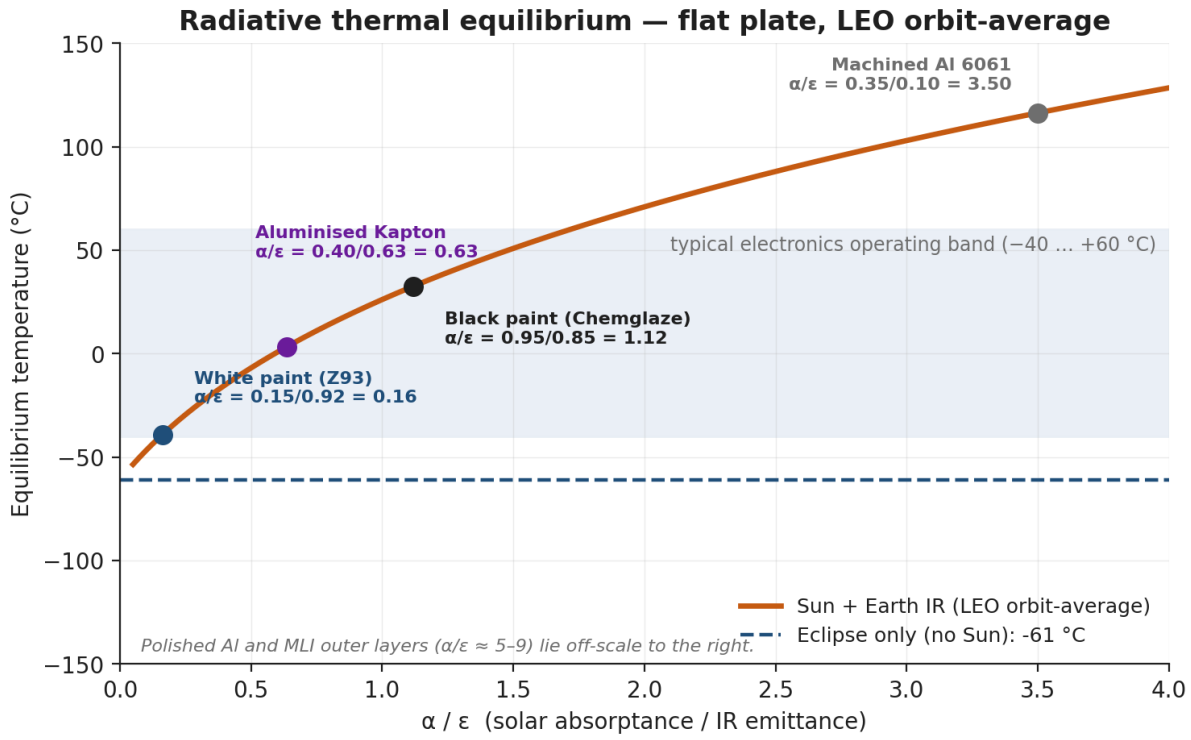


Figure TH.1 — Equilibrium temperature vs α/ε for a flat plate at LEO, with common surface finishes marked.

Common pitfalls

> **Common mistake.** Using α and ε at room temperature for IR calculations — at -100 °C, ε of many materials shifts. Cite the source data temperature.
>
> **Watch out.** MLI is rated by effective ε *not by base material. Use* ε ≈ 0.03 for 10-layer MLI; less for fewer layers.

References

- ECSS-E-ST-31C — *Thermal control* — https://ecss.nl/
- Gilmore, *Spacecraft Thermal Control Handbook*, Vol. I & II.
- SMAD
- ESATAN-TMS — https://www.esatan-tms.com/.

Structures Engineer

You make sure the spacecraft survives launch and on-orbit, and that everyone else's hardware is mechanically interfaced.

Decisions you own

1. **Primary structure topology** — monolithic, panel-and-rib, or 3D-printed.
2. **Material selection** — Al, Ti, CFRP, sandwich.
3. **Mass-budget closure with margin.**
4. **Launch-load verification path** — analysis (FEA), test, or both.

5. **Mechanical interfaces** — fastener pattern, deployer compatibility (ISIPOD / EXOpod / CSD).

What you need to know first

- **Launcher environment** (qualification levels for sine, random vibration, shock).
- **Component mass and CG** — every subsystem provides this.
- **Cal Poly CDS Rev 14** envelope for CubeSats.

Formulas, charts, rules of thumb

Margin of safety:

$$\mathrm{MoS} = \frac{\sigma_{\mathrm{allow}}}{\mathrm{FoS} \cdot \sigma_{\mathrm{applied}}} - 1$$

For Al 7075-T6 yield: σ_yield = 503 MPa, FoS_yield = 1.25.

Modal analysis — first mode > 100 Hz typical CubeSat requirement.

Common pitfalls

> **Watch out.** Forgetting fastener and adhesive mass — they typically add 5–10 % on top of the structural mass.
>
> **Common mistake.** Sizing for static loads and missing the dynamic amplification factor (random vibration). Apply Q-factor (typically 5–10) at resonance.

References

- ECSS-E-ST-32C — *Structures* — https://ecss.nl/.
- Sarafin, *Spacecraft Structures and Mechanisms*.
- Cal Poly CDS Rev 14 — https://www.cubesat.org/cds-announcement.

Propulsion Engineer

You own the Δv. (Even if your CubeSat has none, you decide that.)

Decisions you own

1. **Need for propulsion** — drag make-up? deorbit? phasing? formation? rendezvous?
2. **Propellant choice** — cold gas / monoprop / bipropellant / electric (Hall, ion, PPT, FEEP).
3. **Thruster sizing** — thrust vs Isp vs power.
4. **Total propellant mass** — Tsiolkovsky.
5. **Plume impingement & contamination plan.**

What you need to know first

- **Δv budget** — sum of all required maneuvers with margin.
- **Mission duration** — drives cumulative Δv for drag-dominated orbits.
- **Available power** — drives EP feasibility (a CubeSat at 20 W cannot run a 200 W Hall thruster).

Formulas, charts, rules of thumb

Tsiolkovsky:

$$\Delta v = I_{sp}\, g_0 \ln \frac{m_0}{m_f}$$

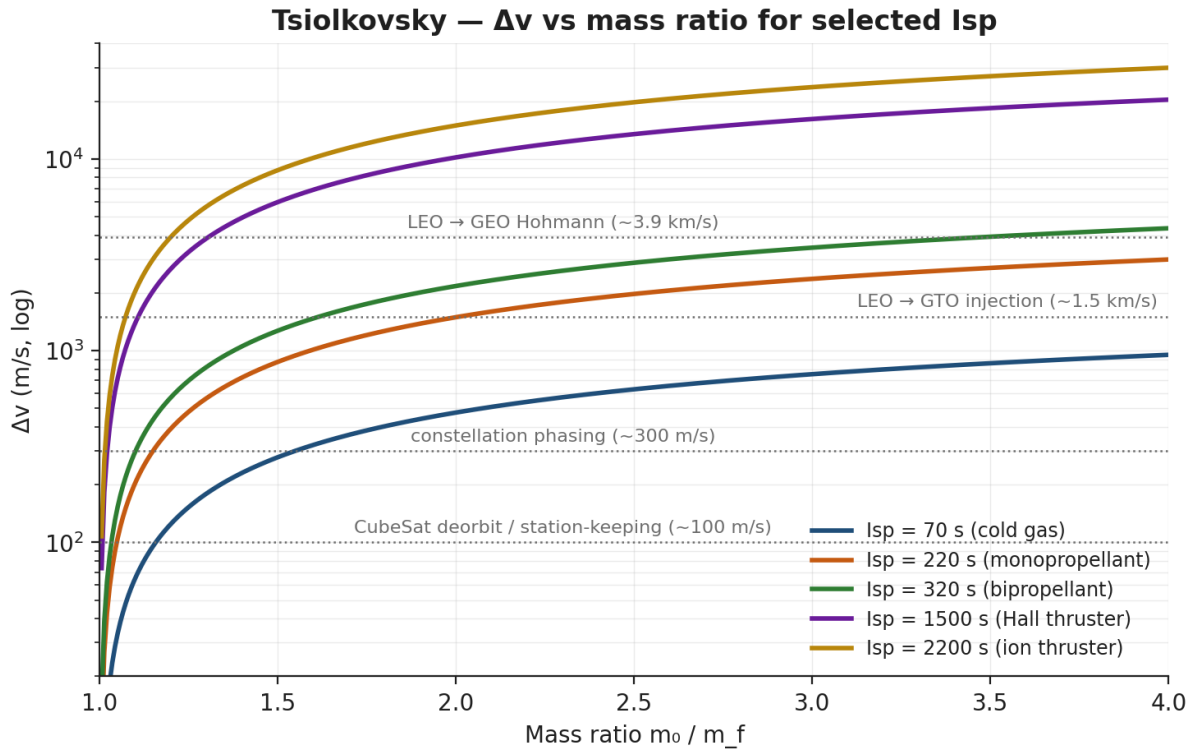


Figure PR.1 — Δv vs mass ratio at five characteristic Isp values. Horizontal lines show typical mission Δv budgets.

Common pitfalls

> **Watch out.** Isp is a function of inlet conditions and nozzle expansion ratio. Quote the value at the operating point you'll actually fly.
>
> **Common mistake.** Confusing total mass (m_0) with dry mass (m_f) when applying Tsiolkovsky.

References

- Sutton & Biblarz, *Rocket Propulsion Elements* (9th ed.).
- ECSS-E-ST-35C — *Propulsion general* — https://ecss.nl/.
- Goebel & Katz, *Fundamentals of Electric Propulsion*.

Communications Engineer

You move the bits — both ways.

Decisions you own

1. **Frequency band** — UHF / S / X / Ka. Drives licensing, antenna size, FSPL.

2. **Modulation and coding.**
3. **Antenna pattern** — high-gain dish vs patch vs deployable.
4. **Ground-station network** — single GS vs network.
5. **Link margin allocation** at each phase.

What you need to know first

- **Daily data volume** from Payload.
- **Latency requirement** — if any.
- **Pointing capability** from AOCS (drives antenna pattern needs).
- **Regulatory constraints** for the chosen band.

Formulas, charts, rules of thumb

Free-space path loss:

$$\mathrm{L_{FS}} = 20\log_{10}\left(4\pi \mathrm{d}/\lambda\right)\ \mathrm{dB}$$

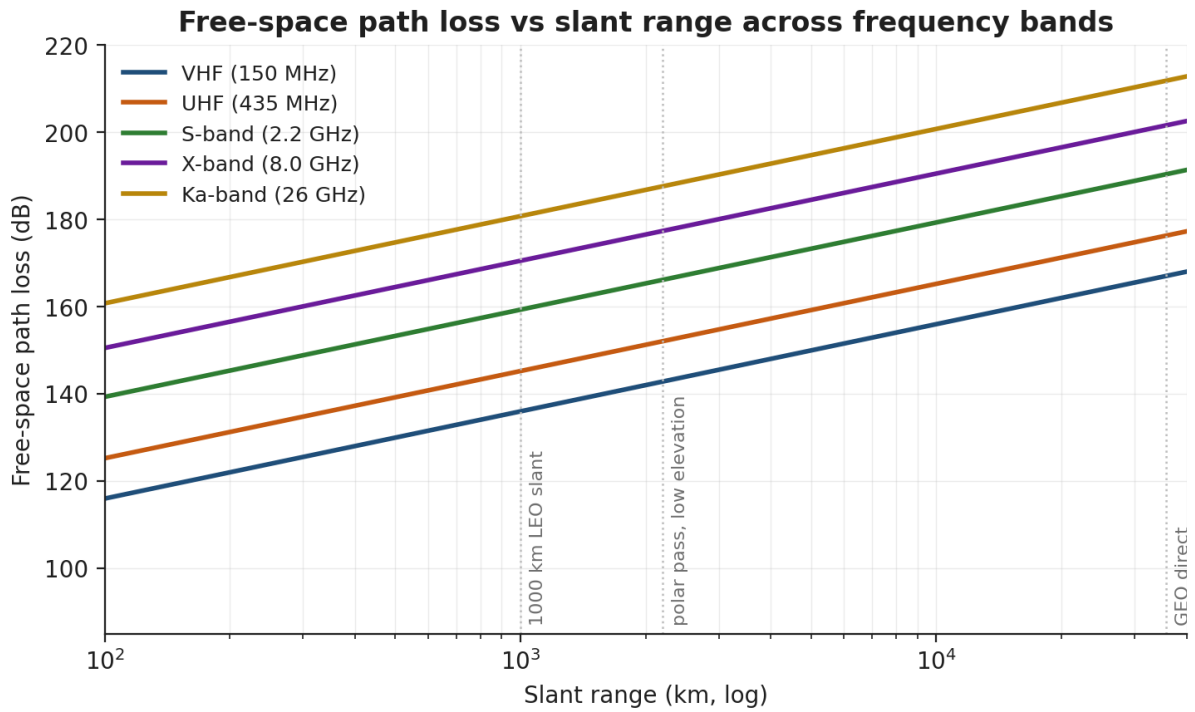


Figure CO.1 — FSPL across bands. At 2.2 GHz (S-band) over a 1500 km slant, FSPL ≈ 163 dB; at 26 GHz (Ka-band), 184 dB.

Link budget closure:

$$\frac{\mathrm{C}}{\mathrm{N_0}} = \mathrm{EIRP} - \mathrm{L_{path}} - \mathrm{L_{atm}} - \mathrm{L_{point}} + \mathrm{G/T} - \mathrm{k}\ [\mathrm{dB\cdot Hz}]$$

with k = -228.6 dBW/(K·Hz).

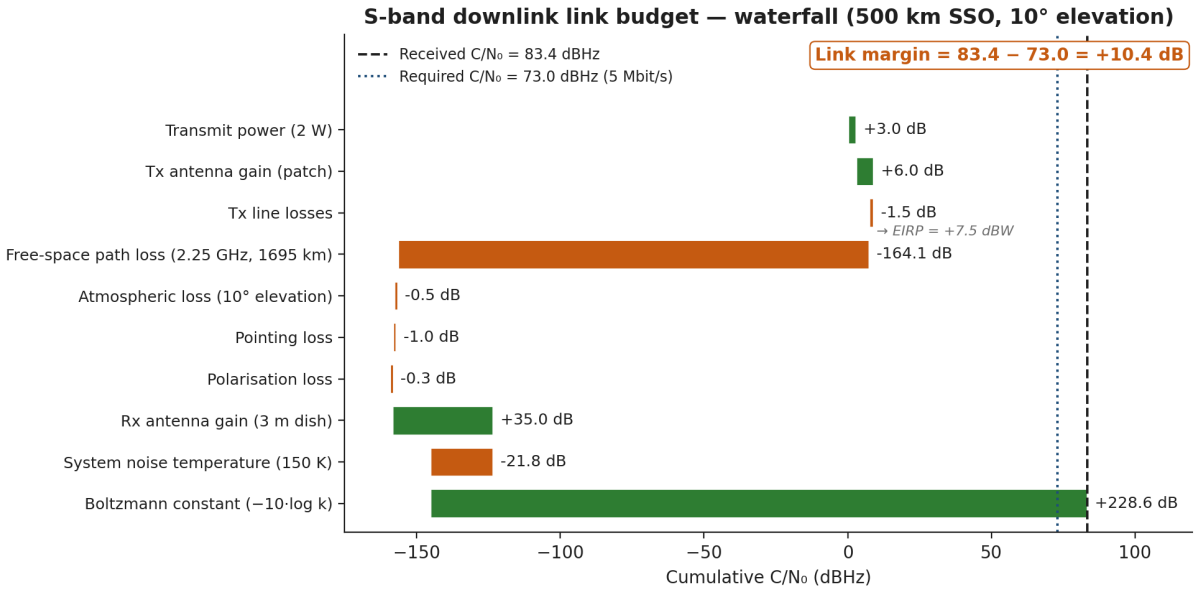


Figure CO.2 — A canonical S-band waterfall. Read it left-to-right: gains add, losses subtract, the bottom-right is required C/N_0 and margin.

Eb/N_0 → BER — Figure CO.3 shows the standard curves.

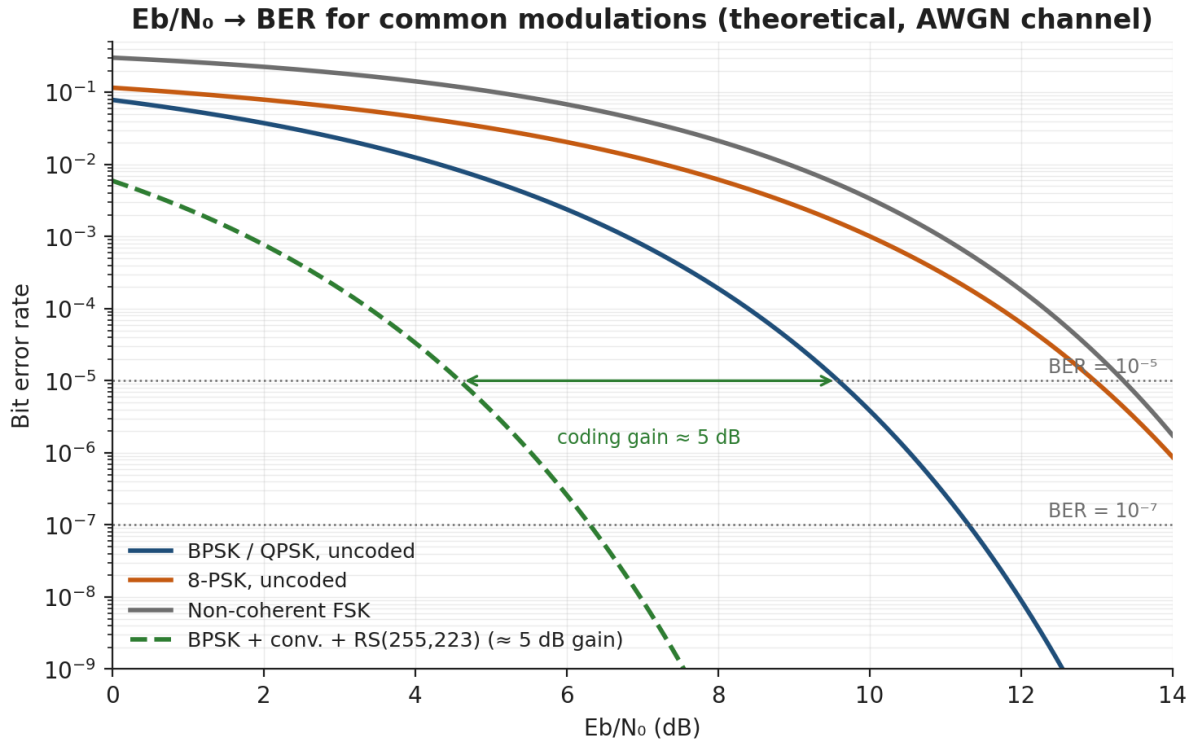


Figure CO.3 — Eb/N_0 vs BER for BPSK, QPSK, 8-PSK, FSK, and a typical concatenated coding scheme.

Antenna patterns — high-gain vs low-gain.

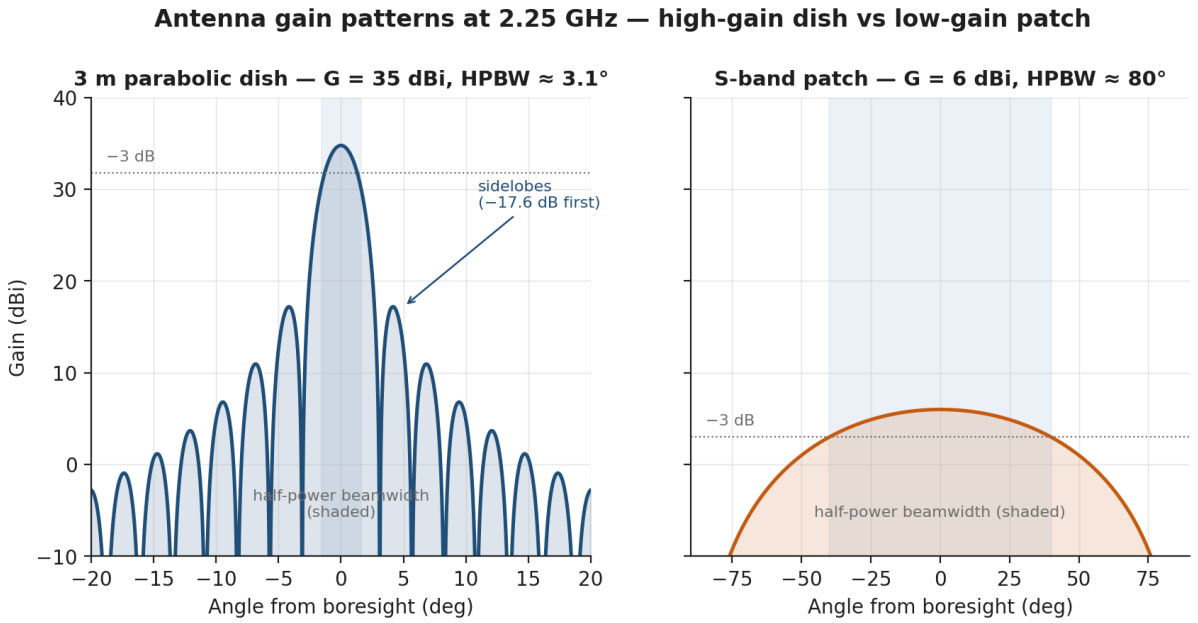


Figure CO.4 — High-gain dish (33 dBi, narrow) vs S-band patch (~6 dBi, wide). Dish needs accurate AOCS pointing; patch tolerates ~30° pointing error.

Common pitfalls

> **Watch out.** Pointing loss is not negligible. For a 1° HPBW dish, a 0.3° pointing error costs ~1 dB; at 0.5° it costs ~3 dB.
>
> **Common mistake.** Forgetting Doppler. A 2.2 GHz S-band link from a 7.5 km/s LEO has ±55 kHz Doppler at horizon — your receiver tracking has to follow.

References

- ECSS-E-ST-50-05C — RF & modulation — https://ecss.nl/.
- Pratt, Bostian & Allnutt, *Satellite Communications* (3rd ed.).
- ITU-R Recommendations — https://www.itu.int/en/ITU-R/.

OBDH / Software Engineer

You own the flight computer, the flight software, the telemetry list, and FDIR.

Decisions you own

1. **OBC architecture** — single SBC or redundant pair; SoC vs discrete.
2. **Bus topology** — CAN / SpaceWire / I²C / SPI.
3. **Memory architecture** — RAM, flash, mass-memory, EDAC scheme.
4. **FSW architecture** — mode manager, ADCS control loop, TM/TC handler, FDIR rules.
5. **PUS service set** — which CCSDS PUS services to implement.

What you need to know first

- **TM/TC volume** from each subsystem.
- **Real-time loop rates** required by AOCS.
- **Storage volume** for the longest no-contact gap (mass-memory sizing).

Formulas, charts, rules of thumb

- **Mass memory sizing:** longest contact gap × peak data rate × margin (1.5–2×).
- **Watchdog timer:** typical 30 s for CubeSat OBC; less for high-criticality FSW threads.
- **EDAC overhead:** Hamming SECDED ~ 12.5 % for 64-bit words.

Common pitfalls

> **Watch out.** Telemetry list overruns are the #1 OBDH late defect. Lock the list at SRR; treat any addition as a change-control item.

References

- CCSDS PUS — https://public.ccsds.org/Pubs/660x0g3.pdf.
- ECSS-E-ST-40C — *Software* — https://ecss.nl/.
- ECSS-Q-ST-80C — *Software product assurance* — https://ecss.nl/.

Operations Engineer

You will fly the spacecraft after the team builds it, taking the design into the operations chapters in Part 2.

Decisions you own

1. **Operations concept** — staffing, automation level, shift structure.
2. **Pass plan** — which procedures, in what order, every contact.
3. **Anomaly response tree** — who calls what, when.
4. **Telecommand / telemetry definition** — coordination with OBDH.

What you need to know first

- **Mission ops modes** from ConOps.
- **Ground-station coverage** from Mission Analyst.
- **Procedure inventory** from the mission's procedure set (LEOP / commissioning / nominal / contingency / emergency).

Common pitfalls

> **Watch out.** Voice-loop discipline is the silent killer. Cross-talk that sounds harmless in training breaks down under real-time pressure.

References

- ECSS-E-ST-70C — *Ground systems and operations* — https://ecss.nl/.

- ECSS-E-ST-70-32C — *Test and operations procedures*.
- NASA *Mission Operations Handbook* — https://nodis3.gsfc.nasa.gov/.

Ground Segment Engineer

You own the architecture between the antenna and the user.

Decisions you own

1. **Ground-station network** — owned, leased, or commercial (Atlas, KSAT, Viasat, AWS Ground Station).
2. **MCS choice** — COSMOS / OpenMCT / Yamcs / commercial.
3. **Archive design** — L0 → L1 → L2 → distribution latency.
4. **Pass-planning tool**.
5. **Cyber-security architecture.**

What you need to know first

- **Daily data volume** — drives bandwidth contracts.
- **Latency requirement** — drives station network density.
- **Mission classification** — defence missions need different architectures than civil.

Common pitfalls

> **Watch out.** AWS Ground Station's pricing model can flip economics between band choices — do the math at the chosen data rate.

References

- CCSDS Mission Operations standards — https://public.ccsds.org/Pubs/.
- KSAT — https://www.ksat.no/.
- AWS Ground Station — https://aws.amazon.com/ground-station/.

Cost & Schedule

You're the team's reality check.

Decisions you own

1. **WBS structure** (NPR 7120.5).
2. **Cost-estimating method** — parametric (SSCM, COMPACT), analogy, bottom-up.
3. **Schedule baseline** — milestones, critical path, float.
4. **Risk-adjusted estimates** — how to roll risk into both cost and schedule.

Common pitfalls

> **Common mistake.** Treating the parametric tool's output as the answer. The output is a starting point; tailor it for your mission and the launch market you're working in.

References

- NPR 7120.5F — https://nodis3.gsfc.nasa.gov/.
- Wertz et al., SME, Ch. 23 (Cost).
- SMAD
- AIAA Cost Modeling — https://www.aiaa.org/.

Compliance / Regulatory Engineer

You own the mission's relationship with the rule-makers: standards tailoring, spectrum, export control, and debris mitigation.

Decisions you own

1. **Applicable-standards matrix.** Which ECSS / NASA standards apply, and how they are tailored for a CubeSat-class mission (ECSS-S-ST-00-02C).
2. **Frequency licensing route.** IARU coordination, national administration filing (e.g. ISED CPC-2-6-02), and ITU notification under RR Article 9.
3. **Export classification.** ITAR / EAR / national equivalent for every component and dataset.
4. **Debris-mitigation compliance.** 25-year IADC, 5-year FCC, casualty-risk assessment, and post-launch COPUOS registration.
5. **Remote-sensing licensing** (e.g. RSSSA filing) where the mission has an imaging capability.

What you need to know first

- **Mission classification** (civil / commercial / defence) — sets the regulatory regime.
- **Frequency bands** chosen by Communications — drives the filing path.
- **Orbit and lifetime** from Mission Analyst — drives debris compliance.

Common pitfalls

> **Watch out.** Licensing lead times routinely exceed hardware lead times. Start the spectrum filing at SRR, not at CDR.
>
> **Common mistake.** Treating export classification as a launch-day problem. A single ITAR-controlled component can block an international launch or a foreign-national team member months earlier.

References

- ECSS-S-ST-00-02C — *Tailoring of standards* — https://ecss.nl/.
- ISED CPC-2-6-02 — spectrum licensing.
- ITU Radio Regulations, Article 9 — https://www.itu.int/en/ITU-R/.
- IADC Space Debris Mitigation Guidelines; FCC orbital-debris rules.

Project Manager

You own schedule, budget, the team, and the gates — the programmatic spine that holds the engineering together.

Decisions you own

1. **Schedule baseline** — milestones, dependencies, and the critical path through SRR / PDR / CDR / FRR.
2. **Budget control** — estimate-at-completion (EAC), estimate-to-complete (ETC), and tracking against the programmatic ceiling.
3. **Resource allocation** — FTE assignment across positions and protection of schedule margin.
4. **Risk acceptance** — owning the risk register and convening risk-review boards (with engineering concurrence).
5. **Review readiness** — when to convene each gate and what evidence package it needs.

What you need to know first

- **WBS and cost model** from Cost & Schedule.
- **Interface-agreement status** between positions (where the schedule risk hides).
- **Stakeholder/customer requirement-change pressure** from the User Representative.

Common pitfalls

> **Common mistake.** Letting early phases consume the schedule margin, leaving none for integration and test where it is most needed.
>
> **Watch out.** Holding a gate review before the engineering products are mature converts a review into a re-work generator. Accepting scope creep without a schedule/cost impact assessment is the same failure in slow motion.

References

- ECSS-M-ST-10C — *Project management* — https://ecss.nl/.
- NPR 7120.5F — https://nodis3.gsfc.nasa.gov/.
- PMI, *PMBOK Guide*.

User Representative

You carry the end-user's needs into the room and keep the design honest about what the mission is actually *for*.

Decisions you own

1. **Data-product requirements** — format, resolution, accuracy, and timeliness of the delivered product.
2. **Service-level agreements** — availability, response time, and data freshness.
3. **User-access method** — API, web portal, or direct download.
4. **Utilisation plan** — who the users are (research / commercial / public) and how each consumes the product.

What you need to know first

- **Mission objective** as the user actually expresses it (a need, not an implementation).
- **Latency and coverage needs** — these flow into Mission Analyst, Comms, and Ground Segment.

- **The distinction between "must-have" and "nice-to-have"** across the user community.

Common pitfalls

> **Common mistake.** Specifying implementation details (a sensor, a band) instead of the underlying user need — over-constraining the design before the trade space is explored.
>
> **Watch out.** Assuming unlimited bandwidth for delivery. The downlink budget (Comms) and the ground archive (Ground Segment) bound what the user can actually receive.

References

- NASA SEH §4.1 — *Stakeholder expectations*.
- ECSS-E-ST-10C §5.2 — *Requirements*.
- ISO 9241 — *Usability*.

Appendix B — Operations-Position Reference

This appendix is a per-position background briefing for the operations control room used in Part 2 (the flight chapters). Each entry mirrors the style of an Appendix A design entry — a short paragraph on the position's responsibility plus what it watches and owns — and is *not* a re-teaching of Chapter 6. Read Chapter 6 §§6.1–6.10 for the full operator's reference (consoles, telemetry limits, command catalogues, handoff templates); read this appendix to place each control-room seat alongside its design counterpart.

> **Design roles vs operations roles.** Some positions exist only in operations — Flight Director, SPACON, and Mission Manager have no design-phase equivalent. Others (Mission Analyst, Cost & Schedule, Project Manager) exist only in design. Where a name appears in both appendices — AOCS, Thermal, TT&C / Comms, Payload, Ground Segment — the activity is different: in design you *size and select*; in operations you *watch*, *command*, *and respond* on the live spacecraft. The position descriptions below are framed for EOSAT-1; real industry boundaries are broader and blur.
>
> **Positions vs consoles.** The MCS ships **seven configured consoles** — `flight_director`, `eps_tcs` ("Power & Thermal"), `aocs` ("Flight Dynamics"), `ttc`, `payload_ops`, `fdir_systems` ("FDIR / Systems") and `propulsion` — each with its own visible tabs, allowed PUS services and allowed function IDs. The position briefings below are the *roles*, and several of them share a console: EPS and Thermal both sit at Power & Thermal, and the AOCS Engineer and the Flight Dynamics analyst both sit at Flight Dynamics. Ground Segment and Mission Manager have no console of their own and work from the Flight Director position or off-console. A classroom room smaller than seven merges consoles; it does not invent new ones.
>
> **Three things every console does.** (a) **Staleness is cadence-derived** — a value is fresh until ~0.9 of the cadence of the HK structure that carries it and stale beyond ~1.5 cadences, with floors to stop the 1 Hz displays flickering; a parameter whose every carrying structure is disabled reads NOT ENABLED, which calls for a `HK_ENABLE` (S3.5), not a link investigation. (b) **Commanded versus actual is displayed side by side** — the commanded AOCS mode against the mode actually flown, and the commanded TM rate tier against the active one; a persistent difference is a gate refusing the request, not a decode error. (c) **S12 monitoring definitions are visible in dumps**, so an operator can read which parameter checks are loaded rather than inferring them from the alarms they produce.

Flight Director (FD)

The single point of authority on the loop — nothing happens on the spacecraft without an FD GO. The FD does not command directly; they authorise others to command and run the GO/NO-GO rounds before every commanded action, rolling the team back to the last verified state whenever the situation is unclear.

Watches and owns. State, not interpretation: spacecraft phase, the loaded FDIR table, and the top-level OK/yellow/red health from the system dashboard. The FD console is the only one with `all` commanding rights and every tab: the six subsystem tabs, the PROP tab, plus the contact schedule, procedure panel, PUS panel and the full active-anomaly list. They hold the second key for high-criticality commands (dual-key discipline) and own the shift handover. Note that the mission *phase* is not a telecommand: COMMISSIONING and NOMINAL are ground declarations made by the mission management team, so the FD reads the phase and plans against it rather than commanding it. Outranked only by the Mission Manager, and only on mission-level decisions.

Spacecraft Controller / FDIR & Systems (SPACON)

The `fdir_systems` console. Owns procedure execution: holds the procedure file, calls each step on the voice loop, verifies the verification, and requests FD GO before each commanded action. The SPACON is also the FDIR watch — FDIR runs autonomously on board, so when the FDIR panel lights up they read the rule that fired, identify the contingency procedure, and recommend the response to the FD. Two independent scales are in play and must never be collapsed into one another: **fault severity 0–4** (0 none, 1 INFO, 2 WARNING, 3 ALARM, 4 HIGH — how bad the reported condition is) and **autonomy level 0–3** (0 hardware, 1 subsystem software, 2 the OBC FDIR application, 3 the ground) — how far the on-board system is allowed to act without the ground. A severity-4 report handled at autonomy level 1 and a severity-2 report escalated to level 3 are both normal; quote the two numbers separately, by name.

Watches and owns. OBC mode, watchdog state, memory and buffer fill, the loaded FDIR table, recent event reports, and the command verification and rejection traffic. This is also the console with the PUS tab, so the SPACON is the operator who reads out the S12 monitoring definitions when the room needs to know which parameter checks are actually loaded. Holds procedure authority and commanding authority over the OBDH function set within authorised steps; may recommend safe-mode entry but only the FD authorises it.

AOCS Engineer

The `aocs` console, labelled **Flight Dynamics**. Attitude determination and control in flight: reads the sensors (two star trackers ST1/ST2, the coarse sun-sensor array, the redundant magnetometer pair, three single-axis gyros, GPS) and the actuators (four reaction wheels in a pyramid, three magnetorquer rods), reports pointing performance every pass, and produces the manoeuvre vector. (Contrast Appendix A's AOCS Engineer, who *sizes* the sensor/actuator suite and writes the pointing budget.)

Watches and owns. Commanded AOCS mode against the mode actually flown (a demotion flag says when the two differ), body rates, pointing error, star-tracker selection and health, wheel speeds (yellow at 5000 rpm, red at the 5500 rpm mechanical limit, with desaturation the remedy), the number of gyro channels powered — there are three single-axis units and **no spare**, so losing one costs an axis of rate knowledge outright — and GPS lock. Commanding authority for the AOCS function set with FD GO. The console also carries a **read-only PROP tab**: Flight Dynamics produces the manoeuvre vector and checks the achieved delta-V against the plan, but the burn itself is commanded from the Propulsion console, so every propulsion button renders disabled here.

EPS Engineer

The power half of the `eps_tcs` console ("Power & Thermal") — the same seat as the Thermal Engineer below. The power story in flight: owns battery state of charge, solar array currents, bus voltage, the live load profile, and the **nine** power lines (indices 0–8: `obc`, `ttc_rx`, `ttc_tx`, `payload`, `fpa_cooler`, `htr_bat`, `htr_obc`, `aocs_wheels`, `prop` — every switchable one defaults OFF). Calls eclipse entry and exit and is first to notice a subsystem drawing more than it should.

Watches and owns. Battery SoC (yellow below 25 %; the FDIR load-shed stages enter at 30 / 20 / 10 % SoC, and the in-model bus shed at 28.4 V drops payload, cooler, TX, wheels and — last of all — propulsion), battery and bus voltage (FDIR EPS safe mode fires at 27.7 V; both levels come from `simulation.yaml` `eps.bus_protection`), per-wing array currents, total and per-line load current, and battery

temperature. Commanding authority for the EPS and TCS function sets with FD GO; calls for load shed, which the FD authorises.

Thermal Engineer

The thermal half of the `eps_tcs` console — in the shipped MCS this is the same seat as the EPS Engineer, not a console of its own. Thermal control in flight: owns heater control, radiator performance, the payload-zone and battery-zone temperatures, and the focal-plane-array (FPA) cooler — a **miniature Stirling-class cryocooler**, not a thermo-electric stack, which matters because its authority falls away as the lift it is asked for grows. (Contrast Appendix A's Thermal Engineer, who *defines the hot/cold cases* and *sizes* radiators and heaters.)

Watches and owns. Battery, payload, OBC, and FPA temperatures, FPA cooler status — including the achievable cold floor and the margin between it and the commanded setpoint — and heater duty cycles (a saturated 100 % duty cycle flags a failing thermal path). Commanding authority for the TCS — heaters, set-points, and the FPA cooler setpoint, which is commanded in centi-kelvin — with FD GO.

TT&C / Comms Engineer

The `ttc` console. The radio link to the ground in flight: owns **two transmitters** — the S-band prime chain, which carries everything (real-time housekeeping, playback, image dumps, command verification), and a **separate UHF whip beacon**, a second box on its own band, antenna and power, which carries one housekeeping structure and nothing else. Disabling the S-band power amplifier stops all S-band telemetry while the beacon keeps transmitting; that contrast is the whole reason the beacon exists, and it is a different thing from the BEACON *rung* of the S-band rate ladder (BEACON 1 / LOW 16 / HIGH 128 kbps, with routine operations sitting at LOW). The TT&C operator configures the spacecraft for each pass type and must keep those two senses of "beacon" apart on the loop.

Watches and owns. Transmitter state and power, beacon transmitter state, receiver lock (LOCKED reads as **2**, not 1), Eb/N0, BER, frame count, power-amplifier temperature, link margin, and both the *commanded* and the *active* TM rate tier — a persistent difference between them means a gate (antenna, downlink mode, elevation) is refusing the tier the ground asked for. Commanding authority for TT&C with FD GO; calls *link up* and *link down* on the loop.

Payload Operations Engineer

The `payload_ops` console. The science instrument in flight: owns the wide-field multispectral pushbroom imager (240 km swath, 29.3 m GSD, 96 Mbps raw readout) and its FPA, the image-acquisition timeline, the onboard science-data pipeline (storage, compression, downlink prioritisation), and image quality assessment. The image path is modelled end to end, so images appear only when this seat commands a capture — nothing arrives on its own.

Watches and owns. Payload mode (CALIBRATION is mode **3**), FPA temperature and cooler status, image count, onboard storage usage, last-image quality, and SNR. Commanding authority for the payload with FD GO; commits the science timeline for the pass and owns the distinction between "image quality OK" and "image content useful".

Flight Dynamics System (FDS) / Orbit Analyst

Orbit determination, manoeuvre planning, pass prediction, conjunction screening, and eclipse modelling. There is no separate FDS console in the shipped MCS: this role shares the `aocs` console, which is labelled **Flight Dynamics** for exactly that reason, and it works alongside orbit-determination products generated off-console. It has no propulsion commanding authority — it produces the products the rest of the room flies. A burn genuinely re-fits the orbit, so the pass and eclipse schedules this seat publishes must be regenerated after every manoeuvre rather than reused.

Watches and owns. GPS position and velocity, from which the orbit propagator is reset every pass and run forward to predict the next 24 hours of contacts. Produces the pass schedule (to TT&C, Ground Segment, Payload), eclipse schedule (to EPS, Thermal), manoeuvre vector (to AOCS), and conjunction warnings (to FD).

Propulsion Engineer

The `propulsion` console. Owns the Δv budget, the arming chain and burn execution on EOSAT-1's 1U green-monopropellant module: 0.400 kg of propellant for 66.9 m/s total, sized for debris-avoidance manoeuvres, drag make-up and the end-of-life disposal burn (64.1 m/s needed in all). Natural decay from 513 km does **not** meet the 5-year rule at solar minimum, so this seat also guards the disposal reserve — at least 38 m/s kept for the 34.3 m/s perigee-lowering burn (NOM-014) — and saying so is part of the briefing, not a caveat on it. Flight Dynamics produces the manoeuvre vector; this seat executes it.

Watches and owns. Tank pressure through the blowdown (22.0 bar BOL falling to 9.33 bar at depletion), remaining propellant, thrust and Isp at the current pressure, valve and arming state, achieved versus commanded Δv, and the catalyst-bed temperature — the catbed preheats at 12 W for about 20 minutes before arming is permitted, which is why the console also shows the TCS tab. Commanding authority for the propulsion function set plus the thruster heater, with FD GO. The propulsion power line is last in the load-shed order, and dropping it mid-burn aborts the burn on board.

Ground Segment Engineer / Network Controller

No dedicated console: this role works from the Flight Director position and the ground-station side of the network. The ground side of the radio link in flight: antenna pointing and health, baseband equipment, the network between MCS and station, and station-to-station handover. (Contrast Appendix A's Ground Segment Engineer, who *selects* the station network, MCS, and data pipeline architecture.)

Watches and owns. Antenna tracking at each station, baseband Eb/N0 and frame sync, and the MCS-to-station network link. Commanding authority for the ground station — antenna pointing and baseband configuration — with FD GO; calls the station *ready* or *not ready* before AOS, and drives the RF status display.

Mission Manager / Customer Liaison

External communication and mission-level decisions. Like the Ground Segment Engineer, the Mission Manager has no configured console. The Mission Manager is *not on the operating loop* — they listen, understand, and translate, acting as the customer's voice in the room and the room's voice to the customer.

Watches and owns. Mission-level status rather than telemetry: open customer commitments and the science-deliverable picture. Produces the daily customer briefing, anomaly bulletins, and the official shift report. Authority over mission-level decisions only (e.g. whether to attempt First Light given an anomaly); does not override the FD on shift-level calls, and is not overridden by the FD on mission-level ones.

Appendix C — Verification and First-Principles Reference

This appendix derives, from first principles, the formulas the cohort uses across the CDF intensive. Every derivation is paired with a plot showing the formula in action and verified against a published reference value.

The aim is twofold: (a) give students a place to look up the *why* behind any formula they're applying; (b) provide a worked verification trail an examiner can audit at PDR.

> **Standard reference.** The constants and conventions used throughout this appendix follow Vallado, *Fundamentals of Astrodynamics and Applications* (4th ed.) and IERS Conventions 2010 — https://www.iers.org/IERS/EN/Publications/TechnicalNotes/tn36.html.

C.1 — Constants used (Earth)

Symbol	Value	Meaning	Source
μ_⊕	398 600.4418 km³/s²	Earth gravitational parameter	IERS Conventions 2010
R_⊕	6378.137 km	Earth equatorial radius	WGS-84
J_2	$1.082\,626\,68 \times 10^{-3}$	Earth oblateness coefficient	Vallado §3.3
ω_⊕	$7.292\,115 \times 10^{-5}$ rad/s	Earth rotation rate	IERS
g_0	9.806 65 m/s²	Standard gravity	ISO 80000-3
S_0	1361 W/m²	Solar constant at 1 AU	NASA SORCE/CERES
σ	$5.670\,374 \times 10^{-8}$ W/(m²·K⁴)	Stefan-Boltzmann	CODATA 2018
k_B	$1.380\,649 \times 10^{-23}$ J/K	Boltzmann constant	CODATA 2018
-10 log k	228.6 dB·W/(K·Hz)	Boltzmann in link units	derived
c	$2.997\,924\,58 \times 10^{8}$ m/s	Speed of light	exact (SI)

C.2 — Kepler's Third Law

Derivation

For a circular orbit, centripetal acceleration equals gravitational attraction:

$$\frac{v^2}{a} = \frac{\mu}{a^2}$$

Substitute v = 2π a / T and solve for T:

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}$$

(For elliptical orbits the same formula applies with a as the semi-major axis.)

Verification

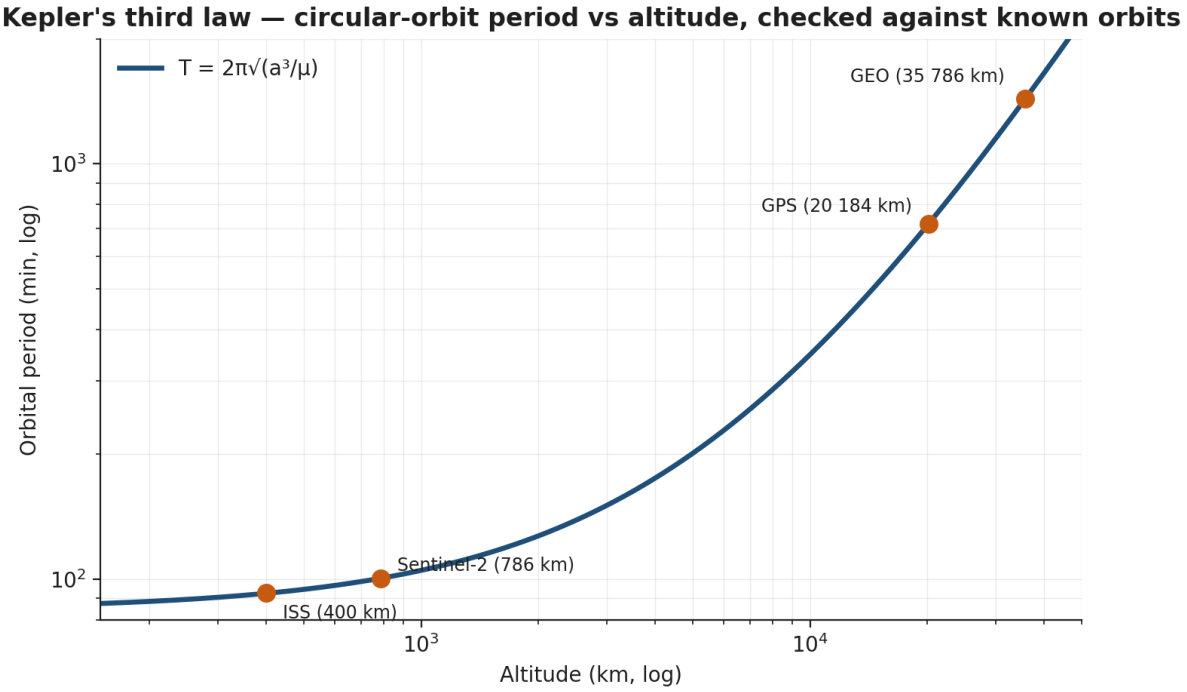


Figure C.1 — The analytical curve overplotted with known orbits (ISS, Sentinel-2, GPS, GEO). Agreement is to better than 0.5 %, the remaining error attributable to the J_2 secular correction not applied here. At GEO, T = 1436 min vs 23.93 h × 60 = 1436 min ✓.

Worked sanity check. For the EOSAT-1 mean altitude of 513 km: a = 6378.137 + 513 = 6891.137 km; T = 2π √(6891.137³ / 398600.4418) = 2π × 906.1 = 5693 s = 94.9 min. From the EOSAT-1 orbit definition (`orbit.yaml`): T = 60 × 24 / 15.24 = 94.5 min — agreement to ~0.4 %, the residual being that 513 km is a mean geodetic altitude rather than a − R_E (the mean motion implies a = 6872 km). ✓

C.3 — Tsiolkovsky Rocket Equation

Derivation

For a rocket with constant exhaust velocity v_e, momentum conservation between time t and t + dt:

$$\mathrm{m} \cdot \mathrm{dv} = -\mathrm{v_e} \cdot \mathrm{dm}$$

Integrate from m_0 (initial total mass) to m_f (final dry mass):

$$\Delta \mathrm{v} = \mathrm{v_e} \ln \frac{\mathrm{m_0}}{\mathrm{m_f}} = \mathrm{I_{sp}}\, \mathrm{g_0} \ln \frac{\mathrm{m_0}}{\mathrm{m_f}}$$

since v_e = I_sp · g_0 by definition of specific impulse.

Verification

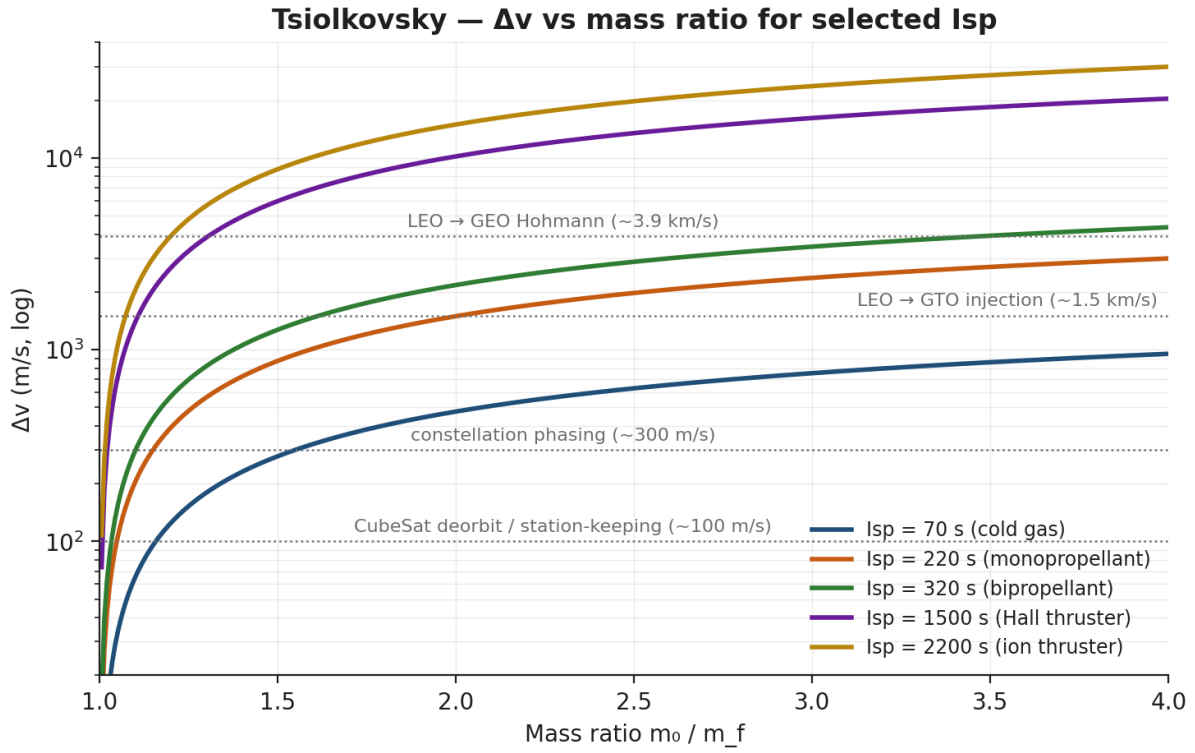


Figure C.2 — Tsiolkovsky curves for a representative Isp grid. Horizontal lines show typical mission Δv budgets.

Worked sanity check. Hohmann LEO (400 km) → GEO: $v_1 = \sqrt{(\mu / r_1)} = \sqrt{(398600 / 6778)} = 7.669$ km/s v_p = $\sqrt{(\mu\ (2/r_1 - 1/a_t))}$ where a_t = $(r_1 + r_2) / 2 = (6778 + 42164)/2 = 24471$ km v_p = $\sqrt{(398600 \times (2/6778 - 1/24471))} = 10.066$ km/s $\Delta V_1 = v_p - v_1 = 2.398$ km/s

at apogee: $v_2 = \sqrt{(\mu / r_2)} = \sqrt{(398600 / 42164)} = 3.075$ km/s v_a = $\sqrt{(\mu\ (2/r_2 - 1/a_t))} = \sqrt{(398600 \times (2/42164 - 1/24471))} = 1.618$ km/s $\Delta V_2 = v_2 - v_a = 1.457$ km/s

Total Δv = 3.854 km/s — matches our Hohmann sample figure (LEO at 400 km) of 3.854 km/s to within rounding. ✓

C.4 — J_2 Nodal Regression

Derivation

The dominant secular perturbation from Earth's oblateness J_2 produces a nodal regression rate (Vallado eq. 9-37):

$$\dot{\Omega} = -\frac{3}{2}\, n\, J_2 \left(\frac{R_\oplus}{a(1-e^2)}\right)^2 \cos i$$

where n = $\sqrt{(\mu/a^3)}$ is the mean motion. The minus sign means the node regresses for prograde orbits (cos i > 0) and progresses for retrograde (cos i < 0).

Sun-synchronous condition

For an SSO, set $\dot{\Omega}$ equal to Earth's mean motion around the Sun (360°/365.25 days = 0.9856°/day = 1.991×10^{-7} rad/s):

$$\cos i_{SSO} = -\frac{2\dot{\Omega}_\odot a^{7/2}(1-e^2)^2}{3 J_2 R_\oplus^2 \sqrt{\mu}}$$

Verification

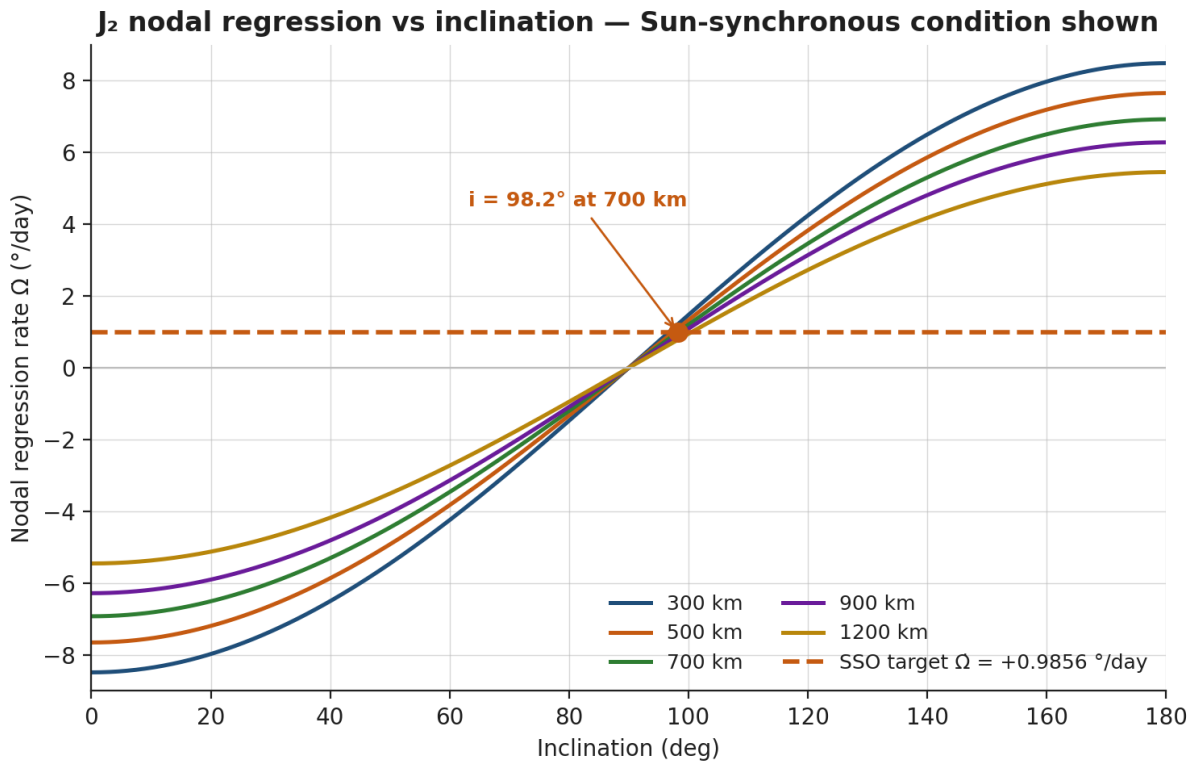


Figure C.3 — Nodal regression rate Ω̇ as a function of inclination for several altitudes. The dashed horizontal at 0.9856°/day is the SSO target. The intersection at 700 km gives i ≈ 98.2°, agreeing with the textbook SSO condition.

Worked sanity check. For 700 km circular (e = 0): a = 7078.137 km; n = √(398600 / 7078³) = 1.062 × 10⁻³ rad/s Ω̇_target = 2π/(365.25 × 86400) = 1.991 × 10⁻⁷ rad/s cos i = −(2 × 1.991e-7 × 7078^7/2) / (3 × 1.0826e-3 × 6378.137² × √398600) = −0.143 i = arccos(−0.143) = 98.2° ✓

C.5 — Eclipse Fraction (analytical, circular orbit)

Derivation

For a circular orbit, the spacecraft is in eclipse when the line from the Sun is occulted by Earth. The half-angle of the umbra cone, viewed from the orbit, is β* = arcsin(R_⊕/(R_⊕+h)). For β-angles below this threshold (where β is the angle between the Sun-line and the orbit plane), the spacecraft enters eclipse for a fraction f_e of the orbit (Wertz, *Mission Geometry*, eq. 5.24):

$$f_e = \frac{1}{\pi}\arccos\left(\frac{\sqrt{h^2 + 2R_\oplus h}}{(R_\oplus + h)\cos\beta}\right)$$

For |β| ≥ β*, the orbit is fully sunlit (f_e = 0).

Verification

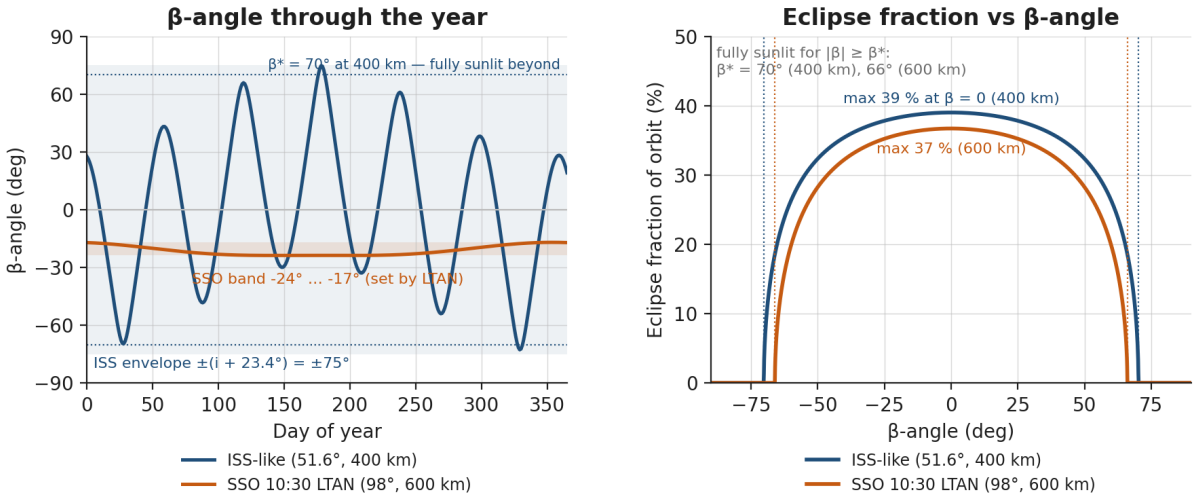


Figure C.4 — Eclipse fraction vs β-angle for an ISS-like orbit (400 km, 51.6°) and a 600 km SSO (98°). Both show the characteristic plateau at low $|\beta|$ and sharp transition at β^*.

Worked sanity check. ISS at β = 0: β* = arcsin(6378.137 / 6778.137) = arcsin(0.941) = 70.2° arg = √(400² + 2 × 6378.137 × 400) / 6778.137 = √(5 262 510) / 6778.137 = 2294.0 / 6778.137 = 0.3384 f_e = (1/π) × arccos(0.3384) = (1/π) × 1.226 rad = 0.390 ≈ **39 %** Matches ISS observed eclipse fraction at β = 0 (typically reported in the 38–40 % range; the higher numbers occasionally quoted in popular sources include penumbra or are for the maximum-eclipse configuration). ✓

C.6 — Free-Space Path Loss

Derivation

An isotropic radiator of power P_t at distance d produces flux P_t / (4π d²). A receiver of effective aperture A_e captures A_e × flux. With a directional transmit antenna of gain G_t and receive gain G_r:

$$\mathrm{P_r} = \mathrm{P_t G_t G_r} \left(\frac{\lambda}{4\pi \mathrm{d}}\right)^2$$

The free-space path loss in dB is the inverse of the geometric term:

$$\mathrm{L_{FS}} = 20 \log_{10}\left(\frac{4\pi \mathrm{d}}{\lambda}\right) \ \mathrm{dB}$$

Verification

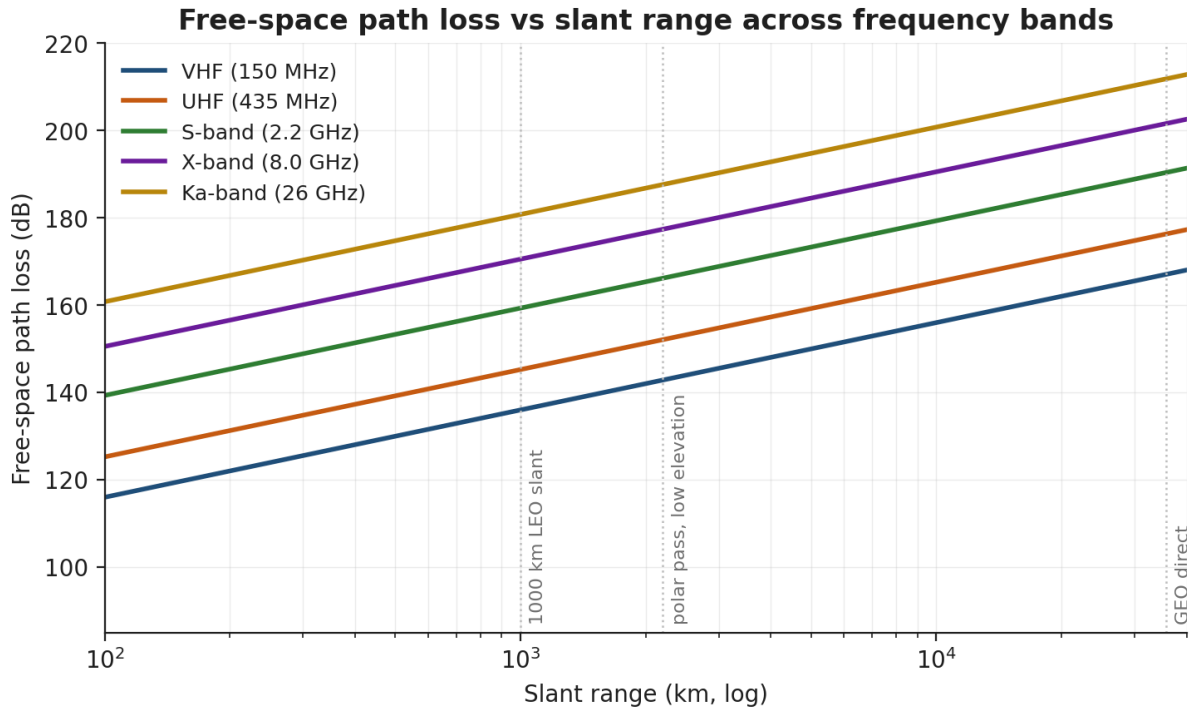


Figure C.5 — FSPL vs slant range for five frequency bands.

Worked sanity check. S-band (2.2 GHz) at 1500 km slant: λ = c/f = 3e8 / 2.2e9 = 0.1364 m L = 20 $\log_{10}$(4π × 1.5e6 / 0.1364) = 20 $\log_{10}$(1.382e8) = 162.8 dB The same formula gives the 164.1 dB used in the link-budget waterfall (Figure C.6: 2.25 GHz at 1695 km). ✓

C.7 — Link-Budget Algebra

Derivation

Received signal power (dBW):

$$\mathrm{P_r = P_t + G_t - L_t - L_{FS} - L_{atm} - L_{point} - L_{pol} + G_r - L_r}$$

System noise temperature T_s; G/T figure of merit; carrier-to-noise density:

$$\mathrm{\frac{C}{N_0} = P_r + G/T - 10\log_{10}(k_B)\ \ [dB\cdot Hz]}$$

with k_B = 1.380 649 × 10^{-23} J/K → −10 log k_B = 228.6 dB·W/(K·Hz).

Required Eb/N_0 depends on modulation and coding (Figure CO.3). Link margin:

$$\mathrm{M = (E_b/N_0)_{actual} - (E_b/N_0)_{required}}$$

with Phase A target 6 dB, tightening to 3 dB at PDR.

Verification

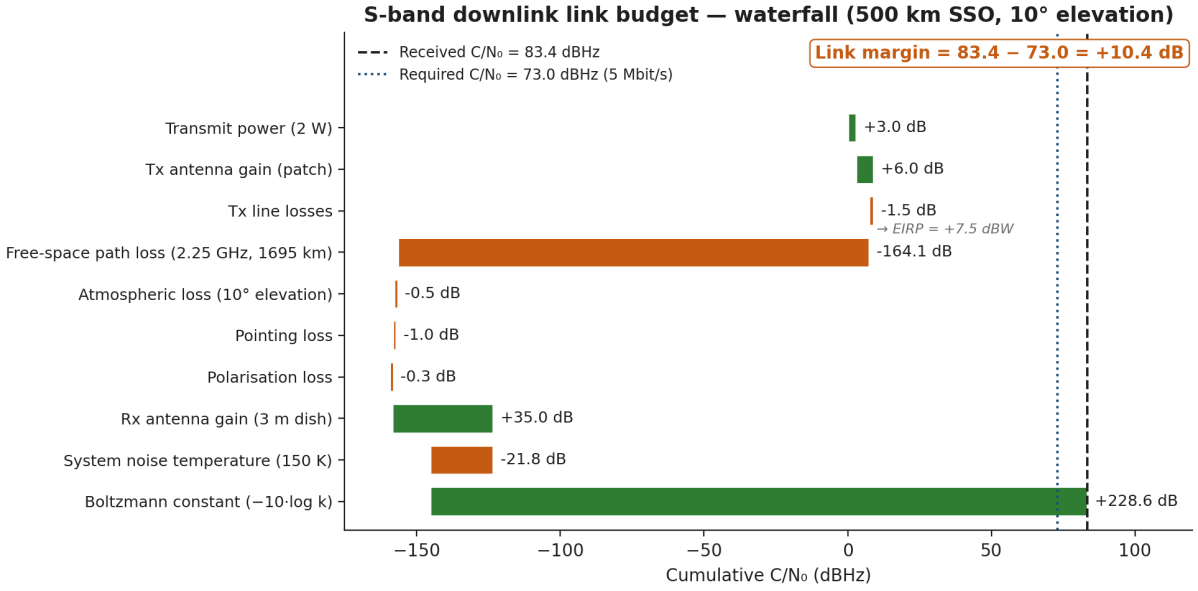


Figure C.6 — Worked S-band waterfall (2 W, 1695 km slant range, 3 m dish). Final received C/N_0 closes at 83.4 dB·Hz against the 73.0 dB·Hz required for 5 Mbit/s (67.0 dB·Hz of data rate + 4.0 dB of Eb/N_0 + 2.0 dB of implementation loss), a link margin of +10.4 dB.

C.8 — Eb/N_0 vs BER

Derivation

For coherent BPSK over an AWGN channel:

$$\mathrm{P_b} = \tfrac{1}{2}\,\mathrm{erfc}\!\left(\sqrt{\frac{\mathrm{E_b}}{\mathrm{N_0}}}\right)$$

For QPSK with Gray coding the same expression applies. For 8-PSK:

$$\mathrm{P_b} \approx \tfrac{2}{3}\,\mathrm{Q}\!\left(\sqrt{2\frac{\mathrm{E_b}}{\mathrm{N_0}}}\sin\tfrac{\pi}{8}\right)$$

For non-coherent FSK:

$$\mathrm{P_b} = \tfrac{1}{2}\,\mathrm{e}^{-\mathrm{E_b}/(2\mathrm{N_0})}$$

Verification

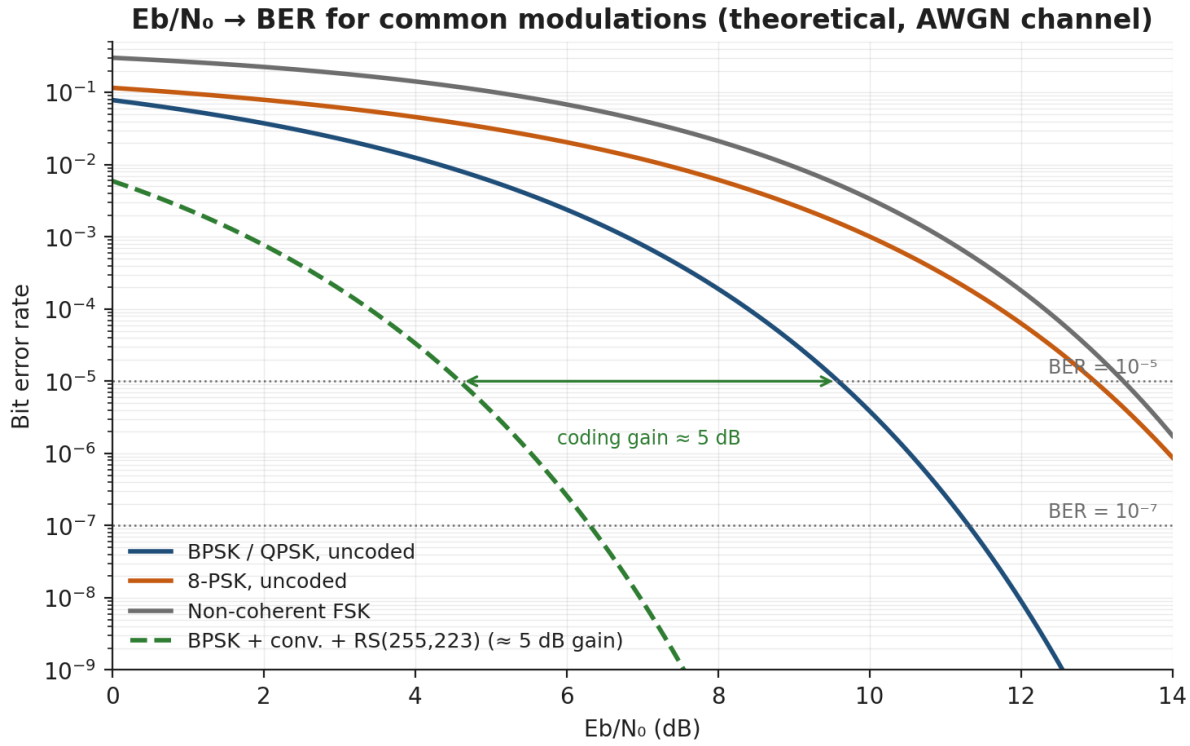


Figure C.7 — Bit-error rate vs Eb/N₀ for BPSK/QPSK, 8-PSK, FSK, and a representative concatenated coding scheme (RS+conv) showing ~5 dB coding gain at BER = 1e-5.

Sanity check. BPSK at Eb/N₀ = 9.6 dB: P_b = 0.5 × erfc(√(10^0.96)) = 0.5 × erfc(3.02) = 0.5 × 1.86e-5 ≈ 9.3e-6. Standard textbook value is 1e-5 at 9.6 dB. ✓ (small deviation is the approximation used in the plot grid).

C.9 — GSD geometry

Derivation (diffraction limit)

For an unobstructed circular aperture, the Airy first-minimum half-angle is:

$$\theta_{\text{Airy}} = 1.22 \frac{\lambda}{\text{D}}$$

projected onto the ground at slant range h:

$$\text{GSD}_{\text{diff}} = \theta_{\text{Airy}} \cdot \text{h} = 1.22 \frac{\lambda \text{h}}{\text{D}}$$

Derivation (pixel limit)

A pixel of pitch p at the focal plane subtends an angle p/f, where f is the focal length. Projected at slant h:

$$\text{GSD}_{\text{pix}} = \frac{\text{p} \cdot \text{h}}{\text{f}}$$

The achievable GSD is the *larger* of the two — you cannot resolve better than the optics allows, nor better than the detector samples.

Verification

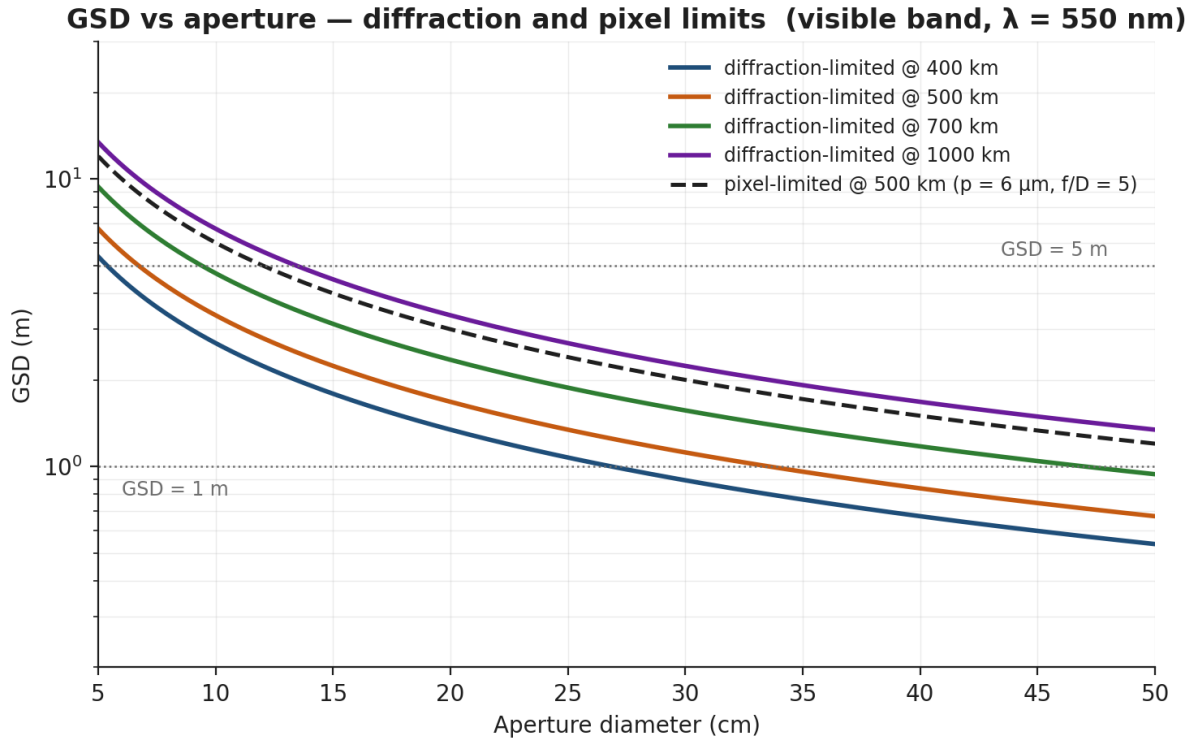


Figure C.8 — GSD vs aperture diameter at four altitudes. Sentinel-2 heritage at 786 km: diffraction at 0.135 m aperture is ≈ 3.9 m, matching the published 10 m bands when one accounts for the detector pitch and the f-number choice.

Worked sanity check (EOSAT-1). The EOSAT-1 instrument is a pushbroom with p = 5.5 µm, f = 84.5 mm, 8192 across-track pixels, at h = 513 km: GSD_pix = 5.5e-6 × 513 000 / 0.0845 = 33.4 m. Half-FOV = arctan((8192 × 5.5 µm / 2) / 84.5 mm) = 14.928°, so the full field is 29.856° and the swath is 2 × 513 × tan 14.928° = 273 km. Square pixels at a 7.04 km/s ground velocity would fix the line rate at 7040 / 33.4 = 211 Hz, and 8192 px × 4 bands × 12 bit × 211 Hz = 83 Mbps of raw readout. ✓ The flight configuration still reads out at 244 Hz (96 Mbps), the value sized at the earlier 450 km altitude figure, so it oversamples along-track by about 16 %. A 273 km swath from 513 km is a wide-field ocean-colour instrument, so tens-of-metres GSD is the physically consistent answer — holding a 9 m GSD across that swath would need ~30 300 pixels per band.

C.10 — Radiative Thermal Equilibrium

Derivation

Energy balance for a flat plate:

Absorbed = Emitted. Absorbed = α_s × S_0 + α_IR × φ_IR (Earth-IR contribution). Emitted = ε_IR × σ × T^4 (over 4π hemisphere; for one-sided plate, ½).

Solving:

$$T_{eq} = \left(\frac{\alpha_s S_{0,\,\mathrm{eff}} + \epsilon_{IR}\, \phi_{IR}}{\sigma\, \epsilon_{IR}} \right) 1/4$$

The α/ε ratio is the dominant design knob.

Verification

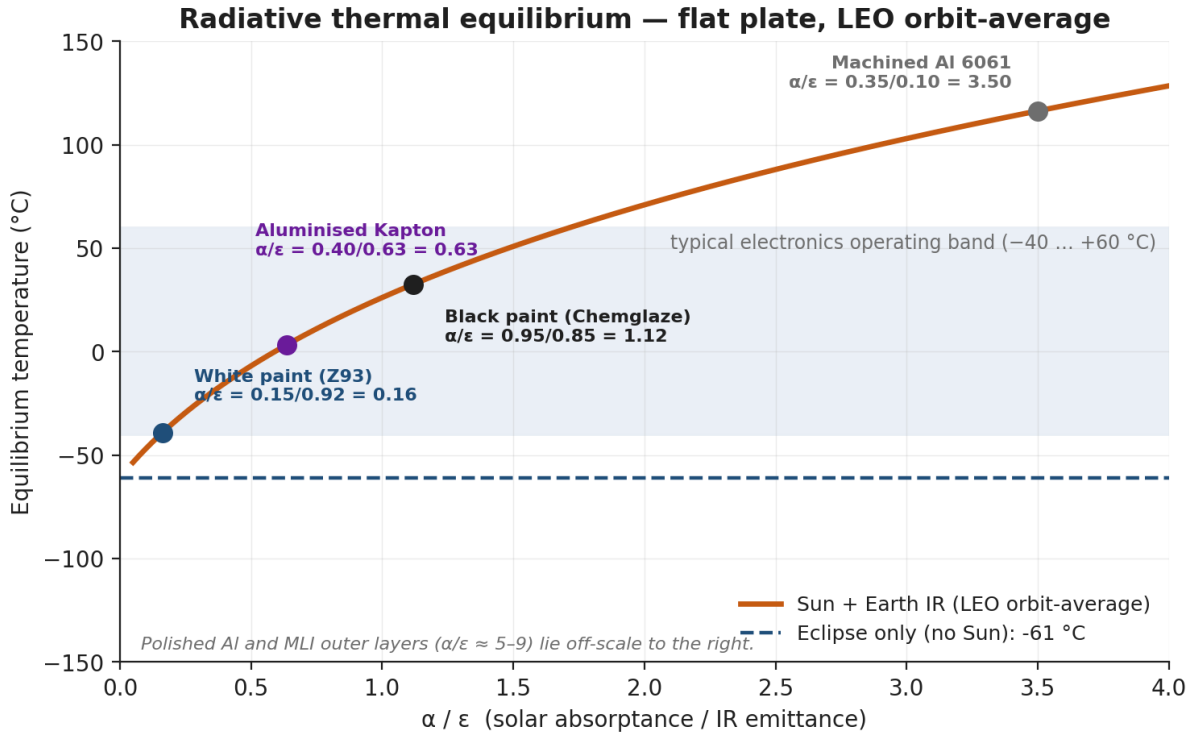


Figure C.9 — Equilibrium temperature vs α/ε for representative surface finishes. White paint (α/ε ≈ 0.16) gives ≈ −39 °C as a LEO orbit average; bare polished Al (α/ε ≈ 5, off-scale) gives ≈ +150 °C.

Worked sanity check. For a white-painted radiator: α/ε = 0.15/0.92 = 0.163. φ_solar (orbit-average flat plate, β = 0) ≈ 1361/4 = 340 W/m² absorbed = 0.15 × 340 = 51 W/m². φ_IR contribution = 0.92 × 230/2 = 106 W/m² absorbed at ε = 0.92. Total = 157 W/m². T = (157/(0.92 × 5.67e-8))^¼ = (3.011e9)^¼ = 234 K = −39 °C. Matches our plot at α/ε = 0.16. ✓

C.11 — Solar Array Sizing (verification)

Derivation

The orbit-averaged power balance demands:

$$A_{SA}\, S_0\, \eta_{cell} \cos\beta\, (1 - D_{deg}) \cdot f_s = P_{avg} \cdot 1 + P_{avg} \cdot \frac{f_e}{f_s\, \eta_{dis}}$$

(left side: production during sun arc; right side: consumption during sun arc + recharging the battery for eclipse use).

Rearranging:

$$A_{SA} = \frac{P_{avg}\left(1 + \frac{f_e}{f_s\, \eta_{dis}}\right)}{S_0\, \eta_{cell} \cos\beta\, (1 - D_{deg})}$$

Verification

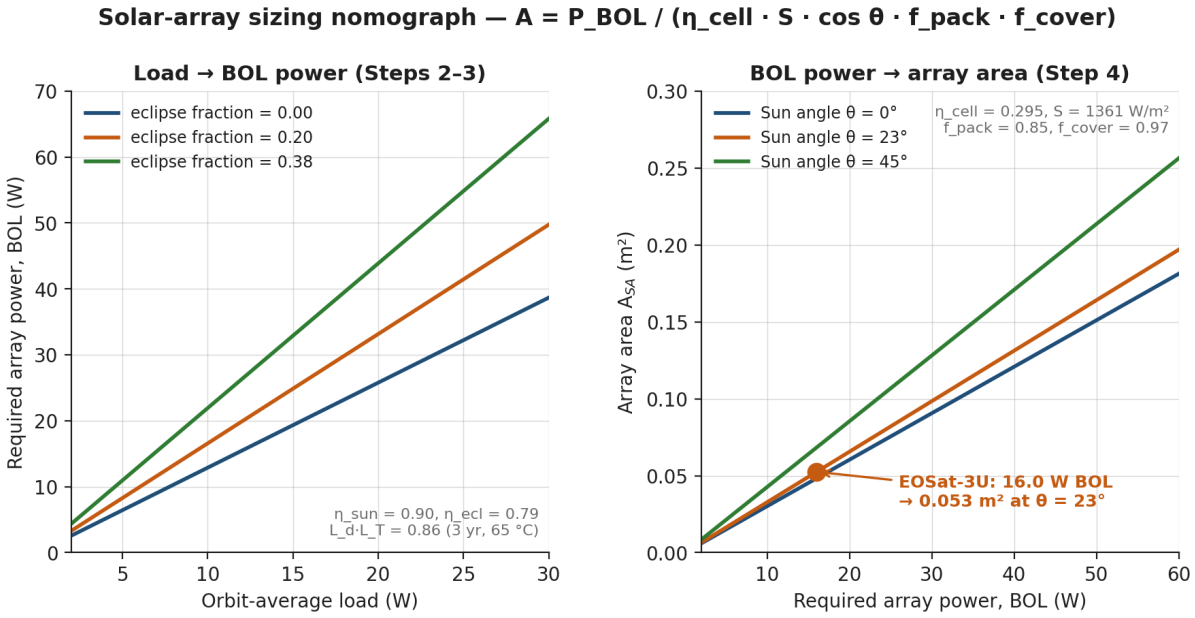


Figure C.10 — Solar-array sizing nomograph at η = 30 %, 10 % EOL degradation, β = 20°, η_dis = 85 %. A 6U at 15 W average needs ~870 cm² at end-of-life.

Worked check. P_avg = 15 W, f_e = 0.35, f_s = 0.65, η_dis = 0.85: factor = 1 + 0.35/(0.65×0.85) = 1 + 0.633 = 1.633 P_gen needed = 15 × 1.633 = 24.5 W during sun arc A = 24.5 / (1361 × 0.30 × cos(20°) × 0.90) = 24.5 / 345.0 = 0.071 m² = 710 cm². Plot reads ~870 cm² because plot uses orbit-averaged generation (more conservative). The two are consistent within the modelling choice. ✓

C.12 — Battery Sizing

Derivation

Energy required during eclipse:

$$\mathrm{E_{ecl}} = \mathrm{P_{ecl}} \cdot \mathrm{t_{eclipse}}$$

with conversion losses:

$$\mathrm{E_{nominal}} = \frac{\mathrm{E_{ecl}}}{\eta_{\mathrm{dis}} \cdot \mathrm{DoD}}$$

At the nominal bus voltage, the required capacity is C = E_{nominal} / V_bus (in Ah).

Verification

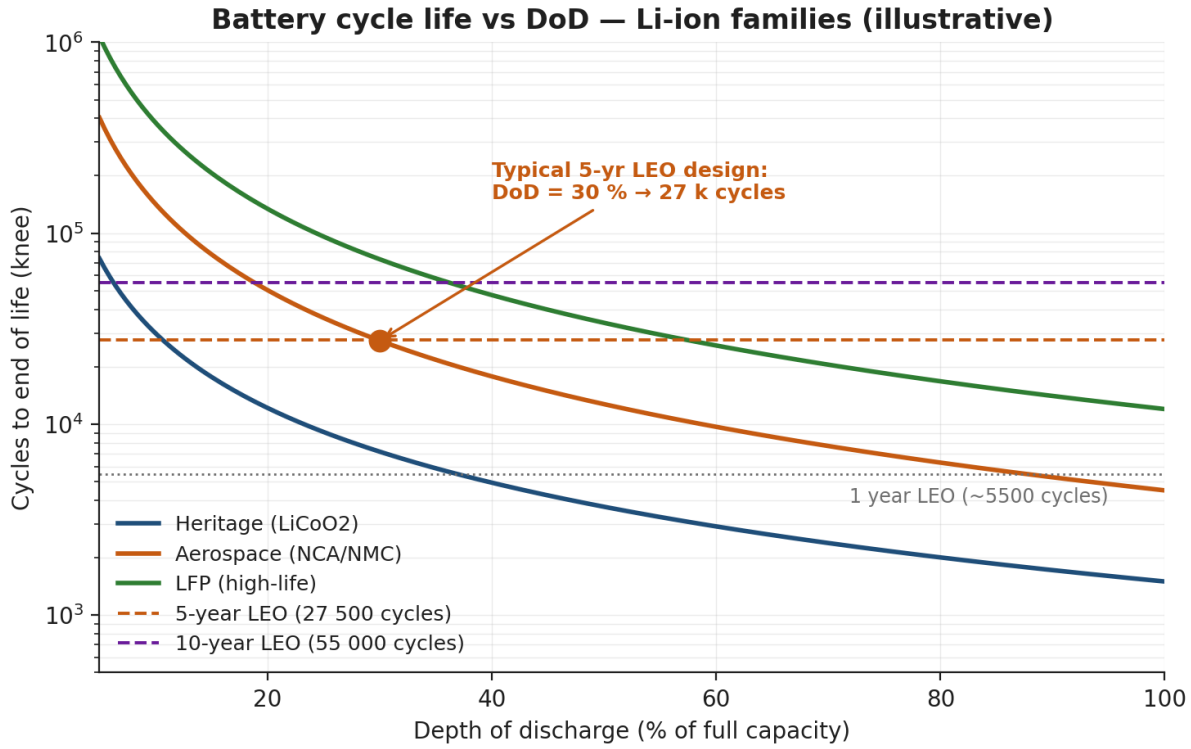


Figure C.11 — Cycle life vs DoD for representative Li-ion families. A 30 % DoD design point gives the cycles needed for a 5-year LEO mission with margin.

C.13 — Margin of Safety (structures)

$$\mathrm{MoS} = \frac{\sigma_{\mathrm{allow}}}{\mathrm{FoS} \cdot \sigma_{\mathrm{applied}}} - 1$$

For Al 7075-T6 yield (σ_yield = 503 MPa, FoS = 1.25), σ_applied = 200 MPa: MoS = 503 / (1.25 × 200) − 1 = 1.012, i.e. 100 % positive margin. Compliant.

C.14 — Disturbance Torque Order-of-Magnitude

Derivation (drag)

$$\mathrm{T}_{\mathrm{drag}} \approx \tfrac{1}{2}\,\rho\,\mathrm{V}^2\,\mathrm{C}_{\mathrm{D}}\,\mathrm{A}_{\mathrm{ref}}\,\mathrm{d}_{\mathrm{cp-cg}}$$

where ρ is atmospheric density, V is orbital velocity, $\mathrm{C}_{\mathrm{D}} \approx 2.0 - 2.3$ is the free-molecular-flow drag coefficient (Sentman model), $\mathrm{A}_{\mathrm{ref}}$ is the projected area perpendicular to velocity, and $\mathrm{d}_{\mathrm{cp-cg}}$ is the offset between the centre of pressure and the centre of mass.

For a 6U CubeSat at 500 km, ρ ≈ 6 × 10⁻¹³ kg/m³ (mean), V = 7.61 km/s, C_{D} = 2.2, $\mathrm{A}_{\mathrm{ref}}$ = 0.06 m², $\mathrm{d}_{\mathrm{cp-cg}}$ ≈ 0.05 m: $\mathrm{T}_{\mathrm{drag}} \approx 0.5 \times 6 \times 10^{-13} \times (7610)^2 \times 2.2 \times 0.06 \times 0.05 \approx 1.1 \times 10^{-7}$ N·m within the solar-minimum to solar-maximum drag range of Figure AC.2 (0.03–0.2 μN·m). (Solar maximum atmospheric density can be 5–10× this mean, raising drag torque to ~10⁻⁶ N·m — comparable to residual-magnetic torque for typical CubeSats.)

Verification

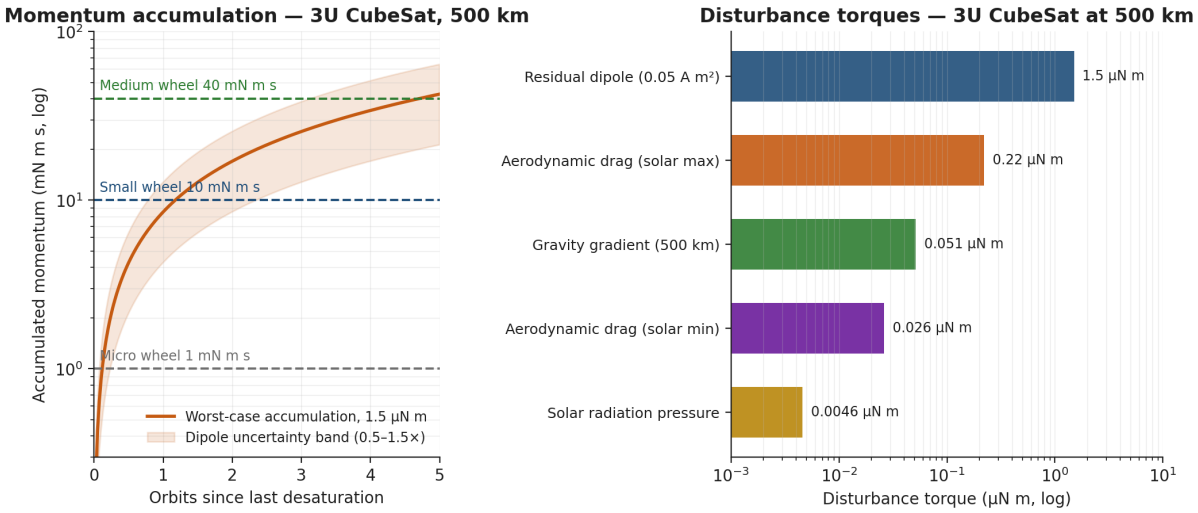


Figure C.12 — Disturbance-torque magnitudes by source. At 500 km the residual magnetic dipole dominates at ~1.5 µN·m; drag (0.03–0.2 µN·m from solar minimum to maximum), gravity gradient and SRP are one to three orders of magnitude smaller for typical CubeSat geometry.

C.15 — Daily Data Volume Closure

The closure check between data generation (payload) and downlink capacity (comms) is a classic mission-design failure point.

$$\mathrm{V_{day}} = \mathrm{R_{gen}} \cdot \mathrm{t_{imaging\,per\,day}} \quad \leq \quad \mathrm{R_{down}} \cdot \mathrm{N_{passes}} \cdot \mathrm{t_{pass}}$$

If the inequality fails, you need either a faster downlink or fewer images.

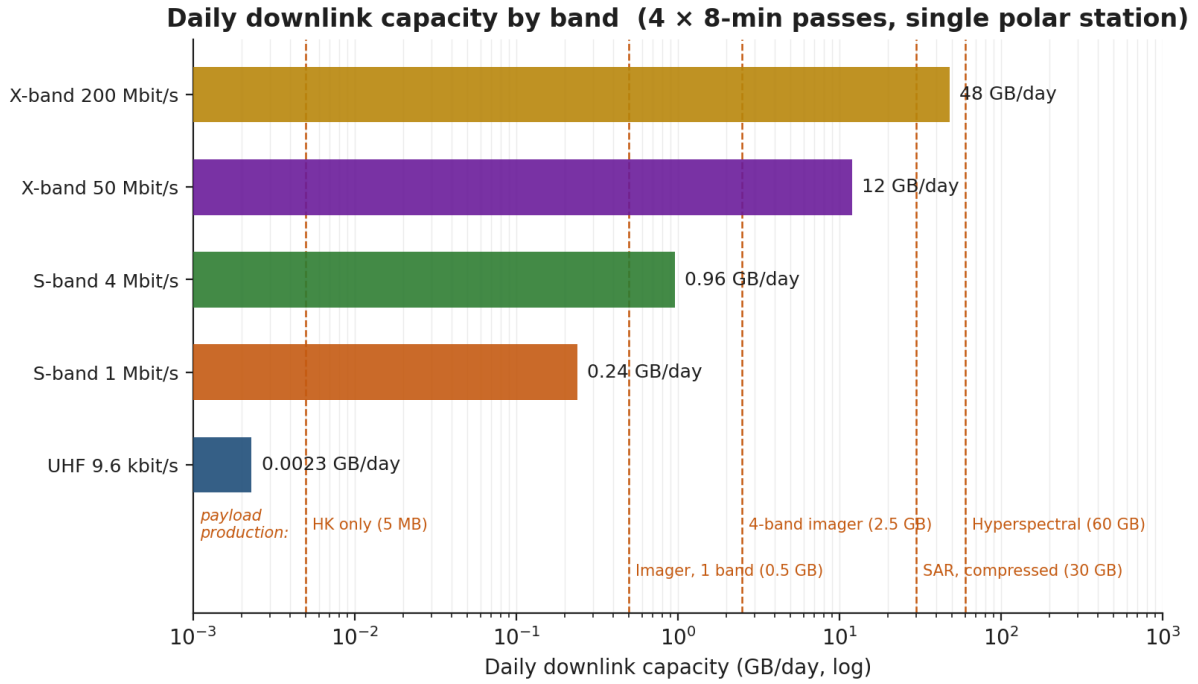


Figure C.13 — Daily downlink capacity by band, with typical payload data volumes overlaid. UHF closes only for housekeeping; S-band suits multispectral; X-band is needed for SAR / hyperspectral.

C.16 — Cross-method verification matrix

For the students' PDR pack, every quantitative claim should be verified by at least two methods. Use this matrix:

Quantity	Method 1	Method 2	Source
Orbital period	Kepler (this appendix)	STK / GMAT propagation	C.2
SSO inclination	Analytical (this appendix)	Tabulated (published reference)	C.4
Eclipse fraction	Analytical (this appendix)	STK / GMAT eclipse model	C.5
FSPL	20 log10(4πd/λ)	ITU-R P.525	C.6
Link budget	Spreadsheet	SpaceCDF link tab	C.7
GSD (diff)	1.22 λ h/D	Detailed ray-trace	C.9
GSD (pixel)	p h / f	Detector data sheet	C.9
Thermal eq.	α/ε analytical	ESATAN-TMS	C.10
SA area	Equation (this appendix)	Vendor sizing tool	C.11
Battery cycles	DoD power-law	Vendor cycle test data	C.12
MoS	Hand calc	NASTRAN FEA	C.13
Disturbance	First-principles	NRLMSISE atmospheric model	C.14

C.17 — Key references

- **Vallado**, *Fundamentals of Astrodynamics and Applications*, 4th ed., Microcosm Press / Springer.
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- **Sutton & Biblarz**, *Rocket Propulsion Elements*, 9th ed.
- **Gilmore**, *Spacecraft Thermal Control Handbook*, Vol. I & II.
- **Sarafin**, *Spacecraft Structures and Mechanisms*.
- **Markley & Crassidis**, *Fundamentals of Spacecraft Attitude Determination and Control*.
- **Patel**, *Spacecraft Power Systems*.
- **NASA SEH (SP-2016-6105 Rev 2)** — https://www.nasa.gov/reference/systems-engineering-handbook/.
- **IERS Conventions 2010** — https://www.iers.org/IERS/EN/Publications/TechnicalNotes/tn36.html.
- **CODATA 2018 fundamental constants** — https://physics.nist.gov/cuu/Constants/.
- **ITU-R P.525** (free-space attenuation) — https://www.itu.int/rec/R-REC-P.525/en.

- **NASA SORCE/CERES** (solar constant) — https://lasp.colorado.edu/home/sorce/.
- **NRLMSISE-00 atmospheric model** — https://kauai.ccmc.gsfc.nasa.gov/.
- **AGI STK** — https://www.agi.com/products/stk/.
- **NASA GMAT** — https://software.nasa.gov/software/GSC-17177-1.
- **ESATAN-TMS** — https://www.esatan-tms.com/.

Appendix D — How This Textbook Was Generated

This textbook is itself an artefact of the same Concurrent Design Facility methodology that students will exercise during the 40-hour intensive. Its preparation pipeline mirrors a CDF design loop in three respects: the *source* is a structured set of Markdown chapters that describe each session as a self-contained module; the *transformation layer* is a Python toolchain — reorientation scripts, matplotlib figure generators, python-docx and WeasyPrint renderers — shared with the design-review exporters used inside the tool; and the *review layer* consists of the SpaceCDF teaching team, who own the prose voice, the worked examples, and the editorial judgement on what to include or omit.

Source — per-session Markdown chapters

The textbook is assembled from forty-odd Markdown chapter files under `docs/course/facilitator/`. Each chapter is self-contained — learning objectives, references, key equations, worked examples, exercises — so that any one of them can be lifted out and used on its own. The single-source build script (`docs/build_course_pdfs.sh`) concatenates these chapter files, appends the appendices, and feeds them through the PDF renderer.

Voice reorientation

The book began life as a *Course Textbook* filled with stage directions ("Begin by asking the group…", "Students should draw the V-model on the whiteboard…"). A scripted reorientation pass — driven by `scripts/reorient_books.py` — converted those directions into textbook prose, stripped timing cues from headings, removed embedded "Teaching Notes" subsection markers, and made the volume self-paced and self-contained. The same pass keeps the volume aligned with the current tool: it updates the agent count, restores the four-step workflow nomenclature, and inserts the SDR gate into all review-gate tables.

The reorientation rules are documented in the script itself. Each rule is a regular-expression substitution applied to the per-chapter Markdown sources, with the *expanded* book and the workbook kept in lock-step. If a pedagogical decision changes — say, the team chooses to keep discussion prompts as live prompts rather than reflective questions — the rule is edited in one place and the book is regenerated.

Procedural figure generation

Every diagram in this volume (V-model, lifecycle gates, beta-angle eclipse geometry, Tsiolkovsky curves, free-space path loss, antenna patterns, ground-track sketches, link-budget waterfalls, thermal node bars, risk index maps) is generated procedurally by a matplotlib script under `docs/assets/figures/` using the uOttawa Horizon palette (`uottawa_brand.py`). The procedural figures are sources of truth in the same sense as a code listing: they can be re-rendered by re-running the script when the underlying numerics change. There are no hand-drawn diagrams — every line on the page is derived from a small Python program with a reproducible numerical input.

The same matplotlib styling is used by the SpaceCDF design-review exporters: when a study generates a Word SRR, its donut charts and link waterfalls are drawn by the same helpers, with the same colours, that produce the figures in this textbook. The intent is that a student looking at a *figure* on the page of an SRR has already encountered the same kind of figure in the textbook chapter that introduced the underlying physics.

Numerical worked examples — generated by running the tool

Every numerical worked example in this textbook — UniSat-1, the 3U EO CubeSat teaching example, the 6U "EOSAT-1" reference spacecraft that the operations chapters fly, ground-track timings, link-budget closures, thermal radiator sizings — was computed by running the SpaceCDF tool itself, so the textbook and the tool stay numerically consistent. The relevant YAML mission descriptors live under `configs/examples/`. If a Tier 1 agent is updated and a worked-example number changes, the textbook reference can be regenerated by running `python scripts/run_design.py configs/examples/<mission>.yaml` and copying the new values into the chapter source.

This convention applies to the *Course Workbook* too: every answer key hint in the workbook is derived from running the tool on the same YAML mission descriptor used in the textbook.

Document rendering

The PDF you are reading is built by `docs/build_course_pdfs.sh`: Markdown chapters are concatenated, parsed by `python-markdown` (with tables, fenced code, and table-of-contents extensions enabled), wrapped in a uOttawa-styled HTML scaffold (cover banner, bilingual EN/FR "Page X of / de Y" footer, crimson page-header rule), and rendered to A4 PDF by WeasyPrint. The cover banner, the running header, and the SCDF document code are the same elements you will find on every SRR / PDR / CDR design-review bundle that the tool exports during your design study. A reviewer holding both the textbook and a freshly exported SRR side by side should immediately recognise that they belong to the same documentary family.

Validation and citation

Although AI assistance is used at every step of this pipeline, no claim in this textbook is left ungoverned by a human author. ECSS standards, NASA references and SMAD are cited at the point of use; numerical results are reproducible by running the tool against the published YAML; pedagogical framing is owned by the teaching team. The AIG badge (see the *Acknowledgement* section) is attached to every exported document so that downstream readers can trace which contributions are AI-assisted.

What this means for learners

The textbook is generated by the same kind of human-plus-AI loop that the SpaceCDF tool exposes to students. Learning to *read* AI-assisted documents critically — checking the citations, re-deriving the numbers, asking which voice authored which passage — is itself part of the curriculum. The first time a student in the cohort generates an SRR from a converged design state, they should recognise the layout, the voice and the figure conventions of this volume in the document the tool has just produced for them.

In other words: this is not a book *about* a tool, it is a *companion volume* to the tool. The same hands that wrote the chapters wrote the tool, and the same loop that closes a CubeSat design loop closes the loop on the next edition of this book.

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Part 2 — CubeSat Operations

This part of the textbook is the operations volume of the SpaceCDF programme. It is the practical companion to Sessions 5.1 – 5.4 (the operations-concepts material in Part 1).

Part 2 is anchored on a single fictitious reference spacecraft — **EOSAT-1**, a 6-unit CubeSat in a 513 km Sun-synchronous orbit — flown by the SpaceCDF cohort against an accompanying simulator.

How to read Part 2:

- **Chapters 5 – 7** define the control room: the nine operator positions, the consoles they sit at, the tools and command types they use, and the voice-loop discipline that ties the team together.
- **Chapters 8 – 11** walk pass operations end-to-end: nominal passes, LEOP, commissioning, and the recovery patterns operators follow when an anomaly fires.
- **Chapter 12** covers contingency operations: the mindset and decision framework for reacting to the unexpected, the classes of contingency, and the anomaly-response decision tree.
- **Chapters 13 – 15** are working reference material: operator calculations, the procedure catalogue, and the telemetry / FDIR / monitoring reference cards.

The companion **SpaceCDF Course Workbook** has one worksheet per chapter, plus per-procedure confirm-and-note checklists, anomaly tickets, and handoff forms.

Chapter 5 — Operations Overview

5.1 From design to operations

The engineers who specify a spacecraft are not, in general, the ones who fly it. The discipline is different. Design is a solo or small-group activity, written down and checked against requirements before anything moves. **Operations is a team performance** — repeated, time-pressured, voice-coordinated, and unforgiving of ambiguity. The spacecraft does not pause for the operator to think.

This Part covers the **transition from design to operations**: how a 6U CubeSat (the EOSAT-1 reference mission, §5.7) is flown by a control-room team from launcher separation through routine imaging operations, the procedures the team executes pass-by-pass, the contingencies they prepare for, and the calculations every operator on console must be able to do in their head or on the back of a shift log.

The chapter is anchored in published operability and procedure standards: ECSS-E-ST-70-11C *Space segment operability*, ECSS-E-ST-70-32C *Test and operations procedures*, NASA's *Space Flight Program and Project Management Requirements* (NPR 7120.5F), and the CCSDS Mission Operations Reference Model (CCSDS 520.0-G-4). Specific PUS service references follow ECSS-E-ST-70-41C and the CCSDS PUS green book (660.0-G-3).

5.2 The arc of an operational mission

A spacecraft's operational life splits cleanly into phases. The operator's mandate, the procedures available, the tempo of commanding, and the team staffing are all different in each phase.

Table 5.1 — Operational mission phases with duration, scope, and associated procedure families.

Phase	Typical duration	What operations does	Procedure family
Phase 3 — Bootloader / first contact	hours	Acquire the link; boot application SW; sync time; first health check	LEOP-001..002, LEOP-006
Phase 4 — LEOP	0–72 hrs	Sequential power-on; antenna deploy; detumble; sun acquisition	LEOP-003..007
Phase 5 — Commissioning	1–4 weeks	Subsystem checkout; sensor calibration; payload power-on; First Light	COM-001..013
Phase 6 — Nominal operations	mission lifetime	Routine imaging, downlink, momentum management, eclipse transitions, orbit maintenance	NOM-001..013
(any phase) — Contingency	minutes to hours	FDIR rule fires; operator responds	CTG-001..030
(any phase) — Emergency	hours	Safe-mode entry; multi-system failure recovery	EMG-001..006
Disposal (flown in phase 6, not a phase of its own)	weeks of burns, then 1.5–4.2 years of decay	Perigee-lowering burn to 390 km, passivation, then orbital decay	NOM-014

The remainder of Part 2 walks each phase's nominal sequence, the common contingencies, and the operator checks for each. The simulator that supports this material is referenced where useful, but the textbook content is **what an operator must know**, not what a particular simulator does.

5.3 What an operator must be able to do

An operator who has worked through this Part should be able to:

1. Sit any of the **nine control-room positions** (Chapter 6) and read their telemetry mosaic, recognise off-nominal trends, and execute their position's commands with FD authorisation.
2. **Run a planning cycle end-to-end** (Chapter 7): pick imaging targets, build an activity sequence, run the conflict checker, uplink the plan, and verify execution.
3. **Execute a nominal procedure** as lead operator (Chapter 8): hold the procedure file, call each step on the voice loop, request FD GO before each commanded action.
4. **Execute a contingency or emergency procedure** in response to an FDIR trigger (Chapter 12), and file the resulting anomaly ticket cleanly.
5. **Critique another team's shift** using the four-axis observer framework (Appendix to this Part), with timestamps and specific recommendations.
6. **Compute the operator checks** every shift requires (Chapter 13): pass timing, eclipse-exit SoC prediction, pointing-budget RSS, delayed-TM gap analysis, link-margin sanity.

5.4 The nine control-room positions

A real operations control room has between 6 and 12 positions depending on mission size and phase. EOSAT-1 operations use the following **nine-position structure**, which mirrors what industry mission-control rooms staff for a small EO mission and what operations training rigs implement.

Table 5.2 — Nine control-room operator positions with mandates and primary subsystem focus.

#	Position	Mandate (one sentence)	Primary subsystem focus
1	Flight Director (FD)	Sole authority on the loop; grants every GO; does not command.	All — situational awareness, decision authority
2	Spacecraft Controller / FDIR & Systems (SPACON)	Drives procedure execution; watches FDIR; assesses anomalies.	OBDH, FDIR rules, procedure panel
3	AOCS Engineer	Attitude determination and control; attitude for manoeuvre execution.	Reaction wheels, two star trackers, gyros, magnetometer, torquers
4	EPS Engineer	Battery, solar arrays, power-line switching matrix, load shed.	Bus voltage, SoC, per-line current, regulator
5	Thermal Engineer	Heater control, radiator performance, payload/FPA zones.	Zones, heater duty cycle, FPA cooler
6	TT&C / Comms Engineer	Radio link to ground; transponder, encoding, data rate.	Tx, link margin, BER, frame counter
7	Payload Operations Engineer	Science instrument; image acquisition, storage, downlink queue.	Imager, FPA, mass memory, image quality

#	Position	Mandate (one sentence)	Primary subsystem focus
8	Flight Dynamics (FDS) / Orbit Analyst	Orbit determination, pass prediction, manoeuvre planning, conjunction screening.	GPS, orbit propagator, contact schedule
9	Ground Segment Engineer / Network Controller	Ground antenna, baseband, radio (RF) status display, network.	Antenna pointing, baseband health, RF chain
—	Mission Manager / Customer Liaison	External communication, customer reporting, mission-level decisions.	(non-console; observer area)

> **The simulator's position list is not the same list.** The MCS defines **seven** console positions in `mcs/positions.yaml`: `flight_director`, `fdir_systems` ("FDIR / Systems"), `eps_tcs` ("Power & Thermal"), `aocs` ("Flight Dynamics"), `ttc`, `payload_ops` and `propulsion`. The nine-position room above maps onto them: EPS and Thermal both sit at `eps_tcs`; the AOCS Engineer and the FDS analyst both work from `aocs`; Ground Segment and Mission Manager have no MCS position at all. The one position the room roster must add that the table above does not name is the **Propulsion Engineer** — EOSAT-1 flies a 1U green monopropellant module (66.9 m/s total), and `propulsion` is the only position whose `allowed_func_ids` include the burn commands (S8 funcs 108–118). In a nine-seat rehearsal the FDS operator normally doubles as Propulsion, because FDS produces the manoeuvre vector that Propulsion executes.

The EPS and Thermal Engineer positions share a single MCS console image — the `eps_tcs` position, with EPS+TCS tabs — in many control rooms (including the EOSAT-1 reference rig); they are seated **adjacent** so commanding authority hands between them by physical proximity and a single voice-loop callout.

The FDS position works from the `aocs` MCS console (labelled *Flight Dynamics*) plus a workstation outside the MCS rig: it ingests GPS telemetry, propagates the orbit forward, produces the pass schedule the rest of the team flies, and computes manoeuvre vectors when needed. The `aocs` position also carries a **read-only** PROP tab: Flight Dynamics can watch a burn but cannot command one.

The Ground Segment Engineer runs from a secondary workstation and is the **only operator who routinely watches the ground-station RF chain end-to-end** — antenna pointing, baseband health, and the radio status display (carrier presence, lock status, sweep, Doppler estimate). They are the voice of the ground network on the loop.

The Mission Manager is *non-console*: they sit at the back of the room, on the loop but not commanding, and own the customer-facing status report and external mission-level decisions.

Chapter 6 walks each position in depth: the mandate, the console, the telemetry to watch, the commands typically issued, the handover content, the cross-position dependencies, and the common failure modes.

5.5 How the team works: passes, shifts, and the voice loop

Operations is structured around **passes**: the orbital periods when the spacecraft is in contact with a ground station. The EOSAT-1 reference orbit (sun-synchronous, 513 km, 97.39°, LTAN 10:00) produces passes of up to about 9 minutes (7.5 minutes on average) over the five polar stations of the teaching baseline (Iqaluit, Troll, Svalbard, Inuvik and O'Higgins), with roughly 8–13 contacts per ground station per day depending on the station's latitude.

Shifts are organised around the contact schedule. A LEOP shift in industry runs **24/7 with three eight-hour shifts** until phase 4 is complete; commissioning typically runs **16-hour days with two shifts**; nominal operations usually run **8/5 with automation** filling in the rest of the day's passes. The pass discipline is the same across all of these: operators are at their consoles, on the voice loop, ready to command, from AOS minus 5 minutes through LOS plus 5 minutes.

Between passes the team **debriefs** the just-finished pass (5-minute hot-wash), **re-plans** the next pass (planning panel), **reads ahead** in the procedure file for the next phase activity, and **rests, hydrates, and eats** — all four are mission-critical in equal measure across a multi-day LEOP.

The **voice loop** is the spine of the room. Every position is on the loop; every commanded action is voice-coupled. Discipline on the loop matters more than any other single operator skill — industry data attributes ~80% of "recoverable made unrecoverable" anomalies to voice-loop failures, not telemetry misreads. Chapter 9 covers loop discipline in depth.

5.6 The control room physical layout

The control room operations room is laid out as a **horseshoe of consoles facing a central display wall**. The Flight Director seat is at the apex of the horseshoe; the eight operator consoles flank it; the Mission Manager and Mission Director sit at the open end of the horseshoe behind the FD. Voice loop hardware (softphone headsets) is at each seat. The display wall shows the master ground-station map and the AOS/LOS countdown.

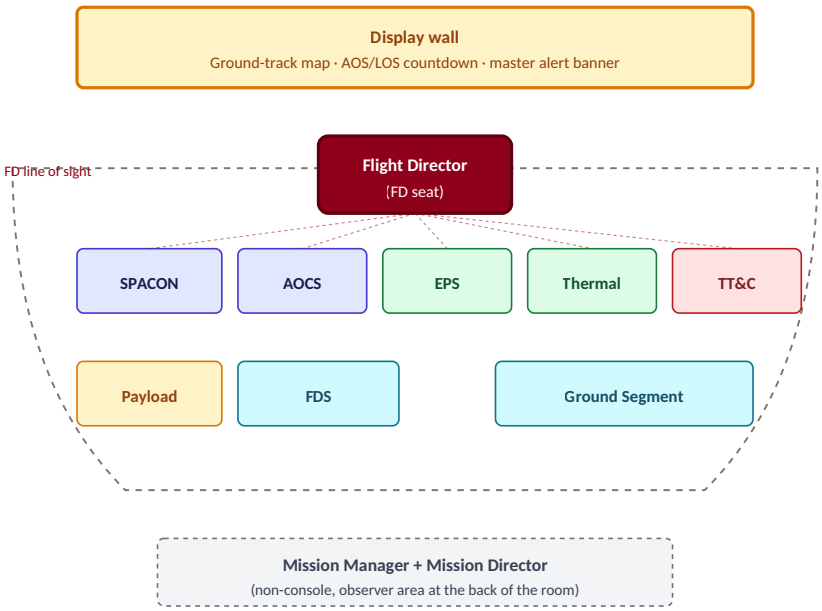


Figure 5.1 — Control-room console layout. The Flight Director seat sits at the apex of a horseshoe with line-of-sight to every operator seat (dashed red lines). Subsystem operators (SPACON, AOCS, EPS, Thermal, TT&C) form the inner ring; Payload, FDS and Ground Segment form the outer ring. The Mission Manager and Mission Director sit at the back, on the loop but not commanding.

The layout matters. A student commanding from the EPS seat must be able to **make eye contact with the FD** before issuing a power-on command, because the GO is voice-and-eye. A student at the Payload seat must be able to **see the AOS countdown** without turning around. These are not arbitrary ergonomics; they

directly affect shift performance, and they are scored on the *Space usage* axis of the peer-review worksheet.

5.7 What we are flying — EOSAT-1 in 600 words

EOSAT-1 is a **6-unit CubeSat** built in the uOttawa SmallSat lab, launched June 2026 into a sun-synchronous orbit at **513 km altitude, 97.39° inclination**, with a 10:00 local time of ascending node, so it passes through Earth's shadow on every orbit. Its mission: ocean-current monitoring along the Canadian coastlines using a four-band multispectral imager (Bands at 443, 560, 665, and 865 nm — blue, green, red, and near-infrared respectively, chosen to maximise sensitivity to dissolved organic carbon and surface chlorophyll).

The spacecraft is **12.0 kg dry** (12.40 kg wet, with 400 g of propellant). The orbit-average load with all nominal systems on is **≈ 22.7 W**; the peak instantaneous load during a payload pass is higher still once the imager, FPA cooler, and transmitter come on. The two deployable wings carry 0.120 m² of cells each and are rated **70.1 W BOL at β = 0**; whole-spacecraft peak generation, including the body panels, is **88.6 W**, and orbit-average generation at end of life is 45.1 W at the worst-generation beta angle of the year (β = −31.2°) — a nominal margin of +22.4 W, or 1.98×. Over the year β stays between −20.8° and −31.2°; because the array drive follows the Sun only to within cos β, the largest |β| gives the least generation, even though its eclipse is the shortest. Storage is a **100 Wh Li-ion pack** (GomSpace NanoPower BPX class, 8S-1P; `eps.yaml` `voltage_nom` 28.8 V, 33.6 V at 100 % SoC, 24.0 V at 0 %, the BPX datasheet values). *(These are flight-part-derived numbers: 12–18 W of orbit-average payload headroom is what a real 6U with deployables delivers. Do not expect the +40 W of slack a mis-sized budget would show.)*

The spacecraft carries **two transmitters**: an S-band prime chain (deployable patch, 2200 MHz downlink / 2025 MHz uplink) for commanding and full-rate telemetry, and a **separate UHF whip beacon** at 437.5 MHz, 1 W, which carries HK SID 11 and nothing else. The distinction matters operationally: switching the S-band PA off stops all S-band telemetry, and the beacon keeps transmitting. It is a second radio, not a mode of the first.

Attitude control is a 4-wheel pyramid with magnetic torquers for momentum management (FINE pointing is flown on three wheels if one is lost, flagged as degraded). Sensors are **two star trackers, ST1 and ST2** (primary attitude determination; the active head is selected with `ST_SELECT`, S8 func 6), coarse sun sensors on all six faces (for safe mode and sun acquisition), one magnetometer (no spare), **three single-axis gyros with no redundant unit**, and a GPS receiver (orbit determination). Pointing performance is read from `aocs.total_pointing_err_3sig` (0x02AA, the RSS of knowledge and control error) and policed on `aocs.att_error` (0x0217), whose alarm bands are ±1.0° yellow and ±2.0° red; measured FINE_POINT performance is 0.021° mean on four wheels and 0.019° on a three-wheel degraded set.

Orbit control is a **1U green monopropellant module**, 400 g of propellant, 66.9 m/s total delta-V, flown for debris avoidance, limited drag make-up and the end-of-life disposal burn. With a ballistic coefficient of 77.9 kg/m², natural decay from 513 km takes 5.0 years at mean solar activity and 14.2 years at solar minimum, which fails the 5-year post-mission disposal rule. The mission therefore ends with NOM-014: a retrograde perigee-lowering burn of 34.3 m/s, flown as six burns of 5.72 m/s, takes the perigee from 513 km to 390 km, after which re-entry takes 1.5 years at mean activity and 4.2 years at solar minimum. The ΔV budget (`prop.yaml`) needs 64.1 m/s — 15.1 m/s of drag make-up over two years at 513 km, 6.1 m/s of orbit and collision-avoidance work, the 34.3 m/s disposal and 8.6 m/s of margin — against 66.9 m/s available (+4.4 %; +2.6 % against the integrated blowdown figure of 65.8 m/s). From mid-mission at least 38 m/s stays reserved for disposal.

These decay times come from the project's drag integration with the CIRA-72 exponential atmosphere, with solar minimum modelled as 0.35 × that table. With NRLMSIS 2.1 and solar minimum held for years, decay

from 390 km takes 7.2 years. The cross-check recorded in `prop.yaml` uses NRLMSIS 2.1 with global-mean density: F10.7 = 140 gives 4.7 years from 513 km, close to the 5.0, but a sustained F10.7 = 70 gives 33 years — an upper bound, since a solar minimum does not last decades. A perigee of 360 km and 42.8 m/s would meet 5 years even under that bound, but the ΔV budget does not close. Two respectable atmosphere models disagree by years: that is the uncertainty a disposal plan has to carry.

The **ground segment** is a five-station polar S-band network (the teaching baseline throughout Part 2):

- **Iqaluit** (Nunavut, 63.75° N): primary, owned by uOttawa, 9 m antenna, G/T 20.0 dB/K.
- **Troll** (Antarctica, 72.01° S): KSAT lease, 7.3 m, G/T 18.5.
- **Svalbard** (78.23° N): 11.3 m, G/T 21.5.
- **Inuvik** (68.32° N): 13.0 m, G/T 22.0.
- **O'Higgins** (63.32° S): 9 m, G/T 20.0.

All five have a 5° minimum elevation mask; the geometry of record is `orbit.yaml`, with RF augmentation in `planning/ground_stations.yaml`.

Mission operations are split into **phases**:

Table 5.3 — EOSAT-1 mission phase plan from launch through disposal with durations and activities.

Phase	Name	Duration	Activity
0	PRE_SEPARATION	(launch)	On the launch vehicle; everything off, no subsystem ticks
1	SEPARATION_TIMER	minutes	Deployer release, tip-off, inhibit timer running
2	INITIAL_POWER_ON	minutes	PDM brings up the unswitchable lines (OBC, TTC RX)
3	BOOTLOADER_OPS	hours	UHF beacon (SID 11 only), OBC bootloader, first AOS
4	LEOP	0–72 hrs	Detumble, deploy antenna and wings, sequential power-on, full HK
5	COMMISSIONING	1–4 weeks	Subsystem checkouts, calibration, First Light
6	NOMINAL	nominal mission	Routine imaging, downlink, health checks

Part 2 covers phases **3, 4, 5, and 6** in operational depth — LEOP, commissioning, and routine nominal operations. Phases 0–2 (launch, separation timer, initial power-on) happen before the first ground contact and are entirely autonomous. There is no disposal phase in the enumeration: end-of-life disposal (NOM-014) is flown in phase 6, which the propulsion arm interlock requires. It is the perigee-lowering burns, then passivation (propellant depleted, propulsion inhibit latched, wheels stopped, charge regulator off, transmitters off), then orbital decay.

The phase counter is `eps.spacecraft_phase` (0x0129), carried in HK SID 4. **Phase transitions are ground declarations** — the mission management team declares the phase with the `SYS_SET_MISSION_PHASE` telecommand (S8 func 120); nothing on board advances it past 4 (LEOP).

The full reference-spacecraft documentation lives in the simulator repo under `configs/eosat1/manual/00..12_*.md`. The most-cited chapters when reading Part 2 are 00 (mission overview), 07 (orbit operations), and 09 (LEOP, `09_leop.md`; the command reference is also numbered 09).

5.8 Reading guide

Operator familiarisation should proceed in this order:

1. **This Chapter (5.0) in full** — the mission, phases, control room.
2. **Chapter 6** — every position, in depth. Read your assigned position first; skim the rest so you know who you're talking to on the loop.
3. **Chapter 7 §§7.1–7.5** — the MCS UI, MPS panels, procedure panel, FDIR panel.
4. **Chapter 8** — pass operations: NOM-001 startup through NOM-012 handover, the routine flow.
5. **Chapter 9** — voice-loop discipline. Read before any shift you're on the loop for.
6. **Chapters 10 and 11** — LEOP and Commissioning sequences, procedure by procedure.
7. **Chapter 12** — contingency operations: the decision framework and anomaly-response decision tree (§12.3). Read before any shift that may face a contingency.
8. **Chapter 13** — operator calculations and checks you'll do every shift.
9. **Chapters 14 and 15** — the procedure catalogue and the telemetry / command / FDIR reference, kept at the console for lookup.
10. **§14.4 and §14.5** — the contingency procedure cards and the emergency procedure cards. Read alongside Chapter 12 before any shift that may face a contingency.
11. **the source-of-truth note** — the source-of-truth reconciliation note — read if you find your simulator giving you values that disagree with this textbook.

External standards cited throughout Part 2. **Executive summary only** is enough for first reading; the vocabulary is used freely throughout.

Table 5.4 — External operability and procedure standards cited throughout Part 2.

Standard	What it covers
ECSS-E-ST-70C	The operator-side of the ground/space contract
ECSS-E-ST-70-11C	What "operability" means for the space segment
ECSS-E-ST-70-32C	What a procedure file looks like
ECSS-E-ST-70-41C	Packet Utilization Standard (PUS) — telemetry/telecommand framing
CCSDS 520.0-G-4	Mission-operations reference model and vocabulary
CCSDS 660.0-G-3	PUS green book — implementation notes
NASA NPR 7120.5F	Space Flight Program and Project Management Requirements

Full URLs are in the References section at the end of this Part.

5.20 Mission Operations Concepts

> **Expected reading before this session.** Open textbook *Concurrent Spacecraft Design* (OER), ch. 18, §§18.1–18.5 (ConOps, operational modes, FDIR architecture, operational procedures, anomaly response); this book, §§5.2–5.5; ECSS-E-ST-70-11C *Space segment operability* — §4 *General requirements* (operability), §5.7 *On-board autonomy* and §5.7.5 *On-board fault management* (FDIR) (clause numbers from the 31 July 2008 issue; the current issue is Rev.1, 15 October 2025); NASA-

> HDBK-1002 *Fault Management Handbook* (Draft 2, 2 April 2012); NASA *Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, Appendix S — *Concept of Operations Annotated Outline*.

Prerequisites: §7.20 and OER ch. 17 (ground segment architecture understood) **References:** ECSS-E-ST-70-11C (Space Segment Operability), ECSS-E-ST-70-32C (Procedures), ECSS-E-ST-70-41C (Packet Utilisation Standard), NPR 7120.5F, NASA Fault Management Handbook (NASA-HDBK-1002)

Learning Objectives

By the end of this session, participants will be able to: 1. Define a Concept of Operations (ConOps) with operational modes and transitions 2. Design an FDIR (Fault Detection, Isolation, Recovery) architecture 3. Write operational procedures for routine and contingency operations 4. Describe anomaly response process with escalation levels 5. Plan spacecraft operational modes and their entry/exit criteria

---

1. Concept of Operations (ConOps)

The ConOps is the bridge between engineering design and operational reality. It describes HOW the system will be used, not just WHAT it can do.

[Source: ECSS-E-ST-70-11C (Space Segment Operability); IEEE 1362 (Concept of Operations Document)]

ConOps Structure

A complete ConOps document covers:

1. **Mission overview** -- Objectives, stakeholders, success criteria
2. **System description** -- Space segment, ground segment, user segment
3. **Operational scenarios** -- Nominal operations, contingency operations
4. **Operational modes** -- Definition, transitions, constraints
5. **Resource management** -- Power, data, propellant budgets over time
6. **Communication plan** -- Contact schedule, data flow, latency requirements
7. **Staffing plan** -- Personnel, shifts, training requirements
8. **Maintenance and logistics** -- Software updates, ground equipment maintenance

Operational Modes

Every spacecraft has a defined set of operational modes. Each mode specifies which subsystems are active, power consumption, and data rates.

Table 5.5 — Spacecraft operational modes with power draw, data rate, duration, and entry conditions.

Mode	Description	Power (W)	Data Rate	Typical Duration	Entry Condition
Safe	Minimum functionality; survival only	5-10	Beacon only (1 bps)	Until ground intervenes	Autonomous fault detection
Detumble	Stop rotation after deployment	8-12	HK only (100 bps)	Minutes to hours	Post-separation; high angular rate
Standby	Sun-pointing; all subsystems ready	12-18	HK telemetry (1 kbps)	Between activities	After detumble; idle periods

Mode	Description	Power (W)	Data Rate	Typical Duration	Entry Condition
Science	Payload operating; attitude controlled	25-40	Science data (variable)	Target-dependent	Scheduled; attitude achieved
Downlink	High-rate data transfer to ground	20-30	Full rate (2-10 Mbps)	During GS contact	GS in view; link established
Manoeuvre	Orbit adjustment or desaturation	15-25	HK only	Minutes	Scheduled; constraints met
Eclipse	Reduced operations during eclipse	8-15	HK only (100 bps)	~35 min (LEO)	Sun not in view

Mode Transition State Machine

Spacecraft mode-transition state machine with nominal (green), standby (amber), and anomaly (red) states and the labelled triggers that move the spacecraft between them.

Spacecraft mode-transition state machine with nominal (green), standby (amber), and anomaly (red) states and the labelled triggers that move the spacecraft between them.

Real Mission Example: PROBA-2 Operations

ESA's PROBA-2 (launched 2009, still operational in 2026) demonstrates mature CubeSat-class operations: - **5 operational modes:** Standby, Science, Manoeuvre, Safe, Off - **Autonomous operations:** 95% of routine activities automated via onboard timeline - **Ground contacts:** 4-6 passes/day via ESA Redu station (S-band) - **Anomaly rate:** ~2-3 per year requiring ground intervention (after initial commissioning) - **Key lesson:** Investing in onboard autonomy dramatically reduces operations cost

[Source: ESA PROBA-2 Operations Team, "PROBA-2: Over a Decade of Operations", SpaceOps 2020]

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2. FDIR Architecture

FDIR (Fault Detection, Isolation, Recovery) is the spacecraft's autonomous ability to detect failures, determine which component failed, and take corrective action -- all without ground intervention.

[Source: NASA Fault Management Handbook (NASA-HDBK-1002); ECSS-E-ST-70-11C section 5.3] [URL: https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002]

Why FDIR is Critical for CubeSats

- **Limited ground contact:** LEO CubeSats are out of view ~85% of the time
- **No crewed intervention:** Unlike ISS, there is no astronaut to "fix" problems
- **Thermal constraints:** A safe mode must maintain thermal limits within minutes
- **Power constraints:** Battery can discharge completely in one eclipse without load shedding

FDIR Hierarchy (Levels)

Table 5.6 — Five-level FDIR hierarchy from hardware watchdogs to ground-commanded recovery.

Level	Detection	Response	Latency	Example
0 -- Hardware	Built-in watchdog	Component reset	Milliseconds	Processor watchdog timer

Level	Detection	Response	Latency	Example
1 -- Unit	Unit-level monitoring	Unit reconfiguration	Seconds	EPS over-current trip
2 -- Subsystem	Subsystem health check	Subsystem fallback mode	Seconds-minutes	Switch to backup star tracker
3 -- System	System-level anomaly detection	Mode change (e.g., enter Safe)	Minutes	Transition to Safe Mode
4 -- Ground	Human-in-the-loop diagnosis	Ground command recovery	Hours-days	Anomaly investigation + patch

FDIR State Machine

FDIR state machine: NORMAL → MONITOR → DETECT → ISOLATE → RECOVER → EVALUATE, with the fault levels that trigger each transition.

FDIR state machine: NORMAL → MONITOR → DETECT → ISOLATE → RECOVER → EVALUATE, with the fault levels that trigger each transition (the same state machine is used as the operational reference in §15.4).

Common CubeSat FDIR Rules

Table 5.7 — Common CubeSat FDIR rules with detection method, threshold, and autonomous response.

Fault	Detection Method	Threshold	Response
Processor lockup	Watchdog timer	No heartbeat for 60 s	Hardware reset (Level 0)
Battery under-voltage	EPS voltage monitor	V_bat < 6.0V	Load shedding; enter Safe Mode
Battery over-temperature	Temperature sensor	T_bat > 45 C	Disable charging; reduce loads
Attitude loss	No attitude solution for 5 min	ADCS health flag timeout	Enter Detumble Mode
Communication loss	No uplink for 48 hr	Timer-based	Reset TTC; revert to beacon mode
Memory corruption	EDAC error count	> 10 uncorrectable errors/hr	Memory scrub; reboot from backup
Solar array current anomaly	Current sensor	I_SA < expected - 30%	Check attitude; enter Standby
Reaction wheel over-speed	Wheel speed monitor	Omega > 5000 RPM	Reduce momentum; desaturation

Key Design Principles

1. **Safe Mode must always work** -- It is the last resort; it must be tested exhaustively
2. **Fail operational, then fail safe** -- Try to continue the mission before giving up
3. **No single fault should cause mission loss** -- Cross-reference with FMECA
4. **Ground override capability** -- Every autonomous action must be commandable from ground
5. **Logging** -- Record all fault events in non-volatile memory for post-analysis

---

3. Operational Procedures

[Source: ECSS-E-ST-70-32C (Procedure Definition Language); ECSS-E-ST-70-01C section 5.6]

Procedure Types

Table 5.8 — Four procedure types distinguished by purpose, execution mode, and example.

Type	Purpose	Execution	Example
Nominal	Routine scheduled operations	Automated (timeline) or manual	Science observation sequence
Contingency	Response to known anomaly types	Semi-automated; operator decision	Safe mode recovery
Emergency	Response to critical unexpected events	Manual; real-time decision	Communication loss recovery
Maintenance	Periodic calibration or updates	Scheduled manual	Magnetometer calibration

Procedure Structure (ECSS-E-ST-70-32C)

Each procedure has: 1. **Identifier** -- Unique procedure number (e.g., NOM-001) 2. **Purpose** -- What the procedure accomplishes 3. **Preconditions** -- System state required before starting 4. **Steps** -- Ordered list of actions, verifications, decisions 5. **Expected results** -- What should happen at each step 6. **Recovery actions** -- What to do if expected results are not observed 7. **Post-conditions** -- System state after successful completion

Example: Science Observation Procedure

```
Procedure: NOM-SCI-001 "Earth Observation Target Acquisition"
Preconditions: Mode = Standby, Battery SoC > 60%, Target in FOV within 10 min

Step 1: Verify ADCS mode = Fine Pointing
  Expected: ADCS status = FINE, pointing error < 0.1 deg
  If not: Wait 60 s, retry. If still not achieved -> ABORT, remain in Standby

Step 2: Command payload power ON
  Expected: Payload HK reports power = ON within 5 s
  If not: Retry power command. If 3 fails -> ABORT, log anomaly

Step 3: Wait for payload thermal stabilisation (90 s)
  Expected: Payload temperature within operational range

Step 4: Command payload to imaging mode
  Expected: Payload status = IMAGING, frame counter incrementing

Step 5: Wait for target acquisition (ground-predicted start time)
  Expected: Imaging starts at predicted time +/- 5 s

Step 6: Collect imagery for scheduled duration

Step 7: Command payload to standby
  Expected: Payload status = STANDBY
```

```
Step 8: Command payload power OFF (if no more targets this orbit)
  Expected: Payload HK reports power = OFF

Post-conditions: Mode = Standby, imagery stored in mass memory
```

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4. Anomaly Response Process

[Source: NASA-HDBK-1002 (Fault Management Handbook); ESA Anomaly Review Board procedures]

Anomaly Escalation Levels

Table 5.9 — Four anomaly severity levels with response time, authority, and example trigger.

Level	Severity	Response Time	Authority	Example
Green	Informational	Next business day	Operations engineer	Minor parameter out of expected range
Yellow	Warning	Within 4 hours	Lead operator	Subsystem performance degraded
Orange	Serious	Within 1 hour	Flight director	Mode change required; mission impact
Red	Critical	Immediate	Project manager + FD	Mission-threatening; safe mode entered

Anomaly Response Workflow

```
1. DETECT: Telemetry alarm or operator observation
2. ASSESS: Is this a known anomaly type? Check procedures library
   -> Known: Execute contingency procedure
   -> Unknown: Continue to step 3
3. DIAGNOSE: Gather additional telemetry, correlate events, form hypothesis
4. PLAN: Develop response plan (may require Anomaly Review Board)
5. EXECUTE: Implement recovery commands (verify each step)
6. VERIFY: Confirm system returned to expected state
7. DOCUMENT: Record anomaly, root cause, response, and lessons learned
```

Anomaly Report Form Fields

Table 5.10 — Standard fields required on a written anomaly report form.

Field	Description
Anomaly ID	Unique identifier (e.g., ANO-2026-042)
Date/Time (UTC)	When the anomaly was detected
Severity	Green / Yellow / Orange / Red
Affected subsystem	EPS, AOCS, TTC, Payload, etc.
Observation	What was observed (TM values, trends)
Diagnosis	Root cause analysis

Field	Description
Action taken	Commands sent, procedures executed
Result	System state after response
Status	Open / Resolved / Under investigation
Lessons learned	Preventive measures for future

Real Mission Example: Kepler Reaction Wheel Failure

NASA's Kepler space telescope lost reaction wheel #2 in July 2012, followed by wheel #4 in May 2013. The anomaly response:

1. **Detection:** Elevated friction torque in wheel #2 HK telemetry
2. **Assessment:** Known degradation signature; monitoring intensified
3. **Failure:** Wheel #2 ceased operation. Mission continued on 3 wheels.
4. **Second failure:** Wheel #4 failed 10 months later. Only 2 wheels remaining -- insufficient for fine pointing.
5. **Recovery:** Engineering team developed the "K2" mission concept using solar radiation pressure as a "virtual third wheel"
6. **Result:** K2 operated for 4+ additional years, discovering 2,700+ exoplanet candidates

Lesson: Creative operational workarounds can save missions. FDIR should be designed with graceful degradation, not just "safe mode or nothing."

[Source: Howell et al., "The K2 Mission: Characterization and Early Results", PASP, 2014]

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5. Operations Concepts Exercise

Instructions

1. **ConOps Editor** in SpaceCDF: - Define all operational modes for your mission (at least 5) - Set entry/exit conditions for each mode - Verify power budget closes in each mode (especially Safe and Eclipse)
2. **FDIR Design:** - Define 5 FDIR rules for your mission (fault, detection, threshold, response) - Verify that Safe Mode power budget is sustainable indefinitely - Check: Can every FDIR response be overridden from ground?
3. **Worksheet 5.2:** - Complete the operational modes table - Write one nominal procedure (science observation or downlink) - Write one contingency procedure (safe mode recovery) - Fill in the FDIR rules table

Discussion Prompts

- "What is the worst anomaly that could happen to your satellite? How would you respond?"
- "If the satellite enters safe mode on Friday evening and the next ground contact is Monday, will it survive?"
- "How much autonomy should the spacecraft have? Where is the line between onboard and ground decision-making?"

Worksheet 5.2 Tasks

1. Complete the operational modes table (5+ modes with power, data rate, entry/exit)
2. Design the FDIR state machine (at least 5 rules)
3. Write one nominal and one contingency procedure
4. Complete the anomaly response form for a hypothetical scenario
5. Estimate operations staffing requirements by mission phase

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Cited References

Table 5.11 — Cited references for the Mission Operations Concepts session with URLs.

#	Reference	URL
1	ECSS-E-ST-70-11C (Space Segment Operability)	https://ecss.nl/standard/ecss-e-st-70-11c-space-segment-operability/
2	ECSS-E-ST-70-32C (Procedures)	https://ecss.nl/standard/ecss-e-st-70-32c-test-and-operations-procedure-language/
3	NASA Fault Management Handbook (NASA-HDBK-1002)	https://standards.nasa.gov/standard/NASA/NASA-HDBK-1002
4	ESA PROBA-2 Operations, SpaceOps 2020	https://www.spaceops.org/
5	Howell et al., K2 Mission, PASP 2014	https://doi.org/10.1086/676406
6	ECSS-E-ST-70-41C (Packet Utilisation Standard)	https://ecss.nl/standard/ecss-e-st-70-41c-telemetry-and-telecommand-packet-utilization/

Session Summary

Table 5.12 — Session-summary key takeaways across operations-concepts topics.

Topic	Key Takeaway
ConOps	Defines HOW the system is operated: modes, transitions, schedules, staffing
Operational modes	Safe, Detumble, Standby, Science, Downlink, Manoeuvre, Eclipse (minimum set)
Mode transitions	Defined by entry/exit conditions; Safe Mode always reachable from any mode
FDIR	5 levels: Hardware -> Unit -> Subsystem -> System -> Ground; Safe Mode is last resort
Procedures	Nominal (routine), Contingency (known faults), Emergency (unknown), Maintenance
Anomaly response	Detect -> Assess -> Diagnose -> Plan -> Execute -> Verify -> Document
Design principles	Safe Mode must always work; fail operational before fail safe; ground override always

5.9 References

- ECSS-E-ST-70-11C — Space segment operability — ecss.nl

- CCSDS 520.0-G-4 — Mission Operations Reference Model — public.ccsds.org
- NASA NPR 7120.5F — NASA Space Flight Program and Project Management Requirements — nodis3.gsfc.nasa.gov

Chapter 6 — The Control Room: Nine Positions in Depth

This chapter is the operator's reference: one section per position. For each, you will find the **mandate**, the **console** (visible tabs and the position's place in the room), the **telemetry to watch** (top mnemonics with limits), the **commands typically issued** (with PUS service and function ID), the **handoff content** (what the outgoing operator must brief their relief), and **cross-position dependencies** (who do you talk to first).

The position descriptions are valid for **EOSAT-1 only**. Real industry positions are broader and the boundaries blur — but the discipline of *one operator, one mandate, one set of telemetry, one set of commands* is universal.

> **In-tool vs off-tool positions.** The MCS defines **seven** console positions in `mcs/positions.yaml`: `flight_director`, `fdir_systems` (the SPACON seat), `eps_tcs` ("Power & Thermal"), `aocs` ("Flight Dynamics"), `ttc`, `payload_ops` and `propulsion`. Each position gets a filtered tab set, a subsystem list and — except for the Flight Director, who has `allowed_commands: all` — an explicit `allowed_func_ids` whitelist, so a button the position may not press renders disabled. Two of the nine room positions are held *off-tool*: the **Mission Manager** tracks the mission programme externally (programme tracker, customer comms, schedule), and the **Ground Segment** position drives the RF simulation tool rather than the MCS — its primary display is the radio page served by `smo-rfsim`, not a position tab. The **Flight Dynamics System** engineer works from orbit-determination software on a separate workstation *and* the `aocs` console. When this chapter says "switch to the EPS tab", treat that as MCS-relative; when it says "FD calls GO/NO-GO", that is a headset action — the MCS has no GO/NO-GO button and no voice-loop feature.

6.1 Flight Director (FD)

Mandate. The FD is the single point of authority on the loop. Nothing happens on the spacecraft without an FD GO. The FD does **not command** — the FD authorises others to command. The FD runs the GO/NO-GO rounds before every commanded action. The FD calls *no-go* and rolls the team back to the last verified state when the situation is unclear.

Authority. Highest in the room. Outranked only by the Mission Manager on mission-level decisions (e.g. "should we attempt First Light during this pass with the star tracker degraded?" — the FD recommends, the Mission Manager decides).

Console. Every tab visible — `overview`, `eps`, `aocs`, `tcs`, `obdh`, `ttc`, `payload`, `prop`, `commanding`, `pus`, `procedures`, `trends`, `manual` — plus the four FD-only displays declared in `mcs/displays.yaml`: the *system dashboard* (a composite health view), the *power monitor*, the *contact schedule* (next AOS/LOS, station, time-to-AOS), the *procedure panel* (progress, current step, state) and the *FDIR panel* (alert level and alarm counts). The FD's two monitors are usually configured: **left** = system dashboard + contact schedule; **right** = current procedure file with the active step highlighted. There is no voice-loop chat in the MCS; the loop is headsets.

Telemetry to watch. The FD does not *interpret* telemetry — that is the subsystem operator's job. The FD watches **state**: which phase the spacecraft is in (`eps.spacecraft_phase`, 0x0129, carried in HK SID 4), the FDIR alert level and alarm counts (`fdir.level` 0x0700, `fdir.active_alarms` 0x0701,

`fdir.critical_alarms` 0x0702, `fdir.high_alarms` 0x0703 — the FDIR panel's four parameters), and the top-level OK/yellow/red state from the system dashboard. If any position calls *yellow*, the FD wants to know within 5 seconds; if any position calls *red*, the FD wants to know on the loop, by name, with the parameter and the value.

Commands typically issued. Zero, *by discipline* — not by tooling. The `flight_director` position is the only one with `allowed_commands: all`, so the MCS will let the FD send anything; the rule that the FD authorises rather than commands is a room convention the team enforces on the loop. There is no FD second-key interlock in the MCS. What the tool does give you is a **criticality flag on each telecommand** in `commands/tc_catalog.yaml` — `nominal`, `caution` or `critical` (`OBC_REBOOT`, `OBC_SWITCH_UNIT`, `PROP_ARM`, `PROP_SET_BURN`, `SYS_SET_MISSION_PHASE`; the sender must press ARM before SEND on these) — and the dual-key discipline of this course is: nothing marked *caution* or *critical* leaves the console without a spoken FD GO, read back by name.

Handoff content. When an FD hands shift to the relieving FD, the brief must cover:

1. **Current phase** and time-to-next-phase boundary.
2. **All active anomalies** and which procedure addresses each.
3. **Open work items** — procedures begun but not completed.
4. **Next pass** — station, AOS time, planned activities, who is the operator-of-record for each major step.
5. **Customer commitments** active in this shift (Mission Manager provides the list).
6. **Any GO authorisations granted but not yet executed** — these are the bear traps.
7. **The voice-loop tone** — is the team tight? tired? distracted? The relieving FD needs the human read.

A formal NOM-012 *Shift Handover* template is provided in the workbook; the target completion time is 6 minutes.

Cross-position dependencies. The FD talks to everyone. By priority order during a pass: SPACON (running the procedure), TT&C (link health), the originating subsystem operator (whoever is about to command), Ground Segment (station health), AOCS (pointing), then EPS/Thermal/Payload/FDS as needed.

Common failure modes (FD-specific). Over-driving the position — telling operators what to command rather than asking for their assessment. Going silent on the loop during a tense moment — operators interpret silence as "FD is processing", and that pause is exactly when wrong commands happen. Granting GO prematurely — *always* poll all relevant positions, even when the answer feels obvious.

6.2 Spacecraft Controller / FDIR & Systems (SPACON)

Mandate. The SPACON owns the **procedure execution**. They hold the procedure file, call each step on the voice loop, verify the verification, and request FD GO before each commanded action. They are also the FDIR watch — when the FDIR panel lights up, the SPACON is the operator who reads the rule, identifies the contingency procedure to invoke, and recommends the response to the FD.

Authority. Procedure authority. Commanding authority within authorised steps, with FD GO. Recommends safe-mode entry; only the FD authorises safe-mode entry.

Console. The `fdir_systems` position: overview, obdh, commanding, pus, procedures, trends, manual. It is the only position other than the FD with the **PUS tab**, and the only one allowed service 12 (on-board monitoring) and service 19 (event-action). The SPACON has two monitors: **left** = procedure panel + FDIR panel; **right** = OBDH telemetry (OBC state, memory, bus health, reboot cause).

Telemetry to watch. OBC mode (`obdh.mode` , 0x0300, SID 4), running image (`obdh.sw_image` , 0x0311 — 0 = bootloader, 1 = application), active OBC and backup status (`obdh.active_obc` 0x030C, `obdh.obc_b_status` 0x030D), bus selection and health (`obdh.active_bus` 0x030E, `obdh.bus_a_status` 0x030F, `obdh.bus_b_status` 0x0310), reboot count and cause (`obdh.reboot_count` 0x030A, `obdh.last_reboot_cause` 0x0316), recent event reports (the S5 event log), and the command counters: received `obdh.tc_rx_count` (0x0304), accepted `obdh.tc_acc_count` (0x0305), rejected `obdh.tc_rej_count` (0x0306) — and, next to it, `obdh.last_tc_error_code` (0x0315), the S1 failure code of the most recent refusal. Read the code, not just the counter: a rejection counter that ticks without a code tells you nothing. Yellow at 5 rejects in any 30 s; red at 10. Note there is **no `fdir_table_id` parameter and no rule-table concept** — see the commanding note below.

Commands typically issued. All procedure-driven commands flow *through* the SPACON. Direct OBDH commands the SPACON owns:

- `OBC_REBOOT` (S8 func_id 52) — restart the OBC.
- `OBC_SWITCH_UNIT` (S8 func_id 53) — switch to redundant OBC.
- `OBC_BOOT_APP` (S8 func_id 55) — boot application software from bootloader.
- `SET_TIME` (S9.1) — synchronise onboard clock; `OBC_GPS_TIME_SYNC` (S8 func_id 80) for the GPS-sourced update.
- `EA_ENABLE` / `EA_DISABLE` (S19.4 / S19.5, argument `ea_id`) — enable or inhibit individual event-action rules. The simulator does not implement rule-table swapping; commissioning-phase isolation is achieved by inhibiting nominal rules while LEOP rules stay enabled, then re-enabling them one at a time once commissioning closes. `EA_ADD` (S19.1) and `EA_DELETE` (S19.2) define and remove rules; `EA_REPORT_DEFS` (S19.8) dumps them.
- `MON_ENABLE_DEF` / `MON_DISABLE_DEF` (S12.3 / S12.4) — the on-board limit monitors, which only this position may command.

Handoff content. Current procedure in progress (step number, last verified step), all FDIR anomalies and their state (triggered/acknowledged/cleared), any event-action rules or S12 monitors manually disabled (and why), command rejection rate trend and the last TC failure code.

Cross-position dependencies. Constantly with FD (every GO request). Constantly with the operator-of-record for the active step (e.g. AOCS engineer if the step is a wheel command). With Ground Segment if a TC is rejected by the spacecraft — sometimes the rejection is uplink (a flipped bit), not onboard.

Common failure modes. Skipping verification steps to stay on schedule. Reading ahead in the procedure and pre-commanding ("priming the next step") — pre-commanding without an explicit FD GO is a procedure violation. Failing to log an anomaly because the team is busy responding to it.

6.3 AOCS Engineer

Mandate. Attitude determination and control. Reads the attitude sensors (two star trackers, coarse sun sensors, gyros, the magnetometer, GPS), the actuators (four reaction wheels, magnetic torquers), and the AOCS state machine. Reports pointing performance every pass. Owns the **attitude** for a manoeuvre — the burn itself belongs to the Propulsion position.

Authority. Commanding authority for the AOCS subsystem with FD GO (`allowed_func_ids` 0–15, 84–91 and 96). Recommends AOCS mode transitions; the FD authorises. **No** propulsion commanding: funcs 108–118 are deliberately absent from this position's whitelist, so every button on its PROP tab is disabled.

Console. The `aocs` position, labelled *Flight Dynamics*: overview, aocs, prop (read-only), commanding, procedures, trends, manual. The AOCS tab shows attitude and reference quaternions, body rates, the pointing-error channels, wheel speeds and temperatures, sensor health, the AOCS mode integer — **and, since the 2026-08b batch, the commanded mode alongside the actual one** (`aocs.commanded_mode` 0x0297, `aocs.mode_demoted` 0x0298), which is how you tell a transient apart from a spacecraft that has quietly abandoned its mission mode.

Telemetry to watch.

Table 6.1 — AOCS telemetry parameters with SIDs and nominal, yellow, and red limit bands.

Mnemonic	Hex id	SID	Nominal	Yellow	Red
`aocs.mode`	0x020F	2	4 (NOMINAL) / 5 (FINE_POINT)	2 (DETUMBLE) / 3 (COARSE_SUN)	0 (OFF) / 1 (SAFE_BOOT)
`aocs.rate_roll` / `rate_pitch` / `rate_yaw`	0x0204–06	2	< 0.01 °/s	±0.5	±2.0
`aocs.att_error`	0x0217	2	< 0.3°	±1.0	±2.0
`aocs.st1_status` / `st2_status`	0x0240 / 0x0243	2	2 (tracking)	1 (boot) / 3 (blind)	0 (off) / 4 (failed)
`aocs.rw1_speed` .. `rw4_speed`	0x0207–0A	2	< 3000 rpm	±5000	±5500 (speed limit)
`aocs.gyro_units_on`	0x0294	2	3	2	0–1 (no rate reference)
`aocs.gps_fix`	0x0274	2	3 (3D+vel)	1–2	0 (no fix) > 10 min
`aocs.h_env_frac`	0x02BA	7	< 50 %	> 70 %	> 90 % (desat now)

(Mode integers per `aocs/aocs_mode_codes.yaml`, which is the single source of truth: 0 = OFF, 1 = SAFE_BOOT, 2 = DETUMBLE, 3 = COARSE_SUN, 4 = NOMINAL, 5 = FINE_POINT, 6 = SLEW, 7 = DESAT, 8 = ECLIPSE, 9 = INERTIAL. Older cards and older printings of this book were systematically **off by one** — check every mode number you read against that file. Note also that the determination block, including the momentum-envelope figure, is HK **SID 7**, which is disabled at boot and must be enabled with `HK_ENABLE`.)

Commands typically issued.

- `AOCS_SET_MODE` (S8 func_id 0) with mode integer.
- `AOCS_ENABLE_WHEEL` / `AOCS_DISABLE_WHEEL` (S8 func_id 3 / 2) with `wheel_idx`.
- `RW_SET_SPEED_BIAS` (S8 func_id 8) with `wheel_idx`, `bias_rpm` — the bias is **added to** the current speed, not an absolute setpoint. `AOCS_RW_RAMP_DOWN` (func_id 14) ramps a wheel to a target speed; `wheel_idx=255` means all four. Wheel speeds (0x0207–0x020A) are **read-only telemetry**: these two commands are the only way to move them from the ground.
- `ST1_POWER` / `ST2_POWER` (S8 func_id 4 / 5) with `on`, and `ST_SELECT` (S8 func_id 6) with `unit` — 1 or 2 — to choose the active head. Two star trackers, two commands.

- `MTQ_ENABLE` (S8 func_id 9) with `enable`.
- `AOCS_DESATURATE` (S8 func_id 1) — runs the momentum dump per NOM-007. `AOCS_SET_DESAT_INHIBIT` (func_id 96) latches dumps off.
- `AOCS_GYRO_CALIBRATION` (S8 func_id 13) — gyro bias calibration is computed **on board**; there is no ground-computed bias upload.
- `AOCS_EST_RESET` (S8 func_id 91) — reset the MEKF; check `aocs.est_source` (0x02C1) afterwards for which sensors it is actually using.
- `AOCS_SET_TARGET_QUAT` (func_id 86) / `AOCS_SET_TARGET_GEO` (func_id 87) / `AOCS_SLEW_TO` (func_id 10) — pointing targets.
- Gyros are powered individually as S2 devices: `S2_AOCS_GYROSCOPE_0/1/2` (S2.1). There are three of them, one per axis, and **no redundant unit**.

Handoff content. Current AOCS mode and time since last transition (`aocs.time_in_mode`, 0x0264), and whether the actual mode matches the commanded one; pointing-error trend over the pass; momentum envelope usage and any wheel approaching the 5500 rpm speed limit; which star tracker is selected (`aocs.st_selected`, 0x0244) and the health of both; whether FINE is running degraded on three wheels (`aocs.fine_degraded`, 0x0299); any desat inhibit still latched.

Cross-position dependencies. With FDS for manoeuvre planning (FDS produces the vector) and with Propulsion, who arms and fires it while AOCS holds the burn attitude. With Payload for pointing during imaging (Payload requests fine-point mode 60 s before imaging). With SPACON for FDIR rules that touch attitude (many of them — momentum saturation, star-tracker loss, gyro disagreement).

Common failure modes. Commanding NOMINAL (mode 4) before detumble has converged — and detumble is genuinely slow: the B-dot law is magnetometer-driven with the dipole saturating at 1.6 A·m² per axis, so a 14 °/s separation tumble takes of order 3.5×10^3 s to damp, passing below the 0.5 °/s exit guard at around 1166 s. Any expectation of a fast detumble is wrong. Mis-reading the mode integer (the off-by-one described above is specifically here). Not noticing momentum building over many orbits. Expecting a desat to finish in five minutes: an honest dump is **20–50 minutes**, it inhibits imaging while it runs, it has a timeout, and it returns the spacecraft to its pre-desat mode.

6.4 EPS Engineer

Mandate. The power story. Owns battery state of charge, solar array currents, the bus voltage, the load profile, and the power-line switching matrix. Calls eclipse entry and exit. First position to recognise that a subsystem is drawing more than it should.

Authority. Commanding authority for EPS subsystem (power-line on/off, charge regulator parameters) with FD GO. Calls for load shed (the FD authorises).

Console. Overview, EPS, TCS, commanding, procedures, trends, manual. Shared `eps_tcs` console image with the Thermal Engineer in the simulator; in the operations programme the two operators sit side-by-side and the EPS Engineer focuses on the *EPS* tab while Thermal focuses on the *TCS* tab.

Telemetry to watch.

Table 6.2 — EPS telemetry parameters with SIDs and nominal, yellow, and red limit bands.

(The pack is 8S — nominally 28.8 V, 33.6 V at 100 % SoC and 24.0 V at 0 % (GomSpace BPX datasheet). The bus is not regulated: it follows the battery up to a 33.4 V cap. That is high for a 6U, where 2S buses are common; it is what `eps.yaml` models, so treat the voltage values below as simulator-specific. Bands are as shipped in `mcs/limits.yaml`.)

Mnemonic	Hex id	SID	Nominal	Yellow	Red
`eps.bat_soc`	0x0101	1	70–95 %	< 25 %	< 15 %
`eps.bat_voltage`	0x0100	1	29.6–33.6 V	< 25.9	< 24.6
`eps.bus_voltage`	0x0105	1	30.9–33.4 V	< 28.4	< 27.7
`eps.sa_a_current` / `sa_b_current`	0x0103 / 0x0104	1	lit-side nominal	low (lit)	≈ 0 (lit)
`eps.bus_load_current`	0x0148	1	sum of the nine lines	rising trend	per-line trip
Per-line current (9 lines, 0–8)	0x0118–0x011F, 0x014B	1	per-line nominal	per-line yellow	per-line red
`eps.bat_temp`	0x0102	1	5–25 °C	< 2 or > 40	< 0 or > 45
`eps.wing_status`	0x0144	1	bits 0+1 set (both deployed)	bit 4/5 (in progress)	neither bit set post-LEOP

Every bus-protection level comes from one named table, `simulation.yaml` `eps.bus_protection`: under-voltage flag and in-model load shed at **28.4 V** (≈ 46 % SoC), FDIR EPS safe mode at **27.7 V** (≈ 39 %), AOCS safe at 26.1 V, the fault cascade at 25.9 V, over-voltage at 34.0 V. The bus yellow and red limits are the first two, and all of them sit above the yellow band on `eps.bat_voltage` (25.9 V) — so the spacecraft starts shedding before the battery-voltage limit alarms. Watch the bus.

Commands typically issued.

- `EPS_POWER_ON` (S8 func_id 19) with `line_index`.
- `EPS_POWER_OFF` (S8 func_id 20) with `line_index`.
- `EPS_RESET_OC_FLAG` (S8 func_id 21) with `line_index` — clear an overcurrent trip.
- `EPS_EMERGENCY_LOAD_SHED` (S8 func_id 25) with `stage` (0–3).
- `EPS_SET_CHARGE_RATE` (S8 func_id 23) — override charge current.
- `EPS_DEPLOY_WING` (S8 func_id 81) with `wing`, and `EPS_WING_STATUS` (func_id 82).

There are **nine power lines, indices 0–8**: 0 obc, 1 ttc_rx, 2 ttc_tx, 3 payload, 4 fpa_cooler, 5 htr_bat, 6 htr_obc, 7 aocs_wheels, 8 prop. Lines 0 and 1 are unswitchable; every switchable line **defaults OFF at power-up**, which is why LEOP-007 exists. EPS bills each line from the actual state of the hardware on it, not from the position of the switch — a line that is on but idle costs almost nothing.

Handoff content. Battery SoC trend across the pass; predicted SoC at next eclipse exit; any line drawing more than nominal; load shed level currently active (if any); battery temperature trend; per-wing currents (`eps.sa_wing_py_current` 0x0146, `eps.sa_wing_my_current` 0x0147) — asymmetry between the two wings is the tell for panel degradation or a seized array drive (`sada_stuck`).

Cross-position dependencies. With Thermal (heaters are the largest controlled load and EPS sees their duty cycle). With Payload (the payload power-on draw is the largest predictable load spike; coordinated with EPS pre-pass). With FDS (eclipse times — EPS uses them for SoC prediction).

Common failure modes. Failing to plan for eclipse depth before commanding a high-load activity. Reading load current yellow but not raising it because "we usually run a bit hot". Forgetting to re-enable the line after a manual load-shed test.

6.5 Thermal Engineer

Mandate. Thermal control. Owns heater control, radiator performance, payload-zone temperatures (critical for FPA), battery-zone temperatures, and the FPA cooler.

Authority. Commanding authority for TCS (heater on/off, heater set-points, FPA cooler enable) with FD GO.

Console. Overview, EPS, TCS, commanding, procedures, trends, manual. Shared `eps_tcs` image (see EPS note above).

Telemetry to watch.

Table 6.3 — Thermal-control telemetry parameters with SIDs and nominal, yellow, and red limit bands.

Mnemonic	Hex id	SID	Nominal	Yellow	Red
`tcs.temp_battery`	0x0407	3	5–25 °C	< 2 or > 40	< 0 or > 45
`tcs.temp_obc`	0x0406	3	10–35 °C	< −10 or > 60	< −20 or > 70
`tcs.temp_fpa`	0x0408	3	−15 (at setpoint)	< −22 or > 14	< −25.5 or > 17.5
`tcs.cooler_fpa`	0x040C	3	ON / OFF (commanded)	(n/a)	UNEXPECTED OFF
`tcs.htr_duty_battery` / `htr_duty_obc`	0x040E / 0x040F	3	10–60 %	> 80 %	100 % saturated
`tcs.htr_total_power`	0x0411	3	< 12 W installed	> 12.5	> 14
`tcs.fpa_cooler_floor_c`	0x0424	(on demand)	below setpoint	within 2 K	above setpoint
`tcs.fpa_cooler_margin_k`	0x0425	(on demand)	> 2 K	0–2 K	≤ 0 (setpoint unreachable)

Commands typically issued.

- `HEATER_BATTERY` / `HEATER_OBC` / `HEATER_THRUSTER` (S8 func_id 40/41/42) with `on=0/1` — per-circuit heater control.
- `HEATER_SET_SETPOINT` (S8 func_id 44) with `circuit`, `on_temp`, `off_temp`; `HEATER_AUTO_MODE` (func_id 45) with `circuit`.
- `EPS_FPA_COOLER` (S8 func_id 17) or `FPA_COOLER` (S8 func_id 43) with `on=0/1`.
- `PAYLOAD_COOLER_SETPOINT` (S8 func_id 36) — argument `setpoint` is a **uint16 in centi-kelvin**, `raw = round((T_degC + 273.15) × 100)`, so −15 °C is 25815 = 0x64D7. The commandable envelope is [−20 °C, 0 °C]; outside it the TC is rejected with S1.8 error 0x0003, not silently clamped.

(Editor's note. Real-mission multispectral imagers run FPA setpoints far colder — often near −50 °C — to suppress dark current. EOSAT-1's cooler is a miniature Stirling-class cryocooler, not a Peltier stack, but it is a small one: its authority is modelled at 23 % of Carnot (`fpa_cooler_carnot_fraction`, 15 W input), so its heat lift collapses as the temperature difference grows and the commandable envelope stops at −20 °C. That is why `fpa_cooler_margin_k` (0x0425) exists: "the cooler is struggling" is a readable state, and a negative margin is the honest answer to an FPA that will not come ready. The wording "Peltier envelope" survives in a few generated files; the machine being modelled is Stirling.)

Handoff content. All zones in band? Any heater at saturated duty cycle (indicates failing thermal path)? FPA cooler state, temperature, and remaining cooling margin — critical for payload commissioning. Any thermal trend with the orbit (entering/exiting eclipse).

Cross-position dependencies. With EPS (heater power, cooler power). With Payload (FPA temperature is a hard constraint for imaging). With AOCS (sun angle affects radiator efficiency — sometimes the team has to coordinate a slight attitude bias for thermal reasons).

Common failure modes. Commanding the FPA cooler on before the payload zone is at the right baseline temperature — wastes cooler energy and can over-cool. Missing a heater stuck-on case (duty cycle 100%) because it produces no telemetry red until the zone is already out of band.

6.6 TT&C / Comms Engineer

Mandate. The radio links to the ground — plural. Owns the S-band transponder (uplink + downlink), encoding/modulation parameters, the link margin, the TM rate ladder, **and the separate UHF beacon transmitter**. Configures the spacecraft for each pass type.

Authority. Commanding authority for TT&C subsystem with FD GO (`allowed_func_ids` 63–79). Calls *link is up* and *link is down* on the loop.

Console. Overview, TTC, commanding, procedures, trends, manual.

Telemetry to watch.

Table 6.4 — TT&C telemetry parameters with SIDs and nominal, yellow, and red limit bands.

Mnemonic	Hex id	SID	Nominal	Yellow	Red
`ttc.tx_active`	0x0525	6	1 (radiating)	(n/a)	0 when expected 1
`ttc.link_status`	0x0501	6	**2 = LOCKED**	1 = ACQUIRING	0 = NO_LINK
`ttc.pa_on`	0x0516	6	1 during pass	(n/a)	0 during pass
`ttc.tx_fwd_power`	0x050D	6	27–33 dBm	drop of > 3 dB	≈ 0
`ttc.carrier_lock` / `bit_sync` / `frame_sync`	0x0510–0x0512	6	1	(n/a)	0 during pass
`ttc.eb_n0`	0x0519	6	> 10 dB	4–10 dB	< 4 dB
`ttc.ber` (log10)	0x050C	6	< −7	−7 to −5	< −8 or > −4
`ttc.cmd_rx_count`	0x0513	6	increasing	stalled	decreasing(!)
`ttc.pa_temp`	0x050F	6	< 55 °C	0–55 band edge	< −10 or > 65

Mnemonic	Hex id	SID	Nominal	Yellow	Red
`ttc.link_margin`	0x0503	6	> 6 dB	3–6 dB	< 3 dB
`ttc.tm_data_rate`	0x0506	6	16000 bps (LOW, routine)	128000 (HIGH, dumps)	< 16000 (beacon lifeline)
`ttc.tm_tier` / `tm_tier_cmd`	0x0539 / 0x053A	6	equal	differ (a gate is refusing)	index off the ladder
`ttc.downlink_mode`	0x053C	6	1 during a dump	0 with HIGH commanded	out of

`link_status` **LOCKED is 2**, not 1. A 1 means carrier lock without frame sync — acquiring, not locked. This is the single most-repeated misreading in the shipped material.

Commands typically issued.

- `EPS_TRANSPONDER_TX` (S8 func_id 18) `on=1` — power the transmit chain.
- `TTC_PA_ON` (S8 func_id 66) / `TTC_PA_OFF` (S8 func_id 67) — power amplifier on/off.
- `TTC_SET_DATA_RATE` (S8 func_id 65) — `rate=0` selects LOW (16 kbps BPSK, conv(7,1/2)+RS — routine operations and the FDIR fallback), `rate=1` selects HIGH (128 kbps QPSK, RS(255,223) — dumps only). The third rung, BEACON (1 kbps), is entered with `TTC_SET_BEACON_MODE`, not selected here. HIGH is *gated*: it needs a deployed antenna, DOWNLINK mode on, and elevation ≥ 10°, and an unmet gate produces an S1.8 rejection (error 0x0007) naming the gate — never a silent clamp.
- `TTC_SET_DOWNLINK_MODE` (S8 func_id 79) — `on=1`; this is the mode gate on the HIGH tier, and without it func 65's HIGH rung is unreachable.
- `TTC_SET_TX_POWER` (S8 func_id 68) — `level`.
- `TTC_SET_MODULATION` (S8 func_id 74) — 0=BPSK, 1=QPSK, 2=8PSK, 3=OQPSK.
- `TTC_SET_BEACON_MODE` (S8 func_id 70) — `on=1` drops the S-band link to the 1 kbps beacon *tier*.
- `TTC_BEACON_TX_ON` / `TTC_BEACON_TX_OFF` (S8 func_id 97 / 98) — the **UHF beacon transmitter**, which is a different box from the beacon tier above. Turning it off is a `caution` command.
- `TTC_DEPLOY_ANTENNA` (S8 func_id 69) — one-shot, in LEOP-007.
- `CONNECTION_TEST` (S17.1) — link test ping.

> **The teaching point of this position.** EOSAT-1 has two transmitters. `TTC_PA_OFF` kills the S-band prime chain and with it every telemetry packet except SID 11 — and the UHF whip beacon at 437.5 MHz, 1 W, keeps radiating SID 11 regardless of the PA, the `ttc_tx` power line, the transponder, or the S-band patch. `ttc.tx_active` (0x0525) is the S-band gate; `ttc.beacon_active` (0x0527) is the beacon's, and the two are independent by construction. When an operator says "we lost telemetry", the first question is *which radio*.

Handoff content. Current link state and configuration; the active and commanded tier and whether they agree; BER trend; any amplifier temperature concern; predicted link margin for next pass given orbit geometry; the beacon transmitter's commanded state; the *standby data rate* (what tier the spacecraft is configured to come up at after the next pass).

Cross-position dependencies. With Ground Segment (ground antenna gain, ground LNA temperature, station status). With Payload (the data rate must match the payload data production rate, or the mass memory either fills up or starves). With FDS (predicted Doppler and range for the next pass).

Common failure modes. Not returning the link to the LOW tier at end of pass — the HIGH tier costs 11 W on the TX line against 5 W at LOW. Commanding HIGH and assuming it took: if the antenna, DOWNLINK mode or the 10° elevation gate is unmet the command is **rejected**, and the giveaway is `ttc.tm_tier` disagreeing with `ttc.tm_tier_cmd`. Confusing the beacon *tier* (`TTC_SET_BEACON_MODE` , still S-band) with the beacon *transmitter* (`TTC_BEACON_TX_ON` , UHF, a different radio). Failing to log the link configuration in the shift log so the next shift doesn't know what it's coming up to.

6.7 Payload Operations Engineer

Mandate. The science instrument. Owns the multispectral imager, the FPA, the image acquisition timeline, the science data pipeline (onboard storage, compression, prioritisation for downlink), and image quality assessment.

Authority. Commanding authority for the payload with FD GO. Commits the science timeline for the pass.

Console. Overview, Payload, commanding, procedures, trends, manual. This position also holds service 2, so the payload devices — FPA (0x0600), cooler (0x0601), calibration lamp (0x0602), shutter (0x0603), compression (0x0604) — are reachable from here.

Telemetry to watch.

Table 6.5 — Payload telemetry parameters with SIDs and nominal, yellow, and red limit bands.

Mnemonic	Hex id	SID	Nominal	Yellow	Red
`payload.mode`	0x0600	5	0 OFF / 1 STANDBY / 2 IMAGING / 3 CALIBRATION	unexpected	FAULT
`payload.fpa_temp`	0x0601	5	−15 (at setpoint)	< −22.5 or > 13.5	< −26 or > 17
`payload.fpa_ready`	0x0610	5	1 before a capture	0 with cooler on	0 at capture time
`payload.cooler_pwr`	0x0602	5	≈ 15 W when cooling	(n/a)	0 W when cooling expected
`payload.image_count`	0x0605	5	increasing per plan	stalled	unexpected drop
`payload.store_used`	0x0604	5	< 80 %	> 90 %	> 95 % (full)
`payload.last_scene_quality`	0x060D	5	high %	falling	corrupt
`payload.snr`	0x0616	5	30–55 dB	20–30 dB	< 20 dB
`payload.checksum_errors`	0x0609	5	0	any	rising

Commands typically issued.

- `EPS_POWER_ON(line_index=3)` (S8 func_id 19) — switch payload power line ON, then `PAYLOAD_SET_MODE(mode=1)` (S8 func_id 26) to bring the payload to standby. There is no single `PAYLOAD_POWER_ON` opcode in the telecommand catalogue.
- `PAYLOAD_SET_MODE` (S8 func_id 26) — 0=OFF, 1=STANDBY, 2=IMAGING, **3=CALIBRATION**. Modes 2 and 3 request the FPA cooler; modes 0 and 1 release it (the cooler is off in STANDBY and is switched on with `FPA_COOLER` to prepare for imaging), and the request drives both the TCS cooler flag and the EPS `fpa_cooler` power line.
- `PAYLOAD_CAPTURE` (S8 func_id 28) with `scene_id`, `lines`.
- `PAYLOAD_START_CALIBRATION` (S8 func_id 37) with `mode`; `PAYLOAD_STOP_CALIBRATION` (func_id 38). Dark calibration no longer requires IMAGING mode.
- `PAYLOAD_SET_SHUTTER` (S8 func_id 119) with `position` — an absolute shutter command; `PAYLOAD_CYCLE_SHUTTER` (func_id 93) exercises it. `PAYLOAD_CAL_LAMP` (func_id 95) with `on`.
- `S13_CREATE_TRANSFER` → `S13_REPORT_STATUS` (S13.1 / S13.9) — canonical large-data downlink workflow; `PAYLOAD_INITIATE_TRANSFER` (S8 func_id 94) stages a scene.
- `PAYLOAD_DELETE_IMAGE` (S8 func_id 30) — purge downlinked image.

Five calibration-refusal failure codes are worth memorising, because they are the ones this position will actually see: **0x0024** refused because the payload is IMAGING, **0x0025** FPA not ready, **0x0026** shutter in the wrong position, **0x0027** lamp in the wrong state, **0x0028** capture attempted with the shutter closed. (0x0021–0x0023 remain the capture-pointing codes.)

The image path is real end to end: **images do not appear on their own.** Nothing is captured that was not commanded, and nothing reaches the ground that was not transferred.

Handoff content. Current payload mode; FPA temperature and whether `payload.fpa_ready` is set (and if not, the cooler margin at 0x064B); storage usage and predicted fill rate; product count and volume (0x0648 / 0x0647); image quality across the shift; outstanding downlink queue.

Cross-position dependencies. With AOCS (need fine-point mode 60 s before imaging). With Thermal (FPA temperature). With TT&C (downlink rate must support the image volume). With FDS (imaging-target geometry — payload may ask FDS for the predicted ground swath at acquisition).

Common failure modes. Commanding imaging before fine-point is established. Failing to downlink before storage saturates. Mis-reading "image quality OK" as "image content useful" — the two are not the same and Payload Ops owns the distinction.

6.8 Flight Dynamics System (FDS) / Orbit Analyst

Mandate. Orbit determination, manoeuvre planning, pass prediction, conjunction screening, eclipse modelling. In the MCS the FDS analyst sits at the `aocs` position, which is labelled *Flight Dynamics* for exactly this reason; in industry the seat is sometimes called the *Orbit Analyst* or *Mission Analyst* depending on the agency.

Authority. No direct commanding authority for the burn. Produces the **manoeuvre vector** and checks the achieved delta-V against the plan; the Propulsion position arms and fires (`PROP_SET_BURN` S8 func 110, `PROP_ARM` 108, `PROP_EXECUTE` 111). Produces the **pass plan** the team flies. NOM-004 (planning) and NOM-013 (execution) both list `flight_director`, `aocs` and `propulsion` as required positions.

Console. The `aocs` MCS position — including a **read-only PROP tab**, on which every command button renders disabled — plus a secondary workstation. Software: `tools/orbit_tools.py` (orbit propagator, AOS/LOS predictor) and the planner.

Telemetry/data to watch. GPS state from `aocs.gps_lat`, `aocs.gps_lon`, `aocs.gps_alt` (0x0210–0x0212), with `aocs.gps_fix` (0x0274), `aocs.gps_pdop` (0x0275) and `aocs.gps_num_sats` (0x0276) as the quality channels. These are **on-demand parameters** — they are not in the periodic SID 2 sweep, so FDS polls them with `GET_PARAM` (S20.3) rather than waiting for housekeeping. The orbit propagator on the FDS workstation is reset every pass from these values and propagated forward to predict the next 24 h of contacts. After a burn, watch `prop.orbit_epoch_version` (0x0B19) and `prop.sma_km` (0x0B1A): the ground segment re-fits the element set, so passes and eclipses really do move — 1 m/s buys about +1.800 km of semi-major axis and shifts the pass 33.5 s earlier at +24 h.

Commands issued. None directly. Outputs:

- Pass schedule (handed to TT&C, Ground Segment, Payload).
- Eclipse schedule (handed to EPS, Thermal).
- Manoeuvre vector (handed to Propulsion; AOCS holds the attitude).
- Conjunction warnings (handed to FD; rarely actionable in a control room training week, but the procedure exists).

Handoff content. Next 24 h pass schedule (printed); next eclipse-entry and eclipse-exit times; any open manoeuvre planning work; any conjunction alert.

Cross-position dependencies. Most cross-talk in the room. FDS produces the timing every other position relies on.

Common failure modes. Using stale GPS data (must refresh every pass). Producing pass times in the wrong reference frame (spacecraft body vs ground-station). Forgetting to flag an upcoming eclipse to EPS/Thermal.

6.9 Ground Segment Engineer / Network Controller

Mandate. The ground side of the radio link. Antenna pointing and health, baseband equipment, network connectivity between MCS and station, station-to-station handover. Real industry titles include *Network Controller*, *Ground Operator*, *Antenna Operator* depending on agency.

Authority. Commanding authority for the ground station (antenna pointing, baseband config) with FD GO. Calls station **ready** or **not ready** before AOS.

Console. Secondary workstation. Software: the planner (full access, including station selection for the next pass), read access to the TTC telemetry tab, and the **radio page served by `smo-rfsim`** — carrier / bit-sync / frame-sync lock LEDs, Doppler estimate, constellation and eye diagram. The GS Engineer is the primary watcher of the ground-station radio chain. The RF bridge runs in one of three modes (`PACKET`, `FRAME`, `RF`, set in `rfsim.yaml`); only `RF` produces the modem-level displays.

Telemetry to watch.

Table 6.6 — Ground-segment items watched by the Ground Segment Engineer with concern indicators.

Item	Where	Nominal	Concern
Antenna / tracking, per station	`ttc.gs_equipment_status` (0x0524), bit 0	bit set = healthy	bit clear pre-AOS

Item	Where	Nominal	Concern
LNA, uplink HPA, reference oscillator	`ttc.gs_equipment_status` bits 1–3	15 = all four healthy	any bit clear
Which station is up	`ttc.active_gs` (0x0523)	1-based index into `orbit.yaml`	0 = out of contact
Baseband — Eb/N0	rfsim radio page	matches spacecraft Tx	mismatch suggests RF chain issue
Baseband — frame sync	rfsim radio page / `ttc.frame_sync` (0x0512)	locked during pass	loss during pass

Commands issued. None on the spacecraft, and none on the ground station either — there is no `GS_*` telecommand in the catalogue. The Ground Segment position works entirely through the ground-side tools:

- Station selection for the next pass (planner).
- RF bridge mode and pipeline configuration (`rfsim.yaml` , radio page).
- Station equipment health is *observed* through `ttc.gs_equipment_status` , and is maintained by the ground station, not by an operator command.

Handoff content. Station readiness for each upcoming pass; any antenna health concern; any network degradation seen during the shift; baseband configuration in use (rate, encoding).

Cross-position dependencies. With TT&C constantly (the spacecraft side and ground side of the same link). With FDS for pass-by-pass station selection.

Common failure modes. Not catching that the antenna is tracking the wrong slot in the schedule. Missing a baseband re-sync that lost a partial pass.

6.10 Mission Manager / Customer Liaison

Mandate. External communication and mission-level decisions. The Mission Manager is **not on the operating loop** — they listen, they understand, they translate. They are the customer's voice in the room and the room's voice to the customer.

Authority. Mission-level decisions only (e.g. *should we attempt First Light tomorrow given today's anomaly?*). They do not override the FD on shift-level decisions; the FD does not override the Mission Manager on mission-level decisions.

Console. None operational. A laptop for shift reporting and customer communications. Sits at the back of the room.

Outputs.

- **Daily customer briefing** — short written summary of the shift, sent end-of-day. Template in workbook.
- **Anomaly bulletin** — if an injected anomaly results in a meaningful loss (deferred First Light, missed imaging target), the Mission Manager writes the customer-facing bulletin.
- **Shift report** — files the shift's official mission log.

Handoff content. Open customer commitments; any pending external communication; mission-level decisions taken or deferred.

Cross-position dependencies. With FD (operational status). With Payload Operations (science deliverable status). Outside the room with the customer (who, in rehearsals, is represented by the Mission Director).

Common failure modes. Drifting into operational discussion on the loop — confuses authority. Not capturing the mission-level trade-off when the FD makes an in-shift decision.

Chapter 7 — Tools, Consoles, and Command Types

This chapter is the operator's tools reference. You will use it on the operations console familiarisation walkthrough and during ongoing operations. Treat it like the cockpit handbook: you don't read it cover to cover; you look up the bit you need *right now*.

7.1 The Mission Control System (MCS) at a glance

The MCS is a single webapp served at `[http://localhost:9090/](http://localhost:9090/)` on the control room. Every operator's console runs an instance, configured for that operator's position (the mcs/positions.yaml` file in the simulator maps each position to the tabs it shows). Two other webapps sit alongside it and are easy to confuse with it: the simulator / instructor console on port **8080** and the Mission Planning System on port **9091** (§7.3).

The MCS has five regions:

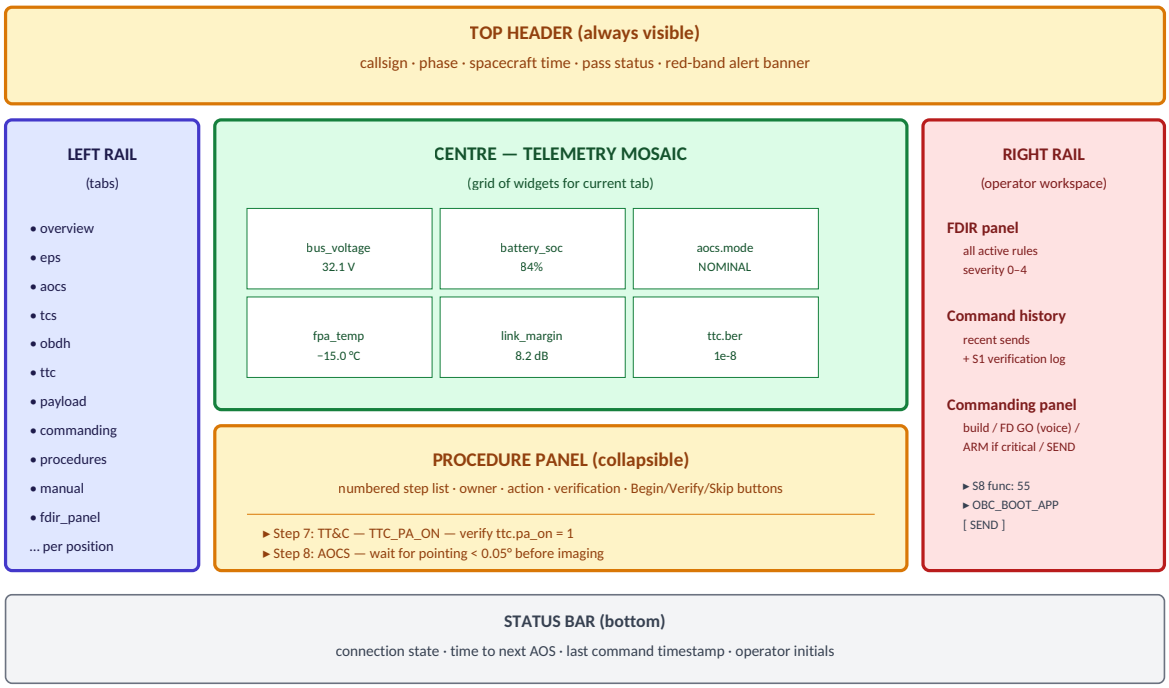


Figure 7.1 — MCS regions. Header is always visible at the top. The left rail lists the position-specific tabs. The centre splits between the telemetry mosaic (top, grid of pinned widgets) and the procedure panel (bottom, collapsible). The right rail is the operator's working zone — FDIR panel, command history, and commanding panel — and is the most-used region during a pass.

Top header

Five fields, always visible:

1. **Callsign.** `EOSAT-1`.
2. **Phase.** Integer 0–6 plus name: 0 PRE_SEPARATION, 1 SEPARATION_TIMER, 2 INITIAL_POWER_ON, 3 BOOTLOADER_OPS, 4 LEOP, 5 COMMISSIONING, 6 NOMINAL. **Phases 5 and 6 are set by telecommand.** The mission management team declares them with `SYS_SET_MISSION_PHASE` (S8

func 120); the instructor console (`set_phase`, range 0–6) and scenario scripts are instructor overrides only; the spacecraft also advances 1→2→3 on its own once the separation timer expires. Do not look for a `SET_PHASE` TC in the catalogue — the mnemonic used to exist as an alias of `OBC_BOOT_APP` (S8 func 55) and was removed in 2026-08 precisely because it invited an operator who wanted a phase advance to reboot the OBC instead.

3. **Spacecraft time.** Onboard clock as `YYYY-MM-DDTHH:MM:SSZ`. Yellow if drift > 1 s vs ground; red if drift > 5 s.
4. **Pass status.** `IN PASS (Svalbard, T-LOS 04:32)` or `BETWEEN PASSES (next: Iqaluit AOS T+18:21)`. The station named is `ttc.active_gs` (0x0523), a 1-based index into the five-station list in `orbit.yaml`; 0 means out of contact.
5. **Alert banner.** Empty when nominal; red bar when any active FDIR rule is severity 3 or 4.

Left rail — tabs

The tab set depends on the operator's position. The full superset (eighteen tabs, visible to FD only) is:

Table 7.1 — Full MCS left-rail tab catalogue with purpose of each tab.

Tab	Use
`system_dashboard`	Composite health view across all subsystems
`power_monitor`	EPS-dedicated trend view
`fdir_panel`	All active rules, all rule definitions
`contact_schedule`	Next 24 h passes per station
`procedure_panel`	Load and run procedures
`overview`	The cross-position health summary
`eps`	EPS telemetry mosaic
`aocs`	AOCS visualiser + telemetry
`tcs`	TCS zones + heater duty cycles
`obdh`	OBC state, watchdog, memory, bus
`ttc`	Transponder, link margin, BER
`payload`	Imager state, FPA, storage
`prop`	Propulsion: tank, valves, arm state, burn progress
`commanding`	Direct command builder
`pus`	Raw PUS service view (advanced)
`procedures`	Procedure file viewer (read-only browser)
`trends`	Long-window parameter trending
`manual`	Spacecraft manual reader

Position tab maps are in Chapter 6 and in `mcs/positions.yaml`.

Centre — telemetry mosaic

A grid of widgets, one per parameter or parameter group. Each widget shows current value, units, limit colouring (green/yellow/ red bands), and a trend sparkline. Click a widget to expand it to a full-screen plot with selectable history window.

The mosaic is **position-configured**: each operator sees the parameters their position cares about by default, but any operator can pin any parameter to their mosaic for the shift. The pin survives a logout/login.

Telemetry status indicators

- **Green** — value within nominal band.
- **Yellow** — value in yellow band; not an FDIR trigger but the operator should note it on the loop.
- **Red** — value in red band; usually an FDIR trigger fires simultaneously.
- **Degraded / Stale** — the value is older than its own downlink cadence allows. The thresholds are **derived per parameter from the HK structure that carries it** (`mcs/staleness.yaml`), not from one global timeout: a value is fresh while its age is under `max(10 s, interval × 0.9)`, degraded while under `max(20 s, interval × 1.5)`, and stale beyond that. This matters because SID 3 (TCS) downlinks every 60 s and SID 11 (beacon) every 30 s — under a flat 10 s rule those displays would read stale most of the time. Stale is *not* the same as red — stale means *I don't know the value*.
- **Not enabled** — every HK structure carrying the parameter is disabled, or has never been seen. This is not a link fault: only SIDs **1 (EPS)** and **11 (Beacon)** are enabled at application boot, so the correct response is `HK_ENABLE` (TC(3,5)), not chasing the radio.
- **Flagged** — operator-marked. Right-click a widget → Flag. The flag stays through the shift and is captured in the shift log.

Procedure panel

Collapsible bottom-of-centre region. Shows the currently loaded procedure as a numbered step list with three columns: **owner** (position responsible for the step), **action** (commanded or observed), **verification** (what telemetry confirms the step). Each step has three buttons: *Begin*, *Verify*, *Mark Skipped*. A step's status is one of `pending`, `in-progress`, `verified`, `skipped`, `failed`.

Procedures auto-log timestamps for each status change. The log appears in the shift log at end-of-shift.

FDIR panel

Right-rail upper region. Lists every monitoring rule (the **46** definitions in `monitoring/s12_definitions.yaml`, each declared `WARNING` or `ALARM`) and every event-action rule (the **32** rules in `monitoring/s19_rules.yaml`) currently active, colour-coded by severity. When a rule triggers, the panel:

1. Shows the rule name and triggering parameter value.
2. Plays a short audio tone when a new alarm appears.
3. Adds a row to the "active anomalies" list.
4. Suggests the canonical contingency procedure. The alarm → procedure linkage is data, not folklore: `mcs/alarm_procedures.yaml` resolves by parameter id first, then by event id, then by subsystem — so an over-current on any of the line 0–7 currents (0x0118–0x011F) points at CTG-012 *Overcurrent Response*.

The SPACON and the FD watch the FDIR panel constantly during a pass. Other operators rely on the audio alert and the FDIR row appearing in their overview tab.

Voice-loop chat

The current MCS has **no** text channel: the voice loop is headsets only, and *low-priority informational* traffic (e.g. "PAYLOAD: storage 42% used") goes on the loop or into the shift log. A text backup channel is being considered for the planned web-architecture work; do not plan a shift around it. Either way, commanding intentions are voice-only.

Commanding panel

Right-rail lower region. The position's command builder. Three input modes:

- **Catalogue command** — pick the mnemonic from `tc_catalog.yaml`; the builder fills the service, function ID and typed fields, and shows the packed hex and its byte count.
- **Direct command** — for ad-hoc commands. Operator picks the service and subtype and types (or pastes) the data hex; *DECODE HEX → FIELDS* checks it against the catalogue.
- **Quick commands** — one-press presets for the position. An entry the position may not send is shown denied, with the owning position named.

Every command goes through:

1. **Authority check.** *SEND* is enabled only if the position holds the commanding authority (`mcs/positions.yaml`); the server refuses the command otherwise.
2. **FD GO.** Operator requests GO on the voice loop. FD polls relevant positions and authorises. The tool neither records nor enforces this; it is room discipline.
3. **ARM.** For the *critical* commands the originating operator must press *ARM* before *SEND* (a critical quick command arms on the first press and executes on a second press within 8 s). There is no FD *Second-Key* button; an FD-side second key is among the ideas for the planned web-architecture work, not a current feature. Other commands skip this step.
4. **Uplink.** The command goes to the ground-station transmitter. The spacecraft answers with the S1 verification chain: **S1.1** acceptance success, **S1.3** execution start, **S1.5** progress, **S1.7** execution complete. A refusal comes back as **S1.2** (acceptance failure), **S1.4** (execution-start failure) or **S1.8** (execution failure), each carrying a 16-bit failure code and, on this spacecraft, an appended human-readable failure notice — so the console prints *why* ("Capture refused: AOCS not in FINE_POINT", 0x0021) rather than a bare number. A refusal also raises the `COMMAND_REJECTED` event 0x030B.

Command outcomes are auto-logged in the command history and the verification log: timestamp, the command itself, the send result and the spacecraft's S1 response. The FD authoriser is not captured; SPACON records the GO in the shift log.

Criticality is declared in the catalogue, not remembered. `commands/tc_catalog.yaml` marks five commands `critical` — `OBC_REBOOT` (52), `OBC_SWITCH_UNIT` (53), `PROP_ARM` (108), `PROP_SET_BURN` (110) and `SYS_SET_MISSION_PHASE` (120) — and fourteen more `caution`. ARM is required for the critical set.

7.2 Position-specific tab sets (cheat sheet)

Table 7.2 — Position-specific MCS tab visibility cheat sheet across the seven configured consoles.

Position key	Display name	Visible tabs
`flight_director`		

Position key	Display name	Visible tabs
	Flight Director	All eighteen (system_dashboard, power_monitor, fdir_panel, contact_schedule, procedure_panel, overview, eps, aocs, tcs, obdh, ttc, payload, prop, commanding, pus, procedures, trends, manual)
`eps_tcs`	Power & Thermal	overview, eps, tcs, commanding, procedures, trends, manual
`aocs`	Flight Dynamics	overview, aocs, prop (read-only), commanding, procedures, trends, manual
`ttc`	TT&C	overview, ttc, commanding, procedures, trends, manual
`payload_ops`	Payload Operations	overview, payload, commanding, procedures, trends, manual
`fdir_systems`	FDIR / Systems	overview, obdh, commanding, pus, procedures, trends, manual
`propulsion`	Propulsion	overview, prop, tcs, commanding, procedures, trends, manual

Note what the file actually contains: **seven** console positions, not nine. EPS and Thermal share one *Power & Thermal* console; there is no separate SPACON, FDS or Ground Segment position in `mcs/positions.yaml` — those are room roles the course assigns on top of the seven configured consoles. Propulsion is a first-class position (the 1U green monoprop is flown hardware).

The `overview` tab is always visible to every position — this is the shared situational awareness.

Commanding authority is gated separately from viewing: `allowed_services` and `allowed_func_ids` decide which buttons are live. Flight Dynamics can *see* the PROP tab but func_ids 108–118 are deliberately absent from its list, so every propulsion button renders disabled for that operator.

Combined view of the OSI 7-layer model alongside the equivalent space-protocol stack (Application → PUS → CCSDS Space Packet → TM/TC Transfer Frame → CLTU/CADU → RF), with PUS service types and packet byte structures.

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7.3 The Mission Planning System (MPS)

The MPS is a separate webapp on its own port, http://localhost:9091/ (a wide-screen layout is at `/wide`). Three positions use it routinely:

- **Payload Operations** — for imaging-target selection and activity scheduling.
- **FDS** — for pass prediction and manoeuvre window planning.
- **Ground Segment** — for station readiness and pass-by-pass resource allocation.

MPS panels

1. **Imaging-target picker.** A map of the world with the pre-loaded ocean-current targets from `planning/imaging_targets.yaml` overlaid as boxes — each target is a lat/lon bounding box (`OCM-001` Gulf Stream, `OCM-002` Kuroshio, `OCM-003` Agulhas, …), with a `priority`, a

`revisit_days` interval and a `min_solar_elevation_deg` gate. Those are the fields you filter on. Targets are not band-specific: the imager's four ocean-colour bands (443, 560, 665, 865 nm) are configured per acquisition with `PAYLOAD_SET_BAND_CONFIG`, not per target.

2. **Activity-type catalogue.** From `planning/activity_types.yaml`. Ten types: `imaging_pass`, `data_dump`, `calibration`, `housekeeping_collection`, `software_upload`, `momentum_desaturation`, `thermal_rebalance`, `load_shedding`, `fdir_recovery`, `orbit_manoeuvre`. Each type has a power profile, a thermal profile, a pointing requirement, and a duration.
3. **Ground-station schedule.** Per-station pass table for the next 24 h over all **five** stations (Iqaluit, Troll, Svalbard, Inuvik, O'Higgins), computed from the orbit propagator. Columns: AOS, max elevation, LOS, duration, station, visibility quality. Max elevation is not cosmetic — the 128 kbps HIGH tier is gated at 10° elevation, so a pass that peaks below 10° can carry commanding and housekeeping but no image dump.
4. **Conflict checker.** When you stage an activity in a pass, the conflict checker validates: slew-time between consecutive activities, power budget (cumulative load vs available), thermal envelope (no zone goes out of band), pointing budget (no activity requested in a pointing mode that can't deliver). Conflicts are red; warnings yellow.
5. **Plan exporter.** Converts the staged plan into the activity sequence file the OBC ingests via S11 time-tagged scheduling — `SCHEDULE_TC` is **S11.4**, with `ENABLE_SCHEDULE` (S11.13), `DISABLE_SCHEDULE` (S11.9), `DELETE_SCHEDULED` (S11.7), `DELETE_ALL_SCHEDULED` (S11.11) and `LIST_SCHEDULE` (S11.17) around it.

The MPS is read-only for most positions; the Payload Ops, FDS, and Ground Segment operators have write access.

7.4 The Procedure Panel — mechanics

Procedures live in the simulator as Markdown files under `configs/eosat1/procedures/{leop,commissioning,nominal,contingency,emergency}/`. They are indexed in `procedures/procedure_index.yaml`. The MCS procedure panel reads from this index.

Procedure file structure

Every shipped procedure card has the same structure — open `procedures/nominal/routine_health_check.md` alongside this and follow along:

```
# {ID}: {Title}
**Subsystem:** {ALL | EPS | AOCS | …}
**Phase:** {LEOP | COMMISSIONING | NOMINAL | …}
**Revision:** {n.n}
**Approved:** {authority}

## Purpose
…

## Prerequisites
- [ ] `ttc.link_status` (0x0501) = 2 (LOCKED)
- [ ] …

## Procedure Steps
```

```
### Step 1 --- {name}
**TC:** `HK_REQUEST(sid=1)` (Service 3, Subtype 27) --- EPS housekeeping
**Verify:** `eps.bat_soc` (0x0101) > 50 % within 10 s
**GO/NO-GO:** {what to do if it fails}

## Off-Nominal Handling
- …

## Post-Conditions
- [ ] …
```

Note the shape of a step: it is not a four-column table but a set of bolded labels — **TC**, **Verify**, **Action**, **Note**, **GO/NO-GO**, **Trend**, **Flag**. Every command is written with its arguments **inside** the backticks and with the service and subtype spelled out beside it, and every verification names both the parameter mnemonic and its hex id. Write your own cards the same way; a command name with the arguments floating outside the code span is the single most common defect found in this courseware.

Machine-runnable cards are flagged in the index. Eight cards are executable end to end (NOM-002, NOM-003, NOM-004, NOM-008, NOM-009, CTG-001, CTG-008, EMG-006); the executor no longer reports success for a procedure that did nothing, and it fails closed on any step type it does not understand. A card that deliberately commands nothing is declared with `advisory: true` and an `advisory_reason` on its index entry — CTG-020 (ground-station antenna failure) is the shipped example.

Running a procedure

1. **Load.** Search the procedure index in the panel; click to load. The procedure becomes the *active procedure* for the shift. Only one active procedure at a time (per FD's policy).
2. **Begin.** The SPACON clicks *Begin step 1*. The step status changes to `in-progress`.
3. **Call.** The step *Owner* position is paged on the voice loop ("AOCS, beginning ST1_POWER").
4. **Request GO.** Owner requests GO. FD polls. FD authorises.
5. **Execute.** Owner clicks the commanding panel's auto-populated TC and first-keys. Second-key if required. Uplink.
6. **Verify.** Owner reads the verification telemetry on the loop. SPACON clicks *Verify step 1* on confirmation.
7. **Repeat** for next step.

A procedure usually has 5–15 steps and takes from a few minutes (LEOP-006 *Time Sync*) to 40 minutes (LEOP-007 *Sequential Power-On*) of real time. The nominal set runs **NOM-001 to NOM-013**; note that `nominal/clock_sync.md` is **NOM-008** and NOM-006 is *Software Upload* — a pairing the older courseware had the wrong way round.

Skip and Hold

- **Skip.** Allowed only if the step's verification confirms the state is already met (e.g. step says "enable battery heater" but the heater is already enabled per telemetry). The skip is logged with the verifying telemetry value.
- **Hold.** Allowed if the operator needs time to assess. The procedure pauses; the team can confer; the FD authorises resumption. Holds are logged with duration.

Anything other than verified-skip or authorised-hold is a **procedure violation** — logged as such in the shift log and discussed at debrief.

7.5 The FDIR Panel — mechanics

FDIR (Fault Detection, Isolation, and Recovery) rules live in `monitoring/s12_definitions.yaml` (monitoring rules — what's watched) and `monitoring/s19_rules.yaml` (event-action rules — what to do when something triggers). The MCS FDIR panel reads from both.

Rule structure

Each rule has:

- **Name** (human-readable, e.g. `eps_overcurrent_payload`).
- **Parameter** monitored (e.g. `eps.payload_line_current_a`).
- **Threshold** and **persistence time** (e.g. `> 0.6 A for > 2 s`).
- **Severity** (level 0 informational → level 4 safe-mode candidate). Do not confuse FDIR **severity** (0–4) with onboard **autonomy level** (0–3); they are different scales and the book never uses one for the other.
- **Action** — for an S19 rule, the `action_func_id` the spacecraft autonomously invokes when its trigger event fires (each rule carries an `ea_id`, a trigger and one function id). S12 monitors have no action; they raise an out-of-limit event and alert the ground.

The two files are the rule list: **32 S19 event-action rules** and **46 S12 monitor definitions**. Read them directly — there is no separate index document.

Severity escalation ladder

Table 7.3 — FDIR severity ladder with default action and required operator response per level.

Level	Meaning	Default action	Operator response
0	Informational	Log only	Note on loop; no action
1	Caution	Log + alert	Discuss in next debrief
2	Warning	Log + alert + autonomous mitigation if defined	Run the recovery procedure
3	Severe	Autonomous + ground alert + flag for next pass	Run recovery + ticket
4	Critical	Safe-mode candidate (autonomous if rule says so)	Confirm safe-mode entry, brief Mission Manager

The FDIR panel colour-codes by severity. A level-3 or level-4 trigger raises the red alert banner in the MCS top header.

Acknowledging vs clearing

- **Acknowledge.** Operator confirms they have seen the rule trigger. Silences the audio alert. Does NOT clear the rule; the parameter is still in violation.
- **Clear.** Available only when the parameter has returned to within band. The rule moves from "active" to "history" in the panel. Always preceded by an anomaly ticket.

Manual override

A FDIR rule can be disabled by the SPACON with FD authorisation (e.g. during a known-bad-sensor calibration where the sensor will read out of band for a planned reason). Disabled rules show on the panel in grey with a *Manual override* tag. Re-enabled through a paired procedure step.

Ground segment architecture: spacecraft RF link to ground station, MOC components (TM/TC server, MPS, FDS, archive), POC, and external interfaces.

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7.6 PUS Service Primer

The spacecraft talks to the ground using the **Packet Utilization Standard** (PUS, ECSS-E-ST-70-41C / CCSDS 660.0-G-3). It defines a set of *services* and *subservices* the spacecraft must implement. Operators don't write PUS frames — the MCS does — but operators do read PUS service IDs in procedure files and in the commanding panel.

EOSAT-1 declares **fifteen** services in `mcs/pus_services.yaml`, and all fifteen appear in `commands/tc_catalog.yaml`. The uplink (TC) subtypes below are the ones the catalogue actually defines; the downlink (TM) column is what comes back.

Table 7.5 — The PUS services implemented on EOSAT-1 with subservices and operator-facing purpose.

Service	TC subtypes	TM subtypes	Purpose	Where you see it
S1	—	1, 2, 3, 4, 5, 7, 8	Telecommand verification	Every TC produces an S1 chain
S2	1, 6	6	Device access	Direct on/off of a named device — commissioning and contingency only
S3	5, 6, 27, 31	25	Housekeeping	`HK_ENABLE` **S3.5**, `HK_DISABLE` S3.6, `HK_REQUEST` **S3.27**, `HK_SET_INTERVAL` S3.31; the HK packet itself is TM(3,25)
S5	5, 6	1, 2, 3, 4	Event reporting	`EVENT_ENABLE` / `EVENT_DISABLE` ; the report subtype **is** the severity
S6	2, 5, 9	6, 10	Memory management	Read/write/checksum OBC memory (advanced)
S8	1	—	Function management	The big one — invoke a function by `func_id`
S9	1, 2	2	Time management	`SET_TIME` S9.1, `REQUEST_TIME` S9.2
S11	4, 7, 9, 11, 13, 17	5, 18	Time-tagged commanding	Onboard schedule; `SCHEDULE_TC` is S11.4
S12	1, 2, 3, 4, 6, 7, 12	9, 13	On-board monitoring	The 46 monitor definitions
S13	1, 3, 5, 9	2, 4, 10		Science-image downlink session

Service	TC subtypes	TM subtypes	Purpose	Where you see it
			Large-data transfer	
S15	1, 2, 9, 11, 13	14	Onboard storage	The four packet stores (HK, event, science, alarm)
S17	1	2	Connection test	Link health ping
S18	1, 2, 3, 4, 6	7	On-board control procedures	OBCP load / start / stop / report
S19	1, 2, 4, 5, 8	128	Event-action	The 32 FDIR autonomous rules; `EA_ENABLE` S19.4, `EA_DISABLE` S19.5
S20	1, 3	2	Parameter management	`SET_PARAM` S20.1, `GET_PARAM` S20.3

Three PUS traps are worth memorising, because they are the ones this course's earlier editions got wrong:

- **`HK_ENABLE` is S3.5 and `HK_REQUEST` is S3.27.** Not subtype 1, and not subtype 25 — 25 is the *downlinked HK packet*, not a request for one.
- **S8 has exactly one subtype: 1.** There is no scheme in which the S8 subtype names the subsystem. The subsystem is implied by the `func_id` block, and the `func_id` is the first octet of the TC data field.
- **Only SIDs 1 (EPS) and 11 (Beacon) are enabled at boot** (`simulation.yaml: default_enabled_sids: [1, 11]` , re-applied by the engine on every application boot). Everything else — AOCS, TCS, OBDH, Payload, TT&C, the extended EPS structure, PROP — needs an explicit `HK_ENABLE` .

The HK structure ID map (`telemetry/hk_structures.yaml`) is:

Table 7.5b — Housekeeping structure IDs, contents and downlink cadence.

SID	Contents	Interval
1	EPS	1 s (enabled at boot)
2	AOCS	4 s
3	TCS	60 s
4	Platform / OBDH	8 s
5	Payload	8 s
6	TT&C	8 s
7	AOCS determination	4 s
8	EPS power management (extended)	8 s
9	PROP	8 s
11	Beacon	30 s (enabled at boot)

You'll meet **S8** most often in procedures — it's the way the ground asks the spacecraft to do anything other than report telemetry or accept a parameter change. Each S8 invocation carries a *function ID*. The EOSAT-1 function ID catalogue is in `commands/tc_catalog.yaml`; the generated, wire-exact reference — every argument with its type, offset, unit, scale, range and a worked hex example — is `manual/09_command_reference.md`. A working subset is in §7.7 below.

7.7 Onboard functions (S8 Function ID — working operator subset)

The EOSAT-1 telecommand database (`commands/tc_catalog.yaml`) defines S8 function IDs **0–98** across all subsystems, a legacy block **100–107** for the EPS/TCS quick-action buttons, the propulsion block **108–118**, and 119 (`PAYLOAD_SET_SHUTTER`). The tables below list every function-ID the operator will meet during a typical operational pass, grouped by the position that owns the commanding authority. These are extracted directly from `tc_catalog.yaml` — the catalogue file is the canonical authority, and `manual/09_command_reference.md` is generated from it.

Arguments belong inside the backticks. The catalogue now carries a typed argument schema — name, type, byte offset, unit, raw ↔ engineering calibration, permitted range and default — so there is a right answer for every argument and no reason to write a bare command name. Write `PAYLOAD_COOLER_SETPOINT(setpoint=-15.0 degC)`, not "PAYLOAD_COOLER_SETPOINT, setpoint minus fifteen". Values outside the declared range are **rejected** with TM(1,8) error 0x0003, never silently clamped.

AOCS (func_id 0–15, 84–91, 96)

Table 7.6a — S8 AOCS function IDs for mode, wheel, sensor, and actuator control.

Func ID	Name	Purpose
0	`AOCS_SET_MODE(mode=…)`	Set AOCS mode. `mode` uint8: 0=OFF, 1=SAFE_BOOT, 2=DETUMBLE, 3=COARSE_SUN, 4=NOMINAL (nadir), 5=FINE_POINT, 6=SLEW, 7=DESAT, 8=ECLIPSE, 9=INERTIAL. Mode 5 is refused with TM(1,8) unless the `aocs.fine_ready` precondition bitmask (0x02C5) is 0
1	`AOCS_DESATURATE`	Start wheel desaturation (no arguments)
2	`AOCS_DISABLE_WHEEL(wheel_idx=0…3)`	Disable a specific reaction wheel
3	`AOCS_ENABLE_WHEEL(wheel_idx=0…3)`	Enable a specific reaction wheel
4	`ST1_POWER(on=0/1)`	Star tracker 1 power on/off
5	`ST2_POWER(on=0/1)`	Star tracker 2 power on/off
6	`ST_SELECT(unit=1/2)`	Select primary star tracker (1=ST1, 2=ST2; 0 is refused) — there are **two** heads
7	*(retired)*	Formerly `MAG_SELECT` . EOSAT-1 flies a **single** magnetometer, so there is nothing to select; func 7 is refused on board with 0x0007
8		

Func ID	Name	Purpose
	`RW_SET_SPEED_BIAS(wheel_idx=…, bias_rpm=… RPM)`	Wheel speed bias, float32. The bias is **added** to the wheel's current speed
9	`MTQ_ENABLE(enable=0/1)`	Enable / disable magnetorquers
10	`AOCS_SLEW_TO(q_x, q_y, q_z, q_w, rate_dps)`	Slew to target quaternion, five float32
11	`AOCS_CHECK_MOMENTUM`	Query wheel momentum status
12	`AOCS_BEGIN_ACQUISITION`	Start automated attitude acquisition
13	`AOCS_GYRO_CALIBRATION`	Run the **on-board** gyro bias calibration. Not a ground-computed bias upload
14	`AOCS_RW_RAMP_DOWN(wheel_idx=…, target_rpm=… RPM)`	Ramp wheel(s) down; `wheel_idx=255` means all four
15	`AOCS_SET_DEADBAND(deadband_deg=… deg)`	Set attitude-error deadband, float32

Wheel speeds themselves are **read-only** telemetry (0x0207–0x020A, temperatures 0x0218–0x021B); funcs 8 and 14 are the only commanding path to them. Fine pointing works on **three** wheels — bit 1 of the `aocs.fine_ready` bitmask 0x02C5 means "fewer than three wheels", not fewer than four.

Later AOCS additions (Flight Dynamics console, funcs 84–91 and 96): 84 `AOCS_GPS_SET_START_MODE`, 85 `AOCS_SET_POINTING_FRAME`, 86 `AOCS_SET_TARGET_QUAT`, 87 `AOCS_SET_TARGET_GEO(lat, lon, alt_m)`, 88 `AOCS_SET_CONTROL_BW(omega_n, zeta)`, 89 `AOCS_FINE_READY_CHECK`, 90 `AOCS_SET_MOM_MGMT(enable, entry_pct, exit_pct)`, 91 `AOCS_EST_RESET(mode=0 cold / 1 warm)`, 96 `AOCS_SET_DESAT_INHIBIT(inhibit=0/1)`.

EPS (func_id 16–25, 81–83)

Table 7.6b — S8 EPS function IDs for power-line control, load shedding, wing deployment, and the cooler/transponder enables that physically live on the EPS distribution bus.

Func ID	Name	Purpose
16	`EPS_PAYLOAD_MODE(mode=0/1/2)`	Payload power mode (0=off, 1=standby, 2=imaging)
17	`EPS_FPA_COOLER(on=0/1)`	Enable / disable the FPA cryocooler line
18	`EPS_TRANSPONDER_TX(on=0/1)`	Enable / disable transponder TX
19	`EPS_POWER_ON(line_index=0…8)`	Switch power line ON
20	`EPS_POWER_OFF(line_index=0…8)`	Switch power line OFF
21	`EPS_RESET_OC_FLAG(line_index=0…8)`	Reset overcurrent trip flag, re-enable line
22	`EPS_SWITCH_LOAD(load_id=…, state=0/1)`	Switch individual power load on/off
23	`EPS_SET_CHARGE_RATE(rate_a=… A)`	Override battery charge rate, float32

Func ID	Name	Purpose
24	`EPS_SOLAR_ARRAY_DRIVE(angle_deg=… deg)`	Drive solar array angle, int16
25	`EPS_EMERGENCY_LOAD_SHED(stage=0…3)`	Immediate load shed
81	`EPS_DEPLOY_WING(wing=0/1/2)`	Burn-wire wing deploy (0=+Y, 1=−Y, 2=both)
82	`EPS_WING_STATUS`	Query wing deployment status (0x0144)
83	`EPS_SET_MODE(mode=…)`	Set EPS mode

There are nine power lines, indices 0–8, not seven: 0 `obc`, 1 `ttc_rx`, 2 `ttc_tx`, 3 `payload`, 4 `fpa_cooler`, 5 `htr_bat`, 6 `htr_obc`, 7 `aocs_wheels`, 8 `prop`. Lines 0 and 1 are non-switchable and always on; **every switchable line defaults OFF**, which is why a freshly booted spacecraft generates telemetry and does nothing else until the operator turns things on. The line-status flags are 0x0110–0x0117 (line 8, prop, is 0x014A) and the per-line currents 0x0118–0x011F (line 8 is 0x014B); wing deployment status is 0x0144 (bit 0 +Y, bit 1 −Y, bits 4/5 in progress) with per-wing currents at 0x0146 (+Y) and 0x0147 (−Y).

Payload (func_id 26–39, 92–95, 119)

Table 7.6c — S8 payload function IDs for imager mode, capture, scene management, calibration, and detector configuration.

Func ID	Name	Purpose
26	`PAYLOAD_SET_MODE(mode=0…3)`	0=OFF, 1=STANDBY, 2=IMAGING, **3=CALIBRATION**. Modes 2/3 request the FPA cooler; modes 0/1 release it (STANDBY runs without the cooler)
27	`PAYLOAD_SET_SCENE(scene_id=…)`	Set scene ID for imaging, uint16
28	`PAYLOAD_CAPTURE(scene_id=…, lines=…)`	Trigger capture — **caution**
29	`PAYLOAD_DOWNLOAD_IMAGE(scene_id=…)`	Queue image for downlink
30	`PAYLOAD_DELETE_IMAGE(scene_id=…)`	Delete image from store — **caution**
31	`PAYLOAD_MARK_BAD_SEGMENT(segment_id=…)`	Mark memory segment unusable — **caution**
32	`PAYLOAD_GET_IMAGE_CATALOG`	Request stored image catalog
33	`PAYLOAD_SET_BAND_CONFIG(mask=…)`	Spectral band enable mask (443/560/665/865 nm)
34	`PAYLOAD_SET_INTEGRATION_TIME(blue_ms, green_ms, red_ms, nir_ms)`	Per-band integration time, uint16 ms each
35	`PAYLOAD_SET_GAIN(gain=… ×)`	Detector gain, uint16
36	`PAYLOAD_COOLER_SETPOINT(setpoint=… degC)`	

Func ID	Name	Purpose
		FPA regulation setpoint — **caution**. Wire field is a **uint16 in centi-kelvin**: `raw = round((T_degC + 273.15) × 100)`, so −15.00 °C is 25815 = `0x64D7`. Engineering range −20…0 °C (raw 25315…27315); outside it the TC is **rejected** with S1.8 / 0x0003, not clamped
37	`PAYLOAD_START_CALIBRATION(mode=0/1/2)`	0=DARK, 1=FLAT, 2=BOTH. Runs from STANDBY **or** CALIBRATION mode — **never** from IMAGING
38	`PAYLOAD_STOP_CALIBRATION`	Abort calibration
39	`PAYLOAD_SET_COMPRESSION(algorithm=…, ratio=… ×)`	Override compression
92	`PAYLOAD_SELECT_FILTER(position=…)`	Filter-wheel position
93	`PAYLOAD_CYCLE_SHUTTER(cycles=… transitions)`	Relative shutter exercise
94	`PAYLOAD_INITIATE_TRANSFER(scene_id=…)`	Start an image transfer
95	`PAYLOAD_CAL_LAMP(on=0/1)`	Calibration lamp
119	`PAYLOAD_SET_SHUTTER(position=0/1)`	Drive the shutter to an **absolute** position (0=CLOSED, 1=OPEN) — **caution**. Idempotent, and the only way to reach a *known* shutter state

Calibration has its own refusal vocabulary, so the console tells you which gate failed rather than printing "invalid in the current mode" five different ways: **0x0024** refused because IMAGING, **0x0025** FPA not thermally ready (0x0610), **0x0026** shutter in the wrong position (0x063C), **0x0027** calibration lamp in the wrong state (0x0615), **0x0028** capture attempted with the shutter closed. Codes 0x0021–0x0023 remain the capture-pointing gates (not FINE_POINT / pointing error > 0.05° / rate > 0.005 °/s). Dark calibration no longer requires IMAGING mode.

TCS (func_id 40–49)

Table 7.6d — S8 TCS function IDs for the three heater circuits, the FPA cooler, thermostat setpoints, and decontamination.

Func ID	Name	Purpose
40	`HEATER_BATTERY(on=0/1)`	Battery heater on/off
41	`HEATER_OBC(on=0/1)`	OBC heater on/off
42	`HEATER_THRUSTER(on=0/1)`	Thruster (catalyst-bed) heater on/off

Func ID	Name	Purpose
43	`FPA_COOLER(on=0/1)`	FPA cooler on/off (TCS-side enable)
44	`HEATER_SET_SETPOINT(circuit=…, on_temp=… degC, off_temp=… degC)`	Thermostat band for a circuit, two float32
45	`HEATER_AUTO_MODE(circuit=…)`	Return heater to autonomous thermostat
46	`TCS_SET_HEATER_DUTY_LIMIT(circuit=…, max_duty_pct=… %)`	Limit heater duty cycle
47	`TCS_DECONTAMINATION_START(target_temp_c=… degC)`	Begin decontamination heating
48	`TCS_DECONTAMINATION_STOP`	Abort decontamination
49	`TCS_GET_THERMAL_MAP`	Request thermal status of all zones

The FPA cooler is a **miniature Stirling-class cryocooler**, not a Peltier stack, and its authority falls away as the temperature difference grows (`fpa_cooler_carnot_fraction` , 0.23). Two telemetry points exist specifically so the operator can see that: `tcs.fpa_cooler_floor_c` (0x0424), the coldest temperature the cooler can actually hold in the present thermal state, and `tcs.fpa_cooler_margin_k` (0x0425), the margin between that floor and the commanded setpoint. A setpoint below the floor is not a fault — it is a request the physics cannot fill.

OBDH / SPACON (func_id 50–62, 80)

Table 7.6e — S8 OBDH function IDs for OBC mode, reboot, redundant selection, watchdog, and time-sync.

Func ID	Name	Purpose
50	`OBC_SET_MODE(mode=0/1/2)`	Set OBC mode (0=nominal, 1=safe, 2=emergency)
51	`OBC_MEMORY_SCRUB`	Trigger manual memory scrub
52	`OBC_REBOOT`	Force OBC reboot (drops to bootloader) — **critical**
53	`OBC_SWITCH_UNIT`	Switch to redundant OBC — **critical**
54	`OBC_SELECT_BUS(bus=0/1)`	Select active CAN bus (0=A, 1=B) — **caution**
55	`OBC_BOOT_APP`	Boot application software from bootloader — **caution**. No arguments
56	`OBC_BOOT_INHIBIT(inhibit=0/1)`	Inhibit auto-boot (stay in bootloader)
57	`OBC_CLEAR_REBOOT_CNT`	Reset reboot counter
58	`OBC_SET_WATCHDOG_PERIOD(period_ticks=…)`	Configure watchdog timeout, uint16

Func ID	Name	Purpose
59	`OBC_WATCHDOG_ENABLE`	Enable watchdog monitoring
60	`OBC_WATCHDOG_DISABLE`	Disable watchdog monitoring
61	`OBC_DIAGNOSTIC`	Request OBC health diagnostic
62	`OBC_ERROR_LOG`	Request recent error-log entries
80	`OBC_GPS_TIME_SYNC`	Sync onboard clock to GPS — **caution** (may invalidate AOCS attitude solution; recover with `AOCS_EST_RESET`, func 91)

`OBC_BOOT_APP` boots the application image and does nothing else. It used to carry a `SET_PHASE` alias; that alias was **removed** in 2026-08, because booting a software image has nothing to do with the mission phase, and the mnemonic tempted operators looking for a phase advance into rebooting the OBC. There is no phase telecommand at all — see §7.1.

TT&C (func_id 63–79, 97–98)

Table 7.6f — S8 TT&C function IDs for transponder switching, data rate, transmit power, modulation, downlink mode, both beacons, and ranging.

Func ID	Name	Purpose
63	`TTC_SWITCH_PRIMARY`	Switch to primary transponder
64	`TTC_SWITCH_REDUNDANT`	Switch to redundant transponder
65	`TTC_SET_DATA_RATE(rate=0/1)`	Select a rung of the TM rate ladder: **0 = LOW (16 kbps, BPSK, conv(7,1/2)+RS) — routine operations**; **1 = HIGH (128 kbps, QPSK, RS(255,223)) — dumps only**. 255 = BEACON (1 kbps) is not selectable here
66	`TTC_PA_ON`	Enable the S-band power amplifier
67	`TTC_PA_OFF`	Disable the S-band power amplifier
68	`TTC_SET_TX_POWER(level=0/1/2)`	**0 = LOW 1.0 W, 1 = NOMINAL 2.0 W, 2 = HIGH 5.0 W** RF — **caution**. 5 W is the ceiling (`pa_max_power_w`); an unknown code is rejected with 0x0003, never clamped
69	`TTC_DEPLOY_ANTENNA`	Deploy spacecraft antenna (burn-wire)
70	`TTC_SET_BEACON_MODE(on=0/1)`	Latch the S-band chain into its 1 kbps beacon rung
71	`TTC_CMD_CHANNEL_START`	Start dedicated command channel (15-min)
72	`TTC_SET_UL_FREQ(freq_mhz=… MHz)`	Set uplink frequency, float32
73	`TTC_SET_DL_FREQ(freq_mhz=… MHz)`	Set downlink frequency, float32

Func ID	Name	Purpose
74	`TTC_SET_MODULATION(mode=0…3)`	0=BPSK, 1=QPSK, 2=8PSK, 3=OQPSK
75	`TTC_SET_RX_GAIN(agc_db=… dB)`	Set receiver AGC target, float32
76	`TTC_RANGING_START`	Begin ranging sequence
77	`TTC_RANGING_STOP`	Stop ranging sequence
78	`TTC_SET_COHERENT_MODE(on=0/1)`	Coherent transponder on/off
79	`TTC_SET_DOWNLINK_MODE(on=0/1)`	Enter / leave DOWNLINK (payload-dump) mode. **This is the gate on the HIGH tier**
97	`TTC_BEACON_TX_ON`	Switch the **UHF whip beacon transmitter** on
98	`TTC_BEACON_TX_OFF`	Switch the UHF beacon transmitter off — **caution**

Three rungs, not two. The TM rate ladder in `subsystems/ttc.yaml` is BEACON / LOW / HIGH = **1 / 16 / 128 kbps**. The legacy argument encoding was preserved verbatim — `rate=0` still means "the low rate" and `rate=1` still means "the high rate" — but those now select 16 kbps and 128 kbps rather than the 1 kbps and 64 kbps of the older configuration. HIGH is refused with S1.8 / 0x0007 unless the antenna is deployed, DOWNLINK mode is on (func 79) and elevation is at least 10°. Watch the two tier parameters side by side: `ttc.tm_tier_cmd` (0x053A) is what you asked for and `ttc.tm_tier` (0x0539) is what is on air after the gates — a demotion shows as a disagreement between them.

Two transmitters, and they fail independently. EOSAT-1 carries an S-band prime chain (2025 MHz up / 2200 MHz down, 2 W nominal, all real-time HK, playback, image dumps and S1 verification) *and* a separate UHF whip beacon (437.5 MHz, 1 W into a dipole, 1 kbps, carrying HK **SID 11 and nothing else**). They are separate radios with separate PAs, antennas and link budgets. Funcs 66/67 and the EPS `ttc_tx` line kill the **S-band** downlink — `ttc.tx_active` (0x0525) goes to 0 and S-band telemetry stops — while the beacon keeps radiating: `ttc.beacon_tx_on` (0x0526) and `ttc.beacon_active` (0x0527) are independent of it by construction. That is what "is the OBC alive?" is answered with when the prime chain is dead. Do not confuse either of them with `ttc.beacon_mode` (0x0521), which is a *rung of the S-band ladder* selected by func 70 — three different things sharing one word.

Propulsion (func_id 108–118)

Table 7.6g — S8 propulsion function IDs. EOSAT-1 flies a 1U green monopropellant module with **66.9 m/s** total delta-V, used for debris avoidance, limited orbit maintenance and the 34.3 m/s end-of-life perigee-lowering burn (NOM-014). These are owned by the Propulsion console.

Func ID	Name	Purpose
108	`PROP_ARM(arm_code=…)`	Arm the propulsion module — **critical**
109	`PROP_DISARM`	Return to SAFED
110	`PROP_SET_BURN(dv_m_s=… m/s, direction=…, mode=…, t_start_cuc=… s)`	Load a burn — **critical**
111	`PROP_EXECUTE`	

Func ID	Name	Purpose
		Execute the loaded, armed burn
112	`PROP_ABORT`	Abort a burn in progress
113	`PROP_ISO_OPEN(branch=…)`	Open a latch-valve branch
114	`PROP_ISO_CLOSE(branch=…)`	Close a latch-valve branch
115	`PROP_SET_LIMITS(att_err_max_deg, p_min_bar, catbed_max_c, t_burn_max_s)`	Set arm interlock limits
116	`PROP_INHIBIT_SET(on=0/1)`	Latch the propulsion inhibit
117	`PROP_INHIBIT_CLR(key=…)`	Clear the inhibit with the key
118	`PROP_GET_STATUS`	Report propulsion status

Arm interlocks each have their own failure code (0x0B01–0x0B13), mirroring the `arm_inhibit_mask` at 0x0B15, so a refusal names the gate rather than returning a generic "not permitted". Propulsion telemetry starts at 0x0B00 and rides HK **SID 9**. A burn re-fits the on-board element set, so passes and eclipses genuinely move afterwards (1 m/s gives about +1.800 km in semi-major axis and −33.5 s in pass time at +24 h) — re-read the contact schedule after every manoeuvre. Simulation speed is clamped to ≤ 10× while a burn runs.

What you will *not* find in the simulator

The textbook used to reference a few "natural" command names that are *not* in `tc_catalog.yaml`. Use the workarounds below:

Old textbook name	Use instead
`OBC_NOOP`	(no heartbeat command; the watchdog handles liveness)
`OBC_WATCHDOG_KICK`	`OBC_WATCHDOG_ENABLE` (59) and `OBC_SET_WATCHDOG_PERIOD` (58)
`FDIR_LOAD_TABLE`	Enable/disable individual rules via `EA_ENABLE` (S19.4) and `EA_DISABLE` (S19.5); see Chapter 15
`FDIR_ENABLE_RULE` / `FDIR_DISABLE_RULE`	`EA_ENABLE` / `EA_DISABLE` (S19.4 / S19.5)
`EPS_LOAD_SHED`	`EPS_EMERGENCY_LOAD_SHED` (25), stage 0–3
`TCS_HEATER_ENABLE` / `TCS_HEATER_DISABLE`	Per-circuit `HEATER_BATTERY` / `HEATER_OBC` / `HEATER_THRUSTER` (40/41/42) with `on=0/1`
`FPA_COOLER_ENABLE`	`EPS_FPA_COOLER` (17) and/or `FPA_COOLER` (43)

Old textbook name	Use instead
`IMU_ENABLE`	The gyros are three **individually powerable devices** — `S2_AOCS_GYROSCOPE_0/1/2` at device ids 0x0206, 0x0207, 0x0208 — commanded with S2.1 `(device_id, state)`. There is no redundant IMU unit, no IMU power line and no cold-boot delay; bias calibration is on-board via `AOCS_GYRO_CALIBRATION` (13)
`STAR_TRACKER_ENABLE`	`ST1_POWER` (4), `ST2_POWER` (5), `ST_SELECT` (6)
`WHEEL_ENABLE` / `WHEEL_DISABLE`	`AOCS_ENABLE_WHEEL` (3) / `AOCS_DISABLE_WHEEL` (2)
`MAGNETOTORQUER_ENABLE`	`MTQ_ENABLE` (9)
`MOMENTUM_DUMP_INITIATE`	`AOCS_DESATURATE` (1)
`TTC_TX_ENABLE` / `TTC_TX_DISABLE`	`EPS_TRANSPONDER_TX` (18) or `EPS_POWER_ON/OFF(line_index=2)` for the `ttc_tx` line, plus `TTC_PA_ON` / `TTC_PA_OFF` (66/67). All of these kill the **S-band** downlink only
`TTC_BEACON_ENABLE`	Decide which beacon you mean. The **UHF whip beacon transmitter** is `TTC_BEACON_TX_ON` / `TTC_BEACON_TX_OFF` (97/98). The **S-band 1 kbps beacon rung** is `TTC_SET_BEACON_MODE` (70)
`PAYLOAD_POWER_ON` / `PAYLOAD_POWER_OFF`	`EPS_POWER_ON(line_index=3)` (19) then `PAYLOAD_SET_MODE` (26)
`PAYLOAD_MODE_SET`	`PAYLOAD_SET_MODE` (26)
`IMAGE_ACQUIRE`	`PAYLOAD_CAPTURE` (28)
`PAYLOAD_CALIBRATE`	`PAYLOAD_START_CALIBRATION(mode=0/1/2)` (37)
`SET_PHASE`	Nothing. The alias was removed; mission phase is declared by mission management only (§7.1)
`SHUTTER_OPEN` / `SHUTTER_CLOSE`	`PAYLOAD_SET_SHUTTER(position=0/1)` (119) — absolute, idempotent
`STORAGE_REQUEST_DOWNLINK`	`S13_CREATE_TRANSFER` (S13.1) then poll `S13_REPORT_STATUS` (S13.9)

The full catalogue, including the legacy quick-action IDs (100–107) that drive the EPS/TCS MCS display buttons, is in `commands/tc_catalog.yaml`, rendered wire-exact in `manual/09_command_reference.md`.

7.8 Direct commands vs procedure-driven commands

Table 7.12 — When to use each command path (procedure-driven, direct, raw PUS) with associated risk.

Path	When to use	Risk
Procedure-driven	Whenever a procedure exists for what you're doing.	Lowest. The auto-populated command has parameters validated.

Path	When to use	Risk
Direct from commanding panel	(a) Ad-hoc actions during anomaly recovery, with FD explicit authorisation; (b) the "fix-it-now" steps you've practised.	Operator typing errors, mis-remembered parameters. Always slower than the procedure path.
Raw PUS frame	Almost never. Reserved for debugging the simulator.	High. No safeties.

The default is *always* procedure-driven. If you find yourself needing a direct command, ask: *is there a procedure I should load first?*

7.9 Keyboard shortcuts

Keyboard shortcuts are not standardised across MCS deployments — they differ between installations and between operator workstations. Refer to your installation's quick-reference card for the local conventions (focus telemetry, focus commanding, command picker, first-key / second-key bindings, mark-time-in-log).

7.10 Where to read more

Table 7.14 — Source files and configuration paths for telemetry, limits, procedures, and command catalogues.

Topic	Source
Subsystem manuals	`configs/eosat1/manual/0*.md`
Generated command reference	`configs/eosat1/manual/09_command_reference.md`
Position role analyses	`configs/eosat1/mcs/role_analysis/*.md`
Position tab / command gating	`configs/eosat1/mcs/positions.yaml`
Telemetry parameters	`configs/eosat1/telemetry/parameters.yaml`
HK structures and SIDs	`configs/eosat1/telemetry/hk_structures.yaml`
Display staleness policy	`configs/eosat1/mcs/staleness.yaml`
Telemetry limits	`configs/eosat1/mcs/limits.yaml`
Alarm → procedure linkage	`configs/eosat1/mcs/alarm_procedures.yaml`
AOCS mode codes (canonical)	`configs/eosat1/aocs/aocs_mode_codes.yaml`
Procedure index	`configs/eosat1/procedures/procedure_index.yaml`
FDIR rules	`configs/eosat1/monitoring/s19_rules.yaml`
Monitoring rules	`configs/eosat1/monitoring/s12_definitions.yaml`
PUS service implementation	`configs/eosat1/mcs/pus_services.yaml`
Command catalogue	`configs/eosat1/commands/tc_catalog.yaml`
Ground-station geometry (canonical)	`configs/eosat1/orbit.yaml`
Ground-station RF augmentation	`configs/eosat1/planning/ground_stations.yaml`

Topic	Source
TM rate ladder	`configs/eosat1/subsystems/ttc.yaml`

External references:

- ECSS-E-ST-70-41C — Packet Utilization Standard — ecss.nl
- CCSDS 660.0-G-3 — PUS green book — public.ccsds.org
- ECSS-E-ST-70-11C — Space segment operability — ecss.nl
- ECSS-E-ST-70-32C — Test and operations procedures — ecss.nl

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7.20 Chapter 20 — Ground Segment & Operations Architecture

> **Expected reading before this session.** Open textbook *Concurrent Spacecraft Design* (OER), ch. 17, §§17.1–17.4 (ground segment architecture, contact analysis, CCSDS protocol stack, operations timeline); this book, §7.1 and §7.3; ECSS-E-ST-70C *Ground systems and operations* (31 July 2008), §4.3 *Ground systems*; CCSDS 130.0-G-4 *Overview of Space Communications Protocols* (Green Book, Issue 4, April 2023, with Editorial Change 1, February 2024), sections 2–3; NASA *State-of-the-Art of Small Spacecraft Technology* (2026 edition, NASA/TP-20260003140), ch. 11 *Ground Data Systems and Mission Operations*.

Prerequisites: Week 2 complete (full design cycle through cost estimation) **References:** ECSS-E-ST-70C (Ground Systems), ECSS-E-ST-70-01C (Spacecraft On-Board Control Procedures), CCSDS 131.0-B-4 (TM Coding), CCSDS 231.0-B-4 (TC Coding), CCSDS 133.0-B-2 (Space Packet Protocol), NASA DSN Handbook (810-005)

Learning Objectives

By the end of this session, participants will be able to: 1. Describe the components of a ground segment architecture (antennas, MCS, FDS, networks) 2. Calculate ground station contact time and data volume per pass 3. Explain the CCSDS protocol stack for space communication 4. Design a ground station network for a given orbit and data downlink requirement 5. Construct a mission operations timeline with key milestones

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1. Ground Segment Architecture

The ground segment is everything on the ground that supports the space mission. It is frequently underestimated in early mission design but accounts for 5-15% of total mission cost and is critical to mission success.

[Source: ECSS-E-ST-70C (Ground Systems and Operations); ECSS-E-ST-70-01C (Spacecraft On-Board Control Procedures)] [URL: https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/]

Ground Segment Components

The ground-segment architecture — the ground-station network, the Mission Control System (MCS), and the Flight Dynamics System (FDS), with their external interfaces — is shown in the **ground-segment architecture figure** in §7.5.

Ground Station Components

Table 7.15 — Ground-station hardware components with their function and typical specification.

Component	Function	Typical Specification
Antenna	Transmit uplink commands, receive downlink telemetry	3-13 m dish (S/X-band); Yagi/turnstile (UHF)
RF front-end	Low-noise amplifier (LNA), power amplifier, filters	LNA NF < 1 dB; PA 50-500 W
Modem/ baseband	Modulation/demodulation, coding/decoding	CCSDS-compliant; BPSK/QPSK/8PSK
Tracking system	Antenna pointing, Doppler tracking, ranging	Autotrack or program-track
Data capture	Record raw baseband data for post-processing	High-speed disk array
Timing	GPS-disciplined oscillator for time synchronisation	< 1 microsecond accuracy

Mission Control System (MCS)

Table 7.16 — Mission Control System functions with descriptions and example tooling.

Function	Description	Tools
Telemetry processing	Decode, decommutate, display, and archive TM packets	COSMOS, YAMCS, EGOS (ESA)
Telecommand generation	Create, validate, encode, and uplink TC packets	Same MCS tools
Pass scheduling	Schedule antenna time, plan contact windows	STK, GMAT, custom schedulers
Monitoring & control	Real-time health monitoring, limit checking, alarming	MCS dashboards
Anomaly management	Detect, diagnose, and resolve anomalies	Procedures + expert judgment

Flight Dynamics System (FDS)

Table 7.17 — Flight Dynamics System functions with descriptions and required inputs.

Function	Description	Inputs
Orbit determination	Compute current orbit from tracking data	Range, Doppler, GPS (if onboard)
Orbit prediction	Propagate orbit forward for pass planning	Current state vector, perturbation models
Manoeuvre planning	Compute delta-V for orbit maintenance	Target orbit, propulsion model
Conjunction assessment	Predict close approaches with debris/other objects	TLE catalogue, own orbit
End-of-life planning		Remaining propellant, target orbit

Function	Description	Inputs
	Compute deorbit manoeuvre or passivation plan	

2. Ground Station Contact Analysis

Understanding how much data can be transferred per pass is fundamental to mission design. The data budget depends on the contact geometry, data rate, and pass duration.

Contact Geometry

For a circular LEO orbit, the visibility of a ground station is determined by the minimum elevation angle constraint.

Maximum Slant Range at Minimum Elevation:

rho = R_E x (sqrt((h/R_E + 1)^2 - cos^2(epsilon)) - sin(epsilon))

Where: - R_E = Earth radius (6371 km) - h = orbital altitude - epsilon = minimum elevation angle (typically 5-10 degrees)

For h = 500 km, epsilon = 5 degrees: rho = 6371 x (sqrt((500/6371 + 1)^2 - cos^2(5)) - sin(5)) rho = 6371 x (sqrt(1.0785^2 - 0.9962^2) - 0.0872) rho = 6371 x (sqrt(1.1632 - 0.9924) - 0.0872) rho = 6371 x (sqrt(0.1708) - 0.0872) rho = 6371 x (0.4133 - 0.0872) rho = 6371 x 0.3261 rho = 2077 km

Maximum Pass Duration (overhead pass):

T_max = 2 x R_E x arccos(R_E x cos(epsilon) / (R_E + h)) / V_ground_track

Simplified approximation for LEO: T_max ~ (2 x rho_max) / V_orbital x (R_E / (R_E + h))

For 500 km SSO: V_orbital = sqrt(mu / (R_E + h)) = sqrt(3.986e14 / 6.871e6) = 7613 m/s T_max ~ 2 x 2077e3 / 7613 x (6371/6871) ~ 505 s ~ 8.4 minutes

Typical average pass duration (not all passes are overhead): ~6 minutes

Data Volume per Pass:

V_data = R_data x T_contact x eta_protocol

Where: - R_data = downlink data rate (bps) - T_contact = contact duration (seconds) - eta_protocol = protocol efficiency (typically 0.85-0.95 for CCSDS)

Example: R = 2 Mbps, T = 360 s (6 min average), eta = 0.9: V_data = 2e6 x 360 x 0.9 = 648 Mbit = 81 MB per pass

Contact Frequency

Table 7.18 — Daily passes and downlink-volume estimates by ground-station latitude for a 500 km SSO.

Orbit	GS Latitude	Passes per Day	Avg Duration	Daily Data Volume (2 Mbps)
500 km SSO	52 N (e.g., Waterloo)	3-5	6 min	243-405 MB
500 km SSO	69 N (e.g., Kiruna)	6-8	7 min	567-756 MB

Orbit	GS Latitude	Passes per Day	Avg Duration	Daily Data Volume (2 Mbps)
500 km, 51.6 (ISS)	52 N	2-4	5 min	135-270 MB
500 km SSO	SatNOGS network (global)	10-15	5 min	675-1013 MB

[Source: STK simulations; validated against SatNOGS observation logs]

Worked Example: Data Budget Closure

Problem: An Earth observation mission generates 500 MB/day of imagery. Is a single ground station at 52 N latitude sufficient with 2 Mbps S-band downlink?

Calculation: - Passes per day: ~4 (500 km SSO from 52 N) - Avg pass duration: 6 min = 360 s - Data per pass: 2e6 bps x 360 s x 0.9 / 8 = 81 MB - Daily capacity: 4 x 81 = 324 MB - Required: 500 MB/day - **Shortfall: 176 MB/day -- budget does not close!**

Solutions: 1. Add a second ground station at high latitude (e.g., Kiruna -> +6 passes -> +486 MB) 2. Increase data rate to X-band (10 Mbps -> 405 MB/pass -> easily sufficient) 3. Reduce data generation (lower duty cycle or reduce image resolution) 4. Add onboard data compression (2:1 lossless -> 250 MB/day required) 5. Use SatNOGS network for additional UHF passes (housekeeping only)

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3. CCSDS Protocol Stack

The Consultative Committee for Space Data Systems (CCSDS) defines the standard protocols for space communication, analogous to TCP/IP for ground networks.

[Source: CCSDS 131.0-B-4 (TM Synchronization and Channel Coding); CCSDS 231.0-B-4 (TC Synchronization and Channel Coding); CCSDS 133.0-B-2 (Space Packet Protocol)] [URL: https://public.ccsds.org/Pubs/131x0b4.pdf (freely available)]

Protocol Layers

Table 7.19 — CCSDS protocol-stack layers mapped to their function and terrestrial analogues.

Layer	CCSDS Standard	Function	Ground Analogy
Application	CCSDS 133.0 (Space Packet)	Mission data and housekeeping packets	HTTP/application data
Network	CCSDS 732.0 (AOS) or 132.0 (TM)	Virtual channels, multiplexing	IP routing
Data Link	CCSDS 131.0 (TM coding) / 231.0 (TC coding)	FEC, frame sync, CRC	Ethernet
Physical	CCSDS 401.0 (RF & Modulation)	Modulation, frequency, power	Physical layer

Telemetry (TM) Frame Structure

The TM/TC Transfer Frame layout — header, data field, and error-control trailer, with the spacecraft ID, virtual-channel ID, and frame-counter fields inside the header — is shown in context in the protocol-stack figure in §7.2, which maps the full protocol stack from the PUS service layer down to the CADU on the RF link.

Forward Error Correction (FEC) Options

Table 7.20 — Forward error-correction code options with rate, coding gain, and use case.

Code	Rate	Coding Gain (dB)	Use Case
Convolutional (7, 1/2)	1/2	5.5	Legacy CubeSats
Reed-Solomon (255, 223)	0.87	3.5	Combined with convolutional
Turbo (rate 1/2)	1/2	7.5	High-performance CubeSats
LDPC (rate 7/8)	7/8	6.0	High data rate, bandwidth efficient

Modern CubeSats typically use LDPC coding for downlink (high data rate) and convolutional + RS for uplink (robustness priority).

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4. Operations Timeline Construction

The operations timeline (or operations concept timeline) defines all major activities from launch to end of life. It is part of the ConOps and drives ground segment design.

Mission Phases (Operations Perspective)

Table 7.21 — Mission operations phases with duration, activities, and staffing model.

Phase	Duration	Key Activities	Staffing
LEOP	0-72 hours	Separation, beacon acquisition, SA/antenna deploy, initial health check	24/7 (3 shifts)
Commissioning	1-4 weeks	Subsystem checkout, ADCS calibration, payload first light	16/7 (2 shifts)
Early Operations	1-3 months	Performance characterisation, procedure refinement	8/5 (1 shift)
Nominal Operations	Mission lifetime	Routine science/service, orbit maintenance	8/5 or automated
Extended Operations	If approved	Degraded mode operations, reduced data	Part-time
End of Life	1-4 weeks	Passivation, deorbit manoeuvre, final telemetry	8/5

Operations Timeline (Gantt-Style)

Operations timeline: mission phases (LEOP, commissioning, early operations, nominal operations) and console staffing levels (24/7, 16/7, 8/5) across the first year, by week from launch.

Operations timeline: mission phases (LEOP, commissioning, early operations, nominal operations) and console staffing levels (24/7, 16/7, 8/5) across the first year, by week from launch.

LEOP Timeline Detail

Table 7.22 — Hourly LEOP timeline for the first 72 hours with activities and success criteria.

Time (after separation)	Activity	Success Criterion
T+0 s	Separation from deployer	Deployment switches released
T+0 to T+30 min	Deployment timer countdown (no RF)	Timer runs; all inhibits clear
T+30 min	Antenna deployment	Beacon detected by ground station
T+30 to T+60 min	Beacon acquisition	Carrier lock; beacon decoded
T+1 hr	First telemetry downlink	Housekeeping data received and validated
T+1 to T+6 hr	Initial health assessment	All subsystems reporting nominal HK
T+6 to T+12 hr	Solar array deployment (if separate)	Power generation confirmed
T+12 to T+24 hr	ADCS initialisation	Attitude determination active
T+24 to T+48 hr	ADCS calibration (magnetometer, sun sensor)	Pointing within coarse spec
T+48 to T+72 hr	Communication chain validation (full duplex)	Uplink and downlink at operational rate

SpaceCDF Operations Planning

SpaceCDF's **ConOps Editor** allows you to: - Define mission phases with start/end conditions - Specify operational modes per phase (safe, nominal, science, downlink) - Set ground contact requirements per phase (frequency, duration) - The system validates that the ground segment design supports the operations concept

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5. Ground Segment Design Exercise

Instructions

1. **Data Budget** tab -- Review the daily data generation vs downlink capacity - Is the data budget closing? If not, identify the bottleneck - How many ground station passes per day does your orbit provide?
2. **ConOps Editor** -- Define mission phases: - LEOP (duration, staffing, contact requirements) - Commissioning (activities, success criteria) - Nominal operations (duty cycle, data volume, contact frequency)
3. On Worksheet 5.1: - Calculate data volume per pass for your mission - Determine number of ground stations needed to close the data budget - Construct a LEOP timeline - Draft a simplified operations timeline (Gantt chart)

Discussion Prompts

- "What happens if your first pass after deployment has no signal?"
- "How would you handle a safe mode entry at 3 AM on a Saturday?"
- "What is the minimum ground segment that could support your mission?"

Worksheet 5.1 Tasks

1. Calculate ground station contact time and data volume per pass
2. Design a ground station network (locations, antenna sizes, data rates)
3. Construct a LEOP timeline (first 72 hours, hourly resolution)
4. Draft a mission operations timeline (Gantt chart, weekly resolution)
5. Estimate ground segment cost (antennas + MCS + FDS + operations staff)

Cited References

Table 7.23 — Cited references for the Ground Segment and Operations Architecture session with URLs.

#	Reference	URL
1	ECSS-E-ST-70C (Ground Systems and Operations)	https://ecss.nl/standard/ecss-e-st-70c-ground-systems-and-operations/
2	CCSDS 131.0-B-4 (TM Synchronization and Channel Coding)	https://public.ccsds.org/Pubs/131x0b4.pdf
3	CCSDS 133.0-B-2 (Space Packet Protocol)	https://public.ccsds.org/Pubs/133x0b2.pdf
4	CCSDS 231.0-B-4 (TC Synchronization and Channel Coding)	https://public.ccsds.org/Pubs/231x0b4.pdf
5	SatNOGS Network	https://network.satnogs.org/
6	NASA DSN Handbook (810-005)	https://deepspace.jpl.nasa.gov/dsndocs/810-005/
7	YAMCS Mission Control Software	https://yamcs.org/

Session Summary

Table 7.24 — Session-summary key takeaways across ground-segment and operations-architecture topics.

Topic	Key Takeaway
Ground segment	Antenna network + MCS + FDS + mission planning + data processing
Contact analysis	Pass duration ~6 min (LEO/SSO); data volume = rate x time x efficiency
Data budget	Must close: daily generation <= daily downlink capacity; add GS or increase rate
CCSDS	Standard protocol stack: Space Packet, AOS/TM frames, FEC coding, RF modulation
Operations timeline	LEOP (24/7) -> Commissioning (16/7) -> Nominal (8/5 or automated)
LEOP	First 72 hours critical: antenna deploy, beacon acquisition, health check
SpaceCDF	ConOps Editor defines phases; Data Budget validates ground segment sizing

7.11 References

- ECSS-E-ST-70C — Ground systems and operations — ecss.nl
- ECSS-E-ST-70-41C — Telemetry and telecommand packet utilization (PUS) — ecss.nl

- CCSDS 132.0-B-3 — TM Space Data Link Protocol — public.ccsds.org
- CCSDS 232.0-B-4 — TC Space Data Link Protocol — public.ccsds.org
- CCSDS 660.0-G-3 — PUS green book — public.ccsds.org

Chapter 8 — Pass Operations (Routine)

This chapter walks the **routine pass cycle** during nominal operations (phase 6) — the post-commissioning steady state that constitutes most of the mission's operational life. A nominal pass runs: pre-pass briefing → AOS → pass startup → in-pass activities (imaging, downlink, health check) → LOS → post-pass debrief → shift handover (if at shift boundary).

The procedures referenced in this chapter are NOM-001 through NOM-013 (Chapter 14 holds the full catalogue). Each procedure is walked step-by-step with the FD call, the operator action, the verification telemetry, and the on-fail cross-reference.

Single ground-pass cycle in swimlane view across FD, FE, payload, and ground-station roles, with AOS/LOS guides and an example FDIR rule firing mid-pass.

Single ground-pass cycle in swimlane view across FD, FE, payload, and ground-station roles, with AOS/LOS guides and an example FDIR rule firing mid-pass.

8.1 Pass cycle overview

A routine pass on a 513 km SSO spacecraft like EOSAT-1 lasts between 3 and 9.4 minutes depending on the maximum elevation of the node crossing — 3–5 min for a 5–10° grazing pass, about 9 min for a near-overhead one, with a measured mean of 7.5 minutes. Across the **five-station** network the spacecraft is in contact with some station on most orbits, often more than once. The operator-side timeline:

Table 8.1 — Pass-cycle activity timeline from pre-pass briefing through post-pass debrief.

Time relative to AOS	Activity	Lead	Procedure
AOS-30 to AOS-15 min	Pre-pass briefing; plan review; ticket review	FD	(none — briefing)
AOS-10 to AOS-5 min	Station readiness check; antenna pre-position; baseband ready	Ground Seg	(NOM-001 step 1)
AOS-2 to AOS	Transmitter pre-arm; carrier acquisition standby	TT&C	(NOM-001 steps 2-3)
AOS	Carrier locks; full HK begins. **`ttc.link_status` (0x0501) = 2 is LOCKED**; 1 is only ACQUIRING (carrier lock without frame sync) and 0 is NO_LINK — do not read a non-zero value as a good link	TT&C + SPACON	NOM-001 (Pass Startup)
AOS+1 min	S17.1 `CONNECTION_TEST`; full HK verification; FDIR sweep	SPACON	NOM-001 steps 4–5
AOS+2 to LOS-2	Planned activities (imaging, downlink)	per plan	NOM-002, NOM-003
LOS-2 min	Pre-LOS health check; pass-end log entry	SPACON	NOM-009
LOS		TT&C	

Time relative to AOS	Activity	Lead	Procedure
	S-band chain back to LOW (16 kbps), DOWNLINK mode off; configuration recorded		NOM-001 close-out
LOS+0 to LOS+15	Hot-wash debrief; ticket close-out; next-pass review	FD	(NOM-012 handover if at shift end)

The mission planning system (MPS) builds the pass plan upstream; the team's job during the pass is to **execute the plan and respond to anomalies**, not to plan in real time.

8.2 Reference spacecraft and pass geometry

The sections that follow assume the EOSAT-1 reference mission (Chapter 5 §5.7), the ground segment (Chapter 7 §7.1), and the control room (Chapter 5 §5.4). Students will have read these chapters. The brief is *check for comprehension* rather than first-presentation.

The four ground-segment elements, in turn:

1. **Ground stations.** Open the MPS station-status panel. Walk all **five** stations — Iqaluit (63.7° N), Inuvik (68.3° N), Svalbard (78.2° N), Troll (72.0° S) and O'Higgins (63.3° S) — and show the per-station next-AOS countdown. Confirm on the MCS which one is live now: `ttc.active_gs` (0x0523) is a 1-based index into that same list.
2. **Baseband.** Open the rfsim baseband health feed. Show the Eb/N0 and frame-sync state, both currently nominal because we are *in* a pass.
3. **Mission Control System (MCS).** Open the FD's MCS dashboard at `:9090`. Walk the regions (top, left, centre, right, procedure).
4. **Flight Dynamics.** Show the FDS workstation. Read the pass list from the planner's contact schedule at `:9091`, and run `python3 tools/pass_gap_analysis.py --config configs/eosat1 --hours 24` for the gap statistics behind it.

The brief closes on **pass geometry**: why polar SSO with polar ground stations gives many short passes per day rather than fewer, longer ones. See Figure 8.1.

8.2.1 Pass geometry — polar SSO over polar stations

A sun-synchronous orbit at 513 km has a period of ~94.5 minutes. EOSAT-1's is a **mid-morning** orbit, not a dawn–dusk one: the local solar time at the ascending node is held near **10:00**, so the spacecraft passes through Earth's shadow on every orbit (33.2 to 35.2 minutes). All five stations — Svalbard (78.2° N), Inuvik (68.3° N), Iqaluit (63.7° N), Troll (72.0° S) and O'Higgins (63.3° S) — sit at high latitude, close to where the ground track converges, so most ascending or descending nodes bring the spacecraft into at least one station footprint.

Result, measured over 48 h from the mission epoch with `tools/pass_gap_analysis.py`: **103 passes in 48 h** — call it ~51 per day — with a **median gap of 33 minutes** and a **worst gap of 79 minutes**. (The two-station network the older courseware assumed gives 40 passes, an 84-minute median gap and a 147-minute worst case; the five-station baseline is what this course uses throughout.) Pass duration runs **3 to 9.4 minutes** and maximum elevation anywhere from the 5° mask to overhead, depending on the geometry of the node crossing. There is still no contact when the spacecraft is over the equator or mid-latitudes — radio-silence segments are normal, just shorter. EPS and Thermal must plan accordingly.

Note the operational consequence of elevation: the 128 kbps HIGH tier is gated at **10° elevation**, so a pass that never rises above 10° is usable for commanding and 16 kbps housekeeping but carries no imagery at all.

Worked example: at 513 km, the geometric horizon-to-horizon transit at maximum elevation (overhead) is

$$t_{transit} = 2 \cdot \arccos\left(\frac{R_E + h_{station}}{R_E + h_{sat}}\right) \cdot \frac{T_{orbit}}{2\pi}$$

With $R_E = 6371$ km, $h_{sat} = 513$ km, $h_{station} \approx 0$, and $T_{orbit} = 5669$ s, the geometric transit is ≈ 11.7 minutes. Adding the 5° elevation mask — $\lambda = \arccos\left(\frac{R_E \cos\varepsilon}{R_E + h}\right) - \varepsilon = 17.8^\circ$, so a 35.6° arc — the usable overhead contact reduces to ≈ 9.3 minutes. With the mask plus realistic mid-elevation passes the measured mean is 7.5 minutes.

Figure 8.1 — Schematic of one EOSAT-1 orbit with the five-station visibility cones (5° elevation mask). Lit and eclipsed segments are shaded.

8.5 Mission Planning System hands-on

Operators work at their workstations with the MPS open and work through the exercise individually.

Exercise objective

Build, validate, and export a 60–90 minute pass plan for the next Iqaluit pass. Plan must include one imaging activity and the associated slew, the downlink for the resulting image, and a routine health check.

Steps

1. **Open MPS** at http://localhost:9091/.
2. **Open the contact schedule.** Find the next Iqaluit pass. Note AOS, max elevation, LOS, duration. If its maximum elevation is under 10°, note that it cannot carry a dump and pick the downlink pass accordingly.
3. **Open the imaging-target picker.** Pick three ocean-current targets (`OCM-…`) visible during the pass; the picker shows visibility per pass and each target's `min_solar_elevation_deg` gate.
4. **Drop a slew + image + slew sequence** into the pass. The conflict checker runs automatically.
5. **Resolve any conflict.** Likely conflicts: slew time too short between targets (reduce to two targets); cumulative power exceeds budget (drop one band); requested downlink needs the HIGH tier on a pass that never clears 10°.
6. **Add a downlink activity** for the resulting image(s) using the same pass or the next pass (whichever has more contact time).
7. **Add an NOM-009 routine health check** at the start of the pass.
8. **Run final conflict check.** Should be green.
9. **Export the plan.** Save the file under `outputs/student_plans/<your-name>_day1.yaml` .
10. **Open the plan in the MCS procedure panel.** Verify the activity sequence appears as expected.

What good looks like

- All conflicts resolved (green status).
- Pass plan duration matches actual pass window minus a 60 s margin at each end.
- Health check is the first activity (NOM-009 step 1 requests the six subsystem HK SIDs with `HK_REQUEST(sid=…)` (S3.27); do it before commanding anything else). Remember that only SIDs

1 and 11 are enabled at boot — if a SID has never been enabled this session you need `HK_ENABLE(sid=…)` (S3.5) first.

- Downlink activity matches the image volume produced (rule of thumb: `image_size_mb` is 4 MB per imaging session; at the HIGH tier's 128 kbps that is 4 × 8 / 0.128 ≈ **250 s**, a little over four minutes — a real bite out of a 7.5-minute average pass, and it needs DOWNLINK mode on and elevation above 10° for its whole duration).

8.6 MCS round-robin

Seven configured consoles (§7.2), 15 minutes each, after a 15-minute briefing: the 13:00–15:00 block exactly. Each student visits all seven during the afternoon. At each station, the seat-holder reads the position's brief aloud (Chapter 6), then completes one position-specific task while watching live telemetry.

Table 8.2 — Round-robin tasks assigned to each control-room position during the familiarisation rotation.

Position	Task
Flight Director	Run a GO/NO-GO round on the live dashboard. Poll three positions for *go for* `CONNECTION_TEST` (S17.1) — there is no NOOP or heartbeat command on this spacecraft; record the responses.
FDIR / Systems	Walk through one FDIR rule on the FDIR panel; explain its trigger, threshold, severity, and the contingency procedure `mcs/alarm_procedures.yaml` points it at.
Flight Dynamics (AOCS)	Read the current attitude error from the visualiser and compare it against both budgets: the NOM-001 nadir-tracking check (`aocs.att_error` 0x0217 < 1.0°) and the far tighter capture gate (FINE_POINT, total pointing error < 0.05°, rate stability < 0.005 °/s).
Power & Thermal	Predict battery SoC at exit of the next eclipse from the contact schedule; then identify which heater is at the highest duty cycle and predict whether it will saturate next eclipse.
TT&C	Command the LOW → HIGH rate transition at AOS: `TTC_SET_DOWNLINK_MODE(on=1)` (79) then `TTC_SET_DATA_RATE(rate=1)` (65). Confirm `ttc.tm_tier` (0x0539) actually reaches 2 and does not sit demoted below `ttc.tm_tier_cmd` (0x053A). Read the link margin after the transition. Then, with FD authorisation, watch what `TTC_PA_OFF` (67) does: S-band telemetry stops (`ttc.tx_active` 0x0525 → 0) while the UHF beacon at `ttc.beacon_active` (0x0527) keeps reporting.
Payload Operations	Schedule a single imaging session via the MPS. Confirm FPA temperature is at setpoint and `FPA ready` (0x0610) = 1 before commanding.
Propulsion	Read the arm interlock mask (`arm_inhibit_mask` , 0x0B15) and name which gate is currently blocking an arm, without commanding anything.

The Mission Manager and Ground Segment roles are not part of the round-robin — they are room roles rather than configured MCS consoles, and are held by the Mission Director or a designated operator throughout the day.

A position-score card is in the workbook (WB-5.1), one per console. The student writes one observation per station.

8.7 Pass-plan team exercise

Teams of 4–6 (Borealis and Austral) work in parallel. Each team produces a written 60–90 minute pass plan for the same upcoming Iqaluit pass. The plan must include:

- Pass timing (AOS – TCA – LOS) with margins.
- Procedures to run, by procedure ID.
- Imaging window, target ID, predicted ground swath, science band(s) chosen, integration time.
- Downlink config: which rung of the ladder (LOW 16 kbps BPSK conv(7,1/2)+RS, or HIGH 128 kbps QPSK RS(255,223)), whether the pass clears the 10° HIGH gate for long enough, and the expected duration.
- Anomaly contingency: **two** plausible failures and the procedure ID to invoke for each.

Plans are read aloud at 16:30. The other team uses the *Technical* peer-review worksheet axis to critique.

8.8 Debrief and this-shift reading

Debrief questions:

1. Which MCS position felt most natural to you? Why?
2. What surprised you about a ground-station pass timing-wise? Did the five-station network behave the way you expected?
3. What's one thing you'd want documented better before the next shift?

this-shift reading: Chapter 9 in full, plus simulator procedures `procedures/nominal/startup.md` (NOM-001) and `procedures/contingency/ttc_link_loss.md` (CTG-003) — approximately 45 minutes.

8.9 References

- ECSS-E-ST-70C — Ground systems and operations — ecss.nl
- CCSDS 520.0-G-4 — Mission Operations Reference Model — public.ccsds.org

Chapter 9 — Voice-Loop Discipline and Procedure Execution

The voice loop is the spine of the operations room. Every position is on the loop; every commanded action is voice-coupled. This chapter covers the **discipline** of operating on the loop — the patterns, the vocabulary, and the protocols that distinguish a team that responds to an anomaly in 90 seconds from one that responds in 5 minutes — and the **procedure-execution rhythm** that maps voice callouts to command actions.

The chapter is reference material, used by every operator before every shift. The drill exercises are in §9.2.

9.1 Why the voice loop matters

Industry data: in roughly 80% of cases where a mission operations team turns a recoverable anomaly into an unrecoverable one, the root cause traces to a voice-loop failure — a callout missed, an ambiguous instruction, a parallel conversation that hid an important update — not a telemetry misread or a wrong command. Voice-loop discipline is the single highest-leverage operator skill, ahead of even telemetry interpretation.

9.2 Voice-loop fundamentals

Voice-loop discipline is a learned skill that distinguishes a team that responds to an anomaly in 90 seconds from one that responds in 5 minutes. The reference standards are:

- ESA *Operations Manual for Mission Control*, especially the loop-discipline chapter
- NASA APPEL voice-procedure materials
- FAA Crew Resource Management (CRM) AC-120-51E — the underlying theory of multi-position crew coordination

Three drills build the skill:

Drill 1 — Addressing pattern

Operators pair up — one as FD, one as a subsystem operator — and run 10 paired callouts using the pattern `[ADDRESSEE], [SPEAKER] — [MESSAGE]`. Examples:

- "AOCS, FD — pointing report."
- "FD, AOCS — pointing 0.28° (NOMINAL), nominal."
- "EPS, FD — eclipse-entry predictor."
- "FD, EPS — eclipse-entry T+04:12, SoC at exit 78%."

Score cross-talk and missed addressing.

Drill 2 — GO / NO-GO with hold

A scenario is set: the team is about to command a payload power-on during a pass. The FD must poll EPS, Thermal, and Payload for GO. Each operator has been briefed on a *secret state* (e.g. EPS sees unexpected load already, Thermal sees FPA still warming up). They must respond honestly: *go*, *no-go* with reason, or *hold* with reason.

The FD makes the decision and explains it on the loop. The secret states are then revealed and the team discusses.

Drill 3 — Anomaly callout

An anomaly is injected off-loop (a parameter goes yellow on one operator's tab). The operator must:

1. Recognise the change.
2. Call it on the loop with the format *"FD, [POSITION] — anomaly, [parameter], [value], [trend]"*.
3. Wait for FD direction.

Time from first telemetry change to first operator callout. The team target is **under 15 seconds**.

Voice-loop reference: hardware, hot-mike, vocabulary, and GO/NO-GO

The fundamentals above are reinforced by a small set of standing conventions every position is expected to know cold. A wallet-card version of the callout vocabulary is in the workbook (WB-5.4-VC).

Hardware

One control-room softphone headset per position, all conferenced into a single channel. The Mission Manager is on the loop but rarely speaks operationally.

Hot-mike rule

Headsets are push-to-talk by default. Hot-mike (always-on) is reserved for the FD during a critical phase (the FD calls "hot mike on" before a complex sequence and "hot mike off" after).

Callout vocabulary

Table 7.4 — Standard voice-loop callout vocabulary with meaning and expected response.

Callout	Meaning	Response
"Comm check 1-2-3"	Test voice loop	"FD copy, comm check loud and clear"
"Standby"	I am working; will respond	"Standby" or wait
"Standby plus 30"	Standby, I will be back in 30 s	(none required)
"Go for X"	Request authorisation to do X	"Go" or "No-go, [reason]" or "Hold, [reason]"
"Hold"	Pause the current action	All positions stop
"Resume"	Continue the held action	(action resumes)
"No-go"	Refuse the request, with reason	(next step is FD's call)
"AOS in N"	Acquisition of signal in N minutes	(no response required)
"AOS"	Spacecraft is in view	(Ground Segment confirms link)
"LOS in N"	Loss of signal in N minutes	(no response required)
"LOS"	Spacecraft is out of view	(operators wrap pending steps)
"Pass closed"	Pass is fully over, decompress	(debrief begins)

Callout	Meaning	Response
"Anomaly, [parameter], [value]"	Out-of-band telemetry	(SPACON confirms FDIR state)
"Mark"	Note the time for the log	(auto-stamps shift log)
"Loop discipline"	FD is calling cross-talk to order	All stop and re-acquire

GO / NO-GO protocol

Before any commanded action that affects spacecraft state:

1. FD polls relevant positions. ("AOCS, EPS, FDIR — go for ST1_POWER?")
2. Each position responds in poll order: "Go" / "No-go, [reason]" / "Hold, [reason]".
3. FD declares: "Go for ST1_POWER" or "Hold on ST1_POWER, [reason]".
4. Owner executes only on FD's explicit GO. Pre-staging the command (first-key) before the GO is fine; uplinking before the GO is a violation.

9.3 Procedure walkthrough — NOM-001 Pass Startup

NOM-001 is the canonical *pass startup* procedure, run at the beginning of every pass during nominal operations. It is presented below in **aircraft-QRH challenge–response format**: the SPACON calls a *challenge*, the named owner gives the *response*, and the *action* column shows the actual telecommand or read. Operators write the actual telemetry value or a tick in the *Note / actual* column.

Source of truth: `procedures/nominal/startup.md` . **Owner positions:** SPACON (lead, holds the procedure file) · TT&C · Ground Segment · FD. **Trigger:** AOS-5 min for any scheduled pass during phase 6. **Pass criterion:** All ten steps verified; pass log entry made. **Estimated duration:** 3–6 minutes.

> **Memory items** (operators do these *before* opening the procedure file, in order of priority):
>
> 1. **Confirm ground station tracking.**
> 2. **Verify spacecraft tx state ON before AOS-2 min.**
> 3. **At AOS, listen for carrier lock callout.**

Pre-checklist calculations (operator does these before NOM-001 step 1)

Table 9.1 — Pre-pass operator calculations with reference section and expected typical answer.

Calculation	Reference	Typical answer
Predicted AOS time	Contact schedule (MPS) — see §13.2	nominal
Eclipse-exit SoC at AOS	§13.3 formula	> 55 %
Current pointing-error RSS	AOCS tab — see §13.4	< 0.4 °
Predicted link margin at TCA	MPS link-budget tab — see §13.6	> 5 dB

NOM-001 checklist

Table 9.2 — NOM-001 Pass Startup challenge-response checklist with verifications and on-fail links.

#	Owner	Challenge	Expected response	Action / TC	On-fail link	☐	Note / actual
1	Ground Seg	"GS, stations in view tracking?"	"FD, GS — stations in view tracking."	Read MPS station-status panel — every station in view shows TRACKING	CTG-020	☐	_____
2	TT&C	"TT&C, tx state pre-AOS?"	"FD, TT&C — `ttc.tx_active` = 1, time-to-AOS T-1:30."	Read TTC tab `tx_active` (0x0525); if it reads 0, `ttc.tx_gate_status` (0x052B) names the gate that is holding the downlink down	Check S11 onboard schedule	☐	_____
3	TT&C	"TT&C, link status?" (at AOS)	"FD, TT&C — `ttc.link_status` 2 LOCKED, RSSI -98 dBm, margin 8.2 dB."	Read TTC tab `link_status` (0x0501), `rssi` (0x0502), `link_margin` (0x0503) within 30 s of AOS	CTG-003	☐	_____
4	SPACON	"SPACON, link test."	"FD, SPACON — S17.1 sent, response within 4 s."	Commanding panel: S17 sub-1; first-key; FD GO; uplink	CTG-019 (no TM)	☐	_____
5	SPACON	"SPACON, full HK?"	"FD, SPACON — SIDs 1, 2, 3, 4, 5, 6 streaming."	FDIR panel + telemetry mosaic visual scan; a missing SID means the line is powered but the SID was never `HK_ENABLE`d (S3.5)	CTG-019	☐	_____
6	AOCS	"AOCS, mode and rates?"	"FD, AOCS — mode 4 NOMINAL, roll and yaw rates < 0.05 °/s, pitch -0.063 °/s."	AOCS tab `aocs.mode` (0x020F), `aocs.rate_roll/pitch/yaw` (0x0204–0x0206)	CTG-002	☐	_____
7	EPS	"EPS, bus and SoC?"	"FD, EPS — SoC 78 %, bus 31.5 V."	EPS tab	CTG-001 (load shed)	☐	_____
8	Thermal	"Thermal, zones?"	"FD, Thermal — all zones green."	TCS tab	CTG-004	☐	_____
9	Payload	"Payload, state vs plan?"	"FD, Payload — mode per plan."	Payload tab	(per pass plan)	☐	_____
10	SPACON	"SPACON, log pass start."	"FD, SPACON — entry logged at AOS+0:34."	Shift log	(none)	☐	_____

Two things in that table trip up new operators. First, **housekeeping does not stream by default**: the power-up enable set is `simulation.yaml` `default_enabled_sids: [1, 11]` — SID 1 (EPS) and SID 11

(the beacon) — and the engine re-applies exactly that set on every application boot. Every other SID (2 AOCS, 3 TCS, 4 OBDH/Platform, 5 Payload, 6 TT&C, 8 EPS-extended, 9 PROP) has to be commanded on with `HK_ENABLE(sid=N)` (**S3.5**), and powering a subsystem's line does not enable its SID. During nominal operations they are already on from commissioning; if one is missing, re-issue the `HK_ENABLE`, do not wait for it. A one-off snapshot of any SID is `HK_REQUEST(sid=N)` (**S3.27**).

Second, in mode 4 (NOMINAL, nadir tracking) the pitch rate is *not* zero: nadir tracking carries the orbit rate about body −Y, so `aocs.rate_pitch` reads −0.063 ± 0.02 °/s. A pitch rate of zero here means the spacecraft is **not** tracking nadir, and is the fault report — not the nominal reading.

> **Pause point — pass-start GO.** FD polls every position after step 10 with: *"All positions, pass start verified — GO for planned activities?"* Each position returns GO / NO-GO / HOLD with reason. The FD declares GO for the pass before any in-pass commanding.

Post-conditions

- Pass log entry created with AOS time, link margin baseline, SoC at AOS, pointing-error RSS at AOS.
- FDIR panel scanned and clean.
- All positions ready to execute the in-pass plan.

Common deviations

- Skipping step 4 (S17 link test). Without it, the team discovers the uplink is broken at the first commanded action ~8 minutes into the pass.
- Logging the pass start at end-of-pass rather than at AOS. The audit trail of when the pass actually began is important.

Machine-runnable cards, advisory cards, and the executor

Most procedure cards are read-and-do documents for a human. A small, explicitly marked subset is also **machine-runnable**: the index entry in `procedures/procedure_index.yaml` carries a `runnable:` link, and the MCS offers the card for execution. Today that is **eight cards** — NOM-002 (Imaging Session), NOM-003 (Data Downlink), NOM-004 (Orbit Maintenance Manoeuvre Planning), NOM-008 (Clock Synchronisation), NOM-009 (Routine Health Check), CTG-001 (Under-Voltage Load Shed), CTG-008 (Star Tracker Failure) and EMG-006 (Thermal Runaway). Three kinds of link exist: `activity_type` (the card maps onto a planner activity), `steps_file` (a canonical YAML step list) and `markdown` (the command steps are derived from the card's own `**TC:**` annotations, which is only legitimate when those annotations are the *whole* of the card's commanding).

A card that deliberately commands nothing — a pure ground-segment procedure, for instance — is declared as such with `advisory: true` and an `advisory_reason` on its index entry. It carries **no** `runnable:` link and the MCS reports it as not runnable. Do not read "advisory" as a weaker kind of runnable: it is a first-class statement that the document has zero telecommands in it.

Three executor behaviours matter at the console:

- **A procedure that did nothing is not a success.** The executor no longer reports success for a run in which no command was actually dispatched. A recovery that silently executed zero steps used to be indistinguishable from one that worked; it is now a failure you can see.
- **`label` and `branch` steps really execute.** The MCS runner addresses steps through a program counter rather than a fixed loop, so a `branch` can jump execution to a labelled step. Two

independent guards keep this finite: a per-arm `max_taken` bound and a whole-procedure step budget.

- **Unknown step types fail closed.** If the runner meets a step type it does not understand, it fails the procedure rather than skipping the step and carrying on. If you see a run fail on an unrecognised step, the card is wrong — the run is not.

9.4 Procedure walkthrough — CTG-003 TTC Link Loss Recovery

CTG-003 fires when `ttc.link_status` (0x0501) sits at 0 for more than 30 s after predicted AOS, or when an established link drops mid-pass. LOCKED is **2**, not 1 — a card or a display that treats 1 as "locked" is wrong. Presented below in **aircraft-QRH problem-first format**: the operator's plain- language symptom is the title, and the page tells them what to expect to see, what could be causing it, what to try, and in what order.

When I see: the link drop mid-pass

First indicators (≤ 3, in order of obviousness): 1. **`ttc.link_status` (0x0501) drops from 2 (LOCKED) to 0** — the most obvious symptom. 2. **`ttc.link_margin` (0x0503) collapses** and `ttc.rssi` (0x0502) falls away with it. 3. **`ttc.tx_gate_status` (0x052B) shows which gate opened** — bit 0 PA on, bit 1 EPS `ttc_tx` line on, bit 2 station in view, bit 3 S-band radiating. 0xDF is transmitting normally.

There is no `ttc_link_loss` FDIR rule to wait for; link loss is diagnosed from telemetry, not from a rule firing. Note also that the **UHF beacon is a separate transmitter**: losing the S-band prime does not necessarily mean the spacecraft is silent. Check `ttc.beacon_active` (0x0527) — SID 11 on the beacon can survive an S-band PA that is off.

Memory items (do BEFORE consulting this page): 1. **Verify ground station is still tracking** (look at MPS station-status panel — if NOT tracking, the spacecraft is *not* the problem). 2. **Note the current spacecraft attitude** — fast slews can drop the link briefly; if attitude is slewing, wait 10 s. 3. **Mark the time** on the loop so post-pass debrief has the exact loss moment.

Cause hypotheses (most → least common): - **RF path margin too low** — sun-shadow boundary, ground- antenna shadowing, mid-pass elevation dip. - **Spacecraft tx amplifier overheat** — `ttc.pa_temp` (0x050F) high; the `ttc_pa_overheat_pa_off` rule turns the PA off. - **Ground-station baseband re-sync** — recoverable in seconds. - **Spacecraft tx forward power dropped** — `ttc.tx_fwd_power` (0x050D) low; `ttc.pa_on` (0x0516) = 0. - **Solar particle event interference** — rare; affects polar passes during space-weather events.

Decision tree: the recovery follows the escalation ladder below.

TTC link-loss recovery escalation ladder: if link_status drops from 2, check whether the ground station is still tracking, then blind-command the primary transponder, switch to the redundant transponder, retry the primary, and verify link quality and subsystem health, escalating to EMG-004 Loss of Communication if none restores the link.

TTC link-loss recovery escalation ladder: if `link_status` drops from 2 (LOCKED), check whether the ground station is still tracking, then blind-command the primary transponder, switch to the redundant transponder, retry the primary, and verify link quality and subsystem health, escalating to EMG-004 Loss of Communication if none restores the link.

CTG-003 checklist (read-and-do)

Table 9.3 — CTG-003 TTC Link Loss recovery read-and-do checklist with verifications and escalation links.

#	Owner	Challenge	Action / TC	Expected response	On-fail link	☐	Note / actual
1	TT&C	Confirm link loss	Read `ttc.link_status` (0x0501), `ttc.rssi` (0x0502), `ttc.link_margin` (0x0503)	`link_status` = 0, margin collapsed	(none)	☐	status _ **margin** ___

#	Owner	Challenge	Action / TC	Expected response	On-fail link	☐	Note / actual
2	Ground Seg	Is it a ground issue?	MPS station-status panel; check antenna tracking, uplink power, Doppler pre-compensation, local RFI/ weather	Antenna pointing correct, baseband nominal	CTG-020	☐	_ _____
3	TT&C	Blind re-acquire on primary	`HK_REQUEST(sid=6)` (S3.27), then wait 30 s	`ttc.link_status` = 2 (LOCKED) within 30 s; `ttc.rssi` > -95 dBm	(none)	☐	status: ____
4	TT&C	Switch to redundant transponder (FD GO)	`TTC_SWITCH_REDUNDANT` (S8.1) — blind command, wait 45 s	`link_status` = 2 within 45 s; `ttc.mode` (0x0500) shows redundant active	(none)	☐	mode: ____
5	TT&C	Retry the primary	`TTC_SWITCH_PRIMARY` (S8.1) — blind command, wait 45 s	`link_status` = 2 within 45 s	EMG-004	☐	status: ____
6	TT&C	Verify link quality	Read `link_status`, `rssi`, `link_margin`, `ttc.mode`	`link_status` = 2, RSSI > -90 dBm, margin > 3.0 dB	(none)	☐	margin ____ dB
7	SPACON	Rule out a spacecraft-side cause	`HK_REQUEST(sid=1)`, `HK_REQUEST(sid=4)`, `HK_REQUEST(sid=6)` (all S3.27)	`eps.bus_voltage` (0x0105) > 30.9 V; `obdh.mode` (0x0300) not safe/ emergency	(per finding)	☐	bus ____ V
8	SPACON	Log recovery	Shift-log entry with every blind command and its timestamp	Log complete	(none)	☐	log ID: __
9	TT&C	File anomaly ticket	(ticket form)	Ticket logged with cause hypothesis	(none)	☐	ticket #: _

> **Exit criterion.** Either: (a) `ttc.link_status` = 2 (LOCKED) with margin > 3 dB; or (b) link cannot be recovered on either transponder and EMG-004 (Loss of Communication) is declared by FD.
>
> **If the link comes back but weak** (RSSI < -95 dBm or margin < 1.5 dB), step the TM rate ladder *down* to LOW (16 kbps) with `TTC_SET_DATA_RATE(rate=0)` (S8.1, func 65). LOW is the routine operations rung: it carries the full housekeeping set, it just cannot dump. The ladder is BEACON / LOW / HIGH = 1 / 16 / 128 kbps.

Ticket-items checklist (must be filled before ticket submission):

- ☐ Time of first indicator
- ☐ Time of recovery (or escalation to EMG-004)
- ☐ Step at which link was recovered (3, 4, 5, or "not recovered")
- ☐ Root-cause hypothesis (RF margin / amplifier / baseband / power / unknown)

- ☐ Recommendation for next pass

Why it's structured this way

The procedure escalates from *least intrusive* (rule out the ground segment, then a single blind housekeeping request on the transponder you already have) to *most intrusive* (a blind transponder switchover, which costs you the configuration you know and 45 s of pass time). The principle is borrowed from medical CPR: try the cheap thing first; only escalate if it doesn't work. The same principle governs most contingency procedures in this textbook.

What operators often miss

- Skipping to step 4 because *"the transponder has obviously failed"*. Steps 2 and 3 are cheap and tell you whether the problem is on the ground or on the spacecraft. Skipping them loses diagnostic information and can leave you on the redundant transponder for no reason.
- Forgetting step 9 (the ticket). Anomalies that "self-resolved during the pass" still get tickets. The accumulated tickets drive next pass's planning.

9.5 Shift handover practice — NOM-012

Source: `procedures/nominal/shift_handover.md`.

NOM-012 is the *outgoing FD briefs incoming FD* protocol. Target completion time: **6 minutes**. A longer handover usually means the outgoing FD didn't keep the shift log clean during the shift.

NOM-012 template (verbatim from procedure)

The outgoing FD reads:

1. **Phase and time.** "Spacecraft is phase 6, mission elapsed time T+ [days]. Next phase boundary: nominal, none planned."
2. **Last completed pass.** "Last pass: Iqaluit AOS T-22:45 LOS T-13:20. Nominal."
3. **Open anomalies.** List each by FDIR rule name, status (active / acknowledged / cleared), and the recovery procedure in progress. "Open anomaly: `eps_payload_load_yellow`, acknowledged, CTG-012 step 3 of 7 complete."
4. **Open work items.** Procedures begun but not finished. "NOM-002 imaging session in progress, two of three targets acquired, third deferred to next pass."
5. **Next pass.** "Next pass: Troll AOS T+05:12. Plan calls for NOM-001 startup, NOM-002 to acquire the deferred target, NOM-003 downlink, NOM-009 health check."
6. **Customer commitments.** "Customer expects Image Set 047 by end-of-day; we are tracking."
7. **GO authorisations outstanding.** "I have authorised AOCS wheel desaturation; AOCS has not yet executed. Recommend incoming FD re-confirm GO before they do."
8. **Voice-loop tone.** "Team is rested, no fatigue concerns."

Incoming FD reads back **two specific items** to confirm comprehension (typically the open anomalies and the GO authorisations outstanding).

Drill

Each team does two handovers. Times under 6 min get a tick; times over 8 min get a debrief note.

9.6 Flight-dynamics workshop

Workshop run by the FDS-seat-holder for the team. Three exercises:

1. **Predict the next three AOS times** from the current orbit. Use `tools/orbit_tools.py predict --hours 24`. Compare to MPS.
2. **Compute pass duration as a function of maximum elevation.** Use the geometric formula in §8.2.1. Verify that the next pass's duration matches the geometric prediction within 30 s.
3. **Explain eclipse.** Compute the eclipse fraction in the next 24 h. Identify the longest eclipse and its duration. Predict what battery SoC will be at exit of that eclipse using:

$$\mathrm{SoC}_{\mathrm{exit}} = \mathrm{SoC}_{\mathrm{entry}} - \frac{\mathrm{P}_{\mathrm{avg}} \cdot \mathrm{t}_{\mathrm{eclipse}}}{\mathrm{E}_{\mathrm{battery}}} \cdot 100\%$$

where $\mathrm{P}_{\mathrm{avg}}$ is the average eclipse-phase load, $\mathrm{t}_{\mathrm{eclipse}}$ is in hours, and $\mathrm{E}_{\mathrm{battery}}$ is the battery energy. EOSAT-1 flies a GomSpace NanoPower BPX 8S-1P, so $\mathrm{E}_{\mathrm{battery}} = 100$ Wh, and the orbit-average load at the *nominal* profile is 22.7 W (`tools/power_budget.py` , sim manual 01_eps §3.2: payload STANDBY with the FPA cooler off; 19.7 W quiescent, 26.4 W with imaging duty-cycled in). Use ~23 W unless the pass plan says otherwise.

Worked, for a ~35 min eclipse: $\mathrm{SoC}_{\mathrm{exit}} = 85\% - \frac{23 \cdot 0.58}{100} \cdot 100\% = 85\% - 13.3\% = 71.7\%$.

Cross-check the answer against `tools/power_budget.py` , which is the same model the closure criterion is judged on.

9.7 Dry-run 1 — nominal pass, no failures

- **Scenario:** BP-1 `nominal_ops.yaml` .
- **Duration:** 90 min real time at 1× sim speed.
- **Procedures expected:** NOM-001 → NOM-002 → NOM-003 → NOM-009 → LOS.
- **Failure injection:** none.
- **Outputs:** operator shift log per position; observer team uses the *Communications* peer-review worksheet only.

9.8 Dry-run 2 — nominal pass with one failure

- **Scenario:** BP-1 plus injection of `eps_overcurrent.yaml` (line = `payload`) at T+30 min into the pass.
- **Expected response:** CTG-012 *Overcurrent Response*.
- **Pass criterion:** anomaly detected within 30 s, isolated within 90 s, CTG-012 executed in full, pass continues to LOS.
- **Outputs:** operator shift log + anomaly ticket; observer team uses the *Technical* peer-review worksheet.

9.9 Voice Loop Master Class (extended)

This section goes beyond the fundamentals in §9.2. The basics will get a team through nominal passes; the master-class material is what carries a team through a hot anomaly with composure intact.

Why voice-loop discipline beats raw skill

As §9.1 noted, voice-loop failure — not telemetry misreads — is the dominant cause of recoverable anomalies becoming unrecoverable. Operators who can interpret telemetry but can't run the loop are dangerous; operators who can run the loop but are slow on telemetry can be coached. Spend the time on the loop.

Three patterns to internalise

Pattern 1: Address-Sender-Message

Always: `[ADDRESSEE], [SENDER] — [MESSAGE].`

Why: the addressee's brain primes on hearing its callsign at the front. By the time the message arrives, they're listening rather than parsing.

Anti-pattern: `"Hey, can you check the battery voltage?"` — nobody knows who's asking, nobody knows who's supposed to answer.

Pattern 2: Standby — Working — Done

Never go silent during an active action. The loop expects to hear at intervals of < 30 seconds. If you're busy:

- `"Standby."` — I'm working it; will respond.
- `"Standby plus 30."` — I'll be back in 30 seconds.
- `"Working it."` — Mid-action.
- `"Done."` — Action complete (always paired with what was done).

Silence is the loop's worst signal. Silence means "I might be gone" or "I haven't heard you" or "I'm confused" — three different things, all bad.

Pattern 3: Read-Back on Critical

For any commanded action that touches spacecraft state, the owner reads back the FD's GO. Pattern:

- FD: `"AOCS, FD — go for ST1_POWER on, then mode 4 NOMINAL."`
- AOCS: `"FD, AOCS — going for ST1_POWER on, then mode 4 NOMINAL."` (then execute)

This is the same protocol airliners use for ATC clearances. It catches mis-hearings, mis-rememberings, and mis-clicks. It costs 1.5 seconds and saves missions.

The callout vocabulary — hot-anomaly extension

(Basic vocabulary in §9.2; here we add the hot-anomaly variants.)

Table 9.4 — Extended hot-anomaly voice-loop callouts with meaning and operational use case.

Callout	Meaning	Use case
"Anomaly, [parameter], [value]"	Out-of-band telemetry	Always say the parameter and value; the FD needs both
"FDIR active, [rule]"	An FDIR rule triggered	SPACON's job; everyone listens for the rule name
"Recovery procedure, [ID]"	Invoking a contingency or emergency procedure	SPACON declares which
"Safe mode candidate"	Anomaly is severity 4	FD's call to actually enter safe mode

Callout	Meaning	Use case
"GO for safe mode"	FD authorises safe-mode entry	Said only by FD; requires Mission Manager acknowledgement
"Loss of [system]"	Subsystem appears to have failed	TT&C for link loss, EPS for power loss, etc.
"Recovered, [system]"	The failure passed	Counterpart to "loss of"
"Holding, [reason]"	Pausing the procedure	Anyone can call HOLD; FD authorises resume
"Continuing, on [step]"	Resuming after hold	SPACON resumes procedure
"Mark, [event]"	Stamp the time + event into the log	Standard event-logging callout
"Working an anomaly"	I'm focused, brief me only if critical	Tells the loop the speaker is in deep focus
"Clear to talk"	I'm available again	Counterpart to "working an anomaly"
"Confidence?"	Asks for an operator's confidence in their reading	FD's tool when telemetry is suspect
"[N] percent / [low/med/high]"	Confidence answer	Standard form

Example loop transcript — anomaly response

The following is a model transcript for a CTG-012 overcurrent response during COM-009. Read this aloud as a team to feel the rhythm.

```
T+24:18  EPS: "FD, EPS — anomaly, payload line current,
              0.92 A trending up. FDIR alert imminent."
T+24:20  Event log: 0x010F PDU_OVERCURRENT (severity HIGH,
                    subsystem EPS) — the EPS trips the payload
                    line and latches its overcurrent flag.
T+24:21  SPACON: "FD, SPACON — event 0x010F PDU overcurrent trip,
                  payload line, severity HIGH. Line opened
                  on board, OC flag latched."
T+24:22  EPS: "FD, EPS — confirming, payload line current 0 A,
              bus stable at 32.1 V. Line is OFF."
T+24:23  FD: "Copy all. SPACON, recovery procedure?"
T+24:24  SPACON: "FD, SPACON — recovery is CTG-012 Overcurrent
                  Response, six steps. The on-board trip did the
                  isolation for us. Step 1 identify — payload
                  line, confirmed. Step 2 per-line current —
                  `eps.line_current_3` reads 0 A. Step 3
                  isolate — Payload confirms no demand."
T+24:26  FD: "Steps 4 and 5 — reset the OC flag and re-enable —
              are next-pass work, not this shift. We are not
              re-energising a line we haven't diagnosed.
              Payload, file the ticket."
T+24:28  Payload: "FD, Payload — going for ticket. Working it."
T+24:35  Payload: "FD, Payload — ticket 003 filed. Root cause
```

```
                  hypothesis: cold-start in-rush exceeded the
                  trip threshold; cooler-side draw spike unlikely.
                  Recommend re-attempt next pass after thermal
                  stabilisation."
T+24:38  Mission Manager: "FD, MM — customer bulletin?"
T+24:39  FD: "MM, FD — yes, draft a bulletin: payload power-on
             deferred to next pass; FPA cool-down delayed by
             one pass; First Light projection shifts from end
             of Pass 4 to start of Pass 5. Standby for FD
             edit."
T+24:42  MM: "FD, MM — drafting, will read for your edit at
             plus 5 minutes."
T+24:43  FD: "All positions — we're holding COM-009 retry until
             next AOS. Until then, run COM-007 OBDH checkout
             early; that's productive work that doesn't depend
             on payload. SPACON, lead. Pass continues."
T+24:45  SPACON: "FD, SPACON — going for COM-007. Loading procedure.
                 OBDH, ready?"
```

Note the rhythm: an indicator at +18, an FDIR alert two seconds later, recognition four seconds after that, a recovery decision in another four seconds. From first indicator to "we're holding COM-009" was 25 seconds. That's competent.

Common loop failures (with corrections)

Failure: Cross-talk

Two people start talking simultaneously. Standard correction is the FD calling `"Loop discipline."` — everyone stops, the FD names who should speak first, that person speaks, then the other.

If the cross-talk is *between FD and an operator*, the FD defers to the operator (the operator usually has the time- critical content; the FD's question can repeat). If between two operators, the FD picks one.

Failure: The Silent FD

The FD doesn't speak for 30+ seconds during an active anomaly. Operators stop knowing whether to act. Correction: the SPACON calls `"FD, status?"` and the FD must respond within 5 seconds with at least *"FD, standby."*

Failure: The Over-driving FD

The FD calls actions rather than asking operators for their assessment. Symptoms: `"AOCS, do X"` rather than `"AOCS, FD — recommend?"`. Operators feel demoted; engagement drops; the FD becomes a single point of cognitive failure.

Correction: the Mission Director intervenes mid- shift, calls a 30-second hold, reminds the FD to *ask, don't tell*.

Failure: Dropped read-back

FD says `"GO for ST1_POWER on"`, AOCS executes without read-back. If the wrong TC was just sent, nobody catches it until the verification fails.

Correction: the Mission Director intervenes immediately. Operator does not execute again without read-back.

Failure: The acronym storm

Operators string together so many acronyms that nobody knows what was said. Symptoms: `"FD, AOCS — STR at FAULT, gyros in DRA, resetting EST for ADQS."` — said at speed, in initials, with no parameter and no command named.

Correction: at the next planning meeting, the team agrees a shorthand library. New operators get briefed before joining the loop.

What to practice

Pair drills, 15 minutes each, before any high-stakes shift:

1. **Read-back drill.** One person plays FD, gives a series of commands. Other read each back.
2. **Loop-discipline drill.** Two people deliberately cross-talk on a scripted scenario; a third calls `"Loop discipline"` and restarts.
3. **Anomaly callout drill.** One person reads a telemetry value; the other has to compose the callout in <5 seconds.
4. **HOLD-and-resume drill.** Run a 90-second procedure with a surprise HOLD call halfway. Practice the resume.

9.10 References

- ECSS-E-ST-70-32C — Test and operations procedures — ecss.nl
- NASA APPEL voice-loop and mission-ops materials — appel.nasa.gov
- FAA AC 120-71 — Standard Operating Procedures — www.faa.gov
- FAA Crew Resource Management AC 120-51E — www.faa.gov

Chapter 10 — LEOP Operations

This chapter covers operations from launcher separation through the end of the **Launch and Early Orbit Phase (LEOP)** — the period where the spacecraft transitions from phase 3 (BOOTLOADER_OPS, just the OBC and the receiver energised and the bootloader image running) through phase 4 (LEOP) and out the far side into phase 5 (COMMISSIONING): the solar wings deployed, the AOCS suite powered, detumble converged into the COARSE_SUN sun hold, time synced and the orbit determined. The payload, the cooler, the wheels and the star trackers stay off; they are commissioning's. Note that the **S-band antenna** deployment is *not* a LEOP step — it is COM-006 step 0 (`TTC_DEPLOY_ANTENNA` , S8.1 func 69), because the LOW rung of the rate ladder does not require it and LEOP never needs HIGH. The phase codes are `eps.spacecraft_phase` (0x0129): 0 pre-separation, 1 separation timer, 2 initial power-on, 3 bootloader ops, 4 LEOP, 5 commissioning, 6 nominal. The final declaration of phase 5 is made at the end of LEOP-003, the last LEOP card, with the flight telecommand `SYS_SET_MISSION_PHASE(phase=5)` (S8 func 120); nothing on board advances the phase past 4 by itself.

LEOP is the highest-stakes phase of the mission. It is the *first* time everything must work, with no opportunity to dry-run. A real LEOP runs **24/7 with three eight-hour shifts**; this textbook covers a compressed presentation suitable for training where the LEOP sequence is split across a sequence of ground-station passes.

10.1 LEOP phase entry conditions

- **Sim checkpoint:** BP-2 *LEOP Day Start* (`scenarios/first_contact.yaml`).
- **Phase at start:** 3 (BOOTLOADER_OPS).
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- **`override_passes`** flag set so visibility holds for the pass.
- **Speed multiplier:** 1×.

10.3 Mission phase brief

The spacecraft has just exited the separation timer (phase 2, initial power-on) and is at phase 3, BOOTLOADER_OPS. The OBC is running the *bootloader*, which emits **SID 11 (Beacon) only** — the enabled-at-power-up set is `default_enabled_sids: [1, 11]` , and in the bootloader only the beacon structure is produced. Neither the S-band antenna nor the solar wings are deployed: `ttc.antenna_deployed` (0x0520) = 0 and `eps.wing_status` (0x0144) = 0x00. The cold-boot line state is: `obc` and `ttc_rx` ON (non-switchable); `ttc_tx` , `htr_bat` , `htr_obc` , `aocs_wheels` , `payload` , `fpa_cooler` and `prop` all **OFF**. The spacecraft is therefore a one-way listener — dark on the downlink until the first accepted telecommand arms the transmitter.

AOCS is in **mode 0 (OFF)** — not detumbling, not even ticking — with residual tumble rates of roughly 1–2 °/s from the separation impulse. Detumble has *not* happened yet; it is commanded in LEOP-007 step 5. Battery SoC is 90 % (`simulation.yaml` `eps.initial.soc_pct`). The first Iqaluit AOS occurs at T+0 in the scenario.

The team's goal across the shift is to drive the spacecraft from phase 3 (bootloader) through phase 4 (LEOP) to the point where the mission management team can declare phase 5 (COMMISSIONING).

10.4 Procedure execution order

The seven LEOP procedures, in execution order, with predicted duration and lead position:

Table 10.1 — LEOP procedure execution order with lead position, duration, and critical step.

#	Procedure	Lead position	Duration (sim)	Critical step
1	LEOP-001 First Acquisition of Signal & OBC Boot	TT&C + SPACON	~6 min	Step 5: `OBC_BOOT_APP`, then step 5a: clear beacon mode
2	LEOP-002 Initial Health Check	SPACON + all positions	~8 min	SIDs 1, 3, 4 and 6 in band; the AOCS, payload, cooler and prop lines confirmed OFF
3	LEOP-004 Solar Array Verification	EPS	~10 min	Step 0: `EPS_DEPLOY_WING(wing=2)`; step 6: wing symmetry
4	LEOP-007 Sequential Power-On	SPACON + EPS + AOCS	one 8-min contact	Steps 3–5: AOCS suite line ON and SID 2 enabled; magnetometer, magnetorquers and GPS on; mode → 2 DETUMBLE
5	LEOP-005 Sun Acquisition	AOCS	detumble-limited	Step 3: the automatic handover to 3 COARSE_SUN — observed, not commanded
6	LEOP-006 Time Synchronisation	SPACON	~5 min	Step 4: `SET_TIME`, residual < 1.0 s; step 6a: SID 12 and GPS discipline
7	LEOP-003 Initial Orbit Determination	FDS + FD	~15 min over two fixes	Step 7: `SYS_SET_MISSION_PHASE(phase=5, flags=0)` — the end of LEOP

Note the ordering. It is the one in `procedures/procedure_index.yaml` and in each card's entry and exit states. The wings are deployed (LEOP-004) straight after the health check, before anything else draws power. LEOP-007 then powers only what the rest of LEOP needs — heaters if their zones need heat, the AOCS suite line, the magnetometer, the magnetorquers and the GPS receiver — and commands DETUMBLE. The payload line and the FPA cooler are commissioning loads (COM-009, COM-010); the reaction wheels stay DISABLED until COM-004, and the star trackers and gyros stay off until COM-003. LEOP-005 commands nothing: the AOCS promotes itself from DETUMBLE to COARSE_SUN and, with no star tracker up, parks there. LEOP-006 sets the clock from the ground first and only then disciplines it to GPS; LEOP-001 deliberately does **not** set the clock (its step 6 only notes that it is unset).

LEOP-003 is a console procedure, not an autonomous one. It reads the GPS solution, hands the refined TLE to the ground tools — nothing is uploaded to the spacecraft — and ends LEOP with the phase-5 declaration, the one telecommand that moves the spacecraft out of phase 4. LEOP-004 is **Solar Array Verification**, not "Solar Array Deployment", and it does *not* describe something that already happened: at cold boot both wings are stowed (`eps.wing_status` 0x0144 = 0x00), and the card's step 0 actually commands `EPS_DEPLOY_WING(wing=2)`. Budget the time for it.

10.5 Per-position playbook for LEOP

Flight Director

- Pre-pass: confirm both teams' position assignments. Run a comm check on the loop.
- Pass: poll for GO on every commanded action. Watch the phase indicator. Authorise progression LEOP-001 → -002 → -004 → -007 → -005 → -006 → -003.
- Pass criterion: at the end of LEOP both wings read deployed on `eps.wing_status` (0x0144), the AOCS holds the sun in mode 3 on the magnetorquers, the clock is referenced, the orbit is determined, and the phase reads 5 after a unanimous poll and the `SYS_SET_MISSION_PHASE(phase=5, flags=0)` telecommand (LEOP-003 step 7). Only the LEOP loads are on: the payload, cooler and prop lines are OFF and the wheels are disabled.
- Pitfalls: rushing into LEOP-007 before LEOP-002 confirms health. Granting GO for a COARSE_SUN command at all — the handover is automatic, set by the physics of the B-dot law, and the mode machine promotes itself when the rates are genuinely down. Waiting for the phase to advance by itself: nothing on board moves it past 4.

SPACON

- Pre-pass: load the LEOP procedure pack into the procedure panel. There is **no LEOP FDIR table and no nominal FDIR table** to swap between: the simulator arms every rule all the time, and individual event-action definitions are turned on and off one at a time with `EA_ENABLE` (S19.4) and `EA_DISABLE` (S19.5). If a rule is going to be inhibited for a step, inhibit that rule by `ea_id` and write the id in the shift log so it gets re-enabled.
- Pass: drive procedure execution. Watch the image transition in `obdh.sw_image` (**0x0311**): 0 = bootloader, 1 = application. There is no `obdh.obc_mode` parameter and no watchdog-kick counter; the reboot evidence is `obdh.uptime` (0x0308), `obdh.reboot_count` (0x030A) and `obdh.last_reboot_cause` (0x0316).
- Critical step: `OBC_BOOT_APP` (LEOP-001 step 5). The OBDH verifies the application CRC over a 10 s window before `sw_image` transitions, so a short telemetry gap there is normal. What is **not** automatic is the housekeeping: application boot re-applies `default_enabled_sids: [1, 11]`, so after the boot you have SID 1 and SID 11 and nothing else until you send `HK_ENABLE` (S3.5) for each of 3, 4 and 6 — then 2 in LEOP-007 and 12 in LEOP-006 (SID 5 waits for the payload line: COM-009 requests it).
- Pitfalls: declaring LEOP-001 complete without the `HK_ENABLE` s. Forgetting step 5a — booting the application does **not** clear beacon mode, and while `ttc.beacon_mode` (0x0521) is 1 every `TTC_SET_DATA_RATE` is refused with S1.8 error `0x0007`.

AOCS

- Pre-pass: review residual tumble rates from the scenario brief (~1–2 °/s, partially decayed from separation). Note that the AOCS model is not even ticking: the line is off and the mode is 0 (OFF).
- Pass: passive until LEOP-007 step 3. Then take the lead: `HK_ENABLE(sid=2)` (S3.5) so that you can see anything at all; at step 4 enable the two DETUMBLE legs — `S2_AOCS_MAGNETOMETER(device_id=521, state=1)` and `MTQ_ENABLE(enable=1)` (S8.1 **func 9**) — and the GPS receiver; at step 5 send `AOCS_SET_MODE(mode=2)` — DETUMBLE — with S8.1 **func 0**, which is exempt from the subsystem mode gate and is therefore the command that boots AOCS out of OFF. The mode command refuses honestly if a leg is missing: `0x0A20` no magnetometer, `0x0A21` no magnetorquers, `0x0004` line off.

- Critical step: LEOP-007 step 5 and the wait that follows it, which is LEOP-005. Detumble is a magnetometer-driven B-dot law, and it is slow (see §10.11). COARSE_SUN is not entered by hand: the mode machine promotes DETUMBLE → COARSE_SUN itself once the rate magnitude has been below 0.5 °/s for 30 consecutive seconds. With no star tracker powered it stays there until commissioning; the wheels stay disabled until COM-004.
- Pitfalls: commanding mode 3 at all. Reading a per-axis rate *increase* as a broken control law — B-dot damps only the rate component perpendicular to the local field, so energy genuinely moves between axes while the total magnitude falls. Judge the law on the RSS of the three rates, never on one axis. Magnetorquer state is in `aocs.mtq_enabled` (0x025E), duties at 0x0258–0x025A.

EPS

- Pre-pass: SoC 90 %; `obc` and `ttc_rx` on and non-switchable; every switchable line OFF. There are **nine** power lines, indices 0–8: 0 obc, 1 ttc_rx, 2 ttc_tx, 3 payload, 4 fpa_cooler, 5 htr_bat, 6 htr_obc, 7 aocs_wheels, 8 prop. `EPS_POWER_ON` / `EPS_POWER_OFF` / `EPS_RESET_OC_FLAG` all take a `line_index` in 0–8.
- Pass: lead LEOP-004 first — each wing draws ~2 A for 30 s while it deploys. In LEOP-007, watch each new line's current draw (`eps.line_current_0` … `_7` , 0x0118–0x011F; line 8, prop, is `eps.line_current_8` , 0x014B) and the line flag (`eps.pl_*` , 0x0110–0x0117). Each new line is its own small commanding event with FD GO. Note that EPS bills from the actual actuator state, not from the line switch position — a heater line that is on but thermostatted off does not draw its rated power.
- Critical: LEOP-007 step 2 is conditional. `htr_bat` (`EPS_POWER_ON(line_index=5)` , S8.1 **func 19**) goes on only if `tcs.temp_battery` is below +5 °C, and `htr_obc` (`line_index=6`) only if `tcs.temp_obc` is below −10 °C (the OBC heater ON point); log which lines were energised, because COM-002 reads that record. Step 6 confirms that the payload, cooler and prop lines are still OFF and that the margin is positive.
- Pitfalls: not catching an unexpected current spike when a heater first energises (the inrush is normal; sustained high draw is not).

Thermal

- Pre-pass: all zones passive (no heaters on); battery zone may be at lower end of band.
- Pass: give EPS the battery and OBC zone temperatures before LEOP-007 step 2 — they decide whether either heater line goes on. If one does, watch its zone's temperature trend and the heater's duty cycle.
- Pitfalls: not noticing that the battery zone takes ~15 minutes to reach band after `htr_bat` first enables. Pre-judging the battery heater as "stuck off".

TT&C

- Pre-pass: confirm beacon reception at the station in view, and confirm the demodulator is on the **BEACON** rung — 1 000 bps BPSK, conv(7,1/2)+RS, 1 kBd. That is the only rate the bootloader transmits.
- Pass: lead LEOP-001 (link acquisition). At step 5a, clear beacon mode with `TTC_SET_BEACON_MODE(on=0)` (S8.1, func 70), then select the **LOW** rung — 16 000 bps, 16 kBd — with `TTC_SET_DATA_RATE(rate=0)` (S8.1, func 65). LOW is the routine operations rate; HIGH (128 kbps) is a dump rate and is not entered in LEOP. Then passive for the rest of the pass.

- Pitfalls: expecting HIGH to be selectable. The gates are, in order: beacon mode off → LOW selectable; antenna deployed (`TTC_DEPLOY_ANTENNA` , func 69) **and** downlink mode on (func 79) **and** elevation ≥ 10° → HIGH selectable. Every unmet gate returns S1.8 `0x0007` with a failure notice that names the gate in words — read the notice, not the code. Note also that clearing beacon *mode* is not the same thing as turning off the UHF beacon transmitter: the beacon is a **separate** transmitter (`TTC_BEACON_TX_ON/OFF` , funcs 97/98), and it keeps radiating SID 11 even when the S-band PA is off.

Payload

- Pre-pass: payload line OFF, payload mode OFF, FPA cooler line OFF.
- Pass: passive for the whole of LEOP. LEOP-007 does not touch the payload: the payload line, STANDBY with `PAYLOAD_SET_MODE(mode=1)` (S8.1, func 26) — not `SET_PAYLOAD_MODE` , which the catalogue maps to a different command — and SID 5 come in COM-009, and the cooler line and `FPA_COOLER(on=1)` (func 43) in COM-010. LEOP-007 step 6 confirms that the payload and cooler lines are still OFF.
- Pitfalls: expecting the FPA to reach an arbitrary cold temperature. The commandable set-point envelope is **−20 °C to 0 °C**, default −15 °C, and readiness (`payload.fpa_ready` , 0x0610) follows the *active* set point, not a fixed number. Also: distracted, not on the loop, missing being polled in a system-wide GO/NO-GO round.

FDS

- Pre-pass: produce the 24-hour pass schedule. Hand to FD, TT&C, Ground Segment.
- Pass: passive until the GPS receiver is switched on in LEOP-007 step 4 (`S2_AOCS_GPS_RECEIVER` , on the AOCS suite line); a cold start can take up to 15 min. SID 12 is enabled in LEOP-006. Then lead LEOP-003: verify the fix, read the state vector, compare with the launcher orbit and send the refined TLE to the ground tools.
- Pitfalls: using the launch-vehicle-provided TLE rather than the fresh GPS-derived state vector once GPS is live. Looking for a command to upload the orbit on board — there is none.

Ground Segment

- Pre-pass: verify Iqaluit antenna is on-target. Verify baseband is in the bootloader-beacon configuration (low-rate, narrow bandwidth).
- Pass: lead LEOP-001 step 1 (confirm antenna tracking, arm Pass Override). At LEOP-001 step **5a**, reconfigure the demodulator from 1 kBd to **16 kBd** *before* expecting the subsystem SIDs — the beacon demodulator will not lock 16 kbps.
- Pitfalls: missing the baseband reconfiguration cue; the team ends up with the spacecraft on the LOW rung and the ground still on the beacon rung, and reads the resulting silence as a spacecraft fault.

Mission Manager

- Pre-pass: brief any external watchers (customer reps observing the LEOP sim).
- Pass: passive listener. Notes any item for the customer bulletin.

The Mission Director picks **one** primary injection per team, plus one optional rolling secondary if the primary is resolved cleanly.

Table 10.2 — LEOP anomaly-injection scenarios by severity with first indicators and recovery procedures.

Severity	Scenario file	What students see first	Recovery procedure
Easy	`ttc_no_tm_at_aos.yaml`	Beacon present, full HK never reports despite OBC_BOOT_APP	CTG-019 No TM at Pass Start
Easy	`eps_overcurrent.yaml`	Sudden overcurrent on a power line during sequential power-on	CTG-012 Overcurrent Response
Medium	`aocs_actuator_stuck.yaml`	RW-1 (index 0) seizes at T+120 s — speed frozen and unresponsive to torque commands, wheel temperature rising	CTG-026 AOCS Actuator Stuck Recovery (CTG-007 Reaction Wheel Anomaly for the general case)
Medium	`aocs_star_tracker_failure.yaml`	Star tracker degrades after sun acquisition	CTG-008 Star Tracker Failure
Hard	`obc_watchdog.yaml`	OBC watchdog reset mid-deploy	CTG-010 OBDH Watchdog Recovery
Hard	`obc_bus_failure.yaml`	CAN bus A drops during sequential power-on	CTG-017 Bus Failure Switchover
Very hard	`gs_antenna_failure.yaml`	Iqaluit antenna degraded mid-pass; failover to Troll	CTG-020 Ground Station Antenna Failure

Each injection has a one-page cue card in Chapter 14 (Procedure Catalogue).

10.7 End-of-pass GO/NO-GO criteria (review)

Table 10.3 — End-of-pass go/no-go criteria scored across the two training teams.

Criterion	Pass 1 (Team A)	Pass 2 (Team B)
OBC application SW running (`obdh.sw_image` 0x0311 = 1)	✓	✓
Beacon mode cleared (`ttc.beacon_mode` 0x0521 = 0), downlink at LOW (`ttc.tm_tier` 0x0539 = 1)	✓	✓
Onboard time set, residual < 1.0 s	✓	✓
SIDs 1–6 enabled and emitting	✓	✓
Both solar wings deployed (`eps.wing_status` 0x0144 = 0x03)	✓	✓
Battery heater line ON (`eps.pl_htr_bat` 0x0115 = 1)	✓	✓
`aocs_wheels` line ON, AOCS in mode 2 DETUMBLE and rates falling	✓	✓
Rate magnitude below the 0.5 °/s exit guard; mode has promoted itself to 3 COARSE_SUN	optional	✓
Anomaly ticket filed for any injection	✓	✓

The Mission Director sits behind the FD seat and coaches by exception only:

- **Voice-loop discipline.** If the team starts cross-talking, stop on loop, name the rule, restart.
- **Procedure compliance.** If a position commands without FD GO, stop, log it, treat as a safety violation.
- **Anomaly response.** If the team is stuck > 5 minutes, drop a hint at the FDIR-panel level (do not name the procedure).
- **Time pressure.** Real-time clock — no pausing the sim for discussion. Pass closes when the orbit closes.

10.9 Debrief structure

1. **Hot-wash.** Each team's FD reads shift-log highlights. No critique yet.
2. **Anomaly walkthrough.** Operator team walks through the anomaly response; observer team challenges.
3. **Peer-review readout.** Each observer team reads its worksheet aloud (Communications + Technical). Operator team responds with one accept and one defend per axis.
4. **Lessons-learned synthesis.** The Mission Director consolidates into a single list. Logged in cohort lessons-learned register.
5. **Brief.** Reading: COM-001 through COM-008 procedure cards.

10.10 Red lines

- **No commanding without FD authorisation.** Violation → the Mission Director calls the shift, team logs the violation, no penalty if learnt-from.
- **No simulator-pause to win an exercise.** The clock runs.
- **No silent FD.** The FD must be on the loop. If over-driven, the Mission Director calls a 30 s hold and resets the position.

10.11 Detailed LEOP Procedure Walkthroughs

This section expands each LEOP procedure into a minute-by-minute, per-position playbook for the LEOP phase. The card-format references in Chapter 14 are for at-console reference; this chapter is the *training* material the team reads on Tuesday evening before LEOP day.

For each procedure we cover:

- **Pre-step setup** — what the team should have ready before starting.
- **Per-step playbook** — every step, with the FD prompt, the owner's expected callout, the verification telemetry, and the *why* (the physics or operational reason the step exists).
- **Common deviations** — what really goes wrong and how to recover without abandoning the procedure.
- **Decision points** — places where the FD has a genuine choice to make.
- **Debrief prompts** — what to look at in post-shift review.

LEOP-001 — First Acquisition of Signal & OBC Boot (deep walkthrough)

LEOP-001 is the most consequential 5–12 minutes of the entire mission. The team has *one* Iqaluit window to take the spacecraft from launcher-separated-and-tumbling to OBC-application-running- with-full-HK.

Miss the window, and the next chance is the next Iqaluit pass (90 minutes later, with a degraded EPS picture and a nervous customer).

Pre-step setup (before AOS-5 min)

The Mission Director confirms with the team:

1. **Position assignments confirmed.** TT&C is the lead operator for this procedure; SPACON drives the procedure panel; FD authorises every commanded action; Ground Segment owns the antenna and baseband.
2. **Procedure loaded.** `procedures/leop/first_acquisition.md` is the active procedure in the panel.
3. **Comm check completed.** All nine positions are on the loop, readable.
4. **Pre-pass briefing.** FD reads the LEOP pre-pass checklist (Workbook WB-5.5-PB) aloud. Each position confirms ready.
5. **Voice-loop language declared.** Callouts in English (standard phraseology, whatever the crew's language). Recorded in shift log.
6. **Anomaly ticket forms blank-printed at each console.**
7. **control room room set up.** FD has line-of-sight to TT&C and SPACON (the two operators most active in this procedure).

At AOS-2 min, FD calls *"Two minutes to AOS. All positions, report ready."* Each position responds in order: *"FD, [POSITION] — ready."* Any *not ready* call here triggers an immediate FD decision: proceed or hold.

Per-step playbook

Step 1 — Configure Ground Station Receiver and Force Pass Override

Table 10.4 — LEOP-001 step 1 detail: ground-station receiver configuration and pass override.

Element	Detail
FD prompt	"Ground Segment, FD — go for GS configure and pass override."
Owner	Ground Segment Engineer
Owner expected callout	"FD, Ground Seg — go for GS configure and pass override." → execute → "FD, Ground Seg — Iqaluit tracking ready, pass override armed."
Action	In MPS: command Iqaluit antenna to predicted AOS az/el; enable auto-tracking on S-band downlink; set the demodulator to the **BEACON** rung (1 000 bps BPSK, conv(7,1/2)+RS, 1 kBd) — the only rate the bootloader transmits. In MCS: enable Pass Override (`pass_override = 1` , parameter 0x05FF).
Verification	GS panel shows TRACKING_READY; MCS top header shows `Pass Override: ARMED` .
Why this step	The spacecraft is silent (no carrier to track). Auto-tracking can't lock onto nothing — we must point the antenna ourselves using the predicted ephemeris. Pass override tells the MCS "treat this as in-contact regardless of orbit math" so we can stage TCs before the spacecraft actually replies.
FD goes / no-goes on	Antenna actually pointing — if the slew is mid-flight the antenna won't be settled yet; wait.

Step 2 — Send First Uplink (Connection Test)

Table 10.5 — LEOP-001 step 2 detail: first uplink via the S17 connection test.

Element	Detail
FD prompt	"TT&C, FD — go for first uplink, S17 connection test."
Owner	TT&C Engineer
Owner expected callout	"FD, TT&C — go for first uplink, S17.1." → execute → "FD, TT&C — S17 sent, awaiting S1 ack and S17.2 report."
Action	In commanding panel: service S17, subservice 1, no payload. First-key. (No second-key required for S17 — it's connectivity test, low-risk.)
Verification	Within 10 s: S1.1 acceptance ack received; within 10 s: S17.2 connection-test report received; `eps.pl_ttc_tx` (**0x0112**) reports 1 — the transmitter is auto-armed by the hold-down timer.
Why this step	We have no idea yet whether the link works end-to-end. S17 is the cheapest, lowest-impact way to find out, and it is on the bootloader allow-list, so it is accepted before the application image is loaded. The acceptance ack confirms ground-to-spacecraft uplink; the connection-test reply confirms spacecraft-to-ground downlink. The **auto-TX hold-down** is the mechanism that makes the cold-boot downlink appear at all: any accepted TC autonomously powers the `ttc_tx` line ON for `simulation.yaml ttc.auto_tx_hold_s` , which EOSAT-1 sets to **300 s (5 minutes)**, re-armed on every subsequent accepted TC. Go five minutes without commanding and the transmitter drops; send any TC to bring it back.
Common deviation	S1 ack received but no S17.2 — uplink works but downlink isn't decoded. Go to Step 3 (carrier-lock check) before declaring failure.

Step 3 — Verify Carrier Lock and Link Quality

Table 10.6 — LEOP-001 step 3 detail: carrier-lock verification and link-quality readings.

Element	Detail
FD prompt	"TT&C, FD — report link quality."
Owner	TT&C
Owner expected callout	"FD, TT&C — link status 2, LOCKED, RSSI minus 102 dBm, margin 5.8 dB, nominal."
Action	Read `ttc.link_status` (0x0501), `ttc.rssi` (0x0502), `ttc.link_margin` (0x0503). Record the measured Doppler offset for Flight Dynamics.
Verification	Within 30 s of step 2: `ttc.link_status` = **2** (LOCKED); `rssi > -110 dBm` ; `link_margin > 3.0 dB` . LOCKED is 2 — 1 is not lock, and a display or a card that says otherwise is wrong.
Why this step	The link could be marginal — a 1 dB margin might work today but degrade next pass. We log baselines now so we have something to compare against if the link degrades later in the mission.
Common deviation	RSSI -111 dBm, margin 2 dB: pass margin gate but log a *low-margin caution* in the shift log. FD makes the call whether to continue at this margin or hold for next pass.

Step 4 — Confirm OBC is in Bootloader

Table 10.7 — LEOP-001 step 4 detail: confirming the OBC is still in bootloader via SID 11.

Element	Detail
FD prompt	"SPACON, FD — go for beacon SID request."
Owner	SPACON
Owner expected callout	"FD, SPACON — go for HK request SID 11." → execute → "FD, SPACON — beacon received, OBC in bootloader (sw_image 0)."
Action	`HK_REQUEST(sid=11)` — **S3.27**. (`HK_ENABLE` is S3.5; neither is S3.1 or S3.25.)
Verification	Beacon packet received; `obdh.sw_image` (**0x0311**) = 0 (BOOTLOADER) within 10 s.
Why this step	We need to confirm the OBC is actually in bootloader mode and not in some unexpected state (e.g. partial app boot from a previous test). SID 11 is the only structure the bootloader emits, and it carries `obdh.sw_image` , so it is both the proof of life and the proof of image. No subsystem SID is emitted periodically yet — and that is expected, not a fault.
Common deviation	`sw_image = 1` already — the OBC has already booted app. This means either the bootloader auto-progressed (unusual; usually requires explicit boot command) or the last test left the OBC in app state. Skip Step 5; verify full HK with Step 5's verification telemetry instead.

Step 5 — Boot OBC into Application Image (the *critical step*)

Table 10.8 — LEOP-001 step 5 detail: dual-keyed OBC application boot with FD second-key.

Element	Detail
FD prompt	"ALL POSITIONS, FD — going for OBC application boot. SPACON, ready?"
	"FD, SPACON — ready."
	"EPS, ready?" → "FD, EPS — ready, bus 32.5 V, SoC 89%."
	"AOCS, ready?" → "FD, AOCS — ready, no commands pending."
	"TT&C, ready?" → "FD, TT&C — ready, link nominal."
	"GO for OBC application boot."
Owner	SPACON (with FD second-key)
Owner expected callout	"FD, SPACON — first-key. FD second-key. Sending OBC_BOOT_APP." → execute → (10 s pause) "FD, SPACON — sw_image 1, application running, SIDs 1 and 11 only. Enabling 3, 4 and 6."
Action	`OBC_BOOT_APP` — Service S8, subtype 1, **func_id 55**. (There is no scheme by which the S8 *subtype* names the subsystem; the subtype is 1 and the func_id identifies the function.) High-criticality TC — dual-key: SPACON first-keys, FD second-keys. Then `HK_ENABLE(sid=3)` , `HK_ENABLE(sid=4)` , `HK_ENABLE(sid=6)` , all **S3.5**.
Verification	Within 5 s: S1.1 ack. S1.7 execution complete follows the ~10 s CRC check. Within 15 s: `obdh.sw_image` (0x0311) = 1 (APPLICATION). Then **SID 1 (EPS) and SID 11 (Beacon) — and only those two** — appear: application boot re-applies `simulation.yaml` `default_enabled_sids: [1, 11]` , so every other SID is disabled at boot. After the

Element	Detail
	three `HK_ENABLE` s, SIDs 3, 4 and 6 appear within one structure interval (SID 3 = 60 s, SID 4 = 8 s, SID 6 = 8 s). SIDs 2 (AOCS) and 5 (Payload) are enabled in LEOP-007 once their lines are powered.
Why this step	The OBC has been running bootloader code only — limited functionality. Booting the application image starts the full mission software including FDIR, scheduler, command dispatcher, and HK collector. The OBDH verifies the application CRC over a 10 s window before the transition. The dual-key requirement reflects that a wrong boot (e.g. corrupted image) could brick the OBC.
Decision point	If S1.1 ack received but S1.7 (execution complete) doesn't arrive within 30 s, the application failed to start. FD must decide: retry from bootloader, or invoke CTG-010 (OBDH watchdog recovery). The retry is fine if the team has time in the pass; CTG-010 is the safer call if pass time is tight.
Common deviation	A SID commanded on *before* the boot is silent afterwards. That is the boot re-applying `[1, 11]` , not a fault: re-issue `HK_ENABLE(sid=N)` after the boot. Second deviation: the team reaches for `TTC_SET_DATA_RATE` here and gets S1.8 `0x0007` . Booting the application does **not** clear beacon mode — that is step 5a.

Step 5a — Clear Beacon Mode and Move to the LOW Rung

Table 10.8a — LEOP-001 step 5a detail: clearing beacon mode and stepping the TM rate ladder to LOW.

Element	Detail
FD prompt	"TT&C, FD — go for beacon-mode clear and LOW rung."
Owner	TT&C (Ground Segment reconfigures the demodulator)
Owner expected callout	"FD, TT&C — going for TTC_SET_BEACON_MODE off." → execute → "FD, TT&C — beacon_mode 0. Going for data rate LOW." → execute → "FD, TT&C — tm_tier 1, 16 000 bps, 16 kBd. Ground Seg, demod to 16 kBd."
Action	`TTC_SET_BEACON_MODE(on=0)` (S8.1, **func 70**), then `TTC_SET_DATA_RATE(rate=0)` (S8.1, **func 65**) for LOW. Ground Segment re-configures the demodulator from 1 kBd to 16 kBd *before* the subsystem SIDs are expected.
Verification	`ttc.beacon_mode` (0x0521) = 0 within 10 s; `ttc.tm_data_rate` (0x0506) = 16000; `ttc.tm_tier` (0x0539) = 1 (LOW); `ttc.tm_symbol_rate` (0x053B) = 16000 Bd. S1.7, **not** S1.8.
Why this step	Beacon mode is the gate on the whole rate ladder: while it is latched, BEACON is the only permitted tier and both LOW and HIGH are refused. It is a separate, deliberate step because the beacon rung is the lifeline — the spacecraft never leaves it by accident. LOW carries the full periodic housekeeping set at under 10 % duty; the BEACON rung at 1 kbps cannot (the shipped HK set is ~1 588 bps of channel, i.e. 159 % of the beacon rung, and the backlog simply grows).
Common deviation	S1.8 `0x0007` on the rate command. Read the failure notice, which names the gate in words — "TM rate is forced to the beacon tier in beacon mode (LOW refused)". The same text appears as S5 event `0x030B COMMAND_REJECTED` . Re-issue the beacon-mode clear.

Step 6 — Synchronise On-Board Time

Table 10.9 — LEOP-001 step 6 detail: onboard time synchronisation and drift verification.

Element	Detail
FD prompt	"SPACON, FD — go for time sync."
Owner	SPACON
Owner expected callout	"FD, SPACON — go for SET_TIME." → execute → "FD, SPACON — time set, drift 0.18 s."
Action	`SET_TIME(cuc_seconds=T_corrected)` — Service S9, subtype 1 — with current ground UTC as CUC seconds. Keep the argument **inside** the backticks: an annotation with the argument written outside them extracts as a `SET_TIME` with no argument, and the wire packs `cuc_seconds = 0` — an epoch reset, not a synchronisation. `SET_TIME` is not on the bootloader allow-list, so this only works after step 5.
Verification	Within 5 s: ack + complete. Within 10 s: `REQUEST_TIME` (**S9.2** — `TIME_REPORT_REQUEST` is only an alias; the catalogue name is `REQUEST_TIME`) returns TM(9,2) with `abs(T_obc − T_ground) < 1.0 s`.
Why this step	The OBC clock is whatever it was at launch — could be off by tens of seconds. Sub-second sync matters because (a) S11 time-tagged commands fire based on OBC clock, (b) telemetry timestamps must align with ground for log correlation, (c) AOCS uses time for orbit propagation.
Common deviation	Drift > 1 s after first set — the ground-to-spacecraft round-trip latency (typically 200-500 ms) wasn't compensated. Re-send with leading offset (FD will know whether to authorise; usually just sends again 10 s later).

Step 7 — Request EPS Housekeeping (Power Sanity Check)

Table 10.10 — LEOP-001 step 7 detail: EPS housekeeping request and power sanity check.

Element	Detail
FD prompt	"SPACON, FD — go for EPS HK." → "EPS, prepare to read."
Owner	SPACON requests; EPS reads
Owner expected callout	SPACON: "FD, SPACON — HK SID 1 received." → EPS: "FD, EPS — bus 32.5 V, SoC 89%, all switchable lines off, draws nominal."
Action	`HK_REQUEST(sid=1)` — S3.27.
Verification	`eps.bus_voltage` (0x0105) ∈ [29.6, 34.6] V; `eps.bat_soc` (0x0101) > 50 %; `eps.power_gen` (0x0107) reported — it will be small, because the spacecraft is tumbling, not sun-pointed, and the wings are still stowed.
Why this step	The EPS state determines whether we can proceed with LEOP-007 sequential power-on. If SoC is too low (< 50%), we're in a power-survival state — load shed first, defer commissioning, recover SoC over the next several orbits.

Step 8 — Log and Report

Table 10.11 — LEOP-001 step 8 detail: logging baselines and reporting completion.

Element	Detail
FD prompt	"All positions, FD — LEOP-001 complete, recording log entries."
Owner	Each position records to their shift log
What is logged	AOS time, override-arming time, time of first accepted TC (S17 from step 2), time of application boot, time-sync residual, EPS sanity-check values, link margin baseline.
Why this step	The cohort archive needs the times to compare against next-cohort runs. The link margin baseline is the reference for every subsequent pass's margin call.

Common deviations and FD playbook

Table 10.12 — LEOP-001 common deviations with their meaning and the FD's recovery action.

Deviation	What it means	FD action
`pass_override` not engaged before step 2	TC will be rejected (no carrier track)	Halt; engage override; re-send S17
S1 ack received but command rejection 0x0004	TC sent to bootloader but TC not on the bootloader allow-list	Identify the TC; if it's pre-OBC_BOOT_APP, defer until step 5 done
Beacon not received at all by step 4	Spacecraft is in a state where SID 11 reporting is off (rare; usually a separation issue)	Hold the procedure; verify spacecraft is actually in the predicted geometry (call FDS); if so, escalate to EMG-004 (Loss of Communication)
`obdh.sw_image` does not transition after OBC_BOOT_APP	The boot-app CRC countdown is 10 s	Wait 30 s, re-request SID 11 (which carries `sw_image`); if still 0, the application CRC is failing — escalate to the OBDH engineer and consider `OBC_BOOT_INHIBIT(0)` then `OBC_REBOOT`
OBC_BOOT_APP returns S1.1 ack but not S1.7 within 30 s	Application failed to start	Wait 60 s; if still no S1.7, invoke CTG-010 (OBDH Watchdog Recovery)
`SET_TIME` rejected with 0x0006	The boot-app transition has not completed (race with the 10 s CRC window)	Wait 5 s and retry
Time sync drift > 5 s after first SET_TIME	Ground time reference wrong, or spacecraft clock crystal damaged	Verify ground reference; if confirmed correct, log as an anomaly and continue (large drift is recoverable later)
SIDs 3, 4 or 6 silent after boot	The `HK_ENABLE`s were sent before the boot, and the boot re-applied `default_enabled_sids: [1, 11]`	Re-issue `HK_ENABLE(sid=N)` (S3.5) for each missing SID *after* the boot
	The auto-TX hold-down expired	Send any TC; acceptance re-arms the 300 s hold and re-energises `ttc_tx`

Deviation	What it means	FD action
Downlink goes quiet ~5 minutes after the last TC		

Decision points worth simulating

1. **At step 5, the EPS reports SoC 38% (below the 50% gate).** FD must decide: boot application anyway (gets us full HK but loses pre-boot margin), or hold and let SoC recover over the next orbit. Real LEOP teams have made both calls; both are defensible.
2. **At step 5, FD calls GO but TT&C reports link margin dropped to 1.5 dB during the boot wait.** The transmitter power-on auto-armed in step 2 may have heated the amplifier and degraded the link. FD must decide: continue with marginal link, or hold and let the amplifier cool. Continuation is the usual call (margin still positive); hold is justified if margin has dropped *fast* (suggests a real degradation).
3. **At step 6, time sync residual is consistently > 1 s after three attempts.** FD must decide: accept the residual (log as anomaly, plan to investigate next shift), or invoke time-sync recovery procedure. The 1 s gate is conservative; up to 5 s is tolerable for most mission activities. Accepting the residual and continuing is usually correct.

Debrief prompts (post-pass)

For the operator team:

- What was your time from AOS to first accepted TC? (Industry benchmark: < 30 s.)
- What was your time from AOS to OBC application running? (Industry benchmark: < 4 min for a clean pass.)
- Was there any GO call that, in retrospect, should have been a HOLD?
- Was there any HOLD that, in retrospect, you should have passed through?
- What baseline values did you log? Are they all on your shift log?

For the observer team:

- Was the voice loop discipline tight during the OBC_BOOT_APP authorisation poll?
- Did the FD get an explicit "ready" from every polled position?
- Did the FD correctly second-key the high-crit TC after the SPACON first-keyed it?
- Was the post-boot wait period silent (good — operators monitoring) or chatty (bad — distracted)?

LEOP-007 — Sequential Power-On (deep walkthrough)

LEOP-007 powers only what the rest of LEOP needs, and nothing more. Its six steps fit one 8-minute contact; the detumble wait that follows is physics rather than workload, and it belongs to LEOP-005. The procedure brings the spacecraft from "wings deployed, every switchable line except `ttc_tx` OFF, AOCS OFF" to "AOCS in DETUMBLE on the magnetometer and magnetorquers, GPS acquiring". Each power-line enable is a small ceremony: FD GO, owner first-key, SPACON verifies, log entry, next line.

Pre-step setup

LEOP-007 starts after LEOP-004 (Solar Array Verification). The team has:

- App SW running (`obdh.sw_image` 0x0311 = 1), beacon mode cleared, downlink on the LOW rung, HK SIDs 1, 3, 4, 6 and 11 streaming.
- Both wings deployed and generating (`eps.wing_status` 0x0144 = 0x03).
- EPS SoC > 60 % and bus 29.6–34.6 V (step 1 verifies both). The additions are modest: +~6 W for `htr_bat` and +~3 W for `htr_obc` if their zones need heat, and +~12 W for the AOCS suite.
- The clock still unset: time synchronisation is LEOP-006, after this card.
- No FDIR table to load: every rule is armed all the time, and individual event-action definitions are inhibited one at a time with `EA_DISABLE` (S19.5) and restored with `EA_ENABLE` (S19.4).

Step-by-step playbook (six steps)

For each line-enable, the pattern is:

1. EPS recommends to FD: *"FD, EPS — recommend enable htr_bat."*
2. FD polls: *"SPACON, AOCS, Thermal — go for htr_bat enable?"*
3. Each responds: *"FD, [POSITION] — go."*
4. FD: *"Go for htr_bat enable."*
5. EPS first-keys (no second-key for routine line-enables).
6. EPS reports current: *"FD, EPS — htr_bat 0.35 A, nominal."*
7. SPACON verifies and logs: *"FD, SPACON — htr_bat verified, log entry."*

The full table:

Table 10.13 — LEOP-007 six-step sequential power-on with owners, preconditions, and watch-fors.

#	Owner	Action	Pre-condition	Expected behaviour	Watch fo
1	EPS	Verify EPS baseline: `HK_REQUEST(sid=1)` (S3.27)	app SW running, wings deployed	`eps.bus_voltage` (0x0105) 29.6–34.6 V; `eps.bat_soc` (0x0101) > 60 %; `eps.power_gen` (0x0107) > 0 W	SoC belo not start
2	EPS + Thermal	Heater lines **as needed**: `EPS_POWER_ON(line_index=5)` (S8.1, **func 19**) only if `tcs.temp_battery` (0x0407) < +5 °C; `EPS_POWER_ON(line_index=6)` only if `tcs.temp_obc` (0x0406) < −10 °C (the OBC heater ON point)	bus stable, SoC > 60 %	`eps.pl_htr_bat` (0x0115) / `eps.pl_htr_obc` (0x0116) = 1 within 5 s for each line energised; +~6 W / +~3 W	energisi does not high curr
3	EPS → AOCS	Energise the AOCS suite line and enable its HK: `EPS_POWER_ON(line_index=7)`, then `HK_ENABLE(sid=2)` (**S3.5**)	step 2 done	`eps.pl_aocs_wheels` (0x0117) = 1; SID 2 emitting within one interval; +~12 W	expectin come up the suite `aocs.` (0x0260) COM-00 poweri enable t
4	AOCS				

#	Owner	Action	Pre-condition	Expected behaviour	Watch fo
		Enable the DETUMBLE legs and the GPS receiver: `S2_AOCS_MAGNETOMETER(device_id=521, state=1)` ; `MTQ_ENABLE(enable=1)` (S8.1, **func 9**); `S2_AOCS_GPS_RECEIVER(device_id=527, state=1)`	SID 2 emitting	`aocs.mtq_enabled` (0x025E) = 1 within 10 s; the GPS time to first fix starts	comman — it is re `0x0A2` magneto (no magn
5	AOCS	Command DETUMBLE: `AOCS_SET_MODE(mode=2)` (S8.1, **func 0**)	both legs enabled	`aocs.mode` (0x020F) = 2 within 10 s; `aocs.mag_enabled` (0x025C) = 1; `aocs.mag_field_total` (0x0277) 5–70 µT; rates (0x0204–0x0206) trending down within 60 s	refused the line zeroed fi magneto dot is bli
6	EPS	Confirm the exit power state: `HK_REQUEST(sid=1)`	DETUMBLE active	`eps.pl_payload` (0x0113), `eps.pl_fpa_cooler` (0x0114), `eps.pl_prop` (0x014A) and `aocs.wheels_enabled` (0x0260) all 0; power margin positive	anything the paylo COM-00 COM-01

After step 6 the AOCS is damping. LEOP-005 observes the automatic handover to COARSE_SUN and LEOP-003 declares phase 5: nothing in LEOP-007 commands mode 3 or the phase.

Why detumble takes so long

Step 5 hands the spacecraft to DETUMBLE, and the wait that follows is the part of LEOP that teams consistently misjudge, so it is worth being precise about the physics.

DETUMBLE is a genuine **B-dot** law driven by the magnetometer, not by the gyros: the on-board control differentiates the *measured* field vector and commands a magnetic dipole

$$\mathbf{m} = -\mathrm{k}\,\dot{\mathbf{B}}, \qquad \mathrm{k} = 5 \times 10^{5}\ \mathrm{A\,m^{2}/(T\,s^{-1})}$$

with each of the three rods saturating at **1.6 A m²**. Because $\dot{\mathbf{B}}_{\mathrm{body}} = -\boldsymbol{\omega} \times \mathbf{B}$, the torque is

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B} = -\mathrm{k}\,|\mathbf{B}|^{2}\,\boldsymbol{\omega}_{\perp}$$

— a pure rate damper that needs no attitude knowledge and no rate sensor. Two consequences follow, and both of them show up on the AOCS tab:

1. **Only the component of $\boldsymbol{\omega}$ perpendicular to $\mathbf{B}$ is damped at any instant.** The field direction sweeps through the body as the spacecraft tumbles and as the orbit carries it through the field, so energy genuinely moves **between axes** while the total falls. Measured from (1.0, 0.8, 0.5) °/s, `rate_roll` is *up*, at 1.05 °/s, at t = 50 s, while the magnitude has already fallen from 1.37 to 1.28 °/s. A single axis rising is not a broken control law. Judge convergence on the RSS of the three rates.

2. **It is slow.** The rate time constant is $\tau = I/(k|\mathbf{B}|^2) = 0.18/(5\times10^5\cdot(3\times10^{-5})^2) \approx 400$ s, and the rods saturate above about 1 °/s, so convergence takes of order 10^3 s, not tens of seconds. Measured on this model from $|\omega_0| = 1.37$ °/s: 1.13 °/s at 100 s, 0.95 at 200 s, 0.63 at 600 s, 0.55 at 900 s, and **below the 0.5 °/s exit guard at t = 1166 s** — about 20 minutes. From a 14 °/s separation tumble it takes **~3.5×10³ s** (the gain sweep in `subsystems/aocs.yaml`, run at 1 s ticks, records 4 322 s to the same guard — roughly three quarters of an orbit).

So: any claim that detumble is a few minutes' work is wrong. The card's own "< 0.5 °/s within 5–15 min" note is optimistic for anything but the mildest tumble; plan for a substantial part of an orbit, and expect the mode promotion to DETUMBLE → COARSE_SUN to happen on its own, thirty seconds after the rate magnitude first stays under the guard.

Note also what the mode does *not* use. B-dot is magnetometer-only, so switching the magnetometer off stops the damping dead while every gyro still reads perfectly — that, not a gyro fault, is DETUMBLE's real failure signature. Raising the gain does not help: the sweep shows $k = 1.5\times10^6$ is *worse* (4 914 s), because a higher gain amplifies differentiated magnetometer noise faster than it adds damping, and $k = 4\times10^6$ does not converge at all.

Decision points worth simulating

1. **Step 2 inrush is 1.2 A (about 3.4× the 0.35 A nominal)** on a heater line the cold battery zone needed. Could be a real subsystem fault, or could be cold-start in-rush. EPS recommends a 60 s hold to watch the trend; FD must decide. If the current drops to nominal within 30 s, continue; if it stays high, invoke CTG-012 (Overcurrent Response).
2. **Step 5: `AOCS_SET_MODE(mode=2)` is refused.** Read the failure code before doing anything. `0x0004` means the AOCS suite line write did not take — re-issue `EPS_POWER_ON(line_index=7)` and check `eps.pl_aocs_wheels` (0x0117). `0x0A20` means no magnetometer and `0x0A21` no magnetorquers: step 4 did not take. Re-sending the mode command unchanged will be refused again for the same reason. There is no redundant magnetometer. The wheels are irrelevant here: they are still disabled, and detumble is done entirely with the magnetorquers.
3. **After step 5 (LEOP-005): rates plateau at 0.6 °/s and stop falling.** Two interpretations: the law is converging but the geometry is momentarily poor (the rate is nearly parallel to **B**, so there is little perpendicular component to damp); or the magnetometer has been lost and B-dot is commanding nothing. Distinguish them: read `aocs.mtq_x/y/z_duty` (0x0258–0x025A) and `aocs.mtq_enabled` (0x025E). Non-zero duties with a stalled rate is geometry — wait a few minutes for the field to sweep. Zero duties is a real fault.
4. **After step 6: no GPS fix yet.** A cold start takes up to 15 minutes from the step 4 power-on. Nothing in LEOP-007 waits for it: LEOP-006 sets the clock from the ground first and disciplines it to GPS only once the fix is in, and LEOP-003 falls back to ground range and Doppler if there is still no fix after 30 minutes. The decision is not whether to wait, but whether anyone has started treating a missing fix as a reason to hold.

Common deviations

Table 10.14 — LEOP-007 common deviations with likely root causes and recovery actions.

Deviation	Likely root cause	Recovery
Step 2 nominal, step 3 draw sustained above expectation	Solar-array current variation masking the line's own step	Wait 30 s in the lit portion; EPS bills from actual actuator state, so compare against the line current, not the bus total
SID 2 never appears after the `aocs_wheels` line goes on	The `HK_ENABLE(sid=2)` was never sent — powering a line does not enable its SID	Send `HK_ENABLE(sid=2)` (S3.5); SID 2 appears within one interval
`AOCS_SET_MODE` refused with 0x0004 after the line-on	The EPS line write did not take	Re-issue `EPS_POWER_ON(line_index=7)`, verify `eps.pl_aocs_wheels` (0x0117) = 1, then retry the mode command
`AOCS_SET_MODE` refused with 0x0A20 or 0x0A21	The magnetometer or magnetorquer leg of step 4 did not take	Re-issue `S2_AOCS_MAGNETOMETER(device_id=521, state=1)` or `MTQ_ENABLE(enable=1)`, then retry
One body-axis rate *increases* during detumble	Expected — B-dot damps only the perpendicular component, so energy moves between axes	Nothing. Judge the RSS, not the axis
Body rates increase on all axes	Magnetorquer polarity inverted	Command `AOCS_SET_MODE(mode=0)` (OFF) immediately and investigate before re-entering DETUMBLE
Detumble makes no progress at all and the rod duties read zero	Magnetometer lost — B-dot is a magnetometer-only law	Power-cycle the magnetometer (single unit, no spare to cross-strap to); the gyros being healthy tells you nothing here

Debrief prompts

- Total LEOP-007 *commanding* duration (it fits one 8-minute contact), reported separately from the detumble wait, which is physics and not the team's to speed up.
- Were all FD GO calls explicitly polled, or did any get a drive-by *"go ahead"*?
- Were all current draws within the expected envelope on first read?
- Did anyone command COARSE_SUN rather than wait for it, or read a single rising axis rate as a control-law failure?

LEOP-006 — Time Synchronisation (the step everyone gets wrong)

LEOP-006 is short, and its steps are mostly mechanical: verify the OBC is responsive (step 1), read the clock (step 2), compute the offset (step 3), send `SET_TIME` (step 4), verify the residual (step 5), confirm the rate (step 6), log (step 7). Step 2 is the one worth reading carefully, because until it was rewritten it made step

3's `delta_t` *unobtainable* rather than merely imprecise. Three things were wrong with it, and all three are the kind of error to hunt everywhere else in the book:

1. **It named an alias.** The card said `TIME_REPORT_REQUEST`. The catalogue command is **`REQUEST_TIME`** (S9.2). The alias resolves on the wire, so the command works — but an operator searching the command panel for the card's mnemonic does not find it.
2. **It asked for a parameter without requesting the structure that carries it.** The step told the operator to read `ttc.range_km` (0x0509) but only ever requested SID 4. 0x0509 lives in **HK SID 6**, so the range needed for the one-way light time never arrived.
3. **Its OWLT figure was wrong by a factor of three.** `ttc.range_km` is the *geometric* slant range and is published whether or not a station is in view: for the EOSAT-1 reference orbit it reads about 6 530 km out of contact and 1 000–2 100 km during a pass — i.e. **3–22 ms**, not "2–8 ms". It is still negligible against the 1.0 s LEOP threshold, which is exactly why you record it and move on. A card that quotes a number the spacecraft never produces teaches the operator not to check.

The shipped step 2 therefore requests three things together:

- `REQUEST_TIME` (S9.2) — the reply is TM(9,2), received within 10 s; record the reported CUC value as `T_obc` and ground UTC at the moment of receipt as `T_ground`.
- `HK_REQUEST(sid=6)` (S3.27) — TT&C housekeeping, **for the slant range**. Compute `OWLT_ms = ttc.range_km (0x0509) / 299792.458 × 1000`; expect 3–22 ms.
- `HK_REQUEST(sid=4)` (S3.27) — Platform housekeeping, **for the clock cross-check**. The OBC clock is also downlinked continuously as `obdh.obc_time` (0x0309). The TM(9,2) value and 0x0309 must agree to within the SID 4 collection interval (8 s). They arrive by different routes — one an explicit time report, the other a housekeeping sample — so a disagreement means the time service and the housekeeping sampler disagree about the clock, which is a *different* fault from the clock being wrong.

The GO criterion for step 2 is all three: a valid TM(9,2) with a plausible CUC value, the slant range recorded, and 0x0309 in agreement.

Two more things carry forward from this card. First, `SET_TIME` in step 4 must be written with its argument **inside** the backticks — `SET_TIME(cuc_seconds=T_corrected)`. Written outside, the extractor produces a `SET_TIME` step with no argument and the wire packs `cuc_seconds = 0`: an epoch reset, not a synchronisation. Second, the LEOP acceptance threshold is 1.0 s and the rate threshold is 100 ppm; fine synchronisation to better than 0.1 s is nominal-operations work and belongs to **NOM-008** (`nominal/clock_sync.md`). NOM-006 is Software Upload.

10.12 References

- ECSS-E-ST-70-11C — Space segment operability — ecss.nl

Chapter 11 — Commissioning Operations

This chapter covers operations from LEOP-complete to **First Light** — the point at which the payload has been powered on, calibrated, and acquired a science-grade image of a target. This is phase 5 (COMMISSIONING) in the mission phase table (§5.7).

Commissioning runs across ~4 ground-station passes in a typical operational schedule, with significant inter-pass work (thermal stabilisation of the focal plane, sensor calibration processing, planning of First Light geometry).

11.1 Commissioning phase entry conditions

- **Sim checkpoint:** BP-3 *Commissioning Day Start* (`scenarios/commissioning_day_start.yaml` — NEW; spec in §5.10).
- **Phase at start:** 4 (LEOP). Phase 5 is COMMISSIONING.
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- **Planned passes during shift:** 4. The teaching baseline is the **full five-station network** (Iqaluit, Troll, Svalbard, Inuvik, O'Higgins), so the shift's four passes are selected from a network that delivers ~51 passes/day with a median gap of 33 min.
- **Speed multiplier:** Mission Director-controlled.
- **Phase advance is a ground declaration.** The mission management team declares the spacecraft phase with `SYS_SET_MISSION_PHASE` (S8 func 120). Nothing on board advances it past 4 (LEOP), so the procedure step that sends it must not be skipped.

11.3 Mission phase brief

The spacecraft has just exited LEOP. Antennas deployed; app SW running; detumble complete (rates < 0.05°/s); time synced; FDIR in LEOP-only configuration. Payload is OFF; FPA at its uncooled equilibrium (+8 to +14 °C). Battery at 90% SoC. The team's shift goal is **First Light** — one acquired and downlinked image — by end of pass 4.

11.4 Procedure execution order across 4 passes

Thirteen commissioning cards are shipped, COM-001 through COM-013. The four-pass arc below runs eleven of them; COM-013 (ADCS Full Commissioning) is a consolidated re-run used in remedial sessions and is not on the nominal timeline.

Table 11.1 — Commissioning procedure execution across four passes with critical activity per pass.

Pass	Procedures	Critical activity
1	COM-001 EPS Checkout; COM-002 TCS Verification; COM-004 AOCS Actuator Checkout; COM-003 AOCS Sensor Calibration	The four wheels are enabled and the spacecraft reaches wheel-controlled NOMINAL (mode 4) for the first time; each wheel is biased in both directions and ramped back to zero; then the gyro bias is calibrated on board in that NOMINAL
2	COM-005 AOCS Mode Transitions; COM-006 TTC Link Verification;	

Pass	Procedures	Critical activity
	COM-007 OBDH Checkout; COM-008 FDIR Configuration	The AOCS mode ladder is exercised up to FINE_POINT; both rungs of the TM rate ladder are baselined; the protection stack (S12, S19, load-shed latch) is inventoried and tested
3	COM-009 Payload Power-On; COM-010 FPA Cooler Activation	Cooler started — 15 W Stirling-class cryocooler drives the FPA from ~+11 °C to the −15 °C setpoint in about ten minutes
4	COM-011 Payload Calibration; COM-012 First Light	Dark and flat sets taken in CALIBRATION mode (3), then one image of an ocean-current target captured and downlinked

End-of-shift target state: phase 5 or later (commissioning complete or deferred) — declared by the mission management team — payload calibrated, one usable image acquired (or formally deferred-with-justification).

11.5 Per-position playbook for commissioning

Flight Director

- Pre-shift: brief the team on the four-pass arc. Confirm the pass criteria for each pass.
- Across passes: enforce the *one procedure at a time* rule. Authorise progression COM-001 → -002 → -003 → -004 in pass 1 and COM-005 → -006 → -007 → -008 in pass 2.
- Pass 3 is the pivot: payload power-on and cooler start. Authorise carefully; at 15 W the FPA cooler is the single largest load on the spacecraft — larger than the transmitter at its HIGH tier (11 W) — and EPS must be ready.
- Pass 4 is the commit point. If pointing is shaky or storage is near full, authorise *formal deferral* rather than imaging anyway.
- Pitfalls: over-running pass 2's COM-005 (mode transitions take time to verify); arriving at pass 3 still mid-COM-008.

SPACON

- Pre-shift: load all 13 commissioning procedures into the procedure index. Be ready to switch active procedure quickly.
- Across passes: track which subsystems are now "commissioned" vs "pending". Once COM-008 loads the nominal FDIR table, several new monitors will arm and trigger if those subsystems are not actually nominal.
- Pitfalls: forgetting to acknowledge an FDIR rule before the next procedure step depends on the parameter being green again (sometimes the rule clears autonomously; sometimes it needs ack).

AOCS

- Pass 1 lead: COM-004 Actuator Checkout. Each of the four wheels is biased with `RW_SET_SPEED_BIAS` and returned to zero with `AOCS_RW_RAMP_DOWN`. Wheel speeds (0x0207–0x020A) are read-only telemetry; there is no `SET_PARAM` path to a wheel speed.
- Pass 2 lead: COM-005 mode transitions NOMINAL (4) → FINE_POINT (5) → NOMINAL (4), INERTIAL (9) and back, the COARSE_SUN (3) hold and DESAT (7). Each commanded transition is a separate TC and a separate verification; the automatic returns are observed, not commanded.
- Pass 4 supporting: imaging requires FINE_POINT; AOCS commands `AOCS_SET_MODE(mode=5)` after `AOCS_FINE_READY_CHECK` reads `aocs.fine_ready` (0x02C5) = 0.

- Pitfalls: not verifying that the pointing-error widget settles after a mode transition (the transition is fast; the *settling* takes 30–60 s, and `aocs.settle_time_s` 0x02C4 reports it).

EPS

- Pass 1 lead: COM-001 EPS Checkout establishes the bus, battery and generation baselines. The power-line matrix has **nine** lines, indices 0–8 (`obc` , `ttc_rx` , `ttc_tx` , `payload` , `fpa_cooler` , `htr_bat` , `htr_obc` , `aocs_wheels` , `prop`); lines 0 and 1 are unswitchable and every switchable line defaults OFF.
- Pass 3 critical: payload power-on (2.4 W in STANDBY) plus cooler start (15 W). Total spacecraft load in that configuration is ~39 W against ~54–71 W of sunlit generation depending on beta. Start it in the lit portion of the orbit.
- Pitfalls: not noticing that battery SoC is dropping faster than the eclipse model predicted (suggests a current leak or underrated solar array).

Thermal

- Pass 1 lead: COM-002 TCS Verification — the heater circuits are exercised individually and their thermostat setpoints read back from `tcs.setpoint_bat` (0x0330) and `tcs.setpoint_obc` (0x0331); duty cycles recorded from `tcs.htr_duty_battery` (0x040E) and `tcs.htr_duty_obc` (0x040F).
- Pass 3 critical: FPA cooler activation. The cool-down is fast: from the uncooled equilibrium (~+11 °C) the detector is inside the −15 °C setpoint band within about ten minutes, initial rate ~4 °C/min falling exponentially. The cooler is a **miniature Stirling-class cryocooler** running at ~23 % of Carnot, not a Peltier stack, and it regulates to the setpoint — it never overshoots below it.
- Pitfalls: not noticing that one heater's duty cycle is *suspiciously low* — could mean the zone is leaking heat from an unexpected source (e.g. inadvertent payload power).

TT&C

- Pass 2 lead: COM-006 TTC Link Verification. Configures the S-band chain (line, PA, beacon-mode release, antenna deploy), then baselines the margin at both operational rungs of the rate ladder, LOW (16 kbps) and HIGH (128 kbps).
- Pass 4 critical: downlink the First Light image. A 4 MiB product is 262 s of channel at HIGH, 328 s with the planner's 20 % protocol allowance — about three-quarters of an average pass.
- Pitfalls: not raising the tier to HIGH before the image is queued, then the downlink takes more than a whole pass at LOW and is interrupted by LOS. Note the HIGH tier is *refused* below 10° elevation; that rejection is designed behaviour, not a fault.

Payload

- Pass 2 supporting: COM-007 OBDH checkout includes a payload- storage check.
- Pass 3 lead: COM-009 power on, COM-010 cooler activation.
- Pass 4 lead: COM-011 Calibration, COM-012 First Light.
- Pitfalls: commanding `PAYLOAD_CAPTURE` before `payload.fpa_ready` (0x0610) sets. The image quality is degraded but the procedure verification passes because the *acquisition* succeeded. Quality assessment is a separate verification step.

FDS

- Pass 1 supporting: confirm orbit accuracy (the GPS fix from LEOP day's last pass is now several orbits old; refresh).
- Pass 4 lead: compute the ground footprint for the chosen ocean-current target — 273 km across-track swath, about 59 km along-track for the standard 2048-line strip. Hand to Payload for `PAYLOAD_CAPTURE` parameters.
- Pitfalls: confusing the swath (273 km, a footprint width) with the ground sample distance (33.4 m, a per-pixel figure). Both are properties of the same instrument and neither substitutes for the other.

Ground Segment

- All passes: lead pre-pass station readiness. Track the four passes; hand over cleanly between stations in the five-station network.
- Pass 3 critical: the cool-down settles in about ten minutes, so it fits comfortably inside a single 7–8 minute pass plus the first minutes of the next. Confirm TM is buffered on board and request the playback with `DUMP_STORE` (S15.9) at next AOS for any of the ramp that fell in the gap.

Mission Manager

- Track First Light status. Prepare customer bulletin: "EOSAT-1 acquired First Light at [time], target [name], image queued for downlink."

Table 11.2 — Commissioning anomaly-injection scenarios by severity with first indicators and recoveries.

Severity	Scenario file	What students see	Recovery
Easy	`eps_overcurrent.yaml` (line= `payload`)	Overcurrent on payload power-on	CTG-012 Overcurrent Response
Easy	`gs_rf_degradation.yaml`	BER climbs mid-pass	CTG-014 BER Anomaly
Medium	`fpa_overtemp.yaml`	FPA cooler can't reach the −15 °C setpoint; `payload.cooler_margin_k` (0x064B) goes ≤ 0	CTG-004 Thermal Exceedance + payload safe
Medium	`payload_corrupt_image.yaml`	First image returns corrupt	CTG-015 Corrupted Image Recovery
Hard	`aocs_sensor_cascade.yaml`	Star tracker degrades; pointing budget broken	CTG-008 Star Tracker Failure → CTG-002 + defer First Light

Severity	Scenario file	What students see	Recovery
Hard	`eps_progressive_load_shed.yaml`	Battery degradation forces progressive load-shed during cooler activation	CTG-001 Under-Voltage Load Shed
Very hard	`aocs_wheel_failure.yaml` followed by `transponder_failure.yaml`	Wheel locks late in pass 3, then transponder fails next pass	CTG-007 → CTG-003

Cue cards in Chapter 14 (Procedure Catalogue).

11.7 Pass criteria (review)

Table 11.3 — End-of-commissioning pass criteria scored across the two training teams.

Criterion	Pass-set 1 (Team A)	Pass-set 2 (Team B)
Phase declared COMMISSIONING (5) by mission management	✓	✓
EPS checkout complete with no faults	✓	✓
AOCS NOMINAL (mode 4) achieved, `aocs.att_error` < 0.5°	✓	✓
FDIR nominal table loaded	✓	✓
Payload powered ON (line 3 energised *and* `payload.mode` = 1)	✓	✓
FPA in the setpoint band [−16.5, −14.5] °C and `payload.fpa_ready` (0x0610) = 1	✓	✓
First Light image acquired (or formally deferred)	✓	✓
Anomaly tickets filed for all injections	✓	✓

The "formally deferred" option is genuine: if the team correctly diagnoses that pointing is insufficient and chooses to defer imaging, they pass. If they image anyway and produce garbage, they fail.

11.8 Debrief

Same structure as the LEOP debrief. Observer worksheets used today are **Team** and **Space usage** (the other two were used Wednesday).

11.9 Detailed Commissioning Procedure Walkthroughs

Day 4 walks the team from "LEOP done, all subsystems running" to "First Light acquired and downlinked". Across four ground passes (compressed to 2.5 hours of real time), eleven of the thirteen shipped commissioning procedures execute in roughly that order. This chapter is the *training* material for the team to read on Wednesday evening.

COM-001 — EPS Checkout (deep walkthrough)

EPS Checkout is the first commissioning step because every subsequent step assumes power is being delivered, regulated, and accounted for correctly.

Pre-step setup

- Spacecraft phase 4 (LEOP complete).
- COARSE_SUN (mode 3) achieved — sun-pointing, the recoverable attitude.
- Minimum 2 orbits of post-LEOP HK archived (so EPS Engineer has trend data).
- `eps.bat_soc` (0x0101) > 70%.

Per-step playbook

Step 1 — Baseline Bus Voltage Measurement

- FD GO not required (read-only).
- EPS: "FD, EPS — bus 32.1 V, following the battery at SoC 84 percent."
- Verification: `eps.bus_voltage` $\in$ [28.4, 34.0] V — the console's yellow band. The bus is not regulated: it follows the battery up to the 33.4 V cap, so a full battery reads near 33.4 V.
- Why: this is the **reference** every subsequent reading compares to. If bus is at 31.6 V at start of commissioning, the team knows that's our normal, and 31.6 V mid-imaging isn't a concern.

Step 2 — Battery Voltage & SoC

- EPS: "FD, EPS — battery 32.1 V, SoC 84 percent."
- Verification: `eps.bat_voltage` (0x0100) in [24.0, 33.6] V, the pack's modelled range; `eps.bat_soc` (0x0101) > 70%.
- Why: the pack is a GomSpace NanoPower BPX 8S-1P — **100 Wh, 3.5 Ah, 28.8 V nominal**, one 0.33 U box of 18650 cells. In the EOSAT-1 its terminal voltage runs from `soc_0_v` = 24.0 V empty to `soc_100_v` = 33.6 V full, 28.8 V nominal — the datasheet values, linear in between, so $\mathrm{V} \approx 24.0 + 9.6 \times \mathrm{SoC}$: at 84 % that is $24.0 + 9.6 \times 0.84 = 32.1$ V. It is an eight-cell series string, not a 4S pack, and it is what makes the ~29–33 V unregulated battery bus. Do not confuse `eps.bat_voltage` (0x0100, the pack) with `eps.bus_voltage` (0x0105, the bus, capped at 33.4 V) — they are different parameters with different ranges.

Step 3 — Solar Array Sunlit Performance

- EPS: "FD, EPS — power gen 58 W, beta 32 degrees, in predicted range."
- Verification: `eps.power_gen` (0x0107) in the sunlit band [40, 75] W.
- Why: the array is 0.198 m² of body-mounted cells over five faces (the −Z face points at nadir, carries the imager aperture and radiator, and has no cells) plus **0.120 m² of cells per deployable wing** on a single-axis drive — 0.438 m² of cells deployed in total. The plane-of-array constant after the full loss stack is 292 W/m² BOL and **267.9 W/m² at three-year EOL**, so the wings alone are 2 × 0.120 × 292 = **70.1 W BOL** (which is exactly the MMA eHaWK 6U-class rating) and whole-spacecraft peak generation is **88.6 W**. Sunlit readings run roughly 54–71 W depending on solar beta; a reading far below that in good geometry means degraded cells or a stuck solar array drive, and a reading above ~89 W means the geometry has been mis-modelled.

Step 4 — Eclipse Discharge Monitoring (this step spans an eclipse)

- EPS: "FD, EPS — eclipse entry T-zero. Power gen dropped to 0.1 W in 18 s, nominal."
- Mid-eclipse: "FD, EPS — at +15 min, bus 31.6 V, SoC 79 percent, drop rate 0.4 percent per minute."
- Eclipse exit: "FD, EPS — at exit, bus 31.0 V, SoC 73 percent, reservoir margin healthy."
- Verification: `eps.power_gen` ~0 W within 30 s of eclipse entry; SoC drop < 1.5%/min; `eps.bus_voltage` > 30.9 V mid-eclipse.

- Why: this is the *survival* test. The orbit is 5669 s (94.5 min) and the eclipse fraction over the year runs 0.351–0.372, so an eclipse is **33.2 to 35.2 minutes**. Against a 100 Wh battery and a ~22.7 W nominal orbit-average load, a 34-minute eclipse costs $22.7 \times 0.567 = 12.9$ Wh, about 12.9 % of SoC, i.e. ~0.4 %/min. The 1.5 %/min verification is a ceiling, not an expectation: a reading near it means a load you did not command. The protection ladder below is monotonic — in-model shed at **28.4 V** (~46 % SoC), FDIR `safe_mode_eps` at **27.7 V** (~39 %), both from `simulation.yaml` `eps.bus_protection`, then the SoC stages at 30 / 20 / 10 %.

Step 5 — Eclipse Exit Charge Recovery

- EPS: "FD, EPS — at exit + 45 s, power gen 52 W, charging, recovery profile nominal."
- At exit + 2 min: "FD, EPS — SoC trending up, will fully recover before next eclipse entry in 60 min."
- Verification: `eps.power_gen` > 35 W within 60 s; SoC rising within 120 s.
- Why: catches a charge-controller fault that would prevent the battery from recovering — would cascade into total power loss in 24 hours. Note the charge is coulombically lossy (0.94) and tapers linearly to zero above 90 % SoC (CV phase), so the last ten points come back far more slowly than the first ten.

Step 6 — Power Budget Verification

- EPS computes: average gen over the orbit, average consumption, end-of-eclipse SoC, margin.
- "FD, EPS — orbit-average generation 45.1 watts, load 22.7 watts, margin plus 22.4 watts, ratio 1.98. End-of-eclipse SoC predicted 73 percent under nominal load. Healthy."
- Verification: positive margin; end-of-eclipse SoC > 50%.
- Why: even one positive orbit isn't proof — we want to predict steady-state. Over the year β runs from −20.8° to −31.2°. The sizing case for generation is EOL at **β = −31.2°**, where orbit-average generation is 45.1 W: the solar array drive follows the Sun only to within cos β, so the largest |β| gives the least generation even though its eclipse is the shortest. Against that:

Profile	Gen (W)	Load (W)	Margin (W)	Ratio
quiescent	45.1	19.7	+25.4	2.29
nominal (payload STANDBY, cooler off)	**45.1**	**22.7**	**+22.4**	**1.98**
nominal_imaging	45.1	26.4	+18.7	1.71
manoeuvre	45.1	25.8	+19.4	1.75
worst (abuse case)	45.1	46.6	−1.4	0.97

(`tools/power_budget.py` , re-run 2026-11-03 with the thermal network and the FPA cooler off in STANDBY, sim manual 01_eps §3.2.)

At β = −20.8°, the deepest discharge (the longest eclipse), generation is 47.0 W and the nominal case closes at 2.05×. At beginning of life the two figures are about 49 W and 51 W. The `worst` profile is deliberately negative — it holds FINE for a whole orbit, runs the cooler continuously, images through eclipse and dumps at HIGH on every pass. **A 6U that could sustain that would need an array no 6U can carry.** The engineering answer is to duty-cycle imaging, and that is the lesson the exercise exists to teach. If margin is positive but small (< 2 W), any unexpected load (heater stuck on) could tip it negative.

Step 7 — Bus Voltage Regulation Under Load Transient

- EPS: "FD, EPS — `eps.pl_htr_bat` (0x0115) reads 0 at entry. Going for load-transient test on the battery-heater line."
- EPS: "FD, EPS — first-key `EPS_POWER_ON(line_index=5)`." (FD second-keys.)
- (1 s later): "FD, EPS — heater line on, total load up about 8 watts, about 0.25 amps at 32 volts. Bus held 32.1 V, transient dip 0.15 V for 80 ms."
- (5 s later): "FD, EPS — first-key `EPS_POWER_OFF(line_index=5)`, restoring the entry state." The line goes back OFF — only because it was OFF at entry.
- Verification: bus stays in [30.2, 34.0] V during the transient and returns to [30.9, 34.0] V; `eps.pl_htr_bat` back to its entry value.
- Why: this is the only commissioning step that *intentionally* perturbs the bus. It does **not** use the payload: the payload line stays OFF until COM-009, and a mode command on a dark line is refused, so it would be no transient at all. The imager itself is a small load anyway — the reference part (Simera Sense TriScape100, cited for power, mass and size only) draws **2.4 W idle and 5.5 W imaging**. The genuinely large step comes later, when the 15 W cooler starts in COM-010. The bus is not regulated — it is the battery, capped at 33.4 V — so a dip of more than 1 V here points at a weak battery or a high-resistance path, and future load steps should be sequenced more carefully.

Decision points

- **Step 3: beta angle differs from prediction by > 10°.** This could mean the spacecraft attitude is wrong (FDS recompute) or the orbit model is wrong (FDS re-OD). A third possibility is a **stuck solar array drive** (`sada_stuck`): the drive angle 0x0133 goes flat, the wings revert to a fixed-wing duty and orbit-average generation falls by roughly 36 %, with a characteristic once-per-orbit ripple on the per-wing currents 0x0146 (+Y) and 0x0147 (−Y). Note 0x0110–0x0113 are power-*line* status flags, not array telemetry. Either way, log before continuing.
- **Step 6: predicted margin is negative.** Do NOT continue with COM-002 etc. Invoke CTG-001 load-shed analysis and defer commissioning until the team can identify the parasitic load.

COM-002 — TCS Verification (deep walkthrough)

Pre-step setup

- COM-001 complete and GO.
- ≥ 1 full orbit of thermal HK archived.
- SoC > 60%. The two heater lines are `htr_bat` (8 W) and `htr_obc` (4 W), and EPS bills them from the *thermostat* state published by TCS, not from the line switch — a line that is ON with the thermostat satisfied costs nothing.

Per-step playbook (4 steps with substantive content)

Step 1 — Request Thermal HK

- Thermal: "FD, Thermal — `HK_REQUEST(sid=3)`. temp_obc 22 C, temp_battery 14 C, FPA at plus 11 C, all nominal."
- Verification: `tcs.temp_obc` (0x0406) ∈ [−10, +50] °C; `tcs.temp_battery` reported and in band; `payload.fpa_temp` (0x0601) reported.
- Why: baseline thermal state before exercising heaters. Note the TCS structure is **SID 3**, on a 60 s cadence — the slowest of the housekeeping set, because thermal time constants are minutes.

Step 2 — Exercise Each Heater Individually

- There are **two** heater circuits on the PDM, not four: `htr_bat` (line 5, 8 W) and `htr_obc` (line 6, 4 W).
- For each: Thermal: "FD, Thermal — go for exercise [heater]." FD GO. Thermal energises the line, enables the heater, watches duty cycle and zone trend, then restores automatic control with `HEATER_AUTO_MODE`. The line **stays ON** afterwards: since 2026-11-02 both heater lines are left powered under thermostat control for the rest of the mission (battery ON +5 / OFF +10 °C, OBC ON −10 / OFF −5 °C), so every later heater command finds its line already powered.
- After the exercise: "FD, Thermal — [heater] duty cycle averaged 35 percent, zone rose 3 C, response nominal."
- Verification: duty cycle stays < 80%; zone reaches setpoint band within expected time.
- Why: catches stuck-on or stuck-off heaters. Also catches shorted heater wires (duty cycle near 100% but no temperature rise).

Step 3 — Thermostatic Control Verification

- Thermal observes `tcs.temp_battery` (0x0407) over a half-orbit while `htr_bat` is on automatic.
- "FD, Thermal — htr_bat cycled 3 times in last 30 min, hysteresis 2.5 C, control law nominal."
- Verification: hysteresis observed; on/off cycles regular.
- Why: thermostatic control catches the heater that ticks on randomly (FET stuck on) vs. the heater that doesn't cycle but holds temperature (a coincidental orbit-eclipse balance).

Step 4 — Radiator Performance in Eclipse

- Thermal watches all zones during one eclipse.
- "FD, Thermal — in eclipse, all zones drifted within band, largest drift OBC zone minus 4 C in 30 min, recovered on Sun-side. No runaway."
- Verification: zones stable; no runaway up or down.
- Why: confirms radiator is dissipating heat as predicted and that heat path from internal components to radiator is intact.

COM-004 — AOCS Actuator Checkout (deep walkthrough)

Four reaction wheels in a pyramid (azimuths 45/135/225/315°, elevation 30° above the XY plane), plus three magnetorquer rods. The card enables the wheels, commands wheel-controlled NOMINAL, commissions each wheel individually and then proves the magnetic momentum-dump path.

Pre-step setup

- COM-001 and COM-002 complete; COM-003 has **not** run yet (ruling Q16). Star trackers and gyros are still off — there are **two** star trackers, ST1 and ST2, selected with `ST_SELECT` (S8 func 6), and COM-003 brings them up after this card.
- `aocs.mode` (0x020F) = 3 (COARSE_SUN), the LEOP hold; step 2 of the card commands NOMINAL once the wheels are enabled.
- Body rates < 0.5 °/s on all axes, read from `aocs.rate_roll` (0x0204), `aocs.rate_pitch` (0x0205), `aocs.rate_yaw` (0x0206). There is no `aocs.body_rates` parameter.
- `eps.bat_soc` (0x0101) > 65%.

The commanding path — read this before the steps

Wheel speeds at **0x0207–0x020A are read-only telemetry.** They cannot be commanded with `SET_PARAM`, and 0x0280–0x0283 are not wheel-speed setpoints. The two commands that move a wheel are:

- `RW_SET_SPEED_BIAS(wheel_idx, bias_rpm)` — Service 8, Subtype 1, **func_id 8**. The bias is **added to the wheel's current speed**, so the Step 3 baseline is the reference for every subsequent check. Biasing +2000 twice does not park the wheel at 2000 rpm; it parks it at baseline +4000.
- `AOCS_RW_RAMP_DOWN(wheel_idx, target_rpm)` — Service 8, Subtype 1, **func_id 14**. `wheel_idx = 255` means *all wheels* and is the emergency stop for this card.

Wheel temperatures are **0x0218–0x021B**. 0x028A–0x028D are the coarse-sun-sensor head health percentages (`aocs.css_head_health_py/my/pz/mz`) and have nothing to do with wheels; reading them for a temperature check is a classic misdiagnosis.

Per-step playbook

Step 1 — Enable the four wheels

- AOCS: "FD, AOCS — first-key `AOCS_ENABLE_WHEEL(wheel_idx=0)`", then `wheel_idx` 1, 2 and 3 (S8 func_id 3), one command per wheel.
- Verification: `aocs.wheels_enabled` (0x0260) = 4 within 10 s.
- Why: the bootloader leaves every wheel DISABLED and nothing else in the catalogue enables one. Do not leave the card with fewer than three.

Step 2 — Command wheel-controlled NOMINAL

- AOCS: "FD, AOCS — first-key `AOCS_SET_MODE(mode=4)`, NOMINAL." FD second-keys.
- Verification: `aocs.mode` (0x020F) = 4 (or 8 in eclipse) within 300 s; `aocs.commanded_mode` (0x0297) = 4.
- Why: nothing promotes the AOCS out of the LEOP COARSE_SUN hold without a star tracker, so the card commands NOMINAL itself. The simulator accepts it without a tracker; knowledge stays on the coarse sun and magnetometer path until COM-003. This is the step that makes COM-003 possible: its 0.10 °/s calibration gate is met in wheel-controlled NOMINAL, never in the magnetorquer-only hold (ruling Q16).

Step 3 — Record baseline wheel speeds

- AOCS: "FD, AOCS — `HK_REQUEST(sid=2)`. RW1 through RW4 reading plus 1180, minus 1210, plus 1195, minus 1205 rpm. Recorded as baseline."
- Why: the newly enabled wheels pick up the control load at once, so non-zero speeds in NOMINAL are expected; this baseline is the reference for every check below.

Steps 4–5 — Each wheel in turn

- AOCS: "FD, AOCS — first-key `RW_SET_SPEED_BIAS(wheel_idx=0, bias_rpm=2000.0)`." FD second-keys.
- Verification: `aocs.rw1_speed` (0x0207) reaches baseline +1800 to +2200 rpm within 60 s; `aocs.att_error` (0x0217) < 5°.
- AOCS: "first-key `RW_SET_SPEED_BIAS(wheel_idx=0, bias_rpm=-4000.0)`" — swings the wheel through zero to baseline −2000.

- AOCS: "first-key `AOCS_RW_RAMP_DOWN(wheel_idx=0, target_rpm=0.0)`" — return to < 100 rpm within 60 s.
- Repeat for `wheel_idx` 1, 2, 3.
- Why: the through-zero swing is the point of the test. Bearing drag and friction asymmetry show up at the zero crossing, not at speed.

Step 6 — Wheel temperatures

- `GET_PARAM(param_id=0x0218)` … `0x021B`, all in [−10, +55] °C.

Step 7 — Momentum desaturation test

- AOCS: "FD, AOCS — first-key `AOCS_DESATURATE`" (S8, func_id 1).
- Verification: `aocs.mode` (0x020F) = 7 (DESAT); `aocs.h_body_mag` (0x02B9) falling toward the 5.0×10^{-4} N·m·s deadband.
- **Expect 20 to 50 minutes**, not five to ten. See §13.7 for why.

Decision points

- **A wheel does not reach its commanded bias.** Check that the wheel line is powered — `eps.pl_aocs_wheels` (0x0117) — and its draw at `eps.line_current_7` (0x011F). If power is nominal and the wheel is unresponsive, flag it failed and continue: **the spacecraft flies FINE_POINT on three wheels.** Any three of the four pyramid wheels still span R^3 (condition number 2.175), and measured 3-wheel FINE holds 0.0188° mean pointing error against 0.0208° on four. What the degraded set loses is the null space — bias regulation and zero-crossing avoidance — so jitter is worse and the mode is flagged with `aocs.fine_degraded` (0x0299).
- **Attitude error exceeds 10° during a wheel test.** Halt immediately with `AOCS_RW_RAMP_DOWN(wheel_idx=255, target_rpm=0.0)`.
- **Desaturation does not reduce momentum.** Verify `aocs.mtq_enabled` (0x025E); MTQ enable is S8 func 9 and the rod duties are at 0x0258–0x025A. A dump is far less effective near the magnetic equator — retry over a pole.

COM-005 — AOCS Mode Transitions (deep walkthrough)

NOMINAL → FINE_POINT and back, then INERTIAL, the COARSE_SUN hold and DESAT. Since ruling Q16 the AOCS has been in NOMINAL since COM-004. Each transition is a separate procedure inside this one.

The mode codes are the ones in `aocs/aocs_mode_codes.yaml`, and nothing else:

Code	Mode	Code	Mode
0	OFF	5	FINE_POINT
1	SAFE_BOOT	6	SLEW
2	DETUMBLE	7	DESAT
3	COARSE_SUN	8	ECLIPSE
4	NOMINAL	9	INERTIAL

Older versions of this book and of several shipped procedure cards were systematically **off by one** against this table — calling COARSE_SUN 2 and NOMINAL 3. Treat any mode number you have not personally checked against the YAML as suspect. There is no mode called SAFE_POINT or NOMINAL_POINT.

Per-step playbook

Step 1 — Confirm starting state

- AOCS: "FD, AOCS — mode 4 NOMINAL, body rates roll minus 0.02, pitch plus 0.01, yaw minus 0.03 deg per second, all below the 0.05 threshold."
- Note: detumble is genuinely magnetometer-driven B-dot, m = −k dB/dt, with the rods saturating at 1.6 A·m² per axis. It converges *slowly* — from a 14 °/s separation tumble the model takes ~4300 s to reach the 0.5 °/s DETUMBLE exit guard. Any claim that detumble is quick is wrong, which is precisely why it is finished before this card starts.

Step 2 — Latch NOMINAL

- AOCS: "FD, AOCS — first-key `AOCS_SET_MODE(mode=4)`, NOMINAL." FD second-keys.
- "FD, AOCS — mode change command accepted."
- (30 s later) "FD, AOCS — nadir acquisition in progress, body rates settling."
- (60 s later) "FD, AOCS — mode 4 NOMINAL confirmed, nadir pointing achieved, `aocs.att_error` 0.42 deg."
- Verification: `aocs.mode` (0x020F) = 4 within 60 s; `aocs.att_error` (0x0217) < 0.5° within 120 s.
- Why: the AOCS has flown NOMINAL since COM-004; the command latches `aocs.commanded_mode` (0x0297) = 4, the return path every automatic transition resolves against. COARSE_SUN (3) remains the *recoverable* attitude — sun on the arrays, low body rates — to fall back to if NOMINAL fails.

Step 3 — Check the FINE preconditions before commanding FINE

- AOCS: "FD, AOCS — `AOCS_FINE_READY_CHECK`" (S8 func 89).
- Verification: `aocs.fine_ready` (0x02C5) = 0 — **every bit clear**. The bitmask is: bit 0 star tracker seeing fewer than 5 stars; **bit 1 fewer than THREE wheels available**; bit 2 MEKF unconverged; bit 3 rates high; bit 4 momentum above 70 % of the envelope; bit 5 estimator in propagate-only; bit 6 slew not idle for 60 s.
- Why bit 1 matters: it reads *fewer than three*, not fewer than four. Three wheels is a flyable FINE configuration; two is not.

Step 4 — Transition to FINE_POINT

- AOCS: "FD, AOCS — first-key `AOCS_SET_MODE(mode=5)`, FINE_POINT." FD second-keys.
- (60 s) "FD, AOCS — mode 5 confirmed, total pointing error 3-sigma 0.020 deg on 0x02AA, inside the 0.05 deg imaging gate."
- Verification: `aocs.mode` (0x020F) = 5 within 30 s; `aocs.total_pointing_err_3sig` (0x02AA) < 0.05° within 300 s.

Step 5 — Verify the pointing budget decomposition

- AOCS: "FD, AOCS — knowledge error 0x02A7 reading 18 arcsec 3-sigma, control error 0x02A8 0.0188 deg, RSS 0.020 deg on 0x02AA. Geolocation error 0x02C7 180 metres, about five ground pixels at 33.4 metre GSD."
- The estimator sources in use are readable as a bitmask at `aocs.est_source` (0x02C1): bit 0 star tracker, bit 1 CSS, bit 2 magnetometer, bit 3 gyro. Gyro bias calibration is an **on-board** activity

(`AOCS_GYRO_CALIBRATION` , S8 func 13), not a ground-computed upload, and the three gyro axes have no redundant unit behind them.

Decision points

- **Step 4: FINE not achieved within 300 s.** Re-read 0x02C5 — it will name the reason. Two paths: wait (the MEKF sometimes just needs more convergence time); or fall back to NOMINAL (mode 4) and invoke AOCS Anomaly. Industry default: wait once, then fall back.

COM-006 — TTC Link Verification

This card was rewritten in Phase 5, and the rewrite is worth understanding rather than skimming: the old version measured a link it had never switched on, and asserted an end state it never commanded.

The rate ladder has three rungs, defined in `subsystems/ttc.yaml` :

Tier	Rate	`rate` argument	Role
BEACON	1 kbps	not selectable by func 65	Bootloader / beacon lifeline
LOW	16 kbps	`rate=0`	**Routine operations** and the safe-mode fallback
HIGH	128 kbps	`rate=1`	Delayed telemetry and image dumps; gated and earned

The legacy binary encoding is preserved: 0 still means "the low rate" and 1 still means "the high rate" — the rates themselves moved. 1 kbps was never an operations rate: the shipped housekeeping set generates 1378.5 bps of PUS packets, which is 1588 bps of channel after RS(255,223) and TM framing, so a 1 kbps link is 159 % subscribed before anything else is sent.

Per-step playbook

Step 0 — Configure the S-band chain (this step is the whole point of the rewrite)

The rate ladder is gated on four things, and until all four are satisfied both operational tiers are refused with S1.8 error `0x0007` :

- `EPS_POWER_ON(line_index=2)` (S8 func 19) — the `ttc_tx` line. Verify `eps.pl_ttc_tx` (0x0112) = 1.
- `TTC_PA_ON` (S8 func 66) — the S-band power amplifier. Verify `ttc.pa_on` (0x0516) = 1.
- `TTC_SET_BEACON_MODE(on=0)` (S8 func 70) — release the forced beacon tier. Verify `ttc.beacon_mode` (0x0521) = 0. Beacon mode **survives an application boot**, so a spacecraft can arrive at this card still latched.
- `TTC_DEPLOY_ANTENNA` (S8 func 69). Verify `ttc.antenna_deployed` (0x0520) = 1. Deployment is one-way burn-wire.

Then confirm the chain is actually radiating: `ttc.tx_active` (0x0525) = 1. If it reads 0, `ttc.tx_gate_status` (0x052B) names the missing permission (bit 0 PA, bit 1 `ttc_tx` line, bit 2 station in view, bit 6 antenna).

- TT&C: "FD, TT&C — S-band chain configured, PA on, beacon tier released, antenna deployed, `tx_active` reads 1."

A teaching point that belongs here. EOSAT-1 has **two transmitters**: the S-band prime chain and a separate UHF whip beacon on the non-switchable `ttc_rx` line. Turning the PA off, or pulling the `ttc_tx` line, stops S-band telemetry — and the beacon carries on. That is why HK SID 11 keeps arriving from a spacecraft

whose S-band chain is completely mis-configured, and it is the single most useful diagnostic in a no-telemetry pass. Do not confuse `ttc.beacon_mode` (0x0521, the *forced TM tier* on the S-band ladder) with `ttc.beacon_tx_on` (0x0526, the UHF *transmitter*).

Step 1 — Baseline link, LOW tier

- `HK_REQUEST(sid=6)` (Service 3, **Subtype 27**) — TT&C housekeeping.
- Verification: `ttc.link_status` (0x0501) = **2** (LOCKED — not 1); `ttc.rssi` (0x0502) > −100 dBm; `ttc.link_margin` (0x0503)

> 6.0 dB.

Step 2a — Baseline each rung of the ladder

- `TTC_SET_DATA_RATE(rate=0)` (S8 func 65) — LOW. Verify `ttc.tm_data_rate` (0x0506) = 16000 and `ttc.tm_tier` (0x0539) = 1. Record margin: expect 55–59 dB at 30° elevation.
- `TTC_SET_DOWNLINK_MODE(on=1)` (S8 func 79) — verify `ttc.downlink_mode` (0x053C) = 1.
- `TTC_SET_DATA_RATE(rate=1)` — HIGH. Verify 0x0506 = 128000, 0x0539 = 2, and `ttc.tm_symbol_rate` (0x053B) = 64000 Bd — the QPSK step is visible on the station spectrum analyser as the same information rate in half the bandwidth. Record margin: expect 45–48 dB at 30°.
- **Compute the step: margin(LOW) − margin(HIGH) must be 10.5 ± 0.5 dB.** That number is not empirical — it is 8× the rate (−9.03 dB) plus the coding change from conv+RS 7.5 dB to RS 6.0 dB (−1.5 dB). A measured set that does not show it indicates a modelling error, not a link problem.
- Then drop back: `TTC_SET_DATA_RATE(rate=0)` **and** `TTC_SET_DOWNLINK_MODE(on=0)`. Both, as two separate commands.

Step 2b — Confirm the HIGH-tier elevation gate

- As the pass descends below 10° on `ttc.contact_elevation` (0x050A), attempt `TTC_SET_DATA_RATE(rate=1)`.
- Verification: the command is **rejected** with an S1.8 execution failure naming the elevation gate, and 0x0506 stays at 16000.
- This is designed behaviour, not an anomaly. A rejected tier is how the spacecraft tells the ground a rule was not satisfied; a silent clamp would hide it. Log the rejection as evidence the gate works.

Step 3 — Record margins

- SPACON logs LOW and HIGH margins per station and elevation. These become the mission link-budget baseline.
- **Read the numbers before you interpret them.** The worst case in the whole shipped matrix is HIGH at Troll at 5° elevation, and it is **33.7 dB**. Every rung closes with tens of dB at every station and every geometry: this link is not power-limited. The 3 dB and 6 dB GO/NO-GO thresholds in the card are *sanity* thresholds — a measurement anywhere near them means something is wrong with the model, the attitude or the ground station, not that the geometry is marginal. By the same token the 10° gate on HIGH is a **policy** gate, not a margin gate.

Step 9 — Safe the link

- `TTC_SET_DATA_RATE(rate=0)`, `TTC_SET_DOWNLINK_MODE(on=0)`, `TTC_SWITCH_PRIMARY` (S8 func 63).

- Verification: `ttc.tm_tier` (0x0539) = 1 **and** `ttc.tm_tier_cmd` (0x053A) = 1 — the active tier and the commanded tier must agree; `ttc.downlink_mode` (0x053C) = 0; `ttc.beacon_mode` (0x0521) = 0.
- The antenna is **not** re-stowed and the PA is **not** switched off. Deployment is irreversible, and a commissioning card that ended by silencing the transmitter would hand the next pass a spacecraft on beacon only.
- The lesson: configuration a card *asserts* but never *commands* is configuration the next shift inherits by luck.

COM-007 — OBDH Checkout

Per-step playbook

Step 1 — Verify OBC mode, CPU and memory

- SPACON: "FD, SPACON — `HK_REQUEST(sid=4)`. `obdh.mode` (0x0300) reads NOMINAL, CPU load 0x0302 at 34 percent, mass-memory usage 0x0303 at 21 percent, EDAC/SEU counter 0x0319 unchanged since boot."
- Note the platform structure is **SID 4**, not SID 3 — the housekeeping SID map is 1 EPS, 2 AOCS, 3 TCS, 4 OBDH/platform, 5 Payload, 6 TT&C, 8 EPS-extended, 9 PROP, 11 Beacon. Requesting the wrong SID and then puzzling over the parameters that come back is one of the more common novice errors.

Step 2 — Check the TC/TM counters

- `GET_PARAM(param_id=0x0304)` `obdh.tc_rx_count`, `0x0305` `obdh.tc_acc_count`, `0x0306` `obdh.tc_rej_count`, `0x0307` `obdh.tm_pkt_count`.
- SPACON: "FD, SPACON — received 214, accepted 214, rejected 0."
- Watch out: `obdh.sw_image` (0x0311) is the *running image* (0 = bootloader, 1 = application), not a rejection counter. Reading it as one is a documented past defect.

Step 3 — Watchdog, then the mode excursion

- `OBC_DIAGNOSTIC` (S8 func 61) and `GET_PARAM(param_id=0x0316)`: watchdog armed, `obdh.last_reboot_cause` = 0. The simulator card does no S6 memory dump; memory health is the error and SEU counters of step 1, read twice.
- `OBC_SET_MODE(mode=1)`, then `OBC_SET_MODE(mode=0)` (S8 func 50).
- SPACON: "FD, SPACON — watchdog armed, no resets; SAFE confirmed, NOMINAL restored, `obdh.mode` reads 0."

Step 4 — Verify bus A/B health

- `obdh.active_bus` (0x030E), `obdh.bus_a_status` (0x030F), `obdh.bus_b_status` (0x0310).
- SPACON: "FD, SPACON — bus A active, both CAN buses report OK, no error counts."

COM-008 — FDIR Configuration

This is one of the most consequential commissioning steps. FDIR on EOSAT-1 is not one mechanism but three, and the operator needs to keep them apart:

- **The FDIR rule set** (`subsystems/fdir.yaml`) — 17 parameter conditions, each with a `level` and a named autonomous action, evaluated every tick.

- **PUS Service 12 on-board monitoring** (`monitoring/s12_definitions.yaml`) — 46 limit definitions, each tagged WARNING or ALARM. These produce violations, not actions.
- **PUS Service 19 event-action rules** (`monitoring/s19_rules.yaml`) — 32 rules, each linking an on-board condition to one S8 function the spacecraft executes with no operator in the loop.

Once these are armed, the spacecraft acts on its own. If any condition is *currently* true (e.g. a sensor reading out of band), the response fires immediately — which is why the pre-load check below is not a formality.

Severity and autonomy are two different scales. Do not conflate them.

- **FDIR severity runs 0–4**: 0 nothing/normal, 1 INFO, 2 WARNING, 3 ALARM, 4 CRITICAL. It answers *how bad is this*.
- **Autonomy level runs 0–3**: 0 unit level (hardware, < 1 s — a PCDU over-current trip); 1 subsystem level (subsystem software, 1–10 s — a heater thermostat); 2 system level (the OBC FDIR application, 10–60 s — a mode transition or load shed); 3 ground level (the flight control team, minutes to hours — manual recovery). It answers *who is allowed to act*.

A severity-4 event handled at autonomy level 3 is a critical fault that only the ground may touch. A severity-1 event handled at autonomy level 0 is a thermostat doing its job. The two numbers are independent, they are reported separately, and an operator who reads one as the other will consistently mis-brief the FD.

Per-step playbook

Step 1 — Pre-load check

- SPACON: "FD, SPACON — confirming all subsystems nominal before arming."
- Each position confirms: EPS, AOCS, Thermal, TT&C, Payload all report OK.

Step 2 — Verify the rule set is active and read the thresholds

- SPACON reads back the master enable and the active rule count, then walks the critical thresholds and confirms they match the configured ladder — under-voltage `safe_mode_eps` at **27.7 V** (`simulation.yaml` `eps.bus_protection.eps_safe_v`; the UV flag and in-model shed are at 28.4 V), battery SoC actions at 20 % (payload power-off) and 15 % (EPS safe mode), sustained attitude error > 5° for 420 s, OBDH temperature > 65 °C, reboot count > 4.
- (15 s wait, watching the FDIR panel) "FD, SPACON — nothing fired in the first 15 seconds; the rule set is happy with the current state."

Step 3 — Inject a benign test condition

- SPACON: "FD, SPACON — going to test that FDIR is actually monitoring. Brief telemetry perturbation on battery temperature."
- (perturbs `tcs.temp_battery` (0x0407) into the caution band and restores it)
- "FD, SPACON — the monitor tripped within 3 seconds at WARNING, no autonomous action. It cleared within 2 seconds of my restoring the value. FDIR is live."

Step 4 — Verify no autonomous action was taken

- SPACON: "FD, SPACON — confirmed: the perturbation stayed below the threshold that carries an action, so nothing was commanded autonomously. Autonomy stayed at level 1. FDIR panel clean. GO COM-009."
- Note also that housekeeping generation is commanded, not automatic: only SIDs **1 and 11** are enabled at power-up, and the engine re-applies exactly that set on application boot. Everything else needs `HK_ENABLE`, which is **Service 3, Subtype 5**. (`HK_REQUEST` is Service 3, Subtype 27. Any reference to S3 subtype 1 or 25 for a housekeeping request is wrong.)

COM-009 — Payload Power-On

Per-step playbook

Pre-step pre-flight check (~5 min before commanding)

- Payload: "FD, Payload — pre-flight: imaging-target catalogue loaded, calibration sets not yet taken, `payload.store_used` (0x0604) at 20 percent of the 20 GB store, `payload.product_count` (0x0648) zero."
- EPS: "FD, EPS — battery 84 percent. The cooler is 15 watts and reaches setpoint in about ten minutes, so the ramp costs roughly 2.5 watt-hours — two and a half percent of the 100 watt-hour pack. Well above the 50 percent red line."
- Thermal: "FD, Thermal — FPA at plus 11 C, its uncooled equilibrium; cooler off, ready to receive the payload heat load."

Step 1 — FD GO

- FD polls all positions: "EPS, Thermal, AOCS, TT&C — go for payload power-on?"
- Each: "FD, [POSITION] — go."
- FD: "Go for payload power-on."

Step 2 — Energise the line, then command the mode

- Payload: "FD, Payload — first-key `EPS_POWER_ON(line_index=3)`, the payload line." FD second-keys (high-crit).
- (2 s) "FD, Payload — accepted. `eps.pl_payload` (0x0113) reads 1."
- Payload: "first-key `PAYLOAD_SET_MODE(mode=1)`, STANDBY."
- (5 s) "FD, Payload — total load up 2.4 watts, about 0.08 amps at 32 volts. Overcurrent trip on this line is 2.5 amps, so we are nowhere near it."
- **The order matters and the card enforces it.** A `PAYLOAD_SET_MODE` without the EPS line energised reverts on the next tick: the payload cannot hold a mode it has no power for. Re-request SID 5 after a further 60 s and confirm the mode has *stayed* at 1 rather than quietly falling back to 0.

Step 3 — Verify mode

- Payload: "FD, Payload — `payload.mode` (0x0600) reads 1 STANDBY, held over two housekeeping cycles. Mode change confirmed."
- The payload mode set is **0 OFF, 1 STANDBY, 2 IMAGING, 3 CALIBRATION**. Mode 3 is new and is a real instrument mode, not a flavour of imaging — see COM-011.

Step 4 — Verify storage and health

- Payload: "FD, Payload — `payload.store_used` (0x0604) at 20 percent of the 20 GB store; `payload.product_count` (0x0648) zero and `payload.product_mb` (0x0647) zero — no products on board. `payload.imager_status` (0x0611) shows no fault bits. Ready for COM-010."

Decision points

- **Line current well above the 2.4 W / 0.08 A expectation but below the 2.5 A trip.** Two reads: cold-start inrush settles in 30 s (wait); or genuinely a stuck-high draw (CTG-012 Overcurrent Response). Wait 30 s first.
- **`payload.imager_status` (0x0611) shows a fault bit.** Record it, command `PAYLOAD_SET_MODE(mode=0)`, and do **not** proceed to cooler activation.

COM-010 — FPA Cooler Activation

This is a fast step, and the book used to say otherwise. The FPA cooler is a **miniature Stirling-class cryocooler**, not a thermo-electric/Peltier stack, and the corrected single-node model settles it in about ten minutes.

- Starting point: the **uncooled equilibrium, +8 to +14 °C**, not the +5 °C ambient the older text assumed.
- Setpoint: **−15 °C** (`payload.yaml fpa_cooler_target_c`). The acceptance band is [−16.5, −14.5] °C.
- Initial rate ≈ **4 °C/min**, falling exponentially. The ramp is first-order, not linear, so a slower reading at T+5 than at T+2 is expected and is not a fault.
- Power: **15 W**, billed on EPS line 4 (`fpa_cooler`).
- **There is no eclipse / day-side dependence in the FPA model.** Any procedure that splits the cool-down slope by illumination is describing a model that does not exist. This card does not need to span an eclipse.

Per-step playbook

Step 1 — Verify FPA at uncooled equilibrium, cooler OFF

- Thermal: "FD, Thermal — `payload.fpa_temp` (0x0601) at plus 11.5 C, inside the plus-8 to plus-14 band. Cooler off, ready for enable. Marking cooldown T-zero."

Step 2 — Enable cooler

- Thermal: "FD, Thermal — first-key `FPA_COOLER(on=1)`, Service 8 Subtype 1, func_id 43." FD second-keys (high-crit, largest single load on the spacecraft).
- Verification, and this is the point of the card: **both** `tcs.cooler_fpa` (0x040C) = 1 *and* `eps.pl_fpa_cooler` (0x0114) = 1. The historical defect was a command path that set the cooler flag without energising the EPS line, so the operator believed the cooler was running and the FPA never got cold. All command paths now energise the line, so if 0x040C and 0x0114 ever disagree again, that is the defect returning.
- (30 s) "FD, Thermal — `eps.power_cons` (0x0106) up 15 watts against baseline, bus holding 32.0 V."

Step 3 — Monitor the ramp (first 5 minutes)

- Thermal samples `payload.fpa_temp` (0x0601) every minute.
- At T+2 min: below +4 °C.
- At T+5 min: below −8 °C.
- "FD, Thermal — on the exponential, rate falling as expected."

Step 4 — Monitor settling (5–15 minutes)

- At T+10 min: in [−16.0, −14.0] °C.
- At T+15 min: in [−16.0, −15.0] °C.
- **The detector never goes below the setpoint band.** A reading colder than −16.5 °C is a control fault, not an over-performing cooler; the loop regulates to the setpoint, it does not run open-loop.

Step 5 — Verify readiness, not temperature

- Verification: `payload.fpa_temp` (0x0601) in [−16.5, −14.5] °C **and** `payload.fpa_ready` (0x0610) = 1.
- The readiness flag, not a remembered temperature, is the gate for imaging. It carries 60 s of hysteresis, so expect it to set about a minute after the temperature arrives.
- Also read `payload.cooler_floor_c` (0x064A) and `payload.cooler_margin_k` (0x064B). Margin should be about +5 K at the −15 °C setpoint against a +12 °C internal sink.
- The setpoint itself is a **uint16 in centi-kelvin** on the wire: −15 °C is 25815 cK = `0x64D7`. Get that unit wrong and the command is refused or the setpoint is nonsense.

Decision points (this shift's anomaly territory)

- **`cooler_margin_k` (0x064B) ≤ 0.** The cooler physically cannot reach the commanded setpoint at the present sink temperature and load. The answer is a **warmer setpoint, not more waiting.** The cooler's authority degrades with lift: it runs at about 23 % of Carnot, so holding −15 °C against a +12 °C sink is a 27 K lift pumping 22.5 W of parasitic heat on 15 W of input — a required COP of 1.50 (about 16 % of Carnot, inside the cooler's 23 %), where a single-stage Peltier would top out near 0.88 at the same lift. Ask for less lift and you get it.
- **Temperature falling far more slowly than 4 °C/min.** Check the cooler draw first: `payload.cooler_pwr` (0x0602) should read 15 W. If it reads zero while `tcs.cooler_fpa` reads 1, check `eps.pl_fpa_cooler` (0x0114) — the line, not the flag.
- **Settling at a warmer temperature than −15 °C.** Continue and accept the warmer setpoint (SNR degraded but usable, logged as an anomaly), or abandon and defer First Light. Real-mission protocol: accept the suboptimal setpoint, acquire one image, and defer only if the image quality is genuinely poor.

COM-011 — Payload Calibration

Per-step playbook

Radiometric calibration is a **shutter-closed / lamp-lit** activity that the instrument performs in its **own mode: CALIBRATION, `payload.mode` = 3.** The old design forced the operator into IMAGING first — which meant opening the shutter and integrating the scene before asking for a dark frame, the one thing a dark frame must not have. Dark calibration no longer requires IMAGING mode.

Five dedicated refusal codes exist and are worth memorising, because each names exactly what is wrong:

Code	Meaning
0x0024	Calibration refused because the payload is IMAGING
0x0025	FPA not ready
0x0026	Shutter in the wrong position
0x0027	Lamp in the wrong state
0x0028	Capture attempted with the shutter closed

(0x0021–0x0023 remain the capture-*pointing* refusal codes and are a different family.)

Step 1 — Verify the pre-calibration state

- Payload: "FD, Payload — `payload.mode` (0x0600) 1 STANDBY, `payload.fpa_temp` (0x0601) minus 15.2 C inside the band, `payload.fpa_ready` (0x0610) reads 1."

Step 2 — Close the shutter

- Payload: "FD, Payload — first-key `PAYLOAD_SET_SHUTTER(position=0)`, CLOSED" (S8 func 119). There is an absolute shutter command; the shutter is not something you nudge.
- Verification: `payload.shutter_pos` (0x063C) = 0.

Step 3 — Capture the dark set

- Payload: "first-key `PAYLOAD_CAL_LAMP(on=0)`" (S8 func 95) — lamp off for a dark frame.
- Payload: "first-key `PAYLOAD_START_CALIBRATION(mode=0)`" (S8 func 37) — the DARK phase.
- Verification: `payload.mode` (0x0600) = **3** (CALIBRATION) and `payload.calibration_active` (0x062A) = 1 within 15 s; then 0x062A returns to 0 within 120 s with `payload.cal_abort_reason` (0x0645) = 0.
- "FD, Payload — dark set complete, `payload.dark_frame_cnt` (0x0642) non-zero, mean dark signal 32 DN on the 12-bit 0–4095 scale, standard deviation under 10 DN. No light leak."

Step 3a — Capture the flat set

- Payload: `PAYLOAD_SET_SHUTTER(position=1)` OPEN, then `PAYLOAD_CAL_LAMP(on=1)` — the **internal** calibration lamp; this is not an over-ocean acquisition.
- Verification: `payload.cal_lamp_on` (0x0615) = 1 and `payload.shutter_pos` (0x063C) = 1.
- Payload: `PAYLOAD_START_CALIBRATION(mode=1)` — the FLAT phase.
- Verification: `payload.cal_valid_mask` (0x0646) = 15 — all four bands valid. Then lamp off.
- Order is free: DARK-then-FLAT and FLAT-then-DARK both work, and `PAYLOAD_START_CALIBRATION(mode=2)` runs both in one command.

Step 4 — Return to STANDBY and prepare for the overpass

- Payload: `PAYLOAD_SET_MODE(mode=1)`, `PAYLOAD_SET_SHUTTER(position=1)`.
- AOCS: `AOCS_FINE_READY_CHECK` (S8 func 89) — confirm `aocs.fine_ready` (0x02C5) = 0 — then `AOCS_SET_MODE(mode=5)`, FINE_POINT.
- Verification: `aocs.mode` (0x020F) = 5 within 30 s; `aocs.att_error` (0x0217) < 0.1° within 300 s.
- "FD, Payload — calibration sets committed. GO COM-012."

COM-012 — First Light (the headline procedure)

Per-step playbook

Step 1 — FD GO for First Light

- FD: "All positions, listen up. We're going for First Light. This is the customer-facing deliverable for Week 3. AOCS, pointing?"
- AOCS: "Mode 5 FINE_POINT, `aocs.total_pointing_err_3sig` (0x02AA) 0.020 deg, well inside the 0.05 deg imaging gate."
- "FD: Thermal, FPA?"
- Thermal: "FPA at −15.2 C, in the −16.5 to −14.5 C band, and `payload.fpa_ready` (0x0610) reads 1."
- "FD: Payload, cal frames?"
- Payload: "Dark and flat both taken, `cal_valid_mask` 15."
- "FD: TT&C, downlink?"
- TT&C: "HIGH tier available above 10 degrees, margin 46 dB at TCA over Iqaluit. A 4 MiB product is 262 seconds of channel at 128 kbps, 328 seconds with the protocol allowance — about three-quarters of the pass."
- "FD: FDS, target?"
- FDS: "Target the Labrador Current outflow zone, scene centre 52.5 N, 51.2 W. Footprint 273 km across-track by 59 km along-track for a 2048-line strip, 33.4 metre GSD. Sun glint angle 32 deg (acceptable), cloud cover 8 percent per latest forecast."
- "FD: Ground Seg, downlink ready?"
- Ground Seg: "All five stations configured; Iqaluit has the next pass and it clears 10 degrees, so the HIGH tier will hold."
- "FD: Mission Manager, customer expecting this?"
- Mission Manager: "Customer is watching the feed. Bulletin prepped."
- "FD: GO FOR FIRST LIGHT."

Step 2 — Capture

- Payload: "FD, Payload — first-key `PAYLOAD_SET_MODE(mode=2)`, IMAGING." FD second-keys. Verify `payload.mode` (0x0600) = 2 within 30 s.
- Payload: "first-key `PAYLOAD_CAPTURE(scene_id=1, lines=2048)`" (S8 func 28).
- (At trigger) "FD, Payload — acquiring."
- (8.4 s later) "FD, Payload — strip complete." The commanded strip length is `lines / line_rate` = 2048 / 244 Hz = **8.4 seconds**, which at the 7.06 km/s ground-track speed is the 59 km along-track footprint FDS quoted.
- Geolocation is stamped **on board** from the ephemeris at shutter open; it is not a commanded field.

Step 3 — Verify image stored

- Payload: "FD, Payload — `payload.last_scene_id` (0x060C) reads 1, `payload.product_count` (0x0648) incremented, product size reported on `payload.product_mb` (0x0647). All four bands present, no CRC errors."
- **A note on the frame size.** The courseware carries a 4 MB raw frame (`image_size_mb`), and the focal-plane geometry — 8192 px × 4 bands × 12 bit × 2048 lines — implies 100.7 MB. That gap is deliberate and documented, not an error to be papered over: at 100.7 MB an image would need about 1.75 hours of effective contact at HIGH (128 kbps), some fourteen average passes, and no plan

would ever close. On board, the CCSDS-123 compressor stores 1.0–2.7 MB of the 4 MB frame depending on scene content.

Step 4 — Queue for downlink

- Payload: "FD, Payload — first-key `S13_CREATE_TRANSFER(scene_id=1)` — Service 13, Subtype 1." FD GO.
- Verification: TM(13,2) returns the transfer id, the total block count and the **crc32 the instrument computed over the product at capture**. Recompute that CRC over the received blocks on the ground: a corrupt product will not match, and that is how a corrupted image is diagnosed rather than guessed at.
- A transfer for a scene that is not in the on-board catalogue is **refused** (S1.8 error 0x0003 plus a TM(13,4) NAK). Use `PAYLOAD_GET_IMAGE_CATALOG` (S8 func 32) to see what is actually on board.

Step 5 — Verify ground reception

- Blocks come down with `S13_DOWNLOAD_SEGMENT` (S13.3) and the session is closed with `S13_END_TRANSFER` (S13.5); progress is polled with `S13_REPORT_STATUS` (S13.9).
- (Next pass) Ground Seg: "FD, Ground Seg — downlink complete, 4 megabytes received in 262 seconds of channel at the HIGH tier, CRC matches the value the instrument reported at capture."
- Note that closing a session whose blocks were **all** sent deletes the product on board and returns its storage (event 0x0614). A partial or aborted session leaves the product where it is — which is exactly the "the pass dropped, re-dump next orbit" case the service exists for.

Step 6 — Quality assessment

- Payload: "FD, Payload — image quality assessment: mean DN per band 1840 / 2102 / 1654 / 891 on the 12-bit 0–4095 scale, contrast within expected range. `aocs.geoloc_err_m` (0x02C7) reads 180 metres at the 0.020 degree pointing error — about five ground pixels at 33.4 metre GSD. IMAGE IS SCIENCE USABLE."
- Mission Manager: "FD, MM — customer bulletin going out: EOSAT-1 First Light achieved at 14:32:18 UTC, target Labrador Current, downlink complete, science-usable image acquired."

Decision points

- **Step 2: `aocs.total_pointing_err_3sig` (0x02AA) drifts to 0.042 deg just before the capture trigger** — still under the 0.05 deg gate, but only just. Two reads: defer one orbit (chase a better attitude); or fire anyway (image quality marginal but usable). Real-mission protocol: if the margin to spec is under 30 %, defer. Here 0.042 / 0.05 = 84 % of budget consumed, i.e. 16 % margin — defer. Check 0x02C5 while you wait: if bit 1 is set the spacecraft is flying FINE on three wheels (`aocs.fine_degraded` 0x0299 = 1) and the extra jitter is explained rather than mysterious.
- **Step 6: image quality assessment shows mean DN at 2900 in band 443 (oversaturated)** — integration time was too long for the actual scene brightness. Result is partially saturated. Decisions: declare First Light *acquired* (the data exists) but flag for re-acquisition next pass with shorter integration. Customer bulletin tells the truth: "First Light achieved; image partially saturated, re-acquisition planned."

11.10 References

- ECSS-E-ST-70-11C — Space segment operability — ecss.nl

Chapter 12 — Contingency Operations

The three operational phases covered so far — nominal passes, LEOP, and commissioning — share a defining property: **you know what is coming.** A nominal pass runs NOM-001 through NOM-013 in an order written days in advance. A LEOP shift follows a procedure arc rehearsed in the simulator until the team can recite it. Commissioning is a four-pass script with named entry and exit conditions. The work is demanding, but the *shape* of the work is known before the pass starts.

A contingency is the opposite. It is the pass where something happens that was not on the timeline. The skill this chapter develops is not "how to run procedure X" — those live in the catalogue (Chapter 14). It is the more general skill of reacting well when the situation in front of you does not match any single card you have.

12.1 What a contingency is

A contingency is any condition the team did not plan to be in at this point in the pass: an FDIR rule fires that should not have, a telemetry parameter walks outside its limit, a link drops, a wheel saturates early, a load appears that nobody commanded. The defining feature is not severity — an FDIR level-1 (caution) can be a contingency and a planned safe-mode rehearsal is not — but *surprise*. The team is now operating off-script.

Here is the apparent paradox that every operations organisation lives with, and that this chapter exists to resolve: **the real contingency almost never looks exactly like the one you trained for, yet maintaining a full set of planned-for contingency cases is precisely what lets you handle the one you did not plan for.** The link-loss ladder you drilled does not anticipate the specific way *this* link fails; the under-voltage card assumes a fault tree that the actual battery does not follow. And yet the operator who has worked those cards reaches for the right reflexes in seconds: safe the affected subsystem, confirm the symptom is real, protect power and attitude, talk on the loop, do not make it worse. The planned cases are not valuable because the real event matches them. They are valuable because rehearsing them builds the *decision structure* — the order of operations, the instinct to safe before diagnosing, the muscle memory of the callouts — that transfers wholesale to a situation no card describes.

This is why a mature operations team keeps and drills a contingency catalogue it knows it will never invoke verbatim. The cards are a training instrument as much as a runbook. An operator who has worked twenty contingency scenarios in the simulator has not memorised twenty fixes; they have internalised one way of thinking about trouble, and that way of thinking is general.

12.2 The contingency mindset and decision framework

When something unexpected appears, the worst outcomes in operations come not from the original fault but from the team's reaction to it — a hurried command that compounds the problem, a diagnosis chased while the battery quietly drains, a recovery attempted before anyone confirmed the symptom was real. The framework below is the same regardless of which subsystem is involved, and it is deliberately ordered. The governing principle is borrowed from medicine: *first, do no harm*. Every action should be reversible if at all possible, and when reversibility is in doubt, the bias is toward doing less, not more.

The first step is to **recognise and confirm**. Is the indication real, or is it a telemetry artefact — a single dropped frame, a sensor glitch, a stale value held over from before LOS? Cross-check the suspect parameter against an independent source: a second sensor, a derived quantity, the frame count, the trend rather than

the instantaneous value. A surprising number of "anomalies" evaporate at this step. Confirming costs seconds; reacting to a phantom costs the pass.

Once the condition is confirmed real, the second step is to **preserve the spacecraft**. Before understanding *why* something is wrong, make sure it cannot get worse. Safe the affected subsystem — take a misbehaving payload to standby, halt a manoeuvre, stop commanding into a fault. Protect the three things that keep the spacecraft alive across the next eclipse and the next gap in coverage: **power** (is the battery draining, is a load uncommanded?), **thermal** (are limits being approached, are heaters available?), and **attitude** (is the spacecraft holding pointing, are the wheels or sun-pointing under control?). Preservation buys time; almost nothing else does.

With the spacecraft safe, the third step is to **isolate the cause**: what changed, and what is affected? The most powerful diagnostic question in operations is "what was the last thing we did, and what was the last thing the spacecraft did autonomously?" Correlate the onset of the symptom with the command log, the FDIR event log, the orbit geometry (eclipse entry, station transition), and any recent procedure step. Distinguish the *primary* fault from its *consequences* — an under-voltage that tripped a load shed that dropped the transmitter is one fault with three symptoms, not three faults.

The fourth step is the **decision**: recover now, defer, or escalate. Recover now if the cause is understood, the recovery is carded or obvious, the spacecraft is stable, and there is pass time to verify the result before LOS. Defer if the condition is stable and benign and the recovery would be better done with analysis, with the right people, or on a later pass — there is no prize for fixing something this pass that will keep until next pass. Escalate to safe mode when the spacecraft is not stable, the cause is not understood, or the team is out of time: safe mode is not a failure of operations, it is the designed-in fallback that keeps the spacecraft alive and pointed at the Sun while the ground works the problem offline. Only the Flight Director authorises safe-mode entry, and the Mission Manager is informed.

The fifth step runs in parallel with all the others: **communicate and log.** Every observation, hypothesis, and action goes on the voice loop as it happens, in the discipline of Chapter 9 — directed callouts, read-backs, FD GO before commanded actions. Every step is timestamped in the shift log, and the anomaly ticket is opened immediately, not written from memory after the pass. A clean log is what lets the next shift, the analysts, and the manufacturer reconstruct what happened.

12.3 Anomaly-response decision tree

The framework in §12.2 maps directly onto how FDIR classifies and routes an event on the reference spacecraft. The decision tree below is the operational expression of that framework: an autonomous trigger is classified by severity, and the severity determines whether the team logs and watches, runs a recovery procedure, or treats the event as a safe-mode candidate for the Flight Director to adjudicate. Read it as the "decide" step of §12.2 rendered as a flow — the same recognise, preserve, isolate, decide logic, keyed on *how bad the condition is*.

Two scales, and they are not the same scale. EOSAT-1 reports an **FDIR severity** on a **0–4** scale (0 nothing, 1 INFO, 2 WARNING, 3 ALARM, 4 CRITICAL) and an **autonomy level** on a **0–3** scale (0 unit/hardware, 1 subsystem software, 2 OBC FDIR application, 3 ground). Severity says how bad the condition is; autonomy level says who is permitted to act on it and how quickly. The decision tree below is keyed on **severity**. Reading a severity as an autonomy level — or the reverse — is one of the most persistent mistakes on console, because both numbers are small integers arriving on the same panel, and both are *plausible* in the other's place. Say which one you mean, every time, on the loop.

Anomaly-response decision tree: an FDIR trigger is classified by severity (0 → log only; 1–2 → recovery procedure; 3–4 → safe-mode candidate, with the Flight Director authorising safe-mode entry or directing a recovery procedure).

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The severity bands here are the same ones tabulated in the FDIR escalation ladder (§15.4): severity 0 is informational and logged, severities 1–2 route to a recovery procedure, and severities 3–4 are severe-to-critical and become safe-mode candidates. The tree decides *which kind of response*; the cards in §14.4 and §14.5 supply the steps. The autonomy level is reported alongside and tells you something different again: whether the spacecraft has already acted without you (levels 0–2) or is waiting for the ground (level 3).

12.4 Classes of contingency

Contingencies cluster into a small number of families. An operator who recognises which family a surprise belongs to has already narrowed the response, because each family has a characteristic preservation move and a set of carded recoveries. The contingency procedure cards live in **Chapter 14 §14.4 (Contingency procedure cards)** and the emergency cards — the ones that approach or enter safe mode — in **§14.5 (Emergency procedure cards)**. The families below are the ones an EOSAT-1 operator should be ready for.

Loss of communications. The link degrades or drops: `ttc.link_status` (0x0501) falls away from **2 = LOCKED**, `ttc.carrier_lock` (0x0510) or `ttc.frame_sync` (0x0512) clears, the TM packet counter stops advancing, or `ttc.ber` (0x050C) climbs. The first preservation move is to confirm the ground station is still tracking and the geometry still supports the link before assuming a spacecraft fault. The single most informative observation is **whether the UHF beacon is still arriving**: the beacon is a separate transmitter on the non-switchable `ttc_rx` line, so HK SID 11 survives an S-band PA that is off, a `ttc_tx` line that has been shed, a failed transponder and an undeployed patch. Beacon present but S-band silent narrows the fault to the S-band chain in one step; beacon absent too is a far more serious finding. The carded response walks the TTC link-loss recovery ladder — passive link test, data-rate reduction to LOW, FD-authorised tx-power increase — escalating to **EMG-004 Loss of Communication** only if none of the rungs restores the link. The spacecraft's own 24-hour command-loss timer is the ultimate backstop: if no valid telecommand arrives for 24 h, it switches to the other transponder, and again every 24 h until a telecommand is accepted (FDIR rule TTC-01, S19 ea 4003; `cmd_loss_switch_timeout_s` in `ttc.yaml`). The switch is blind — the spare is cold — and with both transponders failed the link stays cut: the timer keeps toggling, harmlessly, and finds a unit that recovers.

Power and EPS anomalies. Bus over-current, bus under-voltage, or an uncommanded load. The preservation move is to protect the battery: shed the least-critical loads first and let sunlit charging recover the state of charge, exactly as carded in the under-voltage load-shed procedure (CTG-001). The shed order is fixed and worth knowing cold: `payload` → `fpa_cooler` → `ttc_tx` → `aocs_wheels` → `prop`, with propulsion deliberately last because dropping that line mid-burn triggers the on-board burn abort. `eps.load_shed_stage` (0x0134) reports how far down the list the EPS has gone. The escalation path runs through progressive load shed (CTG-023) to **EMG-002 Total Power Failure** if voltage continues to fall. Power anomalies are the ones least tolerant of "diagnose first" — the battery does not wait.

AOCS and pointing anomalies. Reaction-wheel saturation, a sensor failure (a sun-sensor head, a star tracker or the magnetometer), or an outright loss of attitude knowledge or control. The preservation move depends on the fault: a saturated wheel needs a desaturation — and a desaturation is a **20 to 50 minute** commitment, not a five-minute one, so it is scheduled, not improvised (§13.7). A failed sensor needs the

estimator reconfigured onto the surviving sensors, using `AOCS_EST_RESET` (S8 func 91) and `aocs.est_source` (0x02C1) rather than any hand-picked source selection. A lost wheel is usually **not** an emergency: FINE_POINT flies on three of the four pyramid wheels, flagged by `aocs.fine_degraded` (0x0299), and the 0x02C5 precondition bit that blocks FINE means "fewer than **three** wheels", not fewer than four. A genuine loss of attitude is a safe-mode candidate because an uncontrolled spacecraft threatens power and thermal within minutes. Sun-pointing safe mode — COARSE_SUN, mode 3 — exists precisely so that, whatever else is wrong, the arrays face the Sun.

Thermal excursions. A subsystem walks toward a temperature limit — most often the payload focal-plane assembly or its cooler. The preservation move is to remove the heat source or restore the cooling loop, and where neither is immediately possible, to shed the offending load. Thermal contingencies usually develop more slowly than power ones, which buys time to isolate the cause, but a runaway cooler or a stuck heater can move quickly.

Payload faults. The instrument misbehaves — it fails to enter a mode, returns corrupt data, draws the wrong power, or will not cool. Because the payload is mission-critical but not survival-critical, the default preservation move is conservative: take the payload to a safe standby state and protect the platform, then investigate. A lost image is recoverable next orbit; a payload fault that drags down the bus is not worth chasing in real time.

FDIR-triggered safe-mode entries. The spacecraft has already acted: an autonomous rule has put the spacecraft into safe mode or is one trigger away from doing so. This is the family where the two scales of §12.3 earn their keep — the *severity* tells you how bad the underlying condition was, and the *autonomy level* tells you that the response has already been taken out of your hands. Here the contingency is partly already handled by design, and the team's job shifts to confirming the entry was appropriate, briefing the Mission Manager, preserving the configuration, and *not* attempting to exit safe mode the same pass. Recovery from a real safe-mode entry is an offline, multi-position activity with explicit FD and Mission Manager authority.

12.5 Per-position contingency responsibilities

A contingency is a whole-room event, but each position has a distinct role drawn from its standing mandate (Chapter 6). What changes in a contingency is tempo and discipline, not who owns what.

The **Flight Director** owns the decision. The FD slows the room down, enforces one-action-at-a-time, gives or withholds GO for every commanded recovery step, and is the only voice that authorises safe-mode entry. The FD's hardest job is resisting the urge to fix quickly; the framework of §12.2 is, in practice, the FD keeping the team to recognise-preserve-isolate-decide rather than jumping to a remedy.

The **SPACON** runs the procedures and keeps the log. In a contingency SPACON loads the relevant contingency or emergency card, drives it step by step against FD GO, acknowledges FDIR rules in the right order, and opens the anomaly ticket immediately so the timeline is captured as it happens rather than reconstructed afterward.

The **AOCS** operator protects attitude: confirming the spacecraft is still holding pointing, watching wheel speeds (0x0207–0x020A, read-only) and body momentum (`aocs.h_body_mag` 0x02B9), executing desaturations, and being first to call when a fault is an attitude-loss safe-mode candidate. The **EPS** operator protects power: watching bus voltage and state of charge, running load sheds, and calling the threshold at which the team must commit to a power emergency. The **Thermal** operator watches the temperature trends of the affected subsystem, manages heaters and the cooler, and advises whether a load can stay on or must be shed.

The **TT&C** operator owns the link: confirming whether a problem is on the link or on the spacecraft, working the link-loss ladder, and protecting the ability to command at all — without the link, no recovery is possible. The **Payload** operator safes the instrument and reports its state, advising the FD on what science is recoverable and what is lost. The **FDS / Flight Dynamics** operator supplies the orbit and geometry context — when eclipse begins, when the next station rises, whether a manoeuvre is implicated — so the team's decisions account for how much time and visibility they actually have.

The **Ground Segment** operator confirms the ground side is healthy and not the cause, executes station or equipment failover if it is, and keeps the data path alive. The **Mission Manager** sits above the console: informed of every contingency, consulted on safe-mode entry, owner of the customer-facing narrative, and the one who decides whether the anomaly changes the mission plan.

12.6 When the contingency is one you did not plan for

Eventually a pass produces something with no matching card. The response is not to improvise freely; it is to reason from the framework and from the *nearest analogous* planned procedure. Ask which family (§12.4) the symptom most resembles, and borrow that family's preservation move even if the diagnosis is uncertain — an unexplained voltage drop is treated as a power contingency and gets a conservative load shed, an unexplained pointing wander is treated as an AOCS contingency and the estimator is watched, because the preservation moves are safe by construction. Then reason from first principles about the actual telemetry: what physically must be true for the spacecraft to show this, and what is the least-irreversible action that tests the leading hypothesis?

Two biases serve the operator well when the situation is genuinely novel. The first is to **bias toward safing and escalation** when uncertain: a spacecraft in safe mode is a spacecraft you still have, and the cost of an unnecessary safe-mode entry is hours of recovery, while the cost of a delayed one can be the mission. The second is to **prefer the reversible action**: hold and observe before commanding, command the smallest step that yields information, and never issue a command whose effect you cannot undo while the cause is still unknown. This is exactly why the planned-for cases matter (§12.1) — they are the rehearsal that makes these reflexes automatic when the real, unrehearsed event finally arrives.

12.7 References

- ECSS-E-ST-70-11C — Space segment operability — ecss.nl
- ECSS-Q-ST-30C — Dependability (FDIR and failure tolerance principles).
- ISO 17666 / ECSS-M-ST-80C — Risk management, as applied to in-flight anomaly handling.
- Contingency and emergency procedure cards: Chapter 14 §14.4 (Contingency) and §14.5 (Emergency); FDIR escalation ladder, §15.4.

Chapter 13 — Operator Calculations and Checks

This chapter covers the calculations and checks every operator on console must be able to perform — without a script, without the planning panel, with a pencil and the back of the shift log. These are the *operator's working math*, distinct from the flight-dynamics analyses the FDS produces.

Six families of operator math are covered:

Table 13.1 — Six families of operator calculations with the trigger for each.

§	Family	When you do it
13.2	Pass-timing arithmetic — AOS / TCA / LOS prediction	Every pre-pass briefing
13.3	Eclipse-exit SoC prediction	Every eclipse-crossing pass
13.4	Pointing-budget RSS	Before every imaging session
13.5	Delayed-TM gap analysis	When a TM packet sequence skips
13.6	Link-margin sanity check	When the link degrades unexpectedly
13.7	Wheel-saturation desat scheduling	When a wheel passes 75% of its speed limit

For each: the formula, a worked example, the typical ranges (so you know when an answer is wrong), and the operational consequence of getting it wrong.

13.2 Pass-timing arithmetic

For a circular orbit at altitude h over a ground station at geodetic latitude ϕ_{gs}, the **geometric AOS-to-LOS transit time** at maximum elevation (overhead pass) is

$$\mathrm{t}_{\mathrm{transit}} = \frac{\mathrm{T}_{\mathrm{orbit}}}{\pi} \arccos\left(\frac{\mathrm{R_E} + \mathrm{h}_{\mathrm{station}}}{\mathrm{R_E} + \mathrm{h}_{\mathrm{sat}}}\right)$$

The orbit period. EOSAT-1's TLE carries a mean motion of 15.24 rev/day, so

$$\mathrm{T}_{\mathrm{orbit}} = \frac{86400}{15.24} = 5669 \text{ s} = 94.5 \text{ min}$$

Worked example — horizon to horizon. With $\mathrm{R_E} = 6371$ km and $\mathrm{h} = 513$ km, $\frac{\mathrm{R_E}}{\mathrm{R_E}+\mathrm{h}} = \frac{6371}{6884} = 0.92548$, and $\arccos(0.92548) = 22.26^\circ = 0.38850$ rad. So

$$\mathrm{t}_{\mathrm{transit}} = \frac{5669}{\pi} \times 0.38850 = 1804.5 \times 0.38850 = 701 \text{ s} = 11.7 \text{ min}$$

With the 5° elevation mask — which is what all five stations actually use — the usable half-angle is

$$\gamma = \arccos\left(\frac{\mathrm{R_E}\cos\varepsilon}{\mathrm{R_E}+\mathrm{h}}\right) - \varepsilon$$
$$= \arccos(0.92196) - 5^\circ = 22.79^\circ - 5^\circ = 17.79^\circ$$

$$\mathrm{t}_{\mathrm{contact}} = \frac{2\gamma}{360^\circ}\,\mathrm{T}_{\mathrm{orbit}} = \frac{35.57}{360} \times 5669 = 560 \text{ s} = 9.3 \text{ min}$$

That is the **overhead maximum**, and it agrees with the mission's measured 565 s longest pass (the altitude varies between 503 and 530 km round the orbit). Real passes are mostly not overhead: the measured **mean pass over the five-station network is 450 s (7.5 min)**. Anyone quoting an eleven-minute working pass is quoting the geometric horizon figure and has forgotten the mask.

Network statistics you should know without looking them up. The teaching baseline is the **full five-station network** — Iqaluit, Troll, Svalbard, Inuvik and O'Higgins, all polar, all on a 5° mask. Measured over 48 h from the mission epoch:

Network	Passes / day	Median gap	Max gap
Iqaluit + Troll only	20.0	84 min	147 min
All five stations	**≈ 51**	**33 min**	**79 min**

That is roughly 23 200 seconds of contact per day, about 27 % of the clock. Every planning exercise in this book assumes the five-station figures.

Operator working rule. Before every pre-pass briefing, read the contact-schedule entry and confirm:

1. AOS time is in the future (not past — sometimes the schedule is stale by an orbit).
2. Maximum elevation is > 10°. This is not about link margin — the margin closes with tens of dB even at 5° — it is because the **HIGH tier is gated at 10°**, so a pass that never clears 10° cannot carry an image dump at all.
3. Pass duration is in family: 7–9.3 min for a good pass, 4–6 min for a low one. A quoted duration above ~9.5 min at this altitude and mask is arithmetically impossible and means the schedule is wrong.

13.3 Eclipse-exit state-of-charge prediction

During eclipse the battery discharges at approximately the orbit- average load $\mathrm{P}_{\mathrm{avg}}$. SoC at eclipse exit is

$$\mathrm{SoC}_{\mathrm{exit}} = \mathrm{SoC}_{\mathrm{entry}} - \frac{\mathrm{P}_{\mathrm{avg}} \cdot \mathrm{t}_{\mathrm{eclipse}}}{\mathrm{E}_{\mathrm{battery}}} \cdot 100\%$$

with $\mathrm{P}_{\mathrm{avg}}$ in W, $\mathrm{t}_{\mathrm{eclipse}}$ in hours, and $\mathrm{E}_{\mathrm{battery}}$ in Whr.

The three numbers you put into it.

- $\mathrm{E}_{\mathrm{battery}} = 100$ **Whr**. The pack is a GomSpace NanoPower BPX 8S-1P: 100 Wh, 3.5 Ah, 28.8 V nominal, ~500 g. One box, one real flight part.
- $\mathrm{t}_{\mathrm{eclipse}}$. The orbit is 5669 s and the eclipse fraction swept over the year of the LTAN 10:00 orbit runs **0.351 to 0.372**, so the eclipse is **33.2 to 35.2 min**. Take 34 min = 0.567 h as the working figure.
- $\mathrm{P}_{\mathrm{avg}} = 22.7$ **W**, the orbit-average load in the nominal profile (`tools/power_budget.py` , sim manual 01_eps §3.2). This is a **6U CubeSat**: 6 W of OBC and PDU, 1.5 W of always-on receive chain, 5 W of transmitter at the LOW tier (11 W at HIGH) while it transmits, 9 W of AOCS in NOMINAL, 2.4 W of payload in standby, and 15 W of cooler only while preparing for and during imaging — the cooler is off in STANDBY. Nothing on this spacecraft draws 70 W.

Worked example. Eclipse entry SoC = 85 %; $\mathrm{P}_{\mathrm{avg}} = 22.7$ W; $\mathrm{t}_{\mathrm{eclipse}} = 34$ min $= 0.567$ h; $\mathrm{E}_{\mathrm{battery}} = 100$ Whr:

$$\mathrm{SoC}_{\mathrm{exit}} = 85\% - \frac{22.7 \times 0.567}{100} \times 100\%$$
$$= 85\% - 12.9\% = 72.1\%$$

That is **12.9 Wh out of a 100 Wh pack**, a discharge rate of about **0.38 %/min** (12.9 / 34). Sanity-check it the other way: 22.7 W for 0.567 h is 12.9 Wh, and 12.9 / 100 = 12.9 %. If your answer is not a low-teens percentage for a nominal orbit, you have used the wrong load or the wrong battery. On the battery curve ($V \approx 24.0 + 9.6 \times \mathrm{SoC}$) that is 32.2 V at entry and $24.0 + 9.6 \times 0.721 = 30.9$ V at exit.

The same arithmetic at the deep end. Take the `worst` profile — FINE held all orbit, cooler running continuously, imaging through eclipse, HIGH-tier dumps every pass — at 46.6 W over a 35.2 min (0.587 h) eclipse: $46.6 \times 0.587 = 27.4$ Wh, so SoC falls 27.4 points, 85 % → 57.6 %. Still survivable for one orbit, but that profile is generation-negative over the orbit as a whole (45.1 W generated against 46.6 W consumed), so it does not recover and each orbit starts lower than the last. That is the shape of a power budget that does not close.

Typical range. Eclipse-exit SoC should land in **70–85 %** during nominal operations. Below 55 % means the eclipse load profile was wrong or there is a parasitic load — file a ticket. The protection ladder below that is monotonic and worth memorising in order: in-model load shed at **28.4 V** bus (~46 % SoC), FDIR `safe_mode_eps` at **27.7 V** (~39 %) — both from the one table `simulation.yaml` `eps.bus_protection` — then the SoC-triggered shed stages at **30 / 20 / 10 %**, with a payload power-off at 20 % and EPS safe mode at 15 %. The operator's job is to pre-empt the first of those, not to watch them fire.

13.4 Pointing-budget RSS

Pointing error components are statistically independent and combine as root-sum-square:

$$\theta_{\mathrm{total}} = \sqrt{\theta_{\mathrm{ST,\,NEA}}^2 + \theta_{\mathrm{ST,\,bias}}^2 + \theta_{\mathrm{align}}^2 + \theta_{\mathrm{control}}^2}$$

The spacecraft does this for you and publishes the answer at `aocs.total_pointing_err_3sig` (**0x02AA**), as the RSS of a *knowledge* term (`aocs.knowledge_err_3sig`, 0x02A7, in **arcsec**, from the MEKF covariance) and a *control* term (`aocs.control_err`, 0x02A8, in **degrees**). Doing the arithmetic by hand is how you check that the published number is believable — and how you know which term to chase when it is not.

The hardware contributions. The star tracker is a Blue Canyon NST-class nano tracker: noise-equivalent angle **6″ 1σ** cross-boresight (40″ about boresight), thermoelastic low-frequency bias **3″ 1σ**, mounting misalignment **15″**. There are two of them (ST1/ST2), selected with `ST_SELECT` (S8 func 6). The gyros are a tactical-grade triad — ARW 0.06 °/√h, bias instability ~0.1 °/h — with **no redundant unit**, and their bias is calibrated **on board** (`AOCS_GYRO_CALIBRATION`, S8 func 13).

Worked example (FINE_POINT, per axis, 3σ).

Term	1σ / bias	3σ, degrees	Square
ST noise-equivalent angle, cross-boresight	6″	18″ = 0.0050°	2.500×10^{-5}
ST thermoelastic bias	3″	9″ = 0.0025°	6.250×10^{-6}
Mounting misalignment	15″	15″ = 0.0042°	1.764×10^{-5}

Term	1σ / bias	3σ, degrees	Square
Control error, measured 3-wheel FINE	—	0.0188°	3.534×10^{-4}
Sum			**4.023×10^{-4}**

$$\theta_{\text{total}} = \sqrt{4.023 \times 10^{-4}} = 0.0201^\circ \approx 0.020^\circ$$

Against the **0.05° gate** that COM-012 enforces on 0x02AA, that is 60 % of budget in hand. Note where the budget actually goes: the control term is 88 % of the sum, and the whole star-tracker chain is almost negligible. **Chasing sensor error when the control term dominates is wasted pass time.**

Turn it into metres. The spacecraft publishes `aocs.geoloc_err_m` (0x02C7) = altitude × total pointing error:

$$513\ \text{km} \times 0.0201^\circ \times \tfrac{\pi}{180} = 513000 \times 3.51 \times 10^{-4} = 180\ \text{m}$$

— about five ground pixels at the instrument's 33.4 m GSD. At the 0.05° gate it would be 448 m, about thirteen pixels.

Operator working rule. Before commanding `AOCS_SET_MODE(mode=5)` for imaging, run `AOCS_FINE_READY_CHECK` (S8 func 89) and confirm `aocs.fine_ready` (0x02C5) reads **0**. Then decompose 0x02AA into 0x02A7 and 0x02A8. If the knowledge term is the larger of the two, something is wrong with the estimator — check `aocs.est_source` (0x02C1) and `aocs.est_innov_rej_count` (0x02C2) — because in a healthy spacecraft the control term dominates by an order of magnitude. If 0x02C5 bit 1 is set with `aocs.fine_degraded` (0x0299) = 1, you are flying FINE on **three** wheels: that is permitted and it holds 0.0188° against 0.0208° on four, but the control deadband is relaxed fourfold and short-term jitter is worse. Three wheels is degraded, not broken; two wheels is broken.

13.5 Delayed-TM gap analysis

Telemetry packets have a sequence counter (CCSDS packet header, 14-bit count). Gaps in the sequence indicate dropped packets — usually link-quality issues, occasionally onboard storage overruns.

Operator working rule. When the SPACON sees the TC accept counter increment but the matching S1.7 execution-complete report is missing for > 15 s, check:

1. Is `obdh.tm_pkt_count` (0x0307) incrementing, and does `ttc.frame_sync` (0x0512) still read 1? If not → the link is fully down, not just delayed. Invoke CTG-003 TTC Link Loss Recovery. Check the UHF beacon first: if SID 11 is still arriving, the fault is in the S-band chain and not in the spacecraft.
2. Compare the *expected* S1 sequence count to the *received* count. Sequence counts are 14-bit and wrap at 16,384.
3. The gap size tells you how many packets were lost. If it's < 5 packets, this is normal end-of-pass fade — log and continue. If it's > 20, the link degraded mid-pass; check `ttc.ber` (0x050C, $\log_{10}$ scale) and `ttc.rssi` (0x0502) trend.
4. Most missed TM is recoverable by requesting **delayed TM playback** from on-board storage at the next pass. That is **`DUMP_STORE(store_id=1)` — PUS Service 15, Subtype 9** — with `STORE_STATUS` (S15.13) to see what is there first. Service 15 is telemetry storage; Service 13 is the large-data transfer used for *image products* and is a different mechanism entirely. The dump comes down on virtual channel 1 and never reaches the live displays: played-back packets do not raise or clear alarms or change live values, so analyse them in the delayed-TM viewer or on the MCS `/dumps` page. Book the playback on a pass that clears 10° so it runs at the HIGH tier: a whole orbit of

stored housekeeping is ~1.1 MB, which is 70 s at 128 kbps but 9 minutes — more than a pass — at 16 kbps.

Worked example of delayed-TM plot interpretation. A typical plot of `seq_count` vs spacecraft time will show:

- A straight upward ramp (counts increasing monotonically).
- Small jitters in the ramp slope as packet rate changes between HK SIDs at different reporting periods. The shipped cadences are **SID 1 (EPS) at 1 s**, SID 2 (AOCS) and SID 7 (AOCS determination) at 4 s, SID 4 (platform), 5 (payload), 6 (TT&C), 8 (EPS extended) and 9 (PROP) at 8 s, **SID 3 (TCS) at 60 s**, and **SID 11 (Beacon) at 30 s**. EPS at 1 s dominates the packet rate, which is why an EPS-heavy gap looks so much wider than a TCS-heavy one for the same wall-clock outage.
- Remember that only SIDs **1 and 11** are enabled at power-up, and the engine re-applies exactly that set on application boot. Everything else needs `HK_ENABLE` (**S3.5**). A "gap" that turns out to be a SID nobody ever enabled is not a link problem.
- A discontinuous jump at the wrap point (16383 → 0) — this is *not* a gap, just modular arithmetic.

A genuine gap shows as a *plateau* (no counts received) followed by a jump in the count where reception resumed. The width of the plateau (in time) is the gap duration; the difference in count across the plateau is the number of packets dropped.

13.6 Link-margin sanity check

Available link margin (dB) from the cheat-sheet form of the link budget:

$$\mathrm{M_{dB}} = \mathrm{EIRP_{dBW}} + \mathrm{G/T_{dB/K}} + 228.6 - \mathrm{FSPL_{dB}} - \mathrm{L_{atm}} - \mathrm{L_{pol}} - \mathrm{L_{impl}} - 10\log_{10}(\mathrm{R_b}) - [(\mathrm{E_b/N_0})_{\mathrm{req}} - \mathrm{G_{coding}}]$$

with $228.6 = -10\log_{10}(\mathrm{k_B})$ and $\mathrm{R_b}$ in bits/s. The shipped model carries no separate pointing-loss term; the losses it books are atmospheric 0.5 dB, polarisation 0.3 dB and implementation 2.0 dB.

The constants, from `subsystems/ttc.yaml`:

- Downlink 2200 MHz, uplink 2025 MHz.
- Spacecraft EIRP **10.0 dBW** reference plus **3.0 dBi** of antenna gain → **13.0 dBW**, quoted at 2 W of RF. That is the radio the power budget describes too: an Endurosat-class S-band transceiver, 24–33 dBm out, 1.2 W idle, 11 W DC at full transmit. **There is no 20 W transmitter on this spacecraft** — 20 W is neither an RF output any CubeSat PA produces nor a DC draw any CubeSat radio takes.
- Required $\mathrm{E_b/N_0}$ **9.6 dB uncoded**, less the tier's coding gain: 7.5 dB at LOW (convolutional + Reed-Solomon), 6.0 dB at HIGH (RS only).
- Station G/T: Svalbard 21.5, Inuvik 22.0, Iqaluit 20.0, O'Higgins 20.0, Troll 18.5 dB/K, de-rated 0.5 dB below 30° elevation and 1.5 dB below 10°.

Worked example — Iqaluit, 30° elevation, LOW tier.

Slant range first:

$$\begin{aligned} \mathrm{d} &= -\mathrm{R_E}\sin\varepsilon + \sqrt{(\mathrm{R_E}\sin\varepsilon)^2 + 2\mathrm{R_E h} + \mathrm{h}^2} \\ &= -3185.5 + \sqrt{1.0147\times10^7 + 5.734\times10^6 + 2.025\times10^5} = 825\ \mathrm{km} \end{aligned}$$

$$\mathrm{FSPL} = 92.45 + 20\log_{10}(825) + 20\log_{10}(2.2) = 92.45 + 58.33 + 6.85 = 157.63\ \mathrm{dB}$$

$$\mathrm{M_{LOW}} = 13.0 + 20.0 + 228.6 - 157.63 - 0.5 - 0.3 - 2.0 - 10\log_{10}(16000) - (9.6 - 7.5)$$

$$= 261.6 - 157.63 - 2.8 - 42.04 - 2.1 = \mathbf{57.0\ dB}$$

Same geometry, HIGH tier. Only two terms move — the rate and the coding gain:

$$\mathrm{M_{HIGH}} = 261.6 - 157.63 - 2.8 - 10\log_{10}(128000) - (9.6 - 6.0)$$

$$= 261.6 - 157.63 - 2.8 - 51.07 - 3.6 = \mathbf{46.5\ dB}$$

The step between the rungs is therefore $57.0 - 46.5 = \mathbf{10.5\ dB}$, and it decomposes exactly: $-10\log_{10}(8) = -9.03$ dB for the 8× rate, plus -1.5 dB for the coding change. The BEACON→LOW step is $-10\log_{10}(16) = -12.04$ dB. **Any measured set that does not show those two steps indicates a modelling error, not a link problem.**

What the numbers mean. The worst case in the whole station × elevation × tier matrix is **HIGH at Troll at 5°, and it is 33.7 dB.** This link is not power-limited anywhere, at any station, in any geometry. Two consequences an operator must internalise:

1. The 3 dB and 6 dB thresholds in the procedures are **sanity** thresholds. A reading anywhere near them does not mean the geometry is marginal; it means the model, the attitude or the ground station is broken. Treat a single-digit margin as an equipment alarm, not a link report.
2. The 10° gate on the HIGH tier is a **policy** gate, not a margin gate. HIGH closes with 33.7 dB at 5°. It is withheld below 10° because bulk downlink is a planned, earned activity and because tracking, Doppler rate and multipath at low elevation make a 64 kBd QPSK dump a poor bet.

Operator working rule. When TT&C calls a margin on the loop, sanity-check it against the baseline table established in COM-006:

1. Predicted margin for the current station, tier and elevation — e.g. 57 dB, Iqaluit, LOW, 30°.
2. Actual margin from `ttc.link_margin` (0x0503).
3. A 1–2 dB difference is normal variation. A > 3 dB discrepancy warrants a diagnostic before LOS.
4. Likely causes: ground-antenna degradation (call Ground Seg); commanded transmitter power dropped — check `ttc.tx_fwd_power` (0x050D, dBm) and the live `ttc.eirp_dbw` (0x052C), **not** `ttc.carrier_lock` (0x0510), which is a 0/1 flag; attitude mispointing reducing antenna gain; ionospheric scintillation (a UHF concern, rare at S-band).
5. If the forward power reads zero, check `ttc.tx_active` (0x0525) and `ttc.tx_gate_status` (0x052B) before suspecting hardware — the chain may simply not have permission to radiate.

13.6.1 Data-volume arithmetic

The link is not power-limited, so the binding constraint is **time**, and that is what the operator actually has to compute.

One image. A 4 MiB raw frame is $4 \times 1024^2 \times 8 = 33\,554\,432$ bits.

	At HIGH (128 kbps)	At LOW (16 kbps)
Raw channel time	**262.1 s**	2097 s
With the planner's 20 % protocol allowance	**327.7 s**	2621 s
As a fraction of the 450 s average pass	**73 %**	582 %

So one image is most of one pass, and the shipped `data_dump` activity is **330 s** long — one image, rounded up, and nothing more. n images need n dumps. **At LOW an image needs more than a whole pass: imagery is not merely slower at LOW, it is impossible.**

Housekeeping is not free. The shipped HK set generates 1378.5 bps of PUS packets, which is **1588 bps of channel** after RS(255,223) and TM framing — 134.5 Mbit, or **16.0 MiB, per day**. Against ~23 300 s of daily contact:

Tier	Contact time needed / day	Share of the network
LOW (16 kbps)	8410 s	**36 %**
HIGH (128 kbps)	1051 s	4.5 %

Thirty-six percent of the entire five-station network, before a single image. That is the number that makes the case for LOW being 16 kbps rather than 8, and the number that stops a student thinking of housekeeping as free.

Daily capacity, for orientation: the raw figure is 23 300 s × 128 kbps ≈ 355 MiB/day; taking about 64 % of it as usable gives ~227 MiB/day if every usable contact second ran at HIGH, ~28 MiB/day if all of it ran at LOW, and **~162 MiB/day** for the realistic blend (HIGH above the 10° gate, LOW below it). Against 8–34 MB/day of generation the mission has margin — but only because dumps are booked onto passes that clear the gate.

Instrument geometry, since FDS will be asked for it. The imager is a wide-field multispectral pushbroom: 8192 px across track at 5.5 µm pitch on an 84.5 mm focal length, giving a 29.856° full field, a **273 km swath** and a **33.4 m GSD** from 513 km. The standard capture is a **2048-line strip**:

$$\frac{2048 \text{ lines}}{244 \text{ Hz}} = 8.4 \text{ s of scan}$$

which at the 7.06 km/s ground-track speed is **59 km along-track**. The second derivation does not agree:

$$2048 \times 33.4 \text{ m} = 68.4 \text{ km along-track}$$

because the 244 Hz line rate was sized at the retired 450 km altitude (7.14 km/s ÷ 29.3 m). At 513 km successive lines are 28.9 m apart along-track against a 33.4 m pixel across-track, so the image is slightly oversampled along-track; a velocity-matched rate would be about 211 Hz. `payload.yaml` records this as a known inconsistency. Running both derivations is still the check to run — here it tells you that the footprint to quote is 59 km. Focal-plane readout is 96 Mbps ($8192 \times 4\ \text{bands} \times 12\ \text{bit} \times 244\ \text{Hz}$), so the physical raw frame is 100.7 MB. The courseware uses **4 MB** for `image_size_mb` instead, deliberately and on the record: at 100.7 MB an image would need ~1.75 h of effective contact at HIGH (~2.2 h with the 20 % protocol allowance) and no plan would ever close. Do not "fix" that gap; know that it is there.

13.7 Wheel-saturation desat scheduling

A reaction wheel's stored angular momentum is $H = I_{wheel}\,\omega$. EOSAT-1 flies four **Sinclair RW-0.03-class** wheels in a pyramid (azimuths 45/135/225/315°, elevation 30° above the XY plane), with $I_{wheel} = 5.2 \times 10^{-5}$ kg·m² and $\omega_{max} = 5500$ rpm:

$$\omega_{max} = 5500 \times \tfrac{2\pi}{60} = 576.0 \text{ rad/s}$$

$$H_{max} = 5.2 \times 10^{-5} \times 576.0 = 0.0300 \text{ N·m·s} = \mathbf{30 \text{ mN·m·s}}$$

which is exactly the datasheet momentum storage — the rotor inertia was derived from it, so the two agree by construction. Because the wheels are in a pyramid, the *body-axis* authority is larger than one wheel: $2\cos 30^\circ \cos 45^\circ \ \mathrm{H} \approx 0.037$ N·m·s about X and Y, and $2\mathrm{H} = 0.060$ N·m·s about Z.

Disturbance torques on a 6U with deployed wings at 513 km, as configured in `aocs.yaml` (the aerodynamic term uses a fixed density sized at 450 km, so it is conservative by about a factor of two):

Source	Magnitude	Character
Gravity gradient	≤ 4.5 × 10^{-7} N·m	cyclic (45° off-nadir case)
Residual magnetic dipole (|m| ≈ 0.011 A·m²)	~5 × 10^{-7} N·m	cyclic
Aerodynamic	~2.3 × 10^{-7} N·m	**secular** in LVLH
Solar radiation pressure	~1.1 × 10^{-7} N·m	zero in eclipse

RSS bound: $\sqrt{4.5^2+5.0^2+2.3^2+1.1^2}\times10^{-7} = 7.2\times10^{-7}$ N·m. Time to fill one wheel from zero at that bound:

$$\mathrm{t_{sat}} = \frac{\mathrm{H_{max}}}{\tau} = \frac{0.0300}{7.2 \times 10^{-7}} = 4.2 \times 10^{4}\ \mathrm{s} = 11.6\ \mathrm{h} \approx \mathbf{7.4\ orbits}$$

Using the body-axis authority about X or Y (0.037 N·m·s) gives ~9 orbits; using only the genuinely **secular** aerodynamic term gives ~23 orbits. The honest statement is therefore *momentum accumulates over a day, not over an orbit* — most of the disturbance is cyclic and unwinds itself. A wheel that climbs to the desat threshold inside one orbit is telling you about a disturbance you have not accounted for.

Momentum, not wheel speed, is the decision variable. A four-wheel pyramid can sit at 5000 rpm on every wheel with **zero** body momentum — that is the null-space direction — and it can sit at 1200 rpm with a large one. The parameters to watch are `aocs.h_body_mag` (**0x02B9**) and the envelope percentage `aocs.h_env_frac` (**0x02BA**). `aocs.fine_ready` (0x02C5) bit 4 sets once the envelope reaches 70 %, and that is the honest "a dump is coming" warning a planner should act on.

Operator working rule. Watch the envelope and the highest-speed wheel together. 75 % of the 5500 rpm limit is **4125 rpm**; NOM-007 *Momentum Management* calls a dump REQUIRED above 4500 rpm on any wheel and OPTIONAL below 3000 rpm on all of them. Schedule the dump on a **high-latitude arc**, and schedule it with real time around it.

How long a dump actually takes.

$$20 \text{ to } 50 \text{ minutes.}$$

Measured on the corrected model from the 0.020 N·m·s reference loading: **0.005 N·m·s in 1521 s (25 min)** and **0.001 N·m·s in 2857 s (48 min)**. The on-board timeout is 3600 s; a dump that hits it raises event 0x020E `AOCS_DESAT_INEFFECTIVE` and sets `aocs.desat_ineffective` (0x029C) rather than parking in DESAT for ever. **Do not intervene before 3600 s** unless a verification step actually fails — the old 600 s deadline aborted every healthy dump.

The older "5–8 minutes" figure was instantaneous-torque arithmetic, $0.020/4.8 \times 10^{-5}$, and it is wrong for a reason worth understanding: $\tau = \mathrm{m} \times \mathrm{B}$ **is always perpendicular to** B, so at any instant only the component of stored momentum perpendicular to the local field can be removed. The rest waits for the field to sweep through the body, which happens at about twice orbit rate. 5–8 minutes is the right answer for the perpendicular *component*; 20–50 minutes is the right answer for a full dump. Running the rods

harder does not help — a bang-bang law was tested and rejected because it nulls the perpendicular component faster, which makes the momentum vector lock onto $\hat{\mathrm{B}}$ and ride along with it.

What the spacecraft does around the dump.

- The command is `AOCS_DESATURATE` (S8 func 1); it puts the spacecraft in **DESAT, mode 7**.
- It is **refused in FINE_POINT** (`DESAT_REFUSED_FINE`) and while `aocs.desat_inhibit` (0x029A) is latched (`DESAT_INHIBITED`). Both refusals are correct behaviour, not anomalies.
- Automatic desaturation is **inhibited while the payload is imaging** (`payload.mode` = 2), so a dump cannot pre-empt a strip. This is why CALIBRATION had to be a new mode value (3) rather than a renaming of IMAGING.
- Attitude error may rise to 2–3° during the dump; the coarse limit is 5°.
- **Do not command a mode afterwards.** DESAT returns to the mode it was entered *from*, and the spacecraft re-promotes itself to FINE_POINT once the 0x02C5 preconditions have held for 30 s. Confirm with `aocs.mode_demoted` (0x0298) = 0 and event 0x020F `AOCS_MODE_RESTORED` . Commanding NOMINAL here overwrites the ground's commanded mode and cancels the re-promotion — the exact defect that once left a spacecraft sun-pointing for six orbits.

Why high latitude, not eclipse. Magnetic dump effectiveness is a **geometry** question, not an illumination one: it depends on the strength and orientation of the local field, so it is best over the poles and worst near the magnetic equator. Solar radiation pressure is the smallest of the four disturbance terms and its absence in eclipse changes nothing material. Book the dump for a polar passage; if wheel speeds are not falling, `aocs.mtq_enabled` (0x025E) and the orbit position are the first two things to check.

13.8 Worked-shift example: putting them all together

A typical pre-pass briefing requires the FD to know, within 30 seconds of opening the contact-schedule entry:

1. **AOS time** (§13.2): "Iqaluit AOS in 6 min 12 s, TCA at AOS+3:45, LOS at AOS+7:30. Max elevation 47°, so we clear the 10° gate for most of the window."
2. **Eclipse interaction** (§13.3): "We exit eclipse at AOS−2:30 with predicted SoC 69 percent. Pass starts in the lit segment."
3. **Pointing** (§13.4): "0x02AA reads 0.020 degrees total, comfortably inside the 0.05 gate. The control term dominates at 0.019; the star-tracker chain is a rounding error. 0x02C5 reads zero — four wheels, MEKF converged."
4. **Link margin** (§13.6): "Predicted 46.5 dB at HIGH over Iqaluit at TCA; measured 46.2. Within the 1–2 dB normal band. HIGH tier available for the dump."
5. **Data** (§13.6.1): "One 4 MiB product queued. 328 seconds with the protocol allowance against a 450-second pass, most of it above 10 degrees — it fits, with little to spare. No second dump this pass."
6. **Wheels** (§13.7): "Envelope at 62 percent on 0x02BA, highest wheel 3850 rpm — 70 percent of the 5500 limit. Below the 4125 threshold. Desat planned for the polar arc two orbits out, not this pass, and we will allow the full 3600-second timeout for it."

If all six answers are clean, the briefing closes in two minutes and the team is ready to operate.

13.9 References

- ISO 8601 — Date and time representations — www.iso.org

Chapter 14 — Procedure Catalogue (Quick-Reference Cards)

This chapter is the operator's pocket reference: **one card per page**, organised by category and colour-coded by procedure family. Each card contains: header (with owning-position icon), entry point, goal, possible negative outcomes / deviations, an optional decision tree (for contingencies), and a step-by-step table with TC/code, confirmation tickbox and notes.

Colour key. Green = nominal · Blue = LEOP · Purple = commissioning · Amber = contingency · Red = emergency.

Position-icon key (header bracket). `[FD]` Flight Director · `[FE]` Flight Engineer · `[GS]` Ground Station Engineer · `[PE]` Payload Engineer · `[SE]` Spacecraft Engineer · `[DH]` Data Handler · `[TT]` TT&C Engineer · `[OB]` OBDH Engineer · `[AOCS]` AOCS Engineer · `[MULTI]` multi-position.

> **Editor's note — paper artefacts referenced by these cards.** The procedure cards reference an *anomaly-ticket form*, a *contingency- active banner*, and an *FD second-key confirmation*. None of these are implemented in the MCS UI. In simulator training, these are tracked on paper: each card carries a `□` confirm box for the FD second-key; the contingency-active state is held in the FD's voice-loop call log; and anomaly tickets are filled on a paper form that the SPACON files at end of shift. Adjust to your installation's conventions if a deployment provides an electronic equivalent.

The 70 procedures in the simulator split as 7 LEOP, 13 commissioning, 14 nominal, 30 contingency, 6 emergency. The cards below cover the ~57 most consequential procedures across the catalogue. Cards marked **"see §9.3 / §9.4 / §9.5"** for NOM-001, CTG-003 and NOM-012 hold the headline procedures; their full read-and-do checklists are in Chapter 9. The remainder of the catalogue (mostly cleanup, scheduled maintenance and infrequent contingencies) is indexed in §14.6 with file references for at-console reading.

> **Editor's note — which ID is on the card.** Every card in this chapter carries the ID of `configs/eosat1/procedures/procedure_index.yaml` exactly, so the ID called on the voice loop is the file the MCS Procedures tab opens. Two nominal IDs moved when the book was reconciled against the simulator: the orbit-manoeuvre **execution** card is **NOM-013** (simulator NOM-005 is HK Configuration) and NOM-006 is Software Upload (simulator NOM-008 is Clock Synchronisation). The contingency and emergency cards were renumbered to the simulator's IDs in October 2026; the numbers used by earlier printings are listed in the historical note at the end of §14.6. The book's two temporary contingency cards, CTG-B01 and CTG-B02, are now simulator cards: **CTG-029** Gyro Failure Recovery and **CTG-030** Telecommand Rejection and Verification Failure. The shipped catalogue runs LEOP-001 to LEOP-007, COM-001 to COM-013, NOM-001 to NOM-014, CTG-001 to CTG-030 (CTG-018 retired in favour of CTG-021) and EMG-001 to EMG-006.

14.1 LEOP procedure cards

> **Editor's note — LEOP running order.** Under the cold-boot initial state the order is **LEOP-001 → LEOP-002 → LEOP-004 → LEOP-007 → LEOP-005 → LEOP-006 → LEOP-003**. The wings are deployed (LEOP-004) before anything else draws power; LEOP-007 powers only the AOCS suite and commands

> DETUMBLE; LEOP-005 observes the automatic handover to COARSE_SUN; LEOP-003 closes LEOP with the phase-5 declaration.

[MULTI] LEOP-001 — First Acquisition of Signal & OBC Boot
Entry point: First pass over Iqaluit after separation timer expiry. At cold boot the spacecraft is a one-way listener: only `obc` and `ttc_rx` are energised, `ttc_tx` is OFF and the OBC is in the bootloader — so there is no carrier to acquire passively and the pass starts with Pass Override engaged. **Goal:** Establish the first command path, exit the OBDH bootloader into the application image, clear beacon mode onto the LOW rung, and leave the platform HK structures the pass needs enabled before LOS. **Possible negative outcomes / deviations:** - No S1.1 to `CONNECTION_TEST` → the auto-TX hold-down only fires on an accepted TC; check pass override and the uplink, retry once, then repoint to Troll - `OBC_BOOT_APP` acks (S1.1) but `obdh.sw_image` does not transition within 30 s → application CRC failing; invoke CTG-010 OBDH Watchdog Recovery / consider `OBC_REBOOT` - Link established, no HK after boot → CTG-019 No Telemetry at Pass Start (blind-commanding path) - `TTC_SET_DATA_RATE` refused with S1.8 error `0x0007` → beacon mode still latched; re-issue `TTC_SET_BEACON_MODE(on=0)` **Decision tree:** - IF S1.1 + S17.2 received AND `obdh.sw_image` = 1 AND SIDs 1, 3, 4, 6, 11 stream → procedure GO, continue to LEOP-002 - IF S1.1 received but no S1.7 to the boot command within 30 s → CTG-010 - IF no accepted TC by AOS+2 min → hold; if none by AOS+5 min → **EMG-004 Loss of Communication** | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Ground Seg: point Iqaluit at predicted AOS; demodulator on the BEACON rung | (MPS station panel) | □ | 1 kbps BPSK, 1 kBd — the only rate the bootloader transmits | | 2 | SPACON: engage Pass Override in the MCS | `pass_override` (0x05FF) = 1 | □ | Cold boot has `ttc_tx` OFF; the override gives the assured uplink | | 3 | SPACON: send first uplink (connection test) | `CONNECTION_TEST` (S17.1) | □ | S1.1 + S17.2 within 10 s; the auto-TX hold-down powers `ttc_tx` for **300 s** (`simulation.yaml` `ttc.auto_tx_hold_s`), re-armed by every accepted TC — five minutes without commanding and the downlink drops | | 4 | TT&C: confirm carrier lock and link quality | `ttc.link_status` (0x0501) = 2 (LOCKED) | □ | `ttc.rssi` (0x0502) > −110 dBm; `ttc.link_margin` (0x0503) > 3.0 dB | | 5 | SPACON: verify spacecraft still in bootloader | `HK_REQUEST(sid=11)` (S3.27); `obdh.sw_image` (0x0311) = 0 | □ | Beacon SID only; no subsystem SIDs yet | | 6 | SPACON: command OBC application boot (dual-keyed) | `OBC_BOOT_APP` (S8 func_id 55) | □ | S1.1 within 5 s; S1.7 after the ~10 s CRC check; `obdh.sw_image` = 1 within 15 s | | 7 | SPACON: enable the platform HK structures | `HK_ENABLE(sid=3)`, `HK_ENABLE(sid=4)`, `HK_ENABLE(sid=6)` (S3.5) | □ | Boot re-applies `default_enabled_sids: [1, 11]`; SIDs 2 + 5 wait for LEOP-007 | | 8 | TT&C: clear beacon mode, then select the LOW rung | `TTC_SET_BEACON_MODE(on=0)` (S8 func_id 70); `TTC_SET_DATA_RATE(rate=0)` (S8 func_id 65) | □ | `ttc.beacon_mode` (0x0521) = 0; 16 kbps, `ttc.tm_tier` (0x0539) = 1; reconfigure the demodulator to 16 kBd | | 9 | SPACON: synchronise onboard time | `SET_TIME(cuc_seconds=T_corrected)` (S9.1) | □ | The clock is at CUC 0 = **J2000** and `obdh.time_ref` (0x0A13) = 0 (NONE) — the bootloader reset it and `OBC_BOOT_APP` does not restore it. Residual < 1.0 s on `REQUEST_TIME` (S9.2); `time_ref` → 1 (GROUND). Full card is LEOP-006 | | 10 | SPACON: EPS sanity check; log completion; hand to LEOP-002 | `HK_REQUEST(sid=1)` (S3.27) | □ | "LEOP-001 complete" |

[MULTI] LEOP-002 — Initial Health Check

Entry point: Immediately after LEOP-001 reports complete; SIDs 1, 3, 4, 6 and 11 streaming. **Goal:** Each subsystem position confirms baseline values are within band before sequential power-on begins; produce a single PASS/FAIL for the FD. **Possible negative outcomes / deviations:** - Any single subsystem reports FAIL → invoke that subsystem's contingency (e.g. CTG-001 for EPS undervoltage, CTG-002 for AOCS) before proceeding - Multiple FAILs → hold LEOP entirely; FD considers EMG-001 safe-mode entry - One subsystem returns DEGRADED rather than FAIL → continue at FD discretion but log a watch-item - A line flag reads 1 for the AOCS, payload, cooler or `prop` line → something was energised early; record it and hold LEOP-004 until the FD has reviewed it **Decision tree:** - IF all positions return PASS → GO LEOP-004 - IF EPS FAIL → CTG-001 first, then re-run LEOP-002 - IF a line is energised early → hold, FD review before LEOP-004 - IF ≥ 2 FAIL → call FD, consider EMG-001 | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: read battery V, SoC, bus V, line currents | `HK_REQUEST(sid=1)` (S3.27) | □ | SoC (0x0101) > 60 %, bus (0x0105) 29.6–34.6 V | | 2 | Thermal: read all zone temperatures | `HK_REQUEST(sid=3)` (S3.27) | □ | `tcs.temp_obc` (0x0406) −10…+50 °C; `tcs.temp_battery` (0x0407) 0…+35 °C | | 3 | TT&C: confirm link parameters | `HK_REQUEST(sid=6)` (S3.27) | □ | `ttc.link_status` (0x0501) = 2; RSSI > −105 dBm; margin > 3.0 dB | | 4 | OBDH: confirm OBC running, no FDIR events | `HK_REQUEST(sid=4)` (S3.27) | □ | `obdh.mode` (0x0300) = 0 or 1; `obdh.cpu_load` (0x0302) < 70 % | | 5 | EPS: confirm the unpowered subsystems are dark | `eps.pl_aocs_wheels` (0x0117), `eps.pl_payload` (0x0113), `eps.pl_fpa_cooler` (0x0114), `eps.pl_prop` (0x014A) = 0 (SID 1) | □ | Do not request SIDs 2 or 5: they are not enabled yet and their units are unpowered, so the line flags are the health check. There is no AOCS mode and no wheel telemetry to read until LEOP-007 and COM-004 | | 6 | SPACON: consolidate single PASS/FAIL on voice loop | (mission log) | □ | "LEOP-002 PASS" or list of FAILs | | 7 | FD: GO/no-GO for LEOP-004 | (FD call) | □ | Recorded in log |

[OB] LEOP-003 — Initial Orbit Determination

Entry point: After LEOP-006 — this is the **last** LEOP card. COARSE_SUN sun hold; clock referenced (`obdh.time_ref` 0x0A13 = 1, 2 or 3); GPS receiver ON since LEOP-007 with `aocs.gps_fix` (0x0274) ≥ 2; HK SID 12 (GPS_NAV) enabled in LEOP-006. **Goal:** Verify the on-board GPS navigation solution, confirm the 513 km / 97.39° sun-synchronous insertion, hand the refined TLE to the ground tools, and declare the end of LEOP by commanding mission phase 5 (COMMISSIONING). **Possible negative outcomes / deviations:** - `aocs.gps_fix` < 2 → confirm the receiver is ON (`S2_AOCS_GPS_RECEIVER`, device 0x020F, LEOP-007); a cold start can take 15 min. After 30 min with no fix, fall back to ground range and Doppler from the station network; the phase declaration may proceed on a ground-tracked orbit with FD approval - SID 12 absent from the downlink → `HK_ENABLE(sid=12)` (S3.5) was missed in LEOP-006; re-issue it - Altitude outside the [495, 540] km check band → check fix quality (`aocs.gps_pdop`, 0x0275) and take a second fix; only outside [400, 550] km is the insertion itself suspect - Inclination more than 0.2° from 97.39° → flag to Flight Dynamics; sun-synchronous precession and lifetime are affected - `SYS_SET_MISSION_PHASE` refused → `0x0C03`: the OBC is back in the bootloader (re-run LEOP-001 step 5 and LEOP-006); `0x0C04`: a backwards declaration without FORCE | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FDS: verify the GPS navigation solution is downlinking | `HK_REQUEST(sid=12)` (S3.27) | ☐ | `aocs.gps_fix` (0x0274) ≥ 2; `aocs.gps_num_sats` (0x0276) ≥ 4; record `aocs.gps_pdop` (0x0275). Position reads zero below fix 2: wait for the fix, do not conclude anything | | 2 | FDS: read the position fix | `GET_PARAM(param_id=0x0210)`, `(param_id=0x0211)`, `(param_id=0x0212)` (S20.3) | ☐ | `aocs.gps_alt` (0x0212) in the [495, 540] km check band | | 3 | FDS: read the ECEF state vector | `GET_PARAM(param_id=0x0231)` … `(param_id=0x0236)` (S20.3) | ☐ | Position magnitude 6700–6950 km, velocity magnitude 7400–7900 m/s; record with the GPS timestamp | | 4 | FDS: compare with the launcher-predicted orbit | (FDS workflow) | ☐ | i within 0.2° of 97.39°, e < 0.01, LTAN within 15 min of 10:00; period against the TLE (15.24 rev/day, 5669 s) | | 5 | FDS: take a second fix 10 min later | `HK_REQUEST(sid=12)` (S3.27) | ☐ | `aocs.gps_alt` within 20 km of step 2; fix still ≥ 2 | | 6 | FDS: distribute the orbit determination report and the refined TLE to the ground tools | (planner, FDS, station network) | ☐ | The refined TLE updates pass prediction on the ground. **Nothing is uploaded to the spacecraft**: there is no command for an on-board orbit upload | | 7 | FD: GO/NO-GO poll, then declare COMMISSIONING | `SYS_SET_MISSION_PHASE(phase=5, flags=0)` (S8 func_id 120) | ☐ | **Critical**: read the argument back. `eps.spacecraft_phase` (0x0129) = 5 within 5 s; event `MISSION_PHASE_CHANGED` (0x0055) carries 4 → 5. This card is the only one that sends it. "LEOP-003 complete": LEOP is over | **Note — the mission phase is an interlock, not a label.** `SYS_SET_MISSION_PHASE` (S8 func_id 120, arguments `phase` 3–6 and `flags` bit 0 = FORCE) gates six mechanisms: which subsystems tick, telecommand acceptance (at phase ≤ 3 only the bootloader subset is accepted), the derived `sat.mode` (0x0001, pinned to SAFE below phase 6), arming of the advanced fault-propagation model, the phase-gated FDIR rules, and the propulsion arm interlock (`min_phase: 6`). Out-of-range is refused with `0x0C01`, phases 0–2 with `0x0C02`, an application phase while the OBC is still in the bootloader with `0x0C03`, and a backwards transition with `0x0C04` unless FORCE is set. This **supersedes** the earlier "phase transition is console-only" ruling: the telecommand exists and is the way it is done.

[FE] LEOP-004 — Solar Array Verification

Entry point: After LEOP-002, in a sun-illuminated portion of the orbit, SoC > 50 % (each wing draws ~2 A for 30 s while deploying). The AOCS is not powered yet, so there is no rate telemetry: deployment is qualified against the launcher tip-off rates. **Goal:** Deploy the +Y and −Y wings, confirm deployment in telemetry, and verify per-wing current consistent with the predicted sun angle. **Possible negative outcomes / deviations:** - A wing does not show DEPLOYED within 45 s → single-wing retry with `EPS_DEPLOY_WING(wing=0)` (+Y) or `(wing=1)` (−Y), FD approval required; assess the single-wing power budget if it fails - Wing currents asymmetric > 30 % → log anomaly; continue LEOP on a reduced power budget - Current oscillating with attitude → expected before detumble; re-verify after LEOP-005 | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: command wing deployment (both) | `EPS_DEPLOY_WING(wing=2)` (S8 func_id 81) | □ | `eps.wing_status` (0x0144) bit 4/5 = deployment in progress | | 2 | EPS: confirm both wings deployed | `eps.wing_status` (0x0144) = 0x03 | □ | bit0 = +Y, bit1 = −Y — 0x0144 is the single deployment truth source | | 3 | EPS: read per-wing currents | `eps.sa_wing_py_current` (0x0146), `eps.sa_wing_my_current` (0x0147) | □ | > 0.1 A each in sunlight; 0x0110–0x0113 are power-LINE flags, not array telemetry | | 4 | EPS: verify wing symmetry and total generation | `eps.power_gen` (0x0107) | □ | Wings within 15 % of each other; > 20 W while tumbling, > 40 W once sun-pointed | | 5 | SPACON: record baseline; report to FD | (mission log) | □ | "LEOP-004 complete" |

[AOCS] LEOP-005 — Sun Acquisition

Entry point: After LEOP-007: AOCS in DETUMBLE (`aocs.mode` 0x020F = 2) on the magnetometer and magnetorquers, SID 2 downlinking, wheels DISABLED, SoC > 40 %. **Goal:** Observe B-dot rate damping and the **automatic** handover to COARSE_SUN (mode 3), then verify a power-positive sun hold. This card commands no mode change. **Possible negative outcomes / deviations:** - Rates not decreasing after 15 min → check `aocs.mtq_enabled` (0x025E) = 1 and `aocs.mag_enabled` (0x025C) = 1; if either has dropped, re-establish it as in LEOP-007 step 4 and re-command DETUMBLE with `AOCS_SET_MODE(mode=2)` (S8 func_id 0) - Rates below 0.5 °/s but no handover after 30 min → do **not** command mode 3; confirm the rates are below the guard on all three axes and sustained, then call the FD - Attitude error not converging in COARSE_SUN → check coarse sun-sensor confidence (`aocs.css_sun_confidence`, 0x0215); in eclipse, wait for the sunlit phase - Rates re-grow → CTG-002 AOCS Anomaly | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: confirm DETUMBLE and record the initial rates | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.mode` (0x020F) = 2; rates (0x0204–0x0206) recorded; 0.5–5 °/s per axis is typical after separation | | 2 | AOCS: monitor rate damping — command nothing | `HK_REQUEST(sid=2)` (S3.27) every 5 min | □ | All rates decreasing; B-dot roughly halves them every 10–15 min. A zeroed `aocs.mag_field_total` (0x0277) means B-dot is blind | | 3 | AOCS: observe the automatic handover to COARSE_SUN | (observe) | □ | Rates below 0.5 °/s for 30 s → `aocs.mode` = 3 by itself, within 1800 s. Do **not** command mode 3 | | 4 | AOCS: verify the sun hold | `HK_REQUEST(sid=2)` (S3.27) every 60 s for 5 min | □ | `aocs.att_error` (0x0217) < 15° within 600 s; `aocs.css_sun_confidence` (0x0215) > 0.5; rates < 0.5 °/s. The hold flies on magnetorquers, so expect degree-level wander | | 5 | EPS: verify power generation in the hold | `HK_REQUEST(sid=1)` (S3.27) | □ | `eps.power_gen` (0x0107) > 40 W; SoC stable or rising | | 6 | SPACON: log the final AOCS state; GO LEOP-006 | (mission log) | □ | "LEOP-005 complete". The spacecraft stays in COARSE_SUN: with no star tracker powered, nothing promotes it, and the wheels stay disabled until COM-004 |

[OB] LEOP-006 — Time Synchronisation

Entry point: After LEOP-005; OBC in the APPLICATION image (`obdh.sw_image` 0x0311 = 1) — `SET_TIME` is not on the bootloader allow-list — and `ttc.link_status` (0x0501) = 2. **Goal:** Synchronise the onboard clock to ground UTC within 1.0 s and verify the clock rate to better than 100 ppm. Fine synchronisation (< 0.1 s) is a nominal-ops activity, NOM-008. **The clock has never been set, not merely drifted.** The OBC runs an independent clock. The bootloader starts it at CUC 0 = the mission epoch **J2000** (2000-01-01T12:00:00 UTC, `mission.yaml` `time_epoch`) and `obdh.time_ref` (0x0A13) reads **0 (NONE)**; `obdh.time_since_sync_s` (0x0A14) reads 4294967295 = never synced. Every packet emitted before this card is stamped in the year 2000, and that is correct behaviour, not a decoding fault. **A bootloader reset returns the clock to J2000 and `OBC_BOOT_APP` does not restore it — so every reboot means re-running this card.** **Possible negative outcomes / deviations:** - Drift ≥ 86 400 s **with `time_ref` = 1 (GROUND)** → a clock that was set and is now days out; HOLD and consult the OBDH engineer. With `time_ref` = 0 (NONE) an offset of years is the *expected* LEOP signature — the clock is counting from J2000 - `SET_TIME` rejected → `SET_TIME` is not on the bootloader allow-list (check `obdh.sw_image` 0x0311 = 1), or the value is outside the onboard validation range; wait 5 s and retry - Residual > 1.0 s after correction → the correction may have missed the PPS edge; re-send with an updated T_corrected - `time_ref` (0x0A13) reads 0 (NONE) again after a successful sync → the OBC rebooted; check `obdh.uptime` (0x0308) and `obdh.reboot_count` (0x030A) and re-run the card - Clock rate error > 100 ppm → flag the OBC oscillator. A healthy unit runs ~**12 ppm** (≈ 1 s/day) with a few ppm of per-unit spread; ~500 ppm (≈ 43 s/day) is the signature of the instructor's `clock_drift` failure injection, not an ageing part | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | SPACON: verify OBC state and link | `HK_REQUEST(sid=4)` (S3.27); `HK_REQUEST(sid=6)` (S3.27) | □ | `obdh.mode` (0x0300) = 0 or 1; margin (0x0503) > 3.0 dB | | 2 | SPACON: request the time report AND the slant range together | `REQUEST_TIME` (S9.2); `HK_REQUEST(sid=6)` (S3.27); `HK_REQUEST(sid=4)` (S3.27) | □ | `REQUEST_TIME` is the catalogue command (`TIME_REPORT_REQUEST` is only an alias). `ttc.range_km` (0x0509) lives in SID 6 — request it or the OWLT never arrives | | 2a | SPACON: read the time reference before reading the clock | `obdh.time_ref` (0x0A13); `obdh.time_since_sync_s` (0x0A14) | □ | 0 NONE / 1 GROUND / 2 GPS / 3 HOLDOVER. Expect 0 and 4294967295 here; anything else means somebody has already set the clock — establish who before continuing | | 3 | SPACON: compute OWLT and drift | OWLT_ms = `ttc.range_km` / 299792.458 × 1000; delta_t = T_obc − (T_ground − OWLT) | □ | T_obc = `obdh.obc_time` (0x0309) + `obdh.obc_time_fine` (0x0A12)/65536, in CUC seconds since J2000. Expect 3–22 ms of OWLT — two orders below the 1.0 s threshold. Cross-check 0x0309 against the TM(9,2) value; they must agree within the 8 s SID 4 interval | | 4 | SPACON: issue the time correction | `SET_TIME(cuc_seconds=T_corrected)` (S9.1) | □ | Argument goes INSIDE the backticks — a bare `SET_TIME` packs cuc_seconds = 0, an epoch reset. Confirm T_corrected is on the **J2000** epoch: a Unix timestamp is ~9.46e8 s too large and is accepted. Event `OBC_TIME_SET` (0x030C) carries the signed jump | | 5 | SPACON: verify residual | `REQUEST_TIME` (S9.2); `HK_REQUEST(sid=4)` (S3.27) | □ | Residual magnitude < 1.0 s; `obdh.time_ref` (0x0A13) now = 1 (GROUND) and `time_since_sync_s` (0x0A14) restarted from 4294967295; `obdh.mode` unchanged | | 6 | SPACON: validate clock rate over two reports | `REQUEST_TIME` (S9.2) | □ | drift_ppm = ((T_obc2 − T_obc1) − (T_gnd2 − T_gnd1))/interval × 1e6; accept < 100 ppm, expect ~12 ppm. Use the longest interval the pass allows — 12 ppm over 60 s is 0.7 ms and unresolvable. Meaningless if `time_ref` = 2 (GPS): that measures the discipline loop, not the oscillator | | 6a | SPACON: enable GPS navigation HK, then discipline the clock to GPS once the receiver has a fix | `HK_ENABLE(sid=12)` (S3.5); `OBC_GPS_TIME_SYNC` (S8 func_id 80) | □ | SID 12 is not enabled by default: without it LEOP-003 runs blind. Sync only with `aocs.gps_fix` (0x0274) ≥ 2; `obdh.time_ref` → 2 (GPS) within 15 s. Ground set first, GPS second: on a clock already within 1 s the GPS jump is milliseconds, far below the 5 s jump that forces the AOCS to SAFE_BOOT (CTG-027). No fix yet → send the sync on the next contact | | 7 | SPACON: log new drift baseline; GO LEOP-003 | (mission log) | □ | "LEOP-006 complete" |

[MULTI] LEOP-007 — Sequential Power-On

Entry point: After LEOP-004: application image running (`obdh.sw_image` = 1), link on LOW, wings deployed and generating, every switchable line except `ttc_tx` OFF, AOCS OFF, spacecraft tumbling at tip-off rates; SoC > 60 %, bus 26–30 V. **Goal:** Power only what the rest of LEOP needs, in dependency order: the heater lines if their zones need heat, the AOCS suite line, the magnetometer, magnetorquers and GPS receiver, then DETUMBLE. The payload, the cooler, the wheels, the star trackers, the gyros and the `prop` line stay off; their commissioning cards bring them up. **Possible negative outcomes / deviations:** - `AOCS_SET_MODE(mode=2)` refused with `0x0004` → the line write did not take; re-issue `EPS_POWER_ON(line_index=7)`, verify `eps.pl_aocs_wheels` (0x0117), retry - Refused with `0x0A20` (no magnetometer) → re-issue `S2_AOCS_MAGNETOMETER(device_id=521, state=1)`; there is no redundant magnetometer - Refused with `0x0A21` (no magnetorquers) → re-issue `MTQ_ENABLE(enable=1)` - Rates increase in DETUMBLE → `AOCS_SET_MODE(mode=0)` and investigate the magnetometer and magnetorquer polarity - SoC below 40 % → stop, switch off the last line powered, wait for SoC > 60 %, resume; bus below 29.6 V → CTG-001 - Per-line overcurrent within 30 s of enable → CTG-012 Overcurrent Response; the line stays off | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: verify the EPS baseline | `HK_REQUEST(sid=1)` (S3.27) | □ | Bus 29.6–34.6 V, SoC > 60 %, `eps.power_gen` > 0 W | | 2 | EPS: energise the heater lines only if their zones need heat | `EPS_POWER_ON(line_index=5)` / `(line_index=6)` (S8 func_id 19) | □ | `htr_bat` only if `tcs.temp_battery` < +5 °C; `htr_obc` only if `tcs.temp_obc` < −10 °C (its heater ON point). Log which lines were energised: COM-002 reads that record | | 3 | EPS: energise the AOCS suite line and enable its housekeeping | `EPS_POWER_ON(line_index=7)` (S8 func_id 19); `HK_ENABLE(sid=2)` (S3.5) | □ | `eps.pl_aocs_wheels` (0x0117) = 1. The line supplies the whole AOCS suite; it does **not** enable the wheels: `aocs.wheels_enabled` (0x0260) stays 0 until COM-004 | | 4 | AOCS: enable the DETUMBLE sensor and actuators, and the GPS receiver | `S2_AOCS_MAGNETOMETER(device_id=521, state=1)`; `MTQ_ENABLE(enable=1)` (S8 func_id 9); `S2_AOCS_GPS_RECEIVER(device_id=527, state=1)` | □ | `aocs.mtq_enabled` (0x025E) = 1. The GPS time to first fix starts now: LEOP-006 and LEOP-003 need it | | 5 | AOCS: command DETUMBLE | `AOCS_SET_MODE(mode=2)` (S8 func_id 0) | □ | `aocs.mode` (0x020F) = 2 within 10 s; `aocs.mag_enabled` (0x025C) = 1; `aocs.mag_field_total` (0x0277) 5–70 µT; rates (0x0204–0x0206) trending down within 60 s | | 6 | EPS: confirm the exit power state; GO LEOP-005 | `HK_REQUEST(sid=1)` (S3.27) | □ | `eps.pl_payload` (0x0113), `eps.pl_fpa_cooler` (0x0114), `eps.pl_prop` (0x014A) and `aocs.wheels_enabled` (0x0260) all 0; power margin positive. "LEOP-007 complete" | **Note.** The spacecraft has **nine** power lines, indices 0–8: `obc`, `ttc_rx`, `ttc_tx`, `payload`, `fpa_cooler`, `htr_bat`, `htr_obc`, `aocs_wheels`, `prop`. There is no IMU line and no AOCS-sensor line — the gyros are S2 devices `S2_AOCS_GYROSCOPE_0/1/2`, powered in COM-003. The payload line is powered in COM-009, the cooler line in COM-010, and the `prop` line only for manoeuvre preparation (NOM-004 / NOM-013) and disposal (NOM-014). The COARSE_SUN transition is automatic and is observed in LEOP-005; the phase-5 declaration is LEOP-003 step 7.

14.2 Commissioning procedure cards

> **Editor's note — commissioning running order (ruling Q16).** COM-001 → COM-002 → **COM-004 → COM-003** → COM-005 → … → COM-010 → COM-012 → COM-011 → COM-013. COM-004 enables the wheels and commands NOMINAL; COM-003 then calibrates the gyros in that wheel-controlled NOMINAL, because its 0.10 °/s gate cannot be met in the wheels-off COARSE_SUN hold. The cards below stay in ID order.

[FE] COM-001 — EPS Power System Checkout
Entry point: Start of commissioning pass 1, spacecraft steady in the COARSE_SUN hold, phase 5, SoC > 70 %. **Goal:** Baseline bus voltage, battery SoC and array performance, characterise the eclipse discharge and recovery, verify a positive orbit-average energy balance, and verify bus regulation under a commanded load transient. **Possible negative outcomes / deviations:** - Bus voltage below 30.9 V → check SoC; below 29.6 V → CTG-001 Under-Voltage Load Shed and, with FD approval, `OBC_SET_MODE(mode=1)` (1 is SAFE) - Generation below 35 W in sunlight with good sun-pointing → read the per-wing currents (0x0146, 0x0147); log the anomaly and assess a single-wing budget - SoC below 50 % at the end of eclipse → reduce load and defer high-power commissioning | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: baseline bus, battery and SoC | `HK_REQUEST(sid=1)` (S3.27); `GET_PARAM(param_id=0x0100)`, `GET_PARAM(param_id=0x0101)` (S20.3) | □ | Bus (0x0105) 28.4–34.0 V — it follows the battery up to the 33.4 V cap; `eps.bat_voltage` (0x0100) **30.7–33.6 V** at SoC > 70 % — the pack is 8S-1P, 100 Wh, 28.8 V nominal, with an OCV running 24.0 V at 0 % to 33.6 V at 100 %. Any 14.4–16.8 V band is a 4S figure and does not belong to this spacecraft | | 2 | EPS: measure sunlit array performance | `HK_REQUEST(sid=1)` (S3.27) | □ | `eps.power_gen` (0x0107) 40–65 W sun-pointed with wings deployed | | 3 | EPS: monitor a full eclipse discharge | `HK_REQUEST(sid=1)` at entry / mid-eclipse | □ | Generation ≈ 0 W; SoC falling < 1.5 %/min; bus > 30.9 V | | 4 | EPS: monitor charge recovery after eclipse exit | `HK_REQUEST(sid=1)` at exit + 5 min | □ | `eps.power_gen` (0x0107) > 35 W within 60 s of exit; `eps.bat_soc` (0x0101) rising | | 5 | EPS: verify the orbit-average energy balance | `GET_PARAM(param_id=0x0108)` (S20.3) | □ | Orbit-average generation > orbit-average consumption; end-of-eclipse SoC > 50 % | | 6 | EPS: bus regulation under a commanded load transient — the battery-heater line, not the payload | `EPS_POWER_ON(line_index=5)` (S8 func_id 19); then `EPS_POWER_OFF(line_index=5)` (func_id 20) only if the line was off at entry | □ | Record `eps.pl_htr_bat` (0x0115) first. Bus stays 30.2–34.0 V, returns to 30.9–34.0 V; line back to its entry state. The payload line is OFF until COM-009, so a payload mode command is no transient | | 7 | SPACON: record baselines for nominal-ops reference | (mission log) | □ | "COM-001 complete" |

[SE] COM-002 — TCS Verification

Entry point: After COM-001 complete and GO; COARSE_SUN hold, phase 5. Record `eps.pl_htr_bat` (0x0115) and `eps.pl_htr_obc` (0x0116) first — LEOP-007 step 2 may have energised one. Whatever the entry state, this card leaves both heater lines ON with the circuits in AUTO (thermostat): the nominal configuration from now on. **Goal:** Verify every thermal zone against its predicted range over a full orbit, exercise each heater circuit with its line powered, read the autonomous battery thermostat thresholds and check the FPA cooler interface before payload commissioning. **Possible negative outcomes / deviations:** - Heater ON but no zone response within 3 min → check the line is really powered (a heater whose line is dark obeys the switch and heats nothing); then CTG-004 Thermal Exceedance; mark the heater suspect - Thermostat thresholds not +5 / +10 °C → raise an anomaly report; this card reads the thresholds, it does not change them - Two heaters share a fault signature → suspect a common bus; coordinate with EPS | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Thermal: read all zones baseline, then through sunlit phase and eclipse | `HK_REQUEST(sid=3)` (S3.27) | □ | OBC (0x0406) −10…+50 °C; battery (0x0407) +5…+35 °C; FPA (0x0408) +8…+14 °C uncooled | | 2 | Thermal: exercise the battery heater with its line powered | `EPS_POWER_ON(line_index=5)` (S8 func_id 19); `HEATER_BATTERY(on=1)` / `(on=0)` (S8 func_id 40); `HEATER_AUTO_MODE(circuit=0)` (func_id 45) | □ | `eps.pl_htr_bat` (0x0115) = 1; battery temp rising or stable within 180 s; line stays ON, circuit on its thermostat (ON +5 / OFF +10 °C) | | 3 | Thermal: exercise the OBC heater with its line powered | `EPS_POWER_ON(line_index=6)`; `HEATER_OBC(on=1)` / `(on=0)` (S8 func_id 41); `HEATER_AUTO_MODE(circuit=1)` (func_id 45) | □ | `eps.pl_htr_obc` (0x0116) = 1; `tcs.temp_obc` stable or rising within 120 s; line stays ON, circuit on its thermostat (ON −10 / OFF −5 °C). Func 42 is the thruster catbed heater and stays untouched outside manoeuvre preparation | | 4 | Thermal: read the autonomous battery thermostat thresholds | `GET_PARAM(param_id=0x0410)`, `GET_PARAM(param_id=0x0411)` (S20.3) | □ | ON at +5 °C, OFF at +10 °C; between them the previous state holds | | 5 | Thermal: FPA cooler interface check (pre-activation) | `GET_PARAM(param_id=0x040C)`, `GET_PARAM(param_id=0x0114)` (S20.3) | □ | `tcs.cooler_fpa` = 0; `eps.pl_fpa_cooler` = 0; record the uncooled FPA baseline for COM-010 | | 6 | SPACON: compile the thermal baseline report | (mission log) | □ | Record the zone profiles and heater responses | | 7 | SPACON: log; GO COM-004 | (mission log) | □ | "COM-002 complete". COM-004 runs before COM-003 (ruling Q16) |

[AOCS] COM-003 — AOCS Sensor Calibration

Entry point: After COM-004 (ruling Q16): all four wheels ENABLED (`aocs.wheels_enabled` 0x0260 = 4), AOCS in commanded NOMINAL on the wheels (`aocs.mode` 0x020F = 4, or 8 in eclipse), star trackers and gyro units OFF, SoC > 60 %, ≥ 20 min of pass remaining. **Goal:** Bring the three gyro units and a star tracker into the determination loop, then run the on-board gyro bias calibration so that the rate channel is compensated before the spacecraft is asked to hold fine pointing. **Why after COM-004.** The calibration is refused while the body rate is more than **0.10 °/s** from the reference (`0x0A04`). In the wheels-off COARSE_SUN hold the magnetorquer-only control hunts at 0.1–0.4 °/s and never meets that gate; wheel-controlled NOMINAL does. The gate stays at 0.10 °/s. **Possible negative outcomes / deviations:** - Refused with `0x0A02` → a gyro unit is off: read `aocs.gyro_units_mask` (0x02D1) and power it with the matching `S2_AOCS_GYROSCOPE_n` command - Refused with `0x0A03` → no star tracker in the loop: wait for `aocs.est_source` (0x02C1) bit 0 (step 4), or recover the tracker (CTG-008) - Refused with `0x0A04` → body rate above 0.10 °/s: wait for the slew, the desaturation or the wheel transient to end (`aocs.mode` ≠ 7) and retry - `AOCS_GYRO_CAL_ABORT` (0x0212) mid-arc → nothing was applied; fix the cause named in `aocs.gyro_cal_reject` (0x02D0) and re-run from step 6 - Estimated bias above 0.5 °/s on any axis → do not accept it; re-run with `duration_s=900`; if it persists, raise an anomaly report and go to CTG-029 - Tracker not acquired within 120 s → status 3 is BLIND (Sun or Moon in the field of view), not failed; wait, or select ST2 with `ST_SELECT(unit=2)` - There is no magnetometer calibration on board and no `MAG_SELECT` (func 7 is refused): the single magnetometer is used as it is | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: enable the AOCS determination housekeeping | `HK_ENABLE(sid=2)`, `HK_ENABLE(sid=7)` (S3.5) | □ | SID 7 carries the calibration channels 0x02C9–0x02D0 every 4 s; without it the calibration runs blind | | 2 | AOCS: power all three gyro units | `S2_AOCS_GYROSCOPE_0(device_id=518, state=1)`, `S2_AOCS_GYROSCOPE_1(device_id=519, state=1)`, `S2_AOCS_GYROSCOPE_2(device_id=520, state=1)` (S2.1) | □ | `aocs.gyro_units_on` (0x0294) = 3; `aocs.gyro_units_mask` (0x02D1) = 7 (bit 0 X, bit 1 Y, bit 2 Z). No redundant unit and no IMU power line | | 3 | AOCS: power both star trackers, select ST1 | `ST1_POWER(on=1)` (func_id 4), `ST2_POWER(on=1)` (func_id 5), `ST_SELECT(unit=1)` (func_id 6) | □ | 1 = ST1, 2 = ST2, 0 is refused. `aocs.st1_status` (0x0240) = 2 and `aocs.st1_num_stars` (0x0241) ≥ 5 within 120 s; `aocs.st2_status` (0x0243) = 2 | | 4 | AOCS: wait for `aocs.est_source` bit 0 — no mode command | (observe) | □ | Stay in the NOMINAL left by COM-004. `aocs.est_source` (0x02C1) bit 0 (star tracker aiding) set within 600 s; `aocs.knowledge_err_3sig` (0x02A7) < 300 arcsec. The `0x0A03` gate is this bit, not the mode | | 5 | AOCS: record the pre-calibration bias | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.gyro_bias_x/y/z` (0x0270–0x0272) and `aocs.gyro_bias_resid` (0x02D2); a few times 1e-5 °/s on a healthy unit | | 6 | AOCS: run the gyro bias calibration ON BOARD | `AOCS_GYRO_CALIBRATION(action=0, duration_s=300)` (S8 func_id 13) | □ | `aocs.gyro_cal_state` (0x02C9) = 1 (RUNNING) within 20 s. Refusal codes `0x0A01`–`0x0A05` are latched in 0x02D0; S1.7 arrives when the arc ends. There is no ground-computed bias upload | | 7 | AOCS: monitor the 300 s arc — no slew, desaturation or mode change | `HK_REQUEST(sid=7)` (S3.27) every 30 s | □ | `aocs.gyro_cal_progress` (0x02CA) ≥ 100 within 330 s | | 8 | AOCS: verify the result and record the bias | `GET_PARAM(param_id=0x02CB)` … `(param_id=0x02CD)` (S20.3) | □ | `aocs.gyro_cal_state` = 2 (COMPLETE), `aocs.gyro_cal_result` (0x02CF) = 1; each axis within ±0.5 °/s | | 9 | AOCS: confirm the compensation took effect | `HK_REQUEST(sid=2)`, `HK_REQUEST(sid=7)` (S3.27) | □ | `aocs.gyro_bias_resid` (0x02D2) < 0.005 °/s; `aocs.gyro_cal_comp_mag` (0x02CE) recorded | | 10 | AOCS: five-minute determination check; log; GO COM-005 | `HK_REQUEST(sid=7)` (S3.27) every 60 s for 5 min | □ | `aocs.knowledge_err_3sig` < 120 arcsec; `aocs.control_err` (0x02A8) < 1.0°. "COM-003 complete" |

[AOCS] COM-004 — AOCS Actuator Checkout

Entry point: Commissioning pass 1, after COM-002 GO and **before** COM-003 (ruling Q16). AOCS in the LEOP COARSE_SUN hold (`aocs.mode` 0x020F = 3, or 8 in eclipse), wheels DISABLED, star trackers and gyros OFF, rates < 0.5 °/s, `eps.pl_aocs_wheels` (0x0117) = 1 from LEOP-007, SoC > 65 %, ≥ 20 min of pass remaining. **Goal:** **Enable the four reaction wheels** — nothing else in the catalogue does — and command wheel-controlled NOMINAL; then spin each wheel up and down against its baseline, read the wheel temperatures and rehearse a magnetic desaturation. COM-003 then calibrates the gyros in this NOMINAL. **This is the card that hands attitude authority to the wheels.** The bootloader brings the spacecraft up with every wheel **DISABLED** and leaves them that way. Powering the `aocs_wheels` EPS line energises the drive electronics only; the separation is deliberate, so that a power-line switch can never silently return control authority to the loop. If this card is skipped, `aocs.wheels_enabled` (0x0260) stays 0, NOMINAL and FINE_POINT have no actuators, and the FDIR rule on `aocs.wheels_enabled < 2` safes the AOCS the moment the mission phase reaches NOMINAL. **Possible negative outcomes / deviations:** - A wheel does not report enabled → retry once; if it stays disabled, continue with the remaining wheels — three are sufficient (`aocs.fine_degraded`, 0x0299) — but **do not leave this card with fewer than three enabled** - A wheel does not reach the commanded bias within 90 s → reduce the test bias to +1000 rpm as a diagnostic; check `eps.pl_aocs_wheels` (0x0117) and `eps.line_current_7` (0x011F); if power is nominal, flag the wheel failed and go to CTG-007 - Wheel temperature > +55 °C during testing → halt that wheel, 10 min cooldown, re-check before resuming - Attitude error > 10° during a wheel test → halt immediately with `AOCS_RW_RAMP_DOWN(wheel_idx=255, target_rpm=0.0)` — 255 is all wheels — and let the attitude recover before resuming - Desaturation does not reduce wheel speeds → check `aocs.mtq_enabled` (0x025E); magnetic authority is weakest near the equator, so retry on a polar passage | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: enable the four reaction wheels, one command per wheel | `AOCS_ENABLE_WHEEL(wheel_idx=0)`, `(wheel_idx=1)`, `(wheel_idx=2)`, `(wheel_idx=3)` (S8 func_id 3) | ☐ | `aocs.wheels_enabled` (0x0260) = 4 and `aocs.rw1_enabled`…`rw4_enabled` (0x0254–0x0257) = 1 within 10 s. Nothing else in the catalogue enables a wheel | | 2 | AOCS: command wheel-controlled NOMINAL | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | ☐ | `aocs.mode` (0x020F) = 4 (or 8 in eclipse) within 300 s; `aocs.commanded_mode` (0x0297) = 4. Accepted without a star tracker: attitude knowledge stays on the coarse sun and magnetometer path until COM-003 | | 3 | AOCS: record baseline wheel speeds | `HK_REQUEST(sid=2)` (S3.27) | ☐ | 0x0207–0x020A; the newly enabled wheels pick up the control load at once, so non-zero speeds are normal — this baseline is the reference for every check below | | 4 | AOCS: spin each wheel up, through zero, and back | `RW_SET_SPEED_BIAS(wheel_idx=N, bias_rpm=2000.0)`, then `(bias_rpm=-4000.0)` (S8 func_id 8) | ☐ | The bias is **added to the wheel's current speed**. Baseline +1800…+2200 rpm within 60 s, then baseline −2200…−1800 rpm within 120 s; `aocs.att_error` (0x0217) < 5° throughout | | 5 | AOCS: return each wheel to zero | `AOCS_RW_RAMP_DOWN(wheel_idx=N, target_rpm=0.0)` (S8 func_id 14) | ☐ | < 100 rpm within 60 s. Speeds 0x0207–0x020A are READ-ONLY — they cannot be set with `SET_PARAM` | | 6 | AOCS: read the wheel temperatures | `GET_PARAM(param_id=0x0218)`…`(0x021B)` (S20.3) | ☐ | −10…+55 °C. 0x028A–0x028D are CSS head health percentages, **not** wheel temperatures | | 7 | AOCS: desaturation rehearsal on the magnetorquers | `AOCS_DESATURATE` (S8 func_id 1) | ☐ | Speeds trending toward 0 rpm; `aocs.att_error` < 5°. Budget the honest **20–50 min** dump window (see CTG-007), not the five to ten a rehearsal note might suggest | | 8 | SPACON: compile the wheel commissioning report; log; GO COM-003 | (mission log) | ☐ | "COM-004 complete"; post-condition `aocs.wheels_enabled` (0x0260) = 4, AOCS in commanded NOMINAL |

[AOCS] COM-005 — AOCS Mode Transitions

Entry point: After COM-003 GO: wheels enabled (`aocs.wheels_enabled` 0x0260 = 4; three is acceptable, degraded), star tracker in the loop, gyros calibrated, AOCS already in commanded NOMINAL (mode 4, or 8 in eclipse) since COM-004. **Goal:** Exercise every **commanded** transition the operations concept uses — NOMINAL → FINE_POINT → NOMINAL, NOMINAL → INERTIAL → NOMINAL, the COARSE_SUN hold, desaturation — and **observe** the automatic ones: the return from the hold, the eclipse handover and the end of DESAT. Command what is commanded, wait and verify what is automatic, and never verify the mode you have just left. **Possible negative outcomes / deviations:** - `AOCS_SET_MODE(mode=5)` refused with `0x0A22` → read `aocs.fine_ready` (0x02C5): a non-zero bitmask names the unmet precondition (bit 0 ST < 5 stars, bit 1 fewer than THREE wheels, bit 2 MEKF unconverged, bit 3 rates high, bit 4 momentum ≥ 70 %, bit 5 propagate-only or eclipse, bit 6 slew not idle 60 s); in eclipse, wait for the sunlit arc - No automatic promotion out of COARSE_SUN within the window → check CSS validity (eclipse blocks it), `aocs.att_error` < 15° and the tracker still tracking; recover the tracker (CTG-008) rather than forcing the mode - DESAT ends with `aocs.desat_ineffective` (0x029C) = 1 → magnetorquer authority could not unload in time; re-run over a polar arc - Rates re-grow on a transition → CTG-002 AOCS Anomaly | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: latch the commanded mode | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | □ | Already flying mode 4 since COM-004; the command latches `aocs.commanded_mode` (0x0297) = 4. Mode 4 or 8; error < 1.0° within 600 s on the sunlit side | | 2 | AOCS: NOMINAL → FINE_POINT through its gate | `AOCS_FINE_READY_CHECK` (S8 func_id 89); `AOCS_SET_MODE(mode=5)` (S8 func_id 0) | □ | `aocs.fine_ready` (0x02C5) = 0 first; then mode 5 within 30 s, error < 0.1° within 300 s | | 3 | AOCS: FINE_POINT → NOMINAL | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | □ | Mode 4 within 30 s; `aocs.commanded_mode` = 4 | | 4 | AOCS: NOMINAL → INERTIAL and back | `AOCS_SET_MODE(mode=9)`, then `(mode=4)` (S8 func_id 0) | □ | Mode 9 within 30 s, then mode 4 within 30 s. INERTIAL is the one operational mode that holds through the terminator | | 5 | AOCS: command the COARSE_SUN hold, then wait for the automatic return | `AOCS_SET_MODE(mode=3)` (S8 func_id 0) | □ | Mode 3 and `aocs.mode_demoted` (0x0298) = 1 within 30 s; then mode 4 by itself within 1800 s and the demotion flag back to 0. Verify the PROMOTED mode, not mode 3 | | 6 | AOCS: observe the eclipse handover in the archive | (archive) | □ | Mode 4 ↔ 8 at the terminator with `aocs.commanded_mode` = 4 throughout; no command | | 7 | AOCS: desaturation, which exits by itself | `AOCS_DESATURATE` (S8 func_id 1) | □ | Mode 7 (DESAT) within 30 s, then back to mode 4 within 900 s without a command | | 8 | SPACON: archive the transition matrix; log; GO COM-006 | (mission log) | □ | Each commanded row carries its TC and verify, each automatic row its trigger and time. "COM-005 complete" |

[TT] COM-006 — TTC Link Verification

Entry point: Commissioning pass 1, after COM-005 GO. **Goal:** Configure the S-band chain for measurement, baseline the link margin on both operational rungs of the TM rate ladder, demonstrate the HIGH-tier gates, exercise the redundant transponder and leave the link safed at LOW. **Possible negative outcomes / deviations:** - `TTC_SET_DATA_RATE(rate=1)` refused with S1.8 error `0x0007` → read the failure notice, which names the unmet gate verbatim (beacon mode latched / antenna stowed / DOWNLINK mode off / elevation < 10°); all four report the same code - Margin at HIGH below prediction → defer high-rate downlinks until investigated; log to ticket - Receiver lock unstable across a transponder switch → CTG-003 TTC Link Loss Recovery; CTG-014 for a BER anomaly | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | TT&C: energise the transmitter line and PA | `EPS_POWER_ON(line_index=2)` (S8 func_id 19); `TTC_PA_ON` (S8 func_id 66) | □ | `eps.pl_ttc_tx` (0x0112) = 1; `ttc.pa_on` (0x0516) = 1 | | 2 | TT&C: release the forced beacon tier and deploy the antenna | `TTC_SET_BEACON_MODE(on=0)` (func_id 70); `TTC_DEPLOY_ANTENNA` (func_id 69) | □ | `ttc.beacon_mode` (0x0521) = 0; `ttc.antenna_deployed` (0x0520) = 1; `ttc.tx_active` (0x0525) = 1 | | 3 | TT&C: baseline the LOW rung | `TTC_SET_DATA_RATE(rate=0)` (S8 func_id 65) | □ | `ttc.tm_data_rate` (0x0506) = 16000 bps, `ttc.tm_tier` (0x0539) = 1; margin 55–59 dB at 30° | | 4 | TT&C: enter DOWNLINK mode, then baseline HIGH | `TTC_SET_DOWNLINK_MODE(on=1)` (func_id 79); `TTC_SET_DATA_RATE(rate=1)` (func_id 65) | □ | 128000 bps, tier 2, `ttc.tm_symbol_rate` (0x053B) = 64000 Bd (QPSK); margin 45–48 dB — a 10.5 ± 0.5 dB step | | 5 | TT&C: demonstrate the elevation gate | `TTC_SET_DATA_RATE(rate=1)` below 10° elevation | □ | REJECTED with S1.8 error `0x0007`; the link stays at LOW | | 6 | TT&C: switch to the redundant transponder and back | `TTC_SWITCH_REDUNDANT` (func_id 64); `TTC_SWITCH_PRIMARY` (func_id 63) | □ | `ttc.link_status` (0x0501) = 2 within 60 s each way; RSSI within 2 dB | | 7 | TT&C: safe the link; log results | `TTC_SET_DATA_RATE(rate=0)`; `TTC_SET_DOWNLINK_MODE(on=0)` | □ | `ttc.tm_tier` = `ttc.tm_tier_cmd` (0x053A) = 1; PA stays ON; "COM-006 complete" | **Note — two transmitters.** The S-band prime and the UHF whip beacon are separate. `TTC_PA_OFF` (func_id 67) stops S-band TM (`ttc.tx_active` → 0) while the beacon keeps emitting SID 11: the spacecraft can be silent on the prime chain and still alive on the beacon. Do not read a dead S-band downlink as a dead spacecraft.

[OB] COM-007 — OBDH Checkout

Entry point: After COM-006: AOCS in commanded NOMINAL, downlink LOW with the antenna deployed, OBC NOMINAL (`obdh.mode` 0x0300 = 0), SoC > 60 %. **Goal:** Verify CPU, memory and packet counters, autonomous HK generation and the watchdog, exercise the OBC mode excursion NOMINAL → SAFE → NOMINAL, and **end in NOMINAL**. **Possible negative outcomes / deviations:** - Watchdog reported disarmed or an unexpected reset cause → CTG-010 OBDH Watchdog Recovery - Memory error or SEU counter incrementing between reads → CTG-016 Memory Segment Failure; defer payload power-on - Mode transition fails → retry once; if NOMINAL is unreachable, continue in SAFE only with FD approval and raise an anomaly | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | OBDH: verify OBC mode | `HK_REQUEST(sid=4)` (S3.27) | □ | `obdh.mode` (0x0300) = 0 (NOMINAL). Encoding: 0 NOMINAL, 1 SAFE, 2 EMERGENCY | | 2 | OBDH: check CPU load | `GET_PARAM(param_id=0x0302)` (S20.3) | □ | `obdh.cpu_load` 10–50 % | | 3 | OBDH: check memory health | `GET_PARAM(param_id=0x0303)`, `GET_PARAM(param_id=0x031E)`, `GET_PARAM(param_id=0x0319)` (S20.3) | □ | `obdh.mmm_used` < 70 %; `obdh.mem_errors` and `obdh.seu_count` baselined and not incrementing between reads | | 4 | OBDH: check TC/TM counters | `GET_PARAM(param_id=0x0304)`, `GET_PARAM(param_id=0x0306)`, `GET_PARAM(param_id=0x0307)` (S20.3) | □ | TC received matches the ground log; TC rejected = 0; TM generated incrementing | | 5 | OBDH: verify autonomous HK generation | `GET_PARAM(param_id=0x0307)` before and after an interval | □ | `obdh.tm_pkt_count` has incremented | | 6 | OBDH: verify the watchdog and last reset cause | `OBC_DIAGNOSTIC` (S8 func_id 61); `GET_PARAM(param_id=0x0316)` (S20.3) | □ | Diagnostic reply: watchdog armed; `obdh.last_reboot_cause` = 0 (none) | | 7 | OBDH: mode excursion NOMINAL → SAFE → NOMINAL | `OBC_SET_MODE(mode=1)`, then `OBC_SET_MODE(mode=0)` (S8 func_id 50) | □ | `obdh.mode` = 1 within 15 s, then 0 within 15 s. The card must not end in SAFE | | 8 | SPACON: record the OBDH baseline; log; GO COM-008 | (mission log) | □ | "COM-007 complete" |

[OB] COM-008 — FDIR Configuration & Protection Functions

Entry point: After COM-007 GO: OBC NOMINAL (`obdh.mode` 0x0300 = 0), AOCS in commanded NOMINAL, downlink LOW, payload and cooler lines OFF, no load-shed latch (`eps.load_shed_stage` 0x0134 = 0), SoC > 70 %, link margin > 6 dB, FD approval for protection-function testing. **Goal:** Confirm the on-board protection stack is configured and alive before the payload flies: the S12 monitoring service and its definitions, the S19 event-action table, the FDIR summary telemetry, and the commanded load-shed ladder with its latch. The card ends in exactly the state it started in. **Possible negative outcomes / deviations:** - TM(12,13) or TM(19,128) empty or missing definitions → reload from the operations database (`MON_ADD_DEF` / `EA_ADD` per the database record) and re-request the report - `eps.load_shed_stage` does not latch at step 4, or stage 0 re-powers a line → a shed-handler regression; record it, escalate, and do not proceed to payload commissioning - `fdir.level` (0x0700) ≠ 0 at step 3 or 5 → read the FDIR journal and clear the underlying cause; never disable the rule to pass the card | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | SPACON: enable and inventory the S12 monitoring service | `MON_ENABLE` (S12.1); `MON_REPORT_DEFS` (S12.12) | □ | TM(12,13) within 10 s: `count`, then one record per definition (`mon_id`, `param_id`, `check_type`, `enabled`, limits). Compare with the operations database: every shipped definition present, none unexpected | | 2 | SPACON: inventory the S19 event-action table | `EA_REPORT_DEFS` (S19.8) | □ | TM(19,128) within 10 s. The momentum-management chain (S12 monitor on `aocs.h_env_frac` 0x02BA → ea_id 2001 → S8 func 1) and the TCS overtemp safing chain (ea 3003) present and enabled | | 3 | SPACON: read the FDIR summary telemetry | `GET_PARAM(param_id=0x0700)`, `GET_PARAM(param_id=0x0701)` (S20.3) | □ | `fdir.level` = 0 and `fdir.active_alarms` = 0: a clean baseline for the functional test | | 4 | EPS: functional test of the commanded load-shed latch | `EPS_EMERGENCY_LOAD_SHED(stage=1)`, then `(stage=0)` (S8 func_id 25) | □ | Record the line flags first. Stage 1: `eps.load_shed_stage` = 1 within 5 s (it LATCHES), event `LOAD_SHED_STAGE_1` (0x0106), `eps.pl_payload` (0x0113) = 0. Stage 0: latch back to 0, and no line re-powered. Never command stage 2 or 3 here: stage 3 sheds `ttc_tx` and drops this downlink | | 5 | SPACON: confirm no residual protection state; log; FD GO COM-009 | `GET_PARAM(param_id=0x0700)` (S20.3); `HK_REQUEST(sid=1)`, `HK_REQUEST(sid=4)` (S3.27) | □ | `fdir.level` = 0; `eps.load_shed_stage` = 0; `obdh.mode` = 0 (NOMINAL) — this card makes no OBC-mode excursion. "COM-008 complete" |

[PE] COM-009 — Payload Power-On

Entry point: Commissioning pass 3, after COM-008 with the nominal FDIR set armed; SoC > 75 %, generation > 40 W. **Goal:** Energise the payload line, bring the imager to STANDBY at the expected current, and confirm it HOLDS there. **Possible negative outcomes / deviations:** - Overcurrent on the payload line within 2 s → CTG-012 Overcurrent Response (autonomous line open); investigate before retry - Payload does not reach STANDBY, or does not hold → CTG-006 Payload Anomaly; cycle the line off/on after 30 s; if recurring, defer COM-010 - Payload temperature rises faster than predicted → cross-check Thermal; possible cooler fault to follow **Decision tree:** - IF power-on clean AND `payload.mode` = 1 held for 60 s AND load increase within envelope → GO COM-010 - IF FDIR opens the line → CTG-012; investigation before any retry - IF mode does not reach STANDBY in 60 s → cycle once, then escalate | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | SPACON: FD GO for payload power-on | (FD call) | □ | Dual-keyed | | 2 | Payload: confirm the payload is OFF | `HK_REQUEST(sid=5)` (S3.27) | □ | `payload.mode` (0x0600) = 0 | | 3 | EPS: energise the payload line | `EPS_POWER_ON(line_index=3)` (S8 func_id 19) | □ | `eps.pl_payload` (0x0113) = 1 | | 4 | Payload: command STANDBY | `PAYLOAD_SET_MODE(mode=1)` (S8 func_id 26) | □ | `payload.mode` = 1 within 60 s, and STILL 1 after a further 60 s | | 5 | EPS: verify the load increase | `GET_PARAM(param_id=0x0106)` (S20.3) — `eps.power_cons` | □ | +2–3 W; bus (0x0105) > 30.9 V | | 6 | Payload: verify imager health and storage | `payload.imager_status` (0x0611); `payload.store_used` (0x0604); `payload.product_count` (0x0648) | □ | No fault bits; store ~20 % used (ships with archive fixtures); product count 0 | | 7 | SPACON: hold ≥ 30 s for transient settle; log; GO COM-010 | (mission log) | □ | "COM-009 complete" |

[SE] COM-010 — FPA Cooler Activation

Entry point: Commissioning pass 3, immediately after COM-009 GO; FPA at the uncooled equilibrium (+8 to +14 °C), payload in STANDBY. **Goal:** Drive the FPA from ambient to the **−15 °C default setpoint** and confirm it is regulating there, with positive cooler margin. *(EOSAT-1 carries a **miniature Stirling-class cryocooler**, not a thermo-electric stack. Its authority degrades with the temperature lift — `fpa_cooler_carnot_fraction` = 0.23 — so the achievable floor depends on the sink temperature and the imaging load. The commandable setpoint envelope is **−20 °C to 0 °C**; a real multispectral imager would push far colder, and this card adopts the flight envelope.)* **Possible negative outcomes / deviations:** - Cool-down stalls above the setpoint → read `payload.cooler_margin_k` (0x064B): ≤ 0 means the cooler physically cannot reach the commanded setpoint at the present sink and load, and the answer is a **warmer setpoint, not more waiting** (`payload.cooler_floor_c`, 0x064A; TCS mirrors at 0x0424 / 0x0425) - Radiator anomaly suspected → coordinate with Thermal; check radiator-zone temperatures - Cooler current climbs → CTG-012 if the line FDIR fires **Decision tree:** - IF `payload.fpa_ready` (0x0610) = 1 within ~15 min → GO COM-011 - IF margin ≤ 0 → FD decides: (a) command a warmer setpoint and continue with degraded SNR, or (b) abandon and re-plan - IF the cooler line FDIR opens → CTG-012; the FPA passively warms; the payload is safe | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Thermal: record the pre-activation state | `HK_REQUEST(sid=3)`, `HK_REQUEST(sid=5)` (S3.27) | □ | `payload.fpa_temp` (0x0601) +8…+14 °C; `payload.mode` = 1 | | 2 | Thermal: enable the FPA cooler | `FPA_COOLER(on=1)` (S8 func_id 43) | □ | `tcs.cooler_fpa` (0x040C) = 1 AND `eps.pl_fpa_cooler` (0x0114) = 1 — func 43 also guarantees the line | | 3 | EPS: verify the cooler power draw | `GET_PARAM(param_id=0x0106)` (S20.3) | □ | +14–16 W; bus > 30.9 V | | 4 | Thermal: monitor the cool-down | `GET_PARAM(param_id=0x0601)` (S20.3) at T+2 / T+5 min | □ | < +4 °C at T+2 (≈ 4 °C/min initially); < −8 °C at T+5 | | 5 | Thermal: confirm the setpoint is reached | `GET_PARAM(param_id=0x0601)` at T+10 / T+15 min | □ | −16.0…−14.0 °C; `payload.fpa_setpoint` (0x0631) = −15.0 | | 6 | Thermal: confirm readiness and margin | `payload.fpa_ready` (0x0610) = 1; `payload.cooler_margin_k` (0x064B) > 0 | □ | Readiness follows the ACTIVE setpoint, not a fixed number | | 7 | Thermal: hold 30 min and check stability, including through eclipse | `GET_PARAM(param_id=0x0601)` (S20.3) | □ | −16.5…−14.5 °C, < 1.0 °C peak-to-peak | | 8 | SPACON: log; GO COM-011 | (mission log) | □ | "COM-010 complete" | **Note.** To command a different setpoint use `PAYLOAD_COOLER_SETPOINT(setpoint=-15)` (S8 func_id 36). The argument is a uint16 in **centi-kelvin** — −15.00 °C encodes as 25815 = 0x64D7 — and a value outside [−20, 0] °C is rejected with S1.8 error `0x0003`.

[PE] COM-011 — Payload Calibration

Entry point: After COM-012, commissioning pass 4: imaging chain verified, payload in STANDBY with the FPA cooler OFF (STANDBY releases it since 2026-11-03); step 1 switches it back on. **Goal:** Acquire a dark-frame set and a flat-field set with the internal lamp, then capture a calibration-site image set in FINE_POINT. **Possible negative outcomes / deviations:** - `PAYLOAD_START_CALIBRATION` refused → read the code: `0x0024` refused because the payload is IMAGING, `0x0025` FPA not ready, `0x0026` shutter in the wrong position, `0x0027` lamp in the wrong state, `0x0028` capture with the shutter closed. Dark calibration no longer requires IMAGING mode — the gate is on the optics - `cal_abort_reason` (0x0645) non-zero → the run did not complete; re-acquire after checking FPA stability - FINE_POINT refused → read `aocs.fine_ready` (0x02C5); defer to the next pass | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Payload: cooler ON first, then verify the pre-calibration state | `FPA_COOLER(on=1)` (S8 func_id 43); `HK_REQUEST(sid=3)`, `HK_REQUEST(sid=5)` (S3.27) | □ | `tcs.cooler_fpa` (0x040C) = 1; mode 1; `payload.fpa_ready` (0x0610) = 1 before step 3; free store > 1000 MB (0x060A − 0x060B) | | 2 | Payload: close the shutter, lamp off | `PAYLOAD_SET_SHUTTER(position=0)` (func_id 119); `PAYLOAD_CAL_LAMP(on=0)` (func_id 95) | □ | `payload.shutter_pos` (0x063C) = 0 | | 3 | Payload: run the DARK phase | `PAYLOAD_START_CALIBRATION(mode=0)` (S8 func_id 37) | □ | `payload.mode` (0x0600) = 3 (CALIBRATION); `payload.calibration_active` (0x062A) 1 → 0; `cal_abort_reason` (0x0645) = 0 | | 4 | Payload: open the shutter, lamp on, run the FLAT phase | `PAYLOAD_SET_SHUTTER(position=1)`; `PAYLOAD_CAL_LAMP(on=1)`; `PAYLOAD_START_CALIBRATION(mode=1)` | □ | `payload.cal_valid_mask` (0x0646) = 15 — all four bands | | 5 | Payload: lamp off, shutter open — no mode command: the sequencer is already back in STANDBY with the cooler running, and a STANDBY command would switch the cooler off | `PAYLOAD_CAL_LAMP(on=0)`; `PAYLOAD_SET_SHUTTER(position=1)` (func_id 119) | □ | Shutter OPEN, mode 1 | | 6 | AOCS: transition to FINE_POINT for the site overpass | `AOCS_FINE_READY_CHECK` (func_id 89); `AOCS_SET_MODE(mode=5)` (func_id 0) | □ | `aocs.fine_ready` (0x02C5) = 0; error < 0.1° | | 7 | Payload: capture the calibration-site set | `PAYLOAD_SET_MODE(mode=2)`; `PAYLOAD_CAPTURE(scene_id=0, lines=2048)` (func_id 28) | □ | `payload.last_scene_id` (0x060C) increments; vary integration time with `PAYLOAD_SET_INTEGRATION_TIME` (func_id 34) — SNR moves ~3 dB per doubling | | 8 | Payload / AOCS: restore standard configuration | `PAYLOAD_SET_MODE(mode=1)`; `AOCS_SET_MODE(mode=4)` | □ | Mode 1; `tcs.cooler_fpa` = 0 (STANDBY releases the cooler); `aocs.mode` = 4 | | 9 | SPACON: log; FD GO COM-012 | (mission log) | □ | "COM-011 complete" |

[MULTI] COM-012 — First Light
Entry point: After COM-010 — COM-012 runs **before** COM-011, because calibration needs a verified imaging chain. Payload STANDBY, cooler running (switched on by COM-010) with `payload.fpa_ready` (0x0610) = 1 at −15 °C, AOCS in commanded NOMINAL, downlink LOW, SoC > 70 %, approaching a known ground target. The headline procedure. **Goal:** Acquire the first science image, verify it is stored, and downlink it over the real Service-13 chain for a first quality assessment. **Possible negative outcomes / deviations:** - `payload.last_scene_quality` (0x060D) poor → CTG-015 Corrupted Image Recovery; re-acquire next pass - Pointing error > 0.1° during integration → image likely smeared; mark and re-acquire - AOCS / payload cascade (star tracker fails late) → CTG-008 → CTG-002, defer First Light, do not attempt repeats this shift - Transfer starts but the ground reports missing frames → CTG-014 BER Anomaly; re-open the transfer **Decision tree:** - IF image acquired AND stored AND the S13 transfer completes with a good checksum → declare **First Light** - IF the product is corrupt once → CTG-015; retry next pass (max 3 attempts per FD policy) - IF cascading AOCS / payload anomaly → defer formally; brief Mission Manager | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: go to FINE_POINT through its gate | `AOCS_FINE_READY_CHECK` (S8 func_id 89); `AOCS_SET_MODE(mode=5)` (S8 func_id 0) | □ | `aocs.mode` = 5 within 30 s; `aocs.total_pointing_err_3sig` (0x02AA) < 0.05° within 300 s; `aocs.att_error` (0x0217) < 0.1° | | 2 | Payload: verify FPA temperature and readiness | `GET_PARAM(param_id=0x0601)` (S20.3); `HK_REQUEST(sid=5)` (S3.27) | □ | `payload.fpa_temp` −16.5…−14.5 °C; `payload.mode` (0x0600) = 1; record `payload.image_count` (0x0605) and `payload.mem_used_mb` (0x060B) as baselines | | 3 | Payload: command IMAGING | `PAYLOAD_SET_MODE(mode=2)` (S8 func_id 26) | □ | `payload.mode` = 2 within 30 s | | 4 | Payload: trigger the acquisition | `PAYLOAD_CAPTURE(scene_id=1, lines=2048)` (S8 func_id 28) | □ | `payload.last_scene_id` (0x060C) = 1 within 30 s; `payload.image_count` incremented. Geolocation is stamped on board, not an argument | | 5 | Payload: confirm the product is stored | `GET_PARAM(param_id=0x0604)`, `GET_PARAM(param_id=0x060B)` (S20.3) | □ | `payload.store_used` increased; `payload.mem_used_mb` up 1.0–4.0 MB on the step 2 baseline (4 MB raw, 1.0–2.7 MB compressed) | | 6 | Payload: return to STANDBY | `PAYLOAD_SET_MODE(mode=1)` (S8 func_id 26) | □ | `payload.mode` = 1 within 15 s; STANDBY switches the cooler off (`tcs.cooler_fpa` (0x040C) = 0) | | 7 | TT&C + Payload: downlink the product over S13 | `TTC_SET_DOWNLINK_MODE(on=1)`; `TTC_SET_DATA_RATE(rate=1)`; `S13_CREATE_TRANSFER(scene_id=1, size_mb=0)`; `S13_END_TRANSFER(transfer_id=T)`; then `TTC_SET_DATA_RATE(rate=0)`, `TTC_SET_DOWNLINK_MODE(on=0)` | □ | `ttc.tm_data_rate` (0x0506) = 128000 bps; `payload.transfer_active` (0x062D) = 1, `payload.transfer_progress` (0x062E) increasing, then 0x062D = 0; CRC-32 on the ground; back to 16000 bps | | 8 | AOCS: release FINE_POINT; FD declares First Light | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | □ | `aocs.mode` = 4 within 30 s; payload STANDBY, FPA still in band. "COM-012 complete" |

[AOCS] COM-013 — ADCS Commissioning Closure & NOMINAL Declaration
Entry point: After COM-011 — the commissioning order ends COM-010, COM-012 First Light, COM-011 Payload Calibration, COM-013. Payload commissioned; AOCS in commanded NOMINAL with wheels enabled and gyros calibrated; downlink back at LOW; OBC NOMINAL; mission phase 5; SoC > 60 %. **Goal:** Close commissioning: read back the consolidated end state against each card's exit state **without re-testing anything**, poll all positions, and declare mission phase 6 (NOMINAL). Detumble, mode transitions, wheel and magnetorquer checks and the gyro calibration belong to LEOP-005, COM-003, COM-004 and COM-005, and are not repeated here. **Possible negative outcomes / deviations:** - Any readback in steps 1–3 fails → do NOT declare phase 6; return to the owning card's off-nominal section and re-establish its exit state first - `SYS_SET_MISSION_PHASE` refused → `0x0C03`: the OBC is in the bootloader (re-run LEOP-001 step 5 and LEOP-006); `0x0C04`: backwards without FORCE - `sat.mode` does not derive 1 after the declaration → an FDIR contingency is latched; read the FDIR journal and clear the cause — there is no TC that sets `sat.mode` | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: consolidated AOCS end-state readback | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.mode` (0x020F) in {4, 5, 8}; `aocs.commanded_mode` (0x0297) = 4, `aocs.mode_demoted` (0x0298) = 0; `aocs.wheels_enabled` (0x0260) = 4; `aocs.gyro_units_mask` (0x02D1) = 7; `aocs.st1_status` (0x0240) = 2 with ≥ 5 stars; `aocs.mag_enabled` (0x025C) = 1; error < 1.0° on the sunlit side | | 2 | SPACON: consolidated platform end-state readback | `HK_REQUEST(sid=1)`, `HK_REQUEST(sid=4)`, `HK_REQUEST(sid=6)` (S3.27) | □ | `obdh.mode` (0x0300) = 0; LOW tier; `eps.load_shed_stage` (0x0134) = 0; bus 30.2–34.0 V; `obdh.time_ref` (0x0A13) ≠ 0 | | 3 | Payload: consolidated payload end-state readback | `HK_REQUEST(sid=5)` (S3.27) | □ | `payload.mode` (0x0600) = 1 (STANDBY); `payload.fpa_ready` (0x0610) = 1; `payload.transfer_active` (0x062D) = 0 | | 4 | FD: poll all positions, then declare NOMINAL operations | `SYS_SET_MISSION_PHASE(phase=6, flags=0)` (S8 func_id 120) | □ | COM-013 is the last commissioning card. `eps.spacecraft_phase` (0x0129) = 6 within 5 s; `sat.mode` (0x0001) = 1 within 10 s; event `MISSION_PHASE_CHANGED` (0x0055) carries 5 → 6 | | 5 | FD: hold NOMINAL through an orbit; hand over; log | (mission log) | □ | "COM-013 complete": commissioning closed. If the now-armed `aocs.att_error` FDIR rule demotes the AOCS, investigate the excursion before re-declaring anything | **Note — nothing on board advances the phase for you.** Below phase 6 the derived spacecraft mode (`sat.mode` 0x0001, and the console header) is pinned to SAFE regardless of spacecraft health, the advanced fault-propagation model and the phase-gated FDIR rules stay disarmed, and the propulsion interlock (`min_phase: 6`) refuses to arm any burn — so **NOM-013 cannot execute until this step has been sent**. Going back the other way is refused with `0x0C04` unless the FORCE flag is set (`flags` bit 0 = 1); if a declaration was premature, re-send with the flag and record why.

14.3 Nominal procedure cards

> **Editor's note — the nominal set.** The simulator ships fourteen nominal procedures. Eleven are carded below; **NOM-005 HK Configuration**, **NOM-008 Clock Synchronisation** and **NOM-011 Data Rate Change** are short, single-position cards and are indexed in §14.7 for at-console reading. Two further nominal activities — **Pass Plan Upload** and **Long-Term Trend Review** — are real parts of the nominal flow but ship no procedure file of their own; they are carded here without a NOM number, and the flow references them explicitly. Cards run in shipped-ID order, except that NOM-013, NOM-014 and the two unnumbered ground cards sit immediately before the shift-handover card that closes the section.
>
> **NOM-008 in one paragraph, because the clock is new.** (It is one of the eight machine-runnable cards — `runnable: {kind: markdown}` — so it can be driven from the MCS Procedures tab.) The OBC runs an independent clock that starts at CUC 0 = **J2000** and free-runs at about **12 ppm** (≈ 1 s/day). NOM-008 requests a fresh HK SID 4 *and* an explicit `REQUEST_TIME` (S9.2) rather than reading the screen, corrects with `SET_TIME(cuc_seconds=…)` (S9.1) to a tighter 0.1 s threshold than LEOP-006's 1.0 s, and logs `obdh.time_ref` (0x0A13) and `obdh.time_since_sync_s` (0x0A14) either side. GPS discipline (`OBC_GPS_TIME_SYNC` , S8 func 80) is the alternative and holds true time exactly while the fix is good — that path is CTG-027.

[FD] NOM-001 — Pass Startup
Entry point: AOS at any of the five network stations (Iqaluit, Troll, Svalbard, Inuvik, O'Higgins); full read-and-do checklist runs from AOS to AOS+1 min. **Goal:** Establish the link, verify full HK, run the cross-position health round and authorise planned activities for the pass. **Possible negative outcomes / deviations:** - No lock at AOS → CTG-003 TTC Link Loss Recovery - Station not tracking → CTG-020 Ground Station Antenna Failure (an advisory card — every action in it is taken on the ground) - No HK after lock → CTG-019 No Telemetry at Pass Start - Any subsystem out-of-band on the health round → run the relevant CTG before authorising planned activities **Decision tree:** - IF `ttc.link_status` = 2 AND link test AND full HK AND all positions OK → GO planned activities - IF locked but the link test fails → CTG-019 - IF EPS OUT-OF-BAND → CTG-001 before any further commanding **Full read-and-do checklist:** see §9.3, Table 9.2. | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Ground Seg: station tracking? | (MPS panel) | □ | TRACKING | | 2 | TT&C: link locked at AOS? | `ttc.link_status` (0x0501) = 2 (LOCKED) | □ | RSSI (0x0502) > −100 dBm; margin (0x0503) > 3.0 dB | | 3 | SPACON: link test | `CONNECTION_TEST` (S17.1) — first-key / FD GO / uplink | □ | S17.2 response < 4 s | | 4 | SPACON: full HK sweep | `HK_REQUEST(sid=1…6)` (S3.27) | □ | 1 EPS, 2 AOCS, 3 TCS, 4 Platform/OBDH, 5 Payload, 6 TT&C — all six within 30 s | | 5 | AOCS: mode and rates | `aocs.mode` (0x020F) = 4 (NOMINAL, nadir) | □ | Roll/yaw < 0.05 °/s; pitch −0.063 ± 0.02 °/s — nadir tracking is a real body rate, not an error | | 6 | EPS: bus and SoC | `HK_REQUEST(sid=1)` (S3.27) | □ | SoC > 50 %, bus > 30.9 V, generation > consumption | | 7 | Thermal: zones | `HK_REQUEST(sid=3)` (S3.27) | □ | All zones green | | 8 | FD: GO planned activities | (mission log) | □ | Reference §9.3 for full callout text |

[PE] NOM-002 — Imaging Session

Entry point: Scheduled imaging window opens (per pass plan); AOCS in or transitioning to FINE_POINT. *(Runnable from the MCS Procedures tab — linked to the `imaging_pass` activity.)* **Goal:** Acquire the planned image(s) and verify they are stored in payload memory. **Possible negative outcomes / deviations:** - FINE_POINT refused or demoted → check `aocs.fine_ready` (0x02C5) and `aocs.mode_demoted` (0x0298); HOLD up to 60 s for convergence - Pointing error ≥ 0.5° mid-session → terminate the session early and re-plan - Store ≥ 95 % → halt imaging; prioritise NOM-003 downlink - Momentum climbing → NOM-007, but note that dumps are automatically inhibited while `payload.mode` = 2 (`aocs.desat_inhibit`, 0x029A) | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: enter and verify FINE_POINT | `AOCS_FINE_READY_CHECK` (func_id 89); `AOCS_SET_MODE(mode=5)` (func_id 0) | ☐ | `aocs.fine_ready` = 0; `aocs.mode` = 5; `aocs.mode_demoted` (0x0298) = 0; error < 0.1° | | 2 | Thermal: prepare the focal plane — cooler ON at least 10 min before the window (STANDBY does not run it) | `FPA_COOLER(on=1)` (S8 func_id 43), skip if 0x040C already reads 1; `HK_REQUEST(sid=3)`, `HK_REQUEST(sid=5)` (S3.27) | ☐ | `tcs.cooler_fpa` (0x040C) = 1; `payload.fpa_ready` (0x0610) = 1 within 600 s (about 6.5 min from the uncooled ≈ +10 °C); `tcs.temp_fpa` (0x0408) < −10.0 °C | | 3 | Payload: verify storage | `HK_REQUEST(sid=5)` (S3.27) | ☐ | `payload.store_used` (0x0604) < 90 % | | 4 | Payload: command IMAGING | `PAYLOAD_SET_MODE(mode=2)` (S8 func_id 26) | ☐ | `payload.mode` (0x0600) = 2 within 15 s | | 5 | Payload: trigger each acquisition | `PAYLOAD_CAPTURE(scene_id=N, lines=2048)` (S8 func_id 28) | ☐ | Time-tag with `SCHEDULE_TC` (S11.4) at each overpass; `payload.last_scene_id` (0x060C) = commanded id | | 6 | Payload: monitor the session | `HK_REQUEST(sid=5)` / `(sid=2)` / `(sid=3)` every 30 s | ☐ | `payload.image_count` (0x0605) incrementing; error < 0.1°; store < 95 % | | 7 | Payload: terminate the session | `PAYLOAD_SET_MODE(mode=1)` (S8 func_id 26) | ☐ | Mode 1; `tcs.cooler_fpa` = 0 — STANDBY releases the cooler; consumption falls (imaging load and the 15 W cooler) | | 8 | SPACON: log; queue for NOM-003 downlink | (mission log) | ☐ | "NOM-002 complete" |

[TT] NOM-003 — Data Downlink

Entry point: AOS with products queued for downlink (typically immediately after NOM-001). *(Runnable — linked to the `data_dump` activity.)* **Goal:** Raise the link to HIGH, walk the product down over S13 and confirm ground delivery, then release the HIGH tier. **Possible negative outcomes / deviations:** - HIGH refused with S1.8 `0x0007` → a gate is unmet: antenna deployed, DOWNLINK mode on, elevation ≥ 10°, beacon mode clear - BER climbs mid-pass → CTG-014; drop back to LOW; re-queue the affected segments - Station failure mid-pass → CTG-020; the remaining queue carries forward to the next pass - CRC mismatch on the reassembled product → re-request the failing segments before closing the transfer | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | TT&C: confirm link quality | `HK_REQUEST(sid=6)` (S3.27) | □ | `ttc.link_status` (0x0501) = 2; margin (0x0503) > 3.0 dB | | 2 | Payload: read the on-board catalogue | `PAYLOAD_GET_IMAGE_CATALOG` (S8 func_id 32) | □ | Select scene ids; `payload.mode` (0x0600) = 1 | | 3 | TT&C: raise the link to HIGH | `TTC_SET_DOWNLINK_MODE(on=1)` (func_id 79); `TTC_SET_DATA_RATE(rate=1)` (func_id 65) | □ | `ttc.tm_data_rate` (0x0506) = 128000 bps; `ttc.tm_tier` (0x0539) = 2 | | 4 | Payload: open the transfer | `S13_CREATE_TRANSFER(scene_id=N, size_mb=0)` (S13.1) | □ | `payload.transfer_active` (0x062D) = 1; block count and CRC-32 recorded | | 5 | Payload: walk the segments, monitor progress | `S13_DOWNLOAD_SEGMENT(transfer_id=T, segment_num=K)` (S13.3); `S13_REPORT_STATUS(transfer_id=T)` (S13.9) | □ | `payload.transfer_progress` (0x062E) increasing; lock held; margin > 2.0 dB | | 6 | Ground Seg: verify reception and the product CRC-32 | (ground tool) | □ | Frame error rate < 1e-6; CHECKSUM OK | | 7 | TT&C: close the transfer and release HIGH | `S13_END_TRANSFER(transfer_id=T)` (S13.5); `TTC_SET_DATA_RATE(rate=0)`; `TTC_SET_DOWNLINK_MODE(on=0)` | □ | Back to 16000 bps; `ttc.downlink_mode` (0x053C) = 0 | | 8 | SPACON: log; close the pass downlink | (mission log) | □ | "NOM-003 complete" |

[AOCS] NOM-004 — Orbit Maintenance Manoeuvre Planning
Entry point: FDS detects altitude drift exceeding the planned envelope, a debris conjunction requires a collision-avoidance manoeuvre, or a scheduled window approaches. *(Runnable — linked to the `orbit_manoeuvre` activity.)* **Goal:** Design and ground-validate the next burn and load the time-tagged sequence; execution is NOM-013. **Possible negative outcomes / deviations:** - Drift exceeds tolerance before the next planned window → bring the manoeuvre forward; coordinate with the payload schedule - Latch-valve fault (`prop.latch_a_state` / `prop.latch_b_state`, 0x0B06 / 0x0B07 = 2) or pressure out of band → defer; CTG-028 Propulsion Anomaly - Plan rejected at safety review → re-plan with a smaller delta-V | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FDS: design the manoeuvre | (FDS tool) | □ | Total budget is 66.9 m/s from the 1U green monoprop — debris avoidance and limited orbit maintenance; from mid-mission keep `prop.dv_remaining` (0x0B0C) ≥ 38 m/s for the NOM-014 disposal | | 2 | Propulsion: enable PROP housekeeping and check health | `HK_ENABLE(sid=9)` (S3.5) | □ | `prop.state` (0x0B00) = 0 (SAFED); `prop.tank_pressure` (0x0B01) in band | | 3 | FD: review and authorise the vector and the execution pass | (FD GO) | □ | Logged | | 4 | SPACON: time-tag the burn sequence | `SCHEDULE_TC(exec_time_cuc=T, tc_data=TC)` (S11.4) ×3 | □ | `PROP_SET_BURN` at T−5 min, `PROP_ARM` at T−2 min, `PROP_EXECUTE` at T−0 | | 5 | SPACON: read back the on-board schedule | `LIST_SCHEDULE` (S11.17) | □ | Three commands at the planned CUC times | | 6 | FDS: brief the predicted pass shift | (FDS tool) | □ | A burn re-fits the element set: 1 m/s gives Δa ≈ +1.800 km and −33.5 s at +24 h, so passes and eclipses really move | | 7 | SPACON: schedule the debrief slot for the NOM-013 result | (mission log) | □ | "NOM-004 complete (plan only)" |

[OB] NOM-006 — Software Upload / Parameter Update

Entry point: Scheduled patch or parameter slot, approved offline by the software/FDIR boards; SoC > 60 %, bus > 32.1 V, CPU load < 50 %. **Goal:** Upload the image or parameter set over S6, verify by checksum, and activate under a controlled reboot. **Possible negative outcomes / deviations:** - Checksum mismatch → re-upload the affected segment - Reboot does not come back within 90 s → CTG-021 OBC Bootloader Recovery - Side-effects on FDIR behaviour → repeat the COM-008 round-trip test after activation | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | OBDH: pre-upload health check | `HK_REQUEST(sid=4)`, `HK_REQUEST(sid=1)` (S3.27) | □ | `obdh.mem_errors` (0x031E) = 0; `obdh.heap_usage` (0x031D) < 60 % | | 2 | OBDH: suspend on-board scheduling and non-critical loads | `DISABLE_SCHEDULE` (S11.9); `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | □ | `payload.mode` (0x0600) = 0 | | 3 | OBDH: upload the segments | `MEM_LOAD(memory_id=1, address=ADDR, data=BLOCK)` (S6.2) | □ | Each segment acknowledged within 10 s | | 4 | OBDH: verify by checksum | `MEM_CHECK(memory_id=1, address=ADDR, length=LEN)` (S6.9) | □ | Bit-for-bit match | | 5 | OBDH: configure the boot source | `OBC_BOOT_APP` (S8 func_id 55) | □ | `obdh.sw_image` (0x0311) still 1 — no unintended reboot yet | | 6 | OBDH: controlled reboot and confirm the new image | `OBC_REBOOT` (S8 func_id 52) | □ | Responding within 90 s; `obdh.sw_version` (0x030B) as expected; `obdh.last_reboot_cause` (0x0316) = 4 (commanded) | | 7 | SPACON: **re-synchronise the clock** before restoring the schedule | `SET_TIME(cuc_seconds=T_corrected)` (S9.1) per NOM-008 | □ | The bootloader reset the clock to **J2000**: `obdh.time_ref` (0x0A13) = 0 (NONE), `obdh.time_since_sync_s` (0x0A14) = 4294967295, `obdh.obc_time` (0x0309) back near 0. Booting the application does not restore it | | 8 | SPACON: restore scheduling; monitor the first orbit | `ENABLE_SCHEDULE` (S11.13) | □ | Remember the boot re-applies `default_enabled_sids: [1, 11]` — re-issue `HK_ENABLE` for the rest | | 9 | SPACON: log | (mission log) | □ | "NOM-006 complete" | **Note — the schedule fires on ONBOARD time.** Any S11 tag uploaded before the reboot was computed against the pre-reboot clock. Re-enabling the schedule (step 8) before the clock is back (step 7) leaves the OBC holding tags that are decades in its own future and that simply never fire. Sync first, then re-arm, then read back with `LIST_SCHEDULE` (S11.17).

[AOCS] NOM-007 — Momentum Management (Desaturation)
Entry point: Stored body momentum above the dump threshold, or a scheduled desat slot. AOCS in NOMINAL (mode 4), `aocs.desat_inhibit` (0x029A) = 0, no imaging in progress. **Goal:** Remove stored body momentum with the magnetorquers and let the spacecraft return automatically to the pre-desat mode. **Possible negative outcomes / deviations:** - Command refused with `DESAT_REFUSED_FINE` → the spacecraft is in FINE_POINT; desaturation is not permitted there - Dump ineffective → if it runs past `desaturation.timeout_s` (3600 s) the spacecraft raises event `0x020E AOCS_DESAT_INEFFECTIVE`; suspect a magnetorquer fault (CTG-007) - Pointing error grows during the dump → expected up to the 5° coarse limit; if it persists, abort and revert | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: assess the momentum state | `GET_PARAM(param_id=0x029B)` (S20.3) — `aocs.h_body_nms` | ☐ | **Body momentum, not wheel speed, is what a magnetic dump removes.** Cross-check wheels (0x0207–0x020A) and the envelope `aocs.h_env_frac` (0x02BA) | | 2 | AOCS: pre-desaturation attitude snapshot | `HK_REQUEST(sid=2)` (S3.27) | ☐ | Error < 1.0°; rates < 0.05 °/s | | 3 | AOCS: plan the window | (FDS / AOCS tool) | ☐ | Avoid imaging windows — imaging asserts the desat inhibit automatically | | 4 | AOCS: command the dump | `AOCS_DESATURATE` (S8 func_id 1) | ☐ | `aocs.mode` (0x020F) = 7 (DESAT) within 10 s | | 5 | AOCS: monitor progress | `HK_REQUEST(sid=2)` every 30 s | ☐ | **A dump takes tens of minutes, not ten:** ~25 min to 0.005 N·m·s and ~48 min to 0.001 N·m·s from a 0.020 N·m·s loading. A flat `h_body` for a few minutes is normal | | 6 | AOCS: confirm the automatic return to the pre-desat mode | `aocs.mode_demoted` (0x0298) = 0 | ☐ | The spacecraft returns itself — no mode command is needed | | 7 | AOCS: post-desaturation verification | `HK_REQUEST(sid=2)` (S3.27) | ☐ | Error < 1.0°; wheels within nominal range; rates < 0.05 °/s | | 8 | SPACON: log; close the ticket if anomaly-triggered | (mission log) | ☐ | "NOM-007 complete" |

[MULTI] NOM-009 — Routine Health Check

Entry point: Every pass startup as part of NOM-001; also commanded ad hoc by FD. *(Runnable — linked to the `housekeeping_collection` activity: a single S3.27 request per structure.)* **Goal:** Each position reads its structure, calls OK on the loop; SPACON confirms no recent FDIR firings. **Possible negative outcomes / deviations:** - Any position returns out-of-band → the relevant CTG before continuing the pass - FDIR rule fired in the last hour but already cleared → log and continue; raise a ticket post-pass - Slow drift not yet flagged → FD authorises a watch-item, not necessarily a CTG | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | SPACON: request all six structures | `HK_REQUEST(sid=1…6)` (S3.27) | □ | All six within 30 s. `HK_REQUEST` is S3 **subtype 27**; `HK_ENABLE` is S3.5 | | 2 | EPS: SoC, bus, margin | SID 1 | □ | SoC (0x0101) > 40 %; bus (0x0105) 30.9–34.0 V; margin > −50 W | | 3 | AOCS: mode, error, rates, wheels | SID 2 | □ | `aocs.mode` (0x020F) = 4 (NOMINAL); error < 1.0°; wheel speeds < 5000 RPM | | 4 | Thermal: zone temperatures and heater states | SID 3 | □ | OBC (0x0406) −10…+50 °C; battery (0x0407) +5…+35 °C; `tcs.cooler_fpa` (0x040C) as expected | | 5 | OBDH: mode, load, uptime, **clock reference** | SID 4 | □ | `obdh.mode` (0x0300) = 0; load < 80 %; no unexpected reboot. `obdh.time_ref` (0x0A13) = 1 (GROUND) or 2 (GPS) — 0 (NONE) means the OBC rebooted and the clock is back at J2000; 3 is HOLDOVER. `obdh.time_since_sync_s` (0x0A14) yellow at 12 h, red at 24 h; at the nominal ~12 ppm the clock gains ≈ 1 s/day, so a sync once a shift is the routine | | 6 | TT&C: lock, RSSI, margin | SID 6 | □ | `ttc.link_status` (0x0501) = 2 (LOCKED); margin > 2.0 dB | | 7 | Payload: mode, FPA, storage | SID 5 | □ | `payload.store_used` (0x0604) < 95 % | | 8 | SPACON: consolidate a single PASS to FD | (voice loop) | □ | "NOM-009 PASS" |

[FE] NOM-010 — Eclipse Transition

Entry point: Approach to the eclipse boundary (typically every orbit); FDS provides the predicted timing. **Goal:** Configure the spacecraft for the eclipse cold/dark phase, monitor the SoC trough and thermal margins, and confirm recovery after exit. **Possible negative outcomes / deviations:** - SoC trough deeper than predicted → CTG-001 load-shed plan; consider deferring imaging in the next eclipse - A zone leaves its band → CTG-004 Thermal Exceedance - Recovery slope shallower than predicted post-exit → flag possible charge-regulator underperformance | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: log the pre-eclipse baseline | `HK_REQUEST(sid=1)` (S3.27) | □ | SoC (0x0101) > 60 %; bus (0x0105) > 32.1 V; battery current (0x0109) recorded | | 2 | Thermal: power both heater lines, then set the eclipse setpoints if margins are tight | `EPS_POWER_ON(line_index=5)` / `(line_index=6)` only if 0x0115 / 0x0116 = 0; `HEATER_SET_SETPOINT(circuit=0, on_temp=3.0, off_temp=8.0)` / `(circuit=1, on_temp=-5.0, off_temp=0.0)` (S8 func_id 44) | □ | `eps.pl_htr_bat` = `eps.pl_htr_obc` = 1; battery > 5 °C, OBC > 0 °C at entry | | 3 | Thermal: release the FPA cooler if it is still on and no imaging is planned | `FPA_COOLER(on=0)` (S8 func_id 43) — skip if 0x040C already reads 0 (STANDBY does not run it) | □ | `tcs.cooler_fpa` (0x040C) = 0; consumption drops ~15 W | | 4 | AOCS: command the eclipse hold | `AOCS_SET_MODE(mode=8)` (S8 func_id 0) | □ | `aocs.mode` (0x020F) = 8 (ECLIPSE) within 15 s; gyro-only propagation | | 5 | EPS: monitor through the eclipse | `HK_REQUEST(sid=1)` every 30 s | □ | `eps.eclipse_flag` (0x0108) = 1; generation ≈ 0 W; bus > 30.9 V; battery current negative | | 6 | Thermal: monitor the zones | `HK_REQUEST(sid=3)` every 60 s | □ | Battery > 0 °C, OBC > −10 °C; heaters cycling | | 7 | EPS: confirm recovery after exit | `HK_REQUEST(sid=1)` (S3.27) | □ | `eps.eclipse_flag` = 0; generation > 25 W within 60 s | | 8 | SPACON: file the eclipse pass note | (mission log) | □ | "NOM-010 complete" |

[AOCS] NOM-013 — Orbit Maintenance Manoeuvre Execution

Entry point: Time-tagged execution per the NOM-004 plan, preferably over a station for direct observation. *(The book previously numbered this card NOM-005; the shipped ID is NOM-013 — simulator NOM-005 is HK Configuration.)* **Goal:** Execute the planned burn and verify the orbit response. **Possible negative outcomes / deviations:** - Arm refused → check `prop.arm_inhibit_mask` (0x0B15), the catbed temperature, and **`eps.spacecraft_phase` (0x0129) = 6**: `prop.yaml` declares `min_phase: 6`, so with the phase left at 4 or 5 no burn can ever be armed. The phase is set by `SYS_SET_MISSION_PHASE(phase=6, flags=0)` (S8 func_id 120) at the end of COM-013. `PROP_ARM` must follow `PROP_SET_BURN` within 300 s - Burn does not start on time → check the propulsion line and the arm state; if there is no thrust, CTG-028 Propulsion Anomaly - Burn under-performs → re-plan a trim manoeuvre under NOM-004 - Attitude drifts off-axis during the burn → standing abort authority applies: `PROP_ABORT` first, ask afterwards | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Propulsion: energise the prop line and preheat the catbed (T−25 min) | `EPS_POWER_ON(line_index=8)` (func_id 19); `HEATER_THRUSTER(on=1)` (func_id 42) | □ | `eps.pl_prop` (0x014A) = 1; catbed 12 W; `tcs.temp_thruster` (0x0409) rising | | 2 | AOCS: hold the burn attitude | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.mode` (0x020F) = 4, or 9 (INERTIAL) with the 180° yaw for a retrograde burn; error < 0.5° | | 3 | Propulsion: load the manoeuvre | `PROP_SET_BURN(dv_m_s=DV, direction=DIR, mode=MODE, t_start_cuc=0)` (S8 func_id 110) | □ | `prop.burn_dv_cmd` (0x0B0E) = planned dV | | 4 | Propulsion: arm | `PROP_ARM(arm_code=0xA5C3)` (S8 func_id 108) | □ | `tcs.temp_thruster` ≥ 180 °C; `prop.state` (0x0B00) = 1 (ARMED); event `0x0800` | | 5 | Propulsion: execute and monitor | `PROP_EXECUTE` (S8 func_id 111) — or the S11 tag fires it | □ | `prop.state` = 2 (FIRING); event `0x0803`. Sim speed is clamped to ≤ 10× during a burn | | 6 | Propulsion: standing abort criteria | `PROP_ABORT` (S8 func_id 112) | □ | After any abort, `prop.burn_status` (0x0B0D) = 4 | | 7 | FDS: post-burn orbit determination | `prop.sma_km` (0x0B1A); `prop.orbit_epoch_version` (0x0B19) | □ | Event `0x0804`; Δa against prediction; new element set means new pass times | | 8 | Propulsion: safe and close out | `PROP_DISARM` (func_id 109); `HEATER_THRUSTER(on=0)` (func_id 42); `EPS_POWER_OFF(line_index=8)` (func_id 20) | □ | `prop.state` = 0; `eps.pl_prop` = 0; `aocs.total_momentum` (0x025B) < 0.02 N·m·s — "NOM-013 complete" |

[FD] NOM-014 — End-of-Life Disposal (Perigee Lowering and Passivation)
Entry point: End of mission declared by the FD; imaging closed and the last science data downlinked. Mission phase 6 (the propulsion arm interlock needs it), `prop.dv_remaining` (0x0B0C) ≥ 38 m/s, propulsion healthy (`prop.state` 0x0B00 = 0, no latched inhibit, tank 6.0–24.0 bar), SoC > 55 %, FDS burn plan for six consecutive daylight burns. **Goal:** Lower the perigee from 513 km to 390 km with a 34.3 m/s retrograde burn, flown as six burns of 5.72 m/s at the same orbital position (which becomes the apogee), so that EOSAT-1 re-enters in about 1.5 years at mean solar activity and 4.2 years at solar minimum; then passivate. Natural decay from 513 km would take 5.0 years at mean activity and 14.2 years at solar minimum. **Possible negative outcomes / deviations:** - A burn aborts → NOM-013 step 5 and CTG-028; the partial ΔV is already in the orbit, and FDS re-plans the remaining burns from the new orbit - `prop.dv_remaining` below the plan → lower the perigee as far as the propellant allows (every 1 m/s retrograde lowers it by about 3.6 km) and report the achieved residual lifetime - Propulsion unusable at end of life → passivate (steps 9–12) and report the non-compliance: natural decay is about 5 years at mean activity and 14 at solar minimum - Solar-minimum lifetime still above 5 years after step 7 → a trim burn (step 6 again, smaller ΔV) before passivation - There is no DISPOSAL phase: the spacecraft stays in phase 6 throughout, because the arm interlock needs it | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FDS: disposal plan; FD authorises | (FDS tool) | □ | Six burn epochs in daylight, ΔV per burn, retrograde quaternion; predicted perigee 513 → ~492 → ~471 → ~451 → ~431 → ~410 → 390 km | | 2 | Propulsion: health check | `HK_ENABLE(sid=9)` (S3.5) | □ | `prop.state` = 0; `prop.tank_pressure` (0x0B01) 6.0–24.0 bar; `prop.dv_remaining` ≥ 38 m/s; record `prop.arm_inhibit_mask` (0x0B15). Otherwise CTG-028 first | | 3 | Propulsion: raise the burn watchdog | `PROP_SET_LIMITS(att_err_max_deg=2.0, p_min_bar=4.0, catbed_max_c=350.0, t_burn_max_s=600.0)` (S8 func_id 115) | □ | Default watchdog is 300 s; an end-of-life 5.72 m/s burn lasts up to ~480 s. The other abort limits keep their defaults | | 4 | Propulsion: power, preheat and open the valves (T−25 min) | `EPS_POWER_ON(line_index=8)` (S8 func_id 19); `HEATER_THRUSTER(on=1)` (S8 func_id 42); `PROP_ISO_OPEN(branch=2)` (S8 func_id 113) | □ | `eps.pl_prop` (0x014A) = 1; `prop.latch_a_state` (0x0B06) = 1; `tcs.temp_thruster` (0x0409) ≥ 180 °C within 1500 s | | 5 | AOCS: retrograde inertial hold | `AOCS_SET_TARGET_QUAT(q_x=QX, q_y=QY, q_z=QZ, q_w=QW)` (S8 func_id 86); `AOCS_SET_MODE(mode=9)` (S8 func_id 0) | □ | 180° yaw so that the +X thrust axis points against the velocity. `aocs.mode` = 9 within 120 s; `aocs.att_error` < 0.5° within 600 s | | 6 | Propulsion: disposal burn — repeat steps 5 and 6 for each of the six burns | `PROP_SET_BURN(dv_m_s=5.72, direction=1, mode=1, t_start_cuc=0)` (S8 func_id 110); `PROP_ARM(arm_code=0xA5C3)` (S8 func_id 108); `PROP_EXECUTE` (S8 func_id 111); `PROP_DISARM` (S8 func_id 109) | □ | `prop.burn_time_cmd` (0x0B10) within 10 % of the FDS prediction; event `PROP_BURN_COMPLETE` (0x0804) within 600 s; `prop.burn_dv_achieved` (0x0B0F) within 2 % of 5.72; `prop.state` back to 0. Return to NOMINAL for the eclipse between burns | | 7 | FDS: confirm the disposal orbit | `HK_REQUEST(sid=12)` (S3.27) over one full orbit | □ | Minimum `aocs.gps_alt` (0x0212) 380–400 km; semi-major axis ~6830 km; residual lifetime < 5 years at solar minimum | | 8 | Propulsion: deplete the propellant | `PROP_SET_BURN(dv_m_s=DVREM, direction=1, mode=0, t_start_cuc=0)` (S8 func_id 110); `PROP_ARM(arm_code=0xA5C3)`; `PROP_EXECUTE` | □ | From the retrograde hold, in daylight; DVREM is the current `prop.dv_remaining`. `prop.propellant_mass` (0x0B02) < 5 g within 900 s; below 5 g the arm is refused (`0x0B03`) and the residue is recorded | | 9 | Propulsion: safe and inhibit | `PROP_DISARM`; `HEATER_THRUSTER(on=0)`; `PROP_INHIBIT_SET(on=1)` (S8 func_id 116); `EPS_POWER_OFF(line_index=8)` (S8 func_id 20) | □ | `prop.state` = 0; `prop.arm_inhibit_mask` bit 9 set; `eps.pl_prop` = 0. There is no tank-venting command: record the residual tank pressure (about 9–10 bar) in the disposal report | | 10 | Payload and AOCS: shut down, stop the wheels | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26); `EPS_POWER_OFF(line_index=3)`; `AOCS_RW_RAMP_DOWN(wheel_idx=255, target_rpm=0)` (S8 func_id 14) | □ | `payload.mode` (0x0600) = 0; `aocs.total_momentum` (0x025B) < 0.01 within 120 s. No kinetic energy is left in the wheels | | 11 | EPS:

passivate the battery | `S2_EPS_CHARGE_REGULATOR(device_id=271, state=0)` (S2.1) | ☐ | The arrays can no longer charge the battery; `eps.bat_soc` (0x0101) falling in sunlight. Send steps 11 and 12 in the same pass | | 12 | TT&C: RF silence — the last commands | `TTC_BEACON_TX_OFF` (S8 func_id 98); `TTC_PA_OFF` (S8 func_id 67); `EPS_TRANSPONDER_TX(on=0)` (S8 func_id 18) | ☐ | Loss of signal is the verification; confirm no carrier at the next two scheduled passes. File the disposal report. "NOM-014 complete" |

[FD] Pass Plan Upload (nominal flow — no shipped procedure file)

Entry point: New 24 h pass plan ready from FDS; uplink slot scheduled in the current or next pass. This activity sits between NOM-001 (pass startup) and the planned activities it schedules — chiefly NOM-002 imaging and NOM-003 downlink. **Goal:** Upload the time-tagged plan and confirm the on-board schedule matches the ground copy. **Possible negative outcomes / deviations:** - Upload rejected → check the authorisation level and re-attempt; a rejected `SCHEDULE_TC` leaves the previous schedule intact - Read-back differs from ground → HOLD; delete the divergent entries with `DELETE_SCHEDULED` (S11.7) before the next scheduled activity executes - The plan covers a window in which hardware is now degraded → re-plan offline before retry | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | FDS: produce the 24 h plan; ground review | (FDS / FD review) | □ | Plan reviewed against the current spacecraft configuration | | 2 | SPACON: upload the time-tagged commands | `SCHEDULE_TC(exec_time_cuc=T, tc_data=TC)` (S11.4), one per activity | □ | All TCs acked | | 3 | SPACON: read back the on-board schedule | `LIST_SCHEDULE` (S11.17) | □ | Entry-by-entry compare against the ground copy | | 4 | SPACON: remove any superseded entries | `DELETE_SCHEDULED` (S11.7) / `DELETE_ALL_SCHEDULED` (S11.11) | □ | Only with FD GO | | 5 | SPACON: log the new active plan | (mission log) | □ | "Pass plan upload complete" |

[FE] Long-Term Trend Review (nominal flow — no shipped procedure file)
Entry point: End-of-shift or weekly slot; passive ground activity, run from the MCS archive with no commanding. Its output feeds the NOM-012 handover briefing. **Goal:** Review long-term trends (battery capacity and cycle count, cooler margin, momentum envelope, link margin and BER) to detect slow degradations before they become contingencies. **Possible negative outcomes / deviations:** - Trend exceeds the yellow threshold → raise a watch-item; brief the Mission Manager - Trend exceeds the red threshold → open a formal anomaly ticket; invoke the matching CTG | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:---------|:--------|:------| | 1 | All positions: pull the weekly trend plots | (MCS archive) | ☐ | No telecommanding — this card commands nothing | | 2 | EPS: battery capacity and cycles | `eps.bat_capacity_wh` (0x010A), `eps.bat_cycles` (0x0121), `eps.sa_age_factor` (0x0123) | ☐ | Compare with the commissioning baseline | | 3 | Payload / Thermal: cooler authority | `payload.cooler_margin_k` (0x064B), `payload.cooler_floor_c` (0x064A) | ☐ | A shrinking margin is the early warning of a warm FPA | | 4 | AOCS: momentum envelope and bearing health | `aocs.h_env_frac` (0x02BA), `aocs.rw1_bearing_health` … (0x0284–0x0287) | ☐ | Rising dump frequency is the trend, not any single dump | | 5 | TT&C: link margin and BER by station | `ttc.link_margin` (0x0503), `ttc.ber` (0x050C) | ☐ | Per-station, per-elevation | | 6 | Each position: flag anything degraded > 5 % vs the commissioning baseline | (review) | ☐ | | | 7 | FD: brief the Mission Manager on watch-items | (briefing / mission log) | ☐ | "Trend review complete" |

[FD] NOM-012 — Shift Handover

Entry point: End of shift; outgoing FD briefs incoming FD on state, open items and next pass. **Goal:** Transfer responsibility for the spacecraft cleanly within ≤ 5 minutes, ensuring no item is lost in translation. **Possible negative outcomes / deviations:** - Open contingency procedure mid-execution → handover must explicitly transfer authority and step status - Incoming FD disagrees with current posture → outgoing FD remains on station until consensus - Critical pass within 15 min of handover → delay handover until after the pass **Note.** This card is **not** advisory: Step 1 is a real six-structure housekeeping sweep and Step 4 a real S11 schedule request. **Full template:** see §9.5. | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Outgoing FD: state spacecraft posture (mode, SoC, key telemetry) | `HK_REQUEST(sid=1…6)` (S3.27), then voice loop | □ | One line | | 2 | Outgoing FD: open anomalies / acknowledged FDIR | (tickets review) | □ | Each item by ID | | 3 | Outgoing FD: open work items / procedures in progress | (mission log) | □ | E.g. "NOM-007 desat scheduled next eclipse" | | 4 | Outgoing FD: next pass plan summary | `LIST_SCHEDULE` (S11.17); (FDS) | □ | AOS time, key activities | | 5 | Incoming FD: read-back state, anomalies, plan | (voice loop) | □ | Verbatim | | 6 | Both FDs: hand off authority on loop | (formal call) | □ | "NOM-012 complete" |

14.4 Contingency procedure cards

[FE] CTG-001 — Under-Voltage Load Shed
Entry point: FDIR rule `eps_bus_undervoltage_payload_off` (ea_id 1001) fires, or `eps.uv_flag` (0x010E) = 1, or `eps.bus_voltage` (0x0105) observed below 28.4 V. **Goal:** Restore the battery from undervoltage by shedding least-critical loads first and waiting for sunlit charging. *(A ~29–33 V bus is unusual on a 6U CubeSat — 2S (8 V) is the more common architecture. EOSAT-1 flies an **8S-1P** 100 Wh pack (GomSpace BPX: 28.8 V nominal, 33.6 V at 100 % SoC, 24.0 V at 0 %) on an unregulated battery bus capped at 33.4 V; treat the voltage values as simulator-specific. The on-board shed starts at the UV level, 28.4 V, `simulation.yaml` `eps.bus_protection.uv_v`.)* *(This is one of the eight machine-runnable cards. Its shipped form is a four-rung branch ladder on a single parameter, `eps.bus_voltage` (0x0105) — run flat it would shed the payload, the cooler, the transmitter **and** the wheels on a spacecraft the first shed had already saved.)* **Possible negative outcomes / deviations:** - Voltage continues to drop after a shed rung → take the next rung down the ladder - Voltage does not recover once every rung has been taken → `OBC_SET_MODE(mode=2)` (EMERGENCY), then EMG-002 Total Power Failure - `eps.sa_a_current` (0x0103) and `eps.sa_b_current` (0x0104) both near zero after eclipse exit → array cause; go to CTG-009 - Recovery achieved but root cause unclear → file anomaly; brief Mission Manager **Decision tree (the shipped ladder — each rung is a real branch step):** - Check the line states first: shed from the ground only what on-board FDIR has not already opened - Shed payload. IF V > 28.4 V → HOLD at this rung and monitor - ELSE shed the FPA cooler. IF V > 27.8 V → HOLD and monitor - ELSE shed the S-band PA. IF V > 27.2 V → HOLD and monitor - ELSE shed the reaction-wheel line (`line_index=7`). IF V > 26.5 V → HOLD and monitor - ELSE last resort → `OBC_SET_MODE(mode=2)` EMERGENCY; brief FD on EMG-002 readiness | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FD: GO for load shed | (FD call) | □ | Dual-keyed | | 2 | EPS: shed priority 1 — payload OFF | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | □ | `payload.mode` (0x0600) = 0; branch on V > 28.4 V | | 3 | EPS: shed priority 2 — FPA cooler OFF | `FPA_COOLER(on=0)` (S8 func_id 43) | □ | `tcs.cooler_fpa` (0x040C) = 0; branch on V > 27.8 V | | 4 | EPS: shed priority 3 — S-band PA OFF | `TTC_PA_OFF` (S8 func_id 67) | □ | S-band TM stops; the UHF beacon (SID 11) keeps transmitting | | 5 | EPS: shed priority 4 — reaction-wheel line OFF | `EPS_POWER_OFF(line_index=7)` (S8 func_id 20) | □ | `eps.pl_aocs_wheels` (0x0117) = 0; loses active attitude control | | 6 | EPS: hold the shed level and monitor recovery | (read SID 1) | □ | Exit gate: V > 32.1 V **and** SoC > 40 % | | 7 | EPS: restore in reverse priority, 5 min apart | wheels line → `ttc_tx` line + `TTC_PA_ON` → `fpa_cooler` line (cooler left OFF) → `payload` line + `PAYLOAD_SET_MODE(mode=1)` | □ | Re-power each line before its unit; bus > 31.5 V and SoC > 35 % after each | | 8 | SPACON: file ticket; brief Mission Manager | (ticket) | □ | "CTG-001 closed" |

[AOCS] CTG-002 — AOCS Anomaly Recovery
Entry point: Any of the shipped card's three triggers — `aocs.att_error` (0x0217) > 5°, any wheel temperature (`aocs.rw1_temp`–`rw4_temp`, 0x0218–0x021B) > 60 °C, or the star tracker reporting BLIND. **Goal:** Bound the anomaly, restore stable three-axis pointing in NOMINAL (mode 4), and identify the root cause so the correct sub-procedure can be chosen. There is no "SAFE_POINT" mode: the safe hold is COARSE_SUN (mode 3), and the rate-damping fallback is DETUMBLE (mode 2). **Possible negative outcomes / deviations:** - COARSE_SUN not achievable → CTG-008 (star tracker), CTG-007 (wheel) or EMG-005 Loss of Attitude - Root cause cascades (e.g. star tracker then gyro) → coordinate cascading CTGs; defer payload activities - Pointing returns within budget after transient → log; do not return to NOMINAL same pass; await trend review **Decision tree:** - IF star tracker faulted → CTG-008 - IF one wheel anomalous → CTG-007; IF two or more → DETUMBLE (mode 2) and EMG-005 - IF `aocs.att_error` exceeds 15° at any step → DETUMBLE (mode 2) immediately - IF `eps.bus_voltage` (0x0105) < 27.7 V during recovery → CTG-005 EPS Safe Mode first - IF transient cleared but root cause unidentified → hold COARSE_SUN one orbit; review | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: read mode, pointing error, wheel speeds and temperatures | `HK_REQUEST(sid=2)` (S3.27) | ☐ | `aocs.mode` 0x020F, `aocs.att_error` 0x0217 | | 2 | AOCS: disable any wheel over temperature or oscillating | `AOCS_DISABLE_WHEEL(wheel_idx=N)` (S8 func_id 2) | ☐ | Speed decays within 30 s | | 3 | AOCS: dump accumulated momentum | `AOCS_DESATURATE` (S8 func_id 1) | ☐ | Active wheels toward 0 rpm; rates < 0.5 °/s | | 4 | AOCS: verify body rates stable | (read SID 2) | ☐ | `rate_roll/pitch/yaw` (0x0204–0x0206) each < 0.1 °/s | | 5 | AOCS: restore nadir pointing | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | ☐ | `aocs.mode` = 4 (NOMINAL); error < 1.0° in 120 s | | 6 | AOCS: on multi-wheel failure, command DETUMBLE instead | `AOCS_SET_MODE(mode=2)` (S8 func_id 0) | ☐ | Rates < 2.0 °/s in 300 s, else EMG-001 | | 7 | SPACON: file ticket; FD decides resume vs hold | (mission log) | ☐ | "CTG-002 hand-off" |

[TT] CTG-003 — TTC Link Loss Recovery

Entry point: `ttc.link_status` (0x0501) = 0 (NO_LINK) persisting > 30 s after predicted AOS, or an established link dropping mid-pass. LOCKED is **2**, not 1. **Goal:** Restore the link by stepping through ground-side checks and the transponder switchover — primary, redundant, primary again — then verify link quality and subsystem health before escalating. **Possible negative outcomes / deviations:** - Link not recovered by end of pass → second consecutive missed pass triggers EMG-004 Loss of Communication - Link back but margin < 3 dB → recovery is not complete; treat as marginal, re-queue downlink work and brief the FD - Ground antenna fault discovered mid-recovery → CTG-020 Ground Station Antenna Failure **Decision tree (abbreviated; see §9.4 for full):** - IF GS not tracking → CTG-020 - IF GS tracking but no carrier → switch transponder (redundant, then back to primary) - IF still no carrier → blind-command CTG-019 path - IF no recovery before LOS AND next pass also missed → EMG-004 **Full read-and-do checklist:** see §9.4, Table 9.3. | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Ground Seg: verify the tracking station (and any other station in view) | (MPS panel) | ☐ | Antenna tracking, uplink power, Doppler pre-compensation; five-station network; if not tracking, CTG-020 | | 2 | TT&C: blind HK request on the primary transponder | `HK_REQUEST(sid=6)` (S3.27) | ☐ | `ttc.link_status` (0x0501) = 2 within 30 s → step 5 | | 3 | TT&C: switch to the redundant transponder (blind) | `TTC_SWITCH_REDUNDANT` (S8 func_id 64) | ☐ | Wait 45 s for power-up and lock → step 5 if locked | | 4 | TT&C: if still nothing, retry the primary (blind) | `TTC_SWITCH_PRIMARY` (S8 func_id 63) | ☐ | Wait 45 s → step 5 if locked, step 6 if not | | 5 | TT&C + all: verify link quality and subsystem health | `HK_REQUEST(sid=6)`, `HK_REQUEST(sid=1)`, `HK_REQUEST(sid=4)` (S3.27) | ☐ | RSSI (0x0502) > −90 dBm; margin (0x0503) > 3.0 dB; record the active unit (`ttc.mode` 0x0500); bus > 30.9 V; `obdh.mode` not safe or emergency | | 6 | SPACON: if not recovered, log every blind command and escalate | (mission log) | ☐ | Notify the Mission Director and FDS; book the next pass; confirm the 24 h no-contact timer is running; two missed passes → EMG-004 |

[SE] CTG-004 — Thermal Exceedance

Entry point: A monitored zone crosses a yellow limit — `tcs.temp_obc` (0x0406) outside −10 / +55 °C, `tcs.temp_battery` (0x0407) outside +2 / +42 °C, `tcs.temp_fpa` (0x0408) outside −45 / −15 °C. **Goal:** Restore zone temperatures to within band by reconfiguring heater thermostats or shedding heat-producing loads. **Possible negative outcomes / deviations:** - Hot exceedance still trending up after action → escalate to EMG-006 Thermal Runaway - Cold exceedance not recovering → check heater health (CTG-012 if the line opened); consider load shedding to reroute power - Payload zone affected → safe the payload (`PAYLOAD_SET_MODE(mode=0)`) before the zone reaches the damage threshold - Battery above +45 °C or OBC above +60 °C → `OBC_SET_MODE(mode=2)` / `mode=1` and escalate immediately | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Thermal: identify affected zone and direction (hot/cold) | `HK_REQUEST(sid=3)` (S3.27) | □ | SID 3 is TCS | | 2 | Thermal (cold): enable the zone heater | `HEATER_BATTERY(on=1)` (func_id 40) / `HEATER_OBC(on=1)` (func_id 41) | □ | `tcs.htr_battery` 0x040A / `tcs.htr_obc` 0x040B = 1 | | 3 | Thermal (hot): remove the heat load | `PAYLOAD_SET_MODE(mode=0)` (func_id 26); battery heater OFF | □ | `payload.mode` (0x0600) = 0 within 10 s | | 4 | Thermal: monitor zone trend for 15 min | (read SID 3) | □ | ≥ 0.2 °C/min back toward band | | 5 | Thermal: if no improvement, retune the thermostat | `HEATER_SET_SETPOINT(circuit=..., on_temp=..., off_temp=...)` (S8 func_id 44) | □ | Circuits 0 battery, 1 OBC, 2 thruster — there is **no** `tcs.htr__setpoint` parameter | | 6 | Thermal: hot case still rising — reorient for dissipation | `AOCS_SET_MODE(mode=3)` (S8 func_id 0) | □ | `aocs.mode` (0x020F) = 3 (COARSE_SUN) | | 7 | Thermal: on recovery, hand the circuit back to the thermostat | `HEATER_AUTO_MODE(circuit=0)` (S8 func_id 45) | □ | Only if heaters were driven manually | | 8 | SPACON: file ticket; if hot zone still rising, EMG-006 | (mission log) | □ | "CTG-004 status" |

[PE] CTG-006 — Payload Anomaly

*(Editor's note: this is the simulator's **CTG-006 Payload Anomaly**, `contingency/payload_anomaly.md`, condensed. The shipped card is framed around the FPA thermal case; this version also covers payload mode-transition faults.)* **Entry point:** Payload mode does not transition as commanded; or `payload.fpa_temp` (0x0601) above −15 °C, or rising faster than 0.5 °C/min with the cooler active. **Goal:** Recover the payload to STANDBY, restore the thermal state, and identify whether the anomaly is software, command sequencing or hardware. **Payload modes:** 0 OFF, 1 STANDBY, 2 IMAGING, **3 CALIBRATION**. **Possible negative outcomes / deviations:** - Payload stuck in IMAGING with no exit → command OFF, then cycle the payload line (CTG-012-like) after FD GO - FPA above +10 °C at any step → payload OFF immediately; no restart without ground authorisation - Cooler will not activate → keep the payload OFF and run CTG-004 for zone assessment - Payload and TCS FPA sensors disagree by more than 5 °C → suspect a sensor; use the lower reading for safety decisions | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Payload: read current mode and FPA temperature | `HK_REQUEST(sid=5)` + `HK_REQUEST(sid=3)` (S3.27) | □ | `payload.mode` 0x0600; 0x0601 vs `tcs.temp_fpa` 0x0408 | | 2 | Payload: command OFF to clear the stuck state | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | □ | 0x0600 = 0 within 10 s | | 3 | Payload / Thermal: confirm the cooler is running | (read SID 3) | □ | `tcs.cooler_fpa` (0x040C) = 1; miniature Stirling-class cryocooler | | 4 | Payload: monitor cooldown for up to 20 min | (read SIDs 3 and 5) | □ | ≥ 0.3 °C/min; settle below −20 °C | | 5 | Payload: restart in STANDBY | `PAYLOAD_SET_MODE(mode=1)` (S8 func_id 26) | □ | 0x0600 = 1; FPA stays below −18 °C | | 6 | Payload: if still stuck, cycle the payload line (FD GO) | `EPS_POWER_OFF(line_index=3)` / `EPS_POWER_ON(line_index=3)` (func_id 20 / 19) | □ | 30–60 s off; line 3 is payload | | 7 | SPACON: file ticket | (mission log) | □ | "CTG-006 closed" |

[AOCS] CTG-007 — Reaction Wheel Anomaly

Entry point: Speed jitter > ±500 rpm over 10 s, unexpected speed divergence, speed > 6500 rpm, or wheel bearing temperature > 60 °C. **Goal:** Isolate the failed wheel, reconfigure to three-wheel control and confirm that NOMINAL (mode 4) nadir pointing remains achievable. EOSAT-1 flies four wheels in a pyramid; **three are sufficient** — a 3-of-4 set still spans R^3 and FINE_POINT runs degraded on it (`aocs.fine_degraded`, 0x0299). **Possible negative outcomes / deviations:** - Pointing budget not achievable with three wheels → COARSE_SUN (mode 3) and defer imaging - Two or more wheels affected → DETUMBLE (mode 2) and EMG-005 Loss of Attitude consideration - Wheel-stuck symptom actually due to bus failure → CTG-017 instead **Decision tree:** - IF exactly one wheel is anomalous → disable it, reconfigure to three wheels, hold NOMINAL with a relaxed 1.5° budget; raise ticket - IF `aocs.att_error` (0x0217) exceeds 10° during the disable → DETUMBLE (mode 2) until rates stabilise - IF NOT recoverable → COARSE_SUN (mode 3); defer imaging; FD considers mission impact | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: identify the affected wheel from speeds and temperatures | `HK_REQUEST(sid=2)` (S3.27) | □ | Speeds 0x0207–0x020A (read-only); temps 0x0218–0x021B | | 2 | AOCS: disable the affected wheel | `AOCS_DISABLE_WHEEL(wheel_idx=N)` (S8 func_id 2) | □ | Speed begins to decay within 10 s; rates stay < 1.0 °/s | | 3 | AOCS: confirm the loop has reconfigured to three wheels — no command | (observe, 15 s) | □ | `aocs.att_error` (0x0217) converging below 2.0° within 60 s. There is no three-wheel mode command and no `aocs.wheel_config` parameter: `AOCS_DISABLE_WHEEL` in step 2 is the whole reconfiguration | | 4 | AOCS: dump momentum on the remaining three | `AOCS_DESATURATE` (S8 func_id 1) | □ | Budget **20–50 min**, not minutes — see the note below | | 5 | AOCS: confirm nadir pointing in three-wheel mode | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | □ | `aocs.mode` (0x020F) = 4; error < 1.5° | | 6 | SPACON: file ticket; brief Mission Manager on agility impact | (mission log) | □ | "CTG-007 closed" | **Plan the dump honestly.** DESAT (mode 7) is a cross-product magnetic dump, $m = k \cdot (h_{body} \times B)/|B|^2$, and it is geometry-limited: the honest dump time is **20–50 minutes**, not the five to fifteen the book used to quote. It does now complete — it stops on a 5e-4 N·m·s deadband against `aocs.h_body_nms` (0x029B) — and it is bounded by a **3600 s timeout**, after which the spacecraft raises AOCS_DESAT_INEFFECTIVE (0x020E) and hands control back rather than parking in DESAT for ever. **Imaging and desat are mutually exclusive:** `aocs.desat_inhibit` (0x029A) is asserted automatically while `payload.mode` is 2 (IMAGING), and can also be latched by the operator with `AOCS_SET_DESAT_INHIBIT(inhibit=…)` (S8 func_id 96) — a dump requested during an imaging block will be refused, with the reason given. On completion the spacecraft resolves its return path against `aocs.commanded_mode` (0x0297): it goes back to the mode the ground last commanded, not to a default. Schedule dumps against that 20–50 min window; do not book an imaging pass on the other side of a ten-minute assumption.

[AOCS] CTG-008 — Star Tracker Failure

Entry point: `aocs.st1_status` (0x0240) = 4 (FAILED) with `aocs.st1_num_stars` (0x0241) = 0; or attitude-determination divergence. Status 3 is BLIND (sun/Moon/Earth-limb dazzle) — wait up to 120 s before declaring a failure. **Goal:** Switch to the redundant star tracker (ST2), and if that fails, fall back to the COARSE_SUN hold. EOSAT-1 carries **two** star trackers, ST1 and ST2. *(This is one of the eight machine-runnable cards; the shipped version branches on `aocs.st2_status` (0x0243). Run flat, it would select a tracker that is not tracking, power off the only other one, and command the degraded fallback — all three, in that order.)* **Possible negative outcomes / deviations:** - ST2 acquires → return to the pre-failure mode; monitor; no star-tracker redundancy remains - ST2 does not reach TRACKING within 120 s → COARSE_SUN (mode 3); fine pointing is lost; payload imaging is not possible - CSS also invalid (eclipse) → DETUMBLE (mode 2) on magnetorquers and wait for sunlit phase; EMG-005 candidate **Decision tree:** - IF ST2 = 2 (TRACKING) after power-on → `ST_SELECT(unit=2)`, then isolate ST1, and the card ends - IF ST2 does not reach TRACKING → skip straight to the COARSE_SUN fallback; do **not** select a tracker that is not tracking | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: confirm the ST1 failure and record body rates | `HK_REQUEST(sid=2)` (S3.27) | □ | 0x0240 = 4, 0x0241 = 0 | | 2 | AOCS: power on ST2 and wait for boot | `ST2_POWER(on=1)` (S8 func_id 5) | □ | 0x0243: 0 OFF → 1 BOOT → 2 TRACKING within 120 s | | 3 | AOCS: select ST2 as primary | `ST_SELECT(unit=2)` (S8 func_id 6) | □ | `aocs.att_error` (0x0217) begins decreasing | | 4 | AOCS: isolate the failed ST1 | `ST1_POWER(on=0)` (S8 func_id 4) | □ | Confirm via `aocs.st1_num_stars` = 0 — 0x0240 stays 4, a failed unit switched off is still failed | | 5 | AOCS: if ST2 never acquired, take the fallback hold | `AOCS_SET_MODE(mode=3)` (S8 func_id 0) | □ | `aocs.mode` (0x020F) = 3 (COARSE_SUN); `aocs.css_valid` (0x0248) = 1 | | 6 | AOCS: if degraded, power the payload down | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | □ | Imaging is not possible in COARSE_SUN | | 7 | SPACON: file ticket; Mission Manager on imaging defer | (mission log) | □ | "CTG-008 closed" — record which arm was taken | **Note on COARSE_SUN:** it is a **hold, not a destination**. It is not in the model's operational-mode set, so commanding it does not overwrite `aocs.commanded_mode` (0x0297); the spacecraft hands itself back to the ground's commanded mode when a tracker recovers. Watch `aocs.mode_demoted` (0x0298) rather than re-commanding by hand.

[FE] CTG-009 — Solar Array Degradation

*(Editor's note: this is the simulator's **CTG-009 Solar Array Degradation**, `contingency/solar_array_degradation.md`, condensed.)* **Entry point:** `eps.sa_a_current` (0x0103) or `eps.sa_b_current` (0x0104) more than 20 % below prediction during a confirmed sunlit phase, or trend review flags slow degradation. **Goal:** Confirm the anomaly, exclude transient causes (attitude, eclipse), and quantify mission impact. **Possible negative outcomes / deviations:** - One wing only → mission still viable with reduced margin; FDS re-plans - Both arrays below 0.2 A in sunlight → CTG-005 EPS Safe Mode path immediately - Degradation faster than 5 % per day → accelerate downlink priorities; plan a reduced-power profile - SoC drops below 20 % during the investigation → interrupt and go to the EPS-safing path | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: quantify the shortfall against prediction | `HK_REQUEST(sid=1)` (S3.27) | □ | 0x0103 / 0x0104 vs predicted; `eps.power_gen` 0x0107 | | 2 | EPS: compare A-to-B ratio over three samples 30 s apart | (read SID 1) | □ | Nominal ratio 0.9–1.1; one-sided = local failure | | 3 | AOCS: rule out attitude as the cause | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.mode` (0x020F) = 4 (NOMINAL) or 3 (COARSE_SUN); `att_error` < 2° | | 4 | EPS: check wing deployment state | `GET_PARAM(param_id=0x0144)` (S20.3) | □ | `eps.wing_status`: bit0 +Y, bit1 −Y — also carried in SID 1 | | 5 | EPS: log per-wing current over a full orbit | (read SID 8) | □ | `eps.sa_wing_py_current` 0x0146, `sa_wing_my_current` 0x0147 (SID 8, 8 s cadence) | | 6 | EPS: shed to match the degraded budget if needed | `PAYLOAD_SET_MODE(mode=1)` then `(mode=0)`; `HEATER_THRUSTER(on=0)` (func_id 42) | □ | Target ≥ 3 W reserve | | 7 | EPS / FDS: re-budget power for the next 24 h | (FDS tool) | □ | Daily trend comparison for 7 days | | 8 | SPACON: file ticket; Mission Manager brief | (mission log) | □ | "CTG-009 closed" |

[OB] CTG-010 — OBDH Watchdog Recovery

Entry point: Watchdog reset event in S5 log, unexpected reboot, or `OBC_BOOT_APP` ack without execution-complete. **Goal:** Re-establish nominal OBC state after a watchdog reboot and recover lost configuration. **Possible negative outcomes / deviations:** - OBC reboots again immediately → suspect SW corruption; consider EMG-003 OBC Reboot path - FDIR table not restored after reboot → re-load nominal table (COM-008 sub-step) - Heater setpoints lost → re-apply per TCS baseline - Hardware-latched power lines preserved but software state lost → re-arm time-tagged schedules (NOM-008) **Decision tree:** - IF reboot clean and stable → CTG-010 recovers; raise ticket - IF `obdh.reboot_count` (0x030A) increments by more than 3 within one hour → boot loop; `OBC_SET_MODE(mode=2)` and EMG-003 - IF table or setpoint loss cannot be restored → defer next pass commitments; FD review | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | SPACON: re-establish link if it dropped | per CTG-003 | □ | RSSI / margin | | 2 | SPACON: confirm the reboot and record the counter | `HK_REQUEST(sid=4)` (S3.27) | □ | `obdh.reboot_count` (0x030A) incremented | | 3 | SPACON: verify OBC image; boot the application if in the bootloader | `obdh.sw_image` (0x0311); `OBC_BOOT_APP` (S8 func_id 55) | □ | S1.1 + S1.7 within 15 s; 0x0311 → 1 | | 4 | SPACON: re-enable the housekeeping structures | `HK_ENABLE(sid=2…6)` (S3.5) | □ | Boot re-applies `default_enabled_sids` = [1, 11] — SIDs 2–6 do **not** return by themselves | | 5 | SPACON: confirm FDIR table identity | (FDIR panel) | □ | Nominal or LEOP-only | | 6 | SPACON: re-apply lost settings (heater setpoints, FDIR overrides) and **re-sync the clock** | `SET_TIME(cuc_seconds=T_corrected)` (S9.1) | □ | The bootloader put the clock back to **J2000**: expect `obdh.time_ref` (0x0A13) = 0 (NONE) and `obdh.time_since_sync_s` (0x0A14) = 4294967295. Run the LEOP-006 / NOM-008 sequence; a reboot always means a re-sync | | 7 | SPACON: re-arm time-tagged schedules — **after** the clock is right | (S11) | □ | NOM-008 Clock Synchronisation. The onboard schedule fires on onboard time, so tags re-armed against a J2000 clock never fire | | 8 | SPACON: restore OBC NOMINAL if it came back in SAFE | `OBC_SET_MODE(mode=0)` (S8 func_id 50) | □ | `obdh.mode` (0x0300) = 0; CPU load < 70 % | | 9 | SPACON: file ticket; FD GO to resume | (mission log) | □ | "CTG-010 closed" |

[FE] CTG-012 — Overcurrent Response

Entry point: An overcurrent trip is visible in `eps.oc_trip_flags` (0x010D); the matching FDIR cascade (`eps_overcurrent_`, keyed on the same bitmask) opens the line autonomously. **Goal:** Acknowledge the autonomous action, identify the affected line, decide transient versus persistent, and reset/re-enable only when the current says it is safe. **Line map (nine lines, indices 0–8):** 0 obc, 1 ttc_rx, 2 ttc_tx, 3 payload, 4 fpa_cooler, 5 htr_bat, 6 htr_obc, 7 aocs_wheels, 8 prop. Lines 0 and 1 are non-switchable; a trip on either is a severe anomaly. Line status flags run 0x0110–0x0117 for lines 0–7 and per-line currents 0x0118–0x011F for lines 0–7. **The prop line breaks the arithmetic:** line 8 is `eps.pl_prop` at **0x014A** and `eps.line_current_8` at **0x014B**, not 0x0118 and 0x0120. 0x0120 is `eps.bat_dod`. Do not extrapolate the block past index 7. **Possible negative outcomes / deviations:** - Current still elevated after the trip → persistent fault; do **not** reset, isolate and escalate - Re-trip immediately after a reset → do not attempt a third reset; `EPS_POWER_OFF` and escalate - Multiple lines affected → suspect upstream bus; CTG-017 Bus Failure Switchover - Critical line affected (OBC, TT&C RX) → escalate to EMG-001 candidate **Decision tree:** - IF FDIR has already opened the line AND per-line current is near 0 A → transient; reset the flag and re-enable - IF per-line current remains elevated → persistent; leave the line open, ticket, escalate - IF the trip coincided with a mode transition or power-on → inrush is the likely cause; treat as transient | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: decode the trip bitmask to identify the line | `HK_REQUEST(sid=1)` (S3.27) | □ | `eps.oc_trip_flags` (0x010D); SID 1 is EPS | | 2 | EPS: read the per-line current for that line | (read SID 1) | □ | 0x0118 + N for N = 0–7; **0x014B for line 8 (prop)**. Near 0 A = LCL opened as expected | | 3 | EPS: coordinate with the owning position on subsystem state | (voice loop) | □ | Rule out hardware damage before any reset | | 4 | EPS: reset the overcurrent flag (FD GO) | `EPS_RESET_OC_FLAG(line_index=N)` (S8 func_id 21) | □ | Flag clears and stays clear over 5 s | | 5 | EPS: re-enable the line | `EPS_POWER_ON(line_index=N)` (S8 func_id 19) | □ | Status (0x0110 + N) = 1 for N = 0–7; **0x014A for line 8**. No re-trip in 30 s | | 6 | EPS: verify the downstream subsystem recovered | `HK_REQUEST` on that subsystem's SID | □ | 2 AOCS, 3 TCS, 5 Payload, 6 TT&C, 9 PROP (`HK_ENABLE(sid=9)` first — SID 9 is not in the default set) | | 7 | SPACON: file anomaly ticket before pass end | (ticket) | □ | Root cause TBD | | 8 | SPACON: brief incoming shift in NOM-012 handover | (mission log) | □ | "CTG-012 closed" |

[FE] CTG-013 — Battery Cell Failure

*(Editor's note: this is the simulator's **CTG-013 Battery Cell Failure**, `contingency/battery_cell_failure.md`, condensed.)* **Entry point:** `eps.bat_voltage` (0x0100) more than 1.0 V below the value expected for the reported `eps.bat_soc` (0x0101), or a sudden voltage drop, or trend review flags degradation. **Goal:** Characterise the anomaly, confirm the battery is thermally safe, and decide whether the remaining capacity supports eclipse survival at the current load. **Possible negative outcomes / deviations:** - Voltage drops further during characterisation → CTG-001 load shed - `eps.bat_temp` (0x0102) above +40 °C → possible internal short; disable the battery heater and shed load; above +50 °C command EMERGENCY and follow EMG-002 - `eps.bat_voltage` below 24.6 V → deep-discharge risk; shed everything but OBC and TT&C RX - Capacity loss exceeds mission margin → brief Mission Manager; FDS re-plans duty cycle **Note:** there is no cell-balance telecommand in `tc_catalog.yaml`, and no cell-level telemetry. The observable set is bus/battery voltage, SoC, temperature, current and `eps.bat_capacity_wh` (0x010A); the operator infers cell state from the voltage-versus-SoC deviation. | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: pull battery telemetry and compute expected V for the reported SoC | `HK_REQUEST(sid=1)` (S3.27) | □ | SID 1 is EPS; SID 8 is the EPS-extended structure | | 2 | EPS / Thermal: cross-check battery temperature | `HK_REQUEST(sid=3)` (S3.27) | □ | `eps.bat_temp` 0x0102 vs `tcs.temp_battery` 0x0407 | | 3 | EPS: if the battery is warm, disable its heater | `HEATER_BATTERY(on=0)` (S8 func_id 40) | □ | `tcs.htr_battery` (0x040A) = 0 | | 4 | EPS: compute the eclipse-survival margin on `eps.bat_capacity_wh` (0x010A) | (operator calc) | □ | Reserve 20 % SoC; eclipse ≈ 35 min | | 5 | EPS: if margin negative, shed payload, then cooler, then PA | `PAYLOAD_SET_MODE(mode=0)` / `FPA_COOLER(on=0)` / `TTC_PA_OFF` | □ | Recompute after each shed | | 6 | EPS / FDS: re-plan duty cycle if capacity reduced | (FDS tool) | □ | New pass plan | | 7 | SPACON: file ticket; brief Mission Manager | (mission log) | □ | "CTG-013 closed" |

[TT] CTG-014 — BER Anomaly

Entry point: `ttc.ber` (0x050C) ≥ −5 — that is, BER worse than 1e-5. The parameter carries the base-10 exponent, so "less negative" is worse. **Goal:** Restore downlink quality by cooling the amplifier, stepping the rate ladder down, or adjusting tx power; re-queue lost segments. **Rate ladder:** BEACON / LOW / HIGH = **1 / 16 / 128 kbps**. There is no "medium" rung. A HIGH → LOW step buys 9.03 dB of Eb/N0 from the `−10·log10(R)` term plus about 1.5 dB of coding gain (QPSK RS(255,223) → BPSK conv(7,1/2)+RS). BEACON is the bootloader lifeline and is not commandable here. **Possible negative outcomes / deviations:** - BER persists once already at LOW → there is no rung below it; go to the CTG-003 Link Loss path - BER specific to one ground station → CTG-020 candidate - Low elevation (`ttc.contact_elevation`, 0x050A < 10°) → degradation is expected geometry, not an anomaly - Recurring across multiple passes → suspect the spacecraft amplifier; raise ticket | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | TT&C: characterise the anomaly | `HK_REQUEST(sid=6)` (S3.27) | □ | 0x050C, `ttc.eb_n0` 0x0519, `link_margin` 0x0503, `rssi` 0x0502 | | 2 | TT&C: check the power amplifier | (read SID 6) | □ | `ttc.pa_temp` (0x050F) < 55 °C; `ttc.tx_fwd_power` 0x050D | | 3 | TT&C: if the PA is hot, cycle it and drop tx power | `TTC_PA_OFF` (67) → 120 s → `TTC_PA_ON` (66); `TTC_SET_TX_POWER(level=0)` (68) | □ | Beacon survives the PA-off window | | 4 | TT&C: step the rate ladder down to LOW | `TTC_SET_DATA_RATE(rate=0)` (S8 func_id 65) | □ | `ttc.tm_data_rate` (0x0506) = 16000; `ttc.tm_tier` (0x0539) = 1 | | 5 | TT&C: if a dump was running, release the gate | `TTC_SET_DOWNLINK_MODE(on=0)` (S8 func_id 79) | □ | `ttc.downlink_mode` (0x053C) = 0 | | 6 | Payload: re-queue affected downlink segments | (re-queue per NOM-003) | □ | | | 7 | SPACON: file ticket | (mission log) | □ | "CTG-014 closed" |

[PE] CTG-015 — Corrupted Image Recovery

Entry point: `payload.checksum_errors` (0x0609) increases above its baseline, or `payload.last_scene_quality` (0x060D) flags the last scene. **Goal:** Establish whether the cause is FPA temperature, a bad memory segment, or a transient upset; delete the failed image, queue a re-acquisition, and only then resume. **Possible negative outcomes / deviations:** - Recurring corruption → likely FPA over-temp; CTG-004 thermal investigation - One-off → likely cosmic ray; replace image and continue - Re-acquisition also fails → defer scene to next orbit; raise ticket **Decision tree:** - IF first occurrence → mark, re-queue, continue - IF second occurrence in same pass → halt imaging; investigate FPA temperature / pointing budget - IF third occurrence → defer to next shift; brief Mission Manager | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Payload: quantify the checksum errors and identify the scene | `HK_REQUEST(sid=5)` (S3.27) | □ | 0x0609 delta; `payload.last_scene_id` 0x060C | | 2 | Payload / Thermal: check FPA temperature | `HK_REQUEST(sid=3)` (S3.27) + SID 5 | □ | `payload.fpa_temp` 0x0601, `tcs.temp_fpa` 0x0408, `payload.fpa_ready` 0x0610 | | 3 | Payload: if the FPA is warm (> −20 °C), abort to STANDBY and cool | `PAYLOAD_SET_MODE(mode=1)` (func_id 26); `FPA_COOLER(on=1)` (func_id 43) | □ | Wait for < −25 °C and `fpa_ready` = 1 | | 4 | Payload: if a memory segment is implicated, hand off to CTG-016 | `payload.mem_segments_bad` (0x0612) | □ | Mark the segment before re-capturing | | 5 | Payload: delete the corrupted image (canonical S8 path) | `PAYLOAD_DELETE_IMAGE(scene_id=X)` (S8 func_id 30) | □ | | | 6 | Payload + AOCS: re-plan acquisition for the next overflight | (FDS) | □ | Per decision tree | | 7 | SPACON: file ticket | (mission log) | □ | "CTG-015 closed" |

[OB] CTG-016 — Memory Segment Failure

*(Editor's note: this is the simulator's **CTG-016 Memory Segment Failure**, `contingency/memory_segment_failure.md`, condensed.)* **Entry point:** `payload.mem_segments_bad` (0x0612) increases above its previous known value, usually alongside a rise in `payload.checksum_errors` (0x0609). **Goal:** Identify the affected memory segment, mark it as unusable, and re-plan imaging against the reduced capacity. **Possible negative outcomes / deviations:** - Bad-segment count rising quickly → suspect radiation latch-up; payload OFF, power-cycle after 60 s - Science data in the affected segment → the images are lost; log the scene IDs and coordinate re-acquisition - More than 10 % of segments bad → escalate to mission planning for a long-term capacity assessment | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Payload: read the bad-segment count and storage figures | `HK_REQUEST(sid=5)` (S3.27) | □ | 0x0612, `payload.mem_total_mb` 0x060A, `mem_used_mb` 0x060B | | 2 | Payload: request the image catalogue to locate the segment | `PAYLOAD_GET_IMAGE_CATALOG` (S8 func_id 32) | □ | Corrupt scenes point at the failed segment | | 3 | Payload: CRC-check any suspect segment | `MEM_CHECK(memory_id, address, length)` (S6.9) | □ | Mismatch confirms the segment is bad | | 4 | Payload: mark the segment unusable | `PAYLOAD_MARK_BAD_SEGMENT(segment_id=N)` (S8 func_id 31) | □ | **N is 0–7** — `num_memory_segments: 8`; an index ≥ 8 is rejected | | 5 | Payload: delete any images stranded in the bad segment | `PAYLOAD_DELETE_IMAGE(scene_id=X)` (S8 func_id 30) | □ | Per affected scene | | 6 | Payload: verify reduced capacity against the imaging plan | `HK_REQUEST(sid=5)` (S3.27) | □ | SID 5 is Payload; SID 4 is OBDH | | 7 | SPACON: file ticket | (mission log) | □ | "CTG-016 closed" |

[OB] CTG-017 — Bus Failure Switchover

Entry point: `obdh.bus_a_status` (0x030F) or `obdh.bus_b_status` (0x0310) = 1 (ERROR), or multiple subsystem HK SIDs simultaneously stop flowing. **Goal:** Switch to redundant bus and restore HK / commanding to affected subsystems. **Possible negative outcomes / deviations:** - Switchover does not restore HK → suspect dual-bus failure; EMG-003 candidate - Bus B fails subsequently → no further redundancy; flag mission constraint to Mission Manager - Switchover causes momentary subsystem reset → expected; verify each subsystem recovers **Decision tree:** - IF only one subsystem affected → check that subsystem first, not the bus - IF multiple subsystems via bus A → switchover - IF bus B already in use → escalation; suspect OBC | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | OBDH: establish which SIDs are missing | `HK_REQUEST(sid=1…6)` (S3.27) | □ | Only SID 4 responding points at the CAN bus | | 2 | OBDH: confirm the bus status words | `HK_REQUEST(sid=4)` (S3.27) | □ | 0x030F / 0x0310 = 1 (ERROR); `obdh.active_bus` 0x030E | | 3 | FD: GO for bus switchover (dual-keyed) | (FD call) | □ | | | 4 | OBDH: command the switchover | `OBC_SELECT_BUS(bus=0\|1)` (S8 func_id 54) | □ | Func 53 is `OBC_SWITCH_UNIT` — that switches the **OBC**, not the bus | | 5 | OBDH: verify the new bus is active; all subsystem HK resumes | (read SIDs 1, 2, 3, 4, 5, 6) | □ | Within 60 s | | 6 | SPACON: file ticket; brief Mission Manager (no further bus redundancy) | (mission log) | □ | "CTG-017 closed" |

[TT] CTG-019 — No Telemetry at Pass Start

Entry point: Expected AOS, no beacon and no HK after carrier expected; or beacon present but full HK never appears after OBC boot. **Goal:** Determine whether the spacecraft is responding (S1 acks present), and proceed via blind commanding if so; otherwise escalate. **Possible negative outcomes / deviations:** - Spacecraft acks commands but no TM → blind-commanding posture; identify next pass as recovery - Neither acks nor TM → second pass also silent → EMG-004 - Geometry was wrong all along → re-check FDS predictions **Decision tree:** - IF S1 acks received → continue with blind commanding; do not invoke EMG-004 yet - IF no acks AND no TM → repeat link test; if still nothing by LOS, prepare EMG-004 for next pass - IF beacon only, no full HK after `OBC_BOOT_APP` → this is usually **not** a fault: the engine re-applies `default_enabled_sids` = [1, 11] on application boot, so SIDs 2–6 must be re-enabled with `HK_ENABLE` (S3.5) before they stream. Send those first, then blind-command per CTG-019 if they still do not appear | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Ground Seg: verify the tracking station and baseband configuration | (MPS panel) | □ | Five-station network; TRACKING | | 2 | FDS: verify orbit / pass prediction | (FDS) | □ | Geometry OK | | 3 | TT&C: connection test (blind) | `CONNECTION_TEST` (S17.1) | □ | Watch for ack only | | 4 | TT&C: walk the RF chain blind — TX line, PA, antenna | `eps.pl_ttc_tx` 0x0112; `ttc.pa_on` 0x0516; `ttc.antenna_deployed` 0x0520 | □ | `EPS_POWER_ON(line_index=2)`, `TTC_PA_ON` (66) as needed | | 5 | SPACON: if acks present, continue procedures by blind commanding | (mission log) | □ | Reduced confirmations | | 6 | SPACON: if silent by LOS, file high-priority ticket; prepare EMG-004 for next pass | (mission log) | □ | "CTG-019 escalation" |

[GS] CTG-020 — Ground Station Antenna Failure

Entry point: Primary antenna (typically Iqaluit) lost during pass — sudden and complete signal loss, `ttc.carrier_lock` (0x0510) → 0, `ttc.link_status` (0x0501) → 0, with spacecraft TT&C telemetry nominal in the last frame received. **Goal:** Recognise this as a ground-side failure, switch to another station of the five-station network for the next pass, and re-plan the affected activities. **This is the one declared advisory card.** In the shipped index CTG-020 carries `advisory: true` with the reason "pure ground-segment: every action is taken on the ground and the spacecraft is unreachable throughout". Its `command_services` list is empty and the MCS reports `runnable = false` for it. Every step below is a ground action — do **not** expect to send a telecommand during this card; that is precisely the point of it. **Possible negative outcomes / deviations:** - Time-critical activity in the current pass lost → re-plan for the next AOS (Troll, Svalbard, Inuvik or O'Higgins) - Alternate station also degraded → EMG-004 candidate over the next two passes - Customer impact → Mission Manager bulletin | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Ground Seg: confirm antenna failure (vs spacecraft TT&C failure) | (baseband / antenna TM) | ☐ | Last-frame `ttc.rssi` 0x0502 and `link_margin` 0x0503 were nominal | | 2 | Ground Seg: read the GS equipment status and active station | (MPS panel) | ☐ | `ttc.gs_equipment_status` 0x0524, `ttc.active_gs` 0x0523 | | 3 | Ground Seg: hand the next pass to an alternate station | (MPS) | ☐ | Schedule update — no telecommand is possible | | 4 | FDS / Mission Planning: re-plan the lost activities into the next contact | (planning tool) | ☐ | Uplink happens on the next pass, not this one | | 5 | FD: brief Mission Manager (customer bulletin) | (briefing) | ☐ | | | 6 | SPACON: file ticket | (mission log) | ☐ | "CTG-020 closed" |

[OB] CTG-021 — OBC Bootloader Recovery

Entry point: All application-APID TM ceases; only the SID 11 beacon arrives on the bootloader APID (0x002, distinct from the application APID 0x001). `obdh.sw_image` (0x0311) = 0. **Goal:** Identify the reboot cause, attempt application boot, and if that fails, escalate to memory reload or unit switchover. **Possible negative outcomes / deviations:** - `OBC_BOOT_APP` issued but no S1.7 within 30 s → set `OBC_BOOT_INHIBIT(inhibit=1)` (func 56), `MEM_CHECK` (S6.9) the application image, then retry - Image CRC bad and pass time permits → reload via `MEM_LOAD` (S6.2), re-verify, then boot - Reload not feasible or boot fails twice → switch to the redundant unit via `OBC_SWITCH_UNIT` (S8 func_id 53), then confirm `obdh.active_obc` (0x030C) - Memory ECC errors clustered → run `OBC_MEMORY_SCRUB` (S8 func_id 51) before the re-boot attempt **Decision tree:** - IF beacon present AND boot succeeds → return to nominal - IF boot fails twice → switch OBC unit (dual-keyed) - IF both units cycle → EMG-001 safe-mode entry | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | OBDH: confirm bootloader state — `obdh.sw_image` (0x0311) = 0 | (read SID 4, or SID 11 if SID 4 does not answer) | ☐ | Bootloader APID only | | 2 | OBDH: record `obdh.last_reboot_cause` (0x0316) and `obdh.reboot_count` (0x030A) | (read SID 4) | ☐ | 1=watchdog 2=memory 3=switchover 4=commanded | | 3 | OBDH: command application boot (dual-keyed) | `OBC_BOOT_APP` (S8 func_id 55) | ☐ | S1.1 + S1.7 within 30 s; 0x0311 → 1 | | 4 | TT&C: clear beacon mode — booting the app does **not** clear it | `TTC_SET_BEACON_MODE(on=0)` (S8 func_id 70) | ☐ | `ttc.beacon_mode` (0x0521) = 0; until then LOW is refused (failure code 0x0007) | | 5 | TT&C: select the routine operations rung | `TTC_SET_DATA_RATE(rate=0)` (S8 func_id 65) | ☐ | LOW = 16 kbps; the beacon demodulator will not lock it | | 6 | OBDH: re-enable the housekeeping structures | `HK_ENABLE(sid=2…6)` (S3.5) | ☐ | Boot re-applies `default_enabled_sids` = [1, 11] | | 7 | OBDH: confirm SIDs 1–6 stream | (read SIDs) | ☐ | 1 EPS, 2 AOCS, 3 TCS, 4 OBDH, 5 Payload, 6 TT&C | | 8 | OBDH: re-synchronise the clock | `SET_TIME(cuc_seconds=T_corrected)` (S9.1) | ☐ | The bootloader reset the clock to **J2000** and booting the application did not restore it: `obdh.time_ref` (0x0A13) = 0 (NONE), `obdh.time_since_sync_s` (0x0A14) = 4294967295, and packets are stamped in the year 2000 until this is done. That is an un-synced OBC, not an epoch bug | | 9 | SPACON: file ticket; brief FD on root cause | (mission log) | ☐ | "CTG-021 closed" |

[OB] CTG-022 — CAN Bus Failure Isolation

Entry point: Housekeeping from one or more subsystems missing > 30 s; `obdh.bus_a_status` (0x030F) or `obdh.bus_b_status` (0x0310) = 1 (ERROR). **Goal:** Identify the failed CAN bus, switch to the redundant bus, verify all subsystems are restored on the new bus. **Possible negative outcomes / deviations:** - Switchover does not restore comms to lost subsystem → that subsystem's bus stub is broken (single-string failure); proceed in degraded mode - Both bus statuses report ERROR → no bus path remains; EMG-001 Emergency Safe Mode - Switchover restores TM but raises new errors → invoke CTG-021 if OBC resets **Decision tree:** - IF Bus A fault AND Bus B nominal → switch to Bus B - IF both buses fault → escalate to EMG-001 - IF switchover succeeds but subsystem still silent → that subsystem is single-string-on-failed-bus | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | OBDH: identify the failed bus and the active one | (read SID 4) | □ | 0x030F / 0x0310 = 1 (ERROR); `obdh.active_bus` 0x030E | | 2 | OBDH: brief FD with the affected-subsystem list | (briefing) | □ | Decision input | | 3 | OBDH: command bus switchover to the redundant bus (dual-keyed) | `OBC_SELECT_BUS(bus=0\|1)` (S8 func_id 54) | □ | FD second-key | | 4 | OBDH: verify all subsystem HK now streaming | (read SIDs 1, 2, 3, 5, 6) | □ | All five restored | | 5 | OBDH: note operating now on single-string bus | (mission log) | □ | "CTG-022 closed, on Bus X single-string" |

[FE] CTG-023 — Progressive Load Shed

Entry point: Battery SoC declining during extended eclipse; power budget negative. **Goal:** Shed loads in priority order at defined SoC thresholds; restore in reverse order once recharging begins. Unlike CTG-001 (voltage-triggered), this is SoC-triggered and adds an explicit FD GO at each rung. **Possible negative outcomes / deviations:** - SoC still dropping after every rung → `OBC_SET_MODE(mode=2)` EMERGENCY, then EMG-002 Total Power Failure - SoC below 10 % → deep-discharge risk; EMERGENCY immediately, brief the Mission Manager on likely capacity loss - Eclipse-exit recovery slower than predicted → hold the shed level one orbit; reassess - Power still not recovering after eclipse exit → check per-panel currents 0x012B–0x0130; if all near zero, CTG-009 **Decision tree (the shipped thresholds):** - IF SoC < 50 % → rung 1, payload OFF - IF SoC < 35 % → rung 2, FPA cooler OFF - IF SoC < 25 % → rung 3, S-band PA OFF (downlink lost; uplink and the UHF beacon survive) - IF SoC < 15 % → rung 4, reaction-wheel line OFF (active attitude control lost) - IF bus voltage still below 28.4 V with everything shed → `OBC_SET_MODE(mode=2)`, then EMG-002 | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: log SoC, bus V, load, and compute the discharge rate | `HK_REQUEST(sid=1)` (S3.27) | □ | Baseline; `eps.eclipse_flag` (0x0108) = 1 | | 2 | FD: authorise rung 1 | (FD GO) | □ | Dual-keyed | | 3 | EPS: rung 1 at SoC < 50 % — payload OFF | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | □ | −45 W imaging / −8 W standby | | 4 | EPS: rung 2 at SoC < 35 % — FPA cooler OFF | `FPA_COOLER(on=0)` (S8 func_id 43) | □ | ≈ −15 W; FPA will warm | | 5 | EPS: rung 3 at SoC < 25 % — S-band PA OFF | `TTC_PA_OFF` (S8 func_id 67) | □ | ≈ −20 W; verification now rests on the beacon and S1 acks | | 6 | EPS: rung 4 at SoC < 15 % — wheel line OFF, damp on torquers | `EPS_POWER_OFF(line_index=7)` (20); `AOCS_SET_MODE(mode=2)` (0) | □ | Mode 2 is DETUMBLE; mode 1 (SAFE_BOOT) actuates nothing | | 7 | EPS: post eclipse exit, restore in reverse order at rising SoC gates | wheels + `AOCS_SET_MODE(mode=3)` > 25 % → `TTC_PA_ON` > 35 % → `FPA_COOLER(on=1)` > 45 % → `PAYLOAD_SET_MODE(mode=1)` > 60 % | □ | 5 min between steps | | 8 | SPACON: file ticket; document time-to-recovery | (mission log) | □ | "CTG-023 closed" |

[FE] CTG-024 — Solar Panel Loss Response

Entry point: Per-panel solar-array current trending to zero or significantly degraded; total `eps.power_gen` reduced. **Five body faces carry cells, not six:** −Z is the nadir face, with the imager aperture and radiator (`mz_area_m2: 0.000`), so `eps.sa_mz_current` (0x0130) reads zero permanently and is never evidence of a panel failure. Panel areas are +X/−X 0.060 m², +Y/−Y 0.030 m², +Z 0.018 m², plus 0.120 m² of cells per deployable wing. **Goal:** Identify the failed panel, recalculate power budget for the new configuration, decide whether the mission proceeds at reduced duty or enters extended safe-state. **Possible negative outcomes / deviations:** - Two or more panels lost → power budget likely negative even when sunlit; invoke CTG-023 pre-emptively - Panel "loss" is actually attitude-driven shadow → no fix needed; verify with attitude - Asymmetric loss creates torque disturbance → AOCS pointing may degrade **Decision tree:** - IF one panel lost, attitude nominal → reduce duty cycle, continue with monitoring - IF two panels lost on the same axis → hold COARSE_SUN (mode 3) until ground analysis - IF panel "loss" recovers when attitude changes → not a real failure, log and continue | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | EPS: confirm the generation shortfall in sunlight | `HK_REQUEST(sid=1)` (S3.27) | □ | `eps.power_gen` 0x0107; `eps.eclipse_flag` (0x0108) = 0 | | 2 | EPS: read all six per-panel currents, three samples 30 s apart | (read SID 1, params 0x012B–0x0130) | □ | +X, −X, +Y, −Y, +Z, −Z; shadowed faces read ~0 A legitimately, and **−Z (0x0130) always reads zero — it has no cells**. The deployable wings are separate: 0x0146 (+Y) / 0x0147 (−Y) | | 3 | AOCS: confirm attitude nominal (rule out shadow) | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.att_error` (0x0217) < 2°; rates < 0.1 °/s | | 4 | EPS: recompute the orbit-averaged budget with N−1 panels | (operator calc) | □ | Project minimum SoC at next eclipse exit | | 5 | EPS: if the projection is below 25 %, shed to match | `PAYLOAD_SET_MODE(mode=1)` then `(mode=0)`; `FPA_COOLER(on=0)` | □ | Recompute after each step | | 6 | FD: brief Mission Manager on the degraded mission profile | (briefing) | □ | Customer notification | | 7 | EPS: if the budget is still negative, hand off to CTG-023 | (decision) | □ | Or continue at reduced duty | | 8 | SPACON: file ticket; mark the panel as PERMANENT_FAULT in the mission log | (mission log) | □ | "CTG-024 closed" |

[AOCS] CTG-025 — AOCS Sensor Loss Recovery

Entry point: A star tracker, a CSS head, or a magnetometer reports FAULT or fails its validity check; `aocs.att_error` (0x0217) rises. **Goal:** Identify which sensor failed, switch to the redundant unit where one exists (star trackers only), assess whether the remaining suite still supports the mission's pointing requirement, and degrade the mode if not. **Redundancy inventory:** two star trackers (ST1/ST2), six CSS heads (+X, −X, +Y, −Y, +Z, −Z), **one magnetometer with no spare** and **three single-axis gyros with no spare**. The magnetometer and the gyro channels are the AOCS sensors with no redundant unit to select; see CTG-029 for the gyros. **Possible negative outcomes / deviations:** - Cumulative sensor loss reduces attitude knowledge below the mission requirement → fall back to COARSE_SUN (mode 3) - Star tracker boresight blocked by sun → status 3 (BLIND); no failure, expected geometry, wait for sun avoidance - One CSS head lost → composite sun vector degraded to roughly 5–10° but usable; two on opposite faces → that axis is lost; three or more → check `aocs.css_valid` (0x0248) - Magnetometer lost (and a power-cycle does not recover it) → DETUMBLE and magnetorquer desaturation are both impossible; momentum will accumulate **Decision tree:** - IF ST1 fault → power on ST2 and select it (`ST2_POWER(on=1)`, then `ST_SELECT(unit=2)`) - IF both ST fault → fall back to COARSE_SUN (mode 3); imaging suspended - IF magnetometer fault → power-cycle the single unit (`S2_AOCS_MAGNETOMETER`, device 0x0209); there is no second magnetometer to switch to - IF no ST and CSS invalid → DETUMBLE (mode 2) - IF cumulative loss exceeds the pointing budget → EMG-005 | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | AOCS: read all sensor status flags | `HK_REQUEST(sid=2)` (S3.27) | □ | 0x0240, 0x0243, `css_valid` 0x0248, `mag_enabled` 0x025C | | 2 | AOCS: if ST1 faulted, power on ST2 and select it | `ST2_POWER(on=1)` (func_id 5); `ST_SELECT(unit=2)` (func_id 6) | □ | Allow 60–120 s for boot to TRACKING | | 3 | AOCS: isolate the failed tracker | `ST1_POWER(on=0)` (S8 func_id 4) | □ | Do not power both if `eps.bat_soc` < 40 % | | 4 | AOCS: if the magnetometer faulted, power-cycle it | `S2_AOCS_MAGNETOMETER(device_id=521, state=0)`, wait 10 s, then `(state=1)` (S2.1, device 0x0209) | □ | `aocs.mag_enabled` (0x025C) = 1 and `aocs.mag_field_total` (0x0277) back in 25 000–65 000 nT; if not, declare the no-mag configuration and escalate — there is no `MAG_SELECT` | | 5 | AOCS: evaluate cumulative knowledge against the pointing requirement | (operator calc) | □ | NOMINAL needs an ST tracking + gyro + valid mag | | 6 | AOCS: select the best available mode | `AOCS_SET_MODE(mode=4\|3\|2)` (S8 func_id 0) | □ | 4 NOMINAL / 3 COARSE_SUN / 2 DETUMBLE | | 7 | AOCS: if degraded below NOMINAL, power the payload down | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | □ | Imaging suspended | | 8 | SPACON: brief FD; file ticket | (mission log) | □ | "CTG-025 closed" |

[AOCS] CTG-026 — AOCS Actuator Stuck Recovery

Entry point: Reaction wheel speed not changing despite torque command; or wheel speed frozen at constant value; or wheel temperature rising. **Goal:** Identify the stuck wheel, isolate it from the control loop, reconfigure to three-wheel control, restore stable attitude. **Possible negative outcomes / deviations:** - Stuck wheel cannot be commanded off → the wheel remains in the pyramid as fixed momentum; offload via the remaining three - Three-wheel control cannot meet the pointing budget → fall back to COARSE_SUN (mode 3) - Cumulative momentum exceeds capacity → desaturate; if the magnetometer is also invalid, magnetorquer dumping is impossible and momentum will build to saturation **Decision tree:** - IF one wheel stuck AND three operational → reconfigure to three wheels; FINE_POINT still runs, degraded (`aocs.fine_degraded`, 0x0299), with a relaxed 1.5° budget - IF two wheels stuck → DETUMBLE (mode 2); two wheels cannot give three-axis pointing - IF the disabled wheel's temperature keeps rising above 70 °C → shed the wheel power line, `EPS_POWER_OFF(line_index=7)` — note this de-energises **all** wheels | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: identify the stuck wheel from two speed samples 30 s apart | `HK_REQUEST(sid=2)` (S3.27) | □ | Speeds 0x0207–0x020A frozen or at 0 rpm | | 2 | AOCS: verify the wheel temperature trend | (read SID 2) | □ | Temps 0x0218–0x021B; rising = bearing seizure | | 3 | AOCS: disable the stuck wheel (dual-keyed) | `AOCS_DISABLE_WHEEL(wheel_idx=N)` (S8 func_id 2) | □ | FD second-key; rates stay < 1.0 °/s | | 4 | AOCS: confirm the loop has reconfigured to three wheels — no command | (observe, 15 s) | □ | There is no `AOCS_3W_MODE` telecommand (S8 func 5 is `ST2_POWER`) and no `aocs.wheel_config` parameter: `AOCS_DISABLE_WHEEL` in step 3 is the whole reconfiguration. Restore later with `AOCS_ENABLE_WHEEL` (func_id 3) | | 5 | AOCS: dump momentum on the surviving three | `AOCS_DESATURATE` (S8 func_id 1) | □ | MTQ duties 0x0258–0x025A non-zero during the dump | | 6 | AOCS: verify attitude stable in the new configuration | `AOCS_SET_MODE(mode=4)` (S8 func_id 0) | □ | `aocs.att_error` (0x0217) < 1.5° — relaxed from 1.0° for three-wheel ops | | 7 | SPACON: file ticket; mark the wheel PERMANENT_FAULT | (mission log) | □ | "CTG-026 closed"; desat now needed every 3–4 orbits, not 6–8 |

[OB] CTG-027 — GPS Time-Sync and AOCS Recovery
Entry point: Time jump > 5 s detected; AOCS auto-transitioned to SAFE_BOOT (mode 1) protectively. **Goal:** Verify GPS 3D fix, execute time-sync, re-walk the AOCS mode ladder to restore the prior operational mode. **Possible negative outcomes / deviations:** - GPS has no 3D fix → cannot time-sync; wait for a fix and re-attempt - Time-sync succeeds but AOCS will not exit SAFE_BOOT → invoke CTG-002 AOCS Anomaly Recovery - PDOP > 5 → fix quality marginal; defer the time-sync **Decision tree:** - IF `aocs.gps_fix` = 3 AND `aocs.gps_pdop` < 5.0 AND ≥ 5 satellites → command time-sync, then the mode ladder - IF the fix is invalid → hold in SAFE_BOOT; wait one orbit; retry - IF time-sync fails after retry → invoke CTG-021 OBC Bootloader Recovery (the clock subsystem may be at fault) **Note:** the GPS parameters belong to **AOCS (SID 2)**, not OBDH. There is no `gps.*` namespace in `parameters.yaml`, and the quality figure is PDOP, not HDOP. The onboard clock itself is read from `obdh.obc_time` (0x0309) plus `obdh.obc_time_fine` (0x0A12) in SID 4, with `obdh.time_ref` (0x0A13) alongside it. **A ground set and a GPS sync are different operations.** `SET_TIME` (S9.1) leaves `time_ref` = 1 (GROUND) and the clock keeps drifting at its drawn ~12 ppm. `OBC_GPS_TIME_SYNC` latches **disciplined** mode: `time_ref` = 2 (GPS), and while the fix holds the clock is pulled to true time every tick and accumulates no drift at all. Losing the fix moves it to 3 (**HOLDOVER**) and raises `OBC_TIME_REF_LOST` (0x030E) — the clock is not wrong yet, it has simply stopped being corrected, and drift resumes from wherever it was. Regaining the fix re-disciplines autonomously and raises `OBC_TIME_SYNC` (0x030D) carrying the jump. Neither command is available in the bootloader. | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: confirm GPS fix and quality | `HK_REQUEST(sid=2)` (S3.27) | □ | `aocs.gps_fix` (0x0274) = 3, `gps_pdop` (0x0275) < 5.0, `gps_num_sats` (0x0276) ≥ 5. The command's own gate is only `gps_fix` ≥ 2; a 3D+vel fix is this card's operational preference | | 2 | AOCS: record the pre-event mode, the reference state and the clock offset | (read SIDs 2 and 4) | □ | `aocs.mode` 0x020F; `obdh.time_ref` (0x0A13) — expect 3 (HOLDOVER) if the fix has just been lost, or 1 (GROUND) if the clock was never disciplined; `obdh.obc_time` 0x0309 vs ground UTC | | 3 | OBDH: command GPS time-sync (caution — invalidates the AOCS attitude solution briefly) | `OBC_GPS_TIME_SYNC` (S8 func_id 80) | □ | 0x0309 steps to the GPS-derived value within 5 s; `obdh.time_ref` (0x0A13) → 2 (GPS); `obdh.time_since_sync_s` (0x0A14) restarts; event `OBC_TIME_SYNC` (0x030D) carries the jump | | 4 | AOCS: verify AOCS is in SAFE_BOOT (mode 1) post-sync | (read SID 2) | □ | Auto-transition (event 0x020F, which the event catalogue currently labels `AOCS_MODE_RESTORED`) on a jump > 5 s, because the attitude solution was computed against a clock that has just moved; expect it after a long holdover or a reboot. If the mode is unchanged, skip step 5 | | 5 | AOCS: walk the ladder back — DETUMBLE → COARSE_SUN → NOMINAL → FINE_POINT | `AOCS_SET_MODE(mode=2)`, then 3, 4, 5 (dual-keyed) | □ | 30 s between; verify 0x020F equals each commanded value | | 6 | AOCS: confirm attitude convergence at the target mode | (read SID 2) | □ | COARSE_SUN < 10°, NOMINAL < 5°, FINE_POINT < 1° | | 7 | SPACON: file ticket; note the time-discrepancy correction | (mission log) | □ | "CTG-027 closed, Δt = X s" |

[PR] CTG-028 — Propulsion Anomaly

*(Editor's note: this is the simulator's **CTG-028 Propulsion Anomaly**, `contingency/propulsion_anomaly.md`, condensed.)* **Entry point:** S12 alarm on `prop.tank_pressure` (0x0B01) — absolute band or the rate/DELTA monitor that catches a slow leak; or event PROP_VALVE_FAULT (0x080A), PROP_LEAK_DETECTED (0x080B), PROP_UNDERPERFORMANCE (0x080C), PROP_TANK_PRESSURE_HIGH (0x0807); or post-burn OD showing achieved ΔV < 90 % of plan (NOM-013 step 6). **Goal:** Isolate the propellant, **bound** the remaining usable ΔV and protect the collision-avoidance reserve. EOSAT-1's 1U green monoprop (66.9 m/s total) exists for debris avoidance, limited orbit maintenance and the 34.3 m/s end-of-life disposal burn (NOM-014), so the objective is not "fix the thruster" — it is "know exactly how much CAM capability remains and keep it safe". **Possible negative outcomes / deviations:** - A branch reporting 2 (fault) did not close — treat the propellant downstream of the stuck valve as lost - Pressure still falling after isolation → the leak is upstream of the latch valves; it cannot be isolated, only bounded - Bounded reserve below ~1 m/s → debris-avoidance capability is effectively lost; the conjunction posture changes - Rising tank pressure that does not respond to heater safing → CTG-004, and prepare EMG-001 (safe mode disarms propulsion by rule 8005) | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Propulsion: bring up the PROP structure and capture full TM | `HK_ENABLE(sid=9)` (S3.5) | ☐ | SID 9 is PROP; SID 5 is Payload | | 2 | Propulsion: read state, pressure trend, mass, temps, valves, leak rate | (read SID 9) | ☐ | 0x0B00, 0x0B01, 0x0B02, 0x0B04/0x0B05, 0x0B06–0x0B08, 0x0B16 | | 3 | Propulsion: close both latch-valve branches | `PROP_ISO_CLOSE(branch=2)` (S8 func_id 114) | ☐ | `prop.latch_a_state` 0x0B06 and `latch_b_state` 0x0B07 = 0 | | 4 | Propulsion: bound the remaining usable ΔV | read `prop.dv_remaining` (0x0B0C), correct for failure mode | ☐ | Report the bounded CAM reserve to the FD | | 5 | Propulsion: latch the ground inhibit while the anomaly is open | `PROP_INHIBIT_SET(on=1)` (S8 func_id 116) | ☐ | `prop.arm_inhibit_mask` (0x0B15) bit 9 set | | 6 | SPACON: file ticket; FD decides on EMG-001 if cascading | (mission log) | ☐ | Clearing the inhibit (`PROP_INHIBIT_CLR`, func 117) needs a closed ticket + FD authority |

[AOCS] CTG-029 — Gyro Failure Recovery

(Simulator card `contingency/gyro_failure.md`. It replaces the book's temporary CTG-B01 "IMU / Gyro Anomaly".) **Entry point:** Event `GYRO_BIAS_HIGH` (0x020B) — the bias still present at the gyro output exceeds 0.05 °/s (`aocs.gyro_bias_resid` 0x02D2); or `aocs.gyro_units_mask` (0x02D1) ≠ 7, a unit is off; or `aocs.gyro_saturated` (0x0295) = 1; or the attitude estimate walking away (`aocs.att_error` 0x0217 growing) while the star tracker is out of the loop. Link up with ≥ 15 min of pass left, application image running, SoC > 40 %, no slew, desaturation or burn in the next 15 min. **Goal:** Identify the affected gyro channel, restore a bias-free rate reference with the on-board bias calibration and, if the channel cannot be recovered, isolate it and continue attitude determination on the remaining sensors. **What the simulator models.** EOSAT-1 carries three single-axis gyros, one per body axis, and **no redundant unit**: S2 devices `S2_AOCS_GYROSCOPE_0/1/2` (device IDs 0x0206, 0x0207, 0x0208 = 518, 519, 520). There is no IMU power line and nothing to switch to. An injected `gyro_bias` fault is a **persistent** offset on one axis or all three; it does not heal with time, and the calibration compensates it in the measurement chain without repairing the unit. A unit switched off is a dead channel. The three bias numbers are different quantities: healthy bias instability is about 3e-5 °/s (0.1 °/h), event 0x020B fires at **0.05 °/s** of residual bias, and the S12 limit on `aocs.gyro_bias_x/y/z` (0x0270–0x0272) is ±0.5 °/s. This card acts on the 0.05 °/s event, the earliest actionable signal. **Possible negative outcomes / deviations:** - All three units off or failed → no rate reference; the filter coasts on the star tracker, sun sensors and magnetometer. Hold NOMINAL if the tracker is valid, otherwise EMG-005 Loss of Attitude - Saturation (`aocs.gyro_saturated` = 1) → the vehicle is rotating faster than the gyro range: an attitude anomaly, not a gyro fault; CTG-002 first - Refused with `0x0A03` → no star tracker in the loop; recover it (CTG-008), wait for `aocs.est_source` (0x02C1) bit 0, retry. Refused with `0x0A04` → not quiescent; wait for the slew or desaturation to end (`aocs.mode` ≠ 6 and ≠ 7) - Bias returns after a good calibration → if it follows `aocs.gyro_temp` (0x0273), re-run after the thermal transient; at constant temperature it is a degrading unit — raise an anomaly report - Instructor removes an injected bias after a calibration → the stored compensation now over-corrects with the opposite sign; re-run step 4 | Step | Action | TC / Code | Confirm | Notes |
|:----:|:-------|:----------|:--------|:------| | 1 | AOCS: collect the gyro and determination telemetry; classify | `HK_REQUEST(sid=2)`, `HK_REQUEST(sid=7)` (S3.27) | □ | Record `aocs.gyro_units_on` (0x0294), `aocs.gyro_units_mask` (0x02D1), `aocs.gyro_saturated` (0x0295), `aocs.gyro_bias_x/y/z` (0x0270–0x0272), `aocs.gyro_bias_resid` (0x02D2), `aocs.gyro_temp` (0x0273). The axis whose bias stands out is the affected channel | | 2 | AOCS: protect the attitude estimate | (read SID 2 / SID 7) | □ | `aocs.est_source` (0x02C1) bit 0 set; `aocs.st1_status` (0x0240) = 2; `aocs.att_error` (0x0217) < 1.0°. Without a tracker in the loop a 0.05 °/s bias walks the estimate tens of degrees in ten minutes: recover the tracker first (CTG-008) | | 3 | AOCS: restore any unit that is off (only if 0x02D1 ≠ 7) | `S2_AOCS_GYROSCOPE_0(device_id=518, state=1)`, `S2_AOCS_GYROSCOPE_1(device_id=519, state=1)`, `S2_AOCS_GYROSCOPE_2(device_id=520, state=1)` (S2.1) | □ | Idempotent. `aocs.gyro_units_mask` = 7 and `aocs.gyro_units_on` = 3 within 20 s; a calibration with a dead channel is refused with `0x0A02` | | 4 | AOCS: run the on-board bias calibration | `AOCS_GYRO_CALIBRATION(action=0, duration_s=300)` (S8 func_id 13) | □ | Hold NOMINAL (mode 4, or 8 in eclipse), rates within 0.10 °/s. `aocs.gyro_cal_state` (0x02C9) = 1 within 20 s; refusal codes `0x0A01`–`0x0A05` in `aocs.gyro_cal_reject` (0x02D0); S1.7 arrives when the arc ends | | 5 | AOCS: monitor the calibration arc | `HK_REQUEST(sid=7)` (S3.27) every 30 s | □ | `aocs.gyro_cal_progress` (0x02CA) ≥ 100 within 330 s. An abort (`AOCS_GYRO_CAL_ABORT`, 0x0212) applies nothing: fix the cause and re-run step 4 | | 6 | AOCS: verify the result and record the bias | `GET_PARAM(param_id=0x02CB)` … `(param_id=0x02CD)` (S20.3) | □ | `aocs.gyro_cal_state` = 2, `aocs.gyro_cal_result` (0x02CF) = 1; each axis within ±0.5 °/s. Above that, do not accept the result: step 7 | | 7 | AOCS: isolate an unrecoverable channel (only if step 6 failed) | `S2_AOCS_GYROSCOPE_0(device_id=518, state=0)` (X shown; 519 Y, 520 Z); `AOCS_EST_RESET(mode=1)` (S8 func_id 91) | □ | Only for a bias above 0.5 °/s after a 900 s calibration, or two aborts in a row with `0x0A06`. The faulty bit clears in 0x02D1; `aocs.est_source` bit 0 still set within 60

s. That axis now relies on the tracker alone until the end of the mission | | 8 | AOCS: confirm the recovery; file the anomaly report | `HK_REQUEST(sid=2)`, `HK_REQUEST(sid=7)` (S3.27) | □ | `aocs.gyro_bias_resid` (0x02D2) < 0.005 °/s; `aocs.gyro_cal_comp_mag` (0x02CE) recorded; `aocs.knowledge_err_3sig` (0x02A7) < 120 arcsec and `aocs.control_err` (0x02A8) < 1.0° within 300 s. If a channel was isolated, check `aocs.fine_ready` (0x02C5) before the next imaging session. "CTG-029 closed" |

[OB] CTG-030 — Telecommand Rejection and Verification Failure
(Simulator card `contingency/tc_rejection.md`. It replaces the book's temporary CTG-B02 "TC Reject Rate Anomaly".) **Entry point:** An S1.2 acceptance failure or an S1.8 execution failure on a telecommand the ground believes is valid; `obdh.tc_rej_count` (0x0306) incrementing, or event `COMMAND_REJECTED` (0x030B); or a telecommand with **no** S1 report at all (verification time-out in the MCS command log). Link expected (inside a predicted pass); MCS command log available. **Goal:** Find why telecommands are refused or not verified, remove the cause and restore clean commanding. The card sends only HK requests and a connection test. **What the simulator reports.** Every refusal carries a 16-bit failure code and a reason text. The code is latched in `obdh.last_tc_error_code` (0x0315), and an on-board refusal also raises `COMMAND_REJECTED` (0x030B), naming the command, the stage and the code. At **acceptance** (S1.2): 0x0001 unknown service, 0x0002 unknown subtype, 0x0003 invalid parameter or length, 0x0004 target subsystem unpowered, 0x0005 uplink not available, 0x0006 not in the bootloader subset. At **execution** (S1.8): 0x0005 invalid in the current mode, 0x0007 not permitted in the current state, and the families 0x0021–0x0028 payload, 0x0A01–0x0A08 gyro calibration, 0x0A20–0x0A22 AOCS mode entry, 0x0B01–0x0B13 propulsion, 0x0C01–0x0C04 mission phase. Two cases are **not** rejections: 0x0005 at acceptance with no `COMMAND_REJECTED` event (the uplink had no command-channel lock, so the on-board software never saw the TC), and no S1 report at all (the TC never arrived, or it is a deferred-completion command — gyro calibration, application boot, burn — whose S1.7 comes when the action ends). Check which before re-sending anything. **Possible negative outcomes / deviations:** - The step 1 request itself gets no reply → a link problem, not a commanding one: CTG-003 TTC Link Loss Recovery, or CTG-019 if there is no telemetry at all - `CONNECTION_TEST` gets no reply → the uplink is not working; CTG-003 - Rejections continue with a code nobody can explain → stop commanding, record the code, reason text and command, escalate to the FD; do not try other commands to work around it - Rejections after an OBC reboot → the OBC restarts in the bootloader (`obdh.sw_image` 0x0311 = 0): every application command is refused `0x0006` until the application is booted again (EMG-003, CTG-021) - Clock-dependent failures → a time-tagged command against a wrong on-board clock runs at the wrong time rather than being rejected; if `obdh.time_ref` (0x0A13) = 0 or `obdh.time_since_sync_s` (0x0A14) is large, resynchronise (NOM-008) before uploading a new schedule | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FDIR/Systems: read the on-board command counters and state | `HK_REQUEST(sid=4)` (S3.27) | □ | Record `obdh.tc_rx_count` (0x0304), `obdh.tc_acc_count` (0x0305), `obdh.tc_rej_count` (0x0306), `obdh.last_tc_error_code` (0x0315), `obdh.sw_image` (0x0311: 0 bootloader, 1 application), `obdh.mode` (0x0300: 0 nominal, 1 safe, 2 emergency), `eps.spacecraft_phase` (0x0129) | | 2 | TT&C: check the uplink chain | `HK_REQUEST(sid=6)` (S3.27) | □ | `ttc.carrier_lock` (0x0510) = 1; `ttc.frame_sync` (0x0512) = 1; record `ttc.cmd_rx_count` (0x0513) and `ttc.beacon_mode` (0x0521). No S1 report and no change in 0x0513 → the TC never reached the spacecraft | | 3 | FDIR/Systems: classify every rejection by code and stage | (MCS command log; event 0x030B text) | □ | 0x0006 with `sw_image` = 0 → boot the application (LEOP-001 step 5, CTG-021). 0x0004 → switch the line on first. 0x0003 → rebuild the TC in the command builder. 0x0001 / 0x0002 → not implemented. 0x0005 / 0x0007 at execution → valid but not allowed now (for example beacon mode: clear it with `TTC_SET_BEACON_MODE`). 0x0Axx → CTG-029; 0x0Bxx → read `prop.arm_inhibit_mask` (0x0B15), NOM-013 or CTG-028; 0x0Cxx → mission-phase guards | | 4 | TT&C: test clean commanding | `CONNECTION_TEST` (S17.1); `HK_REQUEST(sid=4)` (S3.27) | □ | S1.1 and S1.7 within 10 s; `obdh.tc_acc_count` incremented, `obdh.tc_rej_count` unchanged | | 5 | FD: re-send the corrected telecommand once | (the corrected TC) | □ | Only after its cause has been removed; never repeat a refused TC unchanged. S1.1 and S1.7 received; `obdh.last_tc_error_code` unchanged. "CTG-030 closed" |

14.5 Emergency procedure cards

Emergency procedures must be read once during operator familiarisation but never invoked without explicit FD authorisation. Any approach to safe-mode entry begins with a voice-loop callout to the FD. The cards below give the canonical structure; the simulator's procedure files hold the full step lists.

Every card below carries the simulator's EMG ID and matches the file of the same ID in `configs/eosat1/procedures/emergency/`. The numbering used by earlier printings is in the historical note at the end of §14.6.

[FD] EMG-001 — Emergency Safe Mode
Entry point: Any two or more of — `eps.bat_soc` (0x0101) < 15 %; `aocs.att_error` (0x0217) > 30°; any body rate > 2 °/s; `obdh.reboot_count` (0x030A) up by ≥ 2 in an hour; `tcs.temp_battery` (0x0407) outside −10 / +45 °C; `ttc.link_status` (0x0501) ≠ 2 for more than a pass; `obdh.mode` (0x0300) already = 2 (EMERGENCY). Or an FD declaration of emergency. **Goal:** Place the spacecraft in the safest sustainable configuration — OBC EMERGENCY, COARSE_SUN sun-pointing, minimum loads — and hold for at least three orbits while ground analysis decides the recovery path. There is no "SAFE_POINT": the sun-safe hold is COARSE_SUN, mode 3. **Possible negative outcomes / deviations:** - Safe-mode entry itself triggers further FDIR → expected; allow autonomous protection to converge - Spacecraft tumbling (rates > 2 °/s) → run EMG-005 Loss of Attitude before continuing - Not power-positive after sun acquisition → EMG-002 Total Power Failure - Recovery from safe mode requires offline analysis; do not exit safe mode the same pass without explicit FD + Mission Manager authority | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FD: declare emergency on voice loop | (FD call) | ☐ | All positions ack | | 2 | OBDH: command OBC EMERGENCY | `OBC_SET_MODE(mode=2)` (S8 func_id 50) | ☐ | `obdh.mode` (0x0300) = 2 within 5 s; autonomous load shed follows | | 3 | Payload: confirm the autonomous shed, command OFF if still on | `PAYLOAD_SET_MODE(mode=0)` (S8 func_id 26) | ☐ | `payload.mode` (0x0600) = 0; `eps.power_cons` (0x0106) < 50 W | | 4 | Thermal: OBC and thruster heaters off | `HEATER_OBC(on=0)` (41); `HEATER_THRUSTER(on=0)` (42) | ☐ | Battery heater stays under thermostat control | | 5 | AOCS: command the sun-safe hold | `AOCS_SET_MODE(mode=3)` (S8 func_id 0) | ☐ | `aocs.mode` (0x020F) = 3 (COARSE_SUN); if tumbling, mode 2 first | | 6 | EPS: confirm power-positive at sun acquisition | (read SID 1) | ☐ | `eps.power_gen` (0x0107) > `power_cons`; arrays > 0.5 A | | 7 | All: hold and assess for a minimum of three orbits (~4.75 h) | (read SIDs 1–6) | ☐ | Exit gate: SoC > 25 %, no new anomalies | | 8 | FD: brief Mission Manager; halt all non-survival activity | (briefing) | ☐ | "EMG-001 active"; staged recovery R+1 to R+5 on SoC gates 30/40/50/55/60 % |

[FE] EMG-002 — Total Power Failure

Entry point: `eps.bat_soc` (0x0101) < 10 % **or** `eps.bus_voltage` (0x0105) < 29.6 V. The undervoltage chain comes from one named table, `simulation.yaml` `eps.bus_protection`: UV flag, in-model automatic load shed and the 0x0103 warning event at **28.4 V** (`uv_v`; restoring above 28.9 V, shedding payload → fpa_cooler → ttc_tx → aocs_wheels → prop), FDIR EPS safe mode and the 0x0104 alarm event at **27.7 V** (`eps_safe_v`), AOCS safe at **26.1 V** (`aocs_safe_v`), and the fault-propagation cascade at **25.9 V** (`cascade_v`) — by then the spacecraft is safing itself. The FDIR SoC ladder is separate and runs at 30 / 20 / 10 %. **Goal:** Reach the minimum survival configuration, get the arrays sun-pointing, and wait for charging. **No load may be re-enabled until SoC exceeds 30 %**, confirmed on three consecutive samples. **Possible negative outcomes / deviations:** - Brown-out reaches the OBC → autonomous reboot likely; CTG-010 / EMG-003 follow-on - Battery does not recover even with all loads shed → mission-loss candidate; multi-orbit re-charge attempt - Rates above 2 °/s → the arrays cannot be pointed; run EMG-005 step 2 first - Solar arrays partially shadowed by attitude → cross with EMG-005 | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FD: declare power emergency | (FD call) | □ | | | 2 | EPS: immediate shed, within 60 s | `PAYLOAD_SET_MODE(mode=0)` (26); `HEATER_OBC(on=0)` (41); `HEATER_THRUSTER(on=0)` (42) | □ | `eps.power_cons` (0x0106) < 50 W within 15 s | | 3 | Thermal: leave the battery heater on thermostat, line energised | `EPS_POWER_ON(line_index=5)` (19); `HEATER_BATTERY(on=1)` (40) | □ | `eps.pl_htr_bat` (0x0115) = 1 — the thermostat needs the line | | 4 | OBDH: command OBC EMERGENCY | `OBC_SET_MODE(mode=2)` (S8 func_id 50) | □ | `obdh.mode` (0x0300) = 2; HK generation drops to 1/60 s | | 5 | AOCS: sun acquisition for maximum generation | `AOCS_SET_MODE(mode=3)` (S8 func_id 0) | □ | COARSE_SUN; arrays > 0.3 A each | | 6 | EPS: monitor recovery through the sunlit phase and the next eclipse | (read SID 1) | □ | Bus must stay > 28.4 V through eclipse; recovery is slow | | 7 | EPS: hold the restore gate until SoC > 30 % | (read SID 1) | □ | Three consecutive samples; then OBC heater first | | 8 | FD: brief Mission Manager; multi-orbit recovery posture | (briefing) | □ | "EMG-002 active"; target ≥ 3 % net SoC gain per orbit |

[OB] EMG-003 — OBC Reboot

Entry point: At least two independent indicators — `obdh.cpu_load` (0x0302) above 90 % sustained; `obdh.tc_rej_count` (0x0306) incrementing on valid commands; `obdh.mem_errors` (0x031E) rising; the OBC not responding to mode commands; or housekeeping stale for more than 30 s. **Goal:** Force a clean commanded reboot, then restore minimum mission state — falling back to the redundant unit only if the reboot does not recover the OBC. **Possible negative outcomes / deviations:** - Reboot brings up the bootloader but the application fails again → CTG-021 OBC Bootloader Recovery; suspect SW corruption - Both OBC units affected → spacecraft autonomous safe-mode will eventually trigger; coordinate with the manufacturer - Reboot causes power transients → cross-monitor EPS for CTG-012 / CTG-001 firings | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | FD: declare emergency; coordinate dual-key authorisation | (FD call) | □ | Requires SoC > 30 % for the boot sequence | | 2 | OBDH: capture pre-reboot state | `HK_REQUEST(sid=4)` (S3.27) | □ | 0x0300, 0x0302, 0x030A, 0x030B, 0x030C, 0x0311, 0x0316, 0x0319, 0x031E | | 3 | Payload / Thermal: suspend payload and shed non-critical loads | `PAYLOAD_SET_MODE(mode=0)` (26); `HEATER_OBC(on=0)` (41) | □ | No memory upload or time-tagged sequence in progress | | 4 | OBDH: command the reboot | `OBC_REBOOT` (S8 func_id 52) | □ | Func 42 is `HEATER_THRUSTER` — commanding 42 here would toggle the catbed heater and leave the OBC running | | 5 | SPACON: boot the application image | `OBC_BOOT_APP` (S8 func_id 55) | □ | `obdh.sw_image` (0x0311) = 1. Booting the application does **not** restore the clock | | 6 | OBDH: if the OBC does not recover, switch units (dual-keyed) | `OBC_SWITCH_UNIT` (S8 func_id 53) | □ | Confirm `obdh.active_obc` (0x030C) changed | | 7 | SPACON: post-recovery, run full CTG-010 cleanup | per CTG-010 | □ | Includes `HK_ENABLE` for SIDs 2–6, clearing beacon mode, and a **clock re-sync**: the reboot returned the OBC to J2000 with `obdh.time_ref` (0x0A13) = 0 (NONE), so re-arm no time-tagged schedule until `SET_TIME` (S9.1) has been accepted | | 8 | FD: brief Mission Manager | (briefing) | □ | "EMG-003 closing" |

[TT] EMG-004 — Loss of Communication

Entry point: No telemetry for two or more consecutive pass windows (~3 h). **Goal:** Escalate in a structured way — ground segment, then blind commanding on each transponder, then autonomous-recovery assessment — with decision gates at 6 h, 12 h, 24 h and 48 h since last contact. **A silent S-band does not mean a silent spacecraft.** EOSAT-1 carries two transmitters: the S-band prime and a separate UHF whip beacon. PA-off stops S-band telemetry while the beacon carries on sending HK SID 11 from the non-switchable `ttc_rx` line. Search for the beacon before concluding the spacecraft is dead. **Possible negative outcomes / deviations:** - Spacecraft autonomous recovery brings it back on the next pass → continue with CTG-010 post-recovery cleanup - Sustained silence beyond 48 h → declare formal loss; coordinate with industry partners on radio listening - Partial recovery (carrier only, no demod) → CTG-014 + CTG-003 path | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Ground Seg: confirm every station, baseband and antenna configuration | (full GS check) | ☐ | Five-station network; receiver threshold to minimum (−130 dBm) | | 2 | TT&C: blind HK request on every visible pass, min elevation 5° | `HK_REQUEST(sid=6)` (S3.27) | ☐ | Prioritise by maximum elevation | | 3 | TT&C: blind switch to the primary transponder, 3× at 10 s spacing | `TTC_SWITCH_PRIMARY` (S8 func_id 63) | ☐ | Best on a pass above 20° elevation | | 4 | TT&C: blind switch to the redundant transponder, 3× at 10 s spacing | `TTC_SWITCH_REDUNDANT` (S8 func_id 64) | ☐ | Every S8 function-perform is subtype 1 | | 5 | TT&C: search for the UHF beacon at every predicted pass | (carrier search) | ☐ | Beacon tier, HK SID 11 — independent of the S-band PA | | 6 | FD: escalate to industry-partner radio listening beyond the 24 h gate | (external comm) | ☐ | | | 7 | FD: brief Mission Manager on customer comms strategy | (briefing) | ☐ | "EMG-004 active" |

[AOCS] EMG-005 — Loss of Attitude

Entry point: Any of `aocs.rate_roll` (0x0204), `rate_pitch` (0x0205) or `rate_yaw` (0x0206) above 2.0 °/s — the spacecraft is tumbling. **Goal:** Damp the rates with the magnetorquers, protect the power posture while the arrays are off-sun, and re-acquire the sun. **Detumble is slow, and that is correct.** DETUMBLE (mode 2) is a genuine magnetometer-driven B-dot law, `m = −k·dB/dt`, with the dipole saturating at 1.6 A·m² per axis. From a 14 °/s tumble it takes on the order of 3.5e3 s to converge, passing below the 0.5 °/s exit guard at around 1166 s. Expect roughly a halving of rate per orbit; plan for two orbits, not two minutes. Any claim of a fast detumble is wrong. **Possible negative outcomes / deviations:** - Power posture worsens during recovery (arrays off-sun) → CTG-001 / EMG-002; below 10 % SoC command `OBC_SET_MODE(mode=2)` - Rates increasing rather than decreasing → suspect magnetic-model error or torquer polarity; command AOCS OFF, wait one orbit, re-attempt - Magnetometer invalid → B-dot is impossible; no detumble path until the magnetometer returns - Thermal exceedances develop during tumble → CTG-004 / EMG-006 | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Payload / Thermal: shed immediately to extend battery endurance | `PAYLOAD_SET_MODE(mode=0)` (26); `HEATER_OBC(on=0)` (41) | □ | `eps.power_cons` (0x0106) < 60 W | | 2 | AOCS: command DETUMBLE (magnetorquers only) | `AOCS_SET_MODE(mode=2)` (S8 func_id 0) | □ | `aocs.mode` (0x020F) = 2; wheels are not required | | 3 | AOCS: monitor rate reduction across orbit 1 | `HK_REQUEST(sid=2)` (S3.27) | □ | All axes < 1.0 °/s by end of first orbit | | 4 | AOCS: monitor across orbit 2 | (read SID 2) | □ | All axes < 0.5 °/s and SoC > 15 % | | 5 | AOCS: confirm detumble complete | (read SID 2) | □ | All rates < 0.1 °/s sustained for 10 min | | 6 | AOCS: command sun acquisition | `AOCS_SET_MODE(mode=3)` (S8 func_id 0) | □ | COARSE_SUN; `aocs.att_error` (0x0217) < 5° within 15 min | | 7 | EPS: confirm power-positive, then hold two orbits | (read SID 1) | □ | Both arrays > 0.5 A; SoC trending up | | 8 | FD: brief Mission Manager; coordinate multi-orbit recovery if needed | (briefing) | □ | "EMG-005 active"; hand to EMG-001 for staged recovery |

[SE] EMG-006 — Thermal Runaway

Entry point: A heater circuit stuck ON with its zone temperature rising uncontrollably, despite CTG-004 actions. **Goal:** Command the heater off, apply a manual override if that fails, and if the heater still does not respond, shed its power line at the EPS — then confirm the rise has been arrested. *(This is one of the eight machine-runnable cards. Its executable arm is the **battery** circuit — circuit 0, EPS line 5 — because a lithium cell above 50 °C is a chemical hazard rather than an inconvenience. The OBC (circuit 1, line 6) and thruster (circuit 2) variants are the same branch structure with substituted arguments. The branch matters: run flat, the card would shed the heater power line even when the heater had already obeyed the OFF command.)* **Possible negative outcomes / deviations:** - Battery zone runaway → loss of capacity / cell damage; may push toward EMG-002 - Payload zone runaway → permanent payload damage if not arrested; safe the payload immediately - Runaway driven by attitude (panel sun-stare) → coordinate with AOCS for a reorientation - Multiple zones runaway → EMG-001 candidate | Step | Action | TC / Code | Confirm | Notes | |:----:|:-------|:----------|:--------|:------| | 1 | Thermal: identify the affected zone and confirm the hot trend | `HK_REQUEST(sid=3)` (S3.27) | □ | `tcs.temp_battery` 0x0407 / `temp_obc` 0x0406 / `temp_thruster` 0x0409 | | 2 | Thermal: command the offending heater OFF | `HEATER_BATTERY(on=0)` (40) / `HEATER_OBC(on=0)` (41) / `HEATER_THRUSTER(on=0)` (42) | □ | Status 0x040A / 0x040B / 0x041B → 0 | | 3 | Thermal: if the status did not clear, force the thermostat band cold | `HEATER_SET_SETPOINT(circuit=..., on_temp=..., off_temp=...)` (S8 func_id 44) | □ | Circuits 0 battery, 1 OBC, 2 thruster | | 4 | EPS: if the heater still will not respond, shed its power line | `EPS_POWER_OFF(line_index=5)` (battery) or `6` (OBC) (S8 func_id 20) | □ | Only on the NO-GO arm; `eps.pl_htr_bat` 0x0115 / `pl_htr_obc` 0x0116 → 0 | | 5 | All positions: shed every other load contributing heat to the zone | (per zone map) | □ | Including the payload if applicable | | 6 | Thermal: monitor the trend; if still rising, escalate to EMG-001 | (read SID 3) | □ | Decision logged | | 7 | Thermal: on recovery, restore the line and hand back to the thermostat | `EPS_POWER_ON(line_index=...)` (19); `HEATER_AUTO_MODE(circuit=...)` (45) | □ | Do not leave the battery without freeze protection | | 8 | FD: brief Mission Manager on potential hardware damage | (briefing) | □ | "EMG-006 active" |

14.6 Procedure full index

See the simulator's `configs/eosat1/procedures/procedure_index.yaml` for the authoritative list of all **70** procedure IDs (69 active; CTG-018 is retired in favour of CTG-021) with file paths: LEOP-001 to LEOP-007, COM-001 to COM-013, NOM-001 to **NOM-014**, CTG-001 to **CTG-030**, and EMG-001 to EMG-006. The following procedures are referenced in pass plans but are not expanded into one-per-page cards in this catalogue because their steps are either routine cleanup (closeout logs, ticket triage, trend export) or are placeholders awaiting simulator content:

- LEOP-only minor procedures: none outstanding (LEOP-001 through LEOP-007 all carded above).
- Commissioning: all thirteen COM cards exist in the tree and all thirteen are now carded above. **COM-004 is AOCS Actuator Checkout, not a magnetometer placeholder** — magnetometer calibration is part of COM-003 AOCS Sensor Calibration — and it is the only card in the whole catalogue that sends `AOCS_ENABLE_WHEEL`. Earlier printings dismissed COM-004 as a placeholder and omitted the card; a spacecraft flown that way reaches nominal operations with all four wheels still disabled.
- Nominal: in the shipped index NOM-006 is Software Upload and **NOM-008 is Clock Synchronisation** (`nominal/clock_sync.md`) — the two are commonly transposed, and any reference to "NOM-006 (clock_sync.md)" is wrong. **NOM-013 Orbit Maintenance Manoeuvre Execution** and **NOM-014 End-of-Life Disposal** close the series. The book's two additional nominal cards, **Pass Plan Upload** and **Long-Term Trend Review**, have no shipped simulator counterpart and are no longer left free-floating: the pass plan upload sits in the nominal flow immediately ahead of the imaging block (its product is what NOM-002 executes), and the long-term trend review is a shift-boundary activity, run at handover and briefed out under NOM-012.
- Contingency low-frequency cards CTG-021 through CTG-027 are fully carded above in §14.4; the simulator procedure files for each are at `contingency/obc_bootloader_recovery.md`, `bus_failure_isolation.md`, `progressive_load_shed.md`, `solar_panel_loss_response.md`, `aocs_sensor_loss_recovery.md`, `aocs_actuator_stuck_recovery.md`, and `gps_time_sync_recovery.md`. The cards in this textbook are condensed; the simulator files carry the full step lists.

Three simulator cards have no card of their own in this book: CTG-005 EPS Safe Mode (its content is reached through CTG-001), CTG-011 OBC Redundancy Switchover (through CTG-017 / CTG-022) and CTG-018 Boot Loader Recovery, which is retired in favour of CTG-021 OBC Bootloader Recovery. The book's two former consolidations, CTG-B01 (IMU / Gyro Anomaly) and CTG-B02 (TC Reject Rate Anomaly), are now the simulator cards **CTG-029** Gyro Failure Recovery and **CTG-030** Telecommand Rejection and Verification Failure.

Eight cards are machine-runnable from the MCS Procedures tab: NOM-002, NOM-003, NOM-004, NOM-008, NOM-009, CTG-001, CTG-008 and EMG-006 (Thermal Runaway). NOM-002/003/004 and NOM-009 are linked to planner activity types (`imaging_pass`, `data_dump`, `orbit_manoeuvre`, `housekeeping_collection`); NOM-008, CTG-001, CTG-008 and the thermal-runaway card carry `runnable: {kind: markdown}` and are executed from their own **TC:** annotations. The executor no longer reports success for a procedure that did nothing, and the MCS runner executes `label` and `branch` steps and fails closed on any step type it does not understand. CTG-020 is the one **declared advisory** card: it carries `advisory: true` with an explicit reason and no `runnable:` link, because every action in it is taken on the ground.

Operators reaching for any placeholder ID must consult the FD and the simulator procedure file directly before any commanding.

Three things in the tree that are newer than every printing of this book. (1) The mission phase is now a **telecommand**, `SYS_SET_MISSION_PHASE` (S8 func_id 120): LEOP-003 step 7 declares COMMISSIONING (5) and COM-013 step 4 declares NOMINAL (6). Any statement elsewhere that the phase is console-only, or that `SET_PHASE` is an alias of `OBC_BOOT_APP`, is superseded — that alias was removed. (2) The OBC carries an **independent clock** starting at J2000, drifting ~12 ppm, GPS-disciplinable, with `obdh.time_ref` (0x0A13) / `obdh.obc_time_fine` (0x0A12) / `obdh.time_since_sync_s` (0x0A14) as its telemetry, and **a bootloader reset returns it to J2000** — so every reboot card in this chapter ends in a re-sync. (3) The **reaction wheels must be explicitly enabled** with `AOCS_ENABLE_WHEEL` (S8 func_id 3) in COM-004; the bootloader leaves them off and nothing else turns them on.

Historical note — numbering in earlier printings. Before October 2026 this book numbered twelve contingency and emergency cards differently from the simulator. Old → current: EMG-002 → EMG-005, EMG-003 → EMG-004, EMG-004 → EMG-003, EMG-005 → EMG-006, EMG-006 → EMG-002; CTG-005 → CTG-013, CTG-006 → CTG-028, CTG-011 → CTG-016, CTG-013 → CTG-009, CTG-016 → CTG-006; the book-only cards CTG-009 (IMU / Gyro) and CTG-018 (TC Reject Rate) became CTG-B01 and CTG-B02, and in October 2026 the simulator cards CTG-029 and CTG-030. Every other ID was already the simulator's.

14.7 References

- ECSS-E-ST-70-32C — Test and operations procedures — ecss.nl
- FAA AC 120-71 — Standard Operating Procedures — www.faa.gov

Chapter 15 — Telemetry, FDIR & Monitoring Reference

This chapter is the watch-list. Print the per-position telemetry tables and bring them to the console; the FDIR escalation ladder goes on the wall.

FDIR state machine: NORMAL → MONITOR → DETECT → ISOLATE → RECOVER → EVALUATE, with example rule names and per-transition time windows.

FDIR state machine: NORMAL → MONITOR → DETECT → ISOLATE → RECOVER → EVALUATE, with example rule names and per-transition time windows.

15.1 Telemetry — compact reference by subsystem

(Full parameters file: `telemetry/parameters.yaml`. Limits file: `mcs/limits.yaml`, which carries 51 yellow/red bands, each with its own `check_interval_s`. Ground limits and on-board S12 monitors are now genuinely live on both sides, so a parameter can be yellow on the console without an on-board monitor having tripped, and vice versa — that is a diagnosis, not a bug.)

The per-position tables already appear in Chapter 6 (each position's *Telemetry to watch* block). Cross-reference rather than duplicate.

Housekeeping structures. Periodic telemetry arrives in housekeeping structures defined in `telemetry/hk_structures.yaml`. The SIDs an operator will meet are:

Table 15.1 — Housekeeping structure IDs, contents and cadence.

SID	Contents	Cadence
1	EPS	1 s
2	AOCS	4 s
3	TCS	60 s
4	OBDH / Platform	8 s
5	Payload	8 s
6	TT&C	8 s
7	AOCS determination & control	4 s
8	EPS power management (extended)	8 s
9	Propulsion	8 s
11	Beacon (bootloader lifeline, UHF)	30 s

`default_enabled_sids` is `[1, 11]`, and the engine re-applies exactly that on application boot. **Everything else has to be switched on with `HK_ENABLE` — Service 3, subtype 5 — naming the SID.** A one-shot report is `HK_REQUEST`, **Service 3, subtype 27**. If a display is blank at the start of a session, the first question is not "have we lost the link" but "has anyone enabled that SID".

Staleness is cadence-derived, not a flat threshold. The MCS joins `hk_structures.yaml` with `mcs/staleness.yaml` and derives, per parameter: fresh while age ≤ max(10 s, interval × 0.9), degraded while age ≤ max(20 s, interval × 1.5), stale beyond that. A TCS parameter on the 60 s SID 3 is therefore allowed to

be 54 s old and still read as current; an EPS parameter on the 1 s SID 1 is not. Parameters that belong to no structure (on-demand / S20 fetch) are aperiodic and use 300 s / 900 s. There is a fourth state, NOT ENABLED, shown when every structure carrying the parameter is disabled — the correct response to which is `HK_ENABLE`, not a link investigation.

Telemetry added since the last edition, worth putting on the watch-list:

Table 15.2 — Recently added parameters.

Parameter	ID	Why it matters
`aocs.gyro_units_on`	0x0294	Number of gyro units powered (S2 devices 0x0206–0x0208). 0 = the estimator has no rate reference and coasts on CSS + magnetometer
`aocs.gyro_saturated`	0x0295	At least one gyro channel is at its measurement-range limit — rate knowledge lost on that axis
`aocs.gyro_bias_injected`	0x0296	Magnitude of an injected gyro-bias fault; 0 in a healthy spacecraft
`aocs.commanded_mode`	0x0297	The mode the ground asked for, shown alongside the actual mode 0x020F — a demotion is now visible rather than inferred
`tcs.fpa_cooler_floor_c`	0x0424	Coldest FPA temperature the cooler can hold at the present sink and load (on-demand)
`tcs.fpa_cooler_margin_k`	0x0425	Commanded setpoint minus achievable floor; **≤ 0 means the setpoint is out of reach** (on-demand)
`prop.state`	0x0B00	0 SAFED, 1 ARMED, 2 FIRING, 3 INHIBIT — head of the propulsion block, carried on HK SID 9
`prop.tank_pressure`	0x0B01	Blowdown 22.00 → 9.33 bar; the delta monitor on this parameter sees a leak before the leak-rate estimate does
`prop.dv_remaining`	0x0B0C	Remaining delta-V on live propellant, out of the 66.9 m/s total
`ttc.link_status`	0x0501	**LOCKED is 2, not 1.** A card or display reading "= 1" for lock is wrong

15.2 S12 monitoring rules (46 in the nominal config)

(Full file: `monitoring/s12_definitions.yaml`. Summary by subsystem below. S12 severity in this file is a *word* — `WARNING` or `ALARM` — which the spacecraft maps onto the report octet as 1 INFO / 2 WARNING / 3 ALARM / 4 HIGH before packing it. Do not read those numbers as the FDIR ladder of §15.4.)

Table 15.3 — Count of S12 monitoring rules per subsystem with representative examples.

Subsystem	Rules	Examples
EPS	9	`eps.bat_soc` 0x0101 < 20 % (WARNING) and < 10 % (ALARM); `eps.bus_voltage` 0x0105 < 28.4 V (WARNING) and < 27.7 V (ALARM); battery over/under-temp; battery overcurrent; solar array A/B low current
AOCS	11	

Subsystem	Rules	Examples
		four wheel-overspeed rules on 0x0207–0x020A at ±5000 rpm; `aocs.h_env_frac` 0x02BA momentum envelope > 70 %; `aocs.att_error` 0x0217 > 1 °; star tracker 1 / 2 blind (0x0240 / 0x0243, ALARM); three gyro-bias rules on 0x0270–0x0272
TCS	5	`tcs.temp_fpa` 0x0408 out of −22…+14 °C (WARNING) and −25.5…+17.5 °C (ALARM); OBC overtemp; battery overtemp; `tcs.htr_total_power` 0x0411 > 12.5 W
TT&C	5	`ttc.link_margin` 0x0503 < 3 dB (ALARM); RSSI low; Eb/N0 low; PA overtemp; `ttc.ber` 0x050C high (log10 scale, so > −5 = worse than 1e-5)
OBDH / Platform	4	`obdh.cpu_load` 0x0302 > 85 %; memory usage high; `obdh.reboot_count` 0x030A > 3 (ALARM); memory errors
Payload	4	FPA temperature; `payload.store_used` 0x0604 > 95 %; `payload.checksum_errors` 0x0609 non-zero; SNR low
PROP	8	`prop.tank_pressure` 0x0B01 band, out-of-band and **rate-of-change** (the delta check is what catches a leak); `prop.catbed_temp` 0x0B05 band and overtemp; propellant low; low thrust; `prop.dv_remaining` 0x0B0C envelope low

Note the shape of the propulsion set: it is the only subsystem whose monitors include a *delta* check (`check_type: 1`). Everything else on this spacecraft is monitored as an absolute band.

15.3 S19 event-action rules (32 defined, 31 enabled)

(Full file: `monitoring/s19_rules.yaml`. Each rule is *triggered by* an S5 event or by the S12 monitor on a parameter, and *executes* one autonomous action with no operator in the loop.)

The `trigger.kind` discriminator is the thing to read first. Event IDs and parameter IDs both live in 0x0000–0x0FFF, so a rule is only unambiguous once you know whether `kind` is `event` or `param`. A rule with `kind: param` fires on the S12 limit violation for that parameter — which means disabling an S12 monitor silently disables the autonomy hanging off it.

Highlights:

Table 15.4 — Representative S19 event-action rules with trigger and autonomous action.

Rule	Trigger	Autonomous action
`eps_bus_undervoltage_payload_off`	`kind: event`, S5 event 0x0102 BATTERY_SOC_CRITICAL	payload off (load shed)
`aocs_momentum_saturation_desaturate`	`kind: param`, S12 monitor on `aocs.h_env_frac` 0x02BA (yellow 70 %, red 90 %)	start wheel desaturation
`aocs_star_tracker_1_blind_switch`	`kind: param`, S12 monitor on `aocs.st1_status` 0x0240	select the other star tracker
`tcs_fpa_overtemp_payload_off`	`kind: param`, S12 monitor on `tcs.temp_fpa` 0x0408	payload off

Rule	Trigger	Autonomous action
`ttc_link_margin_critical_boost_tx`	`kind: param`, S12 monitor on `ttc.link_margin` 0x0503 (ALARM below 3 dB)	raise TX power a step
`obc_reboot_event`	`kind: event`, S5 event 0x0301 OBC_REBOOT	post-reboot recovery
`payload_checksum_errors_verify`	`kind: param`, S12 monitor on `payload.checksum_errors` 0x0609	verify the stored product
`prop_abort_on_low_tank_pressure`	`kind: param`, S12 monitor on `prop.tank_pressure` 0x0B01	abort the burn
`prop_disarm_on_safe_mode`	`kind: event`, safe-mode entry	disarm propulsion

Rule `aocs_attitude_error_safe_mode` (ea_id 2004) is present but `enabled: false` — the one disabled rule of the thirty-two. Do not plan a shift on the assumption that a large attitude error will put the spacecraft into safe mode by itself.

15.4 FDIR escalation ladder

Table 15.5 — FDIR escalation ladder with default action and required operator response per severity level.

Level	Meaning	Default action	Operator response
0	Informational	Log only	Note on loop
1	Caution	Log + alert	Discuss in debrief
2	Warning	Log + alert + autonomous mitigation if defined	Run recovery procedure
3	Severe	Autonomous + ground alert + flag for next pass	Run recovery + ticket
4	Critical	Safe-mode candidate (autonomous if rule says so)	Confirm safe-mode entry, brief Mission Manager

Severity is not autonomy level. Two separate numbers live in the FDIR configuration and operators conflate them constantly:

- **FDIR severity, 0–4**, is the table above: *how bad is this*.
- **Recovery escalation level, 0–3**, in `fdir/fault_propagation.yaml`, is *how hard has the spacecraft tried so far*. `initial_level` is 1, `max_attempts` is 3, and each domain has a level 1 / 2 / 3 recovery: for EPS, load-shed stage 1 → 2 → 3; for AOCS, safe-mode recovery → AOCS reset → emergency power down; for TT&C, reduce TX power → antenna switch → emergency shutdown. The escalation delays are 30 s after level 1 and 60 s after level 2; level 3 is final and escalates no further.

A severity-4 event may sit at escalation level 1, and a severity-2 event may reach level 3. Always say which number you mean on the loop.

Load-shedding stages have their own thresholds, also in `fdir/fault_propagation.yaml`: stage 1 "Power Conservation" at 30 % SoC, stage 2 "Payload Offline" at 20 %, stage 3 "Survival Mode" at 10 %. Loads shed by `EPS_SET_MODE` are restored by the operator, not automatically.

15.5 Command Reference — full list by subsystem

This is the complete telecommand set an operator can issue on the reference spacecraft, grouped by subsystem and listed by command mnemonic and purpose. It complements the telemetry, monitoring, and FDIR references above so that Chapter 15 is a single operator lookup. The underlying PUS S8 *function-ID* numbering (and the parameter fields each command takes) is given in §7.7; the names below are the same mnemonics used on the commanding panel and in the procedure catalogue (Chapter 14).

AOCS — attitude & orbit control

Table 15.6 — AOCS — attitude & orbit control commands.

Command	Purpose
`AOCS_SET_MODE`	Set AOCS mode (0=OFF, 1=SAFE_BOOT, 2=DETUMBLE, 3=COARSE_SUN, 4=NOMINAL, 5=FINE_POINT, 6=SLEW, 7=DESAT, 8=ECLIPSE, 9=INERTIAL). Mode 5 is refused with TM(1,8) unless the `aocs.fine_ready` precondition bitmask 0x02C5 reads 0
`AOCS_DESATURATE`	Start wheel desaturation. A dump runs 20–50 min, inhibits imaging while it runs, has a timeout, and returns to the pre-desat mode
`AOCS_DISABLE_WHEEL`	Disable specific reaction wheel (`wheel_idx` 0–3)
`AOCS_ENABLE_WHEEL`	Enable specific reaction wheel
`ST1_POWER`	Star tracker 1 power on/off
`ST2_POWER`	Star tracker 2 power on/off
`ST_SELECT`	Select primary star tracker (two heads are flown, ST1 and ST2)
`RW_SET_SPEED_BIAS`	Set reaction wheel speed bias — the bias is **added to** the wheel's current speed. Wheel speeds 0x0207–0x020A are read-only; this is the commanding path
`MTQ_ENABLE`	Enable / disable magnetorquers (readback `aocs.mtq_enabled` 0x025E)
`AOCS_SLEW_TO`	Execute slew to target quaternion
`AOCS_CHECK_MOMENTUM`	Query wheel momentum status
`AOCS_BEGIN_ACQUISITION`	Start automated attitude acquisition
`AOCS_GYRO_CALIBRATION`	Reset the on-board gyro bias estimate. Bias calibration is done on board — there is no ground-computed bias upload
`AOCS_RW_RAMP_DOWN`	Gracefully ramp wheel(s) down (`wheel_idx` 255 = all wheels)
`AOCS_SET_DEADBAND`	Set the FINE-point control deadband

Command	Purpose
AOCS_SET_POINTING_FRAME	Select the reference frame for NOMINAL / FINE pointing
AOCS_SET_TARGET_QUAT	Set the inertial-hold target quaternion (used by INERTIAL, mode 9)
AOCS_SET_TARGET_GEO	Set a ground target for off-nadir target tracking
AOCS_FINE_READY_CHECK	Report the FINE-entry precondition bitmask 0x02C5; 0 = ready. Bit 1 means "fewer than **three** wheels" — FINE runs on three
AOCS_SET_MOM_MGMT	Configure autonomous momentum management (auto-desat thresholds)
AOCS_SET_DESAT_INHIBIT	Inhibit / allow desaturation — a dump drops fine pointing, so it is inhibited over an imaging window
AOCS_EST_RESET	Reinitialise the attitude estimator (e.g. after a time jump). Estimator source is readable at `aocs.est_source` 0x02C1
AOCS_GPS_SET_START_MODE	Select GPS receiver start mode (cold / warm / hot)

EPS — electrical power

Table 15.7 — EPS — electrical power commands.

Command	Purpose
EPS_PAYLOAD_MODE	Set payload power mode (0=off, 1=standby, 2=imaging)
EPS_FPA_COOLER	Enable / disable FPA cooler
EPS_TRANSPONDER_TX	Enable / disable transponder TX
EPS_POWER_ON	Switch power line ON (`line_index` 0–8). Nine lines, indices 0–8: obc, ttc_rx, ttc_tx, payload, fpa_cooler, htr_bat, htr_obc, aocs_wheels, prop. Every switchable line defaults OFF
EPS_POWER_OFF	Switch power line OFF (`line_index` 0–8)
EPS_RESET_OC_FLAG	Reset overcurrent trip flag, re-enable line (`line_index` 0–8)
EPS_SET_MODE	Set EPS operating mode. SAFE sheds the two least-essential loads, EMERGENCY sheds all of them; loads are restored by the operator, not automatically
EPS_SWITCH_LOAD	Switch individual power load on/off
EPS_SET_CHARGE_RATE	Override battery charge rate
EPS_SOLAR_ARRAY_DRIVE	Drive solar array angle
EPS_EMERGENCY_LOAD_SHED	Immediate load shed (stage 0–3)

Payload — imager & data

Table 15.8 — Payload — imager & data commands.

Command	Purpose
PAYLOAD_SET_MODE	

Command	Purpose
	0=off, 1=standby, 2=imaging, **3=calibration**. Modes 2 and 3 request the FPA cooler; modes 0 and 1 release it (STANDBY runs without the cooler)
`PAYLOAD_SET_SCENE`	*Deprecated* — pre-set the next scene ID. Superseded by `PAYLOAD_CAPTURE` 's own argument
`PAYLOAD_CAPTURE`	Trigger one image acquisition. The *where* is set by the *when*: schedule the capture over the target. Refused with the shutter closed
`PAYLOAD_DOWNLOAD_IMAGE`	Report one stored image product
`PAYLOAD_DELETE_IMAGE`	Delete specific image from store
`PAYLOAD_MARK_BAD_SEGMENT`	Mark memory segment as unusable
`PAYLOAD_GET_IMAGE_CATALOG`	Request stored image catalog
`PAYLOAD_SET_BAND_CONFIG`	Set spectral band enable mask
`PAYLOAD_SET_INTEGRATION_TIME`	Per-band detector integration time
`PAYLOAD_SET_GAIN`	Set detector gain
`PAYLOAD_COOLER_SETPOINT`	FPA regulation setpoint. The argument is a **uint16 in centi-kelvin**: raw = round((T °C + 273.15) × 100), so −15.00 °C is 25815 = 0x64D7. Envelope −20…0 °C; outside it the TC is **rejected** with S1.8 error 0x0003, never clamped
`PAYLOAD_START_CALIBRATION`	Begin a radiometric calibration sequence (dark frame / flat field / both). Requires payload mode 1 STANDBY or 3 CALIBRATION — never 0 OFF and never 2 IMAGING. Dark cal no longer needs IMAGING mode
`PAYLOAD_STOP_CALIBRATION`	Abort calibration
`PAYLOAD_SET_COMPRESSION`	Select compression algorithm and ratio override
`PAYLOAD_SELECT_FILTER`	Rotate the filter wheel to a position (2 s per 90° step)
`PAYLOAD_CYCLE_SHUTTER`	Run a shutter mechanism exercise — a health test, not a way to reach a known position
`PAYLOAD_SET_SHUTTER`	Drive the shutter to an **absolute** position. This is what a dark frame and a capture pre-check need
`PAYLOAD_CAL_LAMP`	Calibration lamp on/off
`PAYLOAD_INITIATE_TRANSFER`	Start the on-board image transfer engine for a stored scene

A refused calibration tells you *why*: five dedicated failure codes, 0x0024 (refused because IMAGING), 0x0025 (FPA not ready), 0x0026 (shutter in the wrong position), 0x0027 (lamp in the wrong state) and 0x0028 (capture with the shutter closed). 0x0021–0x0023 remain the capture-pointing codes. Read the code before re-sending the command.

TCS — thermal control

Table 15.9 — TCS — thermal control commands.

Command	Purpose
`HEATER_BATTERY`	Battery heater on/off
`HEATER_OBC`	OBC heater on/off
`HEATER_THRUSTER`	Thruster heater on/off
`FPA_COOLER`	FPA cooler on/off (TCS-side enable). The cooler is a miniature Stirling-class cryocooler: its authority degrades as the temperature lift grows, so check `tcs.fpa_cooler_margin_k` 0x0425 before blaming the machine
`HEATER_SET_SETPOINT`	Set thermostat on/off temps for a circuit
`HEATER_AUTO_MODE`	Return heater to autonomous thermostat
`TCS_SET_HEATER_DUTY_LIMIT`	Limit heater duty cycle
`TCS_DECONTAMINATION_START`	Begin payload decontamination heating
`TCS_DECONTAMINATION_STOP`	Abort decontamination
`TCS_GET_THERMAL_MAP`	Request thermal status of all zones

OBDH — on-board data handling

Table 15.10 — OBDH — on-board data handling commands.

Command	Purpose
`OBC_SET_MODE`	Set OBC mode (0=nominal, 1=safe, 2=emergency)
`OBC_MEMORY_SCRUB`	Trigger manual memory scrub
`OBC_REBOOT`	Force OBC reboot (drops to bootloader) — **critical**
`OBC_SWITCH_UNIT`	Switch to redundant OBC — **critical**
`OBC_SELECT_BUS`	Select active CAN bus — **caution**
`OBC_BOOT_APP`	Boot application software from bootloader — **caution**
`OBC_BOOT_INHIBIT`	Inhibit auto-boot (stay in bootloader)
`OBC_CLEAR_REBOOT_CNT`	Reset reboot counter
`OBC_SET_WATCHDOG_PERIOD`	Configure watchdog timeout
`OBC_WATCHDOG_ENABLE`	Enable watchdog monitoring
`OBC_WATCHDOG_DISABLE`	Disable watchdog monitoring
`OBC_DIAGNOSTIC`	Request OBC health diagnostic
`OBC_ERROR_LOG`	Request recent error-log entries
`OBC_GPS_TIME_SYNC`	Sync onboard clock to GPS — **caution** (may invalidate AOCS attitude solution)

TT&C — telemetry, tracking & command

Table 15.11 — TT&C — telemetry, tracking & command commands.

Command	Purpose
`TTC_SWITCH_PRIMARY`	Switch to primary transponder
`TTC_SWITCH_REDUNDANT`	Switch to redundant transponder
`TTC_SET_DATA_RATE`	Select a rung of the TM rate ladder. The legacy encoding is preserved: `rate=0` still means "the low rate" and `rate=1` still means "the high rate" — but they now select **LOW 16 kbps** and **HIGH 128 kbps**. The third tier, BEACON 1 kbps, is entered with `TTC_SET_BEACON_MODE`, not here. HIGH requires a deployed antenna, DOWNLINK mode and elevation ≥ 10°, and is **rejected** with S1.8 otherwise, never silently clamped
`TTC_SET_DOWNLINK_MODE`	Enter or leave DOWNLINK / payload-dump mode — the gate on the HIGH tier
`TTC_PA_ON`	Enable power amplifier
`TTC_PA_OFF`	Disable power amplifier. **This stops S-band TM but the UHF beacon survives** — see below
`TTC_SET_TX_POWER`	0=LOW 1.0 W RF, 1=NOMINAL 2.0 W, 2=HIGH 5.0 W (the ceiling). Unknown codes are rejected with S1.8 error 0x0003, never clamped
`TTC_DEPLOY_ANTENNA`	Deploy spacecraft antenna (burn-wire)
`TTC_SET_BEACON_MODE`	Enable / disable low-rate beacon
`TTC_CMD_CHANNEL_START`	Start dedicated command channel (15-min)
`TTC_SET_UL_FREQ`	Set uplink frequency
`TTC_SET_DL_FREQ`	Set downlink frequency
`TTC_SET_MODULATION`	0=BPSK, 1=QPSK, 2=8PSK, 3=OQPSK
`TTC_SET_RX_GAIN`	Set receiver AGC target
`TTC_RANGING_START`	Begin ranging sequence
`TTC_RANGING_STOP`	Stop ranging sequence
`TTC_SET_COHERENT_MODE`	Coherent transponder on/off
`TTC_BEACON_TX_ON`	Switch the UHF beacon **transmitter** on
`TTC_BEACON_TX_OFF`	Switch the UHF beacon transmitter off — caution, this removes the lifeline

EOSAT-1 flies **two transmitters**: an S-band prime chain and a separate UHF whip beacon at 437.5 MHz, 1 W into a dipole. The beacon emits HK SID 11 and only SID 11, whenever the S-band chain is not radiating. That is the teaching point behind `TTC_PA_OFF`: pull the PA, or the `ttc_tx` power line, and S-band telemetry stops while the beacon keeps arriving every 30 s. `ttc.pa_on` 0x0516 rides in SID 11 precisely so that "no telemetry" becomes a one-command diagnosis rather than a mystery.

PROP — propulsion

EOSAT-1 carries a 1U green monopropellant module with 66.9 m/s of total delta-V, flown for debris avoidance, limited orbit maintenance and the end-of-life disposal burn (NOM-014). Arming is deliberately two-step, and a burn re-fits the element set, so passes and eclipses genuinely move afterwards.

Table 15.12 — PROP — propulsion commands.

Command	Purpose
`PROP_SET_BURN`	Load a manoeuvre: delta-V magnitude, direction and cut-off mode
`PROP_ARM`	SAFED → ARMED. Requires a `PROP_SET_BURN` within the preceding 300 s
`PROP_DISARM`	ARMED → SAFED; closes the flow-control valve, leaves latch valves as they are
`PROP_EXECUTE`	ARMED → FIRING. Rejected with S1.8 when not ARMED
`PROP_ABORT`	FIRING → SAFED immediately; closes the FCV and latches an abort reason
`PROP_ISO_OPEN`	Open latch-valve branch(es)
`PROP_ISO_CLOSE`	Close latch-valve branch(es)
`PROP_SET_LIMITS`	Reconfigure the on-board burn-abort thresholds
`PROP_INHIBIT_SET`	Ground-latched propulsion inhibit
`PROP_INHIBIT_CLR`	Clear the ground inhibit and any FDIR-latched inhibit
`PROP_GET_STATUS`	On-demand propulsion status report

Mechanisms & deployables

Table 15.13 — Mechanisms & deployables commands.

Command	Purpose
`OBC_GPS_TIME_SYNC`	GPS time sync (see OBDH table)
`EPS_DEPLOY_WING`	Deploy solar wing (+Y, −Y, both) via burn-wire
`EPS_WING_STATUS`	Query wing deployment status. Readback is `eps.wing_status` 0x0144 (bit 0 = +Y deployed, bit 1 = −Y, bits 4/5 = deployment in progress); per-wing currents are 0x0146 (+Y) and 0x0147 (−Y)
`TTC_DEPLOY_ANTENNA`	Antenna deploy (see TT&C table)

15.6 References

- ECSS-E-ST-70-41C — Telemetry and telecommand packet utilization (PUS) — ecss.nl
- CCSDS 132.0-B-3 — TM Space Data Link Protocol — public.ccsds.org
- CCSDS 660.0-G-3 — PUS green book — public.ccsds.org

Chapter 16 — Per-Position Deep Briefs (case studies)

This chapter expands each of the nine positions with a multi-shift case study — a realistic narrative of how the position behaves across a sequence of shifts. Read as supplementary material before the simulation days.

16.1 FD case study — managing the LEOP Day Three rebound

The team's first LEOP pass goes badly. It takes them four minutes to get the OBC into the application image; the clock offset measured in LEOP-006 comes out at 3.2 seconds, well over the card's 1.0 second acceptance threshold; and nobody manages to close out the time-sync verification before LOS. SoC at LOS is 78%. The team is fragile. On the five-station network the next pass is about 35 minutes away — median gap is 33 minutes, worst case 79.

A weak FD lets the team stew, posts a tense pre-pass briefing, and the second pass goes worse. A strong FD does this:

1. **Calls a 10-minute hot-wash** immediately after LOS. Asks each position one specific positive and one specific improvement.
2. **Names what was actually fine.** `OBC_BOOT_APP` itself completes in about ten seconds — the four minutes were the team waiting on the 30-second beacon cadence for confirmation, which is the bootloader working as designed, not a slow loader. The 3.2 s offset is exactly the thing LEOP-006 exists to remove: `SET_TIME` was uplinked, and the number that matters is the post-sync residual, not the pre-sync one. And the link never came close to being a problem — margin on the LOW tier runs in the forties of dB at every one of the five stations.
3. **Names one specific thing to change next pass.** "AOCS, we never ran the gyro bias calibration. Next pass, send `AOCS_GYRO_CALIBRATION` as your first action and read 0x0270–0x0272 back."
4. **Brings the team back to physical task.** "Take 20 minutes, eat, hydrate, walk outside if you can. We reconvene at AOS−10 min for pre-pass briefing."

The second pass succeeds. The team learns that the LEOP isn't about perfection — it's about adapting. That lesson sticks for the rest of their careers.

16.2 SPACON case study — three procedures in parallel

It's mid-Day 4. The team is running COM-009 (payload power-on) when EPS calls an overcurrent. CTG-012 starts. While SPACON is mid-CTG-012, the FDIR panel surfaces a *second* rule — the star tracker is degrading.

A weak SPACON tries to do both — gets confused, drops a step in CTG-012, misses the star tracker degradation entirely.

A strong SPACON does this:

1. **Calls priority.** "FD, SPACON — two anomalies. CTG-012 underway. Star tracker degrading, FDIR severity 2, not yet 3, and still at escalation level 1. Recommend continuing CTG-012; AOCS lead on star tracker monitoring until severity 3 fires."
2. **Hands the second anomaly off** explicitly. "AOCS, watch `aocs.st1_status` 0x0240. If it goes to 3 — blind — that is an ALARM monitor and the S19 rule will select ST2 without asking us. Call it loud when it happens."
3. **Completes the first procedure.** Files the ticket.

4. **Returns to the second.** "FD, SPACON — CTG-012 ticket filed. AOCS, status?"

The discipline is: **one procedure at a time per operator; explicit handoffs; FD knows where everyone is.**

16.3 AOCS case study — the pointing-budget conversation

It's late on Day 4. Total pointing error (`aocs.total_pointing_err_3sig`, 0x02AA) is running at 0.042° against a 0.021° baseline. Payload wants to image now. Mission Manager wants the customer bulletin.

A weak AOCS engineer says "pointing is degraded but probably fine for imaging" and lets the FD decide based on no real information.

A strong AOCS engineer does this:

1. **Decomposes the budget.** "FD, AOCS — 0x02AA is 0.042 deg total, and that number is the RSS of exactly two terms: knowledge error 0x02A7, currently 65 arcsec — 0.018 deg — and control error 0x02A8, currently 0.038 deg. Knowledge is nominal. The control term is the whole story, and `aocs.h_env_frac` 0x02BA has climbed to 74 % of the momentum envelope across the last three orbits. We're past the 70 % yellow."
2. **Predicts.** "At the present rate 0x02BA reaches the 90 % red inside the hour, control error goes to about 0.048 and total to 0.051 deg. At 513 km that is a geolocation error (0x02C7) of roughly 460 m, against a 33.4 m ground sample — more than a dozen pixels. That is not a science-grade product."
3. **Recommends.** "FD, AOCS — recommend we command the desat now and let it run across the pass gap. A dump takes 20 to 50 minutes and inhibits imaging while it runs, so 'desat this pass and image next pass' is the only version of this that fits. If you image now you'll get one usable frame, and it will be the *last* usable frame until we dump."

This kind of operator is rare. They're the ones the FD trusts to make the close call.

16.4 EPS case study — the parasitic load mystery

Day 4 begins with SoC trending 2% below predicted at each eclipse exit. By mid-shift, SoC is at 38% — heading toward the stage-1 load-shed gate at 30%.

A weak EPS engineer ignores it, hopes it's a transient.

A strong EPS engineer does this:

1. **Measures.** Reads every one of the nine line currents (0x0118–0x011F and 0x014B; the bus total is 0x0148), totals them. They sum to 23.2 W. Predicted orbit-average nominal is 22.7 W. Difference: 0.5 W steady-state extra draw — small against the +22.4 W nominal margin, but unexplained, and that is worth chasing.
2. **Hunts the line.** Compares each line current to nominal: most are within 5%, but the `htr_bat` duty cycle is at 18% when predicted 8%. Extra draw: ~0.4 W. Most of the anomaly accounted for.
3. **Hypothesises.** The battery zone has lost some thermal insulation, so the heater works harder. Coordinates with Thermal: "Thermal, EPS — `htr_bat` duty doubled and `tcs.htr_total_power` 0x0411 is up with it. Hypothesis is insulation degradation. Could you verify?"
4. **Reports to FD.** "FD, EPS — parasitic load identified, approximately 0.4 W extra on the `htr_bat` line. Net SoC drift 2% per orbit. Recommend we accept this and continue commissioning; the load isn't dangerous, just inefficient. File ticket for next-shift investigation."

The mission continues. Next shift figures out the root cause (an MLI sheet partly detached during launch vibration). Without this operator's careful work, nobody would have known the right question to ask.

16.5 Thermal case study — the cooler that won't cool

COM-010 starts well. FPA at +11 C, its uncooled equilibrium, cooler enabled, drawing its rated 15 W. At T+2 min the FPA is at +4 C, on profile. Then the ramp flattens. At T+5 it is only at −1 C; the profile predicted −8 C. At T+10, when it should be regulating at −15 C, it reads −3 C.

A weak Thermal engineer waits, hoping the cooler will catch up. By T+20 the FPA is at −4 C and still not ready. They escalate to the FD too late; First Light is at risk.

A strong Thermal engineer does this:

1. **Catches the deviation early.** At T+5, when the profile says −8 C and reality is −1 C, calls it. "FD, Thermal — FPA cool-down running 7 C behind profile at the T+5 checkpoint. Watching."
2. **Diagnoses with the right parameter.** "FD, Thermal — checking. Cooler power draw still 15 W per spec, so the machine is running. Requesting the on-demand pair: `tcs.fpa_cooler_floor_c` 0x0424 reads −6.5 C and `tcs.fpa_cooler_margin_k` 0x0425 reads −8.5 (setpoint −15 minus floor −6.5). Margin is negative. The setpoint is out of reach at the present sink temperature and detector load — this is not a failed cooler, it is a cooler out of authority."
3. **Explains why, and coordinates with AOCS.** "This is a Stirling-class machine, not a Peltier stack: its efficiency is a fraction of Carnot, so authority collapses as the lift grows. The lever is the sink, not the cooler. AOCS, Thermal — could you verify the radiator is in shadow? If we can drop the sink a few kelvin the floor comes down with it."
4. **Predicts and recommends.** "FD, Thermal — at the present floor we will not reach −10 C at all, and the cool-down has flattened out a few kelvin above that floor. Recommend either relaxing the setpoint toward the achievable floor — `PAYLOAD_COOLER_SETPOINT` is a hard envelope of −20 to 0 C and it will be *rejected*, not clamped, outside that — or deferring First Light to an orbit with better radiator geometry. What I will not do is keep waiting for a cooler that is already telling us it cannot get there."

Crisp diagnosis turns a "the cooler is broken" panic into a "we're operating in a corner of the envelope we didn't model well — defer and re-attempt" operational decision.

16.6 TT&C case study — the BER that wasn't

Mid-Day 4, during the HIGH-tier downlink test in COM-006 (TT&C Link Verification) — antenna deployed, DOWNLINK mode entered, elevation well above the 10° policy gate, 128 kbps. TT&C sees BER (`ttc.ber` 0x050C) tick up briefly to 5e-6, then settle back to 1e-7. Total duration of the BER spike: 8 seconds. Neither value trips the on-board monitor, which sits at 1e-5.

A weak TT&C engineer treats the spike as noise and continues. Two passes later the same spike happens, and the team realises they've been missing a pattern.

A strong TT&C engineer does this:

1. **Logs the spike.** Time, duration, peak BER, link margin at the time. Even though it self-cleared.
2. **Hypothesises.** "FD, TT&C — saw a brief BER spike. Link margin never moved; the shipped budget closes HIGH at Troll at 5 degrees with 33.7 dB — we only plan it above 10 as a ground rule — so this cannot be a margin-limited link. That points at multipath or an antenna-pattern effect rather than

geometry. The spacecraft is at elevation 33 degrees and azimuth 117 from Iqaluit right now — that's a corner of the antenna pattern that's notoriously low gain."
3. **Recommends.** "FD, TT&C — recommend we don't downlink critical data when the spacecraft passes through azimuth 115 to 120 from Iqaluit. I'll flag it in the planning panel for next shift."

The team avoids a recurring data-loss issue for the rest of the mission.

16.7 Payload case study — the image that wasn't science

First Light succeeds. Quality assessment passes — mean DN nominal, contrast nominal, geo-registration error 165 m, about five pixels at the imager's 33.4 m ground sample distance and inside budget. The strip is the full 273 km swath. Mission Manager drafts the bulletin.

A weak Payload operator celebrates and moves on. They miss that the image's actual *content* — once a science colleague looks at it — shows no usable ocean-current signal because the cloud cover was 60% (the forecast said 8%).

A strong Payload operator does this:

1. **Reads the histogram.** "FD, Payload — image quality passes, but cloud cover in the histogram looks heavy. Recommend I pull up the band-1 image and review."
2. **Identifies the actual content.** "FD, Payload — confirmed, approximately 60 percent cloud cover. We have a technically- valid First Light image but no usable ocean-current signal."
3. **Coordinates honesty in the bulletin.** "Mission Manager, Payload — recommend the bulletin says 'First Light acquired, image technically nominal, science extraction deferred due to unexpectedly heavy cloud cover; re-acquisition planned for next clear pass.'"
4. **Plans the re-acquisition.** Goes to FDS for the next pass's cloud forecast over the target area.

This operator turns a "technical success, customer disappointment" moment into a "honest report, planned recovery, customer respects the team" moment.

16.8 FDS case study — the manoeuvre that wasn't needed

Day 4, morning. The orbit propagator predicts the spacecraft will fly the imaging target at 14.6 degrees off-nadir at TCA — inside the planner's 14.93 degree roll limit, but only just.

A weak FDS engineer plans a small manoeuvre to centre the target. Burns propellant unnecessarily out of a total budget of 66.9 m/s, and moves every pass and eclipse in the plan while doing it.

A strong FDS engineer does this:

1. **Verifies the propagator.** "FD, FDS — propagator says we're 14.6 degrees off-nadir at TCA, right on the roll limit. Let me cross-check against the last GPS fix."
2. **Re-runs OD.** Pulls the most recent GPS state vector, re-propagates. Result: 4.2 degrees off-nadir.
3. **Reports.** "FD, FDS — propagator was using stale OD data. Re-OD shows we'll be 4.2 degrees off-nadir — comfortably inside the 14.93 degree roll limit. No manoeuvre needed. Worth saying out loud: a burn here would have cost us propellant *and* re-fitted the element set, so every pass and eclipse in tomorrow's plan would have shifted. One metre per second moves the semi-major axis by 1.8 km and pulls the timeline 33 seconds at plus twenty-four hours."

Saves a manoeuvre, saves propellant, saves the plan, and avoids a slew that would have perturbed AOCS for the imaging window.

16.9 Ground Segment case study — the failover that worked

Day 3, LEOP. Mid-pass over Iqaluit, the station's measured G/T drops by 3 dB. Margin is still enormous — this link closes with tens of dB everywhere — so nothing on the spacecraft complains. A weak Ground Segment engineer waits, hopes it recovers.

A strong Ground Segment engineer does this:

1. **Identifies.** "FD, Ground Seg — Iqaluit G/T dropped 3 dB. Margin is fine, but the *trend* isn't. Investigating."
2. **Hypothesises.** "FD, Ground Seg — most likely an RF chain issue at the station. Checking baseband logs… yes, LNA temperature climbed 8 degrees above nominal in the last minute. LNA is degrading."
3. **Recommends.** "FD, Ground Seg — recommend we finish this pass on the LOW tier, 16 kbps, instead of HIGH at 128 — that's `TTC_SET_DATA_RATE` with rate 0 — and drop out of DOWNLINK mode, then plan all critical activities for the next Troll pass while this LNA gets serviced."
4. **Coordinates handover.** With TT&C: rate change. With FDS: re-confirms Troll pass timing.

Mission continues with no loss.

16.10 Mission Manager case study — the bulletin that mattered

Day 4 ends with a deferred First Light. Customer is watching.

A weak Mission Manager writes: "First Light deferred due to technical anomaly. Recovery planned for next shift."

A strong Mission Manager writes:

> "EOSAT-1 commissioning Day 4 update.
>
> Today the team brought EOSAT-1 from LEOP-complete to the verge of First Light. Across the shift's ground passes — the five-station network gives us one roughly every 33 minutes — the team executed 11 of 12 planned commissioning procedures. The twelfth procedure (First Light, image acquisition) was formally deferred due to an in-shift star-tracker degradation that broke the pointing budget required for science-grade imaging.
>
> The team's decision to defer rather than image with a degraded pointing budget reflects exactly the operations-discipline we aim for: no image is worth getting if it's not science-grade.
>
> Recovery plan: tomorrow morning's first pass will exercise the star tracker's reset procedure. If reset succeeds, First Light will be re-attempted in the second pass. If reset fails, the spacecraft carries a second star tracker and the team will switch to it; gyro-propagated attitude estimation, which our analyses show supports usable (though not optimal) imagery, remains the fallback. We will keep you informed.
>
> Best regards, [Mission Manager]"

The bulletin tells the truth, names the discipline involved, and gives the customer real information about the next 24 hours. The customer learns to trust this team.

SpaceCDF — A Course Textbook

What this volume covers

The SpaceCDF Course Textbook is the companion volume to the University of Ottawa's Concurrent Design Facility programme. It is organised in two complementary parts.

Part 1 — CubeSat Design walks the lifecycle of a small spacecraft from mission need through to integration test. Across twenty chapters it covers the systems-engineering framework (the V-model, the 17 NASA SEH technical processes, the ECSS lifecycle), the Canadian space ecosystem and regulatory environment, international standards for space systems, requirements engineering and trade studies, orbit and mission architecture, the eight major subsystems (EPS, AOCS, TCS, OBDH, TT&C, payload, propulsion, structures), and the V&V / risk / cost framework that closes a preliminary design review.

Part 2 — CubeSat Operations turns the design over to a control room. It documents the **nine operator positions**, the **mission control consoles** they sit at, the **voice-loop discipline** that binds the team, the **nominal procedures** flown pass-by-pass, the **LEOP and commissioning sequences**, the **operator calculations** every shift requires, and the **anomaly response cue cards** that turn a fault into a recoverable contingency. Procedures are presented in aircraft-QRH challenge-response format with operator-write-in space for actual telemetry values.

The volume is published with two companion books:

- **The Course Workbook** — exercises, worksheets, and operational drills that the student completes alongside each chapter.
- **The Instructor Book** — simulator setup, anomaly scenarios, breakpoints, and instructor protocols (not for trainees on console).

Three appendices close the volume: an **engineering-position reference** for the design phase, **position quick-reference cards** for the operations phase, and a **verification & first- principles reference** that re-derives every formula in the body of the textbook from first principles, with cross-method sanity checks suitable for the worked verification trail at PDR.

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Claude, an AI authoring assistant from Anthropic, is credited as co-author for content drafting, technical consolidation, figure generation, and structural editing across the volume. All factual content and technical claims were reviewed and approved by Dr Newland. The textbook draws on decades of published systems-engineering, mission-operations, and astrodynamics literature, refined across multiple cohorts of the SpaceCDF and OrbitNorth programmes.

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